AUTOMATIC CONTROL SYSTEMS

(WITH MATLAB PROGRAMS)

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KATSON BOOKS

To
My Parents

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Properties of Laplace Transforms

1.	$\mathcal{L}\pm\left[\frac{d}{dt} f(t)\right] = sF(s) - f(0\pm)$
2.	$\mathcal{L}\pm\left[\frac{d^n}{dt^n} f(t)\right] = s^n F(s) - \sum_{K=1}^{n} s^{n-K} f^{(K-1)}(0\pm)$ where $f^{(K-1)}(t) = \frac{d^{K-1}}{dt^{K-1}} f(t)$
3.	$\mathcal{L}\pm\left[\int f(t)\,dt\right] = \frac{F(s)}{s} + \frac{1}{s}\left[\int f(t)\,dt\right]_{t=0\pm}$
4.	$\mathcal{L}\left[\int_0^t f(t)\,dt\right] = \frac{F(s)}{s}$
5.	$\mathcal{L}\left[e^{-at} f(t)\right] = F(s+a)$
6.	$\mathcal{L}[t\, f(t)] = \frac{-dF(s)}{ds}$
7.	$\mathcal{L}\left[t^n f(t)\right] = (-1)^n \frac{d^n}{ds^n} F(s)$, $n = 1, 2, 3, 4 \ldots$
8.	$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \int_s^{\infty} F(s)\,ds$ if $\lim_{t\to 0} \frac{1}{t} f(t)$ exists
9.	$\mathcal{L}\left[f\left(\frac{t}{a}\right)\right] = aF(as)$
10.	$\mathcal{L}[f_1(t) * f_2(t)] = F_1(s) \cdot F_2(s)$ (complex multiplication)
11.	Intial value Theorem $\lim_{t\to 0} f(t) = \lim_{s\to\infty} sF(s)$
12.	Final value theorem $\lim_{t\to\infty} sF(s) = \lim_{s\to 0} sF(s)$

Laplace Transform Pairs

S. No.	$f(t)$	$F(s)$
1.	Unit Impulse	1
2.	Unit Step	$1/s$
3.	t	$1/s^2$
4.	$\frac{t^{n-1}}{\lfloor n-1}$, $n = 1, 2, 3, ...$	$1/s^n$
5.	e^{-at}	$\frac{1}{s+a}$
6.	$t\,e^{-at}$	$\frac{1}{(s+a)^2}$
7.	$\frac{t^{n-1}e^{-at}}{\lfloor n-1}$, $n = 1, 2, 3, ...$	$\frac{1}{(s+a)^n}$
8.	$\frac{e^{-at}-e^{-bt}}{b-a}$	$\frac{1}{(s+a)(s+b)}$
9.	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
10.	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
11.	$e^{-at}\sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
12.	$e^{-at}\cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
13.	$\frac{\omega_n}{\sqrt{1-\xi^2}}e^{-\xi\omega_n t}\sin \omega_n\sqrt{1-\xi^2}\,t$	$\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$
14.	$1-\frac{1}{\sqrt{1-\xi^2}}e^{-\xi\omega_n t}\sin\left(\omega_n\sqrt{1-\xi^2}\,t+\tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}\right)$	$\frac{\omega_n^2}{s(s^2+2\xi\omega_n s+\omega_n^2)}$
15.	$\frac{1}{\sqrt{1-\xi^2}}e^{-\xi\omega_n t}\sin\left(\omega_n\sqrt{1-\xi^2}\,t-\tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}\right)$	$\frac{s}{s^2+2\xi\omega_n s+\omega_n^2}$
16.	$\frac{1}{ab}\left[1+\frac{1}{a-b}\left(be^{-at}-ae^{-bt}\right)\right]$	$\frac{1}{s(s+a)(s+b)}$

Chapter 1

Input-Output Relationship

1.1. INTRODUCTION

The control system is very important for all engineers. The first significant control device was James Watt's flyball governer. This was invented in 1767 to keep the speed of the engine constant by regulating the supply of the steam to the engine.

In control system the behaviour of the system is described by the differential equations. Minorsky, in 1922 showed that how to determine the stability from the differential equations describing the systems. The differential equations may be ordinary differential equations or the difference equations. The control system can be classified as open loop control system and closed loop control systems.

1.2. OPEN LOOP CONTROL SYSTEM

The open loop control system is also known as control system without feedback or non feedback control systems. In open loop systems the control action is independent of the desired output. In this system the output is not compared with the reference input.

The component of the open loop systems are controller and controlled process. The controller may be amplifier, filter etc depends upon the system. An input is applied to the controller and the output of the controller gives to the controlled process & we get the output (desired).

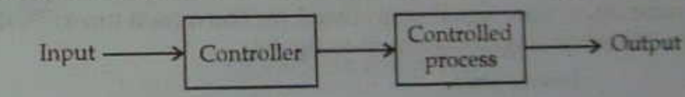


Fig. 1.1.

Examples :

1. Automatic washing machine is the example of the open loop systems. In the machine the operating time is set manually. After the completion of set time the machine will stops, with the result we may or may not get the desired (output) amount of cleanliness of washed cloths because there is no feedback is provided to the machine for desired output.
2. Immersion rod is another example of open loop system. The rod heats the water but how much heating is required is not sense by the rod because of no feedback to the rod.
3. A field control d.c. motor is the example of open loop system.

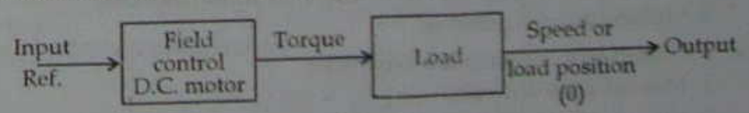


Fig. 1.2.

4. For automatic control of traffic the lamps of three different colours (red, yellow and green) are used. The time for each lamp is fixed. The operation of each lamp does not depends upon the density of the traffic but depends upon the fixed time. Thus, we can say that the control system which operates on the time basis is open loop system.

Advantages : 1. open loop control systems are simple.
2. open loop control systems are economical.
3. less maintenance is required and not difficult.
4. proper calibration is not a problem.

Disadvantages : 1. Open loop systems are inaccurate.
2. These are not reliable.
3. These are slow.
4. Optimization is not possible.

1.3. CLOSED LOOP CONTROL SYSTEM

Closed loop control systems are also known as feedback control systems. In closed loop control systems the control action is dependent on the desired output. If any system having one or more feedback paths forming a closed loop system.

In closed loop systems the output is compared with the reference input and error signal is produced. The error signal is fed to the controller to reduce the error and desired output is obtained.

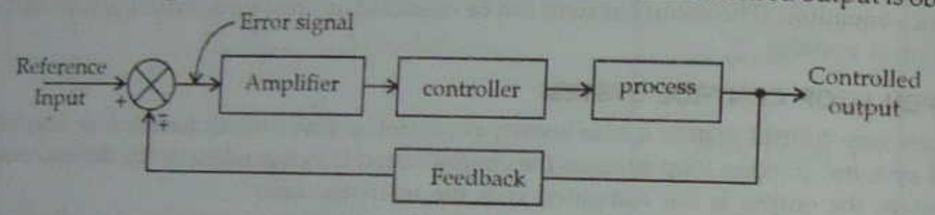


Fig. 1.3.

Example : In a room we need to regulate the temperature & humidity for comfortable living. Air conditioners are provided with thermostat. By measuring the actual room temperature & compared it with desired temperature, an error signal is produced, the thermostat turns ON the compressor or OFF the compressor. The block diagram is shown in fig. 1.4.

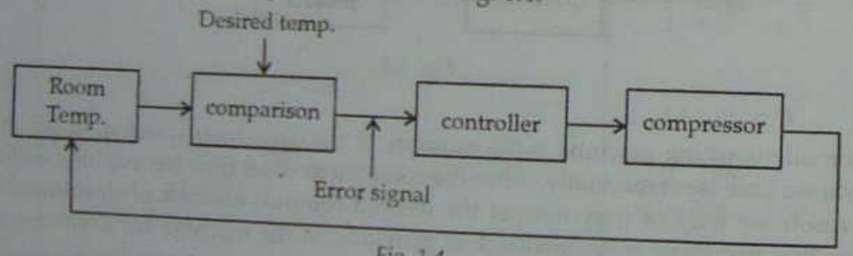


Fig. 1.4.

Advantages : 1. These systems are more reliable.
2. Closed loop systems are faster.
3. A number of variables can be handled simultaneously.
4. Optimization is possible.

Disadvantages : 1. Closed loop systems are expensive
2. Maintenance difficult.
3. Complicated installation.

1.4. COMPARISON BETWEEN OPEN LOOP & CLOSED LOOP

Table 1.1

S.No	Open Loop system	Closed Loop System
1.	These are not reliable	These are reliable
2.	It is easier to build	It is difficult to build.
3.	If calibration is good, they perform accurately.	They are accurate because of feedback
4.	Open loop systems are generally more stable.	These are less stable.
5.	Optimization is not possible.	Optimization is possible.

1.5. ELEMENTS OR COMPONENTS OF CLOSED LOOP SYSTEMS

The various components of closed loop system are shown in fig. 1.5

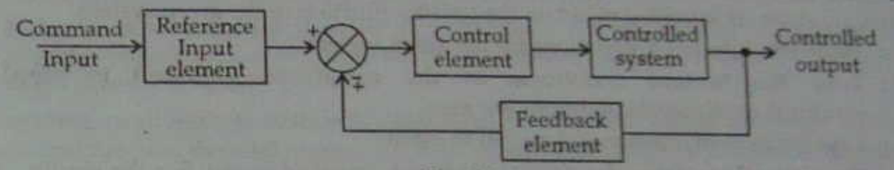


Fig. 1.5.

Command : The command is the externally produced input and independent of the feedback control system.

Reference input element : This produces the standard signals proportional to the command.

Error Detector : The error detector receives the measured signal and compare it with reference input. The difference of two signals produces the error signal.

Control element : This regulates the output according to the signal obtained from error detector.

Controlled system : This represents what we are controlling by the feedback loop.

Feedback element : This element fed back the output to the error detector for comparison with the reference input.

TRANSFER FUNCTION

1.6. TRANSFER FUNCTION FOR SINGLE INPUT SINGLE OUTPUT SYSTEM

The transfer function is defined as the ratio of Laplace transform of the output to the Laplace transform of input with all initial conditions are zero.

Consider a linear system having input $r(t)$ & $c(t)$ is the output of the system, the input-output relation can be described by the following n^{th}-order differential equation :

$$a_n\frac{d^nC(t)}{dt^n}+a_{n-1}\frac{d^{n-1}c(t)}{dt^{n-1}}+.....+a_1\frac{dc(t)}{dt}+a_0c(t)=b_m\frac{d^mr(t)}{dt^m}+b_{m-1}\frac{d^{m-1}r(t)}{dt^{m-1}}+....+b_1\frac{dr(t)}{dt}+b_0r(t) \quad ...(1.1)$$

Where 'a' & 'b' are constants.

Take the laplace transform of equation (1.1)

$$(a_ns^n+a_{n-1}s^{n-1}+.....+a_1s+a_0)\,C(s)=(b_ms^m+b_{m-1}s^{m-1}+.....+b_1s+b_0)R(s) \quad ...(1.2)$$

We can define the transfer function as

$$G(s)=\frac{C(s)}{R(s)}$$

$$G(s)=\frac{C(s)}{R(s)}=\frac{(b_ms^m+b_{m-1}s^{m-1}+.....+b_1s+b_0)}{(a_ns^n+a_{n-1}s^{n-1}+.....+a_1s+a_0)} \quad ...(1.3)$$

In equation (1.3), if the order of the denominator polynomial is greater than the order of the numerator polynomial then the transfer function is said to be STRICTLY PROPER. If the order of both polynomials are same, then the transfer function is PROPER. The transfer function is said to be IMPROPER, if the order of numerator polynomial is greater than the order of denominator polynomial.

Consider the block diagram of open loop control system fig. (1.6) where $R(s)$ & $C(s)$ are the laplace transform of input and output respectively, then the transfer function $G(s)$ can be expressed as

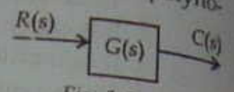


Fig. 1.6

$$G(s) = \frac{C(s)}{R(s)} = \frac{\pounds c(t)}{\pounds r(t)} \qquad ...(1.4)$$

1.7. PROCEDURE

The following steps are involved to obtain the transfer function of the given system.

Step 1 : Write the differential equations for the given system.

Step 2 : Take the laplace transform of the equations obtained in step1, with assumption; all initial conditions are zero.

Step 3 : Take the ratio of transformed output to input

Step 4 : The ratio of transformed output to the input, obtained in step 3 is the required transfer function of the given system.

1.8. CHARACTERISTIC EQUATION OF A TRANSFER FUNCTION

The characteristic equation of a linear system can be obtained by equating the denominator polynomial of the transfer function to zero. Thus, the characteristic equation of the transfer function of equation 1.3 will be

$$a_n s^n + a_{n-1} s^{n-1} + ... + a_1 s + a_0 = 0 \qquad ...(1.5)$$

1.9. POLES AND ZEROS OF A TRANSFER FUNCTION

Consider the equation 1.3, the numerator & denominator can be factored in m and n terms respectively, then the equation 1.3 can be expressed as

$$\frac{C(s)}{R(s)} = G(s) = \frac{K(s+Z_1)(s+Z_2)\left(as^2+bs+c\right)}{(s+p_1)(s+p_2)\left(As^2+Bs+C\right)} \qquad ...(1.6)$$

Where $K = \frac{b_m}{a_n}$ is known as the gain factor, s is the complex frequency

POLES : The poles of $G(s)$ are those values of 's' which make $G(s)$ tend to infinity. For example in equation 1.6 we have poles at $s = -p_1$, $s = -p_2$ and a pair of poles at

$$s = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \qquad ...(1.7)$$

ZEROS : The zeros of $G(s)$ are those values of 's' which make $G(s)$ tend to zero. For example in eq. 1.6 we have zeros at $s_1 = -Z_1$, $s_2 = -Z_2$ and a pair of zeros at

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad ...(1.8)$$

If either poles or zeros coincide, then such type of poles or zeros are called multiple poles or multiple zeros, otherwise they are known as simple poles or simple zeros. Multiple poles are due to the repetitive factor in denominator and multiple zeros are due to the repetitive factor in numerator of a transfer function.

For example, consider the transfer function

$$G(s) = \frac{50(s+3)}{s(s+2)(s+4)^2} \qquad ...(1.9)$$

The above transfer function having the simple poles at $s = 0$, $s = -2$, multiple poles at $s = -4$ i.e. the pole of order 2 at $s = -4$ and simple zero at $s = -3$.

The above mentioned poles and zeros are of finite values. If we consider the entire 's' plane including infinity then two cases arises.

1. If the no. of zeros are less than the no of poles i.e. $Z < P$ then the value of the transfer function becomes zero for $s \to \infty$. Hence we can say that there are zeros at infinity and the order of such zeros is $P - Z$. For example in equation 1.9 there are four finite poles at $s = 0, -2, -4$ and -4; there is one finite zero at $s = -3$ but there are three zeros at infinity $P-Z = 4-1 = 3$. Therefore, the function has a total of four poles and four zeros in the entire s-plane including infinity.
2. If the no. of poles are less than the no. of zeros $P < Z$ then the value of the transfer function becomes infinity for $s \to \infty$. Hence we can say there are poles at infinity ($s \to \infty$) and the order of the poles will be Z-P.

Therefore, in addition to finite poles and zeros, if we consider poles and zeros at infinity, then for a rational function the total number of zeros will be equal to the total number of poles.

The pole is represented by 'X' and zero by 'O'. These symbols are used to locate the poles and zeros on s-plane. The pole-zero plot of eq^n 1.9 is shown in fig. 1.7.

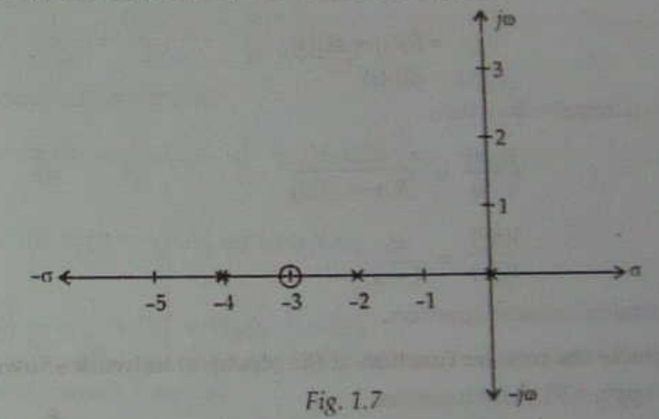


Fig. 1.7

1.10. IMPULSE RESPONSE

From equation 1.4

$$C(s) = R(s)\, G(s) \qquad ...(1.10)$$

Suppose, a system is subjected to a unit impulse, then the output will be

$$C(s) = G(s) \qquad ...(1.11)$$

because the Laplace transform of unit impulse function is unity.

The inverse laplace of eq 1.11 will be

$$C(t) = g(t) \qquad ...(1.12)$$

where $g(t)$ is the unit impulse response of a system.

Therefore inverse laplace of $G(s)$ is called impulse response or the transfer function of a system is the laplace transform of its impulse response.

Practically, it is not possible to generate a true impulse. A pulse with less duration than the time constant of the system can be considered as an impulse & denoted by $\delta(t)$.

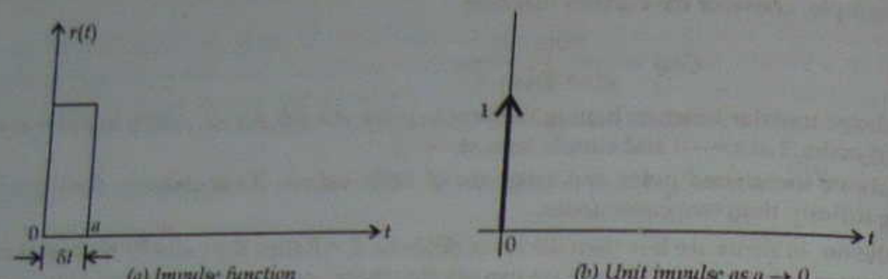


Fig. 1.8.

Example 1.1. Find the transfer function of the given network

Solution : Step 1 : Apply *KVL* in mesh (1)

$$V_i = Ri + L\frac{di}{dt} \qquad ...(1.13)$$

Apply *KVL* in mesh (2)

$$V_0 = L\frac{di}{dt} \qquad ...(1.14)$$

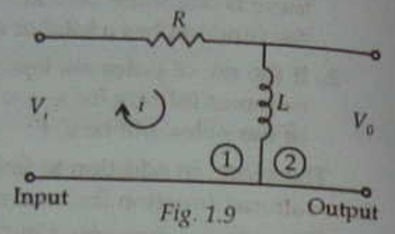


Fig. 1.9

Step 2 : Take laplace transform of equation 1.13 & 1.14 with assumption that all initial conditions are zero.

$$V_i(s) = RI(s) + sLI(s) \qquad ...(1.15)$$

$$V_0(s) = sLI(s) \qquad ...(1.16)$$

Step 3 : Calculation of transfer function

$$\frac{V_0(s)}{V_i(s)} = \frac{sLI(s)}{(R+sL)I(s)}$$

$$\frac{V_0(s)}{V_i(s)} = \frac{sL}{R+sL} \qquad ...(1.17)$$

Eq. 1.17 is the required transfer function.

Example 1.2. Determine the transfer function of the electrical network shown in fig. 1.10.

Solution : Step 1 : Apply *KVL* in both meshes

$$E_i = Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt \qquad ...(1.18)$$

$$E_0 = \frac{1}{C}\int i\,dt \qquad ...(1.19)$$

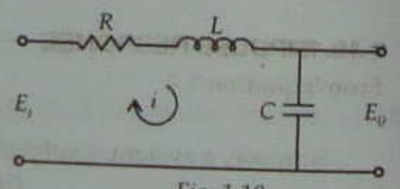


Fig. 1.10.

Step 2 : Take laplace transform of eqn 1.18 & 1.19

$$E_i(s) = RI(s) + sLI(s) + \frac{1}{Cs}I(s)$$

$$= I(s)\left[R + sL + \frac{1}{Cs}\right]$$

$$E_i(s) = I(s)\left[\frac{RCs + s^2LC + 1}{Cs}\right] \qquad ...(1.20)$$

$$E_0(s) = \frac{1}{Cs}I(s) \qquad ...(1.21)$$

Step 3 : Determination of transfer function

$$\frac{E_0(s)}{E_i(s)} = \frac{I(s)}{Cs}\cdot\frac{Cs}{I(s)\left[s^2Lc + SRC + 1\right]}$$

$$\frac{E_0(s)}{E_i(s)} = \frac{1}{S^2LC + SRC + 1} \qquad \textbf{Ans.} \qquad ...(1.22)$$

Example 1.3. Obtain the transfer function $\frac{V_2(s)}{V_1(s)}$ for fig. 1.11.

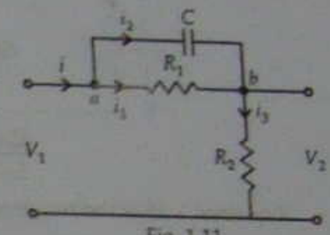


Fig. 1.11.

Solution : Step 1 : *KCL* at node '*a*'

$$i = i_1 + i_2 \qquad ...(1.23)$$

$$i_1 = \frac{V_1 - V_2}{R_1}$$

$$i_2 = C\frac{d}{dt}(V_1 - V_2)$$

$$i = i_3 = \frac{V_2}{R_2}$$

Put all these values in eq^n 1.23

$$\frac{V_2}{R_2} = \frac{V_1 - V_2}{R_1} + C\frac{d}{dt}(V_1 - V_2) \qquad ...(1.24)$$

Step 2 : Take laplace transform of *eq^n* 1.24

$$\frac{V_2(s)}{R_2} = \frac{1}{R_1}V_1(s) - \frac{1}{R_1}V_2(s) + Cs\,V_1(s) - Cs\,V_2(s)$$

$$\frac{V_2(s)}{R_2} + \frac{1}{R_1}V_2(S) + Cs\,V_2(s) = \frac{1}{R_1}V_1(s) + Cs\,V_1(s)$$

$$V_2(s)\left[\frac{1}{R_1} + \frac{1}{R_2} + Cs\right] = V_1(s)\left[\frac{1}{R_1} + Cs\right]$$

Step 3 : Determination of transfer function

$$V_2(s)\left[\frac{R_1 + R_2 + R_1R_2Cs}{R_1R_2}\right] = V_1(s)\left[\frac{1 + R_1Cs}{R_1}\right]$$

$$\frac{V_2(s)}{V_1(s)} = \frac{R_2 + R_1R_2Cs}{R_1 + R_2 + R_1R_2Cs} \qquad \textbf{Ans.} \qquad ...(1.25)$$

Example 1.4. Find the transfer function of lag network shown in fig. 1.12.

Solution : Step 1 : Apply *KVL* in both meshes

$$e_i(t) = R_1i(t) + R_2i(t) + \frac{1}{C}\int i(t)\,dt \qquad ...(1.26)$$

$$e_0(t) = R_2i(t) + \frac{1}{C}\int i(t)\,dt \qquad ...(1.27)$$

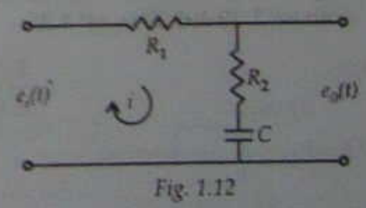


Fig. 1.12

Step 2 : Laplace transform of *eq*^{n} (1.26) & (1.27)

$$E_i(s) = \left[R_1 + R_2 + \frac{1}{Cs}\right] I(s)$$

$$E_0(s) = \left[R_2 + \frac{1}{Cs}\right] I(s)$$

Step 3 : Calculation of transfer function

$$E_0(s) = \frac{\left[R_2 + \frac{1}{Cs}\right] I(s)}{\left[\frac{R_1Cs + R_2Cs + 1}{Cs}\right] I(s)}$$

$$\frac{E_0(s)}{E_i(s)} = \frac{1 + R_2Cs}{1 + R_1Cs + R_2Cs} \qquad ...(1.28)$$

Equation 1.28 is the required transfer function.

Example 1.5. Determine the transfer function of fig 1.13.

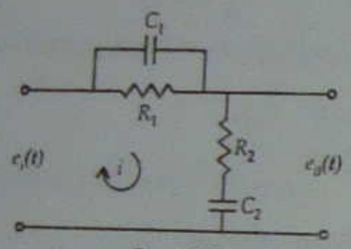


Fig. 1.13.

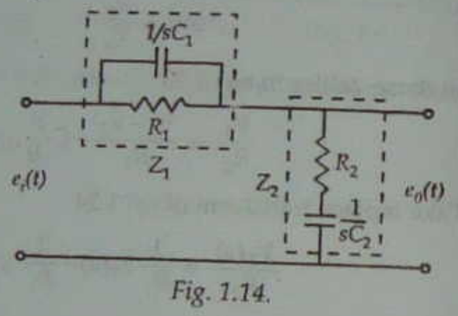


Fig. 1.14.

Solution : Step 1 : calculation of Z_1 :

$$Z_1 = \frac{R_1 \frac{1}{sC_1}}{R_1 + \frac{1}{sC_1}} = \frac{R_1}{R_1C_1s + 1} \qquad ...(1.29)$$

Step 2 : Calculation of Z_2 :

$$Z_2 = R_2 + \frac{1}{sC_2} = \frac{R_2C_2S + 1}{sC_2} \qquad ...(1.30)$$

Step 3 : Calculation of transfer function in terms of Z_1 & Z_2

$$\frac{E_0(s)}{E_i(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)} \qquad ...(1.31)$$

Step 4 : Calculation of transfer function in terms of R_1, R_2, C_1 & C_2 Put the values of Z_1 (s) & $Z_2(s)$ from *eq*^{n} 1.29 & 1.30 in *eq*^{n} 1.31

$$\frac{E_0(s)}{E_i(s)} = \frac{(1 + R_2C_2S)/Sc_2}{\frac{R_1}{C_1R_1S + 1} + \frac{R_2C_2S + 1}{SC_2}}$$

$$\frac{E_0(s)}{E_i(s)} = \frac{(1 + R_1C_1S)(1 + R_2C_2S)}{(1 + R_1C_1S)(1 + R_2C_2S) + R_1C_2S} \qquad ...(1.32)$$

The above *eq*^{n} is the required transfer function of the given circuit.

Example 1.6. Determine the transfer function of given transformer coupled circuit (fig. 2.9).

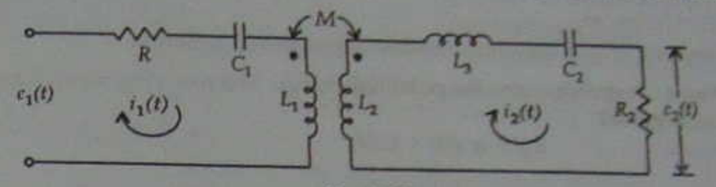


Fig. 1.15.

Solution : Apply *KVL* in both meshes

$$e_i(t) = Ri_1(t) + \frac{1}{C_1}\int i_1\,dt + L_1\frac{di_1(t)}{dt} - M\frac{di_2(t)}{dt} \qquad ...(1.33)$$

$$e_2(t) = R_2i_2(t) \qquad ...(1.34)$$

$$0 = R_2i_2(t) + (L_3 + L_2)\frac{d}{dt}i_2(t) + \frac{1}{C_2}\int i_2(t)dt - M\frac{d}{dt}i_1(t) \qquad ...(1.35)$$

Take Laplace transform of *eq*^{n} 1.33, 1.34 & 1.35

$$E_1(s) = I_1(s)\left[R + \frac{1}{C_1S} + SL_1\right] - SMI_2(s) \qquad ...(1.36)$$

$$E_2(s) = R_2I_2(s) \qquad ...(1.37)$$

$$0 = I_2(s)[R_2 + s(L_2 + L_3) + 1/sC_2] - sMI_1(s) \qquad ...(1.38)$$

Solving the equations 1.36, 1.37 & 1.38. the required transfer function.

$$G(s) = \frac{S^3R_2C_1C_2M}{\left[SR_2C_2 + S^2C_2(L_2 + L_3) + 1\right]\left[S^2L_1C_1 + SC_1R_1 + 1\right] - M^2S^4C_1C_2} \quad \textbf{Ans.}$$

MECHANICAL SYSTEM

1.11. TRANSLATIONAL SYSTEMS

The motion takes place along a straight line is known as translational motion. There are three types of forces that resists motion.

1. Intertia Force : Consider a body of mass 'M' & acceleration 'a', then according to Newton's second law of motion the inertia force will be equal to the product of mass 'M' & acceleration 'a'.

$$F_M(t) = Ma(t) \qquad ...(1.39)$$

In terms of velocity the *eq*^{n} (1.39) becomes

$$F_M(t) = M\frac{dv(t)}{dt} \qquad ...(1.40)$$

In terms of displacement the *eq*^{n} (1.39) can be expressed as

$$F_M(t) = M\frac{d^2x(t)}{dt^2} \qquad ...(1.41)$$

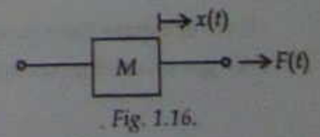


Fig. 1.16.

2. Damping Force : For viscous friction we assume that the damping force is proportional to the velocity.

$$F_D(t) = Bv(t) = B\frac{d}{dt}x(t) \qquad ...(1.42)$$

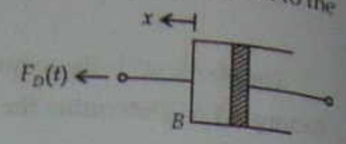


Fig. 1.17.

Where, B = Damping coefficient

unit of B = N/m/sec.

We can represent 'B' by a dashpot, consists of piston and cylinder.

3. Spring Force : A spring stores the potential energy. The restoring force of a spring is proportional to the displacement.

$$F_K(t) \propto x(t) = Kx(t)$$

$$F_K(t) = K\int v(t)dt \qquad ...(1.43)$$

Where 'K' = spring constant or stiffness

unit of K = N/m

The stiffness of a spring can be defined as restoring force per unit displacement.

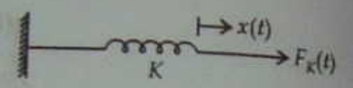


Fig. 1.18.

1.12. ROTATIONAL SYSTEM

The rotational motion of a body can be defined as the motion of a body about a fixed axis. There are three types of torques resists the rotational motion.

1. Inertia Torque : Inertia (J) is the property of an element that stores the kinetic energy of rotational motion. The inertia Torque T_I is the product of moment of Inertia J and angular acceleration $\alpha(t)$

$$T_I(t) = J\alpha(t)$$

$$T_I(t) = J\frac{d}{dt}\omega(t)$$

$$T_I(t) = J\frac{d^2}{dt^2}\theta(t) \qquad ...(1.44)$$

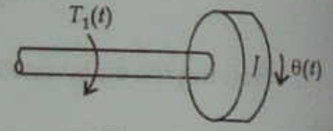


Fig. 1.19.

Where $\omega(t)$ = angular velocity

$\theta(t)$ = angular displacement

unit of Torque = N-m

2. Damping Torque : The damping Torque $T_D(t)$ is the product of damping coefficient B and angular velocity ω.

$$T_D(t) = Bw(t)$$

$$T_D(t) = B\frac{d}{dt}\theta(t) \qquad ...(1.45)$$

3. Spring Torque : Spring torque $T_\theta(t)$ is the product of torsional stiffness and angular displacement.

$$T_\theta(t) = K\theta(t)$$

unit of K = N-m/rad. ...(1.46)

If we compare the equations

We get an analogous system. The analogous quantities are tabulated as follows :

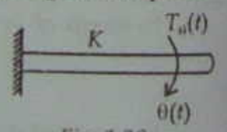


Fig. 1.20.

Table 1.2.

S.No.	Translational	Rotational
1.	Force, F	Torque, T
2.	Acceleration, a	angular acceleration, α
3.	Velocity, v	angular velocity, ω
4.	Displacement, x	angular displacement, θ
5.	Mass, M	Moment of inertia, J
6.	Damping coefficient B	Rotational damping coeff., B
7.	Stiffness	torsional stiffness

1.13. D'ALEMBERT'S PRINCIPLE

This principle states that "for any body, the algebraic sum of externally applied forces and the forces resisting motion in any given direction is zero".

D' Alembert principle is useful in writing the equation of motion of mechanical system. Consider, a system shown in fig. 1.21. consisting of a mass M, spring & dashpot.

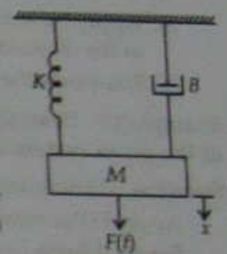


Fig. 1.21.

First choose a reference direction. All the forces in the direction of reference direction considered as positive & the forces opposite to the reference direction taken as negative.

External Force : $F(t)$

Resisting Forces : *a.* Inertia Force $F_m(t) = -M\frac{d^2}{dt^2}x(t)$

b. Damping Force $F_D(t) = -B\frac{d}{dt}x(t)$

c. spring Force $F_K(t) = -Kx(t)$

According to D' Alembert's principle

$$F(t) + F_m(t) + F_D(t) + F_K(t) = 0$$

$$F(t) - M\frac{d^2}{dt^2}x(t) - B\frac{d}{dt}x(t) - Kx(t) = 0$$

or, $$F(t) = M\frac{d^2}{dt^2}x(t) + B\frac{d}{dt}x(t) + Kx(t) \qquad ...(1.47)$$

Consider a rotational system shown in fig. 1.22.

External Torque : $T(t)$

Resisting Torque : *a.* Inertia Torque $T_I(t) = -J\frac{d\omega(t)}{dt}$

b. Damping Torque $T_D(t) = -B\frac{d}{dt}\theta(t)$

c. spring Torque $T_K(t) = -K\theta$

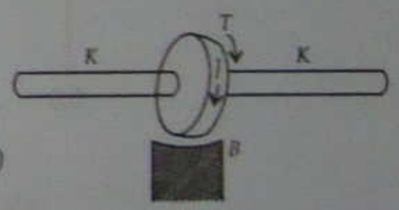


Fig. 1.22.

According to D'Alembert principle

$$T(t) + T_I + T_D + T_K = 0$$

$$T(t) - J\frac{d}{dt}\omega(t) - B\frac{d}{dt}\theta(t) - K\theta(t) = 0$$

or, $$T(t) = J\frac{d}{dt}\omega(t) + B\frac{d}{dt}\theta(t) + K\theta(t) \quad \text{...(1.48)}$$

So, D'Alembert principle for rotational motion is

"For any body, the algebaric sum of externally applied torques and the torques resisting rotation about any axis is zero".

1.14. PROCEDURE OF WRITING THE MODELS OF MECHANICAL SYSTEM

1. Assume system is in equilibrium.
2. Assume that the system is given same arbitrary displacement if no. of distributing forces are present.
3. Draw the free body diagram of forces exerted on each mass in the system.
4. Apply Newton's law of motion to each diagram, using the convention that any force acting in the direction of assume displacement is positive.
5. Rearrange the equations in suitable form to be solved by any means.

Example 1.7. Draw the free body diagram and write the diffrential equation of the given system shown in fig. 1.23.

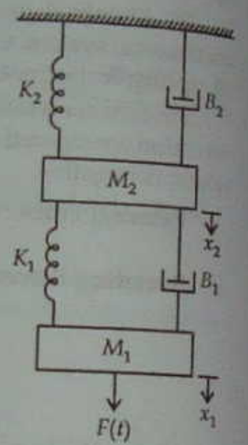


Fig. 1.23

Solution : Differential equation For M_1 :

Apply D'Alembert principle

External force : $F(t)$

Resisting forces :

1. Inertia force $F_M = -M_1\frac{d^2}{dt^2}x_1$
2. Damping force $F_D = -B_1\frac{d}{dt}(x_1 - x_2)$
3. Spring force $F_K = K_1(x_1 - x_2)$

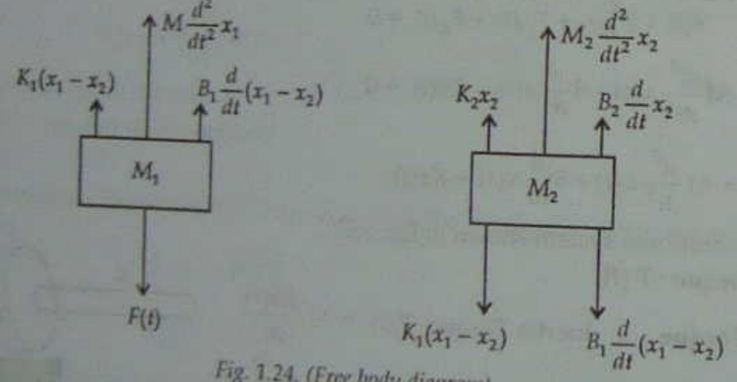


Fig. 1.24. (Free body diagram)

$$\therefore \quad F(t) = M_1\frac{d^2}{dt^2}x_1 + B_1\frac{d}{dt}(x_1 - x_2) + K_1(x_1 - x_2) \quad \text{...(1.49)}$$

Similarly for mass M_2 :

$$K(x_1 - x_2) + B_1\frac{d}{dt}(x_1 - x_2) = M_2\frac{d^2x_2}{dt^2} + B_2\frac{dx_2}{dt} + K_2x_2 \quad \text{...(1.50)}$$

Equation (1.49) & (1.50) are the required equations.

Example 1.8. Write the differential equations describing the dynamics of the system shown in fig. 1.25 and find the ratio $\frac{X_2(s)}{F(s)}$.

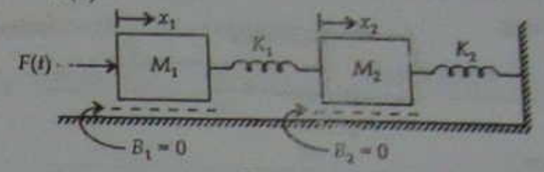


Fig. 1.25

Solution : Free body diag. For mass M_1 :

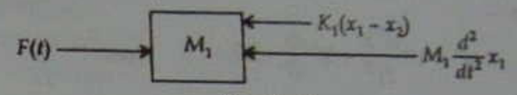


Fig. 1.26(a)

Free body diag. For mass M_2 :

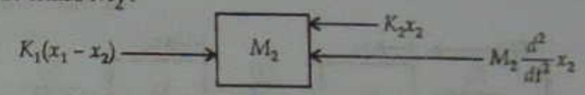


Fig. 1.26(b)

Differential equation for mass M_1

$$F(t) = M_1\frac{d^2}{dt^2}x_1 + K_1(x_1 - x_2) \quad \text{...(1.51)}$$

$$K_1(x_1 - x_2) = K_2x_2 + M_2\frac{d^2}{dt^2}x_2 \quad \text{...(1.52)}$$

Take laplace transform of eq^n (1.51), assume initial conditions zero.

$$F(t) = M_1\frac{d^2}{dt^2}x_1 + K_1x_1 - K_1x_2$$

$$F(s) = M_1S^2X_1(s) + K_1X_1(s) - K_1X_2(s) \quad \text{...(1.53)}$$

laplace transform of equation (1.52)

$$K_1X_1 - K_1X_2 = K_2X_2 + M_2\frac{d^2}{dt^2}X_2$$

$$K_1X_1(s) - K_1X_2(s) = K_2X_2(s) + M_2s^2X_2(s)$$

$$X_1(s) = \frac{X_2(s)}{K_1}[s^2M_2 + K_1 + K_2] \quad \text{...(1.54)}$$

Put the value of $X_1(s)$ in eq^n (1.53)

$$F(s) = \frac{X_2(s)}{K_1}[s^2M_2 + K_1 + K_2][s^2M_1 + K_1] - K_1X_2(s)$$

$$\frac{X_2(s)}{F(s)} = \frac{K_1}{(S^2M_2 + K_1 + K_2)(K_1 + s^2M_1) - K_1^2} \quad \text{Ans.}$$

Example 1.9. A mass spring system under equilibrium condition is shown in fig. 1.27. Drive the system equation where $M = 10$ kg. $B = 30$ N/m/sec. & $K = 20$ N/m.

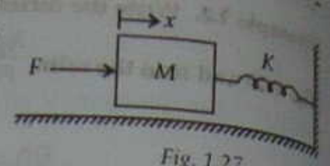


Fig. 1.27

Solution : Free body diag.

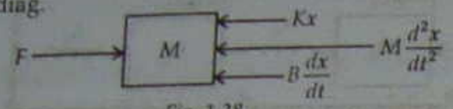


Fig. 1.28

Differential equation

$$F = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx$$

Put the given values

$$F = 10\frac{d^2x}{dt^2} + 30\frac{dx}{dt} + 20x \quad \textbf{Ans.} \qquad \text{....(1.55)}$$

Example 1.10. Drive the system equations & find the value of $X_2(s)/F(s)$ for the system shown in fig. 1.29. *(R.M.L., University, Faizabad, 2003)*

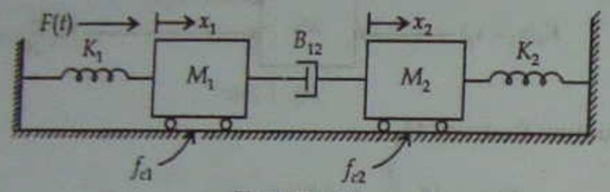
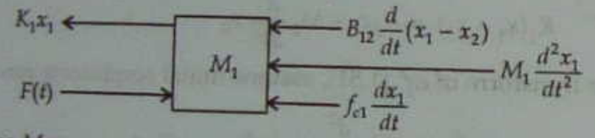
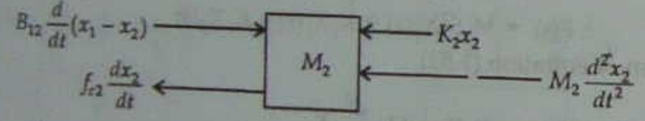


Fig. 1.29.

Solution : Free body diag. for mass M_1:



Free body diag. For mass M_2 :



System equation for mass M_1:

$$F(t) = M_1\frac{d^2x_1}{dt^2} + fc_1\frac{dx_1}{dt} + B_{12}\frac{d}{dt}(x_1 - x_2) + K_1x_1 \qquad \text{.....(1.56)}$$

System equation for mass M_2 :

$$B_{12}\frac{d}{dt}(x_1 - x_2) = M_2\frac{d^2x_2}{dt^2} + fc_2\frac{dx_2}{dt} + K_2x_2 \qquad \text{....(1.57)}$$

Laplace transform of equation 1.56

$$F(s) = X_1(s)[s^2M_1 + sB_{12} + sfc_1 + K_1] - B_{12}\,sX_2(s) \qquad \text{.....(1.58)}$$

Laplace transform of equation 1.57

$$B_{12}sX_1(s) = (s^2M_2 + B_{12}s + sfc_2 + K_2)\,X_2(s)$$

$$\therefore \quad X_1(s) = \frac{X_2(s)(s^2M_2 + B_{12}s + sfc_2 + K_2)}{B_{12}s} \qquad \text{....(1.59)}$$

Put the value of $X_1(s)$ for eqn (1.59) in equation (1.58)

$$F(s) = \frac{X_2(s)\left[S^2M_2 + B_{12}S + sfc_2 + K_2\right]\left[S^2M_1 + SB_{12} + sfc_1 + K_1\right]}{B_{12}S} - B_{12}SX_2(s)$$

$$\therefore \quad \frac{X_2(s)}{F(s)} = \frac{SB_{12}}{(s^2M_1 + sB_{12} + sfc_1 + K_1)(s^2M_2 + sB_{12} + sfc_2 + K_2) - s^2B_{12}^2} \quad \textbf{Ans.}$$

1.15. ANALOGOUS SYSTEM

Consider a series *RLC* circuit

Apply kirchhoff's voltage law

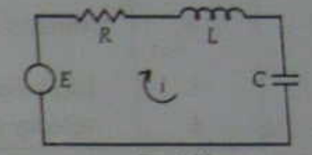


Fig. 1.30

$$E = Ri + L\frac{di}{dt} + \frac{1}{C}\int idt \qquad \text{.....(1.60)}$$

In terms of charge *eq*^{n} 1.60 becomes

$$E = R\frac{dq}{dt} + L\frac{d^2q}{dt^2} + \frac{1}{C}q \qquad \text{....(1.61)}$$

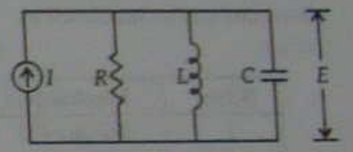


Fig. 1.31.

Now consider a parallal RLC circuit

Now apply Kirchhoff's current law

$$I = \frac{E}{R} + \frac{1}{L}\int Edt + C\frac{dE}{dt} \qquad \text{....(1.62)}$$

In terms of magnetic flux linkage, the *eq*^{n}. 1.62 becomes

Since $\phi = \int Edt$

$$I = \frac{1}{R}\left(\frac{d\phi}{dt}\right) + \frac{1}{L}\phi + C\frac{d^2\phi}{dt^2} \qquad \text{....(1.63)}$$

Now, compare the equation (1.47) with the equation (1.61), both equations are differential equations of same order *i.e.* identical such type of systems whose differential equations are in the same form are called analogous systems.

On comparison the eqn. 1.47 with eqn. 1.61 we can see that in mechanical system the force *F* is analogous to voltage *E* in electrical system, such type of analogy is known as force-voltage analogy.

From equation 1.47 & 1.61, mass (*M*) is analogous to inductance (*L*), coeff. of viscous friction (*B*) is analogous to resistance (*R*), spring stiffness (*K*) is analogous to $\frac{1}{C}$ & so on.

Table 1.3.

S.No.	Mechanical Translational System	Electrical System
1.	Force (F)	Voltage *E*
2.	Mass (*M*)	Inductance (*L*)
3.	Stiffness (*K*) (Elastance, 1/*K*)	Reciprocal of capacitance ($^1/_C$) Capacitance (*C*)
4.	Damping coeff. (*B*)	Resistance *R*
5.	Displacement (*x*)	charge (*q*)

Now compare the equation (1.47) with eqn (1.63), since the force F is analogous to the current source, such type of analogy is known as force current analogy. The analogous quantities are tabulated as.

Table 1.4.

S.No.	Mechanical Translational System	Electrical system
1.	Force (F)	Current (I)
2.	Mass (M)	Capacitance (c)
3.	Damping coeff. (B)	Reciprocal of resistance ($1/R$) *i.e.* conductance G
4.	Stiffness (K) (Elastance, $1/K$)	Reciprocal of Inductance ($1/L$) Inductance (L)
5.	Displacement (x)	Flux linkage (ϕ)
6.	Velocity ($\dot{x}$)	Voltage(E)

Now consider the rotational system & compare the *eq*^{n}. 1.48 with 1.61. On comparison the analogous quantities can be tabulated as

Table 1.5 : Torque-voltage analogy

S.No.	Mechanical Rotational System	Electrical system
1.	Torque (T)	Voltage (E)
2.	Moment of Inertia (J)	Inductance (L)
3.	Damping coeff. (B)	Resistance (R)
4.	Stiffness (K) (Elastance, $1/K$)	Reciprocal of capacitance ($^1/_c$) capacitance (C)
5.	Angular displacement (θ)	Charge (q)
6.	Angular Velocity (w)	current(i)

On comparison the eqn. 1.48 with eqn 1.63 we get Torque current analogy and can be tabulated as

Table 1.6

S.No.	Mechanical Rotational System	Electrical system
1.	Torque (T)	current (I)
2.	Moment of Inertia (J)	capacitance (c)
3.	Damping coeff. (B)	Reciprocal of Resistance ($^1/_R$) *i.e.* conductance (G)
4.	Stiffness (K) (Elastance, $1/_K$)	Reciprocal of Inductance ($^1/_L$) Inductance (L)
5.	Angular displacement (θ)	Flux linkage (ϕ)

Following Cheng*, the rule for drawing the analogous circuit can be stated as

(*a*) For force-Voltage analogy : "Each junction in the mechanical system corresponds to a closed loop which consists of electrical excitation sources & passive elements analogous to the mechanical driving sources and passive elements connected to the junction. All points on a rigid mass are considered as the junction."

(*b*) For force-current analogy : "Each junction in the mechanical system corresponds to a node which joins electrical excitation sources and passive elements analogous to the mechanical driving sources and passive elements connected to the junction . All points on a rigid mass are considered as the same junction and one terminal of the capacitance analogous to a mass is always connected to the ground."

The reason for connecting one terminal of capacitance analogous to a mass is always connected to the ground is that the vetocity (or displacement) of a mass is always referred to the earth.

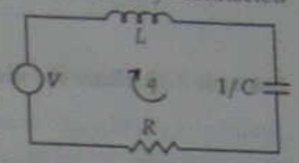


Fig. 1.32

Consider the system shown in fig. 1.21, corresponding to x in mechanical system, we will have one loop in f-v analogous electrical circuit. The circuit consists of a voltage source, inductance $L(M)$, resistance $R(B)$, the reciprocal of capacitance (c) and charge $q(x)$.

The equation for the mechanical system shown in fig. 1.21 is

$$f = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx \qquad \text{....(1.64)}$$

For the electrical circuit the equation will be

$$V = L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q \qquad \text{....(1.65)}$$

Since, $q = \int i\,dt$ the equation (1.65) will becomes

$$V = L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt \qquad \text{....(1.66)}$$

The equation (1.66) is the voltage equation for series RLC circuit.

Series RLC circuit, is the analogous electrical circuit of the mechanical system shown in fig. 1.21.

Example 1.11. Draw the analogous electrical network of the given system. Use *f-V* analogy.

Solution : Now corresponding x_1, x_2, x_3 we have three loops in electrical circuit.

First loop consists of a voltage source $V(f)$, Inductance $L_1(M_1)$ resistance R_1 (B_1). The resistance R_1 is common in between first & second loop.

Second loop consists of inductance $L_2(M_2)$, resistance $R_2(B_2)$ and $\frac{1}{C_2}(K_2)$.

Resistance R_2 & $\frac{1}{C_2}$ are common in loop second & third loop.

Similarly, third loop consists of Inductance $L_3(M_3)$, $\frac{1}{C_3}(K_3)$ & $R_3(B_3)$.

q_1, q_2 & q_3 are analogous quantities of x_1, x_2 & x_3 respectively.

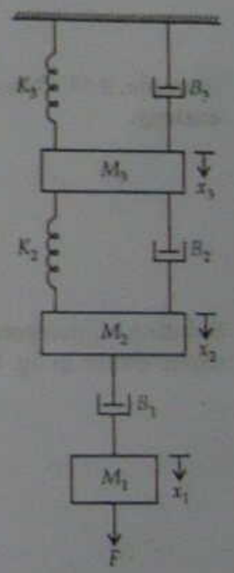


Fig. 1.33.

* From David K. Cheng, Analysis of linear systems., Addison-Wesley

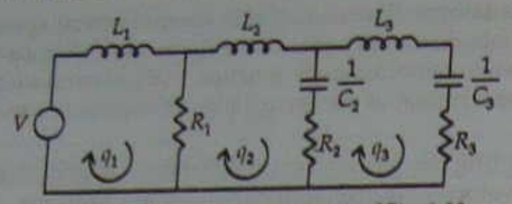


Fig. 1.34. Analogous electrical circuit of Fig. 1.33.

Example 1.12. Draw the electrical analogous circuit of the system shown in Fig 1.35.

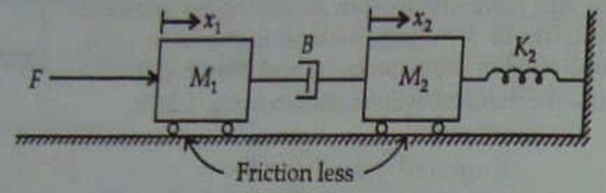


Fig. 1.35.

Solution : By inspection & with the help of table we can draw the electrical circuit shown in fig 1.36.

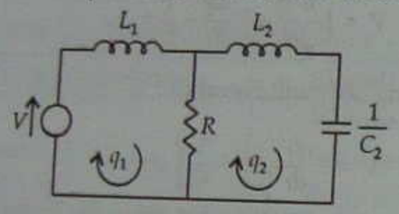


Fig. 1.36.

Example. 1.13. Draw the analogous electrical circuit of the system shown in Fig. 1.37. Use *f*-V analogy.

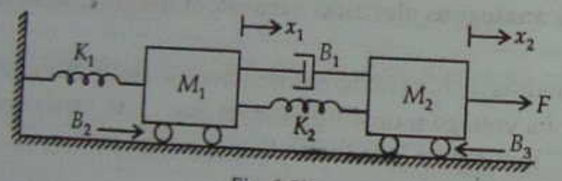


Fig. 1.37.

Solution : Corresponding x_1 & x_2, We have two loops in electrical circuit. The analogous electrical circuit shown in fig. 1.38.

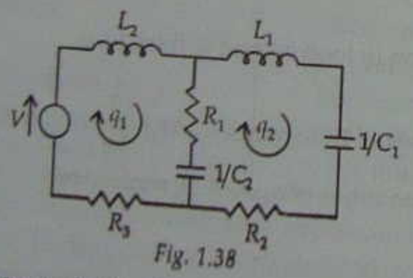


Fig. 1.38

Example 1.14. Draw the analogous electrical circuit of the given system shown in Fig. 1.37 use *f-i* analogy. Assume frictionless wheels.

Solution : Since, wheels are frictionless, $B_2 = 0$ & $B_3 = 0$

Corresponding x_1 & x_2, we have two independent nodes. The analogous electrical circuit shown in Fig. 1.39.

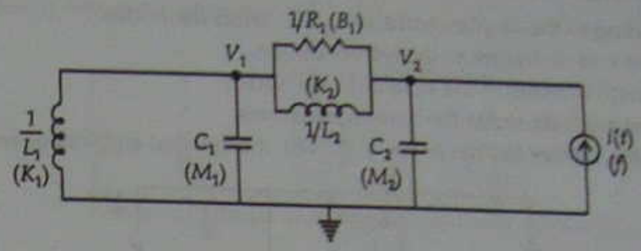


Fig. 1.39.

Example 1.15. Draw the analogous electrical circuit of the system shown in Fig. 1.35 use *f-i* analogy.

Solution : Corresponding x_1 & x_2 we have two independent nodes. The analogous circuit shown in Fig. 1.40.

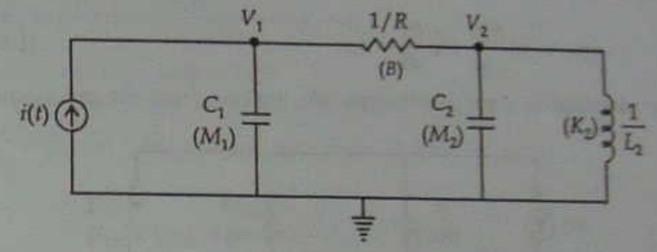


Fig. 1.40.

Example 1.16. Draw the analogous electrical circuit of a rotational mechanical system shown in Fig 1.41. use *f–i* analogy.

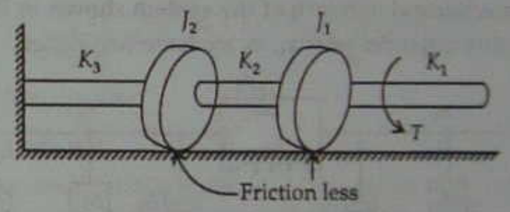


Fig. 1.41.

Solution : The analogous circuit shown in fig. 1.42.

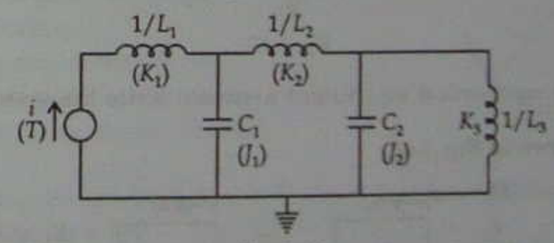


Fig. 1.42.

1.16. MECHANICAL EQUIVALENT NETWORK

The equivalent mechanical network is useful to determine the transfer function of the system. The following procedure is applied to draw the mechanical equivalent network.

Step 1 : Draw a reference line. Consider the fixed points as reference.

Step 2 : Corresponding to the displacement x_1, x_2....., select the nodes.

Step 3 : Connect one end of masses to the reference line.

Step 4 : Connect other elements of the system to the nodes.

Step 5 : Apply nodal analysis, write the system equations.

Consider a simple system shown in fig. 1.21. The mechanical equivalent network is shown in fig. 1.43.

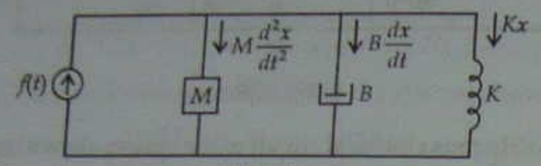


Fig. 1.43.

Apply the nodal analysis the system equation will be

$$f(t) = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx \qquad ...(1.67)$$

With the help of mechanical equivalent network, we can draw the analogous electrical circuit (shown in fig 1.44).

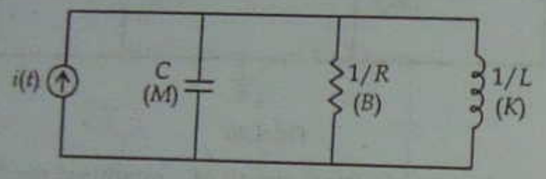


Fig. 1.44.

Example 1.17. Draw the mechanical network of the system shown in Fig. 1.37.

Solution : Since, there are two displacements x_1, x_2 we have two nodes.

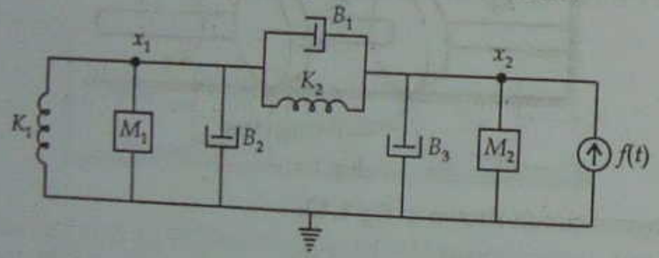


Fig. 1.45.

Example 1.18. Draw the mechanical equivalent network, write the system equations and find $\frac{F(s)}{X_2(s)}$ of the system shown in Fig. 1.46.

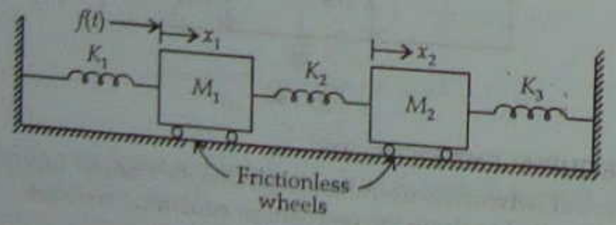


Fig. 1.46.

Solution : Mechanical network is shown in fig. 1.47.

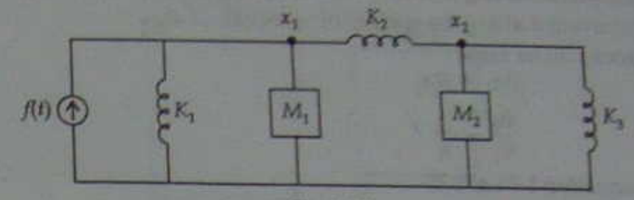


Fig. 1.47.

Apply nodal analysis at x_1 & x_2

$$f(t) = M_1\frac{d^2x_1}{dt^2} + K_1x_1 + K_2(x_1 - x_2) \qquad ...(1.68)$$

$$K_2(x_2 - x_1) + M_2\frac{d^2x_2}{dt^2} + K_3x_2 = 0 \qquad ...(1.69)$$

Laplace transform of eqn. 1.68 & 1.69

$$F(s) = S^2M_1X_1(s) + (K_1 + K_2)X_1(s) - K_2X_2(s) \qquad ...(1.70)$$

$$S^2M_2X_2(s) + (K_2 + K_3)X_2(s) - K_2X_1(s) = 0 \qquad ...(1.71)$$

Form *eq*^{n} 1.71

$$X_1(s) = \frac{s^2M_2 + K_2 + K_3}{K_2}X_2(s) \qquad ...(1.72)$$

Put the value of $X_1(s)$ from 1.72 in *eq*^{n} 1.70 & solve for $\frac{F_2(s)}{X_2(s)}$

$$\frac{F_2(s)}{X_2(s)} = \frac{(s^2M_1 + K_1 + K_2)(s^2M_2 + K_2 + K_3) - K^2_2}{K_2} \qquad ...(1.73)$$

Eq^n. 1.73 is the required answer.

1.17. MECHANICAL COUPLING

In fig. 1.48 two wheels are mechanically coupled. At the points A and B wheels experiences equal and opposite forces and the linear velocity at the point of contact will be same.

Let ω_1 & ω_2 = angular velocities

T_1 & T_2 = Torque

r_1 & r_2 = radius of the wheels

θ_1 & θ_2 = angular displacements

N_1 & N_2 = No. of teeths on wheels

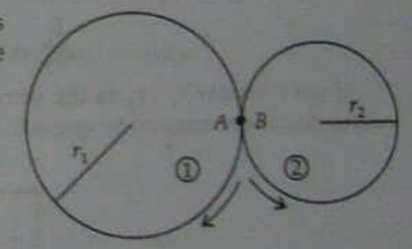


Fig. 1.48.

In Ideal case work done by wheel (1) will be equal to the work done by wheel (2)

Work done by wheel (1) = $T_1\theta_1$

Work done by wheel (2) = $T_2\theta_2$

$$\therefore \quad T_1\theta_1 = T_2\theta_2$$

$$\frac{T_1}{T_2} = \frac{\theta_2}{\theta_1} \qquad ...(1.74)$$

Linear distance travelled along the surface of wheel (1) $= \theta_1 r_1$

Linear distance travelled along the surface of wheel(2) $= \theta_2 r_2$

The linear distance will be same

$$\therefore \quad \theta_1 r_1 = \theta_2 r_2$$

$$\frac{\theta_1}{\theta_2} = \frac{r_2}{r_1} \qquad \text{....(1.75)}$$

Combining the equation 1.74 & 1.75

$$\frac{T_1}{T_2} = \frac{\theta_2}{\theta_1} = \frac{r_1}{r_2} \qquad \text{...(1.76)}$$

No. of teeth on wheels α radius of wheels

$$N_1 \propto r_1 \qquad \text{...(1.77)}$$

$$N_2 \propto r_2 \qquad \text{...(1.78)}$$

combining the equations 1.76, 1.77 & 1.78

$$\frac{T_1}{T_2} = \frac{\theta_2}{\theta_1} = \frac{r_1}{r_2} = \frac{N_1}{N_2} \qquad \text{....(1.79)}$$

Since,

$$\frac{\theta_2}{\theta_1} = \frac{r_1}{r_2}$$

$$\theta_2 = \frac{r_1}{r_2}\theta_1 \qquad \text{...(1.80)}$$

Differentiate eq^n 1.80

$$\dot{\theta}_2 = \frac{r_1}{r_2}\dot{\theta}_1$$

But

$$\dot{\theta}_2 = \omega_2$$

$$\dot{\theta}_1 = \omega_1$$

$$\therefore \quad \frac{\omega_2}{\omega_1} = \frac{r_1}{r_2}$$

Combining the eq^n 1.79 & 1.81 ...(1.81)

$$\frac{T_1}{T_2} = \frac{\theta_2}{\theta_1} = \frac{r_1}{r_2} = \frac{N_1}{N_2} = \frac{\omega_2}{\omega_1} \qquad \text{...(1.82)}$$

If we consider $r_1 : r_2$ as the turn ratio of an ideal transformer then angular velocity will be analogous to current and torque will be analogous to voltage. This is *f-V* analogy.

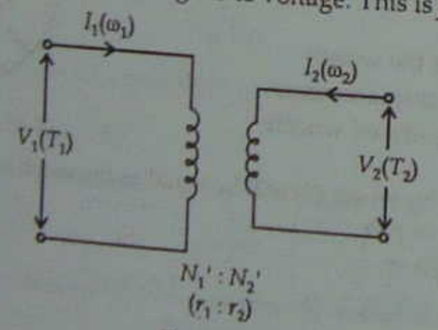
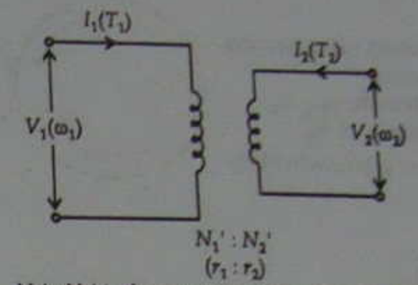


Fig. 1.49.

In case of *f-i* analogy



$N_1' : N_2'$ is the turn ratio of the transformer

Fig. 1.50.

Example 1.19.

(a) Find the ratio of $r_1 : r_2$

(*b*) If $\omega_1 = 10$ rad/sec. Calculate ω_2

(*c*) If torque for wheel (1) is 5 Nm. What will be the torque for wheel (2)

(*d*) If the displacement of wheel (1) is 20°. What is the displacement of wheel (2).

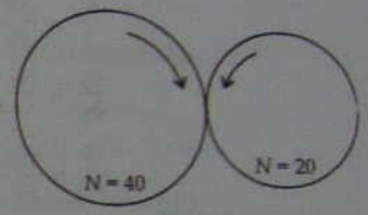


Fig. 1.51.

Solution : We know that

$$\frac{r_1}{r_2} = \frac{N_1}{N_2}$$

$$\frac{r_1}{r_2} = \frac{40}{20} = \frac{2}{1}$$

$$\therefore \quad r_1 : r_2 :: 2 : 1 \qquad \textbf{Ans.}$$

(*b*) From eq^n 1.82

$$\frac{N_1}{N_2} = \frac{\omega_2}{\omega_1}$$

$$\frac{40}{20} = \frac{\omega_2}{10} \qquad \therefore \ \omega_2 = 20 \text{ rad/sec.} \quad \textbf{Ans.}$$

(*c*) from eq$_n$ 1.82

$$\frac{T_1}{T_2} = \frac{N_1}{N_2}$$

$$\frac{5}{T_2} = \frac{40}{20} \qquad \therefore \ T_2 = 2.5 \text{ Nm} \quad \textbf{Ans.}$$

(*d*) from eq^n 1.82

$$\frac{\theta_2}{\theta_1} = \frac{N_1}{N_2}$$

$$\frac{\theta_2}{20} = \frac{40}{20} \qquad \therefore \ \theta_2 = 40° \quad \textbf{Ans.}$$

Example 1.20.

(a) If θ_1 = 3rad. (clockwise) calculate the displacement of wheel 2, 3 and 4.

(b) If ω_1 = 15 rad/sec. Calculate ω_3 & ω_4.

(c) If T_1 = 10 Nm. Calculate T_2 & T_3.

(d) If angular acceleration ($\ddot{\theta}_1$) for wheel (1) is 4 rad/sec^2, calculate $\ddot{\theta}_3$.

(e) Find $r_1 : r_2$ & $r_1 : r_3$

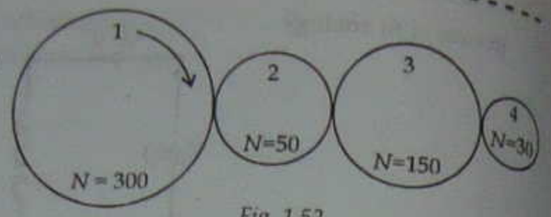


Fig. 1.52.

Solution :

(a) $$\frac{\theta_2}{\theta_1} = \frac{N_1}{N_2}$$

$$\frac{\theta_2}{3} = \frac{300}{50} \qquad \therefore \theta_2 = 18 \text{ rad}$$

$$\frac{\theta_2}{\theta_3} = \frac{N_3}{N_2}$$

$$\frac{18}{\theta_3} = \frac{150}{50} \qquad \theta_3 = 6 \text{ rad.}$$

$$\frac{\theta_3}{\theta_4} = \frac{N_4}{N_3}$$

$$\frac{6}{\theta_4} = \frac{30}{150} \qquad \theta_4 = 30 \text{ rad}$$

(b) $$\frac{\omega_3}{\omega_1} = \frac{N_1}{N_3}$$

$$\frac{\omega_3}{15} = \frac{300}{150} \qquad \therefore \omega = 30 \text{ rad/sec.}$$

$$\frac{\omega_4}{\omega_1} = \frac{N_1}{N_4}$$

$$\omega_4 = \frac{300}{30} \times 15 \qquad \therefore \omega_4 = 150 \text{ rad/sec.}$$

(c) $$\frac{T_1}{T_2} = \frac{N_1}{N_2} \text{ or } \frac{10}{T_2} = \frac{300}{50} \qquad \therefore T_2 = 5/3 \text{ Nm}$$

$$\frac{T_1}{T_3} = \frac{N_1}{N_3} \text{ or } \frac{10}{T_3} = \frac{300}{150} \qquad \therefore T_3 = 5 \text{ Nm}$$

(d) $$\frac{\ddot{\theta}_3}{\ddot{\theta}_1} = \frac{N_1}{N_3}$$

$$\frac{\ddot{\theta}_3}{4} = \frac{300}{150} \qquad \ddot{\theta}_3 = 8 \text{ rad/sec}^2 \quad \text{Ans.}$$

(e) $$\frac{r_1}{r_2} = \frac{N_1}{N_2}$$

$$\frac{r_1}{r_2} = \frac{300}{50}$$

$$r_1 : r_2 = 6 : 1 \qquad \text{Ans.}$$

$$\frac{r_1}{r_3} = \frac{N_1}{N_3}$$

$$\frac{r_1}{r_3} = \frac{300}{150}$$

$$r_1 : r_3 = 2 : 1 \qquad \text{Ans.}$$

Example 1.21. Draw the *f-V* and *f-i* analogy of the mech. system shown in figure 1.53.

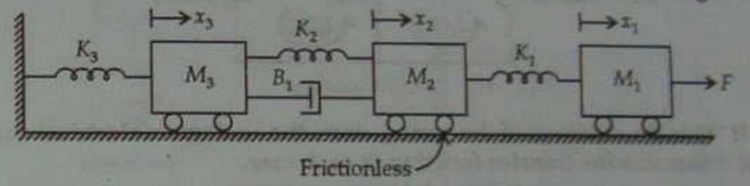


Fig. 1.53.

Solution : Corresponding the displacements x_1, x_2 and x_3 there are three meshes. The first mesh containing a voltage source $V(F)$ and Inductance $L_1(M_1)$, $C_1(1/K_1)$ will be common between two meshes. Similarly $C_2(1/K_2)$ and $R_1(B_1)$ will be common between second and third meshes. The *f-V* & *f-i* analogous circuit are shown in fig. 1.54 and 1.55 respectively.

f-V analogy :

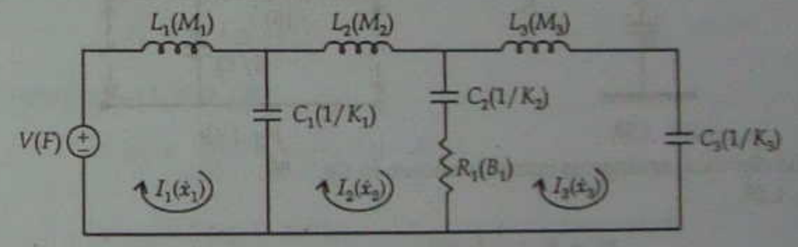


Fig. 1.54.

f-i analogy :

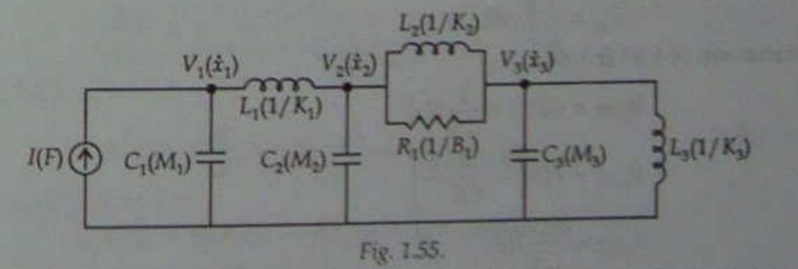


Fig. 1.55.

Example 1.22. Draw the electrical analogous circuit of the given mechanical system shown in fig. 1.56. Use *f-V* analogy.

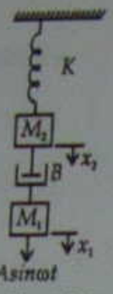


Fig. 1.56.

Solution :

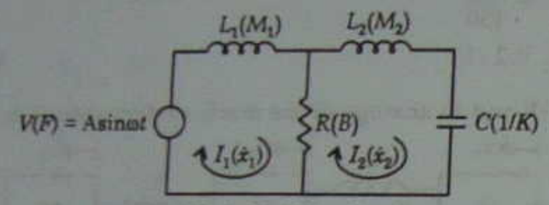


Fig. 1.57. (f-V analogy)

Example 1.23. If 'K' is the stiffness of the spring, draw the analogous electrical circuit based on *f-V* analogy and determine the transfer function in each case.

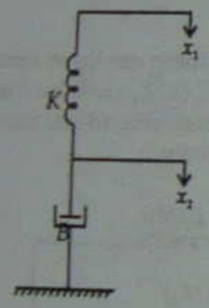


Fig. 1.58.

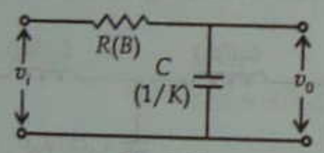


Fig. 1.59.

Solution : The electrical analogous circuit is shown in fig. 1.59.

From fig. 1.59. :

$$V_i = Ri + \frac{1}{C}\int i\,dt \qquad ...(1.83)$$

$$V_0 = \frac{1}{C}\int i\,dt \qquad ...(1.84)$$

Laplace transform of 1.83 & 1.84

$$V_i(s) = RI(s) + \frac{1}{CS}I(s)$$

$$V_i(s) = I(s)\left[\frac{1+RCS}{CS}\right] \qquad ...(1.85)$$

$$V_0(s) = \frac{1}{CS}I(s)$$

$$\frac{V_0(s)}{V_i(s)} = \frac{\frac{1}{CS}}{\frac{1+RCS}{CS}} = \frac{1}{1+RCS} \quad \textbf{Ans.}$$

From fig. 1.58 :

$$B\dot{x}_2 = K(x_1 - x_2) \qquad ...(1.86)$$

Laplace transform of eq^n 1.86

$$SBX_2(s) = KX_1(s) - KX_2(s)$$

$$X_2(s)[SB + K] = KX_1(s)$$

$$\therefore \frac{X_2(s)}{X_1(s)} = \frac{K}{K+SB} \quad \textbf{Ans.} \qquad ...(1.87)$$

Example 1.24. Draw the electrical analogous circuit (use *f-V*) analogy and derive their transfer functions.

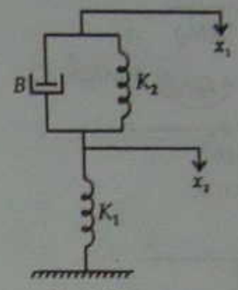


Fig. 1.60.

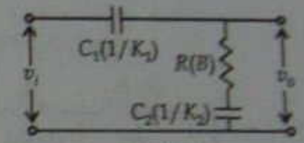


Fig. 1.61.

Solution : Electrical analogous circuit is shown in fig. 1.61.

From fig. 1.61

$$V_i = \frac{1}{C_1}\int i\,dt + \frac{1}{C_2}\int i\,dt + Ri \qquad ...(1.88)$$

$$V_0 = Ri + \frac{1}{C_2}\int i\,dt \qquad ...(1.89)$$

Laplace transform of 1.88 & 1.89

$$V_i(s) = \frac{1}{C_1 S}I(s) + \frac{1}{C_2 S}I(s) + RI(s) \qquad ...(1.90)$$

$$V_0(s) = RI(s) + \frac{1}{C_2 S}I(s)$$

$$\frac{V_0(s)}{V_i(s)} = \frac{(1+RC_2S)C_1}{C_1 + C_2 + C_1C_2RS} \qquad ...(1.91)$$

from fig. 1.60 :

$$K_1x_2 = B\frac{d}{dt}(x_1 - x_2) + K_2(x_1 - x_2) \qquad ...(1.92)$$

Laplace transform of 1.92

$$K_1X_2(s) = SBX_1(s) - SBX_2(s) + K_2X_1(s)\ K_2X_2(s)$$

$$\frac{X_2(s)}{X_1(s)} = \frac{SB + K_2}{K_1 + K_2 + SB} \qquad ...(1.93)$$

Example 1.25. Draw the electrical analogous circuit, use *f-V* and *f-i* analogy.

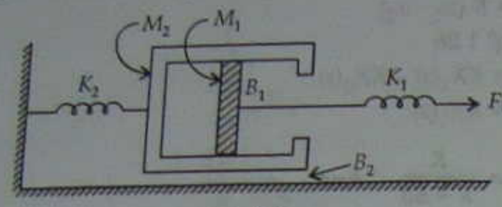


Fig. 1.62.

Solution : Required circuit shown in fig. 1.63 & 1.64.

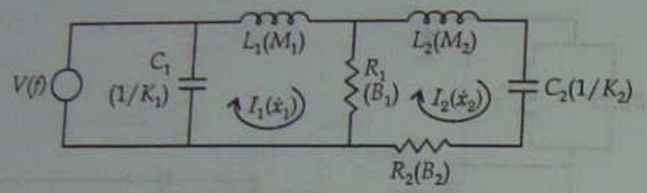


Fig. 1.63. f-V analogous elect. ckt.

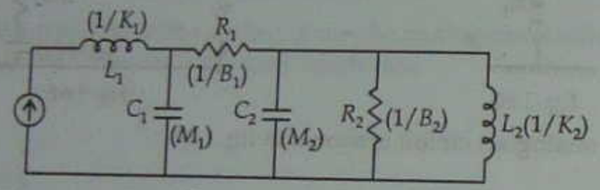
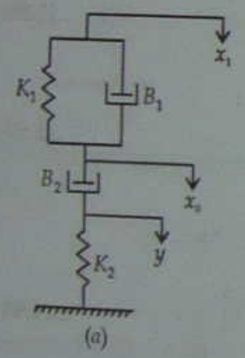


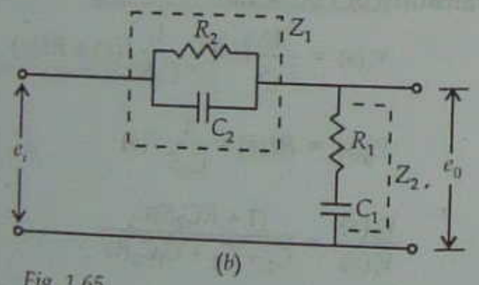
Fig. 1.64. f-i analogous elect. ckt.

Example 1.26. Show that the system shown in fig. 1.65(a), 1.65(b) are analogous system.

(RML Univ, Faizabad, 2002)



(a)



(b)

Fig. 1.65.

Solution : From fig. 1.65a :

$$B_1\frac{d}{dt}(x_i - x_0) + K_1(x_i - x_0) = B_2\frac{d}{dt}(x_0 - y) \quad ...(1.94)$$

$$B_2\frac{d}{dt}(x_0 - y) = K_2 y \quad ...(1.95)$$

Laplace transform of 1.94 & 1.95

$$B_1SX_1(s) - B_1SX_i(s) + K_1X_i(s) - K_iX_0(s) = SB_2X_0(s) - SB_2y(s) \quad ...(1.96)$$

$$SB_2X_0(s) - SB_2y(s) = K_2y(s) \quad ...(1.97)$$

from (1.97) find $y(s)$ and put in (1.96)

$$B_1SX_i(s) - B_1SX_0(s) + K_1X_i(s) - K_1X_0(s) = SB_2X_0(s) - SB_2\cdot\frac{SB_2x_0(s)}{K_2 + SB_2}$$

$$\therefore \quad \frac{X_0(s)}{X_i(s)} = \frac{(SB_1 + K_1)(K_2 + SB_2)}{SB_1K_2 + S^2B_1B_2 + K_1K_2 + SB_2K_1 + SB_2K_2} \quad ...(1.98)$$

using f-v analogy (from table 1.3) the eq^n (1.98) becomes

$$\frac{e_0(s)}{e_i(s)} = \frac{(1 + R_1C_1s)(1 + R_2C_2S)}{S^2R_1R_2C_1C_2 + SR_1C_1 + SR_2C_2 + SR_2C_1} \quad ...(1.99)$$

from fig (1.65b) :

$$Z_1 = \frac{R_2}{1 + R_2C_2S}$$

$$Z_2 = R_1 + \frac{1}{C_1S} = \frac{1 + R_1C_1S}{C_1S}$$

$$\frac{e_0(t)}{e_i(t)} = \frac{Z_2}{Z_1 + Z_2}$$

$$\frac{e_0(s)}{e_i(s)} = \frac{\frac{1 + R_1C_1S}{C_1S}}{\frac{1 + R_1C_1S}{C_1S} + \frac{R_2}{1 + R_2C_2S}} = \frac{(1 + R_1C_1S)(1 + R_2C_2S)}{S^2R_1R_2C_1C_2 + SR_1C_1 + SR_2C_2 + SR_2C_1} \quad ...(1.100)$$

Since eq^n (1.99) & eq^n (1.100) are same, hence the given systems are analogous.

$$B_2\frac{d}{dt}(x_0 - y) = K_2y \quad ...(1.95)$$

Laplace transform of 1.94 & 1.95

$$B_1SX_i(s) - B_1SX_0(s) + K_1X_i(s) - K_1X_0(s) = SB_2X_0(s) - SB_2y(s) \quad ...(1.96)$$

$$SB_2X_0(s) - SB_2y(s) = K_2y(s) \quad ...(1.97)$$

from (1.97) find $y(s)$ and put in (1.96)

$$B_1SX_i(s) - B_1SX_0(s) + K_1X_i(s) - K_1X_0(s) = SB_2X_0(s) - SB_2\cdot\frac{SB_2x_0(s)}{K_2 + SB_2}$$

$$\therefore \quad \frac{X_0(s)}{X_i(s)} = \frac{(SB_1 + K_1)(K_2 + SB_2)}{SB_1K_2 + S^2B_1B_2 + K_1K_2 + SB_2K_1 + SB_2K_2} \quad ...(1.98)$$

using f-v analogy (from table 1.3) the eq^n (1.98) becomes

$$\frac{e_0(s)}{e_i(s)} = \frac{(1 + R_1C_1s)(1 + R_2C_2S)}{S^2R_1R_2C_1C_2 + SR_1C_1 + SR_2C_2 + SR_2C_1} \quad ...(1.99)$$

from fig (1.65b) :

$$Z_1 = \frac{R_2}{1 + R_2C_2S}$$

$$Z_2 = R_1 + \frac{1}{C_1S} = \frac{1 + R_1C_1S}{C_1S}$$

$$\frac{e_0(t)}{e_i(t)} = \frac{Z_2}{Z_1 + Z_2}$$

$$\frac{e_0(s)}{e_i(s)} = \frac{\frac{1+R_1C_1S}{C_1S}}{\frac{1+R_1C_1S}{C_1S} + \frac{R_2}{1+R_2C_2S}} = \frac{(1+R_1C_1S)(1+R_2C_2S)}{S^2R_1R_2C_1C_2 + SR_1C_1 + SR_2C_2 + SR_2C_1} \quad ...(1.100)$$

Since eq^n (1.99) & eq^n (1.100) are same, hence the given systems are analogous.

BLOCK DIAGRAM

1.18. BLOCK DIAGRAM REPRESENTATION

Any system can be described by a set of differential equations or can be represented by the schematic diagram containing all components and their connections. But for the complicated systems these two methods are not suitable. The block diagram representation is the combination of above two methods. A block may represent a single component or a group of components, but each block is completely characterized by a transfer function. The transfer function is an expression which relates output to input in *s*-domain. Transfer function does not give any information about the internal structure of the system. Once we determine the transfer function, then we can represent the system by the block diagram. Block diagrams are single line diagram, that is the flow of system variables from one bolck to another block is represented by a single line. Fig 1.66 shows the block diagram representation of a system.

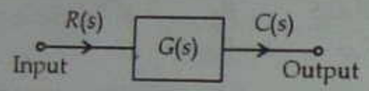


Fig. 1.66.

Where, $R(s)$ = Input
$C(s)$ = output
$G(s)$ = transfer function

Then, the system can be represented as

$$C(s) = R(s).\ G(s) \quad ...(1.101)$$

The flow of system variables from one block to another block is represented by the arrow. In addition to this, the sum of the signals or the difference of the signals are represented by a summing point (fig. 1.67*a*). Application of one input source to two or more blocks is represented by a take off point (fig 1.67*b*).

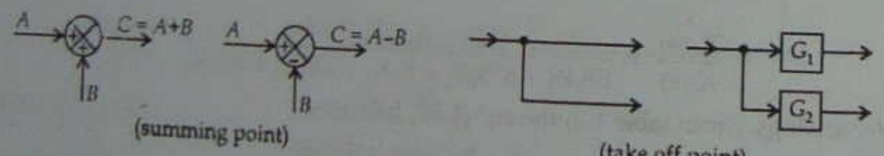


Fig. 1.67.

1.19. HOW TO DRAW THE BLOCK DIAGRAM

Consider a simple *R-L* circuit shown in fig 1.68

Apply *KVL*

$$V_i = Ri + L\frac{di}{dt} \quad ...(1.102)$$

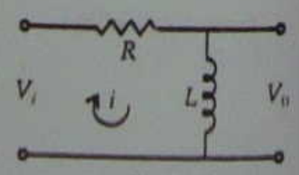


Fig. 1.68.

$$V_0 = L\frac{di}{dt} \quad ...(1.103)$$

Laplace transform of 1.102 & 1.103 with initial condition zero

$$V_i(s) = I(s)\ R + SL\ I(s)$$

$$V_i(s) = I(s)\ (R + SL) \quad ...(1.104)$$

$$V_0(s) = SL\ I(s) \quad ...(1.105)$$

From 1.104 & 1.105

$$\frac{V_0(s)}{V_i(s)} = \frac{sL}{R+sL} \quad ...(1.106)$$

Form fig 1.68 $$i = \frac{V_i - V_0}{R} \quad ...(1.107)$$

$$V_0 = L\frac{di}{dt} \quad ...(1.108)$$

laplace transform of 1.107 & 1.108

$$I(s) = \frac{1}{R}\ [V_i(s) - V_0(s)] \quad ...(1.109)$$

$$V_0(s) = SL\ I(s) \quad ...(1.110)$$

For right hand side of eq^n 1.109 we use a summing point.

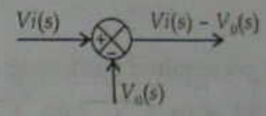


Fig. 1.69

The output of the summing point is given to block and output of the block is $I(s)$ as per eq^n 1.109.

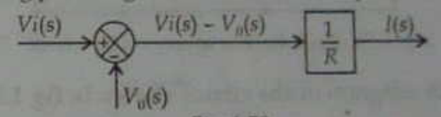


Fig. 1.70.

Form eqn 1.110 the output of block I(s) is given to another block containing the element SL & the output of the second block is V_0

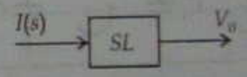


Fig. 1.71.

Combining the fig 1.70 & 1.71 we get required block diag.

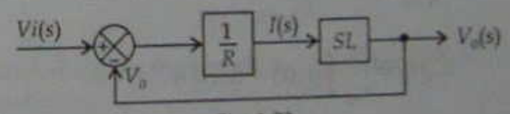


Fig. 1.72.

Example 1.27. Draw the block diagram of series RLC circuit, where V_i & V_0 are the input and output voltages.

Solution : The transformed network in S-domain is shown in fig. 1.74.

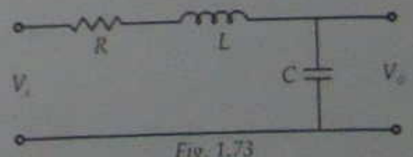


Fig. 1.73

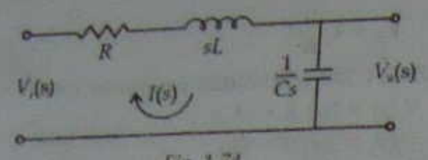


Fig. 1.74

from 1.74 : $$I(s) = \frac{V_i(s) - V_0(s)}{R + sL} \quad ...(1.111)$$

$$V_0(s) = \frac{1}{Cs} I(s) \quad ...(1.112)$$

For *R.H.S* of eq^n 1.111, we require a summing point

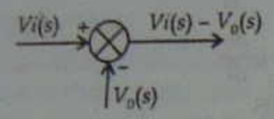


Fig. 1.75.

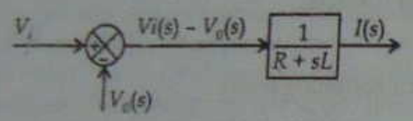


Fig. 1.76.

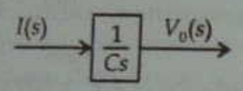


Fig. 1.77.

combining the fig 1.76 & 1.77 we get required block diag.

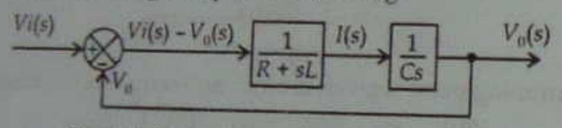


Fig. 1.78. Block diagram of series RLC circuit

Example 1.28. Draw the block diagram of the circuit shown in fig 1.79.

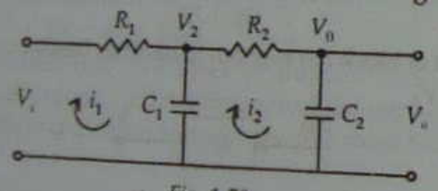


Fig. 1.79.

Solution :

$$i_1(t) = \frac{V_i(t) - V_2(t)}{R_1} \quad ...(1.113a)$$

$$V_2(t) = \frac{1}{C_1}\int [i_1(t) - i_2(t)]dt \quad ...(1.113b)$$

$$i_2(t) = \frac{1}{R_2}[V_2(t) - V_0(t)] \quad ...(1.113c)$$

$$V_0(t) = \frac{1}{C_2}\int i_2 dt \quad ...(1.113d)$$

Take the laplace transform of above eq^{ns}

$$I_1(s) = \frac{1}{R_1}[V_i(s) - V_2(s)] \quad ...(1.113e)$$

$$V_2(s) = \frac{1}{sC_1}[I_1(s) - I_2(s)] \quad ...(1.113f)$$

$$I_2(s) = \frac{1}{R_2}[V_2(s) - V_0(s)] \quad ...(1.113g)$$

$$V_0(s) = \frac{1}{sC_2} I_2(s) \quad ...(1.113h)$$

From eq^n (5)

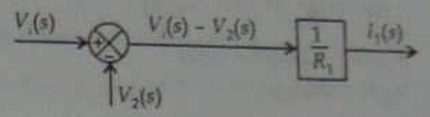


Fig. 1.80(a)

From eq^n (6)

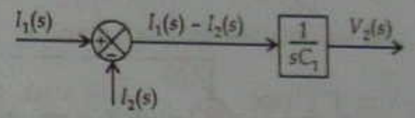


Fig. 1.80(b)

From eq^n (7)

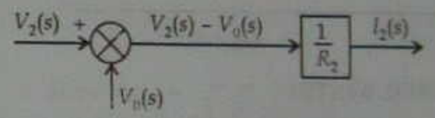


Fig. 1.80(c)

From eq^n (8)

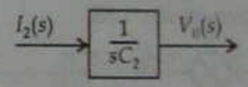


Fig. 1.80(d)

combining all the fig. 1.80*a*, 1.80*b*, 1.80*c*. & 1.80*d*

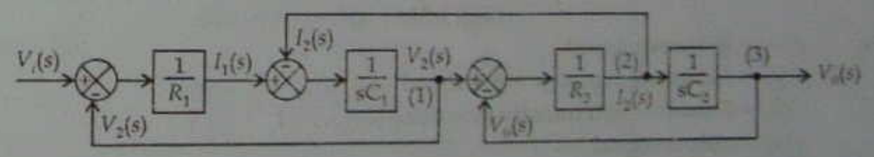


Fig. 1.80(e)

Example 1.29. Draw the block diag. for the circuit shown in fig 1.81, where V_i and i_L are the input and output variables respectively.

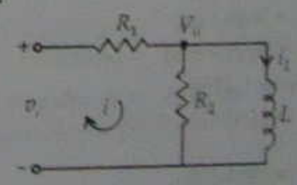


Fig. 1.81.

Solution : From fig. 1.81

$$i = \frac{V_i - V_0}{R_1} \quad \text{...(1.114a)}$$

$$i - i_L = \frac{V_0}{R_2} \quad \text{...(1.114b)}$$

$$V_0 = L\frac{di_L}{dt} \quad \text{...(1.114c)}$$

laplace transform of above eq^{ns}

$$I(s) = \frac{1}{R_1}[V_i(s) - V_0(s)] \quad \text{...(1.114d)}$$

$$V_0(s) = R_2[I(s) - I_L(s)] \quad \text{...(1.114e)}$$

$$V_0(s) = SLI_L(s)$$

$$I_L(s) = \frac{1}{SL}V_0(s) \quad \text{...(1.114f)}$$

From 1.114*d*, 1.114*e* and 1.114*f*, we can draw the block diag.

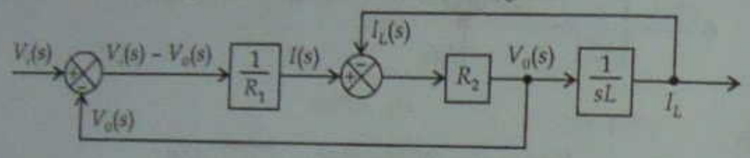


Fig. 1.82.

1.20. CLOSED LOOP CONTROL SYSTEM

A closed loop system is one in which output is fed back into an error detector and compared with the reference input. The feedback may be negative or positive.

Consider a closed loop system shown in fig. 1.83

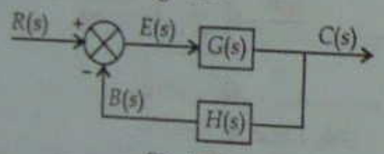


Fig. 1.83

where, $R(s)$ = Reference input
$E(s)$ = Actuating signal or error signal
$G(s)$ = Forward path transfer function
$C(s)$ = output signal
$H(s)$ = Feedback transfer function
$B(s)$ = Feedback signal

From fig 1.83

$$C(s) = G(s).E(s) \quad \text{...(1.115a)}$$

$$B(s) = H(s).C(s) \quad \text{...(1.115b)}$$

$$E(s) = R(s) - B(s) \quad \text{...(1.115c)}$$

Put the value of $C(s)$ from 1.115*a* in 1.115*b*

$$B(s) = H(s).G(s).E(s)$$

$$\frac{B(s)}{E(s)} = G(s).H(s)$$

$\frac{B(s)}{E(s)}$ = open loop transfer function = $G(s)\,H(s)$

Put the value of $E(s)$ from 1.115*c* in 1.115*a* ...(1.115*d*)

$$C(s) = G(s)[R(s) - B(s)]$$

$$C(s) = R(s).G(s) - G(s).B(s)$$

Put the value of $B(s)$ from 1.115*b* in 1.115*e* ...(1.115*e*)

$$C(s) = R(s)\,G(s) - G(s)\,H(s).\,C(s)$$

or, $C(s)[1 + G(s)\,H(s)] = R(s)\,G(s)$

or, $$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)\,H(s)} \quad \text{...(1.115f)}$$

$\frac{C(s)}{R(s)}$ = M(s) = closed loop transfer function = $\frac{G(s)}{1 + G(s)\,H(s)}$

If the feedback is positive, then equation 1.115*f* becomes .

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)\,H(s)} \quad \text{...(1.115g)}$$

From 1.115*a* put the value of $C(s)$ in eq^{n} 1.115*f*

$$\frac{G(s)\,E(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

or, $$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)} \quad \text{...(1.115h)}$$

$$\frac{E(s)}{R(s)} = \text{Error ratio} = \frac{1}{1 + G(s)H(s)}$$

For positive feedback

$$\frac{E(s)}{R(s)} = \frac{1}{1 - G(s)H(s)} \quad \text{...(1.115i)}$$

Put the value of $C(s)$ from 1.115*a* in 1.115*b*

$$B(s) = H(s).G(s).E(s)$$

Put the value of $E(s)$ from 1.115*c* in above eq^{n}

$$B(s) = H(s).G(s)[R(s) - B(s)] \quad \text{...(1.115j)}$$

from 1.115*j*

$$\frac{B(s)}{R(s)} = \frac{G(s)\,H(s)}{1 + G(s)\,H(s)} \quad \text{...(1.115k)}$$

$\frac{B(s)}{R(s)}$ = Primary feedback ratio = $\frac{G(s)\,H(s)}{1 + G(s)\,H(s)}$

For positive feedback

$$\frac{B(s)}{R(s)} = \frac{G(s)\,H(s)}{1 - G(s)\,H(s)} \quad \text{...(1.115l)}$$

A unity feedback control system is shown in fig. 1.84.

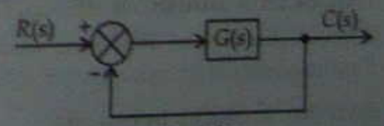


Fig. 1.84.

For unity feedback control system $H(s) = 1$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} \quad \text{For negative feedback}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1-G(s)} \quad \text{For positive feedback}$$

1.21. MULTIINPUT-MULTIOUTPUT SYSTEM (MIMO)

When two or more inputs act on a system, each input can be treated independently of the others. Complete output can be obtained by adding the effect of each input.

Consider a two input linear system shown in fig. 1.85.

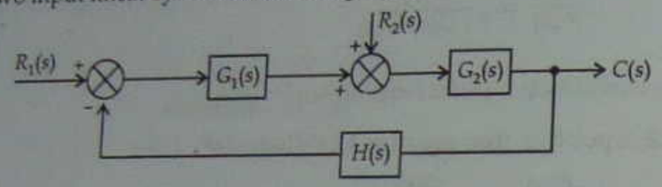


Fig. 1.85.

Step1 : Put $R_1(s) = 0$

$$\frac{C_2(s)}{R_2(s)} = \frac{G_2(s)}{1+G_1(s)G_2(s)H(s)} \qquad ...(1.116a)$$

Step 2: Put $R_2(s) = 0$

$$\frac{C_1(s)}{R_1(s)} = \frac{G_1(s)G_2(s)}{1+G_1(s)G_2(s)H(s)} \qquad ...(1.116b)$$

Hence, $C(s) = C_1(s) + C_2(s)$

$$C(s) = \frac{G_1(s)G_2(s)R_1(s)}{1+G_1(s)G_2(s)H(s)} + \frac{G_2(s)R_2(s)}{1+G_1(s)G_2(s)H(s)}$$

$$C(s) = \frac{G_2(s)}{1+G_1(s)G_2(s)H(s)}[R_1(s)G_1(s) + R_2(s)] \qquad ...(1.116c)$$

MIMO system can be expressed in matrix form

$$\begin{bmatrix} C_1(s) \\ C_2(s) \\ \vdots \\ C_n(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{12}(s) & \cdots \\ \vdots & & \\ \vdots & & \\ G_{nm}(s) & & \end{bmatrix} \begin{bmatrix} R_1(s) \\ R_2(s) \\ \vdots \\ R_r(s) \end{bmatrix}$$

where, $C(s) = G(s)\,R(s)$

$G(s)$ = matrix transfer function

1.22. BLOCK DIAGRAM REDUCTION

When a number of blocks are connected, the overall transfer function can be obtained by block diagram reduction technique. The following rules are associated with the block reduction technique.

Rule No. 1. Blocks in cascade

When two or more blocks are in cascade, the resultant block is a product of the individual block transfer function. Consider the two blocks are in cascade shown in fig. 1.86.

Fig. 1.86

From fig. 1.86

$$\frac{C_1(s)}{R(s)} = G_1(s) \qquad ...(1.117a)$$

$$\frac{C(s)}{C_1(s)} = G_2(s) \qquad ...(1.117b)$$

From (1.117*a*) and (1.117*b*)

$$\frac{C_1(s)}{R(s)}\,\frac{C(s)}{C_1(s)} = \frac{C(s)}{R(s)} = G_1(s)\,.\,G_2(s)$$

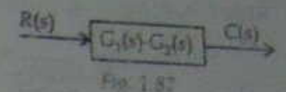


Fig. 1.87

The equivalent diagram is shown in fig 1.87.

Rule No 2. Blocks in Parallel

When two or more blocks are connected in parallel as shown in fig. (1.88*a*), the resultant block is the sum of individual block transfer function.

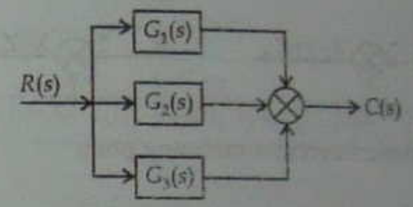


Fig. 1.88(a)

from fig. 1.88*a*

$$C(s) = R(s)\,G_1(s) + R(s)\,G_2(s) + R(s)\,G_3(s)$$
$$= R(s)\,[G_1(s) + G_2(s) + G_3(s)]$$

$$\therefore \quad \frac{C(s)}{R(s)} = G_1(s) + G_2(s) + G_3(s)$$

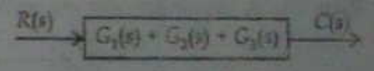


Fig. 1.88(b)

The equivalent diag. is shown in fig. 1.88*b*

Rule No 3. Moving a take off point ahead of a block

If a take off point is moved ahead of a block, a block with same transfer function will introduce in the branch of a take off point, as shown in fig 1.88(*c*)

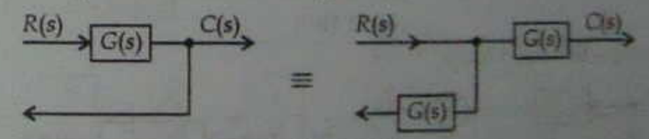


Fig. 1.88(c)

Rule No. 4. Moving a take off point after the block

If a take off point is moved after the block, a block with the reciprocal of the transfer function is introduced in the branch of a take off point as shown in fig. 1.88(*d*).

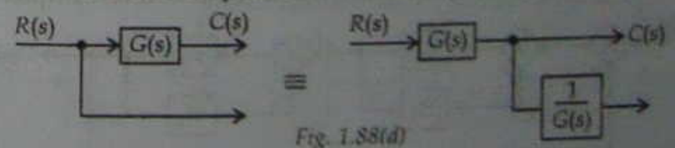


Fig. 1.88(d)

Rule No. 5. Moving a summing point beyond a block

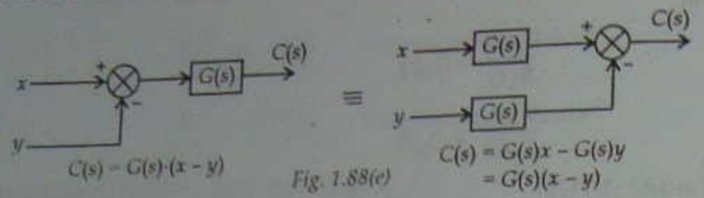


Fig. 1.88(e)

Rule No. 6. Moving a summing point ahead of a block

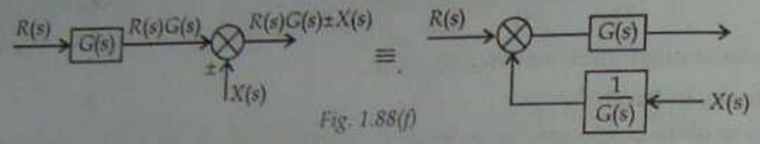


Fig. 1.88(f)

Rule No. 7. Interchanging two summing points

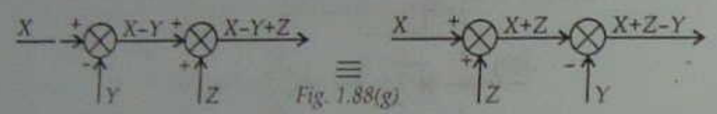


Fig. 1.88(g)

Rule No. 8. Moving a take off point beyond a summing point

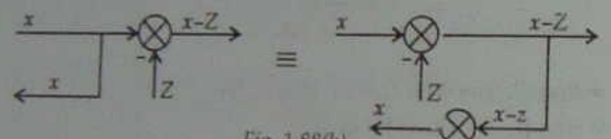


Fig. 1.88(h)

Rule No. 9. Moving a take off point ahead of a summing point

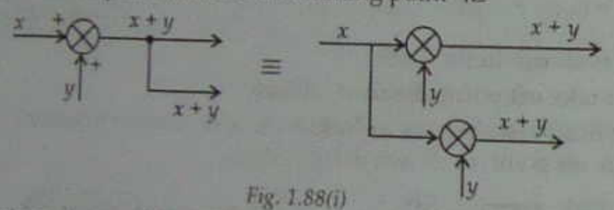


Fig. 1.88(i)

Rule No. 10. Eliminating a forward loop

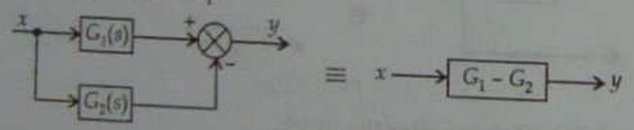


Fig. 1.88(j)

Example 1.30. Derive the transfer function using block reduction technique.

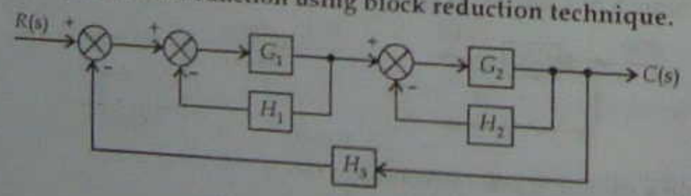


Fig. 1.89

Solution : Step 1: There are two internal closed loops, remove these loops by using the eqn 1.115*f*.

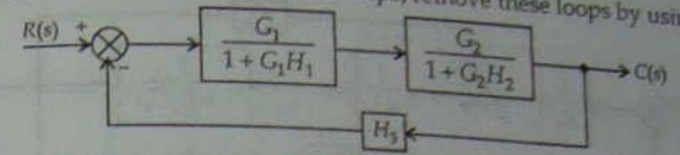


Fig. 1.89(a)

Step 2: Two blocks are in cascade, use the rule no. 1.

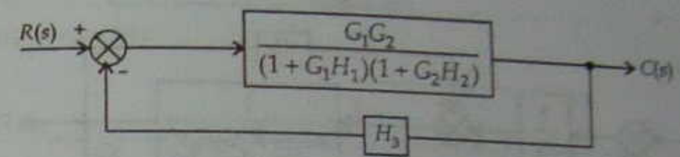
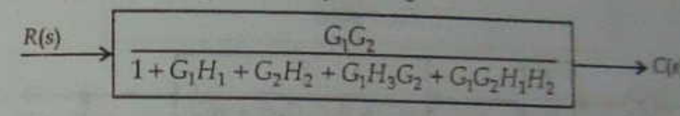


Fig. 1.89(b)

Step 3: Fig 1.89*b* is a closed loop, again use eqn 1.115f



$$\therefore \quad \frac{C(s)}{R(s)} = \frac{G_1G_2}{1 + G_1H_1 + G_2H_2 + G_1H_3G_2 + G_1G_2H_1H_2}$$ **Ans.**

Example 1.31. Find the overall transfer function of the system shown in fig. 1.80e.

Solution : Step 1 : Shift the pick off point beyond the block $1/sC_2$

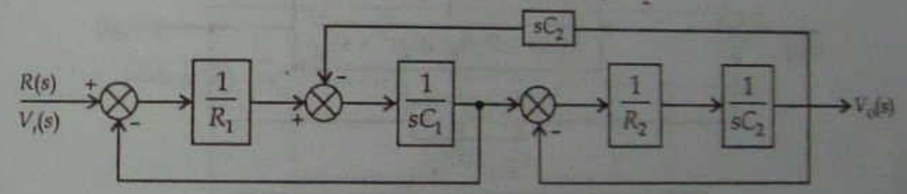


Fig. 1.90(a)

Step 2 : Two blocks are in cascade

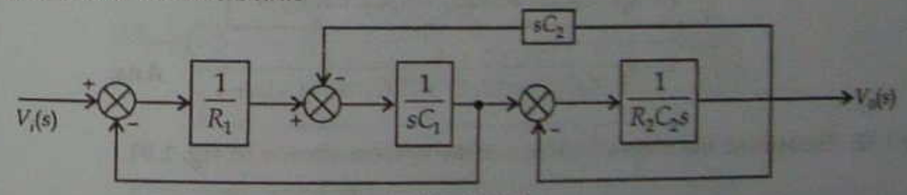


Fig. 1.90(b)

Step 3 :

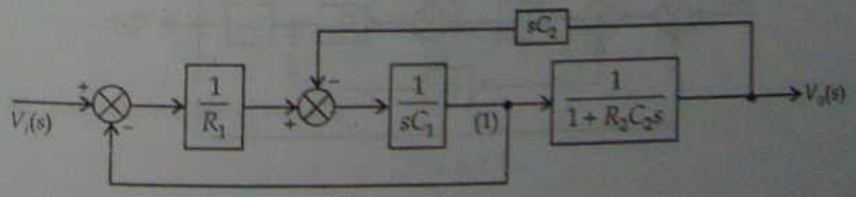


Fig. 1.90(c)

Step 4 :

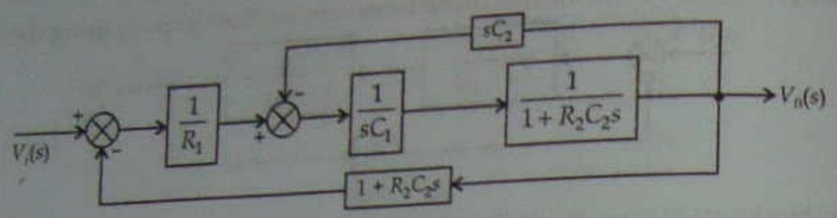


Fig. 1.90(d)

Step 5 :

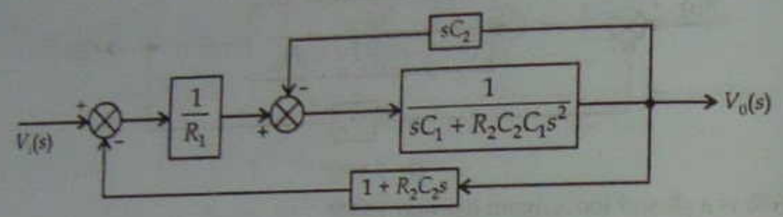


Fig. 1.90(e)

Step 6 :

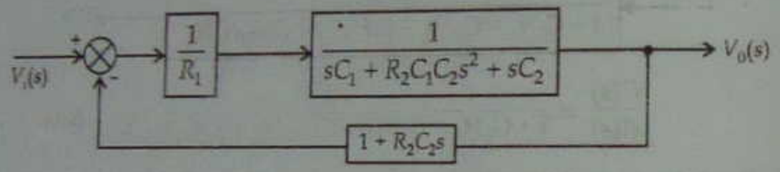


Fig. 1.90(f)

Step 7 :

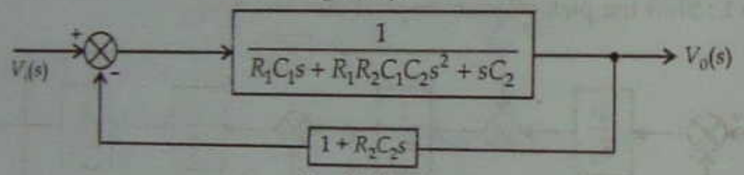
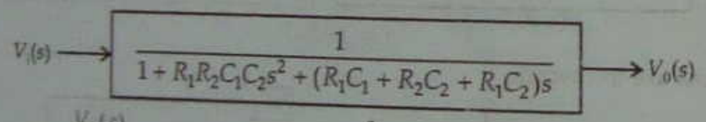


Fig. 1.90(g)



$$\therefore \quad \frac{V_0(s)}{V_i(s)} = \frac{1}{S^2R_1R_2C_1C_2 + (R_1C_1 + R_2C_2 + R_1C_2)s + 1} \quad \text{Ans.}$$

Example 1.32. Determine the ratio $C(s)/R(s)$ for the system shown in fig. 1.91.

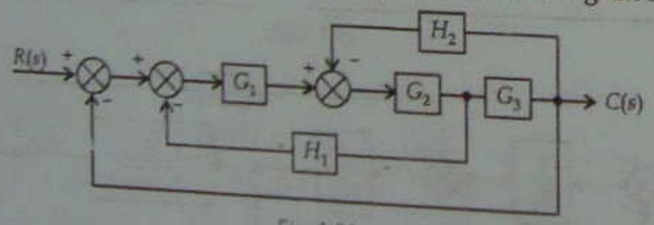
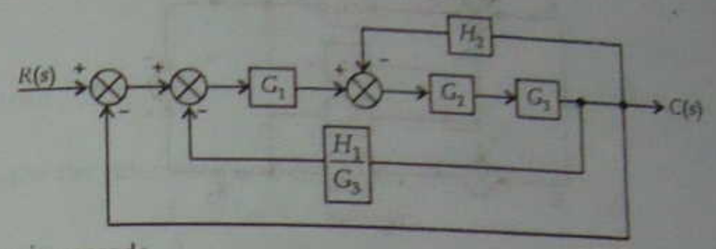
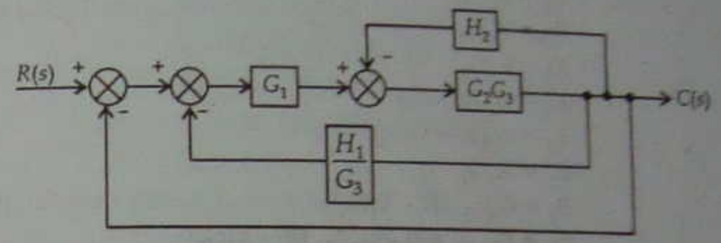


Fig. 1.91.

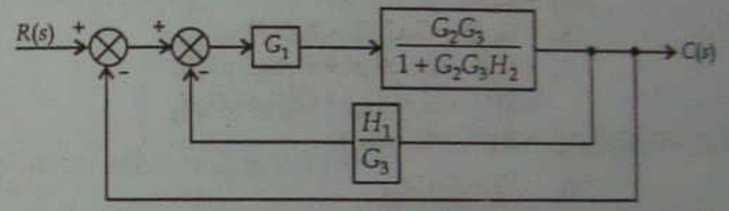
Solution : Step 1 : Shift the takeoff point beyond the block G_3



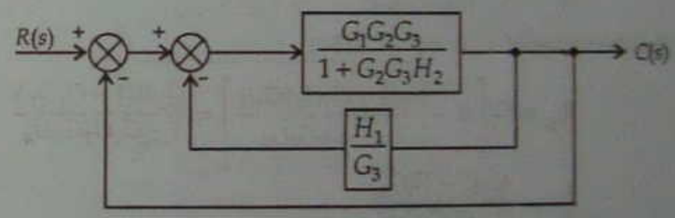
Step 2 : G_2 & G_3 are in cascade



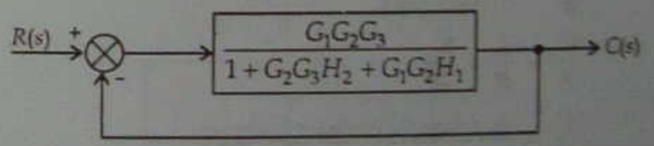
Step 3 :



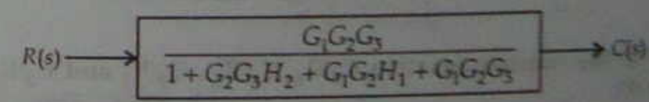
Step 4 :



Step 5 :



Step 6 :



$$\frac{C(s)}{R(s)} = \frac{G_1G_2G_3}{1 + G_2G_3H_2 + G_1G_2H_1 + G_1G_2G_3} \quad \text{Ans.}$$

Example 1.33. Determine the ratio C/R.

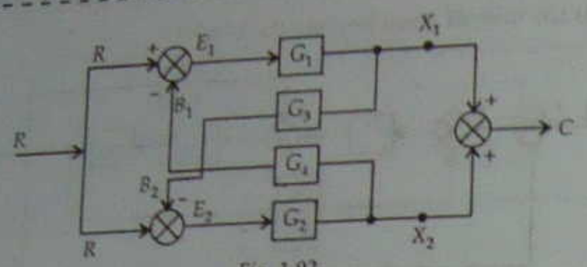


Fig. 1.92.

Solution :

$$E_2 = R - B_2 \quad ...(1.118a)$$

$$B_2 = X_1 G_3 \quad ...(1.118b)$$

$$X_2 = E_2 G_2$$

$$X_2 = G_2 (R - B_2)$$

$$X_2 = G_2(R - X_1 G_3) \quad ...(1.118c)$$

$$B_1 = X_2 G_4$$

$$B_1 = G_4 G_2 (R - X_1 G_3) \quad ...(1.118d)$$

$$E_1 = R - B_1 = R - G_2 G_4 (R - X_1 G_3) \quad ...(1.118e)$$

$$X_1 = E_1 G_1$$

$$X_1 = G_1 [R - G_2 G_4 (R - X_1 G_3)] \quad ...(1.118f)$$

$$\therefore \quad X_1 = G_1 R - G_1 G_2 G_4 R + G_1 G_2 G_3 G_4 X_1 \quad ...(1.118g)$$

$$= R(G_1 - G_1 G_2 G_4) + G_1 G_2 G_3 G_4 X_1$$

$$\therefore \quad X_1(1 - G_1 G_2 G_3 G_4) = R (G_1 - G_1 G_2 G_4)$$

$$\frac{X_1}{R} = \frac{G_1 - G_1 G_2 G_4}{1 - G_1 G_2 G_3 G_4} \quad ...(1.118h)$$

from eq^n (1.118*h*) put the value of x_1 in eq^n (1.118*c*)

$$X_2 = G_2\left[R - \frac{G_1(1 - G_2 G_4) G_3 R}{1 - G_1 G_2 G_3 G_4}\right] = \frac{G_2 R(1 - G_1 G_3)}{1 - G_1 G_2 G_3 G_4} \quad ...(1.118i)$$

$$\therefore \quad \frac{x_2}{R} = \frac{G_2(1 - G_1 G_3)}{1 - G_1 G_2 G_3 G_4}$$

$$\therefore \quad \frac{C}{R} = \frac{X_1}{R} + \frac{X_2}{R}$$

$$\frac{C}{R} = \frac{G_1 + G_2 - G_1 G_2 G_4 - G_1 G_2 G_3}{1 - G_1 G_2 G_3 G_4}$$

Ans.

Example 1.34. Determine the transfer function C_1/R_1, C_2/R_2, C_1/R_2 and C_2/R_1 from the block diagram shown in fig. 1.93.

(R.M.L. Univ. 2002, Control System)

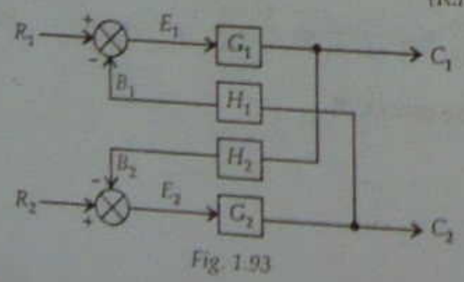


Fig. 1.93

Solution :

$$E_2 = R_2 - B_2$$

$$B_2 = C_1 H_2 \quad ...(1.119a)$$

$$C_2 = E_2 G_2 \quad ...(1.119b)$$

$$...(1.119c)$$

put the value of E_2 from 1.119*a* in 1.119*c*

$$C_2 = G_2(R_2 - B_2) \quad ...(1.119d)$$

from (1.119*b*) put the value of B_2 in (1.119*d*)

$$C_2 = G_2(R_2 - C_1 H_2) \quad ...(1.119e)$$

$$B_1 = C_2 H_1 \quad ...(1.119f)$$

put the value of C_2 from (1.119*e*.) in (1.119*f*)

$$B_1 = G_2 H_1 (R_2 - C_1 H_2) \quad ...(1.119g)$$

$$E_1 = R_1 - B_1$$

$$E_1 = R_1 - G_2 H_1 (R_2 - C_1 H_2) \quad ...(1.119h)$$

$$\text{Also,} \quad C_1 = E_1 G_1 . \quad ...(1.119i)$$

from (1.119*h*) put the value of E_1 in (1.119*i*)

$$C_1 = G_1 [R_1 - G_2 H_1 (R_2 - C_1 H_2)] \quad ...(1.119j)$$

$$C_1(1 - G_1 G_2 H_1 H_2) = G_1 R_1 - G_1 G_2 H_1 R_2 \quad ...(1.119K)$$

When $R_1 = 0$ from (1.119K)

$$\boxed{\frac{C_1}{R_2} = \frac{-G_1 G_2 H_1}{1 - G_1 G_2 H_1 H_2}} \quad \textbf{Ans.} \quad ...(1.119l)$$

from (1.119*l*) put the value of C_1 in (1.119*e*)

$$C_2 = G_2\left[R_2 + \frac{G_1 G_2 H_1 H_2 R_2}{1 - G_1 G_2 H_1 H_2}\right]$$

$$\therefore \quad \boxed{\frac{C_2}{R_2} = \frac{G_2}{1 - G_1 G_2 H_1 H_2}} \quad \textbf{Ans.} \quad ...(1.119m)$$

When $R_2 = 0$, (1.119K) becomes

$$C_1 = G_1 [R_1 + C_1 G_2 H_1 H_2]$$

$$C_1 = G_1 R_1 + C_1 G_1 G_2 H_1 H_2$$

$$\therefore \quad \boxed{\frac{C_1}{R_1} = \frac{G_1}{1 - G_1 G_2 H_1 H_2}} \quad \textbf{Ans.} \quad ...(1.119n)$$

When $R_2 = 0$, (1.119*e*) becomes

$$C_2 = -G_2 C_1 H_2$$

from (1.119*n*) put the value of C_1

$$\therefore \quad C_2 = \frac{-G_1 G_2 R_1 H_2}{1 - G_1 G_2 H_1 H_2}$$

$$\boxed{\frac{C_2}{R_1} = \frac{-G_1 G_2 H_2}{1 - G_1 G_2 H_1 H_2}} \quad \textbf{Ans.} \quad ...(1.119n')$$

Example 1.35. Determine the ratio $C(s)/R(s)$.

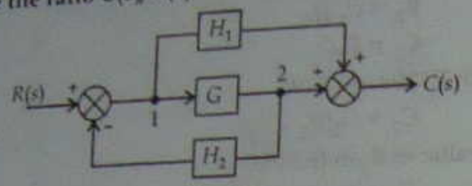
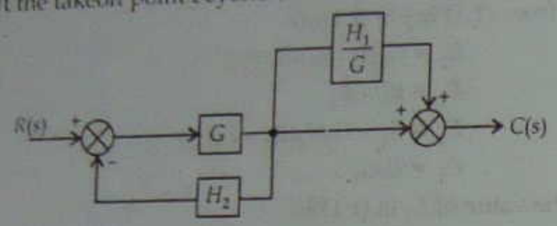
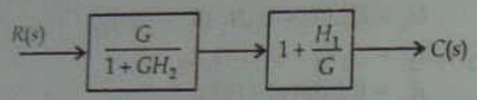


Fig. 1.94.

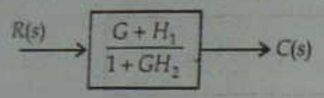
Solution : Step 1 : Shift the takeoff point beyond block G.



Step 2 :



Step 3 :



$$\therefore \quad \frac{C(s)}{R(s)} = \frac{G + H_1}{1 + GH_2}$$ **Ans.**

Example 1.36. Find the ratio $C(s)/R(s)$ of the system shown in fig. 1.95.

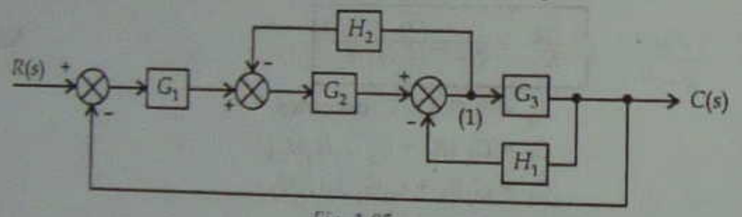
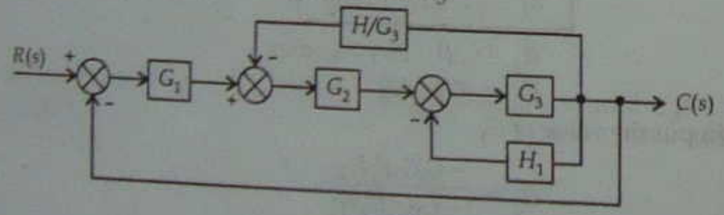
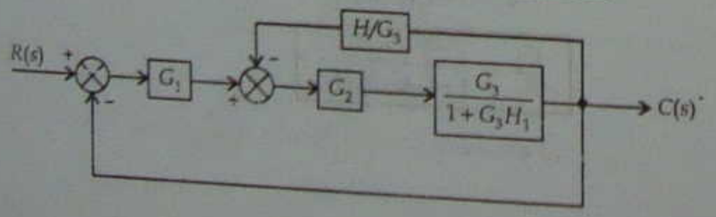


Fig. 1.95

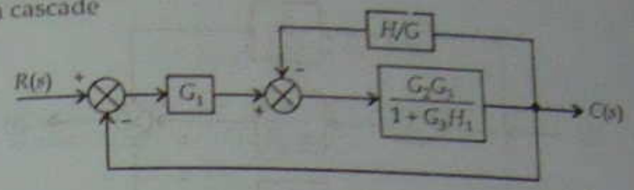
Solution : Step 1 : Shift the take off point beyond block G_3.



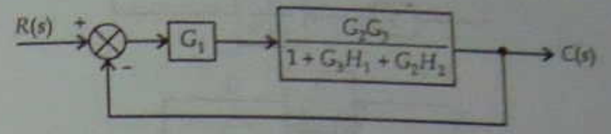
Step 2 :



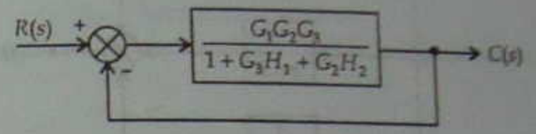
Step 3 : Blocks are in cascade



Step 4 :



Step 5 :



$$\therefore \quad \frac{C(s)}{R(s)} = \frac{G_1G_2G_3}{1 + G_3H_1 + G_2H_2 + G_1G_2G_3}$$ **Ans.**

Example 1.37 Determine the transfer matrix MIMO system of example 1.34.

Solution :

$$G_{11} = \left.\frac{C_1}{R_1}\right|_{R_2 = 0}, \quad G_{21} = \left.\frac{C_2}{R_1}\right|_{R_2 = 0}$$

$$G_{12} = \left.\frac{C_1}{R_2}\right|_{R_1 = 0}, \quad G_{22} = \left.\frac{C_2}{R_2}\right|_{R_1 = 0}$$

$$\therefore \quad G(s) = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$$

$$= \begin{bmatrix} G_1 & -G_1G_2H_1 \\ -G_1G_2H_2 & G_2 \end{bmatrix} \frac{1}{1 - G_1G_2H_1H_2}$$ **Ans.**

Example 1.38. Find the transfer function of the system shown in fig. 1.96.

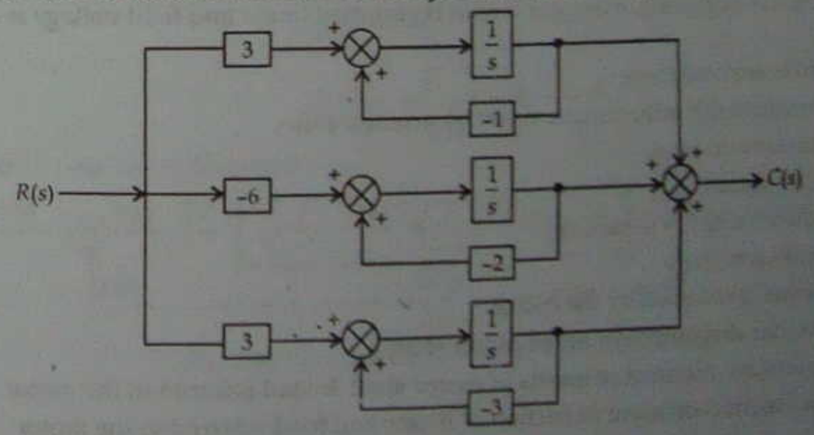
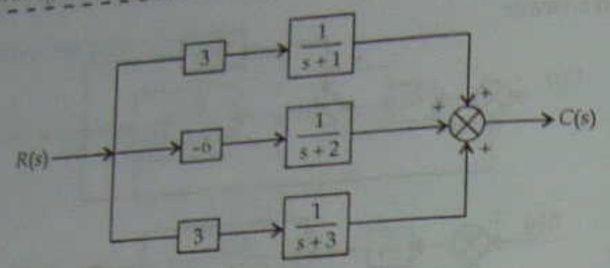
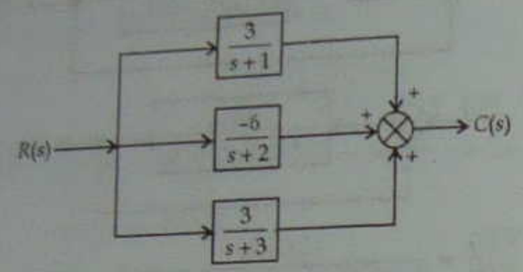


Fig. 1.96.

Solution : Step 1 :



Step 2 :



Step 3 :

$$\frac{C(s)}{R(s)} = \frac{3}{s+1} + \frac{-6}{s+2} + \frac{3}{s+3}$$

$$\frac{C(s)}{R(s)} = \frac{6}{s^3+6s^2+11s+6}$$ **Ans.**

1.23. BLOCK DIAGRAM AND TRANSFER FUNCTION OF D.C MOTOR

1.23.1 Armature Controlled d.c. Motor

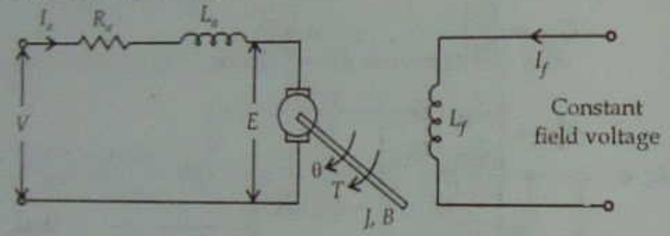


Fig. 1.97.

Consider the armature controlled d.c. motor and assume that the demagnetizing effect of armature reaction is neglected, magnetic circuit is assumed linear and field voltage is constant *i.e.* I_f = constant.

Let R_a = Armature resistance

L_a = Armature self inductance caused by armature flux

i_a = armature current

i_f = field current

E = Induced e.m.f in armature

V = Applied voltage

T = Torque developed by the motor

θ = Angular displacement of the motor shaft

J = Equivalent moment of inertia of motor shaft & load referred to the motor

B = equivalent coefficient of friction of motor and load referred to the motor

Apply *KVL* in armature circuit

$$V = R_a i_a + L\frac{di_a}{dt} + E \quad \text{...(1.120)}$$

Since, field current I_f is constant, the flux ϕ will be constant

When armature is rotating, an e.m.f is induced

$$E \propto \phi\,\omega$$

$$E = K_b\omega$$

or,
$$E = K_b\frac{d\theta}{dt} \quad \text{...(1.121)}$$

Where, ω = angular velocity

K_b = back e.m.f constant

Now, the torque T delivered by the motor will be the product of armature current and flux

$$T \propto \phi\, i_a$$

$$T = K\, i_a \quad \text{...(1.122)}$$

where K = motor torque constant

The dynamic equation with moment of inertia & coeff. of friction will be

$$T = J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt} \quad \text{...(1.123)}$$

Take the laplace transform of equations 1.120, 1.121, 1.122 & 1.123

$$V(s) - E(s) = I_a(s)\,(R_a + SL_a)$$

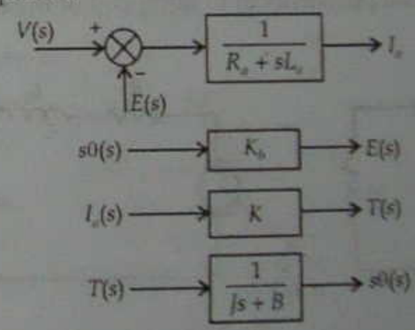
$$E(s) = K_b\, S\theta(s)$$

$$T(s) = K\, I_a(s)$$

$$T(s) = (s^2J + sB)\,\theta(s)$$

$$T(s) = (SJ + B)\,S\theta(s)$$

The block diag. for each equation



Combine all four block diagrams

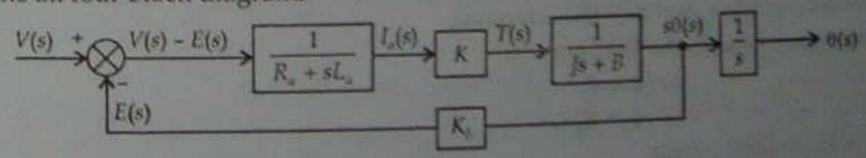
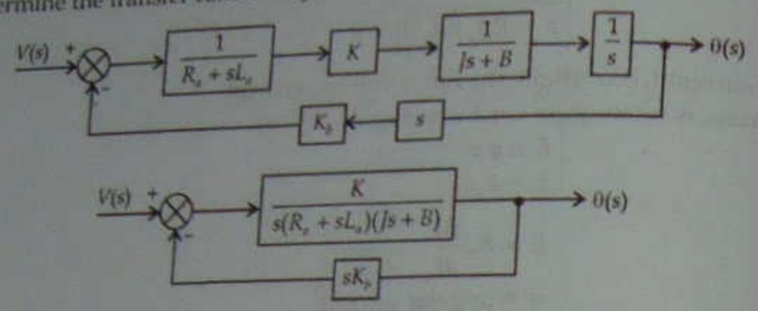


Fig. 1.98. Block diagram of armature controlled d.c motor

Now determine the transfer function by block reduction method.



$$\frac{\theta(s)}{V(s)} = \frac{K}{(Ra + sLa)(Js + B)s + KK_b s} \quad \text{...(1.124)}$$

Equation 1.124 can be written as

$$\frac{\theta(s)}{V(s)} = \frac{K}{R_a\left(1 + s\frac{L_a}{R_a}\right)sB\left(1 + s\cdot\frac{J}{B}\right) + KK_b s}$$

Put $\frac{L_a}{R_a} = \tau_a$ time constant of armature circuit

$\frac{J}{B} = \tau_m$ = mechanical time constant

equation 1.124 becomes

$$\frac{\theta(s)}{V(s)} = \frac{K}{sR_aB(1 + s\tau_a)(1 + s\tau_m) + KK_b s} \quad \text{...(1.125)}$$

From the block diag. 1.98 it is clear that it is a closed loop system. The effect of the back e.m.f is represented by the feedback signal proportional to the speed of the motor.

1.23.2. Field Control d.c. Motor

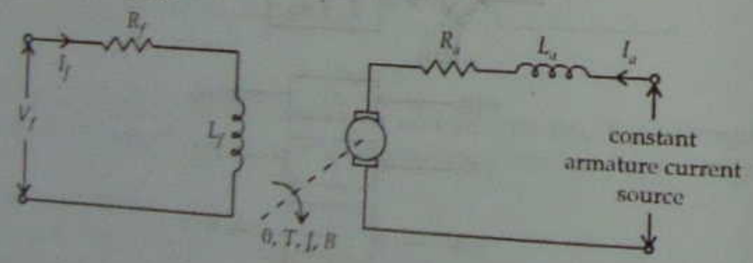


Fig. 1.99

1. A Constant current I_a is fed to the armature.
2. Flux is proprotional to the field current.

$$\phi \alpha I_f$$
$$\phi = K_f I_f$$

3. Apply *KVL* in field circuit

$$V_f = R_f I_f + L_f \frac{d}{dt} I_f \quad \text{...(1.126)}$$

...(1.127)

4. Torque developed by the motor is proportional to the flux and armature current.

$$T \alpha \phi I_a$$

From eq^n 1.126 put the value of ϕ

$$T = K' K_f I_a I_f$$
$$T = KK_f I_f$$

where $K = K' I_a$ ...(1.128)

Dynamic equation of torque in terms of J & B

$$T = J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt} \quad \text{...(1.129)}$$

Laplace transformation of eq^n (1.127), (1.128) & (1.129)

$$V_f(s) = R_f I_f(s) + sL_f I_f(s)$$
$$= I_f(s)\,[R_f + sL_f]$$
$$I_f = \frac{V_f(s)}{R_f + sL_f} \quad \text{...(1.130)}$$
$$T(s) = KK_f I_f(s) \quad \text{...(1.131)}$$
$$T(s) = \theta(s)\,[S^2J + sB] \quad \text{...(1.132)}$$

Put the value of $I_f(s)$ from (1.130) in (1.131)

$$T(s) = K \cdot K_f \frac{V_f(s)}{R_f + sL_f} \quad \text{...(1.133)}$$

From equation (1.132) & (1.133)

$$\theta(s)\,[s^2J + sB] = K \cdot K_f \frac{V_f(s)}{R_f + sL_f}$$

$$\therefore \quad \frac{\theta(s)}{V_f(s)} = \frac{KK_f}{s(sJ + B)(R_f + sL_f)} \quad \text{...(1.134)}$$

Equation 1.134 can be written as

$$\frac{\theta(s)}{V_f(s)} = \frac{KK_f}{R_fBs\left(1 + s\frac{J}{B}\right)\left(1 + s\frac{L_f}{R_f}\right)}$$

$$\frac{\theta(s)}{V_f(s)} = \frac{KK_f}{R_fBs(1 + s\tau_m)(1 + s\tau_f)}$$

where $\tau_m = \frac{J}{B}$ = mechanical time constant

$\tau_f = \frac{L_f}{R_f}$ = time constant for field circuit

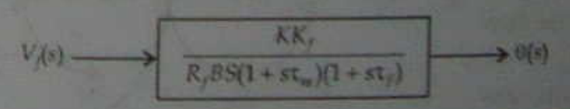
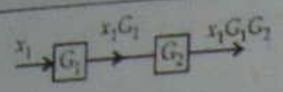
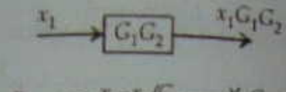
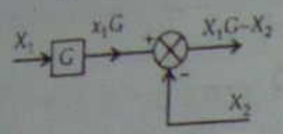
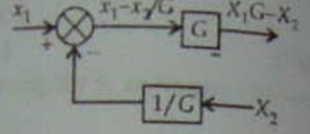
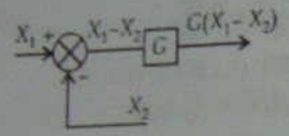
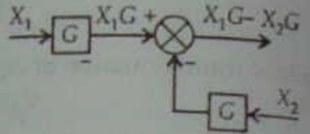
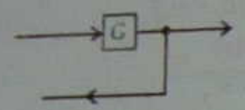
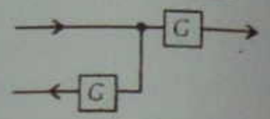
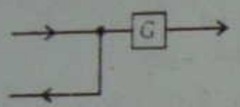
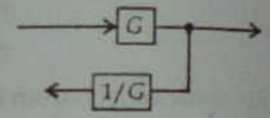
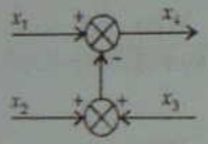
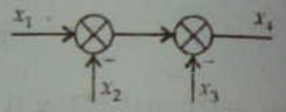
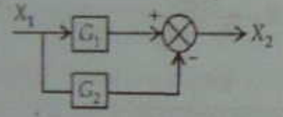
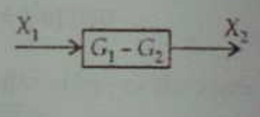
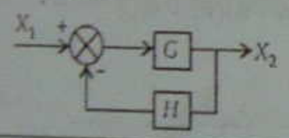
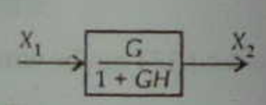


Fig. 1.100. Block diagram of field controlled d.c. motor.

Rule	Original diagram	Equivalent diagram
1. Blocks in cascade		
2. Moving a summing point ahead of a block		
3. Moving a summing point beyond a block		
4. Moving a take-off point ahead of a block		
5. Moving a take-off point beyond a block		
6. Rearrangement of summing point		
7. Eliminating a forward loop		
8. Eliminating feedback loop		

1.24. SIGNAL FLOW GRAPH

The process of block diagram reduction technique is time consuming because at every stage modified block diagram is to be redrawn. A simple method was developed by S.J. Mason which is known as signal flow graph. This method is very simple and does not require any reduction technique. Signal flow graph is applicable to the linear systems.

A signal flow graph is a diagram which represents a set of simultaneous equations. Signal flow graph consists of nodes and these nodes are connected by a directed line called branches. Every branch of signal flow graph having an arrow, which reprsents the flow of signal.

The following terms are associated with the signal flow graph.

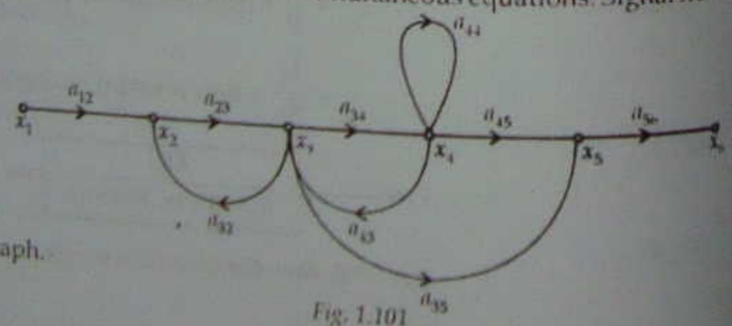


Fig. 1.101

1. Input node or source node : An input node is a node which has only outgoing branches. *e.g.* x_1 is the input node.
2. Output node or sink node : An output node is a node that has only one or more incoming branches.*e.g.* x_6 is the output node.
3. Mixed nodes : A node having incoming and outgoing branches is known as mixed nodes. *e.g.* x_2, x_3, x_4, and x_5 are the mixed nodes.
4. Transmittance : Transmittance also known as transfer function, which is normally written on the branch near the arrow.*e.g.* a_{12}, a_{23} *etc.*
5. Forward path : Forward path is a path which originates from the input node and terminates at the output node and along which no node is traversed more than once.
   e.g. in fig 1.101 there are two forward paths.
   1. x_1 to x_2 to x_3 to x_4 to x_5 to x_6
   2. x_1 to x_2 to x_3 to x_5 to x_6
6. Loop : Loop is a path that originates and terminates on the same node and along which no other node is traversed more than once.
   e.g. x_2 to x_3 to x_2
   x_3 to x_4 to x_3
7. Self loop : It is a path which originates and terminates on the same node. *e.g.* x_4 to x_4
8. Path gain : The product of the branch gains along the path is called path gain. *e.g.* The gain of the path x_1 to x_2 to x_3 to x_4 to x_5 to x_6 is $a_{12}\,a_{23}\,a_{34}\,a_{45}\,a_{56}$
9. Loop gain : The gain of the loop is known as loop gain. *e.g.* the gain of the loop x_2 to x_3 to x_2 is $a_{23}\,a_{32}$.
10. Non-touching loops : Non touching loops having no common nodes branch and paths. *e.g.* The loops x_2 to x_3 to x_2, and x_4 to x_4 are non touching loops.

1.25. CONSTRUCTION OF SIGNAL FLOW GRAPH FROM EQUATIONS

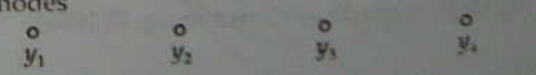
consider the following sets of equations

$$y_2 = t_{21}y_1 + t_{23}y_3$$
$$y_3 = t_{32}y_2 + t_{33}y_3 + t_{31}y_1$$
$$y_4 = t_{43}y_3 + t_{42}y_2$$
$$y_5 = t_{54}y_4$$
$$y_6 = t_{65}y_5 + t_{64}y_4$$

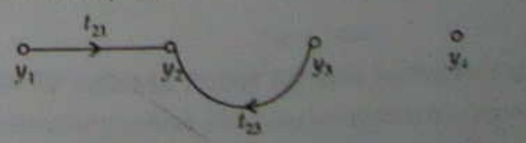
where y_1 is the input and y_6 is the output.

First of all draw the nodes. In the given example there are six nodes. From the first equation it is clear that the y_2 is the sum of two signals. Similarly, y_3 is the sum of three signals & so on. Insert the branches with proper transmittance to connect the nodes.

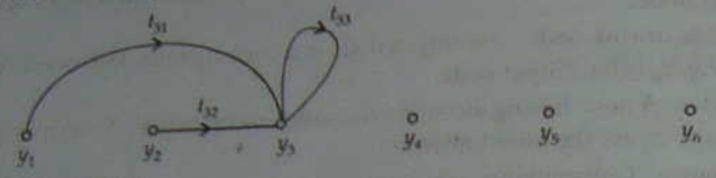
Step 1 : Draw the nodes



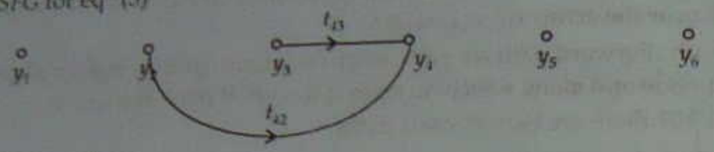
Step 2 : Draw the SFG for eq^n (1)



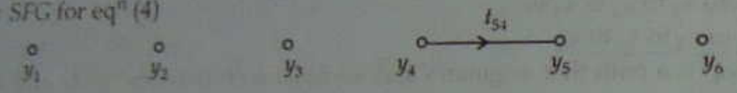
Step 3 : Draw the *SFG* for eq^{n} (2)



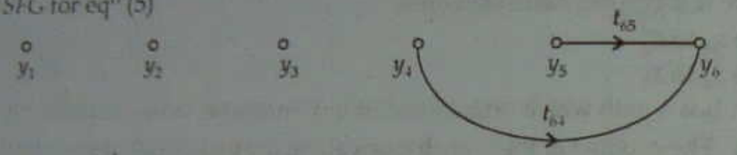
Step 4 : Draw *SFG* for eq^{n} (3)



Step 5 : Draw *SFG* for eq^{n} (4)



Step 6 : Draw *SFG* for eq^{n} (5)



Step 7 : Draw the complete signal flow graph with the help of abvoe graphs.

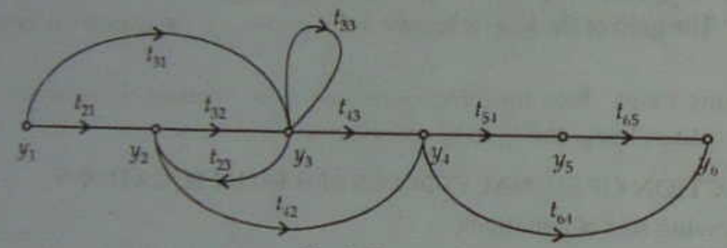


Fig. 1.102.

1.26. SIGNAL FLOW GRAPH FOR DIFFERENTIAL EQUATIONS

Consider the following defferential equation

$$y''' + 3y'' + 5y' + 2y = x \qquad ...(1.135)$$

Step 1 : Solve the eqn 1.135 for the highest order

$$y''' = x - 3y'' - 5y' - 2y$$

Step 2 : Consider the left hand term (highest order derivative) as dependent variable and all other terms on right hand side as independent variables.

construct the branches of signal flow graph as shown in fig. (1.103*a*).

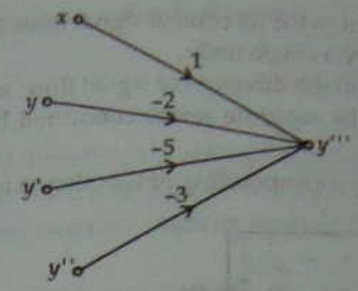


Fig. 1.103 (a)

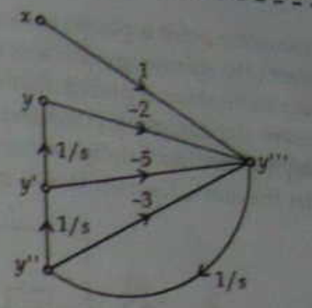


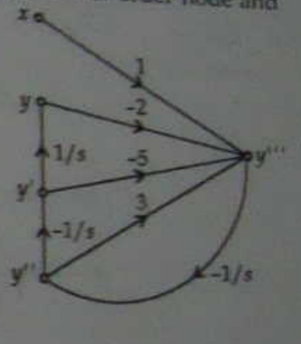
Fig. 1.103(b)

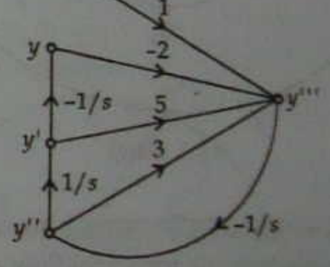
Step 3 : Connect the nodes of highest order derivative to the node whose order is lower than this and so on. The flow of the signal will be from higher node to the lower order node and transmittance will be $^{1}/_{s}$ as shown in fig. 1.103*b*.

Step 4 : Reverse the sign of a branch connecting the p^{th} node to the q^{th} node of a signal flow graph without disturbing the transfer function.

Consider the fig. 1.103*b*, reverse the sign of the branch connecting y''' to y'', it is necessary to reverse the sign of all remaining branches entering as well as leaving the q*th* node.

Similarly, reverse the sign of branch connecting y'' to y'





By reversing the sign, we have already reverse the sign of branch connecting y' to y and therefore further reversal of sign is not required.

Step 5 : Redraw the signal flow graph (SFG)

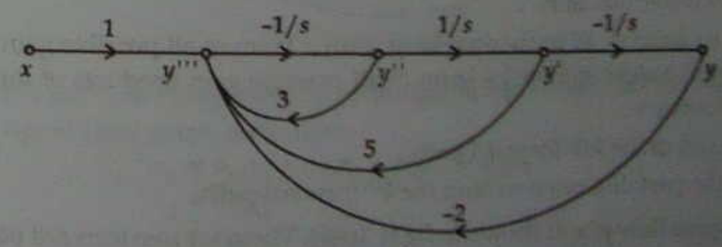


Fig. 1.103(c)

1.27. CONSTRUCTION OF SIGNAL FLOW GRAPH FROM BLOCK DIAGRAM

Rules 1. All variables, summing points and take off points are represented by nodes.

2. If a summing point is placed before a take off point in the dirction of signal flow, in such case represent the summing point and take off point by a single node.
3. If a summing point is placed after a takeoff point in the direction of signal flow, in such case, represent the summing point and takeoff point by separate nodes connected by a branch having transmittance unity.

Consider the block diagram shown in fig 1.104*a*, the corresponding *SFG* is shown in fig 1.104*b*.

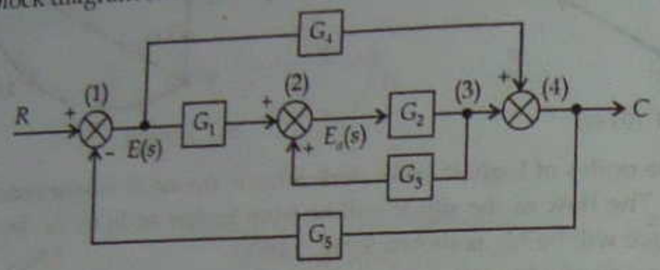


Fig. 1.104(a)

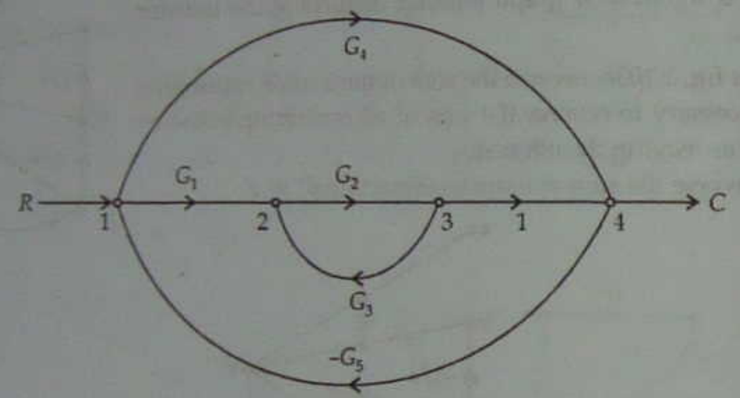


Fig. 1.104(b)

1.28. MASON'S GAIN FORMULA

The overall transmittance or graph transmittance between the source node and sink node is given by Mason's gain formula. Mason's gain formula for signal flow graph is as follows .

$$T = \frac{\sum g_k \Delta_k}{\Delta} \qquad ...(1.136)$$

where, T = transfer function

Δ = 1 – [sum of all individual loop gain] + [sum of all possible gain products of two non-touching loops] – [sum of all possible gain products of three non-touching loops] + ...

g_k = gain of the kth forward path

Δ_k = the part of Δ not touching the k^{th} forward path.

Consider the signal flow graph shown in fig (1.104*b*). There are two forward paths (i) 1-2-3-4 (*ii*) 1-4 therefore gain of the two paths will be

$$g_1 = G_1G_2$$
$$g_2 = G_4$$

There are three individual loops

$$L_1 = G_2G_3 \qquad (2\text{-}3\text{-}2)$$
$$L_2 = -G_1G_2G_5 \qquad (1\text{-}2\text{-}3\text{-}4\text{-}1)$$
$$L_3 = -G_4G_5 \qquad (1\text{-}4\text{-}1)$$

Since all three loops touching the forward path g_1, therefore $\Delta_1 = 1\text{-}0 = 1$ The first loop L_1 do nc touch the forward path g_2, therefore $\Delta_2 = 1 - G_2G_3$

There are two non touching loops L_1 & L_3

$$\therefore \quad L_1L_3 = -G_2G_3G_4G_5$$

Apply mason's gain formula

$$T = \frac{C}{R} = \frac{g_1\Delta_1 + g_2\Delta_2}{\Delta}$$

$$T = \frac{G_1G_2 + G_4(1 - G_2G_3)}{1 - (G_2G_3 - G_1G_2G_5 - G_4G_5) - (-G_2G_3G_4G_5)}$$

$$\frac{C}{R} = \frac{G_1G_2 + G_4 - G_2G_3G_4}{1 - G_2G_3 + G_1G_2G_5 + G_4G_5 - G_2G_3G_4G_5}$$

Note : where $\Delta = 1 - (L_1 + L_2 + L_3) + (L_1L_3)$

Example 1.39. Draw the signal flow graph for the following set of equations.

$$x_2 = x_1 + ax_5$$
$$x_3 = bx_2 + cx_4$$
$$x_4 = dx_2 + ex_3$$
$$x_5 = fx_4 + gx_3$$
$$x_6 = x_5$$

Solution : The required signal flow graph is

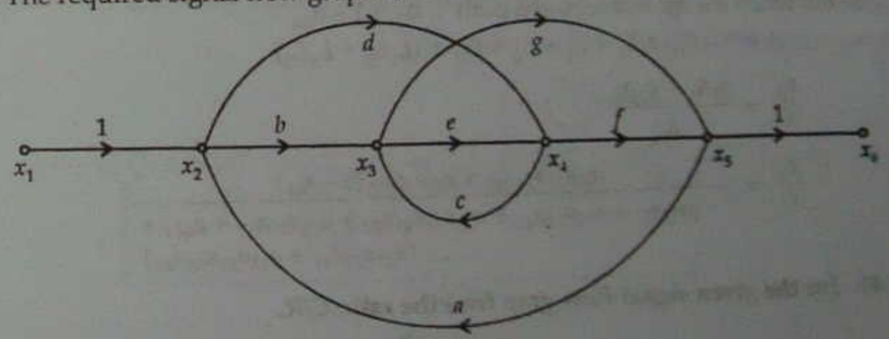


Fig. 1.105.

Example 1.40. For the system represented by the given equations find the transfer function x_5/x_1 by the help of signal flow graph technique.

$$x_2 = a_{12}x_1 + a_{32}x_3 + a_{42}x_4 + a_{52}x_5$$
$$x_3 = a_{23}x_2$$
$$x_4 = a_{34}x_3 + a_{44}x_4$$
$$x_5 = a_{35}x_3 + a_{45}x_4$$

where x_1 is the input variable and x_5 is the output variable. *(R.M.L. University Faizabad, 2002)*

Solution : There are two forward paths (1) x_1 to x_2 to x_3 to x_4 + to x_5

(2) x_1 to x_2 to x_3 to x_5

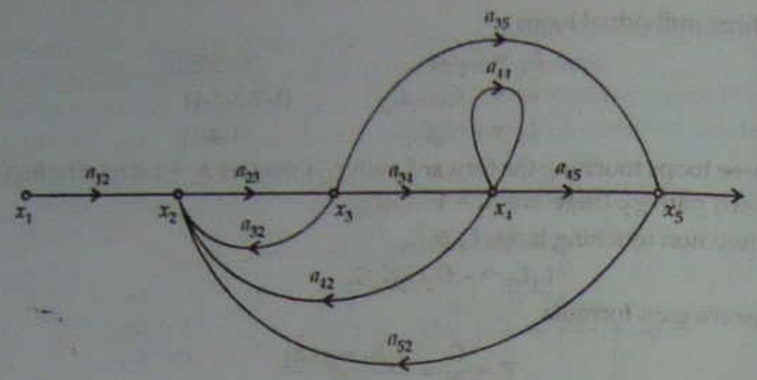


Fig. 1.106.

The gain of the paths $g_1 = a_{12}\, a_{23}\, a_{34}\, a_{45}$

$g_2 = a_{12}\, a_{23}\, a_{35}$

Gain of individual loops

$L_1 = a_{23}\, a_{32}$

$L_2 = a_{23}\, a_{34}\, a_{42}$

$L_3 = a_{23}\, a_{34}\, a_{45}\, a_{52}$

$L_4 = a_{23}\, a_{35}\, a_{52}$

$L_5 = a_{44}$

Gain of two nontouching loops

$L_1 L_5 = a_{23}\, a_{32}\, a_{44}$

$L_4 L_5 = a_{23}\, a_{35}\, a_{52}\, a_{44}$

Since, all the loops touches the forward path (1) $\therefore \Delta_1 = 1 - 0 = 1$

loop L_5 do not touch the second forward path $\therefore \Delta_2 = 1 - a_{44}$

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5) + (L_1 L_5 + L_4 L_5)$$

$$\therefore \quad \frac{x_5}{x_1} = \frac{g_1\Delta_1 + g_2\Delta_2}{\Delta}$$

$$\frac{x_5}{x_1} = \frac{a_{12}a_{23}a_{34}a_{45} + a_{12}a_{23}a_{35}(1 - a_{44})}{1 - (a_{23}a_{32} + a_{23}a_{34}a_{42} + a_{23}a_{34}a_{45}a_{52} + a_{23}a_{35}a_{52} + a_{44}) + (a_{23}a_{32}a_{44} + a_{23}a_{35}a_{52}a_{44})}$$

Example 1.41. For the given signal flow grap find the ratio *C/R*.

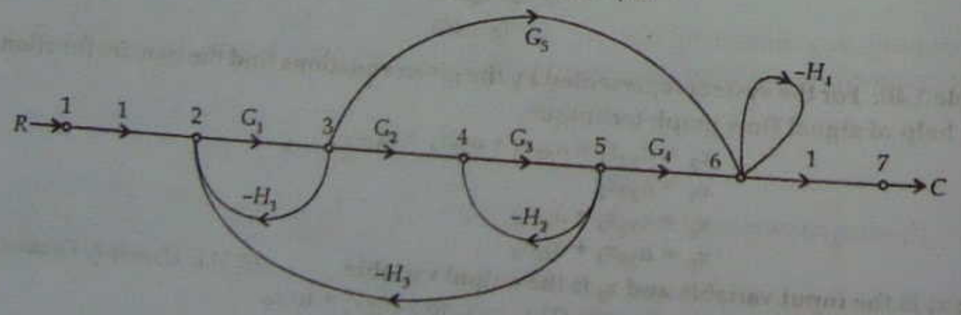


Fig. 1.107.

The gain of the forward paths $g_1 = G_1G_2G_3G_4$

$g_2 = G_1G_5$

Individual loops $L_1 = -G_1H_1$

$L_2 = -G_3H_2$

$L_3 = -G_1G_2G_3H_3$

$L_4 = -H_4$

Two non-touching loop $L_1L_2 = G_1H_1G_3H_2$

$L_1L_4 = G_1H_1H_4$

$L_2L_4 = G_3H_2H_4$

$L_3L_4 = G_1G_2G_3H_3H_4$

Three non-touching loops $L_1L_2L_4 = -G_1H_1G_3H_2H_4$

$\Delta_1 = 1 - 0$

$\Delta_2 = 1 + G_3H_2$

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (L_1L_2 + L_1L_4 + L_2L_4 + L_3L_4) - (L_1L_2L_4)$$

$$\therefore \quad \frac{C}{R} = \frac{g_1\Delta_1 + g_2\Delta_2}{\Delta}$$

$$\frac{C}{R} = \frac{G_1G_2G_3G_4 + G_1G_5(1 + G_3H_2)}{1 + G_1H_1 + G_3H_2 + G_1G_2G_3H_3 + H_4 + G_1G_3H_1H_2 + G_1H_1H_4 + G_3H_2H_4 + G_1G_2G_3H_3H_4 + G_1G_5H_1H_2H_4}$$

Example 1.42. Obtain signal flow graph representation for a system whose block diagram is given below and using Mason's gain formula determine the ratio $\frac{C}{R}$.

(R.M.L. Univ. Faizabad 2001 Linear system theory)

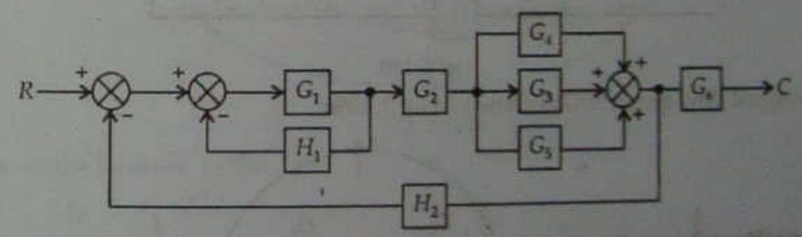


Fig. 1.108.

Solution :

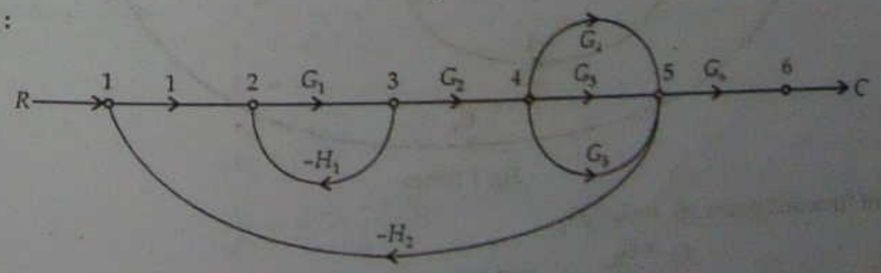


Fig. 1.108(a)

The signal flow graph is shown in fig 1.108*a*. Consider the SFG

$$g_1 = G_1G_2G_3G_6 \qquad \Delta_1 = 1 - 0 = 1$$
$$g_2 = G_1G_2G_4G_6 \qquad \Delta_2 = 1 - 0 = 1$$
$$g_3 = G_1G_2G_5G_6 \qquad \Delta_3 = 1 - 0 = 1$$

Individual loops

$$L_1 = -G_1H_1$$
$$L_2 = -G_1G_2G_3H_2$$
$$L_3 = -G_1G_2G_4H_2$$
$$L_4 = -G_1G_2G_5H_2$$

Two non touching loops = None

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4)$$

$$\therefore \quad \frac{C}{R} = \frac{g_1\Delta_1 + g_2\Delta_2}{\Delta}$$

$$\frac{C}{R} = \frac{G_1G_2G_3G_6 + G_1G_2G_4G_6 + G_1G_2G_5G_6}{1 + G_1H_1 + G_1G_2G_3H_2 + G_1G_2G_4H_2 + G_1G_2G_5H_2}$$

Example 1.43. Obtain the Transfer function for $\frac{C(s)}{R(s)}$ given block diagram using block diag reduction technique or signal flow graph (Mason's gain formula).

(R.M.L. Univ. Faizabad 2001, Control System)

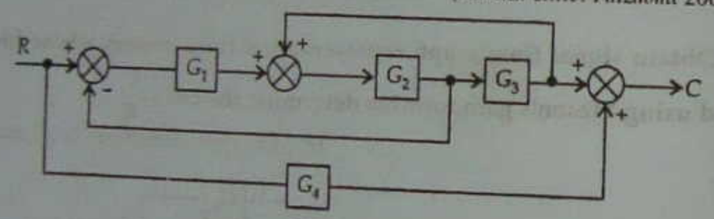


Fig. 1.109.

Solution : By signal flow graph SFG is shown in fig 1.109*a*.

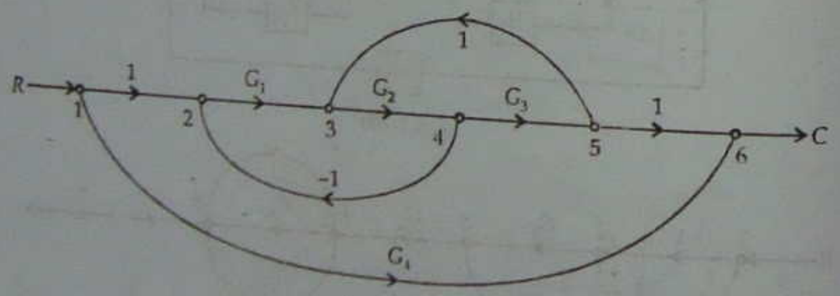


Fig. 1.109(a)

Gain of forward paths : $g_1 = G_1G_2G_3$

$g_2 = G_4$

$\Delta_1 = 1$

$\Delta_2 = 1 + G_1G_2 - G_2G_3$

Individual loops : $L_1 = -G_1G_2$

$L_2 = G_2G_3$

Two non-touching loops : None

$$\therefore \quad \frac{C}{R} = \frac{g_1\Delta_1 + g_2\Delta_2}{\Delta}$$

$$\frac{C}{R} = \frac{G_1G_2G_3 + G_4(1 + G_1G_2 - G_2G_3)}{1 + G_1G_2 - G_2G_3}$$

Example 1.44. Draw the signal flow graph and determine $\frac{C}{R}$ for the block diag. shown in fig 1.110.

(KNIT, Sultanpur, 99-2000)

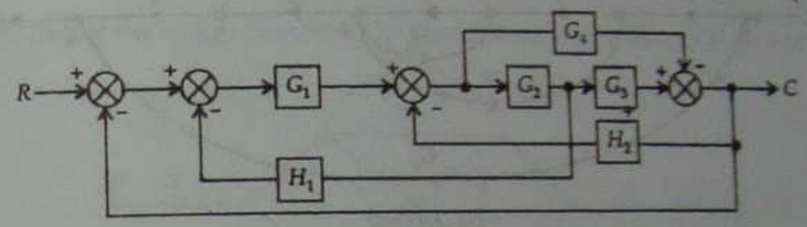


Fig. 1.110.

Solution : The *SFG* is shown in fig 1.110*a*

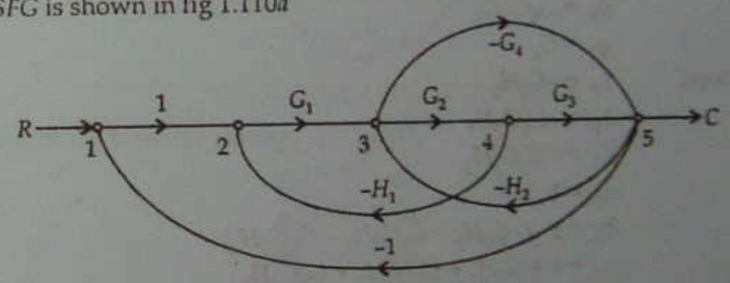


Fig. 1.110a

The gain of the forward paths

$$g_1 = G_1G_2G_3 \qquad \Delta_1 = 1$$
$$g_2 = -G_1G_4 \qquad \Delta_2 = 1$$

Individual loops $L_1 = -G_1G_2H_1$

$L_2 = -G_2G_3H_2$

$L_3 = -G_1G_2G_3$

$L_4 = G_1G_4$

$L_5 = G_4H_2$

$$\frac{C}{R} = \frac{g_1\Delta_1 + g_2\Delta_2}{\Delta}$$

$$\frac{C}{R} = \frac{G_1G_2G_3 - G_1G_4}{1 + G_1G_2H_1 + G_2G_3H_2 + G_1G_2G_3 - G_1G_4 - G_4H_2}$$

Example 1.45. Obtain the transfer function C/R of the block diagram shown in fig. 1.111. (KNIT, Sultanpur)

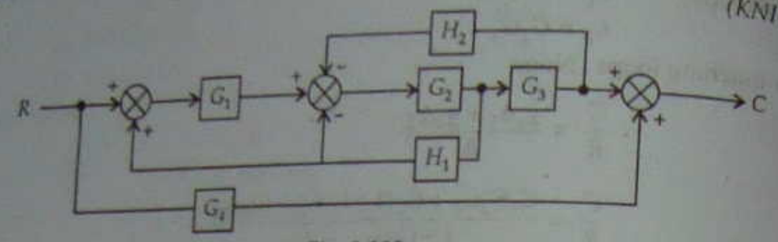


Fig. 1.111.

Solution :

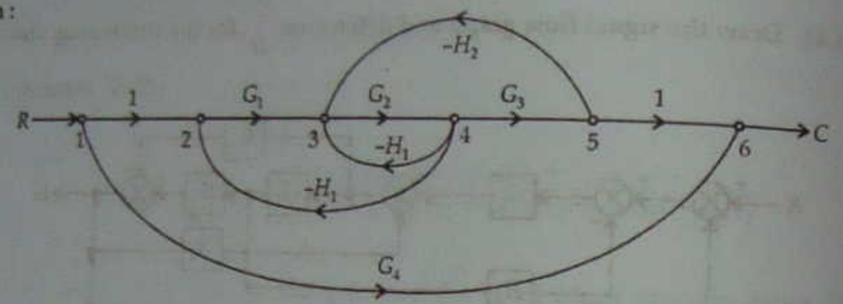


Fig. 1.111(a)

The gain of the forward paths

$$g_1 = G_1G_2G_3$$
$$g_2 = G_4$$

Individual loops

$$L_1 = -G_2H_1$$
$$L_2 = -G_1G_2H_1$$
$$L_3 = -G_2G_3H_2$$
$$\Delta_1 = 1$$
$$\Delta_2 = 1 - (-G_2H_1 - G_1G_2H_1 - G_2G_3H_2)$$
$$\Delta_2 = 1 + G_2H_1 + G_1G_2H_1 + G_2G_3H_2$$
$$\Delta = 1 - (L_1 + L_2 + L_3)$$
$$\Delta = 1 + G_2H_1 + G_1G_2H_1 + G_2G_3H_2$$
$$\frac{C}{R} = \frac{G_1G_2G_3 + G_4(1 + G_2H_1 + G_1G_2H_1 + G_2G_3H_2)}{1 + G_2H_1 + G_1G_2H_1 + G_2H_2G_3}$$

Example 1.46. Draw the signal flow graph for the network shown in fig 1.112, take V_3 as output node.

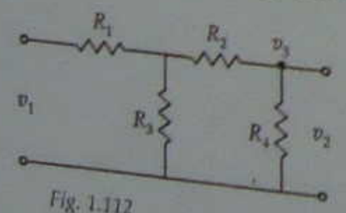


Fig. 1.112

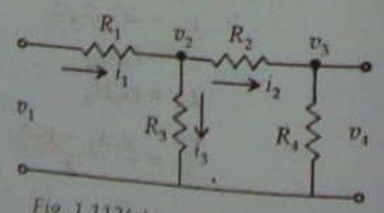


Fig. 1.112(a)

Solution : From 1.112*a* :

$$i_1 = \frac{V_1}{R_1} - \frac{V_2}{R_1}$$
$$V_2 = i_3R_3 = R_3(i_1 - i_2) = R_3i_1 - R_3i_2$$
$$i_2 = \frac{V_2}{R_2} - \frac{V_3}{R_2}$$
$$V_3 = R_4i_2$$

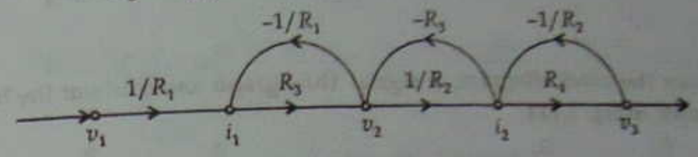


Fig. 1.112(b) Signal flow graph.

Example 1.47. Obtain the transfer function C/R from the signal flow graph shown in fig. 1.113.

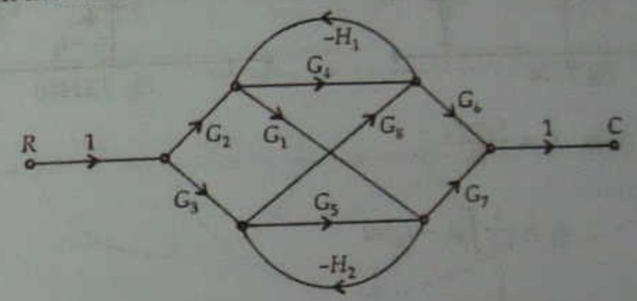


Fig. 1.113.

Solution : The gain of the forward paths $g_1 = G_2G_4G_6$

$$g_2 = G_3G_5G_7$$
$$g_3 = G_2G_1G_7$$
$$g_4 = G_3G_8G_6$$
$$g_5 = -G_2G_1H_2G_8G_6$$
$$g_6 = -G_3G_8H_1G_1G_7$$

Individual loops :

$$L_1 = -G_4H_1$$
$$L_2 = -G_5H_2$$
$$L_3 = H_1G_1H_2G_8$$

Two non touching loops :

$$L_1L_2 = G_4H_1G_5H_2$$
$$\Delta_1 = 1 + G_5H_2$$
$$\Delta_2 = 1 + G_4H_1$$
$$\Delta_3 = 1$$
$$\Delta_4 = 1$$
$$\Delta_5 = 1$$
$$\Delta_6 = 1$$

1.29. BLOCK DIAGRAM FROM SIGNAL FLOW GRAPH

Step 1 : For given signal flow graph, write the system equations.
Step 2 : At each node consider the incoming branches only.
Step 3 : Add all incoming signals algebraically at a node.
Step 4 : For + or – sign in system equations use a summing point
Step 5 : For the gain of each branch of signal flow graph draw the block having the same transfer function as the gain of the branch.

Consider the following examples :

Example 1.49. Draw the block diagram from the given signal flow graph.

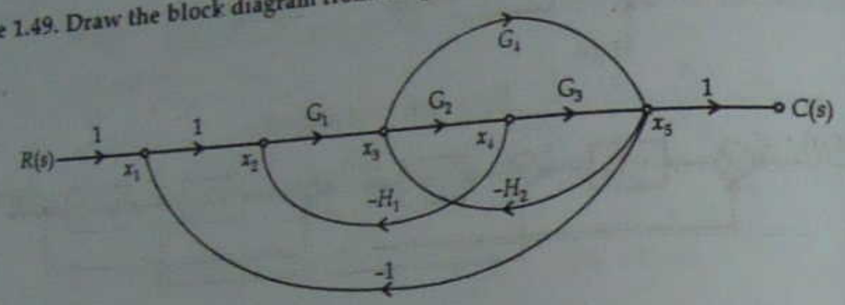


Fig. 1.115.

Solution : The system equations are

at node x_1, the incoming branches are from $R(s)$ and x_5

$$x_1 = 1\,R(s) - 1 \cdot x_5$$

at node x_2, there are two incoming branches

$$x_2 = 1 \cdot x_1 - H_1 x_4$$

at node x_3, there are two incoming branches

$$x_3 = G_1 x_2 - H_2 x_5$$

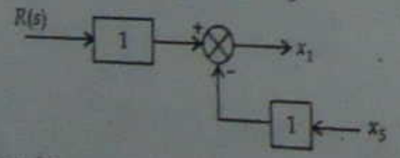
Similarly at node x_4 and x_5 the system equations are

$$x_4 = G_2 x_3$$

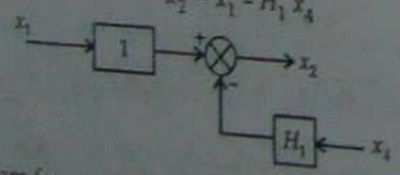
$$x_5 = G_3 x_4 + G_4 x_3$$

Draw the block diagram for each system equation

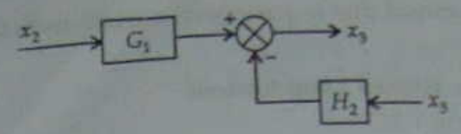
For $x_1 = R(s) - x_5$



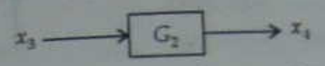
Block diagram for $x_2 = x_1 - H_1 x_4$



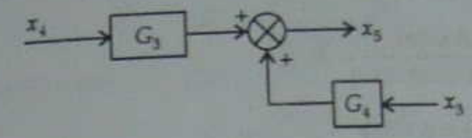
Block diagram for $x_3 = G_1 x_2 - H_2 x_5$



Block diagram for $x_4 = G_2 x_3$



Block diagram for $x_5 = G_3 x_4 + G_4 x_3$



Combining all above block diagrams

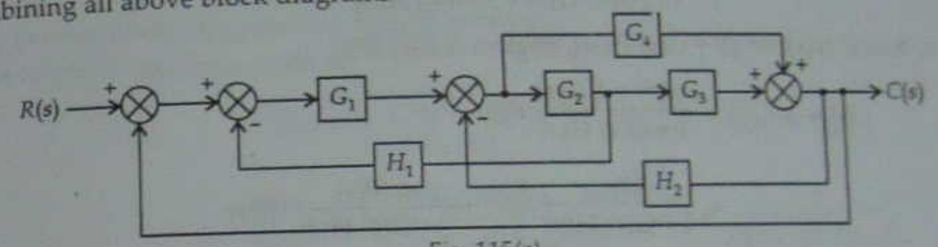


Fig. 115(a)

1.30. EFFECT OF PARAMETER VARIATIONS

In control systems, the feed back reduces the error, also reduces the sensitivity of the system to parameter variations. The parameter may vary due to some change in conditions. The variation in parameter affects the performance of the system. So, it is necessary to make the system in sensitive to such parameter variations.

1.30.1 Effect of Parameter Variations in Open Loop Control System

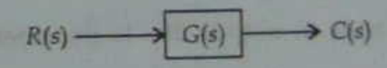


Fig. 1.116

The open loop control system is shown in fig 1.116.. From fig 1.116

$$\frac{C(s)}{R(s)} = G(s)$$

or, $$C(s) = G(s) \cdot R(s) \quad \ldots(1.137)$$

Let $\Delta G(s)$ = Change in $G(s)$ due to parameter variations

$\Delta C(s)$ = Corresponding change in output

From equation 1.137

$$C(s) + \Delta C(s) = [G(s) + \Delta G(s)]\, R(s)$$

$$C(s) + \Delta C(s) = G(s)\, R(s) + \Delta G(s)\, R(s)$$

Since, $$G(s)\, R(s) = C(s)$$

$$C(s) + \Delta C(s) = C(s) + \Delta G(s)\, R(s)$$

or, $$\Delta C(s) = \Delta G(s)\, R(s) \quad \ldots(1.138)$$

Equation 1.138 gives the change in output due to parameter variations in G(s) in open loop system.

1.30.2. Effect of Parameter Variations in Closed Loop System

The closed loop system is shown in fig 1.117

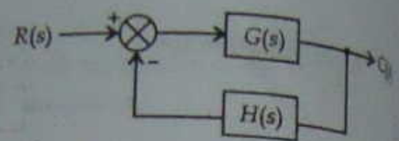


Fig. 1.117

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)\,H(s)} \qquad ...(1.139)$$

or, $$C(s) = \frac{G(s)}{1+G(s)\,H(s)}\,R(s)$$

$$C(s)+\Delta C(s) = \frac{G(s)+\Delta G(s)}{1+G(s)+\Delta G(s)\,H(s)}\cdot R(s)$$

$$= \frac{G(s)+\Delta G(s)}{1+\left[G(s)\,H(s)+\Delta G(s)\right]H(s)}\cdot R(s)$$

Since, $\Delta G(s)\,H(s) << [1 + G(s)\,H(s)]$, neglect $\Delta G(s)\,H(s)$

$$\therefore \quad C(s)+\Delta C(s) = \frac{G(s)+\Delta G(s)}{1+G(s)\,H(s)}\,R(s)$$

$$= \frac{G(s)}{1+G(s)\,H(s)}\,R(s) + \frac{\Delta G(s)}{1+G(s)\,H(s)}\cdot R(s)$$

Since, $$C(s) = \frac{G(s)}{1+G(s)\,H(s)}\,R(s)$$

$$C(s)+\Delta C(s) = C(s) + \frac{\Delta G(s)}{1+G(s)\,H(s)}\,R(s)$$

or, $$\Delta C(s) = \frac{\Delta G(s)}{1+G(s)\,H(s)}\,R(s) \qquad ...(1.140)$$

Equation 1.140 gives the change in output due to parameter variations in $G(s)$ in a closed loop system.

Generally $[G(s)\,H(s)] >> 1$

from equation 1.140 it is clear that the change in output is reduced due to parameter variations in G(s) by $[1 + G(s)\,H(s)]$. But in open loop system there is no reduction because of no feedback.

1.30.3 Effect of Feedback on Sensitivity

The parameters of any control system changes with the change in environment conditions. Also these parameters cannot be constant throughout the life. These parameter variations affects the performance of the system. For example, the resistance of the winding of a motor changes due to the change in temperature during its operation.

So, a control system should be insensitive to the parameter variations. Let P is a gain parameter that may vary due to the variations in parameters 'R' of the system. The sensitivity of the system parameter P to the parameter R is

$$S_P^R = \frac{d(\ln R)}{d(\ln P)} = \frac{\frac{1}{R}}{\frac{1}{F}}\frac{\partial R}{\partial P} = \frac{\partial\, R/R}{\partial\, P/P}$$

In general 'R' may be the output variable and 'P' may be the gain, the feedback factor etc.

let $T(s)$ = Overall transfer function

$G(s)$ = Forward path transfer function is varying

Then, sensitivity will be

$$S_G^T = \frac{\partial T(s)/T(s)}{\partial G(s)/G(s)} \qquad ...(1.141)$$

For open loop system $T(s) = G(s)$

$$\therefore \quad S_G^T = \frac{\partial\, G(s)/G(s)}{\partial\, G(s)/G(s)} = 1$$

Thus, the sensitivity of open loop system is unity.

Sensitivity of closed loop system:

$$T(s) = \frac{G(s)}{1+G(s)\,H(s)} \qquad ...(1.142)$$

$$\frac{\partial T(s)}{\partial G(s)} = \frac{\left[1+G(s)\,H(s)\right]\cdot 1 - G(s)\,H(s)}{\left[1+G(s)\,H(s)\right]^2} = \frac{1}{\left[1+G(s)\,H(s)\right]^2}$$

Sensitivity is given by from eqn. 1.141

$$S_G^T = \frac{G(s)}{T(s)}\cdot\frac{\partial T(s)}{\partial G(s)}$$

put the values of $T(s)$ & $\partial T(s)/\partial G(s)$

$$= \frac{G(s)}{\frac{G(s)}{1+G(s)\,H(s)}}\cdot\frac{1}{\left[1+G(s)\,H(s)\right]^2}$$

$$S_G^T = \frac{1}{1+G(s)\,H(s)} \qquad ...(1.143)$$

From equation (1.143) the sensitivity is reduced due to the feedback by a factor $1/1 + G(s)\,H(s)$ as compared to open loop system.

Sensitivity due to the variation in $H(s)$:

from eqn 1.142

$$\frac{\partial T(s)}{\partial H(s)} = -\frac{\left[G(s)\right]^2}{\left[1+G(s)\,H(s)\right]^2}$$

$$\therefore \quad S_H^T = \frac{H(s)}{T(s)}\cdot\frac{\partial T(s)}{\partial H(s)} = \frac{H(s)}{\frac{G(s)}{1+G(s)\,H(s)}}\cdot\frac{-\left[G(s)\right]^2}{\left[1+G(s)\,H(s)\right]^2}$$

$$S_H^T = \frac{-G(s)\,H(s)}{1+G(s)\,H(s)} \qquad ....(1.144)$$

From equation 1.143 & 1.142 it is clear that the closed loop system is more sensitive to variations in feedback path parameters than variations in forward path variations.

1.30.4 Effect of Feedback on Overall Gain

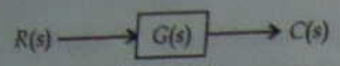


Fig. 1.118

The overall transfer function of open loop system shown in fig 1.118 is

$$\frac{C(s)}{R(s)} = G(s)$$

The overall transfer function of closed loop system shown in fig 1.117 is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)\,H(s)}$$

For negative feedback the gain $G(s)$ is reduced by a factor $\frac{1}{1+G(s)\,H(s)}$. So due to negative feedback overall gain of the system reduces.

1.30.5. Effect of Feedback on Stability

Consider the open loop system with overall transfer function

$$G(s) = \frac{K}{s+T}$$

The pole is located at $s = -T$

Now, consider closed loop system with unity negative feedback. then overall transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{K}{s+(T+K)}$$

Now, closed loop pole is located at $s = -(T+K)$

Thus, feedback controls the time response by adjusting the location of the poles. The stability depends upon the location of poles. Thus we can say the feedback affects the stability. Feedback can improve the stability or may be harmful to stability if it is not properly design and apply.

1.31. THERMAL SYSTEMS

In thermal systems, there is transfer of heat from one substance to another substance. The thermal system can be analysed in terms of resistance and capacitance. Let us consider a simple thermal system shown in fig. 1.119.

1.31.1. Heat Transfer System

Suppose, there is no heat store in the insulation and water having uniform temperature.

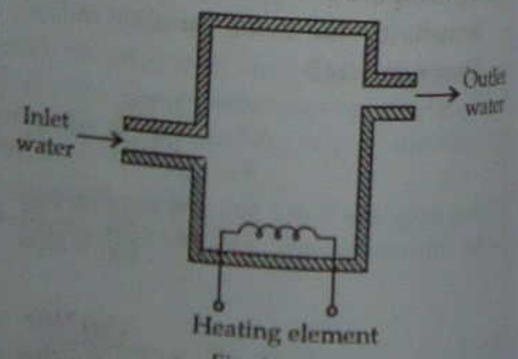


Fig. 1.119

Let, θ_i = temperature of inlet water (°C)
θ_o = temperature of outlet water(°C)
θ = temperature of the surroundings
q = rate of heat flow from heating elements (J/S)

q_i = rate of heat flow to the water
q_t = rate of heat flow through tank insulation
C = thermal capacitance (J/°C)
R = thermal resistance (°C/J-S^{-1})

Rate of heat flow for the water in tank

$$q_i = C\frac{d\theta_0}{dt} \qquad ...(1.145)$$

The rate of heat flow from water to the surrounding through insulation

$$q_t = \frac{\theta_0 - \theta}{R} \qquad ...(1.146)$$

$$q = q_i + q_t \qquad ...(1.147)$$

$$q = C\frac{d\theta_0}{dt} + \frac{\theta_0 - \theta}{R} \qquad ...(1.148)$$

neglect the term $\frac{\theta}{R}$

$$q = C\frac{d\theta_0}{dt} + \frac{\theta_0}{R} \qquad ...(1.149)$$

Take laplace transform of (1.149)

$$Q(s) = sC\,\theta_o(s) + \theta_o(s)$$

$$\text{Transfer function} = \frac{\theta_0(s)}{Q(s)}$$

$$\frac{\theta_0(s)}{Q(s)} = \frac{R}{1+sCR} \qquad ...(1.150)$$

$\therefore$ Time constant of thermal system is RC.

1.31.2. Thermometer

Consider the fig.1.120 A thermometer is immersed in a tub containing water.

Let θi = temperature of the water in tub.
θ_o = temperature indicated by thermometer

then, rate of heat flow to the thermometer

$$\frac{dq}{dt} = \frac{\theta_i - \theta_0}{R} \qquad ...(1.151)$$

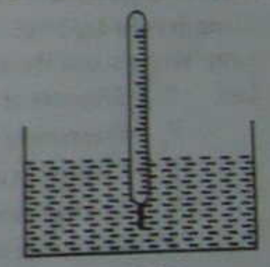


Fig. 1.120.

The indicated temperature rises at the rate

$$\frac{d\theta_0}{dt} = \frac{1}{C}\frac{dq}{dt}$$

$$\therefore \quad \frac{d\theta_0}{dt} = \frac{1}{C}\left(\frac{\theta_i - \theta_0}{R}\right) \qquad ...(1.152)$$

Take the laplace transform

$$s\,\theta_o(s) = \frac{1}{RC}\left[\theta_i(s) - \theta_0(s)\right]$$

$$\therefore \quad \frac{\theta_0(s)}{\theta_i(s)} = \frac{1}{1+sRC} \qquad \text{...(1.153)}$$

Let $\theta_i(s)$ = unit step input

$$\theta_o(s) = \frac{1}{s(1+sRC)}$$

Take inverse laplace

$$\therefore \quad \theta_o(t) = 1 - e^{-t/RC} \qquad \text{...(1.154)}$$

$\therefore$ Time constant $= RC$

Example 1.49. A thermometer has a time constant of 15.33 sec. It is quickly taken from a temperature O°c to a water bath having a temp. 100°c. What temperature will be indicated after 60 sec ?

Solution : The thermometer is subjected to a step input of 100°C

$$\therefore \quad \theta_i(s) = \frac{100}{S}$$

$\therefore$ from equation 1.153 $\quad \theta_0(s) = \frac{1}{1+sRC}\theta_i(s)$

$$\theta_o(s) = \frac{100}{s(1+sRC)}$$

$$= \frac{100}{s(1+sT)}$$

where T is the time constt

$$\theta_0(t) = 100\,(1 - e^{-t/T})$$

$$\theta_0(t) = 100\,(1 - e^{-60/15.33})$$

$$\theta_o(t) = 98°C$$

Temperature indicated by thermometer after 60 sec. = 98°C.

1.32. PNEUMATIC SYSTEM

Consider the fig 1.121. In fig.1.121 a source is supplying air and the air is stored in vessel.

Let, P_i = Pressure of air of the source N/m^2

P_o = Pressure of air in the vessel N/m^2

ΔP_i = Change in air pressure of source

ΔP_0 = Change in air pressure of vessel

R = Resistance to air flow into the vessel

C = Capacitance of the vessel

q = Rate of flow of air due to the differential pressure

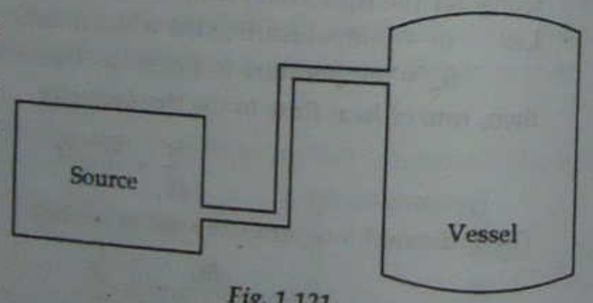


Fig. 1.121.

$$\therefore \quad q = \frac{\Delta P_i - \Delta P_0}{R} \qquad \text{...(1.155)}$$

The volume of air stored in vessel increases the pressure inside the vessel.

$\therefore$ Volume $V = C\Delta P_0$ ...(1.156)

$\therefore$ Rate of storage in vessel $= \frac{dV}{dt} = C\frac{d\Delta P_0}{dt}$ ...(1.157)

Volume of air in flow $V = \int q\,dt$

$$= \int \frac{\Delta P_i - \Delta P_0}{R}\,dt \qquad \text{...(1.158)}$$

Rate of air in flow $= \frac{dv}{dt} = \frac{\Delta P_i - \Delta P_0}{R}$ ...(1.159)

Rate of air in flow = Rate of storage of air in vessel

$$C\frac{d\Delta P_0}{dt} = \frac{\Delta P_i - \Delta P_0}{R} \qquad \text{...(1.160)}$$

Take laplace transform

$$sC\Delta P_0 = \frac{1}{R}[\Delta P_i(s) - \Delta P_0(s)]$$

$$\frac{\Delta P_0}{\Delta P_i} = \frac{1}{1+RCs} \qquad \text{...(1.161)}$$

Time constant $= RC$

1.33. TRANSFER FUNCTION OF MOVING COIL VOLTMETER

Let R = Resistance of coil

$e(t)$ = Voltage to be measured

$i(t)$ = Current flowing through the coil

$\theta(s)$ = Deflection

K_s = spring constant

J = Moment of Inertia

K_b = Back e.m.f. constant

K_T = Torque constant

T_d = Torque Developed

e_b = Back e.m.f.

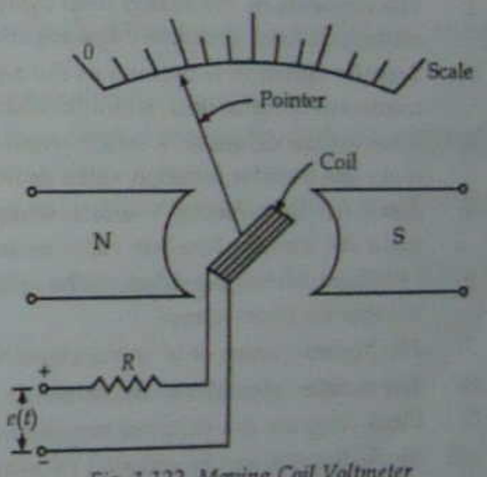


Fig. 1.122. Moving Coil Voltmeter

Deflecting Torque $T_d \alpha\, i(t)$

$$T_d = K_T\, i(t)$$

$$e_b = K_b \frac{d}{dt}\theta(t)$$

$$T_d = K_s\,\theta(t) + J\frac{d^2}{dt^2}\theta(t)$$

$$e(t) = i(t)\,R + e_b(t)$$

$$e(t) = i(t)\,R + K_b\frac{d}{dt}\theta(t)$$

Laplace transform

$$T_d(s) = K_T I(s)$$
$$T_d(s) = K_s \theta(s) + s^2 J \theta(s)$$
$$E(s) = I(s) R + K_b s\theta(s)$$
$$K_T I(s) = K_s \theta(s) + s^2 J \theta(s)$$
$$I(s) = \theta(s)\left[\frac{K_s + s^2 J}{K_T}\right]$$
$$E(s) = R\theta(s)\left[\frac{K_s + s^2 J}{K_T}\right] + K_b s\theta(s)$$
$$\therefore \quad \frac{\theta(s)}{E(s)} = \frac{K_T}{JRs^2 + K_b K_T s + K_s R} \quad \textbf{Ans.}$$

SUMMARY

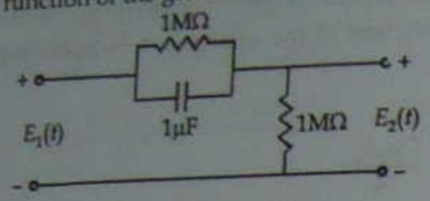
1. The open loop control system is without feedback. In open loop system the control action is independent of the desired output.
   Closed loop control systems are known as feedback control systems. In closed loop control systems the control action is dependent on the desired output.
2. The elements of the closed loop systems are command, reference input, error detector, control element, controlled system and feedback element.
3. Transfer function is defined as the ratio of laplace transform of the output to the laplace transform of input with all initial conditions are zero.
4. Poles are the values of 's' which when substituted in the denominator of a transfer function make the transfer function value as infinity.
5. Zeros are the values of 's' which when substituted in the numerator of a transfer function make the transfer function value as zero.
6. The characteristic equation can be obtained by equating the denominator polynomial of the transfer function to zero.
7. The highest power of 's' in the characteristic equation is called the order of the system.
8. The number of polses at the origin defines the type of system.
9. Block diagram is a pictorial representation of the complicated control system.
10. Block diagram can be reduced by reduction rules.
11. The reduction of blocks is possible by the following sequence :
    (*i*) Reduce the series blocks.
    (*ii*) Reduce the parallel blocks.
    (*iii*) Reduce the minor feedback loop.
    (*iv*) Shift the summing point to the left and take off point to the right, as far as possible.
    Note : There should be no take off point or summing point between the blocks during the reduction of series blocks.
12. Signal flow graph is applicable to the linear systems. In signal flow graph the variables of the system are represented by separate nodes. The node which has only outgoing branches

is known as input node. During construction of signal flow graph represent all variables, summing points and take off points by separate nodes. If a summing point is placed before a take off point represent summing and take off point by a single node. If summing point is placed after take off point, represent summing point and take off point by separate nodes.

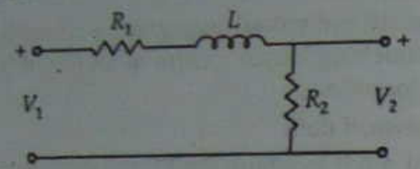
13. The resultant transfer function of the system can be obtained from signal flow graph by Mason's gain formula.
    Mason's gain formula is
    $$T = \frac{\Sigma g_k \Delta_K}{\Delta}$$
    where T = transfer function
    Δ = 1 – (sum of all individual loop gain) + (sum of all possible gain products of two non-touching loops) – (sum of all possible gain products of three non-touching loops) + ...
    g_K = gain of forward path
    Δ_K = the part of Δ not touching the K^{th} forward path.
14. The motion takes place along a straight line is known as translational motion.
15. Rotational motion of a body is the motion about a fixed axis.
16. The main elements of any mechanical system with translational motions are mass, spring and friction. The mass M cannot store energy, hence cannot cause any change in displacement. Spring stores the potential energy which causes in displacement if it is between two moving surfaces. But cannot cause change in displacement, if its one end is connected to a rigid support.
17. The elements of rotational systems are inertia (J), damping coefficient (B) and torsional stiffness K.
18. For every mechanical system, there is analogous electrical system.
19. D'Alembert's principle states that, the algebraic sum of externally applied forces and the forces resisting motion in any given direction is zero.
20. For mechanical network, analogous electrical network can be obtained by using f–v and f–i analogy.
21. **Force – Voltage analogy :** In this method force is analogous to voltage. Similarly
    $M \to L, K \to \frac{1}{C}, B \to R$, displacement x charge q
    Force-current analogy : In this method force is analogous to current
    $M \to C, B \to \frac{1}{R}, K \to \frac{1}{L}, X \to \phi$.
    Mechanical Rotational system : *a*. Force-voltage analogy :
    $T \to E, J \to L, B \to R, K \to \frac{1}{C}, \theta \to q$
    b. Force-current analogy:
    $$T \to I, J \to C, B \to \frac{1}{R}, K \to \frac{1}{L}, \theta \to \phi$$
    Mechanical Coupling: $\frac{T_1}{T_2} = \frac{\theta_2}{\theta_1} = \frac{r_1}{r_2} = \frac{\omega_2}{\omega_1} = \frac{N_1}{N_2}$

EXERCISE

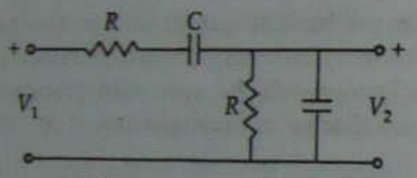
1.1. Determine the transfer function of the given circuit.



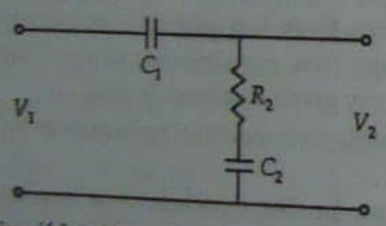
1.2. Determine the transfer function.



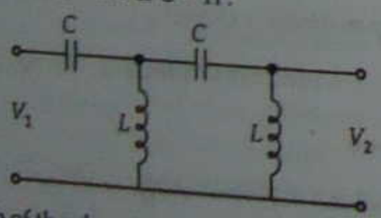
1.3. Derive the transfer function $\frac{V_2(s)}{V_1(s)}$.



1.4. Determine the transfer function $\frac{V_2(s)}{V_1(s)}$.



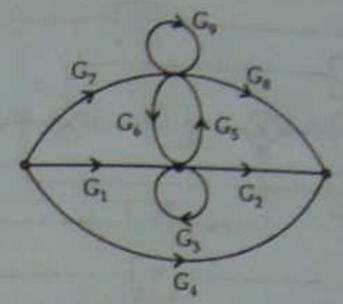
1.5. Find out the transfer function if $L = 1H$ & $C = 1F$.



1.6. Draw the signal flow graph of the electrical circuit shown in fig. 1.79, example 1.28 and find out the transfer function by using the Mason's gain formula.

1.7. Draw the signal flow graph of the block diag. Shown in fig. 1.91 example 1.32 and determine the transfer function.

1.8. Using mason's gain formula, determine the ratio $\frac{C}{R}$.



1.9. Draw the signal flow graph for the following equations

(i) $\frac{d^2y}{dt^2} + 10\frac{dy}{dt} - 5y = 3x$

(ii) $y''' + 6y'' + 5y' + 3y + 2y = x$

(iii) $\frac{d^2y}{dt^2} + 3y = x$

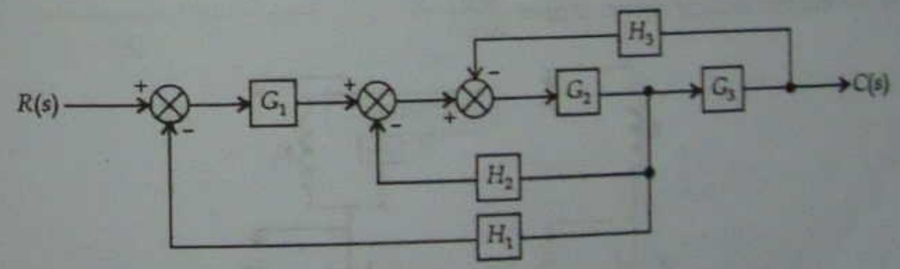
1.10. Draw the signal flow graph for the following equations

(i) $x_2 + 5x_3 - 2x_1 = 0$
$x_3 + 2x_4 - 4x_2 = 0$
$x_4 - 8x_3 = 0$

(ii) $x_2 = 4x_1 + 2x_3 + 2x_2$
$x_3 = 6x_1 + 5x_2 + 2x_3$
$x_4 = 2x_2 + 2x_3$

(iii) $x_1 - x_2 - 2x_3 - 5x_4 = 0$
$2x_2 - 3x_3 - 5x_4 = 0$
$7x_1 - 3x_3 - 2x_4 = 0$

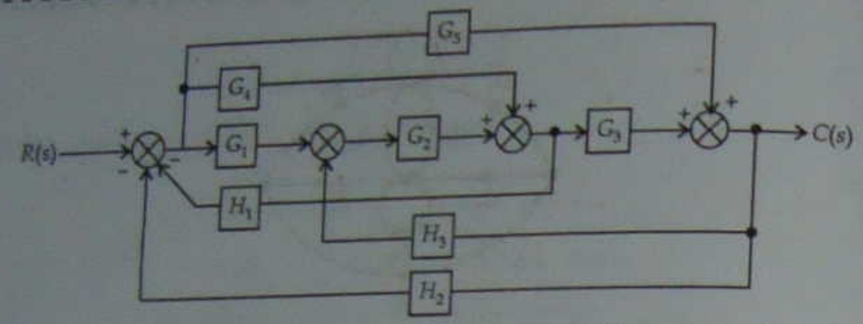
1.11. Findout the transfer function by block reduction method



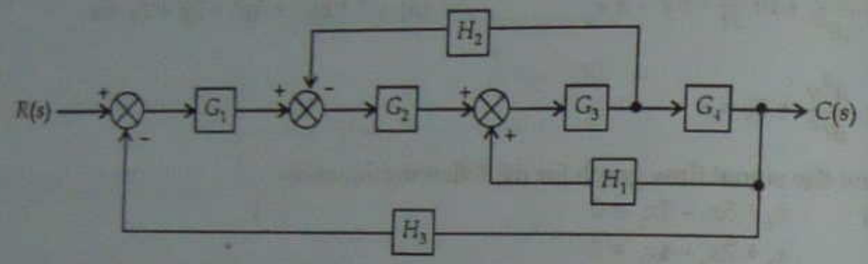
1.12 For the system shown in fig. obtain $\frac{C(s)}{R(s)}$ by

(i) Block diagram reduction technique
(ii) Signal flow graph

(KNIT, Sultanpur 98-99)

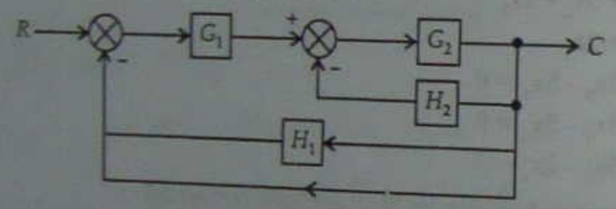


1.13. Determine the overall transfer function relating C and R for the system whose block diag. is given below by block reduction method. *(R.M.L. University Faizabad LST 2001)*

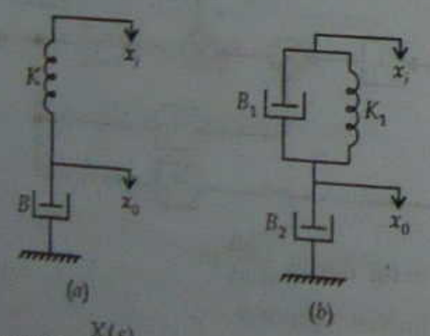


1.14. Determine the transfer function C/R for the system shown in fig. 1.108 (example 1.42) by block reduction technique.

1.15. Determine the transfer function of the given system

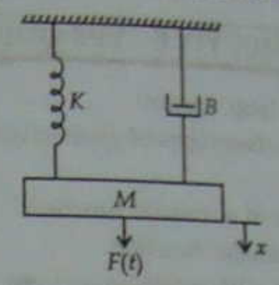


1.16. Draw the electric analog circuit & find $\frac{X_0(s)}{Xi(s)}$

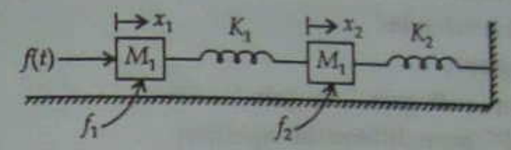


1.17. Determine the system equation $\frac{X(s)}{F(s)}$ where $M = 10$ kg, $B = 30$ N/m/s, $K = 20$ N/m.

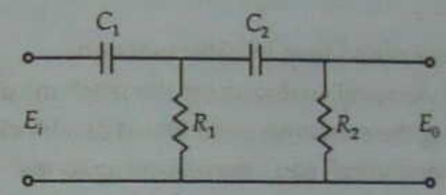
(R.M.L. University Faizabad 2001)



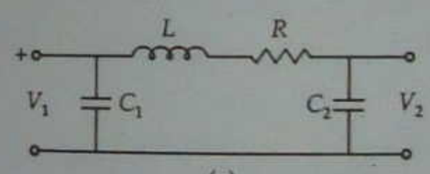
1.18. Draw the free body diag. & determine $\frac{X_2(S)}{F(S)}$ for the system shown.



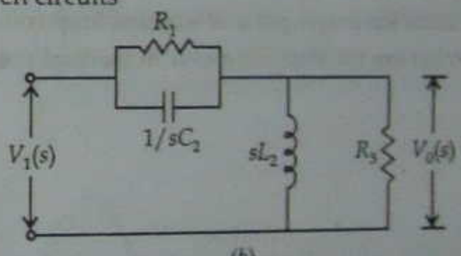
1.19. Draw the block diagram and determine the transfer function.



1.20. Determine the transfer function of the given circuits



(a)



(b)

ANSWERS

1.1. $\frac{s+1}{s+2}$

1.2. $\frac{V_2(s)}{V_1(s)} = \frac{R_2}{(R_1+R_2)+sL}$

1.3. $\frac{V_2(s)}{V_1(s)} = \frac{SRC}{S^2R^2C^2+3SRC+1}$

1.4. $\frac{V_2(s)}{V_1(s)} = \frac{C_1+RC_1C_2s}{C_1+C_2+RC_1C_2s}$

1.5. $\frac{V_2(s)}{V_1(s)} = \frac{s^4}{s^4+3s^2+1}$

1.11. $\frac{C(s)}{R(s)} = \frac{G_1G_2G_3}{1+G_2H_2+G_2G_3H_3+G_1G_2H_1}$

1.15. $\frac{C(s)}{R(s)} = \frac{G_1G_2}{1+G_1G_2+G_2H_2+G_1G_2H_1}$

1.17. $\frac{X_2(s)}{F(s)} = \frac{1}{10s^2+30s+20s}$

1.18. $\frac{X_2(s)}{F(s)} = \frac{K_1}{(s^2M_2+sf_2+K_1+K_2)(s^2M_1+sf_1+K_1)-K_1^2}$

SEMI-OBJECTIVE TYPE QUESTIONS

(*i*) Define open loop and closed loop system. (Articles 1.2 and 1.3)
(*ii*) Give the advantages and disadvantages of open loop system. (Article 1.2)
(*iii*) Define transfer function. (Article 1.6)
(*iv*) Define characteristic equation of a transfer function. (Article 1.8)
(*v*) Define poles and zeros of a transfer function. (Article 1.9)
(*vi*) Compare translational system with rotational system. (Table 1.2)
(*vii*) Define D'Alembert Principle. (Article 1.13)
(*viii*) Write the time constant of thermometer. (Article 1.31.2)
(*ix*) What is Mason's gain formula? (Article 1.28)
(*x*) Define signal flow graph. (Article 1.24)
(*xi*) Define loop, self loop, path gain, loop gain for SFG. (Article 1.24)
(*xii*) How to construct SFG from differential equation. (Article 1.26)
(*xiii*) What is the effect of feedback on stability. (Article 1.30.5)
(*xiv*) Define MIMO system. (Article 1.21)
(*xv*) Derive the expression for closed loop transfer function. (Article 1.20)
(*xvi*) Name the two types of electrical analogies for the mechanical system. (Article 1.15)
(*xvii*) State the rule for shifting the summing point ahead of a block. (Article 1.22)
(*xviii*) Define the terms "transmittance" and "non-touching loops" with respect to a signal flow graph. (Article 1.24)
(*xix*) Name the major parts of a closed loop control system. (Article 1.5)
(*xx*) What are the basic elements in thermal system? (Article 1.31)

□□□

Chapter 2

Time Domain Analysis

2.1. INTRODUCTION

Any system containing energy storing element like inductor, capacitor, mass and inertia etc. posses certain energy. These energy storing elements are the part of the control system and cannot be avoided. If the energy state of the system is disturbed then it takes a certain time to change from one state to another state. This disturbance sometimes occurs at input, sometimes occurs at output and sometimes at both ends. The time required to change from one state to another state is known as transient time and the values of current and voltages during this period is called transient response. These transient may have oscillations which may be either sustained or decaying in nature. This will depend upon the parameters of the system. For any system we obtain a linear differential equation. The solution of linear differential equation gives the response of the system.

Thus, the time response of a control system is divided into two parts (*a*) transient response (*b*) steady state response.

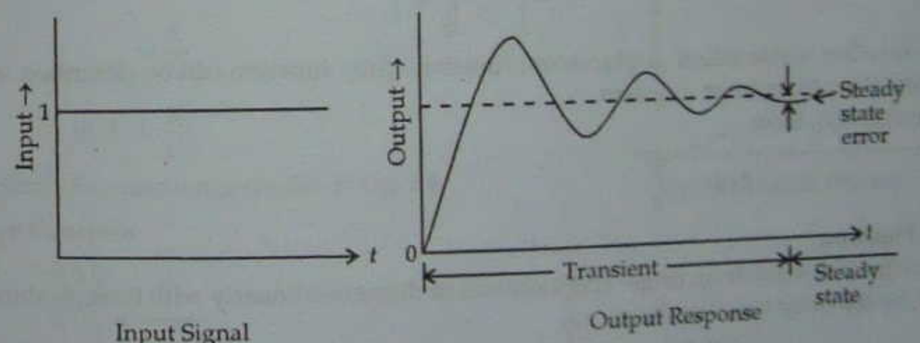


Fig. 2.1.

From the fig. 2.1 it is clear that the transient response is the part of the response which goes to zero as time increases and steady state response is the part of the total response after transient has died. If the steady state response of the output does not match with the input then the system has steady state error.

2.2. TEST INPUT SIGNALS FOR TRANSIENT ANALYSIS

For the analysis of time response of a control system, the following input signals are used.

1. Step Function

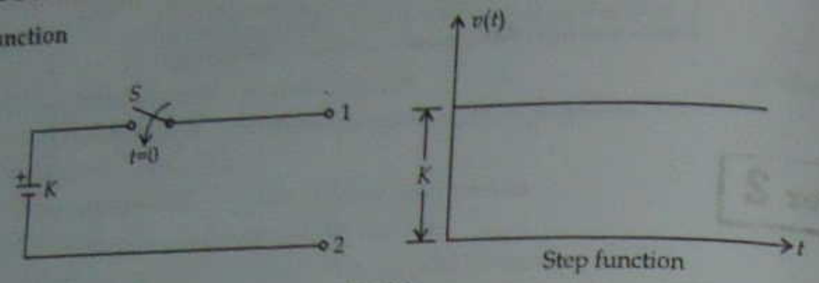


Fig. 2.2

Consider an independent voltage source is in series with a switch 's'. When switch was open the voltage at terminals 1 – 2 is zero. Mathematically

$$V(t) = 0 \qquad -\infty < t < 0$$

When switch is closed at $t = 0$

$$V(t) = K \qquad 0 < t < \infty$$

Combining above two equations

$$V(t) = 0 \quad ; \quad -\infty < t < 0$$
$$= K \quad ; \quad 0 < t < \infty$$

A unit step function is denoted by $u(t)$ and is defined as

$$u(t) = 0 \quad ; \quad t \le 0$$
$$= 1 \quad ; \quad 0 \le t$$

Laplace Transform : let $f(t)$ be defined in the interval $0 \le t \le \infty$ Laplace transform is obtained by multiplying $f(t)$ by e^{-st} & integrate between the limit 0 *to* ∞.

$$\mathcal{L}f(t) = \int_0^\infty f(t)e^{-st} = \int_0^\infty u(t)e^{-st}\,dt$$

$$= \int_0^\infty 1.e^{-st}\,dt = \left[\frac{e^{-st}}{-s}\right]_0^\infty = \frac{1}{s}$$

Step function is also called displacement function . Step function can be described as sudden application of input signal to a system.

If input is $R(s)$, then

$$R(S) = \frac{1}{s}$$

2. Ramp Function

Ramp function starts from origin and increases or decreases linearly with time, as shown in fig 2.3 let $r(t)$ be the ramp function then

$$r(t) = 0 \quad ; \quad t < 0$$
$$= Kt \quad ; \quad t > 0$$

where 'K' is the slope of the line. For positive value of ' K' the slope is upward and the slope is downwards for negative slope.

Laplace Transform :

$$\mathcal{L}r(t) = \int_0^\infty r(t)e^{-st}\,dt = \int_0^\infty Kt\,e^{-st}\,dt = \frac{K}{s^2}$$

$$R(S) = \frac{K}{s^2}$$

For, unit ramp $K = 1$

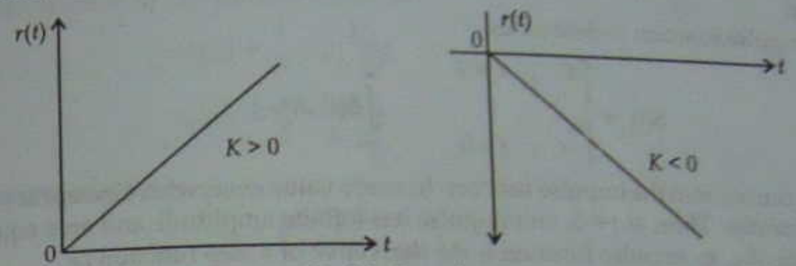


Fig. 2.3. Ramp function

3. Parabolic Function

The value of $r(t)$ is zero for $t < 0$ and is quadratic function of time for $t > 0$. The parabolic function is defined as

$$r(t) = 0 \quad ; \quad t < 0$$
$$= \frac{Kt^2}{2} \quad ; \quad t > 0$$

where 'K' is the constant. For unit parabolic function $K = 1$. The unit parabolic function is defined as

$$r(t) = 0 \quad ; \quad t < 0$$
$$= \frac{t^2}{2} \quad ; \quad t > 0$$

Lalpace Transform :

$$\mathcal{L}r(t) = \int_0^\infty r(t)\,e^{-st}\,dt = \int_0^\infty \frac{Kt^2}{2}e^{-st}\,dt$$

$$= \frac{K}{s^3}$$

$$R(S) = \frac{K}{s^3}$$

The parabolic function is shown in fig. 2.4

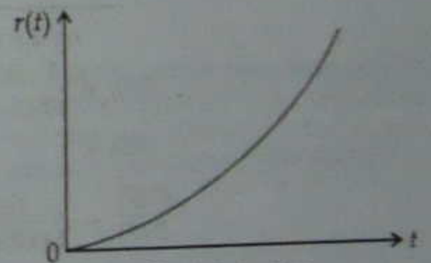
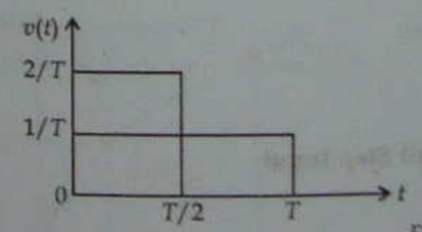


Fig. 2.4. Parabolic function

4. Impulse Function



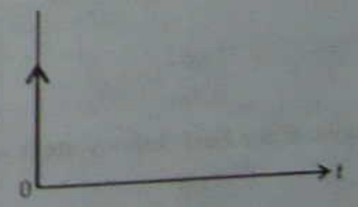


Fig. 2.5.

Consider the fig. 2.5, the first pulse has a width T and height $\frac{1}{T}$ such that area of the pulse is $T \times \frac{1}{T} = 1$. If we halve the duration and double the amplitude we get second pulse. The area under

the second pulse is also unity. We can say the duration of the pulse approaches zero, the amplitude approaches infinity but the area of the pulse is unity. The pulse for which the duration tends to zero and amplitude tends to infinity is called the impulse function. Impulse function is also known as delta function.

A unit impulse function is defined as

$$\delta(t) = \begin{cases} 0 & ; \ t \neq 0 \\ \infty & ; \ t = 0 \end{cases} \qquad \int_{-\infty}^{\infty} \delta(t)\, dt = 1$$

Thus, we can say that the impulse function has zero value everywhere except at $t = 0$, where the amplitude is infinite. Thus, at $t = 0$, unit impulse has infinite amplitude and area equal to unity.

Mathematically, an impulse function is the derivative of a step function *i.e.*

$$\delta(t) = u(t)$$

$$\pounds\, \delta(t) = \pounds \frac{d}{dt}[u(t)]$$

$$= s\,\pounds \text{ (unit step function)}$$

$$= s \cdot \frac{1}{s} = 1$$

2.3. TIME RESPONSE OF A FIRST ORDER SYSTEM

Consider a first order system with unity feedback as shown in fig. 2.6.

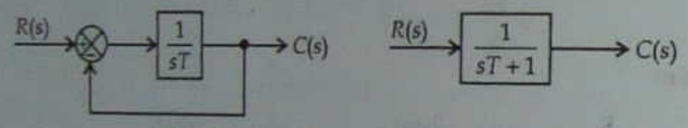


Fig. 2.6. Block diagram of first order system

$$G(S) = \frac{1}{sT}$$

$$H(S) = 1$$

$$\frac{C(S)}{R(S)} = \frac{G(s)}{1+G(s)H(s)} = \frac{1/ST}{1+\frac{1}{ST}\cdot 1}$$

$$= \frac{1/ST}{(ST+1)/ST} = \frac{1}{ST+1}$$

$$\therefore \quad \frac{C(S)}{R(S)} = \frac{1}{ST+1} \qquad ...(2.1)$$

2.3.1. Response of the First Order System with Unit Step Input

$$\frac{C(S)}{R(S)} = \frac{1}{ST+1}$$

$$\therefore \quad C(S) = \frac{1}{ST+1} R(S) \qquad ...(2.2)$$

Since, input is the unit step function

$$\therefore \quad R(S) = \frac{1}{S}$$

Put the value of $R(S)$ in equation (2.2)

$$C(S) = \frac{1}{S(ST+1)} \qquad ...(2.3)$$

After partial fraction *eq*^{n} 2.3 can be written as

$$C(S) = \frac{1}{S} - \frac{T}{1+ST} \qquad ...(2.4)$$

or,

$$C(S) = \frac{1}{S} - \frac{1}{S+\frac{1}{T}} \qquad ...(2.5)$$

Inverse lapiace of *eq*^{n} 2.5

$$\pounds^{-1} C(S) = \pounds^{-1} \frac{1}{S} - \pounds^{-1} \frac{1}{S+1/T}$$

$$C(t) = 1 - e^{-t/T} \qquad ...(2.6)$$

When $t = T$ in equation 2.6

$$C(t) = 1 - e^{-T/T}$$

$$= 1 - e^{-1} = 0.632 \text{ or } 63.2\ \%$$

Where, 'T' is known as time constant and it is defined as the time required for the signal to attain 63.2% of final or steady state value. Time constant indicates how fast the system reaches the final value. Smaller the time constant, faster is the system response. A large time constant corresponds to a sluggish system (slow moving).

Since, the output increases exponentially from zero to final value the slope of the curve at $t = 0$ is

$$\left.\frac{dc}{dt}\right|_{t=0} = \frac{1}{T} e^{-t/T}$$

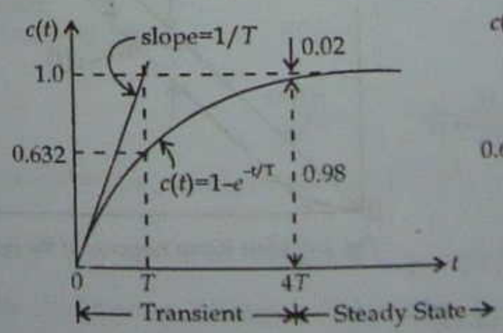
at $t = 0$

$$\frac{dc}{dt} = \frac{1}{T} e^{\circ} = \frac{1}{T}$$

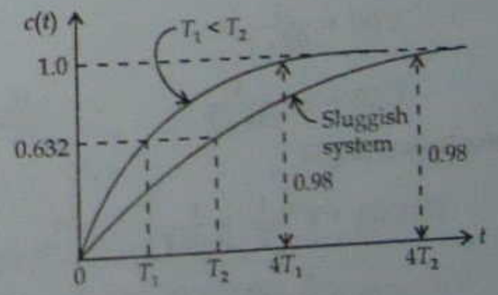
From the exponential curve it is clear that, the magnitude of the output response equals to 63.2% of final value in one time constant (T). In two time constant the magnitude of output reaches 86.4% of final value and approximately 98% in four time constant ($4T$). When the actual output reaches within 2% of the desired output it is said that steady state has reached. The time $4\,T$ is known as setling time (t_s).

The error is given by $\quad e(t) = r(t) - c(t) = 1 - (1 - e^{-t/T}) = e^{-t/T}$

Steady state error $= \lim_{t \to \infty} e^{-t/T} = 0$



Exponential curve
Fig. 2.7.



Exponential curves for two systems having diffferent time constants
Fig. 2.8.

2.3.2. Response of the First Order System with Unit Ramp Function

Since, input is the unit ramp $R(S) = \frac{1}{S^2}$

Put the value of $R(S)$ in equation 2.2

$$C(S) = \frac{1}{ST+1} \cdot \frac{1}{S^2}$$

$$C(S) = \frac{1}{S^2(1+ST)} \quad ...(2.7)$$

After partial fraction eq^n 2.7 can be written as

$$C(S) = \frac{1-ST}{S^2} + \frac{T^2}{1+ST}$$

or,
$$C(S) = \frac{1}{S^2} - \frac{T}{S} + T \cdot \frac{1}{S+\frac{1}{T}} \quad ...(2.8)$$

Inverse laplace of equation 2.8

$$\mathcal{L}^{-1} C(S) = \mathcal{L}^{-1} \frac{1}{S^2} - \mathcal{L}^{-1} \frac{T}{S} + \mathcal{L}^{-1} T \cdot \frac{1}{S+1/T}$$

$$C(t) = t - T + T.e^{-t/T} \quad \text{For} \quad t \geq 0 \quad ...(2.9)$$

The error signal $e(t)$ will be

$$e(t) = r(t) - c(t)$$
$$= t - [t - T + Te^{-t/T}]$$
$$e(t) = T(1 - e^{-t/T}) \quad ...(2.10)$$

$$\text{Steady state error} = \lim_{t \to \infty} (T - T e^{-t/T})$$
$$= T \quad ...(2.11)$$

The steady state error is equal to 'T', where 'T' is the time constant of the system. For smaller time constant steady state error will be small and the speed of the response will increase.

2.3.3. Response of the First Order System with Unit Impulse Function

Input is unit impulse function *i.e.* $R(S) = 1$

Put the value of R(S) in equation 2.2

$$C(S) = \frac{1}{ST+1} \cdot R(S)$$

$$C(S) = \frac{1}{ST+1} \cdot 1$$

or,
$$C(S) = \frac{1}{T} \cdot \frac{1}{S+1/T} \quad ...(2.12)$$

Inverse laplace of 2.12

$$\mathcal{L}^{-1} C(S) = \mathcal{L}^{-1} \left[\frac{1}{T} \cdot \frac{1}{S+1/T} \right]$$

$$C(t) = \frac{1}{T} e^{-t/T} \quad ...(2.13)$$

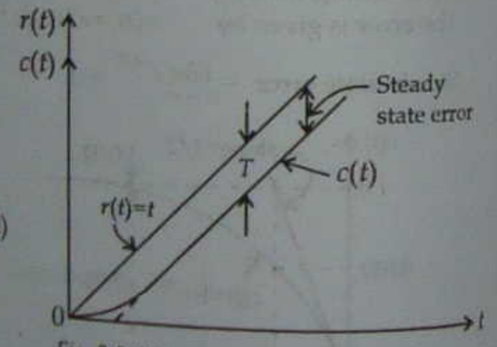


Fig. 2.9. Unit Ramp response of the system

The curve for eq^n (2.13) is shown in fig. (2.10)

Compare the equations 2.6, 2.9 and 2.13, that is

For unit ramp input $R(S) = \frac{1}{S^2}$ $C(t) = t - T + T e^{-t/T}$

For unit step input $R(S) = \frac{1}{S}$ $C(t) = 1 - e^{-t/T}$

For unit impulse $R(S) = 1$ $C(t) = \frac{1}{T} e^{-t/T}$

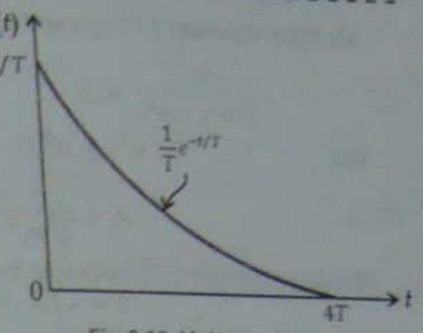


Fig. 2.10. Unit impulse response of the system

From above three equations, it is clear that the unit step input is the derivative of unit ramp input and unit impulse input is the derivative of unit step input. Thus, the response to the derivative of an input signal can be obtained by differentiating the response of the system to the original signal. This is the property of the linear time-invariant systems.

2.4. TIME RESPONSE OF SECOND ORDER SYSTEM

The block diagram of second order control system is shown in fig. 2.11.

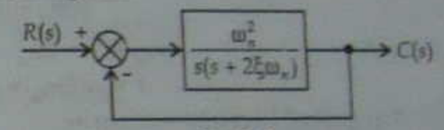


Fig. 2.11.

$$\frac{C(S)}{R(S)} = \frac{G(S)}{1 + G(S)H(S)}$$

$$G(S) = \frac{\omega_n^2}{s(s+2\xi\omega_n)}, \qquad H(S) = 1$$

$$\therefore \quad \frac{C(S)}{R(S)} = \frac{\omega^2_n / s(s+2\xi\omega_n)}{1 + \frac{\omega_n^2}{s(s+2\xi\omega_n)}}$$

$$\frac{C(S)}{R(S)} = \frac{w_n^2}{s^2 + 2\xi w_n s + w^2_n} \quad ...(2.14)$$

2.4.1. Time Response of Second Order System with Unit Step Input

From equation (2.14)

$$C(S) = \frac{w_n^2}{S^2 + 2\xi w_n S + w^2_n} . R(S)$$

For unit step input $R(S) = \frac{1}{s}$

$$\therefore \quad c(s) = \frac{1}{s} \cdot \frac{\omega^2_n}{s^2 + 2\xi\omega_n s + \omega^2_n} \quad ...(2.15)$$

Replace $s^2 + 2\xi\omega_n s + \omega^2_n$ by $(S + \xi\omega_n)^2 + \omega^2_n (1 - \xi^2)$

$$c(S) = \frac{1}{s} \cdot \frac{\omega^2_n}{(s+\xi\omega_n)^2 + \omega^2_n (1-\xi^2)} \quad ...(2.16)$$

Break the Equation 2.16 by partial fraction, put $\omega^2_d = \omega^2_n (1 - \xi^2)$

$$\frac{\omega_n^2}{s[(s+\xi\omega_n)^2 + \omega^2_d]} = \frac{A}{s} + \frac{B}{(s+\xi\omega_n)^2 + \omega_d^2} \quad ...(2.17)$$

Multiply equation 2.17 by s and put s = 0

$$\frac{\omega_n^2}{\xi^2\omega_n^2+\omega_d^2} = A$$

But $\omega_d^2 = \omega_n^2(1-\xi^2) = \omega_n^2 - \omega_n^2\xi^2$

$$A = \frac{\omega_n^2}{\xi^2\omega_n^2+\omega_n^2-\xi^2\omega_n^2} = 1 \quad \therefore \quad \boxed{A = 1}$$

Multiply equation 2.17 both side by $[(S+\xi\omega_n)^2+\omega_d^2]$ and put

$$S = -\xi\omega_n + j\omega_d$$

$$\frac{w^2_n}{s} = B$$

$$B = \frac{\omega_n^2}{-\xi\omega_n - j\omega_d} = \frac{-\omega_n^2}{\xi\omega_n + j\omega_d}$$

$$= \frac{-\omega_n^2(\xi\omega_n - j\omega_d)}{(\xi\omega_n + j\omega_d)(\xi\omega_n - j\omega_d)} = -(\xi\omega_n - j\omega_d)$$

put $-j\omega_d = s + \xi\omega_n$

$$B = -(\xi\omega_n + s + \xi\omega_n) = -(S + 2\xi\omega_n)$$

Equation 2.16 can be written as

$$C(S) = \frac{1}{s} - \frac{s+2\xi\omega_n}{(s+\xi\omega_n)^2+w^2_d} = \frac{1}{s} - \frac{s+\xi\omega_n+\xi\omega_n}{(s+\xi\omega_n)^2+\omega^2_d}$$

or, $$C(s) = \frac{1}{s} - \left[\frac{s+\xi\omega_n}{(s+\xi w_n)^2+\omega^2_d} + \frac{\xi\omega_n}{\omega_d}\cdot\frac{\omega_d}{(s+\xi\omega_n)^2+\omega^2_d}\right] \quad ...(2.18)$$

Laplace inverse of equation (2.18)

$$C(t) = 1 - \left[e^{-\xi\omega_n t}\cdot\cos\omega_d t + \frac{\xi\omega_n}{\omega d}\, e^{-\xi\omega_n t}\sin\omega_d t\right] \quad ...(2.19)$$

Put $\omega_d = \omega_n\sqrt{1-\xi^2}$

$$C(t) = 1 - e^{-\xi\omega_n t}\left[\cos\omega_d t + \frac{\xi}{\sqrt{1-\xi^2}}\cdot\sin\omega_d t\right]$$

$$= 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sqrt{1-\xi^2}\cdot\cos\omega_d t + \xi\sin\omega_d t\right]$$

Put $\sqrt{1-\xi^2} = \sin\phi$

$$\cos\phi = \xi,\ \tan\phi = \frac{\sqrt{1-\xi^2}}{\xi}$$

$$C(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sin\phi\cdot\cos\omega_d t + \cos\phi\sin\omega_d t\right] = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\sin(\omega_d t + \phi)$$

Put the values of ω_d & ϕ *i.e.*

$$\omega_d = \omega_n\sqrt{1-\xi^2}\ ,\ \phi = \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}$$

$$C(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}\right] \quad ...(2.20)$$

The error signal for the system

$$e(t) = r(t) - c(t)$$

$$\therefore \quad e(t) = \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}\right] \quad ...(2.21)$$

The steady state value of $c(t)$

$$e_{ss} = \lim_{t\to\infty} c(t) = 1$$

Therefore at steady state there is no error between input & output.

ω_n = natural frequency of oscillation or undamped natural frequency

ω_d = damped frequency of oscillation

$\xi\omega_n$ = damping factor or actual damping or damping coefficient.

(*a*) Underdamped case ($0 < \xi < 1$)

From the expression 2.20, it is clear that the time constant is $1/\xi\omega_n$ and the response having damped oscillation with overshoot and undershoot. Such response is known as underdamped response. The response is shown in fig. 2.12.

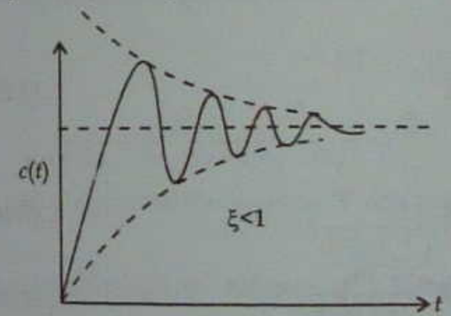


Fig. 2.12.

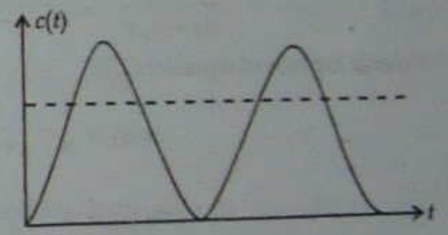


Fig. 2.13. Sustained oscillation

(*b*) When $\xi = 0$, undamped case

The expression 2.20 will be

$$C(t) = 1 - \sin(\omega_n t + \pi/2) \quad ...(2.22)$$

$$C(t) = 1 - \cos\omega_n t$$

Thus at ω_n the system will oscillate (with $\xi = 0$). The damped frequency always less than the undamped frequency (ω_n) because of factor ξ. If the system having certain value of ξ then it is not possible to measure undamped natural frequency experimentally. The observed frequency is the damped frequency (ω_d) which is equal to $\omega_n\sqrt{1-\xi^2}$. The response is shown in fig. 2.13.

(*c*) $\xi = 1$ critically damped case :

Put $\xi = 1$ in equation 2.15

$$C(S) = \frac{1}{s} \cdot \frac{\omega^2_n}{s^2 + 2\omega_n s + \omega^2_n}$$

or,
$$C(S) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s+\omega_n)^2} \quad \ldots(2.23)$$

Break the equation 2.23 by partial fraction

$$\frac{\omega_n^2}{s(s+\omega_n)^2} = \frac{A}{s} + \frac{B}{s+\omega_n} + \frac{C}{(s+\omega_n)^2} \quad \ldots(2.24)$$

Multiply both the sides by $(s+\omega_n)^2$ and put $s = -\omega_n$

$$\frac{\omega^2_n}{s} = \frac{A(s+\omega_n)^2}{s} + B(s+\omega_n) + C \quad \ldots(2.25)$$

$$\boxed{C = -\omega_n}$$

Multiply both the sides by 's' and put $s = 0$

$$\frac{\omega^2_n}{(s+\omega_n)^2} = A + \frac{Bs}{s+\omega_n} + \frac{cs}{(s+\omega_n)^2}$$

$$\boxed{A = 1}$$

Differentiate eqn 2.25 & put $s = -w_n$

$$\frac{-\omega_n^2}{s^2} = A\,\frac{s+\omega_n}{s} + B$$

$$\boxed{B = -1}$$

$$\therefore \quad \frac{w^2_n}{s(s+\omega_n)^2} = \frac{1}{s} - \frac{1}{s+w_n} - \frac{\omega_n}{(s+\omega_n)^2} \quad \ldots(2.25)$$

Inverse laplace of equation 2.25

$$c(s) = \frac{1}{s} - \frac{1}{s+\omega_n} - \frac{\omega_n}{(s+\omega_n)^2} \quad \ldots(2.26)$$

$$\mathcal{L}^{-1}c(s) = \mathcal{L}^{-1}\frac{1}{s} - \mathcal{L}^{-1}\frac{1}{s+\omega_n} - \mathcal{L}^{-1}\frac{\omega_n}{(s+\omega_n)^2} \quad \ldots(2.27)$$

$$\mathcal{L}^{-1}\frac{1}{s} = 1$$

$$\mathcal{L}^{-1}\frac{1}{s+\omega_n} = e^{-\omega_n t}$$

Inverse laplace of $\frac{\omega_n}{(s+\omega_n)^2}$ can be obtained by the method of residues.

$$\mathcal{L}^{-1}\frac{\omega_n}{(s+\omega_n)^2} = \frac{1}{\lfloor 2-1}\frac{d}{ds}\left[(s+\omega_n)e^{st}\frac{\omega_n}{(s+\omega_n)^2}\right]_{s=-\omega_n}$$

$$= \omega_n [e^{st}.t]_{s=-\omega n}$$

$$= t\omega_n e^{-\omega_n t}$$

Put all values in eq^n. (2.27)

$$\therefore \quad \mathcal{L}^{-1}c(s) = c(t) = 1 - e^{-\omega_n t} - t\,\omega_n e^{-\omega_n t}$$

$$= 1 - e^{-\omega_n t}(1 + \omega_n t) \quad \ldots(2.28)$$

From the expression 2.20, it is clear that $\xi\omega_n$ is the actual damping for $\xi = 1$, the actual damping $= \omega_n$. This actual damping when $\xi = 1$ is known as CRITICAL DAMPING. At the value of critical damping the oscillations just disappeared. The ratio of actual damping to the critical damping is known as damping ratio. i.e.

$$\frac{\text{Actual damping}}{\text{critical damping}} = \frac{\xi\omega_n}{\omega_n} = \xi$$

From eq^n 2.28, the time constant of the system $= 1/\xi\omega_n$
The response of the equation 2.28 is shown in fig 2.14.

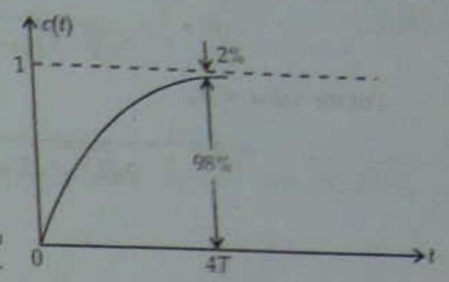


Fig. 2.14. Critically damped response

(d) $\xi > 1$ overdamped case

For $\xi > 1$, eq^n 2.16 can be written as

$$c(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s+\xi\omega_n)^2 - \omega^2_n(\xi^2 - 1)} \quad \ldots(2.29)$$

put
$$\omega^2_d = \omega_n^2(\xi^2 - 1)$$

$$c(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s+\xi\omega_n)^2 - \omega^2_d} \quad \ldots(2.30)$$

Break the equation (2.30) by partial fraction
Equation 2.30 can be written as

$$c(s) = \frac{\omega^2_n}{s(s+\xi\omega_n + \omega_d)(s+\xi\omega_n - \omega_d)}$$

$$\frac{\omega^2_n}{s(s+\xi\omega_n+\omega_d)(s+\xi\omega_n-\omega_d)} = \frac{A}{s} + \frac{B}{s+\xi\omega_n+\omega_d} + \frac{C}{s+\xi\omega_n-\omega_d} \quad \ldots(2.31)$$

Multiply both the sides by 's' and put $s = 0$

$$\boxed{A = 1}$$

Multiply both the sides of eq^n 2.31 by $s + \xi\omega_n + \omega_d$ and put $S = -\xi\omega_n - \omega_d$

$$\frac{\omega_n^2}{s(s+\xi\omega_n-\omega_d)} = B$$

$$\therefore \quad B = \frac{\omega_n^2}{(-\xi\omega_n-\omega_d)(-\xi\omega_n-\omega_d+\xi\omega_n-\omega_d)} = \frac{\omega_n^2}{(\xi\omega_n+\omega_d)(2\omega_d)}$$

Put $\omega^2_d = \omega^2_n(\xi^2 - 1)$ and simplify for B

$$B = \frac{1}{2\xi\sqrt{\xi^2-1} + 2(\xi^2-1)} = \frac{1}{2\xi\sqrt{\xi^2-1} + 2\sqrt{\xi^2-1}\sqrt{\xi^2-1}}$$

$$= \frac{1}{2\sqrt{\xi^2-1}\left[\xi + \sqrt{\xi^2-1}\right]}$$

Similarly, $$C = \frac{-1}{2\sqrt{\xi^2-1}\left[\xi-\sqrt{\xi^2-1}\right]}$$

Put the values of A, B and C in 2.31

$$c(s) = \frac{1}{s} + \frac{1}{2\sqrt{\xi^2-1}\left[\xi+\sqrt{\xi^2-1}\right]\left[s+\xi\omega_n+\omega_d\right]} - \frac{1}{2\sqrt{\xi^2-1}\left[\xi-\sqrt{\xi^2-1}\right]\left[s+\xi\omega_n-\omega_d\right]}$$

Put the value of ω_d

$$c(s) = \frac{1}{s} + \frac{1}{2\sqrt{\xi^2-1}\left[\xi+\sqrt{\xi^2-1}\right]\left[s+\xi\omega_n+\omega_n\sqrt{\xi^2-1}\right]} - \frac{1}{2\sqrt{\xi^2-1}\left[\xi-\sqrt{\xi^2-1}\right]\left[s+\xi\omega_n-\omega_n\sqrt{\xi^2-1}\right]}$$

$$= \frac{1}{s} + \frac{1}{2\sqrt{\xi^2-1}\left[\xi+\sqrt{\xi^2-1}\right]\left[s+\omega_n\left(\xi+\sqrt{\xi^2-1}\right)\right]} - \frac{1}{2\sqrt{\xi^2-1}\left[\xi-\sqrt{\xi^2-1}\right]\left[s+\omega_n\left(\xi-\sqrt{\xi^2-1}\right)\right]} \quad ...(2.32)$$

Inverse laplace of (2.32)

$$c(t) = 1 + \frac{e^{-\left(\xi+\sqrt{\xi^2-1}\right)\omega_n t}}{2\sqrt{\xi^2-1}\left(\xi+\sqrt{\xi^2-1}\right)} - \frac{e^{-\left(\xi-\sqrt{\xi^2-1}\right)\omega_n t}}{2\sqrt{\xi^2-1}\left(\xi-\sqrt{\xi^2-1}\right)} \quad ...(2.33)$$

From equation 2.33 we get two time constant

$$T_1 = \frac{1}{\left(\xi+\sqrt{\xi^2-1}\right)\omega_n}, \qquad T_2 = \frac{1}{\left(\xi-\sqrt{\xi^2-1}\right)\omega_n}$$

From eqn 2.33 we observe that when ξ is greater than one there are two exponential term, the first term has a time constant T_1 which is smaller than the time constant of other exponential term (having time constant T_2), in other words we can say that the first exponential term decaying much faster than other exponential term. So, for time response we can neglect it, then

$$C(t) = 1 - \frac{e^{-\left(\xi-\sqrt{\xi^2-1}\right)\omega_n t}}{2\sqrt{\xi^2-1}\left(\xi-\sqrt{\xi^2-1}\right)} \quad ...(2.34)$$

and, time constant

$$= \frac{1}{(\xi-\sqrt{\xi^2-1})\omega_n} \quad ...(2.35)$$

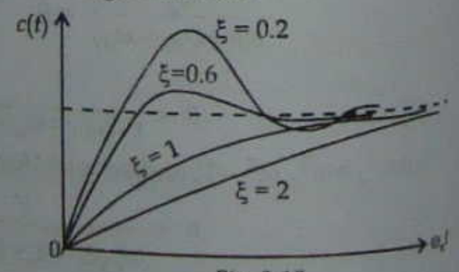


Fig. 2.15.

For different values of ξ the curves of $C(t)$ is shown in fig. 2.15.
Form the curve it is clear that the overdamped systems are sluggish.

2.4.2. Location of Roots of Characteristic Equation and Time Response

The characteristic equation of

$$\frac{C(S)}{R(S)} = \frac{\omega^2_n}{s^2+2\xi\omega_n s+\omega^2_n}$$

is $$s^2 + 2\xi\omega_n s + \omega^2_n = 0 \quad ...(2.36)$$

The roots of eqn 2.36 $$S_1 = -\xi\omega_n + j\omega_n\sqrt{1-\xi^2}$$

$$S_2 = -\xi\omega_n - j\omega_n\sqrt{1-\xi^2}$$

The real part of the roots $(-\xi\omega_n)$ represents the damping and imaginary part $\left(\omega_n\sqrt{1-\xi^2}\right)$ represents the damped frequency.

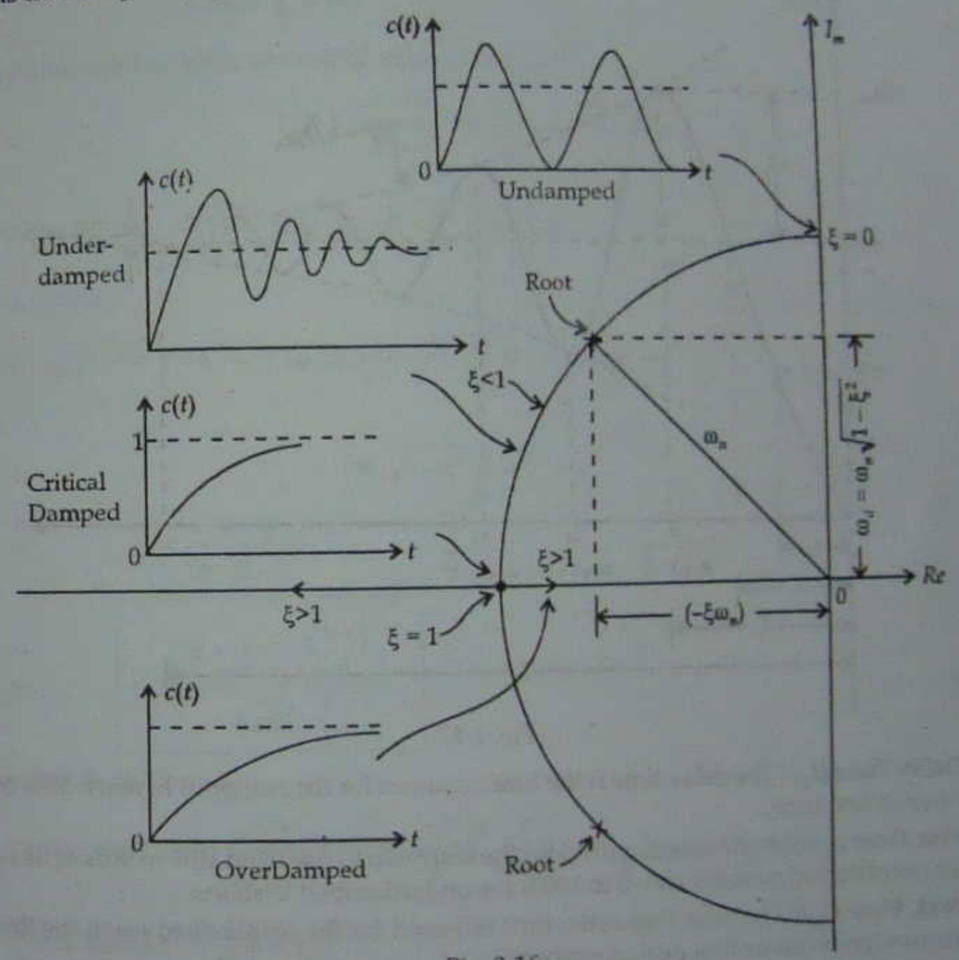


Fig. 2.16.

From the fig. 2.16

1. ω_n is the distance of root from origin.
2. $\cos\theta = \frac{\xi\omega_n}{\omega_n} = \xi$ *i.e* $\theta = \cos^{-1}\xi$ (when the roots are in left-half of s-plane).
3. ω_d is the imaginary part of the root and is known as damped frequency or conditional frequency.

2.5. TRANSIENT RESPONSE SPECIFICATIONS OF SECOND ORDER SYSTEM

The performance of a control system are express in terms of the transient response to a unit step input because it is easy to generate. The transient response of a control system to a unit step input depends upon the initial conditions. Consider a second order system with unit step input and the system initially at rest *i.e* all initial conditions are zero. The following are the common transient response characteristics.

1. Delay time (t_d)
2. Rise time (t_r)
3. Peak time (t_p)
4. Maximum overshoot (M_p)
5. Settling time (t_s)
6. Steady state error (e_{ss})

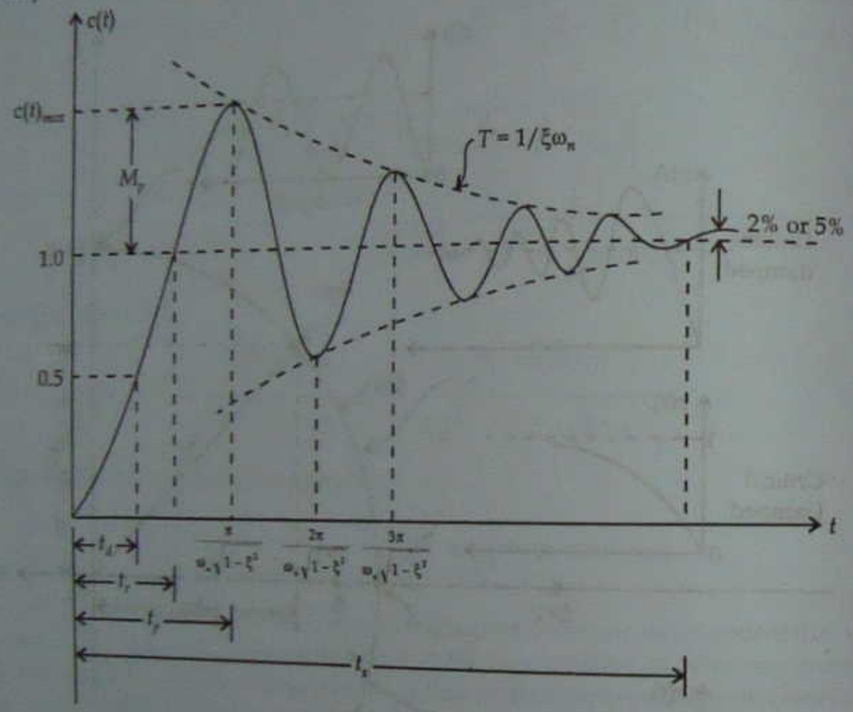


Fig. 2.17.

1. **Delay Time (t_d)** : The delay time is the time required for the response to reach 50% of the final value in first time.
2. **Rise Time (t_r)** : It is the time required for the response to rise from 10% to 90% of its final value for overdamped systems and 0 to 100% for underdamped systems.
3. **Peak Time (t_p)** : The peak time is the time required for the response to reach the first peak of the time response or first peak overshoot.
4. **Maximum Ocershoot (M_p)** : It is the normalized difference between the peak of the time response and steady output. The maximum percent overshoot is defined by

$$\text{Maximum percent overshoot} = \frac{C(t_p) - C(\infty)}{C(\infty)} \times 100$$

5. **Settling Time (t_s)** : The settling time is the time required for the response to reach and stay within the specified range (2% to 5%) of its final value.
6. **Steady State Error (e_{ss})** : It is the difference between actual output and desired output as time 't' tends to infinity.

$$e_{ss} = \lim_{t \to \infty} [r(t) - C(t)]$$

Expression for Rise Time(t_r) :

From the expression 2.20

$$C(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right]$$

$$\text{where } \phi = \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}$$

let response reaches 100% of desired value. Put $c(t) = 1$

$$1 = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right]$$

or, $$\frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right] = 0$$

Since, $e^{-\xi\omega n t} \neq 0$

$\therefore \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right] = 0$, or $\sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right] = \sin n\pi$

Put $n = 1$

$$\therefore \left(\omega_n\sqrt{1-\xi^2}\right)t_r + \phi = \pi$$

or, $$t_r = \frac{\pi - \phi}{\omega_n\sqrt{1-\xi^2}}$$

or, $$\boxed{t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}}{\omega_n\sqrt{1-\xi^2}}} \quad \ldots(2.37)$$

Expression for Peak Time t_p :

since, $$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right]$$

For maximum $\frac{dc(t)}{dt} = 0$

$$\frac{dc(t)}{dt} = \frac{-e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \cos\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right]\omega_n\sqrt{1-\xi^2} + \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right]\frac{\xi\omega_n}{\sqrt{1-\xi^2}}e^{-\xi\omega_n t} \quad \ldots(2.38)$$

Since $e^{-\xi\omega_n t} \neq 0$

Eq^n 2.38 can be writen as

$$\cos\left[\left(\omega_n\sqrt{1-\xi^2}\right)t+\phi\right]\sqrt{1-\xi^2} = \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t+\phi\right]\xi \quad \text{...(2.39)}$$

Put $\sqrt{1-\xi^2} = \sin\phi \quad \& \quad \xi = \cos\phi$

Eq^n 2.39 becomes

$$\cos\left[\left(\omega_n\sqrt{1-\xi^2}\right)t+\phi\right]\sin\phi = \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t+\phi\right]\cos\phi$$

or, $$\sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t+\phi\right]\cos\phi - \cos\left[\left(\omega_n\sqrt{1-\xi^2}\right)t+\phi\right]\sin\phi = 0$$

or, $$\sin\left[\omega_n\sqrt{1-\xi^2}\right]t = 0$$

the time to various peaks

$$\left(\omega_n\sqrt{1-\xi^2}\right)t_p = n\pi$$

where $n = 0, 1, 2, 3, \ldots$

Maximum overshoot identified by putting $n = 1$, therefore the peak time to the first overshoot

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$$

$$\boxed{t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}} \quad \text{...(2.40)}$$

The first minimum (undershoot) occurs at $n = 2$

$$t_{min} = \frac{2\pi}{\omega_n\sqrt{1-\xi^2}} \quad \text{...(2.41)}$$

Expression for Maximum Overshoot : M_p

$$C(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t+\phi\right] \quad \text{...(2.42)}$$

Maximum overshoot occurs at peak time i.e. $t = t_p$

Put $t = t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$ in eq^n 2.42

$$C(t) = 1 - \frac{e^{-\xi\omega_n\frac{\pi}{\omega_n\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}}\sin\left[\omega_n\sqrt{1-\xi^2}\cdot\frac{\pi}{\omega_n\sqrt{1-\xi^2}}+\phi\right]$$

$$= 1 - \frac{e^{\frac{\xi\pi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}}\sin(\pi+\phi) \quad \text{...(2.43)}$$

Since $\phi = \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}$ then $\sin\phi = \sqrt{1-\xi^2}$

and $\sin(\pi+\phi) = -\sin\phi$

$$\therefore \quad C(t) = 1 + \frac{e^{-\pi\xi/\sqrt{1-\xi^2}}}{\sqrt{1-\xi^2}}(\sin\phi)$$

$$C(t)_{max} = 1 + e^{\frac{\pi\xi}{\sqrt{1-\xi^2}}}$$

$$\therefore \quad M_p = C(t)_{max} - 1$$

$$M_p = e^{-\pi\xi/\sqrt{1-\xi^2}}$$

$$\boxed{\%\, M_p = e^{\frac{\pi\xi}{\sqrt{1-\xi^2}}} \times 100} \quad \text{...(2.44)}$$

Settling Time : t_s :

As shown in the fig 2.17, the curves for $1 \pm \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}$ are the envelope curves of the transient response for unit step input. The time constant of these envelope curves is $\frac{1}{\xi\omega_n}$. The speed of the decay depends upon the time constant. The settling time for a second order system is approximately four times the time constant $(1/\xi\omega_n)$

$$\therefore \quad t_s = \frac{4}{\xi\omega_n} \quad \text{...(2.45)}$$

For over damped system, the settling time t_s becomes large because of sluggish start. From 2.45, the settling time is inversly proportional to the product of ξ and ω_n. So, for permissible maximum overshoot, the value of ξ is known therefore the settling time can be determined by undamped natural frequency ω_n.

Example 2.1. When a second order control system is subjected to a unit step input, the values of $\xi = 0.5$ and $\omega_n = 6$ rad/sec. Determine the rise time, peak time, settling time and peak overshoot.

Solution : Given that $\xi = 0.5$

$\omega_n = 6$ rad/sec.

1. RISE TIME :

$$t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}}{\omega_n\sqrt{1-\xi^2}}$$

$$t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-(0.5)^2}}{0.5}}{6\sqrt{1-(0.5)^2}} = \frac{3.14-1.047}{5.19} = 0.403 \text{ sec.}$$

2. PEAK TIME :

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$$

$$t_p = \frac{\pi}{6\sqrt{1-(0.5)^2}} = 0.605 \text{ sec.}$$

3.SETTLING TIME :

$$t_s = \frac{4}{\xi\omega_n}$$

$$t_s = \frac{4}{0.5\times 6} = 1.33 \text{ sec.}$$ (for 2%

4. MAXIMUM OVERSHOOT :

$$M_p = e^{\frac{-\pi\xi}{\sqrt{1-\xi^2}}} \times 100 = e^{\frac{\pi\times 0.5}{\sqrt{1-(0.5)^2}}} \times 100$$

$$M_p = 0.163 \times 100 = 16.3\%$$

2.6. TIME RESPONSE OF SECOND ORDER SYSTEM WITH UNIT IMPULSE INPUT

Since, input is unit impulse R(S) = 1

$$\therefore \quad C(s) = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} \quad ...(2.46)$$

Break the equation 2.46 by partial fraction

$$\frac{\omega^2_n}{s^2+2\xi\omega_n s+\omega_n^2} = \frac{K_1}{s-s_1} + \frac{K_1^*}{s-s_2} \quad ...(2.47)$$

characteristic eq^n $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

Roots of the characteristic equation $s_1 = -\xi\omega_n + j\omega_n\sqrt{1-\xi^2}$

$$s_2 = -\xi\omega_n - j\omega_n\sqrt{1-\xi^2}$$

Put $\omega_d = \omega_n\sqrt{1-\xi^2}$

$\therefore$ $s_1 = -\xi\omega_n + j\omega_d$

$s_2 = -\xi\omega_n - j\omega_d$

From equation (2.47)

$$K_1 = \left.\frac{\omega_n^2}{s-s_2}\right|_{s=s_1} = \frac{\omega_n^2}{-\xi\omega_n + j\omega_d + \xi\omega_n + j\omega_d} = \frac{\omega_n^2}{2j\omega_d}$$

$$K_1^* = \frac{\omega_n^2}{-j2\omega_d}$$

Therefore equation (2.47) becomes

$$c(s) = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} = \frac{\omega_n^2}{j2\omega_d}\cdot\frac{1}{s-s_1} - \frac{\omega_n^2}{j2\omega_d}\cdot\frac{1}{s-s_2} \quad ...(2.48)$$

Inverse laplace of eq^n 2.48

$$c(t) = \frac{\omega_n^2}{j2\omega_d}[e^{s_1 t} - e^{s_2 t}]$$

$$= \frac{\omega_n^2}{j2\omega_d}[e^{(-\xi\omega_n+j\omega_d)t} - e^{(-\xi\omega_n-j\omega_d)t}] = \frac{\omega_n^2}{j2\omega_d}[e^{-\xi\omega_n t}\cdot e^{j\omega_d t} - e^{-\xi\omega_n t}\cdot e^{-j\omega_d t}]$$

$$= \frac{\omega_n^2 e^{-\xi\omega_n t}}{j2\omega_d}[e^{j\omega_d t} - e^{-j\omega_d t}] = \frac{\omega_n^2 e^{-\xi\omega_n t}}{j2\omega_d}[\cos\omega_d t + j\sin\omega_d t - \cos\omega_d t + j\sin\omega_d t]$$

$$= \frac{\omega_n^2 e^{-\xi\omega_n t}}{j2\omega_d}[j2\sin\omega_d t] = \frac{\omega_n^2 e^{-\xi\omega_n t}}{\omega_d}\sin\omega_d t \quad ...(2.49)$$

Put the value of ω_d in eq^n 2.49

$$C(t) = \frac{\omega_n^2 e^{-\xi\omega_n t}}{\omega_n\sqrt{1-\xi^2}}\sin\left(\omega_n\sqrt{1-\xi^2}\right)t$$

$$C(t) = \frac{\omega_n}{\sqrt{1-\xi^2}}e^{-\xi\omega_n t}\sin\left(\omega_n\sqrt{1-\xi^2}\right)t \quad ...(2.50)$$

(a) $\xi < 1$:

$$C(t) = \frac{\omega_n}{\sqrt{1-\xi^2}}e^{-\xi\omega_n t}\sin\left(\omega_n\sqrt{1-\xi^2}\right)t \quad ...(2.51)$$

(b) $\xi = 1$:

$$C(t) = \lim_{\xi\to 1}\frac{\omega_n}{\sqrt{1-\xi^2}}e^{-\xi\omega_n t}\sin\left(\omega_n\sqrt{1-\xi^2}\right)t$$

$$C(t) = t.e^{-\omega_n t}.\omega_n^2 \quad ...(2.52)$$

(c) $\xi > 1$:

Eq^n. 2.46 can be written

$$C(S) = \frac{\omega^2_n}{(s+\xi\omega_n)^2 - \omega^2_n(\xi^2-1)} \quad ...(2.53)$$

Break the eq^n. 2.53 by partial fraction put $\omega^2_d = \omega^2_n(\xi^2-1)$

$$\frac{\omega_n^2}{(s+\xi\omega_n)^2-\omega^2_d} = \frac{\omega^2_n}{(s+\xi\omega_n+\omega_d)(s+\xi\omega_n-\omega_d)}$$

$$= \frac{A}{(s+\zeta\omega_n+\omega_d)} + \frac{B}{(s+\xi\omega_n-\omega_d)} \quad ...(2.54)$$

From (2.54)

$$A = \left.\frac{\omega_n^2}{s+\xi\omega_n-\omega_d}\right|_{s=-\xi\omega_n-\omega_d}$$

$$A = \frac{-\omega_n}{2\sqrt{\xi^2-1}}$$

Similarly, $B = \frac{\omega_n}{2\sqrt{\xi^2-1}}$

$$C(S) = \frac{\omega_n^2}{(s+\xi\omega_n)^2 - \omega_d^2}$$

$$= \frac{-\omega_n}{2\sqrt{\xi^2-1}} \cdot \frac{1}{s+\xi\omega_n+\omega_d} + \frac{\omega_n}{2\sqrt{\xi^2-1}} \cdot \frac{1}{s+\xi\omega_n-\omega_d} \quad ...(2.55)$$

Take inverse laplace

$$C(t) = \frac{\omega_n}{2\sqrt{\xi^2-1}}\left[-e^{-(\xi\omega_n+\omega_d)t} + e^{-(\xi\omega_n-\omega_d)t}\right]$$

$$\therefore \quad C(t) = \frac{\omega_n}{2\sqrt{\xi^2-1}}\left[e^{-\left(\xi\omega_n-\omega_n\sqrt{\xi^2-1}\right)t} - e^{-\left(\xi\omega_n+\omega_n\sqrt{\xi^2-1}\right)t}\right]$$

$$C(t) = \frac{\omega_n}{2\sqrt{\xi^2-1}}\left[e^{-\left(\xi-\sqrt{\xi^2-1}\right)\omega_n t} - e^{-\left(\xi+\sqrt{\xi^2-1}\right)\omega_n t}\right] \quad ...(2.56)$$

let the maximum overshoot for the unit impulse response of the underdamped system occurs at $t = t_p$

$$\frac{dC(t)}{dt} = 0$$

Differentiate eq^n. 2.51 put $\omega_d = \omega_n\sqrt{1-\xi^2}$

$$\frac{\omega_n}{\sqrt{1-\xi^2}}\left[e^{-\xi\omega_n t}\cos\omega_d t\,\omega_d + \sin\omega_d t.\, e^{-\xi\omega_n t}(-\xi\omega_n)\right] = 0$$

$e^{-\xi\omega_n t} \neq 0$

$$\omega_d \cos\omega_d t = \xi\omega_n \sin\omega_d t$$

$$\tan\omega_d t = \frac{\omega_d}{\xi\omega_n} = \frac{\omega_n\sqrt{1-\xi^2}}{\xi\omega_n} = \frac{\sqrt{1-\xi^2}}{\xi}$$

$$\omega_d t = \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}$$

$$t = \frac{\tan^{-1}\dfrac{\sqrt{1-\xi^2}}{\xi}}{\omega_d}$$

$$\boxed{t_p = \frac{\tan^{-1}\dfrac{\sqrt{1-\xi^2}}{\xi}}{\omega_n\sqrt{1-\xi^2}}} \quad ...(2.57)$$

Expression for maximum overshoot.

Since $\tan\omega_d t = \dfrac{\sqrt{1-\xi^2}}{\xi}$, then $\sin\omega_d t = \sqrt{1-\xi^2}$

$$C(t) = \frac{\omega_n}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t_p} \sin\omega_d t$$

$$C(t) = \frac{\omega_n}{\sqrt{1-\xi^2}} e^{-\xi\omega_n \frac{\tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}}{\omega_n\sqrt{1-\xi^2}}} \sqrt{1-\xi^2}$$

$$\boxed{C(t) = \omega_n e^{-\frac{\xi}{\sqrt{1-\xi^2}}\tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}}} \quad ...(2.58)$$

2.7. TIME RESPONSE OF SECOND ORDER SYSTEM WITH UNIT RAMP INPUT

$$\frac{C(S)}{R(S)} = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} \quad ...(2.59)$$

Input $\quad R(S) = \dfrac{1}{s^2}$

$$\therefore \quad C(S) = \frac{\omega_n^2}{s^2(s^2+2\xi\omega_n s+\omega_n^2)} \quad ...(2.60)$$

Break the eq^n. 2.60 by partial fraction

$$\frac{\omega_n^2}{s^2(s^2+2\xi\omega_n s+\omega_n^2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{cs+D}{s^2+2\xi\omega_n s+\omega_n^2} \quad ...(2.61)$$

$$\therefore \quad \omega_n^2 = A\,s(s^2+2\xi\omega_n s+\omega_n^2) + B(s^2+2\xi\omega_n s+\omega_n^2) + (cs+D)s^2$$

Equating the coefficient of s^3 :

$$A + C = 0 \quad ...(2.62)$$

Equating the coefficient of s^2 :

$$2\xi\omega_n A + B + D = 0 \quad ...(2.63)$$

Equating the coefficient of s :

$$A\omega_n^2 + 2\xi\omega_n B = 0 \quad ...(2.64)$$

Equating the absolute terms :

$$B\omega_n^2 = \omega_n^2 \quad ...(2.65)$$

From equations (2.62) , (2.63) , (2.64) , & (2.65)

$$A = -\frac{2\xi}{\omega_n}, C = \frac{2\xi}{\omega_n}$$

$$B = 1, D = -1 + 4\xi^2$$

Put all these values in eq^n. (2.61)

$$C(S) = \frac{-2\xi}{\omega_n s} + \frac{1}{s^2} + \frac{\frac{2\xi}{\omega_n}s + (4\xi^2-1)}{s^2+2\xi\omega_n s+\omega_n^2} \quad ...(2.66)$$

$$= \frac{1}{s^2} - \frac{2\xi}{\omega_n s} + \frac{2\xi}{\omega_n}\cdot\frac{(s+\xi\omega_n-\xi\omega_n)}{(s+\xi\omega_n)^2+\omega_n^2(1-\xi^2)} + \frac{4\xi^2-1}{(s+\xi\omega_n)^2+\omega_n(1-\xi^2)}$$

$$= \frac{1}{s^2} - \frac{2\xi}{\omega_n s} + \frac{2\xi}{\omega_n}\cdot\frac{(s+\xi\omega_n)}{(s+\xi\omega_n)^2+\omega^2{}_n(1-\xi^2)} - \frac{2\xi}{\omega_n}\cdot\frac{\xi\omega_n}{(s+\xi\omega_n)^2+\omega_n^2(1-\xi^2)} + \frac{4\xi^2-1}{(s+\xi\omega_n)^2+\omega_n(1-\xi^2)}$$

$$C(S) = \frac{1}{s^2} - \frac{2\xi}{\omega_n s} + \frac{2\xi}{\omega_n} \cdot \frac{(s+\xi\omega_n)}{(s+\xi\omega_n)^2 + \omega_n^2(1-\xi^2)} + \frac{2\xi^2 - 1}{(s+\xi\omega_n)^2 + \omega_n^2(1-\xi^2)}$$

Put $\omega_d^2 = \omega_n^2(1-\xi^2)$

$$C(S) = \frac{1}{s^2} - \frac{2\xi}{\omega_n s} + \frac{2\xi}{\omega_n} \frac{(s+\xi\omega_n)}{(s+\xi\omega_n)^2 + \omega_d^2} + \frac{2\xi^2 - 1}{(s+\xi\omega_n)^2 + \omega_d^2} \cdot \frac{\omega_d}{\omega_d}$$

$$C(S) = \frac{1}{s^2} - \frac{2\xi}{\omega_n s} + \frac{2\xi}{\omega_n} \frac{s+\xi\omega_n}{(s+\xi\omega_n)^2 + \omega_d^2} + \frac{2\xi^2 - 1}{\omega_d} \cdot \frac{\omega_d}{(s+\xi\omega_n)^2 + \omega^2 d} \quad ...(2.67)$$

Inverse laplace of eq^n (2.67)

$$C(t) = t - \frac{2\xi}{\omega_n} + \frac{2\xi}{\omega_n} \cdot e^{-\xi\omega_n t} \cos\omega_d t + \frac{2\xi^2 - 1}{\omega_d} \cdot e^{-\xi\omega_n t} \sin\omega_d t \quad ...(2.68)$$

$$= t - \frac{2\xi}{\omega_n} + \frac{2\xi\sqrt{1-\xi^2}}{\omega_n\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \cos\omega_d t + \frac{2\xi^2 - 1}{\omega_n\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \sin\omega_d t$$

$$= t - \frac{2\xi}{\omega_n} + \frac{e^{-\xi\omega_n t}}{\omega_n\sqrt{1-\xi^2}} \left[2\xi\sqrt{1-\xi^2}\cos\omega_d t + (2\xi^2 - 1)\sin\omega_d t\right]$$

Put $2\xi\sqrt{1-\xi^2} = \sin\phi$

$2\xi^2 - 1 = \cos\phi$

$$C(t) = t - \frac{2\xi}{\omega_n} + \frac{e^{-\xi\omega_n t}}{\omega_n\sqrt{1-\xi^2}} \left[\sin\phi\cos\omega_d t + \cos\phi\sin\omega_d t\right]$$

$$= t - \frac{2\xi}{\omega_n} + \frac{e^{-\xi\omega_n t}}{\omega_n\sqrt{1-\xi^2}} \sin(\omega_d t + \phi)$$

$$C(t) = t - \frac{2\xi}{\omega_n} + \frac{e^{-\xi\omega_n t}}{\omega_n\sqrt{1-\xi^2}} \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right] \quad ...(2.69)$$

Where, $\phi = \cos^{-1}(2\xi^2 - 1)$

or, $\phi = \tan^{-1}\dfrac{2\xi\sqrt{1-\xi^2}}{2\xi^2 - 1}$

Error signal

$e(t) = r(t) - C(t)$

$r(t) = t$

$$\therefore \quad e(t) = \frac{2\xi}{\omega_n} - \frac{e^{-\xi\omega_n t}}{\omega_n\sqrt{1-\xi^2}} \sin\left[\left(\omega_n\sqrt{1-\xi^2}\right)t + \phi\right] \quad ...(2.70)$$

$$e_{ss} = \lim_{t\to\infty} e(t) = \frac{2\xi}{\omega_n}$$

$\therefore$ steady state error $e_{ss} = \dfrac{2\xi}{\omega_n}$ ...(2.71)

2.8. TIME RESPONSE OF HIGHER ORDER CONTROL SYSTEM

The input, output and transfer function are given by

$$C(s) = R(s) \cdot T(s)$$

Where, $C(S)$ = output (laplace transform) of the system

$R(S)$ = Laplace transform of input

$T(S)$ = Transfer function of the system

$$T(S) = \frac{K(s^m + a_1 s^{m-1} + \ldots + a_m)}{(s^n + b_1 s^{n-1} + \ldots b_n)}$$

$$\therefore \quad C(S) = R(s) \cdot \frac{K(s^m + a_1 s^{m-1} + \ldots + a_m)}{s^n + b_1 s^{n-1} + \ldots + b_n}$$

Apply Heaviside's partial fraction to obtain the partial fraction.

$$C(S) = \frac{K_1}{(s+p_1)} + \frac{K_2}{(s+p_2)} + \ldots + \frac{K_n}{(s+p_n)}$$

For time response take inverse laplace of each part of above eq^n.

$$\mathcal{L}^{-1} C(S) = \mathcal{L}^{-1}\frac{K_1}{s+p_1} + \mathcal{L}^{-1}\frac{K_2}{s+p_2} + \ldots + \mathcal{L}^{-1}\frac{K_n}{(s+p_n)}$$

$$C(t) = K_1 e^{-p_1 t} + K_2 e^{-p_2 t} + \ldots + K_n e^{-p_n t}$$

This is the required result.

Example 2.2. The closed loop transfer function is given by

$$G(s) = \frac{s(s^2 + 9s + 19)}{s^3 + 7s^2 + 14s + 8}$$

Determine the response of the system when a unit step is applied at input.

Solution : $C(S) = R(S)\,G(S)$

$$C(S) = \frac{1}{s} \cdot \frac{s(s^2 + 9s + 19)}{s^3 + 7s^2 + 14s + 8} = \frac{s^2 + 9s + 19}{s^3 + 7s^2 + 14s + 8}$$

Break the equation by partial fraction

$$C(S) = \frac{11}{3(s+1)} - \frac{5}{2(s+2)} - \frac{1}{6(s+4)}$$

Take inverse Laplace of each term, we get

$$C(t) = \frac{11}{3}e^{-t} - \frac{5}{2}e^{-2t} - \frac{1}{6}e^{-4t}$$ **Ans.**

Example 2.3. A second order system has a transfer function given by

$$G(s) = \frac{25}{s^2 + 8s + 25}$$

If the system initially at rest is subjected to a unit step input at t = 0, the second peak in the response will occur at *(GATE 1991)*

(a) π sec. *(b)* $\pi/3$ sec. *(c)* $2\pi/3$ sec. *(d)* $\pi/2$ sec.

Solution : $R(S) = \dfrac{1}{s}$

$$C(S) = R(s) \cdot G(s)$$

$$C(S) = \frac{25}{s(s^2+8s+25)} \quad \text{...(2.72)}$$

compare the eq^n 2.72 with

$$s^2 + 2\xi\omega_n s + \omega^2_n = 0$$

$$2\xi\omega_n = 8$$

$$\omega_n = \sqrt{25} = 5 \text{ rad/sec.}$$

$$\therefore \quad \xi = 0.8$$

Required time $$= \frac{3\pi}{\omega_n\sqrt{1-\xi^2}} = \frac{3\pi}{5\sqrt{1-(0.8)^2}} = \pi \text{ sec.}$$ **Ans.**

Example 2.4. Consider the system as shown in fig. 2.18 Determine the value of a such that the damping ratio is 0.5. Also obtain the values of rise time and maximum overshoot M_P in its step response. *(GATE 1998)*

Solution :

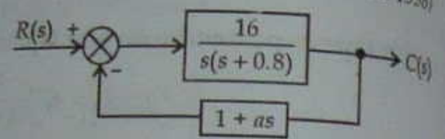


Fig. 2.18.

$$\frac{C(S)}{R(S)} = \frac{\frac{16}{s(s+0.8)}}{1+\frac{16(1+as)}{s(s+0.8)}}$$

$$= \frac{16}{s^2+(0.8+16a)s+16} \quad \text{...(2.73)}$$

Characteristic equation of 2.73

$$s^2+(0.8+16a)s+16 = 0 \quad \text{...(2.74)}$$

compare the equation (2.74) with $s^2 + 2\xi\omega_n s + \omega^2_n = 0$

$$\omega_n^2 = 16 \qquad \therefore \omega_n = 4 \text{ rad/sec.}$$

$$2\xi\omega_n = 0.8 + 16a$$

$$2 \times 0.5 \times 4 = 0.8 + 16a$$

$$\therefore \quad \boxed{a = 0.2} \quad \textbf{Ans.}$$

Rise time $$t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}}{\omega_n\sqrt{1-\zeta^2}}.$$

$$t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-(0.5)^2}}{0.5}}{4\sqrt{1-(0.5)^2}}$$

$$t_r = 0.605 \text{ sec.} \quad \textbf{Ans.}$$

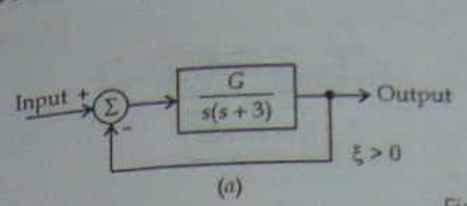
Masimum overshoot $$M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = e^{\frac{-\pi(0.5)}{\sqrt{1-(0.5)^2}}} \times 100 = 16.31\% \quad \textbf{Ans.}$$

Example 2.5. The block diagram of a feedback system is shown in fig. 2.19(*a*)

(*a*) Find closed loop transfer function

(*b*) Find the minimum value of *G* for which the step response of the system would exhibit an overshoot as shown in fig 2.19(*b*).

(*c*) For *G* equal to twice this minimum value, find the time period '*T*' indicated in fig. 2.19(*b*) *(GATE 2000)*



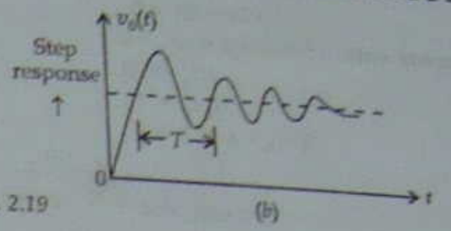


Fig. 2.19

Solution : Closed Loop Transfer Function

(a) $$\frac{C(S)}{R(S)} = \frac{G/s(s+3)}{1+\frac{G}{s(s+3)}}$$

$$\frac{C(S)}{R(S)} = \frac{G}{s^2+3s+G} \quad \textbf{Ans.}$$

(b) Characteristic equation $s^2 + 3s + G = 0$

compare with $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$

$$2\zeta\omega_n = 3$$

$$\omega_n = \sqrt{G}$$

$$\zeta = 0.6 \text{ (assume)}$$

$$2 \times 0.6\sqrt{G} = 3$$

$$\therefore \quad G = 6.25$$

$$G' = 2G = 2 \times 6.25 = 12.5$$

$$\omega_n = \sqrt{12.5} = 3.53 \text{ rad/sec.}$$

$$\zeta = \frac{3}{2\omega_n} = \frac{3}{2\times 3.53} = 0.424$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 3.53\sqrt{1-(0.424)^2} = 3.197$$

$$\frac{2\pi}{T} = 3.197$$

$$\therefore \quad T = 1.96 \text{ sec.} \quad \textbf{Ans.}$$

Example 2.6. The open loop transfer function of a servo system with unity feedback is given by

$$G(S) = \frac{10}{(s+2)(s+5)}$$

Determine the damping ratio, undamped natural frequency of oscillation. What is the percentage overshoot of the response to a unit step input. *(R.M.I. University, Faizabad 2001)*

Solution : Given that

$$G(S) = \frac{10}{(s+2)(s+5)}$$

$$H(S) = 1$$

characteristic equation $= 1 + G(S)\,H(S) = 0$

$$1+\frac{10}{(s+2)(s+5)} = 0$$

or, $s^2 + 7s + 20 = 0$

compare with $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$

$\omega^2_n = 20$

$2\zeta\omega_n = 7$

$2 \times \zeta \times 4.472 = 7$

$\therefore \boxed{\omega_n = 4.472 \text{ rad/sec.}}$

$\boxed{\zeta = 0.7826}$

$$M_p = e^{\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$M_p = e^{\frac{\pi(0.7826)}{\sqrt{1-(0.7826)^2}}} \times 100$$

$\boxed{M_p = 1.92\%}$

Example 2.7. Show that the system transfer function $Y(s)/X(s)$ has a zero in the right half s-plane. Obtain $y(t)$ when $x(t)$ is a unit step for the system shown in fig. 2.20.

(KNIT Sultanpur, 1999-2000)

Solution : The transfer function is given by

$$\frac{Y(S)}{X(S)} = \frac{6}{s+2} - \frac{4}{s+1}$$

(Blocks in parallel)

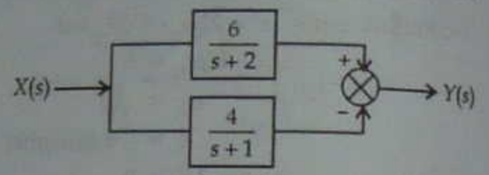


Fig. 2.20.

$$\frac{Y(S)}{X(S)} = \frac{2(s-1)}{(s+1)(s+2)}$$

From above equation it is clear that there is a zero at s = 1 on right half of s-plane.

$X(t) = 1$

$$\therefore \quad X(S) = \frac{1}{s}$$

$$\therefore \quad Y(S) = \left[\frac{6}{s+2} - \frac{4}{s+1}\right]\frac{1}{s} = \frac{6}{s(s+2)} - \frac{4}{s(s+1)}$$

Break the eq^n. by partial fraction

$$Y(S) = 6\left[\frac{1}{2s} - \frac{1}{2(s+2)}\right] - 4\left[\frac{1}{s} - \frac{1}{s+1}\right] = \frac{3}{s} - \frac{3}{s+2} - \frac{4}{s} + \frac{4}{s+1}$$

$$Y(S) = -\frac{1}{s} - \frac{3}{s+2} + \frac{4}{s+1}$$

Inverse laplace of above eq^n.

$$\mathcal{L}^{-1}Y(S) = \mathcal{L}^{-1}(-1/s) + \mathcal{L}^{-1}(-3/s+2) + \mathcal{L}^{-1}(4/s+1)$$

$$= -1 - 3e^{-2t} + 4e^{-t}$$

$$y(t) = 4e^{-t} - 3e^{-2t} - 1 \qquad \textbf{Ans.}$$

Example 2.8. The open loop transfer function of a unity feedback system is given by

$$G(S) = \frac{K}{s(1+sT)}$$

where 'K' and 'T' are positive constants. By what factor should the amplifier gain be reduced so that the peak overshoot of unit step response of the system is reduced from 75% to 25%.

(KNIT Sultanpur, 1998-99)

Solution :

$$\frac{C(S)}{R(S)} = \frac{G(s)}{1+G(s)H(s)}$$

$$= \frac{\frac{K}{s(1+sT)}}{1+\frac{K}{s(1+sT)}\cdot 1} = \frac{K}{s^2T+s+K} = \frac{K/T}{s^2+\frac{1}{T}s+\frac{K}{T}}$$

Characteristic equation : $s^2 + \frac{1}{T}s + \frac{K}{T} = 0$

compare with : $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$

$$2\zeta\omega_n = \frac{1}{T} \qquad \therefore \quad \zeta = \frac{1}{2T\omega_n}$$

$$\omega^2_n = \frac{K}{T} \qquad \therefore \quad \omega_n = \sqrt{\frac{K}{T}}$$

$$\therefore \quad \zeta = \frac{1}{2T\sqrt{\frac{K}{T}}} = \frac{1}{2\sqrt{KT}}$$

Given that $M_{p_1} = e^{\frac{-\pi\zeta_1}{\sqrt{1-\zeta^2}}} = 0.75$

$M_{p_2} = e^{\frac{-\pi\zeta_2}{\sqrt{1-\zeta^2}}} = 0.25$

$e^{\frac{-\pi\zeta_1}{\sqrt{1-\zeta^2}}} = 0.75 \quad \therefore \frac{\pi\zeta_1}{\sqrt{1-\zeta_1^2}} = 0.287$

$e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} = 0.25 \qquad \frac{\pi\zeta_2}{\sqrt{1-\zeta^2_2}} = 1.386$

$$\frac{\pi\zeta_1}{\sqrt{1-\zeta_1^2}} \cdot \frac{\sqrt{1-\zeta_2^2}}{\pi\zeta_2} = \frac{0.287}{1.386} = 0.207$$

$$\frac{\zeta_1}{\zeta_2}\frac{\sqrt{1-\zeta_2^2}}{\sqrt{1-\zeta^2_1}} = 0.207$$

$$\frac{1}{2\sqrt{K_1T}} \cdot 2\sqrt{K_2T}\,\frac{\sqrt{1-\left(\frac{1}{2\sqrt{K_2T}}\right)^2}}{\sqrt{1-\left(\frac{1}{2\sqrt{K_1T}}\right)^2}} = 0.207$$

Solve the above equation we get

$\boxed{K_1 = 20\,K_2}$ **Ans.**

Example 2.9. For given series RLC circuit, determine the undamped natural frequency and damping ratio of the circuit. Assume initial conditions are zero. $R = 1K$, $L = 10$ mH, $C = 0.01\mu f$

Solution : Find the transfer function and compare the characteristic equation with $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$ we get

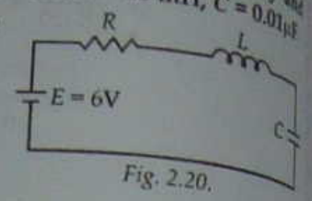


Fig. 2.20.

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} = 0.5$$

$$\omega_n = \frac{1}{\sqrt{LC}} = 10^5 \text{ rad/sec.} \quad \textbf{Ans.}$$

Example 2.10. Determine the value of 'K' and 'H' of the closed loop system so that the maximum overshoot in unit step response is 25% and the peak time is 2 sec. Assume $J = 1$ Kg -m^2.

Solution :

$$\frac{C(S)}{R(S)} = \frac{K}{Js^2 + KHs + K}$$

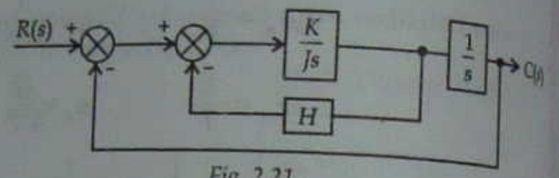


Fig. 2.21.

Put $J = 1$

$$\frac{C(S)}{R(S)} = \frac{K}{s^2 + KHs + K}$$

characteristic equation $s^2 + KHs + K = 0$, compare with $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$

$$\omega_n = \sqrt{K}$$
$$2\zeta\omega_n = KH$$

Given that $M_P = 0.25$ i.e. $\frac{-\pi\zeta}{e^{\sqrt{1-\zeta^2}}} = 0.25$

$\therefore \quad \zeta = 0.404$

Peak time $t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

$$\omega_n\sqrt{1-\zeta^2} = \frac{\pi}{t_p} = \frac{\pi}{2}$$

$$\omega_n\sqrt{1-(0.404)^2} = \frac{\pi}{2}$$

$\therefore \quad \omega_n = 1.72$ rad/sec.

$$\omega_n = \sqrt{K}$$

$\therefore \quad K = \omega^2_n = (1.72)^2 = 2.95\ Nm.$

$$2\zeta\omega_n = KH$$

$H = 0.471$ $\quad \therefore \quad H = \frac{2\zeta\omega_n}{K} = \frac{2\times0.404\times1.72}{2.95}$

$K = 2.95\ Nm$
$H = 0.471$ **Ans.**

Example 2.11. A second order control system is represented by a transfer function given below.

$$\frac{Q_0(S)}{T(S)} = \frac{1}{Js^2 + fs + K}$$

where $Q_0(S)$ is the proportional output and 'T' is the input torque A step input 10 Nm is applied to the system and test results are given blow.

(a) $M_P = 6\%$ (b) $t_p = 1$ sec. (c) The steady state error of the output is 0.5 radian. Determine the value of J, K and f. *(R.M.L. University Faizabad 2002)*

Solution : Given $\quad T(S) = \frac{10}{s}$

$$Q_0(S) = \frac{10}{s(Js^2 + fs + K)}$$

Steady state value $= \lim_{s\to0} s\cdot Q_0(s)$

$$0.5 = \lim_{s\to0} s\cdot\frac{10}{s(Js^2 + fs + K)}$$

$$0.5 = \frac{10}{k} \quad \therefore \quad \boxed{K = 20}$$

$$M_P = 6\% = 0.06 \text{ (given)}$$

$$M_P = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100$$

$$0.06 = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \quad \therefore \quad \boxed{\zeta = 0.667}$$

$t_p = 1$ sec (given)

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

or $\quad 1 = \frac{\pi}{\omega_n\sqrt{1-(0.667)^2}}$

$$\omega_n = 4.21 \text{ rad/sec.}$$

characteristic eq^n.

$$Js^2 + fs + K = 0$$

$$s^2 + \frac{f}{J}s + \frac{K}{J} = 0$$ compare with $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$

$$\omega^2_n = \frac{K}{J} \quad \therefore J = \frac{K}{\omega_n^2} = \frac{20}{(4.21)^2} = 1.128 \quad \boxed{J = 1.128}$$

$$2\zeta\omega_n = \frac{f}{J} \quad \therefore f = 2\zeta\omega_n J = 2\times0.667\times4.21\times1.128$$

$$= 6.34 \quad \boxed{f = 6.34}$$

Example 2.12. A feedback system is described by the following transfer function

$$G(S) = \frac{12}{s^2 + 4s + 16},\ H(s) = Ks$$

The damping factor of the system is 0.8. Determine the overshoot of the system, and the value of K. *(Pune university)*

Solution :

$$\frac{C(S)}{R(S)} = \frac{G(S)}{1 + G(S)H(S)}$$

$$\frac{C(S)}{R(S)} = \frac{16}{s^2 + (4+16K)s + 16}$$

characteristic eq^n. $s^2 + (4 + 16K)s + 16 = 0$, compare with

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\omega_n^2 = 16 \qquad \therefore \omega_n = 4 \text{ rad/sec.}$$

$$2\zeta\omega_n = 4 + 16K$$

$$2 \times 0.8 \times 4 = 4 + 16K \qquad \therefore K = 0.15$$

$$\frac{c(s)}{R(s)} = \frac{16}{s^2 + (4+16\times 0.15)s + 16}$$

$$\therefore \quad \omega_n^2 = 16$$

$$2\zeta\omega_n = 6.4 \qquad \therefore \zeta = 0.8$$

$$M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = e^{\frac{-\pi 0.8}{\sqrt{1-(0.8)^2}}} \times 100$$

$$M_p = 1.5\% \qquad \textbf{Ans.}$$

Example 2.13. The open loop transfer function of a unity feedback control system is given by

$$G(s) = \frac{K}{s(1+sT)}$$

By what factor the amplifier gain K should be multiplied so that the damping ratio is increased from 0.3 to 0.9.

Solution :

$$\frac{c(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{K/s(1+sT)}{1+\frac{K}{s(1+sT)}\cdot 1}$$

$$\frac{c(s)}{R(s)} = \frac{K}{s^2T+s+K} = \frac{K/T}{s^2+\frac{s}{T}+\frac{K}{T}}$$

characteristic equation $s^2 + \frac{1}{T}s + \frac{K}{T} = 0$ compare with $s^2 + 2\zeta\omega_n s + \omega^2{}_i = 0$

$$2\zeta\omega_n = \frac{1}{T}$$

$$\omega_n^2 = \frac{K}{T} \qquad \therefore \omega_n = \sqrt{\frac{K}{T}}$$

$$2\zeta\sqrt{\frac{K}{T}} = \frac{1}{T} \quad \text{or} \qquad \zeta = \frac{1}{2T}\sqrt{\frac{T}{K}} = \frac{1}{2\sqrt{KT}}$$

Damping ratio $\zeta_1 = 0.3$; $\zeta_2 = 0.9$

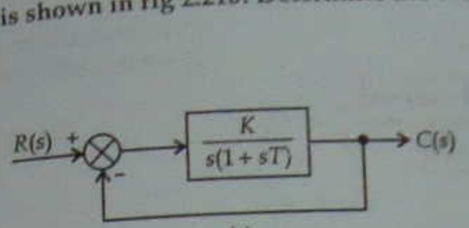
$$\zeta_1 = \frac{1}{2\sqrt{K_1 T}}, \quad \zeta_2 = \frac{1}{2\sqrt{K_2 T}}$$

$$\frac{\zeta_1}{\zeta_2} = \frac{1}{2\sqrt{K_1 T}} \cdot 2\sqrt{K_2 T} = \frac{\sqrt{K_2}}{\sqrt{K_1}} \quad \text{or,} \quad \frac{K_2}{K_1} = \left(\frac{\zeta_1}{\zeta_2}\right)^2$$

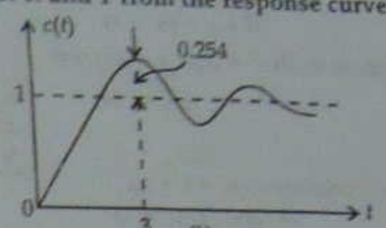
$$\frac{K_2}{K_1} = \left(\frac{0.3}{0.9}\right)^2 = \frac{1}{9} \quad \text{or} \quad 9K_2 = K_1 \qquad \boxed{K_1 = 9K_2}$$

Hence, the gain K_1 at which $\zeta = 0.3$ should be multiplied by 1/9 to increase the damping ratio from 0.3 to 0.9.

Example 2.14. The system shown in fig. 2.21*a* when subjected to a unit step input, the output response is shown in fig 2.21*b*. Determine the value of K and T from the response curve.



(*a*)



(*b*)

Fig. 2.21.

Solution :

$$\frac{C(S)}{R(S)} = \frac{\frac{K}{s(1+sT)}}{1+\frac{K}{s(1+sT)}}$$

$$\frac{C(S)}{R(S)} = \frac{K}{s^2T+s+K} = \frac{K/T}{s^2+\frac{1}{T}s+\frac{K}{T}}$$

characteristic eq^n.

$$s^2 + \frac{1}{T}s + \frac{K}{T} = 0$$

compare with $s^2 + 2\zeta\omega_n s + \omega^2{}_n = 0$

$$\omega_n = \sqrt{\frac{K}{T}} \quad \text{and} \quad 2\zeta\omega_n = \frac{1}{T}$$

$M_p = 0.254$ (given)

$$M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$0.254 = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \qquad \therefore \zeta = 0.4$$

$t_p = 3$ sec. (given)

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} \text{ or } 3 = \frac{\pi}{\omega_n\sqrt{1-(0.4)^2}} \therefore \omega_n = 1.14$$

$$2\zeta\omega_n = \frac{1}{T} \quad \text{or,} \quad T = \frac{1}{2\zeta\omega_n} = \frac{1}{2\times 0.4\times 1.14} = 1.09 \qquad \boxed{T = 1.09}$$

$$\omega_n = \sqrt{\frac{K}{T}} \quad \text{or,} \quad \omega_n^2 = \frac{K}{T} \text{ or } K = \omega_n^2 T = 1.42 \qquad \boxed{K = 1.42}$$

Example 2.15. A system having a forword path transfer function $G(S) = 16/s(s+1)$ and unity feedback. Determine the value of undamped natural frequency, damping ratio. If tachometer feedback is introduced, the feedback path transfer function becomes $(1 + Ks)$. What should be the value of K to obtain damping ratio of 0.6. Also calculate the percentage peak overshoot, first undershoot, t_p and settling time within the 2% of final value.

Solution : The characteristic equation with unity feedback

$$1 + G(S) = 0 \qquad \therefore H(S) = 1$$

$$1 + \frac{16}{s(s+1)} = 0$$

or $$s^2 + s + 16 = 0$$

compare with $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$

$$\omega_n^2 = 16 \qquad \boxed{\omega_n = 4 \text{ rad/sec}}$$

$$2\zeta\omega_n = 1 \qquad \boxed{\zeta = 0.125}$$

Now, consider $H(s) = 1 + Ks$

$$1 + G(S)\,H(S) = 0$$

$$1 + \frac{16}{s(s+1)}(1+Ks) = 0$$

or, $$s^2 + (1 + 16K)s + 16 = 0$$

$$\therefore \quad \omega_n^2 = 16 \qquad \omega_n = 4 \text{ rad/sec.}$$

$$2\zeta\omega_n = 1 + 16K$$

$$2 \times 0.6 \times 4 = 1 + 16K \qquad \therefore \boxed{K = 0.2375}$$

$$M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

When $\zeta = 0.125, M_p = e^{\frac{-\pi(0.125)}{\sqrt{1-(0.125)^2}}} \times 100 = 67.3\%.$

When $\zeta = 0.6, M_p = e^{\frac{-\pi(0.6)}{\sqrt{1-(0.6)^2}}} \times 100 = 9.48\%.$

First undershoot $= e^{-2\pi\zeta/\sqrt{1-\zeta^2}} = e^{-2\pi(0.6/\sqrt{1-(0.6)^2}} = 0.009$

Peak time :

When $\zeta = 0.125$

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{4\sqrt{1-(0.125)^2}} = 0.79 \text{ sec.}$$

when $\zeta = 0.6$

$$t_p = \frac{\pi}{4\sqrt{1-(0.6)^2}} = 0.98 \text{ sec.}$$

Settling time t_s :

$$t_s = \frac{4}{\zeta\omega_n}$$

when $\zeta = 0.125, t_s = \frac{4}{0.125 \times 4} = 8$ sec.

when $\zeta = 0.6, t_s = \frac{4}{4 \times 0.6} = 1.67$ sec.

Example 2.16. In example 2.13, by what factor the time constant 'T' should be multiplied so that the damping ratio is reduced from 0.8 to 0.2.

Solution :

$$\zeta_1 = 0.8$$
$$\zeta_2 = 0.2$$

$$\zeta_1 = \frac{1}{2\sqrt{T_1 K}}, \quad \zeta_2 = \frac{1}{2\sqrt{T_2 K}}$$

T_1 = time constant for ζ_1
T_2 = time constant for ζ_2

$$\frac{\zeta_1}{\zeta_2} = \frac{0.8}{0.2} = 4 = \sqrt{\frac{T_2}{T_1}}$$

$$\frac{T_2}{T_1} = 16$$

$$\boxed{T_2 = 16\,T_1}$$

Hence, original time constant T_1 should be multiplied by 16, then the damping ratio becomes 0.2.

Example 2.17. Consider the system shown in fig. 2.22. Determine the value of k such that the damping ratio ζ is 0.5. Then obtain the rise time (t_r), peak time (t_p), maximum overshoot M_p, and settling time t_s in the unit step response.

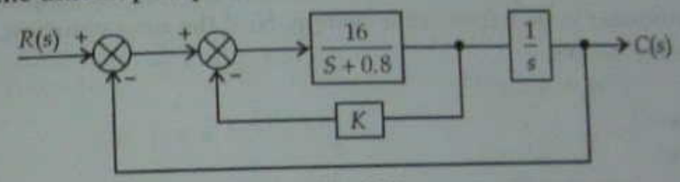


Fig. 2.22.

Solution : Step 1. Find $\frac{c(s)}{R(s)}$ by block reduction method.

$$\frac{c(s)}{R(s)} = \frac{16}{s^2 + (0.8+16k)s + 16}$$

Characteristic equation $s^2 + (0.8 + 16k)\,s + 16 = 0$, compare with $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$

$$\omega_n^2 = 16 \quad \therefore \omega_n = 4 \text{ rad/sec.}$$

$$2\zeta\omega_n = 0.8 + 16k$$

Step 2 : $$2 \times 0.5 \times 4 = 0.8 + 16k \qquad \boxed{\therefore k = 0.2}$$

Rise time $$(t_r) = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi - \tan^{-1}\frac{\sqrt{1-0.5^2}}{0.5}}{4\sqrt{1-0.5^2}} = 0.605 \text{ sec.}$$

Peak time $$(t_p) = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{4\sqrt{1-0.5^2}} = 0.906 \text{ sec.}$$

Max^m. overshoot $$(M_p) = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = e^{\frac{-\pi 0.5}{\sqrt{1-0.5^2}}} \times 100 = 16.3\%$$

Settling time $$(t_s) = \frac{4}{\zeta\omega_n} = \frac{4}{0.5 \times 4} = 2 \text{ sec.}$$ **Ans.**

Example 2.18. A system shown in fig. 2.23, determine the value of K and k such that the system has a damping ratio ξ of 0.7 and an undamped natural frequency w_n of 4 rad/sec.

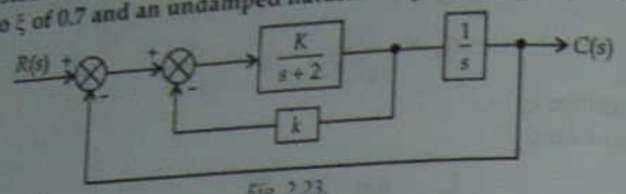


Fig. 2.23.

Solution : Find $\frac{c(s)}{R(s)}$ by block reduction method.

$$\frac{c(s)}{R(s)} = \frac{K}{s^2+(2+Kk)s+K}$$

characteristic equation $s^2 + (2 + Kk)s + K = 0$ compare with $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$

$$\omega^2_n = K \quad \therefore K = 16 \qquad \boxed{K = 16}$$

$$2\zeta\omega_n = 2 + Kk$$

$$2 \times 0.7 \times 4 = 2 + 16k \quad \therefore \quad \boxed{k = 0.225}$$

Example 2.19. A thermometer requires 1 min. to indicate 98% of the response to a step input. Assuming the thermometer to be a first order system, find the time constant.

Solution : For first order system

$$c(t) = 1 - e^{-t/T}$$

T = time constant = ?

t = 1 min. = 60 sec.

$$c(t) = 0.98$$

$$0.98 = 1 - e^{-60/T} \quad \therefore T = 15.33 \text{ sec.}$$

Time constant = 15.33 sec.

Example 2.20. Consider the fig. 2.24. The damping ratio of the system is 0.158 and the undampet natural frequency is 3.16 rad/sec. To improve the relative stability (fig. 2.24b) we employ tachometer feedback. Determine the value of K_h so that the damping ratio of the system is 0.5.

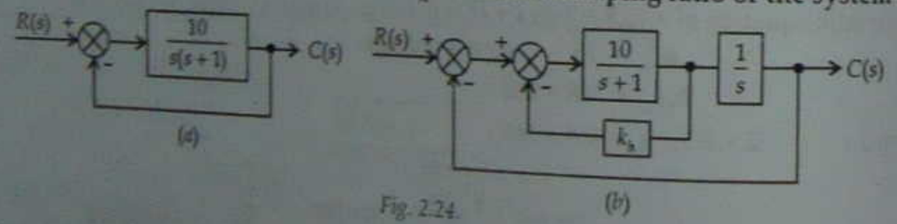


Fig. 2.24.

Solution : From fig. 2.24 (*b*) we can determine $\frac{c(s)}{R(s)}$ by block reduction method.

$$\frac{c(s)}{R(s)} = \frac{10}{s^2+(1+10K_h)s+10}$$

characteristic equation $s^2 + (1 + 10\,K_h)s + 10 = 0$ compare with $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$

$$2\zeta\omega_n = 1 + 10\,K_h$$

$$\omega^2_n = 10 \quad \therefore \omega_n = 3.16 \text{ rad/sec.}$$

$$2 \times 0.5 \times 3.16 = 1 + 10\,K_h$$

$$K_h = 0.216 \qquad \boxed{K_h = 0.216}$$

Example 2.21. Obtain the unit step response of a unity feedback system whose openloop transfer function is

$$G(s) = \frac{4}{s(s+5)}$$

Solution :

$$\frac{c(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{4}{s(s+5)}}{1+\frac{4}{s(s+5)} \cdot 1}$$

$$c(s) = \frac{4}{s^2+5s+4}$$

$$\therefore \quad c(s) = \frac{4}{s^2+5s+4} R(s)$$

Given $R(s) = \frac{1}{s}$

$$\therefore \quad c(s) = \frac{4}{s(s^2+5s+4)} = \frac{4}{s(s+1)(s+4)}$$

Break the equation by partial fraction

$$\frac{4}{s(s+1)(s+4)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+4}$$

$$A = 1, B = -\frac{4}{3}, C = \frac{1}{3}$$

$$c(s) = \frac{1}{s} - \frac{4}{3(s+1)} + \frac{1}{3(s+4)}$$

Inverse laplace of above eqn.

$$\mathcal{L}^{-1} c(s) = \mathcal{L}^{-1} 1/s - \frac{4}{3}\mathcal{L}^{-1}\frac{1}{s+1} + \frac{1}{3}\mathcal{L}^{-1}\frac{1}{s+4}$$

$$c(t) = 1 - \frac{4}{3}e^{-t} + \frac{1}{3}e^{-4t} \qquad \textbf{Ans.}$$

Example 2.22. Consider the unit-step response of a unity-feedback control system whose open loop transfer function is

$$G(s) = \frac{1}{s(s+1)}$$

Obtain the rise time, peak time, maximum overshoot and settling time.

Solution : Characteristic eq^n. $1 + G(s)\,H(s) = 0$

$$1 + \frac{1}{s(s+1)} \cdot 1 = 0$$

$$s^2 + s + 1 = 0$$

Compare with $s^2 + 2\xi\omega_n s + \omega^2_n = 0$

$$\omega^2_n = 1 \qquad \therefore \omega_n = 1 \text{ rad/sec.}$$

$$2\zeta\omega_n = 1 \qquad \therefore \zeta = 0.5$$

Rise time $$t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi - \tan^{-1}\frac{\sqrt{1-0.5^2}}{0.5}}{1\sqrt{1-0.5^2}} = 2.41 \text{ sec.}$$

Peak time $$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{1\sqrt{1-0.5^2}} = 3.625 \text{ sec.}$$

Maximum overshoot $$M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$= e^{\frac{\pi 0.5}{\sqrt{1-0.5^2}}} \times 100 = 16.31\%$$

Settling time $$t_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.5\times 1} = 8 \text{ sec.}$$

Example 2.23. Consider the closed loop system given by

$$\frac{c(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Determine the values of ζ and ω_n so that the system responds to a step input with approximately 5% overshoot and with a settling time of 2 sec.

Solution : Given that

$$t_s = 2 \text{ sec.}$$
$$M_p = 5\% = 0.05$$
$$M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$$
$$0.05 = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \quad \therefore \quad \boxed{\zeta = 0.689}$$
$$t_s = \frac{4}{\zeta\omega_n} \quad \text{or,} \quad \omega_n = \frac{4}{\zeta t_s}$$
$$\omega_n = \frac{4}{0.689\times 2} = 2.9 \text{ rad/sec.} \qquad \boxed{\omega_n = 2.9 \text{ rad/sec.}}$$

Example 2.24. Fig. 2.25 is a block diagram of a space-vehicle attitude control system. Assuming the time constant T of the controller to be 3 sec. and the ratio of torque to inertia K/J to be $\frac{2}{9}$ rad/sec^2 find the damping ratio of the system.

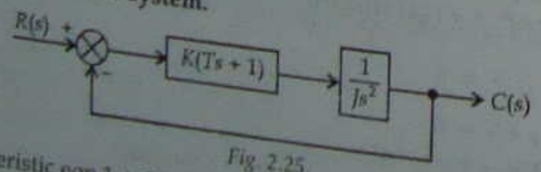


Fig. 2.25.

Solution : The characteristic eqn $1 + G(s)\,H(s) = 0$

$$1 + \frac{K(Ts+1)}{Js^2} = 0$$

$$Js^2 + 3Ks + K = 0$$

or,
$$s^2 + 3\frac{K}{J}s + \frac{K}{J} = 0$$

or,

compare with $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$

$$\therefore \quad \omega_n^2 = \frac{K}{J}$$
$$2\zeta\omega_n = 3\frac{K}{J}$$
$$\omega_n^2 = \frac{2}{9}$$
$$\boxed{\omega_n = 0.471 \text{ rad/sec}}$$
$$2 \times \zeta \times 0.471 = 3\cdot\frac{2}{9}$$
$$\therefore \quad \boxed{\zeta = 0.707} \quad \textbf{Ans.}$$

Example 2.25. Measurement conducted on a servomechanism show the system response to be

$$c(t) = 1 + 0.2\,e^{-60t} - 1.2\,e^{-10t}$$

when subjected to a unit step input.

(a) Obtain the expression for the closed loop transfer function.

(b) Determine the undamped natural frequency and damping ratio of the system.

Solution :
$$c(t) = 1 + 0.2\,e^{-60t} - 1.2\,e^{-10t}$$
$$c(s) = \frac{1}{s} + \frac{0.2}{s+60} - \frac{1.2}{(s+10)}$$
$$c(s) = \frac{600}{s(s+10)(s+60)}$$

(a) Transfer function $$= \frac{c(s)}{R(s)} = \frac{\frac{600}{s(s+10)(s+60)}}{1/s} = \frac{600}{(s+10)(s+60)}$$

(b) Characterisc equation $= s^2 + 70\,s + 600$, compare with $s^2 + 2\xi\omega_n s + \omega^2_n = 0$

compare with $2\zeta\omega_n = 70$

$\omega^2_n = 600$

$\therefore \quad \omega_n = 24.49$ rad/sec.

$2 \times \zeta \times 24.49 = 600$

$\therefore \quad \zeta = 1.42$

$$\frac{c(s)}{R(s)} = \frac{600}{(s+10)(s+60)}$$

$\omega_n = 24.49$ rad/sec.

$\zeta = 1.42$ **Ans.**

2.9. EFFECT OF FEEDBACK ON TIME CONSTANT OF A CONTROL SYSTEM

Consider the open loop transfer function as shown in fig. 2.26.

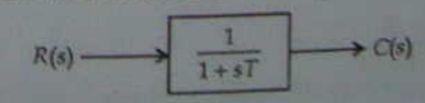


Fig. 2.26.

$$\frac{C(s)}{R(s)} = \frac{1}{1+sT} \quad \text{...(2.76)}$$

Let $R(s) = \frac{1}{s}$

$$\therefore \quad C(s) = \frac{1}{1+sT} R(C)$$

$$C(s) = \frac{1}{1+sT} \cdot \frac{1}{s} \quad \text{...(2.76)}$$

Take inverse laplace of equation 2.76

$$\mathcal{L}^{-1} C(s) = \mathcal{L}^{-1}\left[\frac{1}{s(1+sT)}\right]$$

$$C(t) = 1 - e^{-t/T} \quad \text{...(2.77)}$$

In equation (2.77), 'T' is known as time constant

Now consider the closed loop system as shown in fig 2.27.

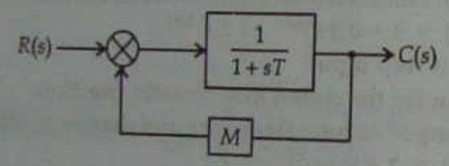


Fig. 2.27.

Suppose $H(s) = M$, then overall transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{\frac{1}{1+sT}}{1+\frac{1}{1+sT}M}$$

$$\frac{C(s)}{R(s)} = \frac{1}{1+sT+M} = \frac{1/T}{s+\left(\frac{1+M}{T}\right)}$$

let $R(s) = \frac{1}{s}$

$$\therefore \quad C(s) = \frac{1/T}{s+\left(\frac{1+M}{T}\right)} \quad \text{...(2.78)}$$

Taking inverse laplace of 2.78

$$\mathcal{L}^{-1} C(s) = \mathcal{L}^{-1}\left[\frac{1/T}{s+\left(\frac{1+M}{T}\right)}\right] = \frac{1}{1+M}\left[T - e^{-\frac{t}{T/1+M}}\right] \quad \text{...(2.79)}$$

Time constant of equation 2.79

$$\text{Time constant} = \frac{T}{1+M}$$

Thus for positive values of M, the time constant $\frac{T}{1+M}$ is less than time constant 'T'. It means that the time constant of closed loop system is less than the time constant of open loop system. Less time constant means response is faster. Therefore feedback improves the time response of the system.

SUMMARY

Time response consists of two parts (*a*) transient response, (*b*) steady state response. Transient response is the part of the response which goes to zero as time increases and steady state response is the part of the total response after transient has died.

Test Input signals :

1. Step function is also known as displacement function. Laplace transform is $\frac{1}{s}$.
2. Ramp function : Ramp function starts from origin and increase or decreases linearly with time. Laplace transform is $\frac{1}{s^2}$
3. Parabolic function : Laplace transform is $\frac{1}{s^3}$.
4. Impulse function: Impulse function has zero value everywhere except at $t = 0$ where the amplitude is infinite.

Step function is the derivative of ramp function.

$$\mathcal{L}\frac{d}{dt}(\text{unit ramp function}) = s \cdot \frac{1}{s^2} = \frac{1}{s} = \text{step function}$$

Impulse function is the derivative of step function

$$\mathcal{L}\frac{d}{dt}(\text{unit step function}) = s \cdot \frac{1}{s} = 1$$

Ramp function is the derivative of parabolic function

$$\mathcal{L}\frac{d}{dt}(\text{parabolic function}) = s \cdot \frac{1}{s^3} = \frac{1}{s^2} = \text{ramp function}$$

Transient response depends upon :

(*a*) Location of closed loop poles in s-plane

(*b*) Order of the system.

Response of first order system with unit step input is given by

$$c(t) = 1 - e^{-t/T}$$

Response of first order system with unit ramp input is given by

$$c(t) = t - T + T e^{-t/T}$$

Response of first order system with unit impulse input is given by

$$c(t) = \frac{1}{T} e^{-t/T}$$

The closed loop transfer function of second order system is

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Where ω_n = natural frequency of oscillations

ζ = damping ratio

Response of second order system with unit step input is given by

$$C(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin\left[\left(\omega_n\sqrt{1-\zeta^2}\right)t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right]$$

The system is under damped for $0 < \zeta < 1$

The system is critical damped for $\zeta = 1$

The system is overdamped for $\zeta > 1$

The system is undamped for $\zeta = 0$

Delay time (t_d) : The delay time is the time required for the response to reach 50% of the final value in first time.

Rise time (t_r) : It is time required for the response to rise from 10% to 90% of its final value for overdamped systems and 0 to 100% for underdamped systems.

Peak time : The peak time is time required for the response to reach the first peak of the time response of first peak overshoot.

Maximum overshoot : It is the normalized difference between the peak of the time response and steady output.

Setting time (t_s) : Setting time is time required for the response to reach and stay within specified range (2% to 5%) of its final value.

Steady state error : Steady state error is the difference between actual output and desired output as time 't' tends to infinity.

$$e_{ss} = \lim_{S\to\infty}[r(t) - c(t)]$$

Rise time is given by

$$t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}}{\omega_n\sqrt{1-\zeta^2}}$$

Peak time is given by

$$t_p = \frac{2\pi}{\omega_n\sqrt{1-\zeta^2}}$$

Maximum overshoot is given by

$$\% M_p = e^{\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

Settling time is given by

$$t_s = \frac{4}{\zeta\omega_n}$$

Response of second order system with unit impulse is given by

$$c(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin\left(\omega_n\sqrt{1-\zeta^2}\right)t$$

Response of second order system with unit ramp input

$$c(t) = t - \frac{2\zeta}{\omega_n} + \frac{e^{-\zeta\omega_n t}}{\omega_n\sqrt{1-\zeta^2}} \sin\left[\left(\omega_n\sqrt{1-\zeta^2}\right)t + \phi\right]$$

EXERCISE

1. A unity feedback system is characterised by an open loop transfer function

$$G(s) = \frac{K}{s(s+20)}$$

Determine the gain K so that the system will have a damping ratio of 0.6. For this value of K calculate settling time, peak overshoot and time to peak overshoot for a unit step input.

2. A feedback system is described by the following transfer function

$$G(s) = \frac{20}{s^2 + 4s + 25}, \quad H(s) = Ks$$

The damping factor of the system is 0.8. Determine the overshoot of the system.

3. The forward path transfer function of a system is given by

$$G(s) = \frac{4}{s(s+6)}$$

Obtain an expression for unit step response of the system. Assume unity feedback system.

4. The open loop transfer function of a servosystem with unity feedback system is given by

$$G(s) = \frac{K}{s(s+2)(s+6)}$$

Determine the damping ratio, undamped natural frequency, maximum overshoot for unit step input and $K = 10$.

5. The open loop transfer function of a unity feedback system is $G(s) = \frac{K}{s(1+sT)}$. By what factor the gain K be reduced so that the overshoot is reduced from 60% to 20%.

6. The block diagram of a position control system with velocity feedback is shown in fig. 2.28. Determine the value of 'a' so that the step response has a maximum overshoot of 20%.

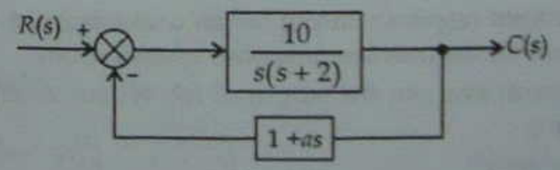


Fig. 2.28.

7. For the closed loop system given by

$$\frac{C(s)}{R(s)} = \frac{w_n^2}{s^2 + 2\xi w_n s + w_n^2}$$

Determine the values of ξ and w_n so that the system responds to a unit step input with 10% overshoot and with settling time of 4 sec.

8. For a unity feedback system with $G(s) = \frac{4}{s(s+2)}$ obtain an expression for unit step response of the system.
9. Determine the time response $y(t)$ for

(a) $Y(s) = \frac{4}{s(s+1)(s+2)}$

(b) $Y(s) = \frac{4}{s(s-1)(s-2)}$

10. What are the time domain quantities which characterise a transient response ? Derive an expression for percentage overshoot of a second order system. *(A.M.U., Aligarh, B.E., 1982-83)*
11. The open loop transfer function of a unity feedback system is

$$G(s) = \frac{K}{s+1}$$

Find the steady state error for an input of a unity feedback system as

$$r(t) = 1 + t + \frac{t^2}{2}$$

(A.M.U., Aligarh, B.E., 1983-84)

SEMI-OBJECTIVE TYPE QUESTIONS

(i) Define transient response.
(ii) Sketch various test input signals for transient analysis. (Article 2.1)
(iii) Write mathematical expressions for various test input signals. (Article 2.2)
(iv) Derive the expression for response of the first order system with unit step input. (Article 2.3)
(v) Define delay time, rise time and peak time. (Article 2.3.1)
(vi) Derive the expression for maximum overshoot (M_p) (Article 2.5)
(vii) For a system having transfer function (Article 2.5)

$$\frac{C(s)}{R(s)} = \frac{64}{s^2 + 5s + 64}$$

determine: (i) ω_n (ii) ξ (iii) ω_d

(**Ans.** $\omega_n = 8$ rad/sec, $\xi = 0.3125$, $\omega_d = 7.6$ rad/sec)

(viii) Draw a typical transient response output for an underdamped second order system and show the locations of all the transient response specifications. (Article 2.5)
(ix) For a first order system find out the output of the system when the input applied to the system is a unit ramp. (Article 2.3.2)
(x) What is steady state error? (Article 2.1)
(xi) What is the effect of feedback on time constant of a control system? (Article 2.9)
(xii) Explain the relationship between damping ratio and % overshoot.
(xiii) What are time domain specifications?

□□□

Chapter 3

Error Analysis

3.1. CLASSIFICATION OF CONTROL SYSTEM

Consider the open loop transfer function

$$G(s)H(s) = \frac{K(1+sT_1)(1+sT_2)\ldots}{s^m(1+sT_a)(1+sT_b)+\ldots} \quad \ldots(3.1)$$

In equation (3.1), the poles are at $s = -\frac{1}{T_a}$, $s = -\frac{1}{T_b}$ ... and zeros are at $s = {}^{-1}/_{T1}$, $s = {}^{-1}/_{T2}$... The equation having a term s^m in denominator, 'm' is the number of poles at the origin.

A system having no pole at origin of the 's' plane, is said to be type '0' (zero) system *i.e.*, $m = 0$

If $m = 1$ *i.e.*, 's', it means the system has a pole at origin of the s-plane and is said to be type '1' (one) system.

A system is called type '2' system if $m = 2$. and so on.

3.2. STEADY STATE ERROR

The steady state error is the difference between the input and output of the system during steady state. For accuracy the steady state error should be minimum.

Consider a closed loop control system shown in fig (3.1)

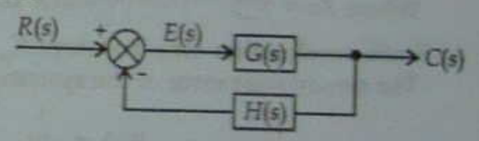


Fig. 3.1.

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

or, $$E(s) = \frac{R(s)}{1+G(s)H(s)} \quad \ldots(3.2)$$

The steady state error of the system is obtained by applying final value theorem.

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} s.E(s) \quad \ldots(3.3)$$

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{R(s)}{1+G(s)H(s)} \quad \ldots(3.4)$$

For unity feedback system $H(s) = 1$

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{R(s)}{1+G(s)} \quad \ldots(3.5)$$

From the equation (3.4) or (3.5) it is clear that the steady state error depends on the input and open loop transfer function.

3.3. STATIC ERROR COEFFICIENTS

(a) Static-Position Error Constant (or Coefficient) K_p

The steady state error is given by equation 3.4

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{R(s)}{1+G(s)H(s)}$$

For unit step input $R(s) = \frac{1}{s}$, the steady state error is given by

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{1}{s} \frac{1}{1+G(s)H(s)} = \frac{1}{1+\lim_{s\to 0} G(s)H(s)}$$

$$e_{ss} = \frac{1}{1+K_p}$$

K_p = static position error constant = $\lim_{s\to 0} G(s)H(s)$ ...(3.6)

(b) Static Velocity Error Constant (or Coefficient) Kv

$$e_{ss} = \lim_{s\to 0} s.R(s) \cdot \frac{1}{1+G(s)H(s)}$$

Steady state error with a unit ramp input is given by $[R(s) = 1/s^2]$

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{1}{s^2} \frac{1}{1+G(s)H(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{1}{s+sG(s)H(s)}$$

$$= \lim_{s\to 0} \frac{1}{sG(s)H(s)}$$

$$e_{ss} = \frac{1}{K_v}$$ ...(3.7)

Where $Kv = \lim_{s\to 0} sG(s)H(s)$ static velocity error coeff.

(c) Static acceleration error constant K_a

The steady state error of the system with unit parabolic input is given by

$$R(s) = \frac{1}{s^3}$$

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{1}{s^3} \cdot \frac{1}{1+G(s)H(s)} = \lim_{s\to 0} \frac{1}{s^2+s^2G(s)H(s)}$$

$$= \lim_{s\to 0} \frac{1}{s^2G(s)H(s)} = \frac{1}{K_a}$$ ...(3.8)

Where $K_a = \lim_{s\to 0} s^2\, G(s)\, H(s)$ = static acceleration constant.

3.4. STEADY STATE ERROR FOR DIFFERENT TYPE OF SYSTEMS

1. (a) Type zero system with unit step input

$$R(s) = \frac{1}{s}$$

From eqn. 3.1

$$G(s)\, H(s) = \frac{K(1+sT_1)(1+sT_2)\ldots}{(1+sT_a)(1+sT_b)\ldots}$$ ...(3.9)

$$K_p = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0} \frac{K(1+sT_1)(1+sT_2)\ldots}{(1+sT_a)(1+sT_b)\ldots} = K$$

From eqn. (3.6)

$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+K} \qquad \boxed{e_{ss} = \frac{1}{1+K}}$$

Hence, for type zero system the static position error constant K_p is finite.

(b) Type '0' system with unit ramp input

$$K_v = \lim_{s\to 0} s \cdot G(s)H(s) = \lim_{s\to 0} s \cdot \frac{K(1+sT_1)(1+sT_2)\ldots}{(1+sT_a)(1+sT_b)\ldots} = 0$$

$$\therefore \quad e_{ss} = \frac{1}{Kv} = \infty \qquad \boxed{e_{ss} = \infty}$$

(c) type '0' system with unit parabolic input

$$Ka = \lim_{s\to 0} s^2 G(s)H(s) = \lim_{s\to 0} s^2 \frac{K(1+sT_1)(1+sT_2)\ldots}{(1+sTa)(1+sT_b)\ldots}$$

$$Ka = 0$$

$$\therefore \quad e_{ss} = \frac{1}{Ka} = \infty \qquad \boxed{e_{ss} = \infty}$$

For type '0' system, the steady state error is infinite for ramp and parabolic inputs. Hence, the ramp and parabolic inputs are not acceptable.

2(a) Type '1' system with unit step input ($m = 1$)

$$G(s)\, H(s) = \frac{K(1+sT_1)(1+sT_2)\ldots}{s(1+sTa)(1+sT_b)\ldots}$$

$$K_P = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0} \frac{K(1+sT_1)(1+sT_2)\ldots}{s(1+sT_a)(1+sT_b)\ldots} = \infty$$

$$e_{ss} = \frac{1}{1+K_p} = 0 \qquad \boxed{e_{ss} = 0}$$

(b) Type '1' system with unit ramp input

$$K_v = \lim_{s\to 0} s \cdot G(s)H(s) = \lim_{s\to 0} s \cdot \frac{K(1+sT_1)(1+sT_2)\ldots}{s(1+sT_a)(1+sT_b)\ldots}$$

$$K_v = K$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{K} \qquad \boxed{e_{ss} = \frac{1}{K}}$$

(c) Type '1' system with unit parabolic input

$$K_a = \lim_{s\to 0} s^2 G(s)H(s)$$

$$= \lim_{s\to 0} s^2 \frac{K(1+sT_1)(1+sT_2)\ldots}{s(1+sT_a)(1+sT_b)\ldots} = 0$$

$$\therefore \quad e_{ss} = \frac{1}{K_a} = \infty \qquad \boxed{e_{ss} = \infty}$$

Hence, from above relations for type '1' system, it is clear that for type '1' system step input and ramp inputs are acceptable and parabolic input is not acceptable.

3 (a) Type '2' system with unit step input

$$G(s)\,H(s) = \frac{K(1+sT_1)(1+sT_2)\ldots}{s^2(1+sT_a)(1+sT_b)\ldots}$$

$$K_p = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0} \frac{K(1+sT_1)(1+sT_2)\ldots}{s^2(1+sT_a)(1+sT_b)\ldots} = \infty$$

$$e_{ss} = \frac{1}{1+K_p} = 0 \qquad \boxed{e_{ss} = 0}$$

(b) **Type '2' system with unit ramp input**

$$K_v = \lim_{s\to 0} sG(s)H(s)$$

$$= \lim_{s\to 0} s \cdot \frac{K(1+sT_1)(1+sT_2)\ldots}{s^2(1+sT_a)(1+sT_b)\ldots}$$

$$= \infty$$

$$\therefore \quad e_{ss} = \frac{1}{K_v} = 0 \qquad \boxed{e_{ss} = 0}$$

(c) Type '2' system with unit parabolic input

$$K_a = \lim_{s\to 0} s^2 G(s)H(s)$$

$$= \lim_{s\to 0} s^2 \cdot \frac{K(1+sT_1)(1+sT_2)\ldots}{s^2(1+sT_a)(1+sT_b)\ldots} = K$$

$$\therefore \quad e_{ss} = \frac{1}{K_a} = \frac{1}{K} \qquad e_{ss} = \boxed{\frac{1}{K}}$$

Hence, for type '2' system all three inputs (step, ramp and parabolic) are acceptable. From table 3.1, the diagonal elements are the finite values of steady state error.

Table 3.1.

	Type '0' system	Type '1' system	Type '2' system
UNIT STEP INPUT	$\frac{1}{1+k}$	0	0
UNIT RAMP INPUT	∞	$\frac{1}{k}$	0
UNIT PARABOLIC INPUT	∞	∞	$\frac{1}{k}$

Example 3.1. The open loop transfer function of unity feedback system is given by

$$G(s) = \frac{50}{(1+0.1s)(s+10)}$$

Determine the static error coefficients K_p, K_v and K_a.

Solution :

$$K_p = \lim_{s\to 0} G(s)H(s)$$

$$= \lim_{s\to 0} \frac{50}{(1+0.1s)(s+10)} = 5$$

$$K_v = \lim_{s\to 0} s\,G(s)H(s) = \lim_{s\to 0} s \cdot \frac{50}{(1+0.1s)(s+10)} = 0$$

$$K_a = \lim_{s\to 0} s^2 \frac{50}{(1+0.1s)(s+10)} = 0$$

Example 3.2. The forward path transfer function of a unity feedback control system is given by

$$G(s) = \frac{5(s^2+2s+100)}{s^2(s+5)(s^2+3s+10)}$$

Determine the step, ramp and parabolic error coefficients. Also determine the type of the system.

Solution :

$$K_P = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0} \frac{5(s^2+2s+100)}{s^2(s+5)(s^2+3s+10)}$$

$$K_P = \infty \qquad \boxed{K_P = \infty}$$

$$K_v = \lim_{s\to 0} sG(s)H(s) = \lim_{s\to 0} s \cdot \frac{5(s^2+2s+100)}{s^2(s+5)(s^2+3s+10)}$$

$$K_v = \infty \qquad \boxed{K_v = \infty}$$

$$K_a = \lim_{s\to 0} s^2 G(s)H(s) = \lim_{s\to 0} s^2 \cdot \frac{5(s^2+2s+100)}{s^2(s+5)(s^2+3s+10)}$$

$$= 10 \qquad \boxed{K_a = 10}$$

In denominator the value of $m = 2$. Hence, the given system is type '2' system.

Example 3.3. The block diagram of an electronic pacemaker is given in fig. 3.2. Determine the steady state error for unit ramp input when K = 400 Also, determine the value of K for which the steady state error to a unit ramp will be 0.02.

Solution : Given that $K = 400$

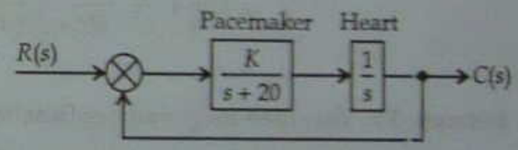


Fig. 3.2

$$R(s) = \frac{1}{s^2}$$

$$H(s) = 1$$

$$\therefore \quad G(s)\,H(s) = \frac{K}{s(s+20)}$$

Steady state error is given by $e_{ss} = \lim_{s\to 0} s \cdot \frac{R(s)}{1+G(s)H(s)}$

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{1}{s^2} \frac{1}{1+\frac{K}{s(s+20)}} = \lim_{s\to 0} \frac{s+20}{s(s+20)+400} = 0.05 \qquad \textbf{Ans.}$$

Now $e_{ss} = 0.02$ (given)

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{1}{s^2} \frac{1}{1+\frac{K}{s(s+20)}}$$

$$0.02 = \lim_{s\to 0} \frac{s+20}{s(s+20)+K}$$

$$0.02 = \frac{20}{K} \qquad \therefore \quad \boxed{K = 1000} \qquad \textbf{Ans.}$$

Example 3.4. For a unity feedback control system the forward path transfer function is given by

$$G(s) = \frac{20}{s(s+2)(s^2+2s+20)}$$

Determine the steady state error of the system. When the inputs are (i) 5 (ii) 5t (iii) $\frac{3t^2}{2}$.

Solution :

(i) $r(t) = 5 \quad \therefore R(s) = \frac{5}{s}$

$$e_{ss} = \lim_{s\to 0} s\cdot E(s) = \lim_{s\to 0} s\cdot \frac{R(s)}{1+G(s)H(s)}$$

$$e_{ss} = \lim_{s\to 0} s\cdot \frac{5}{s}\cdot \frac{1}{1+\frac{20}{s(s+2)(s^2+2s+20)}} = \lim_{s\to 0} \frac{5s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$\therefore \quad e_{ss} = 0$

(ii) $R(s) = \frac{5}{s^2}$

$$e_{ss} = \lim_{s\to 0} s\cdot \frac{5}{s^2}\cdot \frac{1}{1+\frac{20}{s(s+2)(s^2+2s+20)}} = \lim_{s\to 0} s\cdot \frac{5}{s^2}\cdot \frac{s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$e_{ss} = 10$

(iii) $R(s) = \frac{3}{s^3}$

$$e_{ss} = \lim_{s\to 0} s\cdot \frac{3}{s^3}\cdot \frac{s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$e_{ss} = \infty$

Example 3.5. The open loop transfer function of a unity feedback system is given by

$$G(s) = \frac{108}{s^2(s+4)(s^2+3s+12)}$$

Find the static error coefficients and steady state error of the system when subjected to an input given by

$$r(t) = 2 + 5t + 2t^2$$

Solution :

$$K_P = \lim_{s\to 0} G(s) = \lim_{s\to 0} \frac{108}{s^2(s+4)(s^2+3s+12)} = \infty$$

$$K_v = \lim_{s\to 0} s\cdot G(s) = \lim_{s\to 0} s\cdot \frac{108}{s^2(s+4)(s^2+3s+12)} = \infty$$

$$K_a = \lim_{s\to 0} s^2 G(s) = \lim_{s\to 0} s^2\cdot \frac{108}{s^2(s+4)(s^2+3s+12)} = \frac{108}{48}$$

$r(t) = 2 + 5t + 2t^2$

$R(s) = \frac{2}{s} + \frac{5}{s^2} + \frac{4}{s^3}$

$$\therefore \quad e_{ss} = \frac{R_1}{1+K_p} + \frac{R_2}{K_v} + \frac{R_3}{K_a}$$

$$= \frac{2}{1+\infty} + \frac{5}{\infty} + \frac{4\times 48}{108} = 1.77 \qquad \boxed{e_{ss} = 1.77} \quad \textbf{Ans.}$$

Example 3.6. Determine the type of the following unity feedback systems for which the forward path transfer function are given

(a) $G(s) = \frac{K}{(s+10)(s+5)}$ (b) $G(s) = \frac{20}{s(1+0.1s)(1+0.05s)}$

(c) $G(s) = \frac{20(s+2)}{s^3(s+2)(s^2+5s+5)}$ (d) $G(s) = \frac{K}{s(s+1)(s+2)}$

Solution : (a) Since in denominator the power of s is zero *i.e* $m = 0$ hence it is type zerosystem.
(b) type one system
(c) $m = 3$, type '3' system
(d) $m = 1$, type '1' system

Example 3.7. A servomechanism is designed to keep a radar antena pointed at a flying aeroplane. If the aeroplane is flying with a velocity of 600 Km/hr, at a range of 2 km. and the maximum tracking error is to be within 0.1°, determine the required velocity error coefficient. *(GATE-1994)*

Solution : The block diag. of servo mechanism is shown in fig. 3.3.

linear velocity = 600 km./hr

r = 2km.

Angular velocity $\omega = \frac{600}{2} = 300$ rad/*hr*.

$= \frac{300}{3600}$ rad/sec. = 0.083 rad/sec.

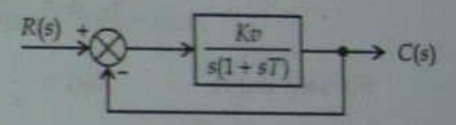


Fig. 3.3.

$$R(s) = \frac{0.083}{s^2}$$

$$e_{ss} = \lim_{s\to 0} s\cdot R(s)\cdot \frac{1}{1+G(s)H(s)} = \lim_{s\to 0} s\cdot \frac{0.083}{s^2}\cdot \frac{1}{1+\frac{K_v}{s(1+sT)}}$$

$$e_{ss} = \lim_{s\to 0} \frac{0.083}{s}\cdot \frac{s(1+sT)}{s(1+sT)+K_v} = \frac{0.083}{K_v} = 0.1° \text{ given}$$

$$0.1 = \frac{0.083}{K_v}$$

$\therefore \quad K_v = 0.83$ per degree

$\therefore$ Required velocity error coefficient $K_v = 0.83$ per degree

Example 3.8. For the system shown in fig. 3.4. Determine K_p and e_{ss} for unit step input.

Solution : $K_p = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0} \frac{1}{(s^2+s+2)(s+1)} = \frac{1}{2}$

$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+\frac{1}{2}} = \frac{1}{3/2} = \frac{2}{3}$$

$\boxed{K_p = 1/2}$ $\boxed{e_{ss} = 2/3}$ **Ans.**

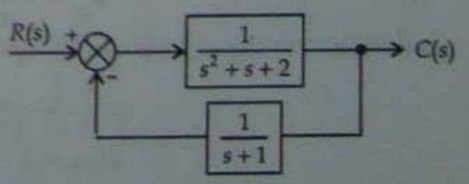


Fig. 3.4.

Example 3.9. For a given system shown in fig. 3.5, determine the actuating signal $E_a(s)$. Also determine the position error constant for unit step input.

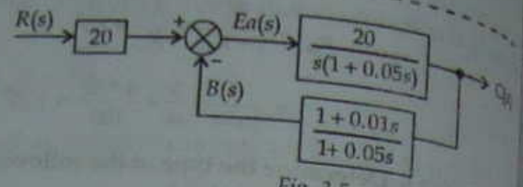


Fig. 3.5.

Solution :

$$E_a(s) = 20\,R(s) - B(s)$$

$$C(s) = \frac{20Ea(s)}{s(1+0.05s)}$$

$$B(s) = C(s)\cdot\frac{1+0.01s}{1+0.05s}$$

$$B(s) = \frac{20(1+0.01s)E_a(s)}{s(1+0.05s)^2}$$

$$\therefore \quad E_a(s) = 20R(s) - \frac{20(1+0.01s)E_a(s)}{s(1+0.05s)^2}$$

$$E_a(s)\left[1+\frac{20(1+0.01s)}{s(1+0.05s)^2}\right] = 20\,R(s)$$

$$\therefore \quad E_a(s) = \frac{20R(s)}{1+\dfrac{20(1+0.01s)}{s(1+0.05s)^2}}$$

$$E_a(s) = \frac{20s(1+0.05s)^2\cdot R(s)}{s(1+0.05s)^2+20(1+0.01s)} \qquad \textbf{Ans.}$$

For step Input $\quad R(s) = \dfrac{1}{s}$

$$E_a(s) = \frac{20(1+0.05s)^2}{s(1+0.05s)^2+20(1+0.01s)}$$

$$\therefore \quad e_{ss} = \lim_{s\to 0} s\cdot E(s) = \lim_{s\to 0} s\cdot\frac{20(1+0.05s)^2}{s(1+0.05s)^2+20(1+0.01s)} = 0$$

But $\quad e_{ss} = \dfrac{1}{1+K_p}$ or, $\quad 0 = \dfrac{1}{1+K_p}$

$$\therefore \quad \boxed{K_p = \infty} \qquad \textbf{Ans.}$$

Example 3.10. Consider a unity feedback control system with the closed loop transfer function

$$\frac{C(s)}{R(s)} = \frac{Ks+b}{s^2+as+b}$$

Determine the open loop transfer function. Show that the steady state error in the unit ramp input response is given by

$$e_{ss} = \frac{a-k}{b}$$

Solution : For closed loop control system

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$H(s) = 1$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)}$$

$$\frac{Ks+b}{s^2+as+b} = \frac{G(s)}{1+G(s)}$$

$$\therefore \quad G(s) = \frac{Ks+b}{s^2+s(a-K)}$$

$\therefore$ Open loop transfer function $G(s) = \dfrac{Ks+b}{s^2+s(a-k)}$ **Ans.**

Steady state error $e_{ss} = \lim_{s\to 0} s\cdot R(s)\cdot\dfrac{1}{1+G(s)H(s)}$

$$e_{ss} = \lim_{s\to 0} s\cdot\frac{1}{s^2}\cdot\frac{1}{1+\dfrac{Ks+b}{s^2+s(a-K)}\cdot 1} = \lim_{s\to 0}\frac{1}{s}\cdot\frac{s^2+s(a-K)}{s^2+s(a-K)+Ks+b}$$

$$= \lim_{s\to 0}\frac{1}{s}\frac{s[s+a-K]}{s^2+s(a-K)+Ks+b} = \frac{a-K}{b}$$

$$e_{ss} = \frac{a-K}{b} \qquad \textbf{Proved.}$$

3.5. DYNAMIC ERROR COEFFICIENTS

For the steady state error, the static error coefficients gives the limited information.

The error function is given by

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)} \qquad \text{...(3.10)}$$

For unity feedback system

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)} \qquad \text{...(3.11)}$$

The equation 3.11 can be expressed in polynomial form (ascending power of 's')

$$\frac{E(s)}{R(s)} = \frac{1}{K_1} + \frac{1}{K_2}s + \frac{1}{K_3}s^2 + \ldots\ldots \qquad \text{...(3.12)}$$

or,
$$E(s) = \frac{1}{K_1}R(s) + \frac{1}{K_2}sR(s) + \frac{1}{K_3}s^2R(s)\ldots. \qquad \text{...(3.13)}$$

Take inverse laplace of eqn. 3.13, the error given by

$$e(t) = \frac{1}{K_1}r(t) + \frac{1}{K_2}\dot{r}(t) + \frac{1}{K_3}\ddot{r}(t) + \ldots. \qquad \text{...(3.14)}$$

Steady state error is given by

$$e_{ss} = \lim_{s\to 0} sE(s)$$

Now, suppose if the input is unit step *i.e.* $R(s) = \dfrac{1}{s}$ then from 3.13

$$e_{ss} = \lim_{s\to 0} s\cdot\left[\frac{1}{K_1}\cdot\frac{1}{s} + \frac{1}{K_2}s\cdot\frac{1}{s} + \frac{1}{K_3}s^2\cdot\frac{1}{s} + \ldots\right]$$

$$e_{ss} = \frac{1}{K_1}$$

Similarly, for other test signal we can find steady state error. K_1, K_2, K_3 ... are known as "D... error coefficients".

Example 3.11. Find the dynamic error coefficients of the unity feedback system whose fo... transfer function is $G(s) = \frac{200}{s(s+5)}$. Find the steady state error of the system for the input ...

Solution : Given that

$$G(s) = \frac{200}{s(s+5)}$$

$$H(s) = 1$$

$$\therefore \quad \frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

$$\frac{E(s)}{R(s)} = \frac{s^2+5s}{s^2+5s+200} = \frac{5s+s^2}{200+5s+s^2}$$

$$200+5s+s^2 \overline{)\,5s+s^2} \quad \frac{0.025s+0.004375s^2-0.0000218s^4}{}$$

$$5s \pm 0.125s^2 \pm 0.025s^3$$

$$0.875\,s^2 - 0.025\,s^3$$

$$0.875\,s^2 \mp 0.022\,s^3 \pm 0.004375\,s^4$$

$$-0.004375\,s^4$$

$$\mp 0.004375\,s^4 \mp 0.0001\,s^5 \mp 0.0000218\,s^6$$

$$0.0001s^5 + 0.0000218\,s^6$$

$$\therefore \quad \frac{E(s)}{R(s)} = 0.025\,s + 0.004375\,s^2 - 0.0000218\,s^4 \ .....$$

or $E(s) = 0.025\,sR(s) + 0.00437\,5s^2\,R(s) - 0.0000218\,s^4\,R(s)$ .....

Take inverse laplace

$$e(t) = 0.025\,\dot{r}(t) + 0.004375\,\ddot{r}(t) - 0.0000218\,\bar{r}(t)$$

Given that $r(t) = 4t^2$

$$\dot{r}(t) = 8t$$

$$\ddot{r}(t) = 8$$

$$\bar{r}(t) = 0$$

$\therefore e(t) = 0.025 \times 8t + 0.004375 \times 8$

$e(t) = 0.2t + 0.035$

The steady state error is given by.

$$\lim_{t\to\infty} e(t) = \lim_{t\to\infty} (0.2t + 0.035)$$

The dynamic error coefficients are given by

$$K_1 = \frac{1}{0.025} = 40$$

$$K_2 = \frac{1}{0.004375} = 228.57$$

$$K_3 = \infty$$

$$K_4 = 45871.55$$

Example 3.12. Find the dynamic error coefficients of the unity feedback system whose forward path transfer function $G(s) = \frac{10}{s(s+1)}$.

Find the steady state error to the input $r(t) = P_0 + P_1 t + P_2 t^2$

Solution :

$$G(s) = \frac{10}{s(s+1)}, \quad H(s) = 1$$

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)} = \frac{s+s^2}{10+s+s^2}$$

$$= 0.1s + 0.09s^2 - 0.019\,s^3 \ ....$$

$$\therefore \quad E(s) = 0.1s\,R(s) + 0.09\,s^2\,R(s) - 0.019s^3\,R(s) \ .....$$

Inverse laplace $e(t) = 0.1\,r(t) + 0.09\,\ddot{r}(t) - 0.019\,\bar{r}(t)$ .....

$r(t) = P_0 + P_1 t + P_2 t^2$

$\dot{r}(t) = P_1 + 2P_2 t$

$\ddot{r}(t) = 2P_2$

$\bar{r}(t) = 0$

$\therefore \quad e(t) = 0.1\,(P_1 + 2P_2 t) + 0.09 \times 2P_2 = 0.1\,(P_1 + 2P_2 t) + 0.18\,P_2$

Steady state error is given by

$$\lim_{t\to\infty} e(t) = \lim_{t\to\infty} 0.1(P_1 + 2P_2 t) + 0.18P_2$$

Dynamic error coeff. are

$K_1 = 10$; $K_2 = 11.1$; $K_3 = -52.63$

SUMMARY

The steady state error is the difference between input and output of the system during steady state. For accuracy steady state error should be minimum.

$$e_{SS} = \lim_{s\to 0} sE(s)$$

$$e_{SS} = \lim_{s\to 0} s \cdot \frac{R(s)}{1+G(s)\,H(s)}$$

$$G(s)\,H(s) = \frac{K(1+sT_1)\,(1+sT_2...}{s^m(1+sT_a)\,(1+sT_b)...}$$

m = type of system

if $m = 0$, system is type zero

$m = 1$, system is type one etc.

K_P = static position error constant

$= \lim_{S\to 0} G(s)\,H(s)$ to be used for step input

K_v = Static velocity coefficient $= \lim_{S\to 0} sG(s)\,H(s)$ to be used for ramp input

K_a = static acceleration error constant

$= \lim_{s\to 0} s^2 G(s)\,H(s)$ to be used for parabolic input

	Type Zero system	Type one system	Type 2 system
Unit step input	$\frac{1}{1+K}$	0	0
Unit ramp input	∞	$\frac{1}{K}$	0
Unit parabolic input	∞	∞	$\frac{1}{K}$

SEMI-OBJECTIVE TYPE QUESTIONS

(i) State how type and order of a control system is determined?
(ii) Define steady-state error. (Artic
(iii) What are static error coefficients? (Artic
(iv) Derive the values of static error coefficients and steady-state error for a type 1 system (Artic
unit step input.
(v) Define D_p, K_v and K_a. (Artic
(vi) What is dynamic error coefficient? (Articles and
(vii) The open loop transfer function of unity feedback system is given by (Artic

$$G(s) = \frac{100}{(1+0.1s)(s+10)}$$

Determine K_p, K_v & K_a
(viii) What is velocity error? (Ans. 10, 0 a
(ix) What is positional error coefficient?
(x) Define static error constant
(xi) Write down the error series giving the generalised error coefficients.

Chapter 4

Frequency Domain Analysis

4.1. INTRODUCTION

The magnitude and phase relationship between sinusoidal input and steady state output of a system is known as frequency response. It is independent of the amplitude and phase of the input signal. If input signal is

$$r(t) = X \sin \omega t$$

the output can be written as

$$c(t) = Y \sin (\omega t + \theta)$$

The main advantage is that the frequency response of a stable open loop system can be obtained experimentally. Apply a sine wave at the input of the system and measure the ratio of the magnitude of output & input as well as the phase difference. Another advantage of this approach is that it is simple and accurate. we can determine experimentally the transfer function of the complicated system. Frequency response tests are not suitable for the systems having large time constant because for each frequency it takes long time to reach the steady state.

$$G(s) = \frac{C(s)}{R(s)} \quad \text{or,} \quad G(j\omega) = \frac{C(j\omega)}{R(j\omega)}$$

$$\text{or,} \quad C(s) = R(s).\, G(s) \quad \text{or,} \quad C(j\omega) = R(j\omega)\, G(j\omega)$$

The output is the product of input and $|G(s)|$

For stable system, the sinusoidal output is of the same frequency as input but the ampitude and phase of the output will be different from the input.

$$\therefore \qquad G(j\omega) = M \angle\phi$$

where M is the ratio of amplitudes of output and input sinusoids and is called the gain or magnitude ratio.

ϕ is angle by which output leads the input. M & ϕ are the function of angular frequcency ω.

4.2. POLAR PLOT

The polar plot of a sinusoidal transfer function $G(j\omega)$ is a plot of the magnitude of $G(j\omega)$ versus the phase angle of $G(j\omega)$ on polar coordinates as 'ω' is varied from zero to infinity. The polar plot, therefore is the locus of vectors $|G(j\omega)| \angle G(j\omega)$ as ω is varied from zero to infinity. Thus we can express the frequency response function $G(j\omega)$ in the polar form $Me^{j\phi(\omega)}$ and plot the vector$Me^{j\phi(\omega)}$ in the G-plane as 'ω' varies from zero to infinity. In polar plots the magnitude of $G(j\omega)$ is plotted as the distance from the origin while the phase angle is measured from positive real axis. Positive phase angle is measured counter clockwise while negative phase angle is measured clockwise from the positive real axis. The polar plot is often called the Nyquist Plot.

The advantage in using a polar plot that it depicts the frequency response characterstics of a system over the entire frequency range in a single plot. The disadvantage is that the plot does not indicate the contributions of each individual factor of the open loop transfer function.

4.3. PROCEDURE TO SKETCH THE POLAR PLOT

Step 1 : Determine the transfer function $G(s)$ of the system.

Step 2 : Put $S = j\omega$ in the transfer function to obtain $G(j\omega)$

Step 3 : At $\omega = 0$ and $\omega = \infty$ calculate $|G(j\omega)|$. by $\lim_{\omega \to 0} |G(j\omega)|$ and $\lim_{\omega \to \infty} |G(j\omega)|$.

Step 4 : Calculate the phase angle of $G(j\omega)$ at $\omega = 0$ and $\omega = \infty$ by $\lim_{\omega \to 0} \angle G(j\omega)$ and $\lim_{\omega \to \infty} \angle G(j\omega)$

Step 5 : Rationalize the function $G(j\omega)$ and separate the real and imaginary parts.

Step 6 : Equate the imaginary part $I_m |G(j\omega)|$ to zero & determine the frequencies at which plot intersects the real axis and caluculate the value $G(j\omega)$ at the point of intersection by substituting the determined value of frequency in the expression of $G(j\omega)$.

Step 7 : Equate the real part $\text{Re}|G(j\omega)|$ to zero and determine the frequencies at which plots intersects the imaginary axis and calculate the value of $G(j\omega)$ at the point of intersection by substituting the determined value of frequency in the rationalized expression of $G(j\omega)$.

Step 8 : Sketch the polar plot with the help of above information.

1. TYPE 'ZERO' SYSTEM

$$G(s) = \frac{K}{(1+ST_1)(1+ST_2)}$$

Step 1 : Put $S = j\omega$

$$G(j\omega) = \frac{K}{(1+j\omega T_1)(1+j\omega T_2)}$$

$$G(j\omega) = \frac{K}{\sqrt{1+(\omega T_1)^2}\sqrt{1+(\omega T_2)^2}} \angle -\tan^{-1}\omega T_1 - \tan^{-1}\omega T_2$$

Step 2 : Taking the limit for the magnitude of $G(j\omega)$.

$$\lim_{\omega \to 0}|G(j\omega)| = \lim_{\omega \to 0} \frac{K}{\sqrt{1+(\omega T_1)^2}\sqrt{1+(\omega T_2)^2}} = K$$

$$\lim_{\omega \to \infty}|G(j\omega)| = \lim_{\omega \to \infty} \frac{K}{\sqrt{1+(\omega T_1)^2}\sqrt{1+(\omega T_2)^2}} = 0$$

Step 3 : Taking the limit for the phase angle of $G(j\omega)$

$$\lim_{\omega \to 0} \angle G(j\omega) = \lim_{\omega \to 0} \angle -\tan^{-1}\omega T_1 - \tan^{-1}\omega T_2 = 0$$

$$\lim_{\omega \to \infty} \angle G(j\omega) = \lim_{\omega \to \infty} \angle -\tan^{-1}\omega T_1 - \tan^{-1}\omega T_2 = -180°$$

Step 4 : Separating the real and imaginary parts of $G(j\omega)$

$$G(j\omega) = \frac{K}{(1+j\omega T_1)(1+j\omega T_2)} \cdot \frac{(1-j\omega T_1)(1-j\omega T_2)}{(1-j\omega T_1)(1-j\omega T_2)}$$

$$G(j\omega) = \frac{K(1-\omega^2 T_1 T_2)}{1+\omega^2 T_1^2+\omega^2 T_2^2+\omega^4 T_1^2 T_2^2} - j\frac{K\omega(T_1+T_2)}{1+\omega^2 T_1^2+\omega^2 T_2^2+\omega^4 T_1^2 T_2^2}$$

Equating the real part to zero.

$$\frac{K(1-\omega^2 T_1 T_2)}{1+\omega^2 T_1^2+\omega^2 T_2^2+\omega^4 T_1 T_2} = 0$$

$$\omega^2 = \frac{1}{T_1 T_2} \quad \text{or} \quad \omega = \frac{1}{\sqrt{T_1 T_2}} \quad \& \quad \omega = \pm\infty$$

$\therefore$ The frequency at which plot intersects the imaginary axis is $\frac{1}{\sqrt{T_1 T_2}}$.

For positive values of frequencies the polar plot intersects the imaginary axis at $\omega = \infty$

$$\omega = \frac{1}{\sqrt{T_1 T_2}} \quad \text{and} \quad \omega = \infty$$

Value of $G(j\omega)$ when

$$\omega = \frac{1}{\sqrt{T_1 T_2}}$$

$$G(j\omega) = 0 - j\frac{K\frac{1}{\sqrt{T_1 T_2}}(T_1+T_2)}{1+\frac{1}{T_1 T_2}T_2^2+\frac{1}{T_1 T_2}T_1^2+\frac{1}{T_1^2 T_2^2}T_1^2 T_2^2}$$

$$= -j\frac{K\frac{T_1+T_2}{\sqrt{T_1 T_2}}}{2+\frac{T_2}{T_1}+\frac{T_1}{T_2}} = -j\frac{K\frac{T_1+T_2}{\sqrt{T_1 T_2}}}{\frac{(T_1+T_2)^2}{T_1 T_2}} = -j\frac{K\sqrt{T_1 T_2}}{T_1+T_2}$$

$$|G(j\omega)| = \frac{K\sqrt{T_1 T_2}}{T_1+T_2} \quad \& \quad \angle G(j\omega) = -90°$$

$\therefore$ When $\omega = \frac{1}{\sqrt{T_1 T_2}}$ $\quad G(j\omega) = \frac{K\sqrt{T_1 T_2}}{T_1+T_2} \angle -90°$

$\omega = \infty$ $\quad G(j\omega) = 0 \angle -180°$

**Alternative :*

Note : Polar plot will intersect -90° axis so put

$$\angle G(j\omega) = -90 \; ; \; -\tan^{-1}\omega T_1 - \tan^{-1}\omega T_2 = -90°$$

taking tan on both side

$$\frac{\omega T_1 + \omega T_2}{1-\omega T_1 \omega T_2} = \infty$$

$\therefore$ $1 - \omega_2 T_1 T_2 = 0$; or $\omega = 1/\sqrt{T_1 T_2}$

Put $\omega = \frac{1}{\sqrt{T_1 T_2}}$ in $\frac{K}{(1+sT_1)(1+sT_2)}$ i.e. $\frac{K}{(1+j\omega T_1)(1+j\omega T_2)}$

& simplify $|G(j\omega)| = \frac{K\sqrt{T_1 T_2}}{T_1+T_2}$

Step 5 : Equating the imaginary part to zero

$$\frac{K\omega(T_1+T_2)}{1+\omega^2T_1^2+\omega^2T_2^2+\omega^4T_1T_2}=0$$

$\therefore \quad \omega = 0 \;\&\; \omega = \pm\infty$

When $\omega = 0 \qquad |G(j\omega)| = K \qquad \angle G(j\omega) = 0°$

$\omega = \infty \qquad |G(j\omega)| = 0 \qquad \angle G(j\omega) = 0°$

The coordinates of the intersection points 'A' and '0'

$$G(s)=\frac{K}{(1+ST_1)(1+ST_2)}$$

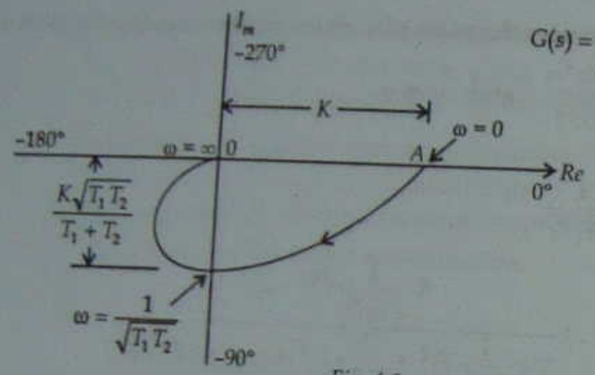


Fig. 4.1.

2. TYPE 'ONE' SYSTEM

$$G(s) = \frac{K}{S(1+ST_1)(1+ST_2)}$$

Step 1 : Put $s = j\omega$

$$G(j\omega) = \frac{K}{j\omega(1+j\omega T_1)(1+(j\omega T_2)}$$

$$G(j\omega) = \frac{K}{\omega\sqrt{1+(\omega T_1)^2}\sqrt{1+(\omega T_2)^2}} \angle -90°-\tan^{-1}\omega T_1-\tan^{-1}\omega T_2$$

Step 2 : Taking the limit for the magnitude of $G(j\omega)$

$$\lim_{\omega\to 0}|G(j\omega)| = \lim_{\omega\to 0}\frac{K}{\omega\sqrt{1+(\omega T_1)^2}\sqrt{1+(\omega T_2^2)}}=\infty$$

$$\lim_{\omega\to \infty}|G(j\omega)| = \lim_{\omega\to \infty}\frac{K}{\omega\sqrt{1+(wT_1)^2}\sqrt{1+(wT_2)^2}}=0$$

Step 3 : Taking the limit for the phase angle of $G(j\omega)$

$$\lim_{\omega\to 0}\angle G(j\omega) = \lim_{\omega\to 0}\angle -90°-\tan^{-1}\omega T_1-\tan^{-1}\omega T_2 = -90°$$

$$\lim_{\omega\to \infty}\angle G(j\omega) = \lim_{\omega\to \infty}\angle -90°-\tan^{-1}\omega T_1-\tan^{-1}\omega T_2 = -270°$$

Step 4 : Separating the real and imaginary parts

$$G(j\omega) = \frac{K}{j\omega(1+j\omega T_1)(1+j\omega T_2)}$$

$$= \frac{-\omega K(T_1+T_2)}{\omega+\omega^3(T_1^2+T_2^2+\omega^2T_1^2T_2^2)}+\frac{j(K\omega^2T_1T_2-K)}{\omega+\omega^3(T_1^2+T_2^2+\omega^2T_1^2T_2^2)} \quad ...(A)$$

Equate the imaginary part equal to zero

$$\frac{K\omega^2T_1T_2-K}{\omega+\omega^3(T_1^2+T_2^2+\omega^2T_1^2T_2^2)}=0$$

$$\therefore \quad \omega = \frac{1}{\sqrt{T_1T_2}} = \frac{1}{\sqrt{T_1T_2}} \;\&\; \omega = \pm\infty$$

The frequency at the point of intersection on real axis is $\frac{1}{\sqrt{T_1T_2}}$ Now calculate the value of $G(j\omega)$ at this point.

Put $\omega = \frac{1}{\sqrt{T_1T_2}}$ in eq$^{n.}$ (A)

$$G(j\omega) = -K\frac{T_1T_2}{T_1+T_2} \qquad \angle G(j\omega) = 0°$$

$$G(j\omega) = \infty \qquad \angle G(j\omega) = 0°$$

Step 5 : Equate the real part to zero

$$\frac{-\omega K(T_1+T_2)}{\omega+\omega^3(T_1^2+T_2^2+\omega^2T_1T_2)}=0$$

$$\therefore \quad \omega = \infty$$

For positive values of frequencies the polar plot intersects the imaginary axis at $\omega = \infty$

$$\therefore \quad G(j\omega) = 0 \angle -270°$$

Polar plot is shown in fig 4.2.

From the polar plot it is clear that in type one system the $j\omega$ term in denominator contributes -90° to the total phase angle. At $\omega = 0$, the magnitude is infinity and phase angle - 90°. At $\omega = \infty$, the magnitude becomes zero & curve converges to origin. At low frequency, the polar plot is asymptotic to a line parallel to negative imaginary axis.

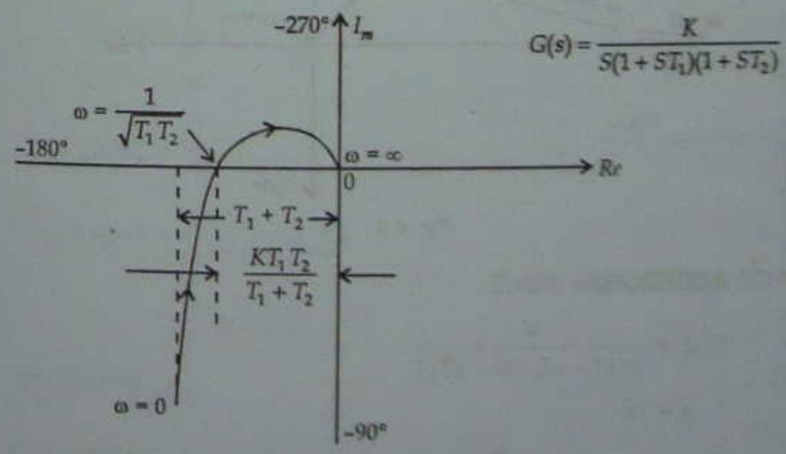


Fig 4.2

3. TYPE 'TWO' SYSTEM

$$G(s) = \frac{K}{s^2(1+sT_1)}$$

Put $s = j\omega$

$$G(j\omega) = \frac{K}{(j\omega)^2(1+j\omega T_1)}$$

$$G(j\omega) = \frac{K}{-\omega^2\sqrt{1+(\omega T_1)^2}} \angle -180° - \tan^{-1}\omega T_1$$

$$\lim_{\omega\to 0}|G(j\omega)| = \lim_{\omega\to 0}\frac{K}{-\omega^2\sqrt{1+(\omega T_1)^2}} = \infty$$

$$\lim_{\omega\to\infty}|G(j\omega)| = \lim_{\omega\to\infty}\frac{K}{-\omega^2\sqrt{1+(\omega T_1)^2}} = 0$$

$$\lim_{\omega\to 0} \angle -180° - \tan^{-1}\omega T_1 = -180°$$

$$\lim_{\omega\to\infty} \angle G(j\omega) = \lim_{\omega\to\infty} \angle -180° - \tan^{-1}\omega T_1 = -270°$$

The presence of S^2 in the denominator contributes a constant – 180° to the angle of $G(j\omega)$ for all frequencies.

The polar plot is a smooth curve whose angle decreases continuously from – 180° to – 270°. The plot is shown in fig.

From the polar plot it is clear that at $\omega = 0$, magnitude is infinity & phase angle – 180°, at $\omega = \infty$ magnitude is zero & at low frequencies the polar is asymptotic to a line parallel to negative real axis.

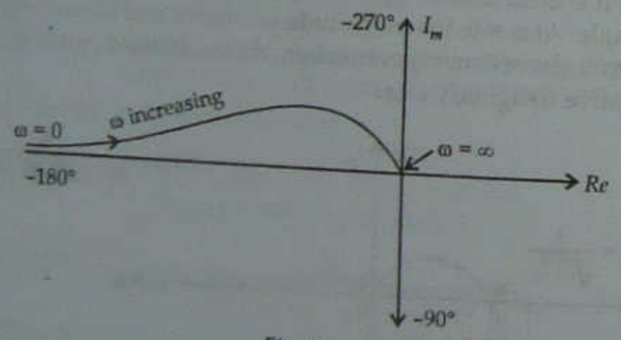


Fig. 4.3.

INTRODUCTION OF ADDITIONAL POLE

$$G(s) = \frac{K}{s^2(1+sT_1)(1+sT_2)}$$

put $s = j\omega$

$$G(j\omega) = \frac{K}{(j\omega)^2(1+j\omega T_1)(1+j\omega T_2)}$$

$$G(j\omega) = \frac{K}{-\omega^2\sqrt{1+(\omega T_1)^2}\sqrt{1+(\omega T_2)^2}} \angle -180° - \tan^{-1}\omega T_1 - \tan^{-1}\omega T_2$$

$$\lim_{\omega\to 0}|G(j\omega)| = \lim_{\omega\to 0}\frac{K}{-\omega^2\sqrt{1+(\omega T_1)^2}\sqrt{1+(\omega T_2)^2}} = \infty$$

$$\lim_{\omega\to\infty}|G(j\omega)| = \lim_{\omega\to\infty}\frac{K}{-\omega^2\sqrt{1+(\omega T_1)^2}\sqrt{1+(\omega T_2)^2}} = 0$$

$$\lim_{\omega\to 0} \angle G(j\omega) = \lim_{\omega\to 0} \angle -180° - \tan^{-1}\omega T_1 - \tan^{-1}\omega T_2 = -180°$$

$$\lim_{\omega\to\infty} \angle G(j\omega) = \lim_{\omega\to\infty} \angle -180° - \tan^{-1}\omega T_1 - \tan^{-1}\omega T_2 = -360°$$

Separate the real and imaginary parts of $G(j\omega)$.

The curve intersects the imaginary axis & 'ω' at this point is $1/\sqrt{T_1T_2}$.

The angle of the curve decreases-continuously from –180° to – 360°.

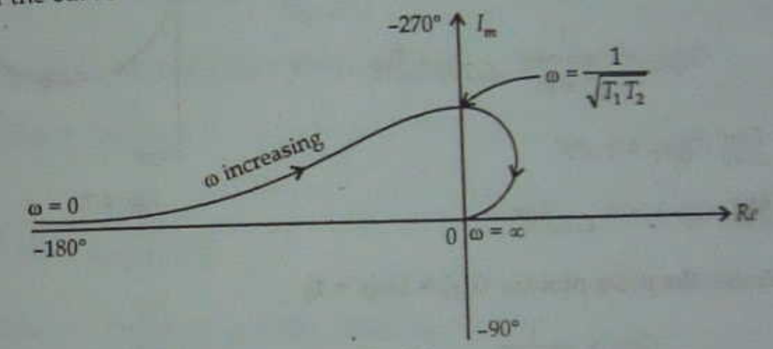


Fig. 4.4.

4.4. POLAR PLOTS OF SOME STANDARD TYPE FUNCTION

1. $G(s) = \frac{1}{s}$

Put $s = j\omega$

$$G(j\omega) = \frac{1}{j\omega}$$

$$G(j\omega) = \frac{1}{\omega} \angle -\tan^{-1}\infty$$

$$= \frac{1}{\omega} \angle -90°$$

$$\lim_{\omega\to 0}|G(j\omega)| = \infty$$

$$\lim_{\omega\to\infty}|G(j\omega)| = 0$$

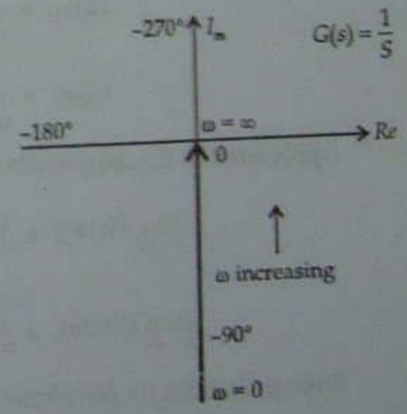


Fig. 4.5.

2. $G(s) = \frac{1}{s^2}$

Put $S = j\omega$

$$G(j\omega) = \frac{1}{(j\omega)^2}$$

or $$G(j\omega) = \frac{1}{-\omega^2} \angle -180°$$

$\omega = 0 \quad G(j\omega) = -\infty$

$\omega = \infty \quad G(j\omega) = 0$

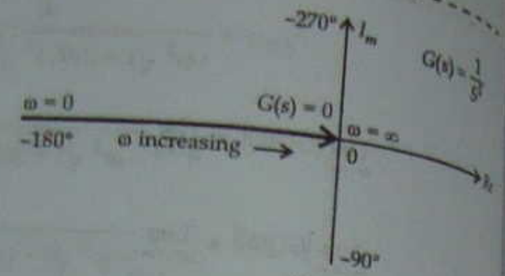


Fig. 4.6.

3. $G(s) = \frac{1}{S+1}$

Put $S = j\omega$

$$G(j\omega) = \frac{1}{1 + j\omega}$$

$$G(j\omega) = \frac{1}{\sqrt{1+\omega^2}} \angle -\tan^{-1}\omega$$

$\lim_{\omega \to 0} G(j\omega) = 1 \angle 0°$

$\lim_{\omega \to \infty} G(j\omega) = 0 \angle -90°$

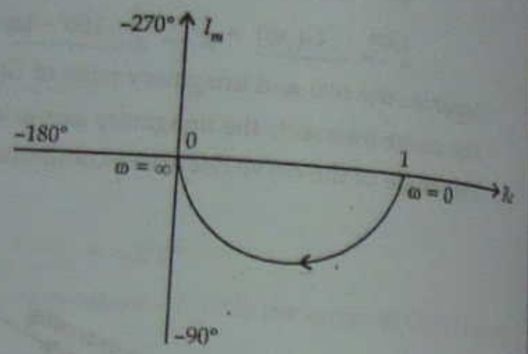


Fig. 4.7.

Example 4.1. Sketch the polar plot for $G(s) = 1/s(s + 1)$

Solution : $$G(s) = \frac{1}{s(s+1)}$$

Put $S = j\omega$

$$G(j\omega) = \frac{1}{j\omega\,(1 + j\omega)}$$

$$G(j\omega) = \frac{1}{\omega\sqrt{1+\omega^2}} \angle -90° - \tan^{-1}\omega$$

Apply limit to the magnitude of $G(j\omega)$

$$\lim_{\omega \to 0} |G(j\omega)| = \lim_{\omega \to 0} \frac{1}{\omega\sqrt{1+\omega^2}} = \infty$$

$$\lim_{\omega \to \infty} |G(j\omega)| = \lim_{\omega \to \infty} \frac{1}{\omega\sqrt{1+\omega^2}} = 0$$

Taking the limit for the phase angle of $G(j\omega)$

$$\lim_{\omega \to 0} \angle G(j\omega) = \lim_{\omega \to 0} \angle -90° - \tan^{-1}\omega = -90°$$

$$\lim_{\omega \to \infty} \angle G(j\omega) = \lim_{\omega \to \infty} \angle -90° - \tan^{-1}\omega = -180°$$

The plot is asymptotic to the vertical line passing through the point (- 1, 0).

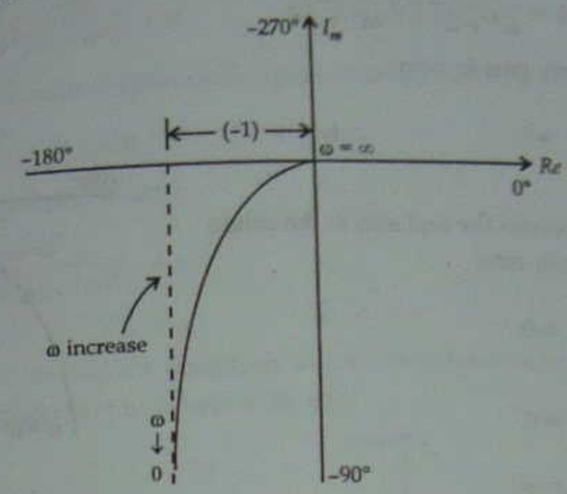


Fig. 4.8.

Example 4.2. Sketch the polar plot of $G(s) = 10/s(s + 1)$.

Solution : $$G(s) = \frac{10}{s(s+1)}$$

Put $s = j\omega$

$$G(j\omega) = \frac{10}{j\omega\,(j\omega + 1)}$$

$$G(j\omega) = \frac{10}{\omega\sqrt{\omega^2+1}} \angle -90° - \tan^{-1}\omega$$

Taking limit for the magnitude of $G(j\omega)$

$$\lim_{\omega \to 0} |G(j\omega)| = \lim_{\omega \to 0} \frac{10}{\omega\sqrt{\omega^2+1}} = \infty$$

$$\lim_{\omega \to \infty} |G(j\omega)| = \lim_{\omega \to \infty} \frac{10}{\omega\sqrt{\omega^2+1}} = 0$$

Taking limit for the phase angle of $G(j\omega)$

$$\lim_{\omega \to 0} \angle G(j\omega) = \lim_{\omega \to 0} \angle -90° - \tan^{1}\omega = -90°$$

$$\lim_{\omega \to \infty} \angle G(j\omega) = \lim_{\omega \to \infty} \angle -90° - \tan^{1}\omega = -90° - 90° = -180°$$

Separate the real and imaginary part of $G(j\omega)$

$$G(j\omega) = \frac{10}{j\omega\,(j\omega+1)} = \frac{}{-\omega^2 + j\omega}$$

$$= \frac{10}{-\omega^2 + j\omega} \frac{-\omega^2 - j\omega}{-\omega^2 - j\omega} = \frac{10(-\omega^2 - j\omega)}{\omega^4 + \omega^2}$$

$$G(j\omega) = \frac{-10\omega^2}{\omega^4+\omega^2} - j\frac{10\omega}{\omega^4+\omega^2} \qquad \text{-(A)}$$

Equating the imaginary part to zero

$$\frac{10\omega}{\omega^4+\omega^2} = 0 \qquad \therefore \omega = \infty$$

Put $\omega = \infty$ in eq^n (A)

$G(j\omega) = 0$, *i.e.*, plot intersect the real axis at the origin

Equating the real part to zero

$$\frac{-10\omega^2}{\omega^4+\omega^2} = 0$$

or, $$\frac{-10}{\omega^2+1} = 0$$

$\therefore \quad \omega = \infty$

Put $\omega = \infty$ in equation (*A*)

$G(j\omega) = 0$ *i.e* the plot intersects the imaginary axis at the origin. The required polar plot is shown in 4.9 Figure.

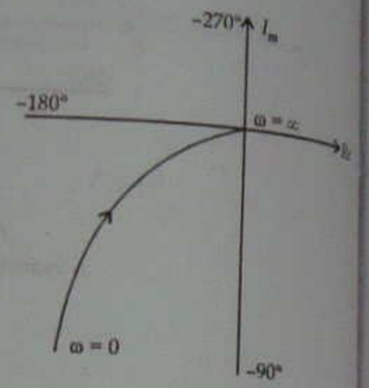


Fig. 4.9.

Example 4.3. Sketch the polar plot for G(s) = 20/s(s + 1)(s + 2) .

Solution : $$G(s) = \frac{20}{s(s+1)(s+2)}$$

Put $s = j\omega$

$\therefore$ $$G(j\omega) = \frac{20}{j\omega\,(1+j\omega)\,(j\omega+2)}$$

$$G(j\omega) = \frac{20}{\omega\sqrt{1+\omega^2}\,\sqrt{4+\omega^2}} \angle -90° - \tan^{-1}\omega - \tan^{-1}\omega/2$$

Taking the limit for the magnitude of $G(j\omega)$

$$\lim_{\omega\to 0}|G(j\omega)| = \lim_{\omega\to 0}\frac{20}{\omega\sqrt{1+\omega^2}\,\sqrt{4+\omega^2}} = \infty$$

$$\lim_{\omega\to \infty}|G(j\omega)| = \lim_{\omega\to \infty}\frac{20}{\omega\sqrt{1+\omega^2}\,\sqrt{4+\omega^2}} = 0$$

Taking the limit for the phase angle of $G(j\omega)$.

$$\lim_{\omega\to 0}\angle G(j\omega) = \lim_{\omega\to 0}\angle -90° - \tan^{-1}\omega - \tan^{-1}\omega/2 = -90°$$

$$\lim_{\omega\to \infty}\angle G(j\omega) = \lim_{\omega\to \infty}\angle -90° - \tan^{-1}\omega - \tan^{-1}\omega/2 = -270°$$

Separating the real and imaginary part of $G(j\omega)$

$$G(j\omega) = \frac{20}{j\omega\,(1+j\omega)(2+j\omega)}$$

$$G(j\omega) = \frac{-60\omega^2}{(\omega^4+\omega^2)(4+\omega^2)} + j\frac{20(\omega^3-2\omega)}{(\omega^4+\omega^2)(4+\omega^2)}$$

Equating the imaginary part to zero

$$\frac{20(\omega^3-2\omega)}{(\omega^4+\omega^2)(4+\omega^2)} = 0 \qquad \therefore \omega = \pm\sqrt{2} \text{ and } \omega = \pm\infty$$

For the positive values of frequencies polar plot intersects the real axis at $\omega = \sqrt{2}$ and $\omega = \infty$ for which $G(j\omega) = \frac{-10}{3}\angle 0°$ and $0\angle 0°$

Equating the real part to zero

$$\frac{-60\,\omega^2}{(\omega^4+\omega^2)(4+\omega^2)} = 0$$

$$\omega = \infty$$

$\therefore$ The polar plot intersect the imaginary axis at $\omega = \infty$ for which $G(j\omega) = 0 \angle -270°$.

The required polar plot is shown in fig. 4.10.

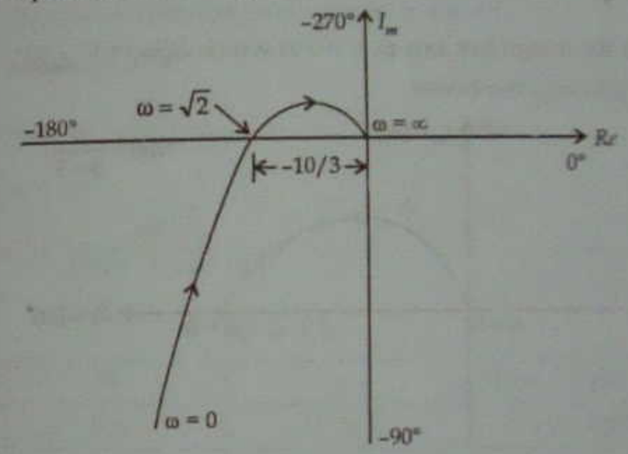


Fig. 4.10.

Example 4.4. Sketch the polar plot for G(s) = s/1 + sT

Solution : Put $s = j\omega$

$$G(j\omega) = \frac{j\omega}{1+j\omega T}$$

$$G(j\omega) = \frac{\omega}{\sqrt{1+\omega^2T^2}} \angle 90° - \tan^{-1}\omega T$$

Taking the limit for the magnitude of $G(j\omega)$

$$\underset{\omega\to 0}{Lim}|G(j\omega)| = \underset{\omega\to 0}{Lim}\frac{\omega}{\sqrt{1+\omega^2T^2}} = 0$$

$$\underset{\omega\to \infty}{Lim}|G(j\omega)| = \underset{\omega\to \infty}{Lim}\frac{\omega}{\sqrt{1+\omega^2T^2}} = \frac{1}{\infty}$$

Taking the limit for the phase angle of $G(j\omega)$

$$\underset{\omega\to 0}{Lim}\angle G(j\omega) = \underset{\omega\to 0}{Lim}\angle 90° - \tan^{-1}\omega T = 90°$$

$$\lim_{\omega \to \infty} \angle G(j\omega) = \lim_{\omega \to \infty} \angle 90° - \tan^{-1} \omega T = 0°$$

Separating the real and imaginary terms of $G(j\omega)$

$$G(j\omega) = \frac{j\omega}{1+j\omega T} \frac{1-j\omega T}{1-j\omega T} = \frac{\omega^2 T}{1+\omega^2 T^2} + j \frac{\omega}{1+\omega^2 T^2}$$

equate the imaginary part to zero

$$\frac{\omega}{1+\omega^2 T^2} = 0 \qquad \therefore \ \omega = \infty$$

Polar plot intersects the real axis at $\omega = \infty$ at which $G(j\omega) = \frac{1}{T} \angle 0°$

Equate the real part to zero

$$\frac{\omega^2 T}{1+\omega^2 T^2} = 0 \qquad \therefore \ \omega = 0$$

Polar plot intersects the imaginary axis at $\omega = 0$ at which $G(j\omega) = 0 \angle 90°$

The required polar plot is given below.

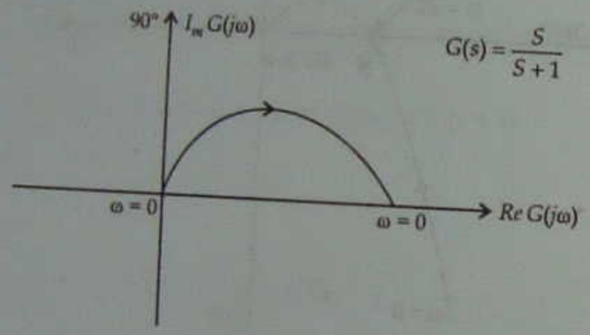


Fig. 4.11.

4.5. ALTERNATIVE METHOD

The other mtehod to draw the polar plot is the tip of the vector $M(\omega)\, e^{j\Phi(\omega)}$ in the G-plane as 'ω' varies from zero to infinity.

The first step is to calculate the magnitude ratio M and phase shift ϕ for different values of 'ω'. Consider the following examples

Example 4.5. Sketch the polar plot for

$$G(s) = \frac{10(s+1)}{s+10}$$

(R.M.L. University Faizabad 2001)

Solution : Put $s = j\omega$

$$\therefore \quad (j\omega) = \frac{10(j\omega+1)}{(j\omega+10)}$$

$$|G(j\omega) = \frac{10\sqrt{1+\omega^2}}{\sqrt{\omega^2+100}}, \quad \angle G(j\omega) = \tan^{-1}\omega - \tan^{-1}\frac{\omega}{10}$$

Table 4.1.

ω	M	ϕ	$M\cos\phi + jM\sin^{\circ}\phi$.
0	1	0°	1 + j0
1	1.41	39.29°	1 . 09 + j 0.89
3.16	3.16	54.9°	1.81 + j 2.58
4	3.83	54.2°	2.24 + j 3.106
7	5.8	46.86°	3.96 + j 4.2
10	7.1	39.3°	5.5 + j4.5
20	8.96	23.7°	8.2 + j 3.6
50	9.81	10.16°	9.67 + j 1.73
100	9.95	5.14°	9.9 + j 0.89
200	9.99	2.58°	9.97 + j 0.45

The corresponding polar plot is shown on the graph paper in fig. 4.12.

Example 4.6. Sketch the polar plot of

$$G(s) = \frac{20s}{(s+1)(s+10)}$$

Solution : Put $s = j\omega$

$$G(j\omega) = \frac{20(j\omega)}{(j\omega+1)(j\omega+10)}$$

Table 4.2.

ω	M	ϕ	$M\cos\phi + jM\sin^{\circ}\phi$.
0.2	0.4	77.56°	0.08 + j0.4
0.5	0.9	60.58	0.44 + j0.78
0.7	1.147	51.0	0.72 + j0.89
1.0	1.41	39.28	1.09 + j0.9
2.0	1.79	15.3	1.72 + j0.47
3.0	1.82	1.74	1.8 + j0.05
4.0	1.8	-7.8°	1.78 - j0.24
5.0	1.78	-15.26°	1.71 - j0.46
10.0	1.41	-39.3	1.09 - j0.9
20.0	0.89	-60.5	0.43 - j0.77
30.0	0.63	-69.5	0.22 - j0.6
40.0	0.48	-74.56	0.12 - j0.5
50.0	0.39	-77.5	0.08 - j0.4
60.0	0.32	-79.5	0.05 - j0.3

Corresponding polar plot is shown in fig. 4.13.

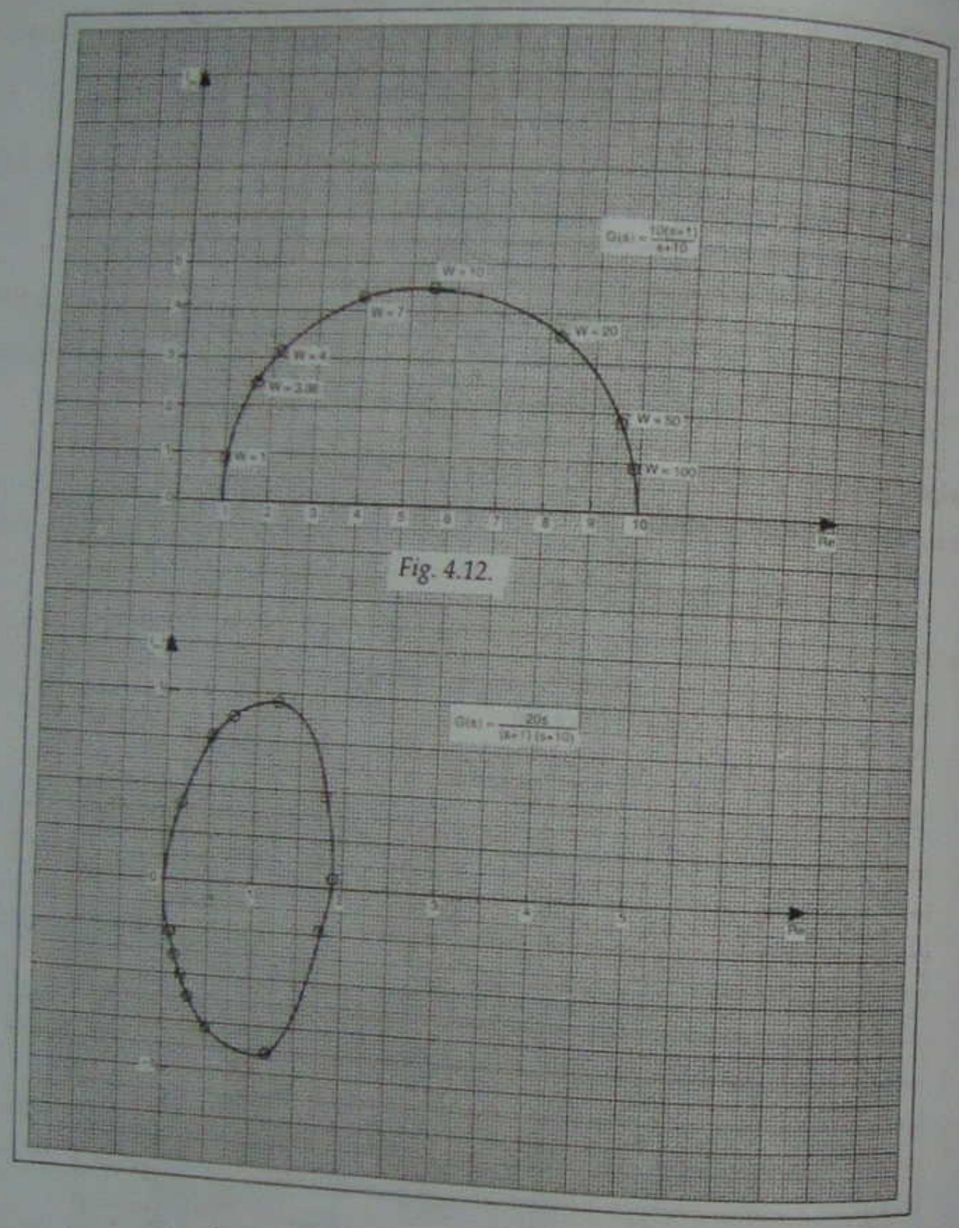


Fig. 4.12.

Fig. 4.13.

4.6. PHASE MARGIN, GAIN MARGIN AND STABILITY ON POLAR PLOT

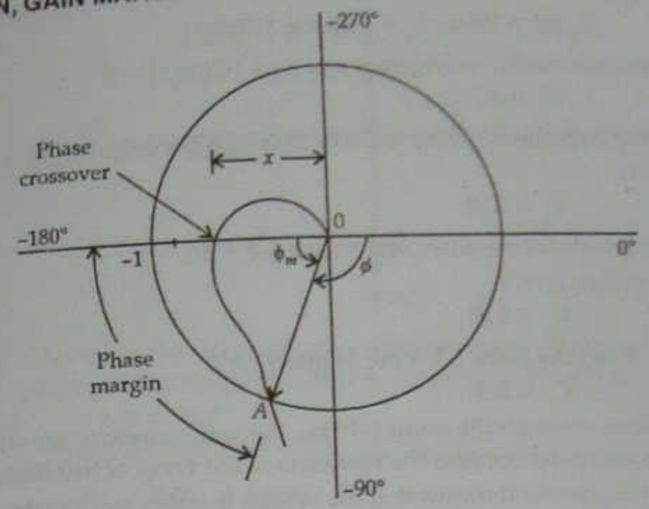


Fig. 4.14.

The stability of control system is measured from the measurement of quantities like 'phase margin' and 'gain margin'. These quantities are useful to design the compensating network for unstable systems, *i.e.* for the improvement of stability of an existing control system.

In above fig. polar plot intersects the negative real axis at a distance of '*x*' from the origin. The frequency where it crosses the negative real axis is called "phase cross-over frequency." The point where the polar plot intersects the negative real axis is called "phase crossover point."

Phase Margin

The phase margin is that amount of additional phase lag at the gain crossover frequency required to bring the system to the verge of instability.

A circle with radius equal to unity is drawn and it intersects the polar plot at point 'A' where the frequency is ω_g. This frequency is called "gain cross over frequency." In other words the gain cross over frequency is the frequency at which $|G(j\omega)|$, the magnitude of the open loop transfer function is unity.

The phase margin is equal to 180° plus the angle of $G(j\omega)$ at the gain crossover point *i.e*

$$\phi_m = 180° + \phi$$

$$\{r - \phi + 180 = 360,\quad r = 360 + \phi - 180 = 180 + \phi\}$$

The angle from the negative real axis to the line OA is the phase margin. The phase margin is positive for $\phi_m > 0$ and negative $\phi_m < 0$. For minimum phase system* to be stable, the phase margin must be positive. For unstable system the gain crossover point would not be found in the third quadrant of $G(j\omega)$ plane. When phase crossover point is to the left of $(-1 + j\,0)$ the phase margin is negative and the closed loop system is unstable.

Gain Margin

The gain margin is the reciprocal of the magnitude $|G(j\omega)|$ at the frequency at which the phase angle is - 180°.

$$\therefore \text{Gain margin (GM) } K_g = \frac{1}{|G(jw_c)|}$$

* minimum phase system is describe in Art 4.9.

In terms of decibles

$$K_g \, dB = 20 \log K_g = -20 \log |G(j\omega_c)|$$

(i) If plot doesnot intersects the negative real axis $|G(j\omega_c)| = 0$

$$\therefore \quad K_g = \infty$$

(ii) If the plot intersects the negative real axis between the origin and $(-1 + j0)$

$$0 < |G(j\omega_c)| < 1$$

$$\therefore \quad K_g > 0 \, db$$

(iii) If the plot intersects the negative real axis at $(-1 + j0)$

$$|G(j\omega_c)| = 1$$

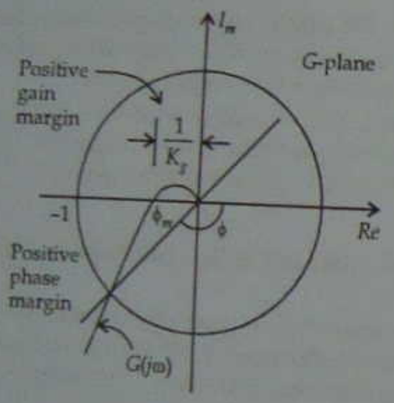
$$\therefore K_g = 0 \, db$$

(iv) If the plot encloses the point $(-1 + j0)$ $|G(j\omega c)| > 1$

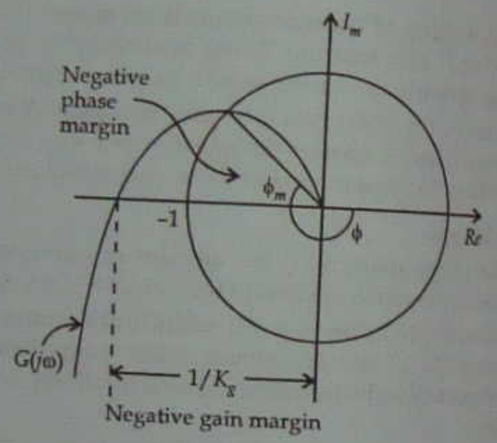
$$\therefore \quad K_g < 0 \, db$$

When the plot passes through the point $(-1+j\omega)$, the gain margin is zero it means that the loop gain can no longer be increased because the system is at the verge of instability.

If the gain margin is positive it means that the system is stable and negative gain margin means the system is unstable.



Stable system
Fig. 4.15.



Unstable stable
Fig. 4.16.

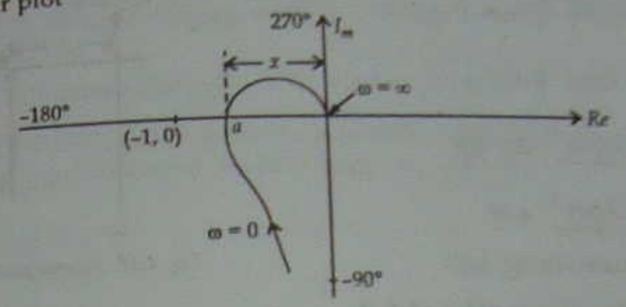
The polar plots can also be used to determine the stability of the system. To determine the stability of the system first draw the polar plot for given transfer function. If the critical point $(-1 + j\omega)$ is within the plot, it means the system is unstable. If the polar plot passes through the pint $(-1 + j\omega)$ the system is marginally stable and if the critical point is outside the polar plot, the system is stable. For example for $G(s)\,H(s) = \dfrac{K}{s(s+1)(s+5)}$ sketch the polar plot and determine the range of K for stability.

$$G(j\omega)\cdot H(j\omega) = \frac{K}{j\omega\,(1+j\omega)\,(5+j\omega)}$$

$\omega = 0$, $\quad |G(j\omega)\,H(j\omega)| = \infty$ and $\angle G(j\omega)\,H(j\omega) = -90°$

$\omega = \infty$, $\quad |G(j\omega)\,H(j\omega)| = 0$ and $\angle G(j\omega)\,H(j\omega) = -270°$

Sketch the polar plot



For stability determine the value of x *i.e.*, distance from origin to 'a'. The point 'a' is called phase crossover point and the frequency at this point 'a' is called phase crossover frequency.

At this point $\quad \phi = -180°$

$$\therefore \quad -180° = -90° - \tan^{-1}\omega - \tan^{-1} 0.2\,\omega$$

$$90° = \tan^{-1}\omega + \tan^{-1} 0.2\omega$$

$$\tan 90° = \tan(\tan^{-1}\omega + \tan^{-1} 0.2\,\omega)$$

$$\infty = \frac{1.2\omega}{1-0.2\omega^2}$$

or, $\quad 1 - 0.2\omega^2 = 0$

or, $\quad \omega = 2.236$ rad/sec.

This is the phase crossover frequency. Now find the magnitude of $G(j\omega)\,H(j\omega)$ at this frequency.

$$|G(j\omega)\,H(j\omega)| = x = \frac{K}{j2.236\,||(1+j2\cdot 236)||5+j2.236|}$$

$$x = \frac{K}{30}$$

$\therefore$ if $K = 30$, $x = 1$ and system will be marginally stable

$K > 30$, $x > 1$ and system will be unstable.

$K < 30$, $x < 1$ and system will be stable.

4.7. INVERSE POLAR PLOT

The inverse polar plot of $G(j\omega)$ is a graph of $\dfrac{1}{G(j\omega)}$ as a function of ω.

For example, if $G(j\omega) = 1/j\omega$ then $G(j\omega)^{-1} = j\omega$

$$G(j\omega)^{-1} = \omega \angle 90°$$

$$\lim_{\omega \to 0} |G(j\omega)^{-1}| = 0$$

$$\lim_{\omega \to \infty} |G(j\omega)^{-1}| = \infty$$

The inverse polar plot is shown in the fig 4.17.

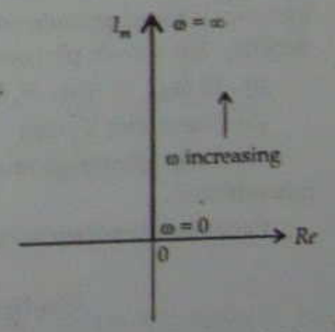


Fig. 4.17. Inverse polar plot of 1/jω

Example 4.7. Sketch the inverse polar plot of $G(j\omega) = j\omega T/1 + j\omega T$

Solution : $\quad G(j\omega)^{-1} = \dfrac{1}{G(j\omega)} = \dfrac{1+j\omega T}{j\omega T} = \dfrac{1}{j\omega} + 1$

$$\lim_{\omega \to 0} |G(j\omega)^{-1}| = \infty$$

$$\lim_{\omega \to \infty} |G(j\omega)^{-1}| = 1$$

$$\lim_{\omega \to 0} \angle G(j\omega)^{-1} = -90°$$

$$\lim_{\omega \to \infty} \angle G(j\omega)^{-1} = 0°$$

The polar plot is shown in fig. 4.18.

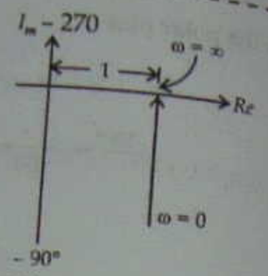


Fig. 4.18. Inverse plot of $j\omega T/1+j\omega T$

Example 4.8. Sketch the inverse polar plot of

$$G(s) = \frac{1+ST}{ST}$$

Solution : $$G(s)^{-1} = \frac{1}{G(s)} = \frac{sT}{1+sT}$$

Put $$s = j\omega$$

$$G(j\omega)^{-1} = \frac{1}{G(j\omega)} = \frac{j\omega T}{1+j\omega T}$$

$$\lim_{\omega \to 0} |G(j\omega)^{-1}| = 0 \qquad \lim_{\omega \to 0} \angle G(j\omega)^{-1} = 90°$$

$$\lim_{\omega \to \infty} |G(j\omega)^{-1}| = 1 \qquad \lim_{\omega \to \infty} \angle G(j\omega)^{-1} = 0°$$

Inverse polar plot is shown in fig. 4.19.

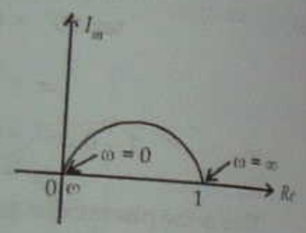


Fig. 4.19. Inverse polar plot of $\frac{1+ST}{ST}$

4.8. BODE PLOT

Bode plot is a graphical representation of the transfer function for determining the stability of the control system. Bode plot consists of two separate plots. One is a plot of the logarithm of the magnitude of a sinusoidal transfer function, the other is a plot of the phase angle, both plots are plotted against the frequency. The curves are drawn on semilog graph paper, using the log scale for frequency and linear scale for magnitude (in decibles) or phase angle (in degrees). The magnitude is represented in decibles. Thus, Bode plot consists of

(*i*) $20 \log_{10} |G(j\omega)|$ $V_s \log\omega$.

(*ii*) Phase shift $V_s \log\omega$

The main advantage of using Bode plot is that multiplication of magnitudes can be converted into addition.

Consider open loop transfer function of a closed loop control system

$$G(s)\,H(s) = \frac{K(1+sT_a)(1+sT_b)....}{s^N(1+sT_1)(1+sT_2)....}$$

Put $$S = j\omega$$

$$G(j\omega)\,H(j\omega) = \frac{K(1+j\omega T_a)(1+j\omega T_b).....}{(j\omega)^N(1+j\omega T_1)(1+j\omega T_2).....}$$

$$20 \log_{10} |G(j\omega)\ H(j\omega)| = \left(20 \log K + 20 \log \sqrt{1+\omega^2 T^2_a} + 20 \log \sqrt{1+\omega^2 T^2_b}\right)....)$$

$$-\left(20N \log\omega + 20 \log \sqrt{1+\omega^2 T_1^2} + 20 \log \sqrt{1+\omega^2 T_2^2}\right)....)$$

Hence, in order to get $|G(j\omega)\ H(j\omega)|$ we will have to obtain the individual plots and adding individual components, the resultant can be obtained. Suppose, $H(s) = 1$

Case 1. The Gain K

$$G(s) = K$$

$$s = j\omega$$

Put $$G(j\omega) = K$$

$$20 log_{10} |G(j\omega)| = 20 log_{10} K \qquad ...(4.1)$$

$$\text{Phase angle } \phi = \angle G(j\omega) = 0° \qquad ...(4.2)$$

From equations 4.1 & 4.2 it is clear that the magnitude is independent of $\log_{10}\omega$ and phase angle always zero. The plots are shown in fig. 4.20.

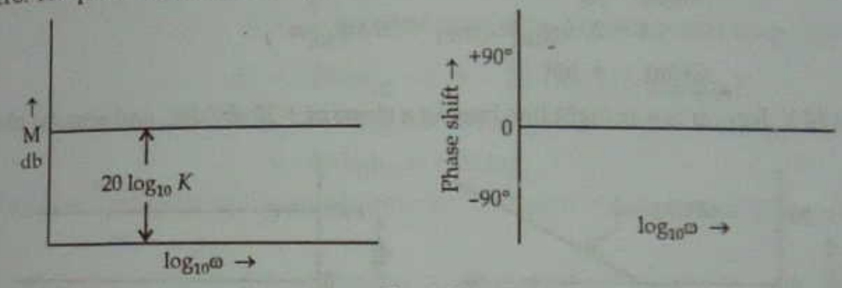


Fig. 4.20.

Case 2 : $$G(s) = \frac{1}{S^N}$$

Put $$s = (j\omega)^N$$

$$\therefore \quad G(j\omega) = \frac{1}{(j\omega)^N}$$

$$20 log_{10} |G(j\omega)| = 20 log_{10} \frac{1}{(j\omega)^N}$$

$$= 20 log_{10} (j\omega)^{-N}$$

$$= -20 N log_{10} (\omega)$$

$$\angle G(j\omega)^N = -90 N°$$

where $N = 1,2,3,$ .....

The plot M Vs $log_{10}\omega$ is a straight line. For $N = 1$ the line has a slope of -20 *db/* decade and angle $-90°$. For $N = 2$, the slope of the line will be -40 *db/* decade and angle will be $-180°$ and so on.

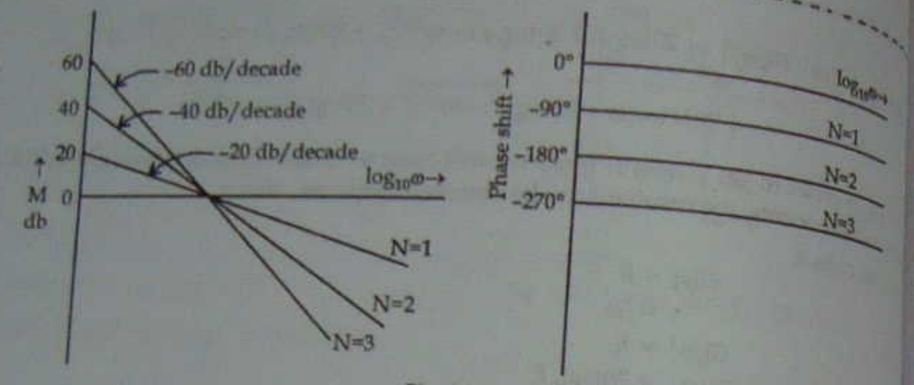


Fig. 4.21.

Case 3 : Put

$$G(s) = S$$
$$S = j\omega$$
$$G(j\omega) = j\omega$$
$$M = 20\log_{10}|G(j\omega)| = 20\log_{10}\omega$$
$$\angle G(j\omega) = +90°$$

The plot M V_s $\log_{10}\omega$ is a straight line having a slope of + 20 *db*/dec. and angular phase sh + 90°.

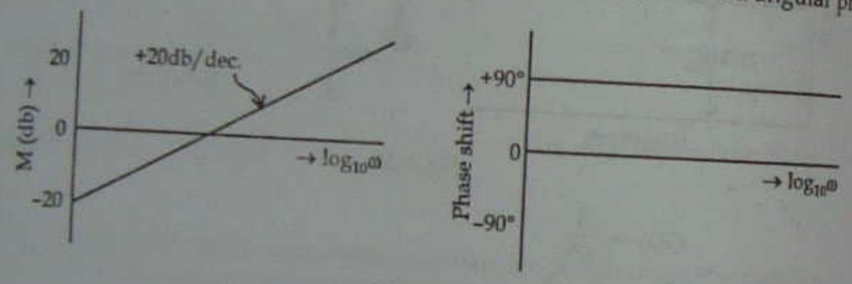


Fig. 4.22.

Case 4 : Put

$$G(s) = \frac{1}{1+sT}$$
$$s = (j\omega)$$

$\therefore$
$$G(j\omega) = \frac{1}{1+j\omega T}$$
$$|G(j\omega)| = \frac{1}{\sqrt{1+\omega^2T^2}}$$
$$20\log_{10}|G(j\omega)| = 20\log_{10}\left[\frac{1}{\sqrt{1+\omega^2T^2}}\right]$$
$$= 20\log_{10}1 - 20\log_{10}\sqrt{1+\omega^2T^2}$$
$$= -20\log_{10}\sqrt{1+\omega^2T^2}$$

$\because \log_{10}1 =$

Put the different values of ω, we will get $|G(j\omega)|$ consider following two cases.

(*a*) For $\omega T << 1$ (very low frequencies)

$$-20\log_{10}\sqrt{1+\omega^2T^2} = -20\log_{10}\sqrt{1} = 0$$

$\therefore$
$$M = 0 \text{ for } \omega T << 1 \text{ or } \omega \leq \frac{1}{T}$$

(*b*) For $\omega T >> 1$ (very high frequencies)

$$-20\log_{10}\sqrt{1+\omega^2T^2} = -20\log_{10}\sqrt{\omega^2T^2}$$
$$= -20\log_{10}\omega T \text{ for } \omega >> 1/T$$

Hence, M *Vs* $\log_{10}\omega$ has two parts

(*i*) One part having $M = 0$ for $\omega << 1/T$

(*ii*) In other part M varies as a straight line with slope of - 20db/decade for $\omega >> \frac{1}{T}$

$\omega = \frac{1}{T}$ is called break frequency or corner frequency

$$M = -20\log_{10}\omega T = -20(\log_{10}\omega + \log_{10}T)$$
$$M = -20\log_{10}\omega - 20\log_{10}T$$
$$= -20\log_{10}\omega + 20\log_{10}1/T \quad \text{...(4.3)}$$

The above two parts of the graph intersect 0 *db* axis is determined by equating the eq^n (4.3) to zero

$$0 = -20\log_{10}\omega + 20\log_{10}1/T$$

$\therefore$ $\omega = 1/T$ is called break frequency.

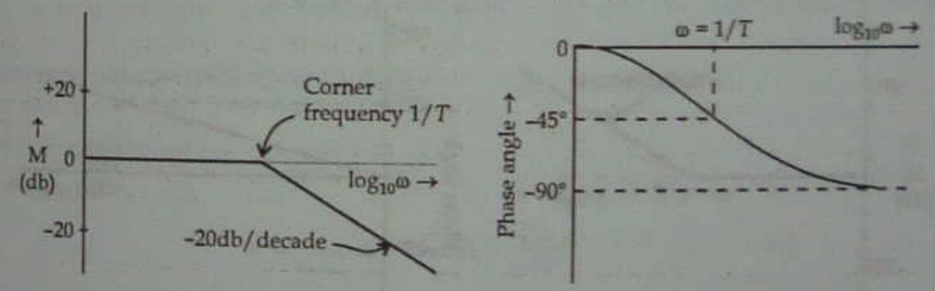


Fig. 4.23.

Case 5 : Put $s = j\omega$

$$G(s) = (1+sT)$$
$$G(j\omega) = (1+j\omega T)$$
$$|G(j\omega)| = \sqrt{1+\omega^2T^2}$$
$$20\log_{10}|G(j\omega)| = 20\log_{10}\sqrt{1+\omega^2T^2}$$

(*i*) When $\omega T << 1$

$$M = 20\log_{10}\sqrt{1} = 0\ db$$

(ii) When $\omega T >> 1$

$$M = 20\log_{10} \omega T$$

$$M = 20\log_{10} \omega T = 20\log_{10} \frac{\omega}{1/T}$$

$$= 20\log_{10} \omega - 20\log_{10} \frac{1}{T}$$

equate the above equation to zero

$$0 = 20\log_{10} \omega - 20\log_{10} \frac{1}{T}$$

$$\therefore \quad \omega = \frac{1}{T} \text{ corner frequency.}$$

Thus, the two parts of the graph intersects the '0' *db* axis at $\omega = \frac{1}{T}$. The second part is a straight line having the slope of + 20 *db*/ decade.

Phase Angle Plot

$$\phi = \angle G(j\omega) = \tan^{-1} \omega T$$

(i) At very low frequencies ωT is very very small

$$\phi = \tan^{-1}(0) = 0°$$

(ii) at $\omega T = 1$

$$\phi = \tan^{-1} 1 = 45°$$

(iii) at very high frequencies

$$\phi = \tan^{-1}(\infty) = 90°$$

Thus, the value of ϕ gradually changes from 0° to 90° as ω increases from 0 to very high values.

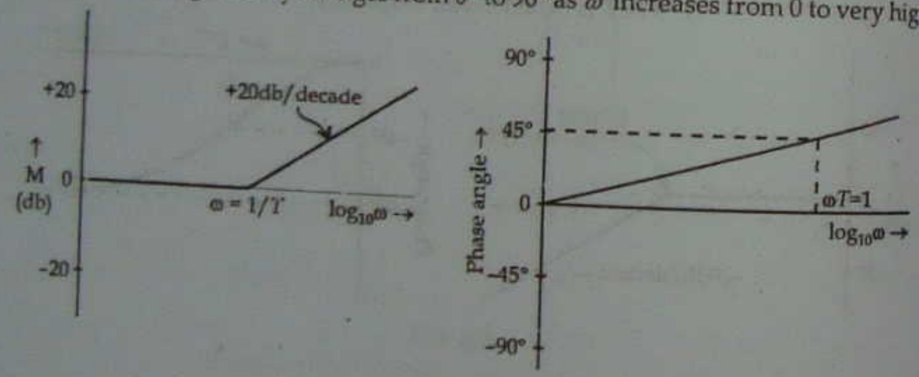


Fig. 4.24.

Case 6 : General second order system

$$C(s) = \frac{\omega^2_n}{s^2 + 2\xi\omega_n s + \omega^2_n}$$

Put $s = j\omega$

$$G(s) = \frac{\omega^2_n}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega^2_n} = \frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega + \omega_n^2}$$

$$G(s) = \frac{w_n^2}{\omega_n^2 - \omega^2 + j2\zeta\omega_n\omega} = \frac{1}{1-\left(\frac{\omega}{\omega_n}\right)^2 + j2\zeta\frac{\omega}{\omega_n}}$$

$$20\log_{10}|G(j\omega)| = 20\log_{10}\left|\frac{1}{1-\left(\frac{\omega}{\omega_n}\right)^2 + j2\zeta\frac{\omega}{\omega_n}}\right|$$

$$= -20\log_{10}\sqrt{\left\{1-\left(\frac{\omega}{\omega_n}\right)^2\right\}^2 + 4\zeta^2\left(\frac{\omega}{\omega_n}\right)^2}$$

Suppose $\frac{\omega}{\omega_n} = u$

$$\therefore \quad 20\log_{10}|G(j\omega)| = M = -20\log_{10}\sqrt{(1-u^2)^2 + 4\zeta^2u^2}$$

Consider the two cases

1. $u << 1 \quad i.e \frac{\omega}{\omega_n} << 1$

$$M = -20\log_{10}\sqrt{1} = 0 \text{ db.}$$

2. $u >> 1 \; i.e \frac{\omega}{\omega_n} >> 1$

$$M = -20\log_{10}\sqrt{(u^2)^2} = -20\log_{10} u^2 = -40\log_{10} u$$

So, it is a straight line having slope of - 40 db/dec. and passing through the point u

Therefore, the asymptotic plot consists of

(i) $M = 0 \qquad u << 1$

(ii) $M = -40\log_{10} u \qquad u >> 1$

Phase Angle Plot

$$\phi = \angle G(j\omega) = -\tan^{-1}\frac{2\xi u}{1-u^2}$$

(i) for small value of u, u^2 is small

$$\therefore \quad \phi = -\tan^{-1} 2\xi u$$

(ii) for large value of u, $u^2 >> 1$

$$\therefore \quad \phi = +\tan^{-1}\frac{2\zeta}{u}$$

(iii) when $u = 1$

$$\phi = -\tan^{-1}\infty = -90°$$

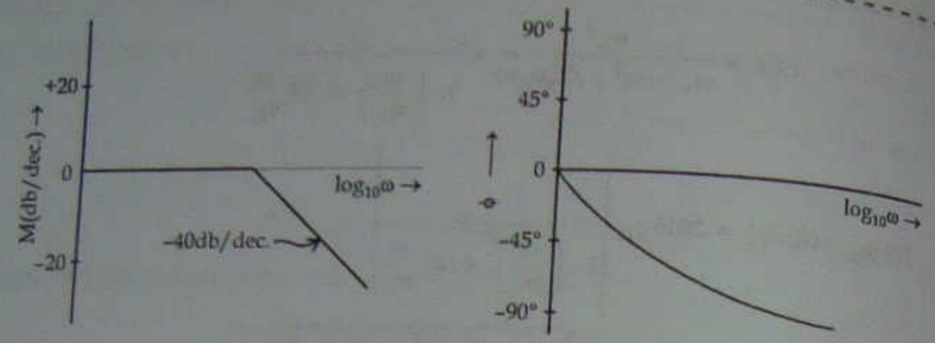


Fig. 4.25.

Initial Slope of Bode Plot

Let $G(s)\,H(s) = \dfrac{K}{S^N}$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{K}{(j\omega)^N}$$

$$20\log_{10}|G(j\omega)\,H(j\omega)| = 20\log_{10}\left|\frac{K}{(j\omega)^N}\right|$$

$$= 20\log_{10} K - 20\,N\log_{10}\omega \qquad \ldots(4.4)$$

1. For $N = 0$ (Type zero system)

$20\log_0 |G(j\omega)\,H(j\omega)| = 20\log_{10} K.$

This is a straight line. The graph is shown in fig. 4.26.

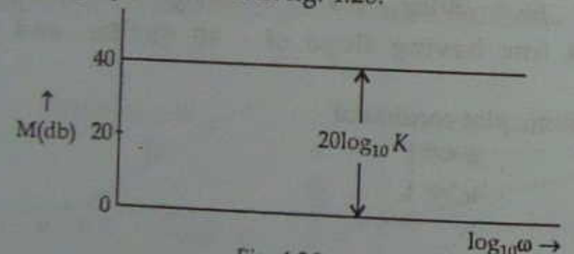


Fig. 4.26.

2. For $N = 1$ (type one system)

Put $N = 1$ in equation (4.4)

$20\log_{10}|G(j\omega)\,H(j\omega)| = 20\log_{10} K - 20\log_{10}\omega$

Intersection with 0 db axis

$$0 = 20\log_{10} K - 20\log_{10}\omega$$

$$\therefore \quad K = \omega$$

locate $\omega = K$ on 0 db axis and at this point draw a line of – 20 db/decade produce it till it intersect the y-axis that will be the starting point on Bode plot.

3. For $N = 2$ (type two system)

Put $N = 2$ in eq^n (4.4)

$$20\log_{10}|G(j\omega)\,H(j\omega)| = 20\log_{10} K - 20.\,2\log_{10}\omega$$

$$= 20\log_{10} K - 40\log_{10}\omega$$

Intersection with 0 db axis

$$0 = 20\log_{10} K - 40\log_{10}\omega$$

$$20\log_{10} K = 40\log_{10}\omega$$

$$20\log_{10} K = 20\log_{10}\omega^2$$

$$\omega^2 = K$$

$$\omega = \sqrt{K}$$

Hence, graph intersect the 0 db axis at $\omega = \sqrt{K}$. Locate $\omega = \sqrt{K}$ on 0 db axis and draw a line – 40 db/dec. and produce it to the y – axis. Graph having the slope of – 40 *db*/ decade is shown in fig. 4.27.

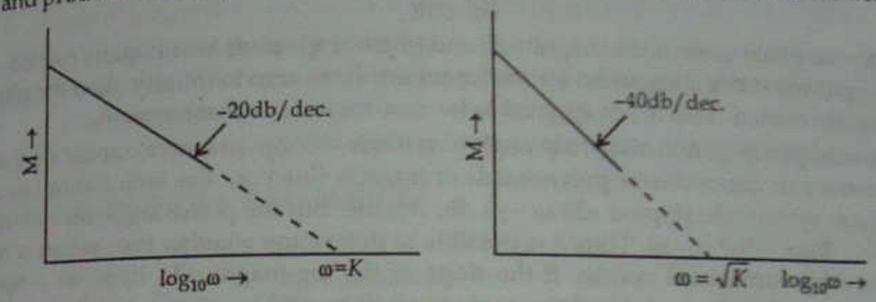


Fig. 4.27.

Table 4.3.

Type of the system N	Initial slope 0db Axis	Intesection with
0	0 db/decade	Parallel to 0 axis
1	– 20 db/dec.	$= K$
2	– 40 db/dec.	$= \sqrt{K}$
3	– 60 db/dec.	$= K^{1/3}$
⋮	⋮	⋮
N	-20N db/dec.	$K^{1/N}$

4.9. MINIMUM PHASE SYSTEMS AND NON-MINIMUM PHASE SYSTEMS

The transfer functions having no poles and zeros in the right half s-plane are called minimum phase transfer functions. Systems with minimum phase transfer functions are called minimum phase systems.

The transfer functions having poles and/or zeros in the right half s-plane are called non-minimum phase transfer functions. Systems with non-minimum phase transfer functions are called non-minimum-phase systems.

let $$G_1(j\omega) = \frac{1 + j\omega T_A}{1 + j\omega T_B} \qquad \ldots(4.5)$$

$$G_2(j\omega) = \frac{1 - j\omega T_A}{1 + j\omega T_B} \qquad \ldots(4.6)$$

The transfer function given by eq^n. 4.5 is a minimum phase transfer function and transfer function given by eq^n 4.6 is non-minimum phase type transfer function.

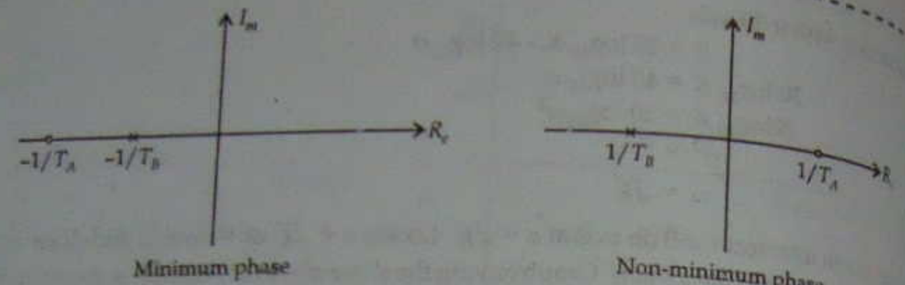


Fig. 4.28.

For minimum phase system, the magnitude and phase angle plots are uniquely related. It means that if the magnitude curve is specified for the frequency from zero to infinity, then the phase angle curve is uniquely related. This is not applicable for non-minimum phase system.

For minimum phase system, the phase angle at $\omega = \infty$ is $-90°\ (q - p)$ where p and q are the degree of the numerator and denominator polynomials of transfer function. For both minimum and non-minimum phase system the slope is $-20\ (q - p)$ db/decade. But the phase angle for non-minimum phase system is differ from $-90°\ (q - p)$. Thus it is possible to determine whether the system is minimum phase or non-minimum phase system. If the slope of the log-magnitude curve as ω approaches infinity is $-20\ (q - p)$ db/dec. and the phase angle is equal to $-90°\ (q - p)$ at $\omega = \infty$ then the system is minimum phase otherwise not.

Non-minimum phase systems are slow in response. In control system excessive phase lag should be avoided.

4.10. PROCEDURE FOR DRAWING THE BODE PLOTS

Consider the transfer function

$$G(s) = \frac{K(1+sT_a)(1+sT_b)\ldots\ldots}{S^N(1+sT_1)(1+sT_2)\ldots\ldots\left[1+2\zeta\left(\frac{\omega}{\omega_n}\right)+\left(j\frac{\omega}{\omega_n}\right)^2\right]}$$

Where N is the number of poles at the origin *i.e.* N defines the type of system.

For type zero system $K = K_p$

For type one system $K = K_v$

For type two system $K = K_a$

In above transfer function put $s = j\omega$

$$G(j\omega) = \frac{K(1+j\omega T_a)(1+j\omega T_b)\ldots..}{(j\omega)^N(1+j\omega T_1)(1+j\omega T_2)\ldots..\left[1+2\zeta(\omega/\omega_n)+(j\omega/\omega_n)^2\right]}$$

$$20\log_{10}|G(j\omega)| = 20\log K + 20\log\sqrt{1+\omega^2T_a^2} + 20\log\sqrt{1+\omega^2T_b^2} + \ldots.. - 20N\log\omega$$

$$-20\log\sqrt{1+\omega^2T_1^2} - 20\log\sqrt{1+\omega^2T_2^2} - 20\log\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2 + 4\xi^2\left(\frac{\omega}{\omega_n}\right)^2}$$

Phase angle

$$\angle G(j\omega) = \tan^{-1}\omega T_a + \tan^{-1}\omega T_b + \ldots. N(90°) - \tan^{-1}\omega T_1 - \tan^{-1}\omega T_2 \ldots. \tan^{-1}\left[\frac{2\zeta\omega\omega_n}{\omega^2{}_n - \omega^2}\right]$$

Step 1 : I dentify the corner frequency.

Step 2 : Draw the asymptotic magnitude plot. The slope will change at each corner frequency by + 20 db / dec. for zero and – 20 db / dec for pole. For complex conjugate pole and zero the slope will change by ∓ 40 db/ decade.

Step 3 : (*i*) For type zero system draw a line upto first (lowest) corner frequency having 0 db dec. slope.

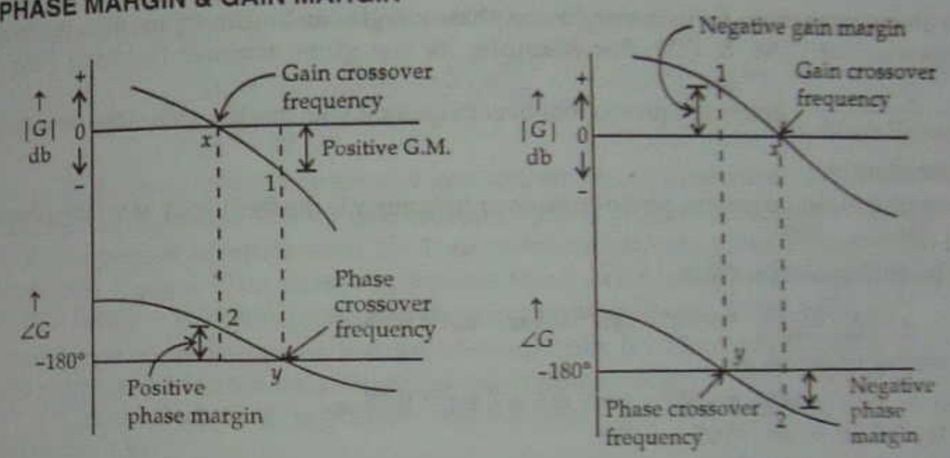
(*ii*) For type one system draw a line having slope – 20 db/ dec. upto $\omega = K$. Mark first (lowest) corner frequency.

(*iii*) For type two system draw the line having slope – 40 db/ dec. upto $\omega = \sqrt{K}$ and so on. Mark first corner frequency.

Step 4 : Draw a line upto second corner frequency by adding the slope of next pole or zero to the previous slope and so on.

Step 5 : Calculate phase angle for different values of ω from the equation 4.10 and join all points.

4.11. PHASE MARGIN & GAIN MARGIN



Stable system *Unstable system*

Fig. 4.29.

Positive gain margin means the system is stable and negative gain margin means the system is unstable. For minimum phase system both phase margin and gain margin must be positive for the system to be stable.

The point at which the magnitude curve crosses the 0db line is the gain crossover frequency. The phase crossover frequency is the point where the phase curve crosses the 180° line.

Gain Margin : Gain margin is defined as the margin in gain allowable by which gain can be increased till system reaches on the verge of instability. Mathematically gain margin is defined as the reciprocal of the magnitude of the $G(j\omega)\ H(j\omega)$ at phase cross-over frequency.

$$\therefore \quad \text{G.M} = \frac{1}{G(j\omega)\,H(j\omega)|_{\omega=\omega_{c_2}}}$$

where ω_{c_2} = phase cross-over frequency.

Generally, G.M is expressed in decibels

$\therefore$ In decibels $\quad G.M = 20 \log \dfrac{1}{|G(j\omega)\,H(j\omega)|_{\omega=\omega_{c_2}}}$

or, $\quad G.M. = -20 \log_{10} |G(j\omega)\,H(j\omega)|_{\omega=\omega_{c_2}}$

Phase Margin : For gain the additional phase lag can be introduced without affecting the magnitude plot. Therefore, phase margin can be defined as the amount of additional phase lag which can be introduced in the system till system reaches on the verge of instability is called as phase margin (P.M.). Mathematically phase margin can be defined as

$$P.M. = \left[\angle G(j\omega)\,H(j\omega)\big|_{\omega=\omega_{c_1}}\right] - (-180°)$$

$$P.M. = 180° + \underline{\angle G(j\omega)\,H(j\omega)}\big|_{\omega=\omega_{c_1}}$$

where ω_{c_1} = Gain cross-over frequency.

The above expressions for gain margin and phase margin can be directly used for mathematical determination of G.M. & P.M. For example, for the given transfer function $G(s)\,H(s) = \dfrac{2}{s(1+0.5s)(1+0.05s)}$ determine phase cross-over frequency, gain margin, gain cross-over frequency and phase margin.

According to definition, the phase cross-over frequency is the frequency at which phase angle of $G(j\omega)\,H(j\omega)$ is – 180°.

For given transfer function

$$\underline{\angle G(j\omega)\,H(j\omega)} = -90° - \tan^{-1} 0.5\,\omega - \tan^{-1} 0.05\,\omega$$

According to definition

$$-180° = -90° - \tan^{-1} 0.5\,\omega - \tan^{-1} 0.05\,\omega$$

or, $\tan^{-1} 0.5\omega + \tan^{-1} 0.05\,\omega = 90°$

taking tangent on both sides

$\tan [\tan^{-1} 0.5\,\omega + \tan^{-1} 0.05\,\omega] = \tan 90°$

$$\frac{0.5\omega + 0.05\omega}{1 - 0.5\omega.0.05\omega} = \infty$$

or, $\quad 1 - 0.025\,\omega^2 = 0$

$$\omega = 6.32 \text{ rad/sec.}$$

This is the phase cross-over frequency.

$\therefore \quad \omega_{c_2} = 6.32$ rad/sec.

Now calculate magnitude at this frequency

$$G(j\omega)\,H(j\omega) = \frac{2}{j6.32\,(1 + j0.5 \times 6.32)\,(1 + j0.05 \times 6.32)}$$

$$|G(j\omega)\,H(j\omega)|_{\omega=\omega_{c_2}} = 0.0918$$

At gain cross-over frequency the magnitude of $G(j\omega)\,H(j\omega)$ is unity.

$$|G(j\omega)\,H(j\omega)| = \frac{2}{\omega\sqrt{1+0.25\omega}\,\sqrt{1+0.0025\omega}}$$

or $\quad 1 = \dfrac{2}{\omega\sqrt{1+0.25\omega}\,\sqrt{1+0.0025\omega}}$

$$\omega = 2 \text{ rad/sec.}$$

$\therefore$ This is the gain cross-over frequency

$$\omega_{c_1} = 2 \text{ rad/sec.}$$

$\therefore$

$$G(j\omega)\,H(j\omega)\big|_{\omega=\omega_{c_1}} = \frac{2}{j2\,(1+j\,1)\,(1+j\,0.1)} = 0.707\ \underline{\angle -140.71°}$$

$$\underline{\angle G(j\omega)\,H(j\omega)}\big|_{\omega=\omega_{c_1}} = -140.71°$$

According to definition

$$P.M. = 180° + \underline{\angle G(j\omega)\,H(j\omega)}\big|_{\omega=\omega_{c_1}} = 180° - 140.71°$$

$$P.M. = 39.28°$$

From Bode's plot, phase margin and gain margin can be determined as follows.

Draw a line downwards from gain crossover point '*x*' till it intersects phase angle plot. This point of intersection is represented by '2' the distance between point '2' and – 180° line is the phase margin. If point '2' on phase angle plot is above – 180° line, phase margin is positive and if point '2' is below – 180° line, phase margin is negative (as shown in fig. 4.29).

From phase cross-over point '*y*' draw a line upwards till it intersects magnitude plot at point '1'. The difference between magnitude corresponding to point '1' and '0' db is gain margin. If point '1' is below 0 db G.M. is positive and if point '1' is above 0 db line G.M. is negative as shown in fig. 4.29.

For positive G.M. and P.M. the system is stable and the gain cross-over frequency<phase cross-over frequency. If gain cross-over frequency > phase cross-over frequency both P.M. & G.M. will be negative and the system is unstable. If gain cross-over frequency = phase crossover frequency, the system is marginally stable.

The transfer function changes with the change of temperature & pressure. The gain of the system is also affected by supply voltage, supply frequency, air pressures. If the gain is high, the gain margin is low and for step response the settling time is large. If the gain is low G.M. & P.M. will be high and settling time will be more i.e., the system response will be sluggish and also the rise time and steady state error will be high.

4.12. DETERMINATION OF K_p, K_v AND K_a FROM BODE PLOT

(a) Static Position Error Constant K_p :

Static position error constant is defined for type zero system and is given by

$$K_p = \lim_{s \to 0} G(s)\,H(s)$$

Put $s = j\omega$

$$K_p = \lim_{j\omega \to 0} G(j\omega)\,H(j\omega)$$

From equation (4.4)

$20 \log_{10} |G(j\omega)\, H(j\omega)| = 20 \log_{10} K \quad \because \; N = 0$

Consider the Bode plot shown in fig. 4.26, the magnitude of $G(j\omega)\, H(j\omega)$ is $20 \log K$ which is K_p

$$\therefore \quad K_p = 20 \log K$$

(b) Static Velocity Error Constant K_v :

Static velocity error constant is defined for type one system and is given by

$$K_v = \lim_{s \to 0} s\, G(s)\, H(s)$$

From equation (4.4) put $N = 1$

$$20 \log_{10} |G(j\omega)\, H(j\omega)| = 20 \log_{10} K - 20 \log_{10} \omega$$

$$0 \text{ db} = 20 \log_{10} K - 20 \log_{10} \omega$$

$$K = \omega$$

This is static velocity error constant

$$K_v = \omega$$

For type one system, the initial slope of Bode plot (magnitude) is – 20 db/decade. So, draw a line having slope – 20 db/dec. which intersect 0 db line. The frequency at which initial slope of a line is – 20db/dec. intersects 0 db line is the value of static velocity error constant (K_v) as shown in fig. 4.27.

(c) Acceleration Error Constant K_a :

Acceleration error constant is defined for type two system and is given by

$$K_a = \lim_{s \to 0} s^2\, G(s)\, H(s)$$

For type two system the initial slope of Bode plot is – 40db/dec.

From equation 4.4.

$$20 \log_{10} |G(j\omega)\, H(j\omega)| = 20 \log_{10} K - 20.2 \log_{10} \omega$$

Extend the line having slope – 40 db/dec. to intersect 0 db line

$$\therefore \quad 0 = 20 \log_{10} K - 40 \log_{10} \omega$$

or, $\omega = \sqrt{K}$

or, $K = \omega^2$

This is the acceleration error constant.

So, the frequency at which initial line of slope -40db/decade intersects 0 db line, gives the value of square root of acceleration constant. as shown in fig. 4.27.

$$\therefore \quad \omega = \sqrt{K_a}$$

or, $K_a = \omega^2$.

4.13. BODE PLOT WITH TRANSPORTATION LAG

Transportation lag is also known as time delay. Due to the various reasons sometimes it is necessary to stop the action for some time. Such time delay is called transportation lag. In many systems such as hydraulic, pneumatic and thermal pure delay occurs. In laplace domain the transportation lag is given by e^{-Ts}, where 'T' is the delay in seconds.

$$\therefore \quad G(s) = e^{-Ts}$$

put $s = j\omega$

$$G(j\omega) = e^{-j\omega T}$$

we know that

$$e^{-j\theta} = \cos\theta - j \sin\theta$$

$$\therefore \quad G(j\omega) = \cos \omega T - j \sin \omega T$$

$$|G(j\omega)| = \sqrt{\cos^2 \omega T + \sin^2 \omega T} = 1$$

Therefore, the magnitude is always equal to unity for all values of ω.

In db $= 20 \log |G(j\omega)| = 20 \log 1 = 0$

In other words we can say that the introduction of time delay has no effect on magnitude plot.

Phase Angle of Transportation Lag

$$\angle G(j\omega) = \tan^{-1}\left[-\frac{\sin \omega T}{\cos \omega T}\right]$$

$$= -\tan^{-1} (\tan \omega T) = -\omega T \text{ (radians)}$$

ω is in radians/sec. and t is in seconds, therefore phase lag is in radians.

$$\angle G(j\omega) = -57.3\, \omega T \text{ degree} \qquad (\because \pi \text{ radians} = 180°)$$

The phase angle linearly varies with the frequency ω.

For $T = 1$

ω	$-57.3\, \omega T$
0.1	– 5.73°
0.5	– 28.65°
1.0	– 114.6°
5.0	– 286.5°
10	– 573°

Example 4.9. Sketch the Bodes plot for the transfer function

$$G(s) = \frac{1000}{(1+0.1s)(1+0.001s)}$$

Determine the *a*. P.M

***b*. Gain margin**

***c*. Stability of the system.**

Solution : Step 1 : Put $s = j\omega$

$$G(j\omega) = \frac{1000}{(1+j0.1\omega)(1+j0.001\omega)}$$

The given transfer function is of type '*o*' system. Therefore the initial slope of the Bode plot is 0 db/decade. The starting point is given by.

$$20 \log_{10} K = 20 \log_{10} 1000 = 60 \text{ db}$$

corner frequencies $\omega_1 = \dfrac{1}{0.1} = 10$ rad/sec.

$$\omega_2 = \frac{1}{0.001} = 1000 \text{ rad/sec.}$$

Step 2 : Mark the starting point 60 db on y-axis and draw a line of slop 0db/decade upto first corner frequency.

Step 3 : From first corner frequency to second corner frequency draw a line with slope (0 – 20 = –20 db/decade).

Step 4 : From second corner frequency to next corner frequency (if given) draw a line having the slope –20 + (–20) = –40 db/decade.

Step 5 : The magnitude plot is complete and now draw the phase plot by calculating the phase at different frequencies (as given in table).

Step 6 : From the bode plot

From the point of intersection of magnitude curve with 0 db axis draw a line on phase curve. This line cuts the phase curve at – 154°

$$\therefore \quad P.M = -154 - (-180)$$
$$= +26°$$

Step 7 : Gain margin $G.M = \infty$

Since, $P.M = +26°$ and gain margin $= \infty$, the system is inherently stable.

Table 4.4

ω	$-\text{Arg}(1 + j0.1\omega)$ $-\tan^{-1} 0.1\omega$	$-\text{Arg}(1 + j0.001\omega)$ $-\tan^{-1} 0.001\,\omega$	Resultant
50	– 78.6°	– 2.86°	– 81.46°
100	– 84.2°	– 5.7°	– 90°
150	– 86.2°	– 8.5°	– 94°
200	– 87.13°	– 11.3°	– 98°
500	– 88.85°	– 26.56°	– 115.4°
800	– 89.28°	– 38.65°	– 127.93°
1000	– 89.48°	– 45°	– 134.42°
2000	– 89.72°	– 63.43°	– 153.15°
3000	– 89.8°	– 71.56°	– 161.36°
5000	– 89.88°	– 78.69°	– 168.57°
8000	– 89.92°	– 82.87°	– 172.79°

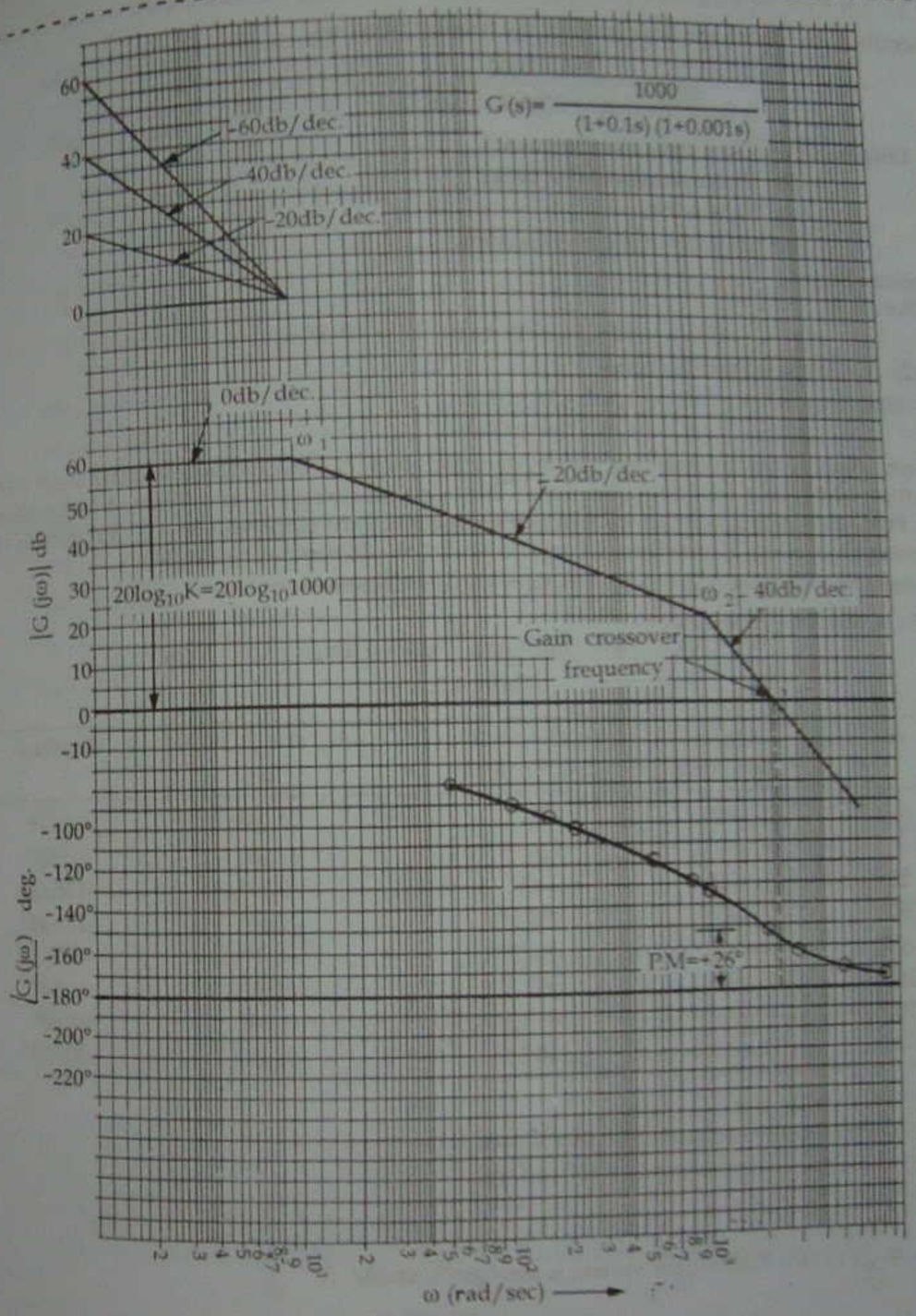


Fig. 4.30.

Example 4.10. Sketch the Bode plot for the transfer function.

$$G(s) = \frac{1000}{s(1+0.1s)(1+0.001s)}$$

Determine (*i*) Gain cross over frequency
(*ii*) Phase crossover frequency
(*iii*) G.M & P.M
(*iv*) Stability of the given system

Solution : Step 1 :

Put $s = j\omega$

$$G(j\omega) = 1000/ (j\omega) (1 + j\,01\omega) (1+ j\,0.001\omega)$$

Step 2 : Draw the magnitude curve

corner frequencies $\omega_1 = 1/0.1 = 10$ rad/sec.

$\omega_2 = 1/0.001 = 1000$ rad/sec.

Initial slope of the curve will be – 20 db/decade due to $^1/_S$ term. On ω – axis mark $\omega = K = 1000$. From this point draw a line having the slope – 20 db/decade to meet y-axis. This will be the starting point. From the starting point to the first corner frequency the slope will be – 20 db/decade, then from corner frequency 10 the slope will be – 20 + (–20) = – 40 db/dec. upto 1000

After 1000, the slope will be – 40 + (–20) = – 60 db/decade.

Step 3 : Draw the phase curve

$$\phi = -90° - \tan^{-1} 0.1\omega - \tan^{-1} 0.001\omega$$

Table 4.5.

ω	$Arg(j\omega)$ $-\tan^{-1} 0.1\omega$	$Arg\,(1 + j\,0.1\omega)$ $-\tan^{-1} 0.001\omega$	$-Arg\,(1 + j\,0.001\omega)$ ϕ	Resultant
1	–90°	–5.7°	–0.06°	–95.7°
5	–90°	–26.5°	–0.28°	–116.5°
10	–90°	–45°	–0.57°	–135.63°
50	–90°	–78.6°	–2.86°	–171.46°
100	–90°	–84.2°	–5.7°	–179.9°
150	–90°	–86.2°	–8.5°	–184°
200	–90°	–87.13°	–11.3°	–188°
500	–90°	–88.85°	–26.56°	–205.41°

Step 4 : From Bode plot

(*i*) Gain cross over frequency = 100 rad/sec.

(*ii*) Phase crossover frequency = 100 rad/sec.

(*iii*) G.M = 0

P.M = 0

(*iv*) Since G.M = P.M = 0, the system is marignally stable

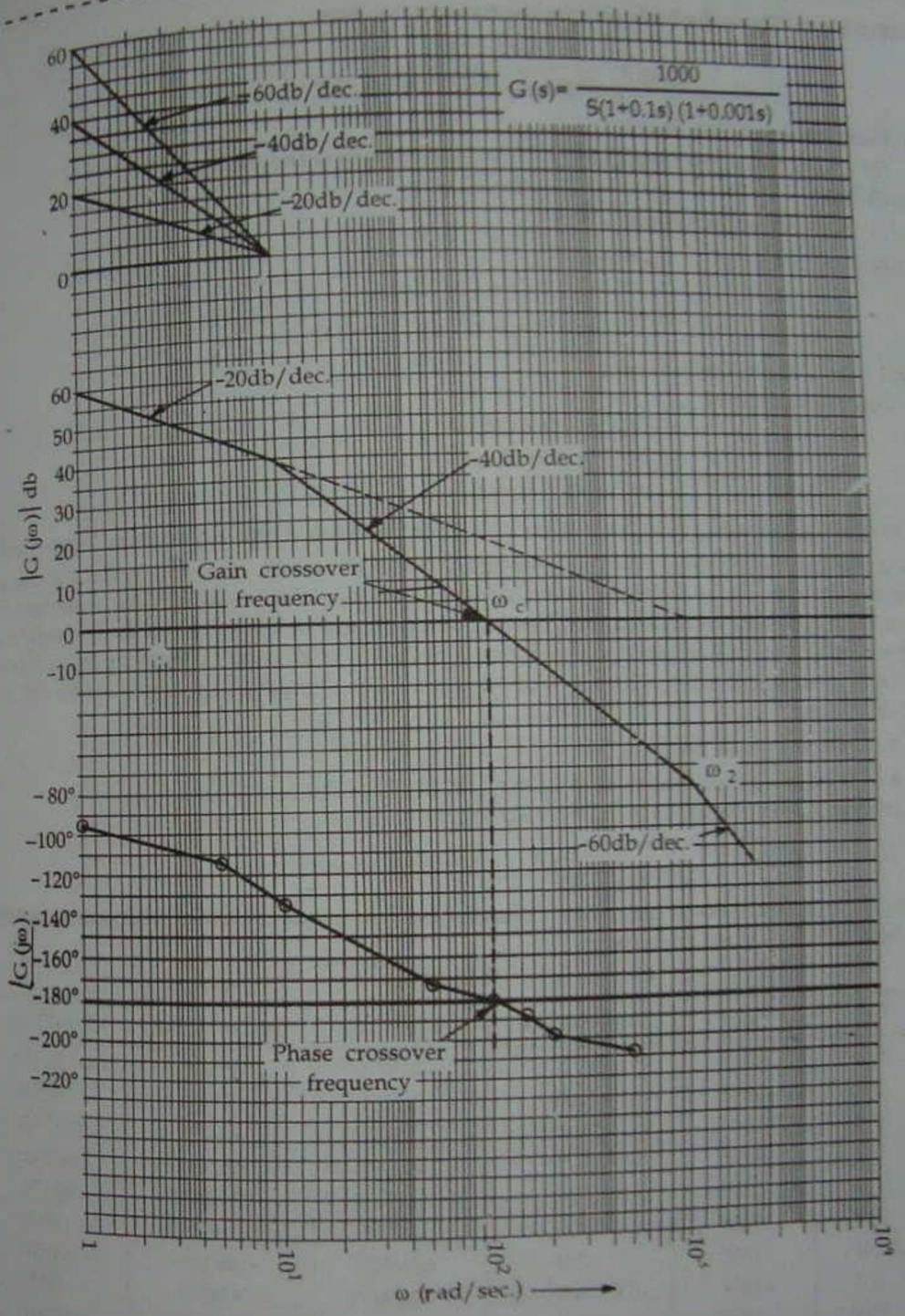


Fig. 4.31.

Example 4.11. Draw the Bode plot for the transfer function

$$G(s) = \frac{16(1+0.5s)}{S^2(1+0.125s)(1+0.1s)}$$

From the graph determine

(*i*) **Phase cross over frequency** (*ii*) **Gain cross over frequency**
(*iii*) **P.M** (*iv*) **G.M**
(*v*) **Stability of the system**

Solution : Step 1 : Put $s = j\omega$

$$G(\omega) = \frac{16(1+j0.5\omega)}{(j\omega)^2(1+j0.125\omega)(1+jo.1\omega)}$$

Step 2 : Draw the magnitude plot

Corner frequencies

$\omega_1 = 1/0.5 = 2$ rad/sec.
$\omega_2 = 1/0.125 = 8$ rad/sec.
$\omega_3 = 1/0.1 = 10$ rad/sec.

At ω-axis mark $\omega = \sqrt{K} = \sqrt{16} = 4$ rad/sec. because this is a type two system. From 4 rad/sec draw a line having the slope of - 40 db/decade to meet the y-axis. This will be the starting point. From the starting point to the first corner frequency the slope will be - 40 db/dec. from first corner frequency (2 rad/sec.) to the second corner frequency (8 rad/sec.) the slope of the line will be -40 (+20) = -20 db/decade From second corner frequency to the third corner frequency (10 rad/sec) the slope of the line will be -20 + (-20) = - 40 db/decade. After 10 rad/sec. the slope will be - 40 + (-20) = -60 db/decade.

Step 3 : Draw the phase diagram

Step 4 : Phase cross over frequency = 6.5 rad/sec (ω_{c2})
Gain cross over frequency = ω_{c1} = 8 rad/sec.

$P.M = -8°$
$G.M = -2$ db

(*v*) Since gain crossover frequency > phase cross over frequency hence the system is un stable. Also both *P.M* & *G.M* are negative.

Table 4.6.

ω	$-(j\omega)^2$	$-\tan^{-1} 0.125\omega$	$-\tan^{-1} 0.1\omega$	$+\tan^{-1} 0.5\omega$	Resultant
0.1	-180°	-0.716°	-0.57°	+2.86°	-178.42°
0.2	-180°	-1.43°	-1.15°	+5.71°	-176.86°
0.5	-180°	-3.57°	-2.86°	+14°	-172.43°
0.8	-180°	-5.71°	-4.57°	+21.8°	-168.48°
1.0	-180°	-7.13°	-5.71°	+26.56°	-166.28°
2.0	-180°	-14°	-11.3°	+ 45°	-160.3°
5.0	-180°	-32°	-26.56°	+ 68.19°	-170.37°
8.0	-180°	-45°	-38.66°	+76°	-187°
10.0	-180°	-51.34°	-45°	+78.69°	-197.65°
20.0	-180°	-68.19°	-63.43°	+ 84.28°	-227.34°
30.0	-180°	- 75°	-71.56°	+86.18°	-240.38°

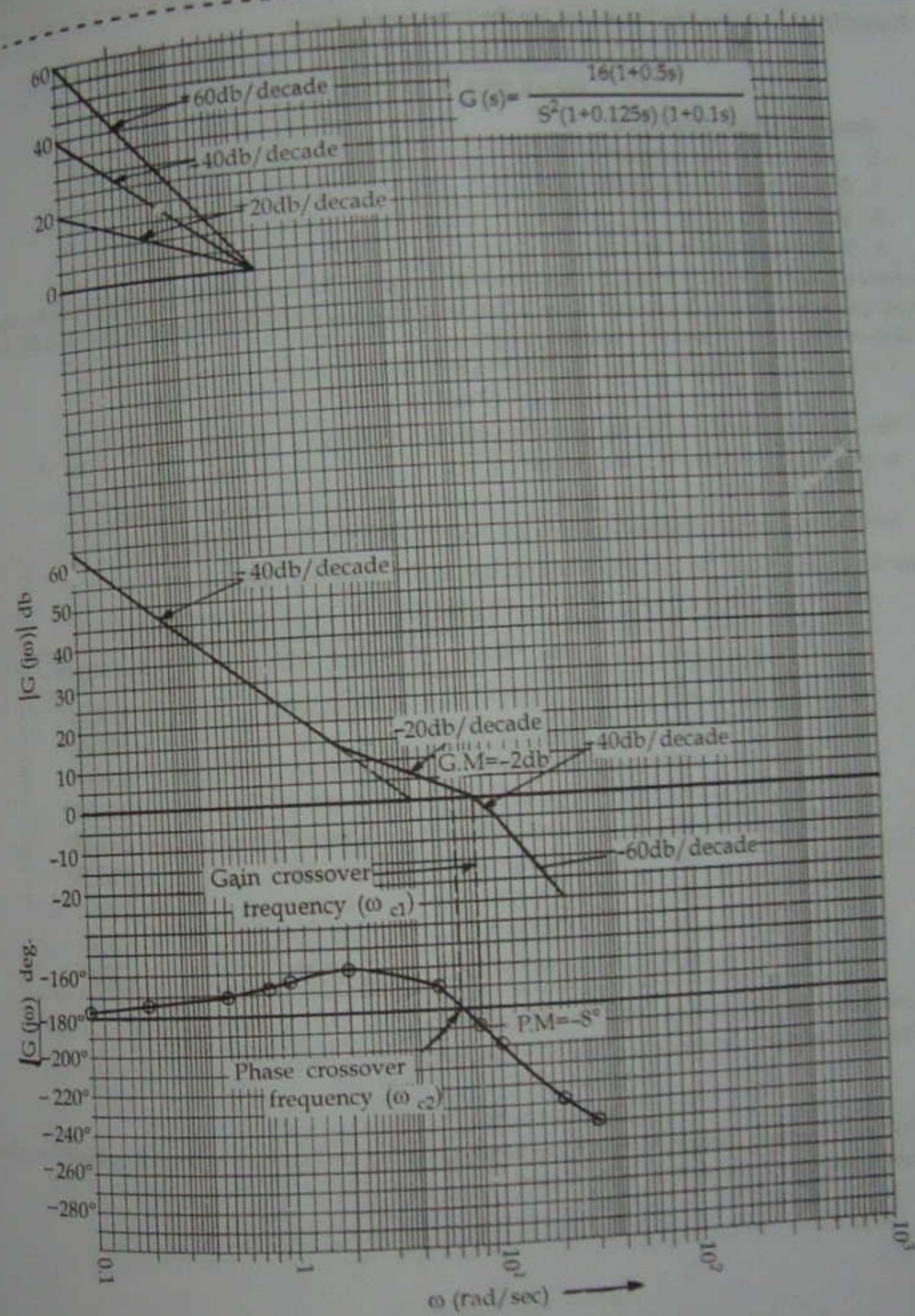


Fig. 4.32.

Example 4.12. Draw the Bode plot for the transfer function

$$G(s) = \frac{50}{s(1+0.25s)(1+0.1s)}$$

From the graph determine

1. **Gain crossover frequency**
2. **Phase crossover frequency**
3. **G.M & P.M.**
4. **Stability of the system.**

Solution : Since this is type one system, the initial slope of the line will be – 20 db/decade. Mark a point on ω-axis at 50 rad/sec and draw a line with slope –20 db/dec. from the point at 50, this line will meet the y-axis. This is the starting point.

corner frequencies are $\omega_1 = 1/0.25 = 4$ rad/sec.

$\omega_2 = 1/0.1 = 10$ rad/sec.

Slope of the line from starting point to I corner freqency = – 20 db/dec.

slope of the line from I corner frequency to II = –20 + (–20)

= – 40 db/dec.

After second corner frequency the slope will be – 40 + (–20) = – 60 db/dec.

Phase Plot : For phase plot calculate the phases at different frequencies.

Table 4.7.

ω	jω	$-\tan^{-1} 0.1\omega$	$-\tan^{-1} 0.25\omega$	Resultant
1	-90°	- 5.71°	- 14.03°	- 109.74°
2	-90°	- 11.3°	- 26.56°	- 127.86°
5	-90°	- 26.56°	- 51.34°	- 168°
10	-90°	- 45°	- 68.19°	- 203.19°
20	-90°	- 63.43°	- 78.69°	- 232.12°
40	-90°	- 76°	- 84.28°	- 250.28°
60	-90°	- 80.53°	- 86.18°	- 256.71°
80	-90°	- 82.87°	- 87.13°	- 260°
100	-90°	- 84.29°	- 87.7°	- 262°

Gain crossover frequency (ω_{c1}) = 13 rad/sec.

Phase crossover frequency (ω_{c2}) = 6.5 rad/sec.

P.M = -36°

G.M = - 13 db

Since, both phase margin & gain margin are negative and gain crossover frequency > phase crossover frequency, the system is un stable.

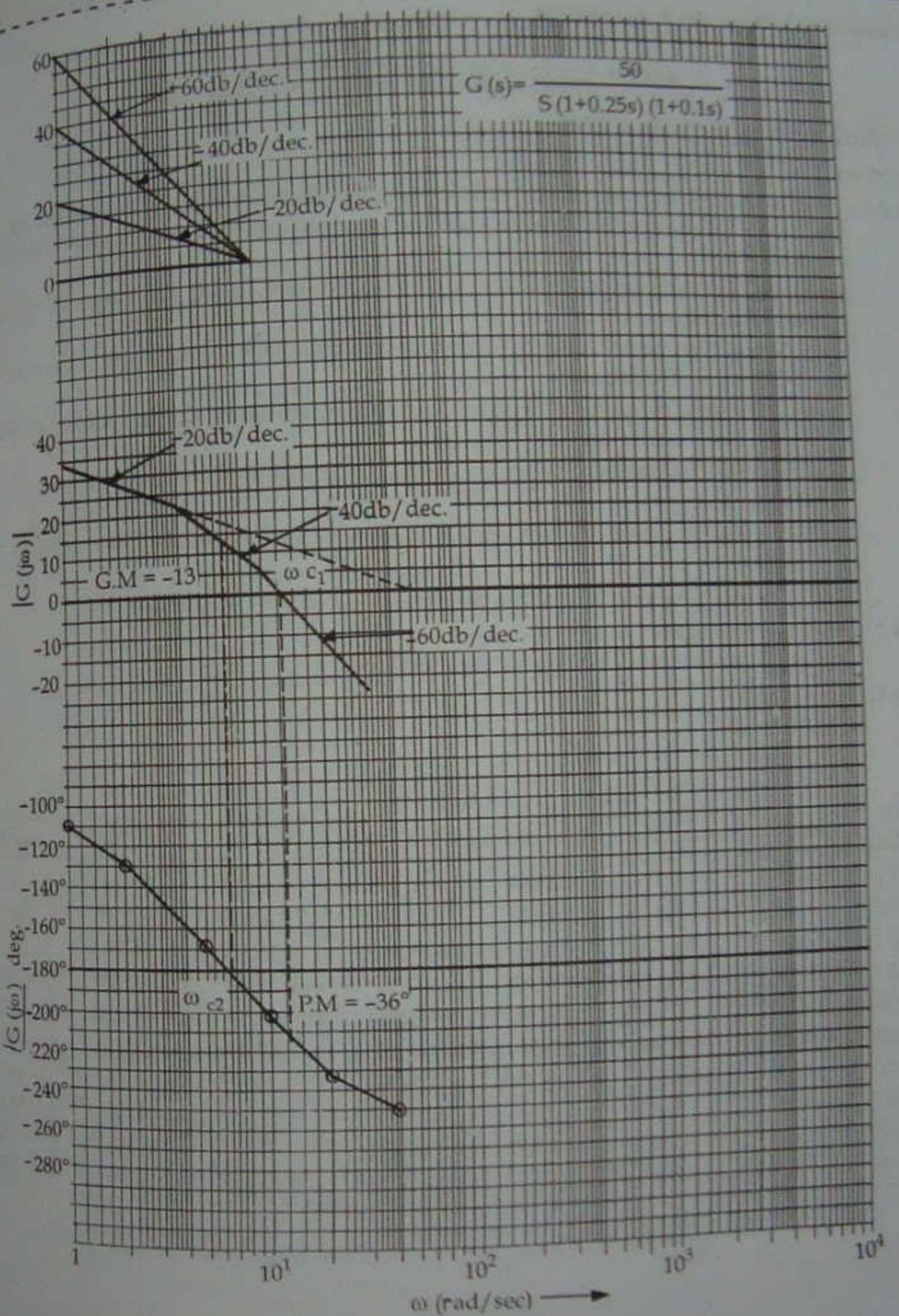


Fig. 4.33.

Example 4.13. Draw the Bode plot for

$$G(s) = \frac{23.7(1 + j\omega)(1 + j0.2\omega)}{(j\omega)(1 + j\,3\omega)(1 + j0.5\omega)(1 + j0.1\omega)}$$

From the plot find G.M & P.M

Solution :

Step 1 : On ω-axis mark the point at 23.7 rad/sec. Since in denominator (jω) term is having pow one, from 23.7 draw a line of slope - 20 db/decade to meet y-axis. This will be the start point.

From the starting point to the I corner frequency (0.33) the slope of the line - 20 db/decade.

From I corner frequency (0.33) to second corner frequency (1) the slope of the line wil 20 + (–20) = – 40 db/decade.

From II corner frequency (1) to III corner frequency (2) the slope of the lin – 40 + (+ 20) = – 20 db/deceade.

From III corner frequency to IV corner frequency, the slope of the line will be – 20 + (-20 – 40 db/decade.

From IV corner frequency (5) to V corner frequency the slope will be – 40 + (+20) = - 20 db decade.

After V corner frequency, the slope will be – 20 + (–20) = – 40 db/dec.

Step 2 : Draw the phase plot.

Values of ϕ at different frequencies are tabulated.

Step 3 : From graph

P.M = + 34°

G.M = ∞

Table 4.8.

ω	$-\tan^{-1}\omega$	$-\tan^{-1}3\omega$	$-\tan^{-1}0.5\omega$	$-\tan^{-1}0.1\omega$	$\tan^{-1}\omega$	$\tan^{-1}0.2\omega$	Resultant
0.1	-90°	-16.7°	-2.86°	-0.57°	+ 5.71°	+1.14°	-103°
0.2	-90°	-31°	-5.71°	-1.14°	+ 11.3°	+2.3°	-114.25°
0.5	-90°	-56.3°	-14.03°	-2.86°	+ 26.56°	+5.71°	-130.92°
0.8	-90°	-67.4°	-21.8°	-4.57°	+ 38.65°	+9.09°	-136.03°
1.0	-90°	-71.56°	-26.56°	-5.71°	+ 45°	+11.3°	-137.5°
2.0	-90°	-80.54°	-45°	-11.3°	+ 63.43°	+ 21.8°	- 141.61°
5.0	-90°	-86.18°	- 68.19°	-26.56°	+ 78.7°	+ 45°	- 147.25°
8.0	-90°	-87.61°	- 76°	- 38.65°	+ 82.87°	+ 58°	- 151.39°
10.0	-90°	- 88°	- 78.7°	- 45°	+ 84.3°	+ 63.4°	- 154.0°
20.0	-90°	- 89°	- 84.3°	- 63.43°	+ 87.13°	+ 76°	- 163.6°

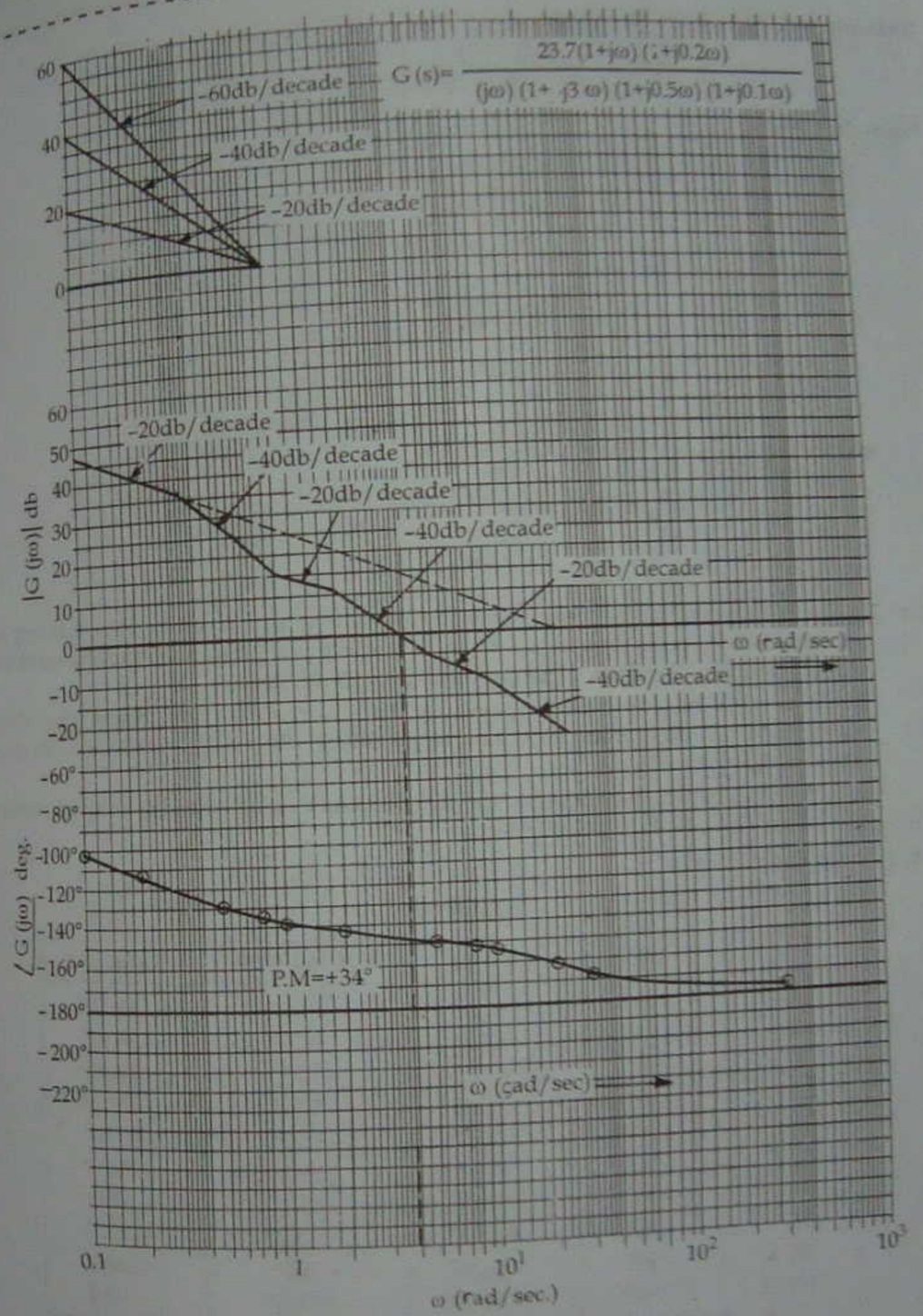


Fig. 4.34.

Example 4.14. Sketch the Bode plot

$$G(s) = \frac{512(s+3)}{s(s^2+16s+256)}$$

Solution : Step 1

$$G(s) = \frac{512\left(\frac{s}{3}+1\right)3}{s\left[\frac{s^2}{256}+\frac{16s}{256}+1\right]256}$$

$$= \frac{6(1+s/3)}{s\left[\frac{s^2}{256}+\frac{16}{256}s+1\right]}$$

Compare $\frac{s^2}{256}+\frac{16}{256}s+1$ With $\frac{s^2}{\omega_n^2}+\frac{2\zeta}{\omega_n}s+1$

$$\omega^2_n = 256 \therefore \omega_n = 16 \text{ rad/sec.}$$

$$\frac{2\zeta}{\omega_n} = \frac{16}{256} \therefore \xi = 0.5$$

Step 2 : On ω-axis mark the point at 6. From this point draw a line having slop - 20 db/decade, meets the *y*-axis. This is the starting point. From the starting point to the corner frequency (3 rad/sec.) the slope will be – 20 db/decade.

Since ω = 3 rad/sec. corresponding to a zero, at this point the slope increased by + 20 db/decade. Hence the slope will be – 20 + (+20) = 0 db/decade upto nex corner frequency *i.e.* 16 rad/sec.

At ω = 16 rad/sec corresponding to the pair of complex poles the slope must be decrease by – 40 db/dec. (0 – 40 = – 40 db/dec.)

Step 3 : Draw the phase plot

Table 4.9

ω	jω	3 + jω	$s^2 + 16s + 256$	Resultant
0.1	– 90°	+ 1.9°	– 0.36°	– 88.46°
0.2	– 90°	+ 3.8°	– 0.72°	– 86.92°
0.5	– 90°	+ 9.5°	– 1.79°	– 82.29°
1.0	– 90°	+ 18.43°	– 3.59°	– 75.16°
2.0	– 90°	+ 33.7°	– 7.23°	– 63.53°
5.0	– 90°	+ 59°	– 19.1°	– 50.1°
10.0	– 90°	+ 73.3°	– 45.7°	– 62.4°
20.0	– 90°	+ 81.5°	– 114.2°	– 122.7°
50.0.	– 90°	+ 86.56°	– 160°	– 163.4
100.0	– 90°	+ 88.28°	– 170.67°	– 172.4

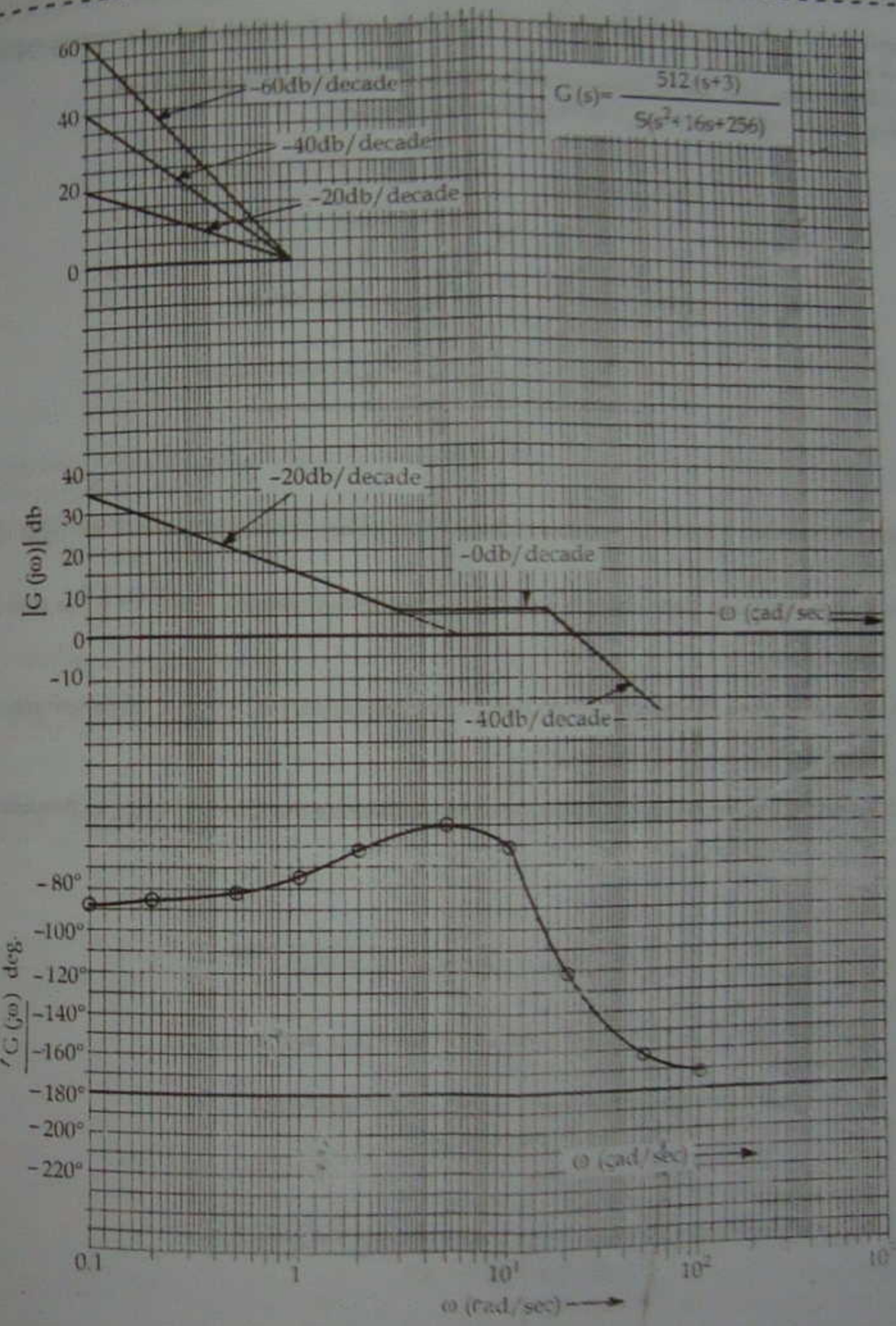


Fig. 4.35.

Example 4.15. The magnitude plot of the open loop transfer function $G(s)$ of a certain system is shown in the fig. 4.36.

(a) Determine $G(s)$ if it is known that the system is of minimum phase type.

(b) Estimate the phase at each of the corner frequencies.

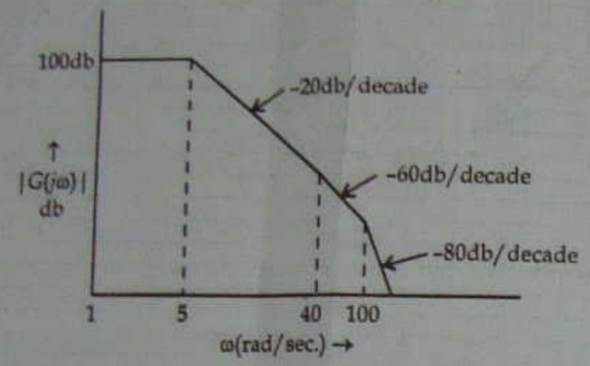


Fig. 4.36.

Solution : Since, the system is of minimum phase type, it has no poles or zeros in the right hand side of s-plane.

a. At $\omega = 5$ rad/sec. slope changes to - 20 db/decade indicating a term $\left(1+\frac{s}{5}\right)$ or (1+0.2s) in denominator.

b. At $\omega = 40$, slope changes to - 60 db/decade indicating a term $\left(1+\frac{s}{40}\right)^2$ in denominator (net slope changes is - 40 db/decade)

c. At $\omega = 100$, slope changes to - 80 db/decade, indicating the term $\left(1+\frac{s}{100}\right)$ in denominator.

d. $20 \log_{10} K = 100$

$\therefore \quad K = 10^5$

$$\therefore \quad G(s) = \frac{10^5}{(1+0.2s)(1+0.025s)^2(1+0.01s)}$$ **Ans.**

(b) At $\omega = 5$

$\phi_5 = -[+\tan^{-1} 0.2 \times 5 + 2(\tan^{-1} 0.025 \times 5) + \tan^{-1} 0.01 \times 5] = -62.11°$

At $\omega = 40$

$\phi_{40} = -[\tan^{-1} 40 \times 0.2 + 2(\tan^{-1} 40 \times 0.025) + \tan^{-1} 40 \times 0.01]$

$= -[82.87 + 90 + 21.8]$

$\phi_{40} = -194.67°$

At $\omega = 100$

$\phi_{100} = -[\tan^{-1} 100 \times 0.2 + (\tan^{-1} 100 \times 0.025)^2 + \tan^{-1} 100 \times 0.01]$

$= -[87.13 + 136.39 + 45]$

$= -268.53°$

Example 4.16. Find the transfer function of the system from the data given on Bode diagram.

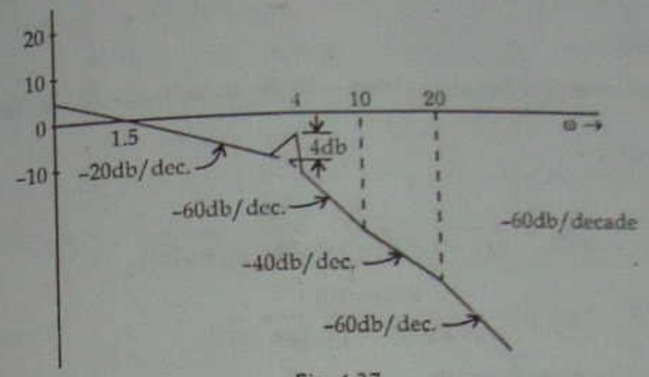


Fig. 4.37.

Solution : First line having a slope of -20 db/decade and it is not passing through $\omega = 1$ rad/sec. it means there is term K/s

At $\omega = 1$ rad/sec. of initial part = $20 \log 1.5 = 3.521$

$\therefore \quad 20 \log K = 3.521$

$K = 1.5$

At $\omega = 4$ rad/sec. there is a term $\left[1+\frac{2\zeta}{\omega_n}s+\frac{s^2}{\omega^2_n}\right]^{-1}$ because the slope changes from - 20 db/decade to - 60 db/decade and also a peak of 4 db is given.

$\omega_n = 4$ rad/sec.

Value of $\left[1+\frac{2\zeta}{\omega_n}s+\frac{s^2}{\omega^2_n}\right]^{-1}$ at $\omega = \omega_n$

Put $s = j\omega$

$$\left[1+j\frac{2\zeta}{\omega_n}\omega-\frac{\omega^2}{\omega^2_n}\right]^{-1} = \left[\sqrt{\left(1-\frac{\omega^2}{\omega^2_n}\right)^2+\left(\frac{2\zeta}{\omega_n}\right)^2}\right]^{-1}$$

$$= \left[\sqrt{\left(1-\frac{4}{4}\right)^2+\left(\frac{2\zeta}{4}\cdot 4\right)^2}\right]^{-1}$$

$$= \frac{1}{2\zeta}$$

$$20 \log \frac{1}{2\zeta} = 4 \qquad \therefore \ \xi = 0.316$$

$$\therefore \left[1+\frac{2\zeta}{\omega_n}s+\frac{s^2}{\omega^2_n}\right]^{-1} = [1 + 0.158s + 0.0625s^2]^{-1}$$

At ω = 10 rad/sec. the slope changes from – 60 db/dec. to – 40 db/dec indicating the

$\left(1+\frac{s}{10}\right)$ in numerator.

At ω = 20 rad/sec, the slope changes from – 40 db/decade to – 60 db/dec. indicating the

$\left(1+\frac{s}{20}\right)$ in denominator

$$\therefore \quad G(s) = \frac{1.5(1+s/10)}{s(1+0.158s+0.0625s^2)(1+s/20)}$$

$$G(s) = \frac{48(s+10)}{s(s^2+2.52s+16)(s+20)}$$

this is the required transfer frnction.

Example 4.17. Find the transfer function of the given Bode diagram.

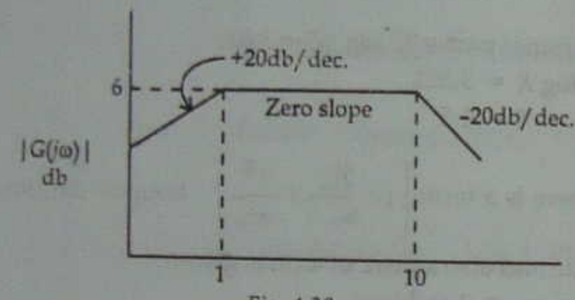


Fig. 4.38.

Solution: 1. First line having a slope of + 20 db/decade. Therefore there is a term 's' in numera

2. At ω = 1, the slope changes to zero, it means there is a term $(1+{}^{s}/_{1}) = (s+1)$ in denomina

3. At ω = 10, the slope changes to – 20 db/decade, this indicates there is a term $(1+{}^{s}/_{10})$ denominator.

4. 20 log K = 6

$$\therefore \quad K = 1.99 \approx 2.0$$

$$\therefore \quad G(s) = \frac{2s}{(s+1)(1+s/10)}$$

$$G(s) = \frac{20s}{(s+1)(s+10)}$$ is the required transfer function.

4.14. FREQUENCY DOMAIN SPECIFICATIONS

(*a*) **Resonant Peak: M_r :** The maximum value of magnitude is known as resonant peak. Th magnitude of resonant peak gives the information about the relative stability of the syste A large value of resonant peak implies undesirable transient response.

(*b*) **Resonant Frequency (ω_r) :** The frequency at which magnitude has maxm. value is known resonant frequency. If ω_r is large, the time response is fast.

(*c*) **Bandwidth :** Bandwidth is defined as the range of frequencies in which the magnitude

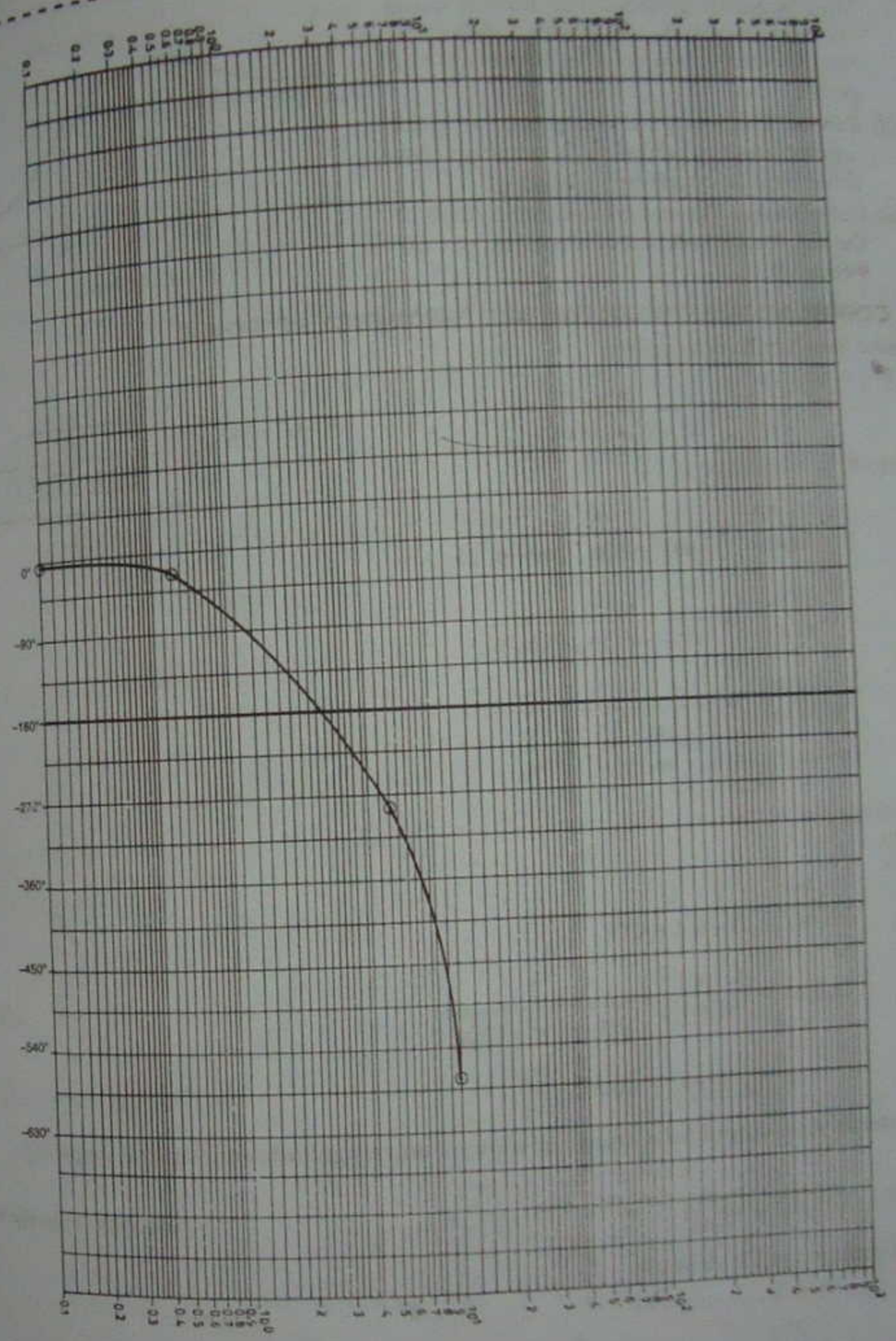


Fig. 4.39.

close loop does not drop – 3 db. In other words we can say that the bandwidth is defined as the band of frequencies lying between – 3 db points.

(*d*) **Cut -off Frequency :** The frequency at which the magnitude is 3 db below its zero frequency value is called cut off frequency (ω_b).

(*e*) **Cut-off rate :** The cut off rate is the slope of the log magnitude curve near the cut off frequency.

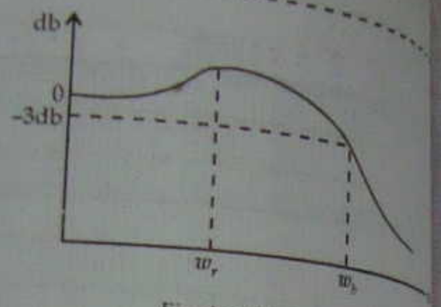


Fig. 4.40. Bode plot

4.15. CORRELATION BETWEEN TIME AND FREQUENCY RESPONSE

Consider the second order system

$$\frac{C(s)}{R(s)} = \frac{\omega^2_n}{s^2 + 2\zeta\omega_n s + \omega^2_n} \qquad ...(4.11)$$

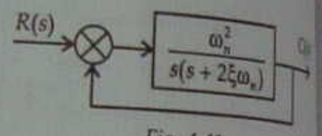


Fig. 4.41.

Put $s = j\omega$ in equation (4.11)

$$\frac{C(j\omega)}{R(j\omega)} = \frac{\omega^2_n}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega^2_n)}$$

$$\frac{C(j\omega)}{R(j\omega)} = \frac{\omega^2_n}{(\omega^2_n - \omega^2) + j\,2\zeta\omega_n\omega}$$

or $$\frac{C(j\omega)}{R(j\omega)} = \frac{1}{\left(1 - \frac{\omega^2}{\omega^2_n}\right) + j\,2\zeta\frac{\omega}{\omega_n}}$$

let $\frac{\omega}{\omega_n} = u$

$$\therefore \quad \frac{C(j\omega)}{R(j\omega)} = \frac{1}{(1-u^2) + j2\zeta u}$$

$$M(j\omega) = \left|\frac{C(j\omega)}{R(j\omega)}\right| = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}} \qquad ...(4.12)$$

$$\theta = \angle M(j\omega) = -\tan^{-1}\frac{2\zeta u}{1-u^2} \qquad ...(4.13)$$

The steady state output for a sinusoidal input of unit magnitude and variable frequency.

$$C(t) = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}} \sin\left[\omega t - \tan^{-1}\frac{2\zeta u}{1-u^2}\right] \qquad \because C(t) = Y\sin(\omega t + \theta)$$

From equation (4.12) & (4.13)

if $u = 0$ $\quad M = 1 \quad \theta = 0°$

$u = 1 \quad M = \frac{1}{2\zeta} \quad \theta = -\pi/2$

$u = \infty \quad M = 0 \quad \theta = -\pi$

The magnitude and phase angle characteristic for a certain value ξ is shown in fig. 4.42.

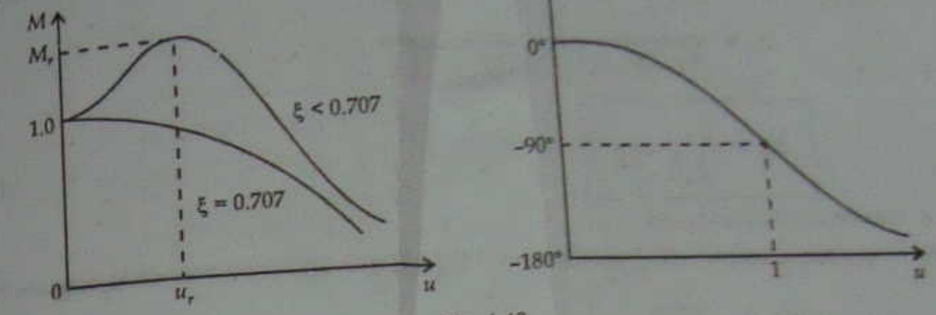


Fig. 4.42.

The frequency at which M has maximum value is known as resonant frequency (ω_r)

$$u_r = \frac{\omega_r}{\omega_n}$$

where u_r = normalized resonant frequency

Differentiate equation (4.12) with respect to u and put $u = u_r$ and equate to zero.

$$\frac{d}{du}M\Big|_{u=u_r} = 0$$

$$\frac{d}{du}M\Big|_{u=u_r} = -\frac{1}{2}\left[(1-u_r^2)^2 + (2\xi u_r)^2\right]^{-3/2}\left[4u^3_r - 4u_r + 8u_r\xi^2\right] = 0$$

or, $\quad 4u_r^3 - 4u_r + 8u_r\xi^2 = 0$

or, $\quad u_r^3 - u_r + 2u_r\xi^2 = 0$

or, $\quad u_r = \sqrt{1-2\zeta^2}$

But $\quad u_r = \frac{\omega_r}{\omega_n}$

$$\therefore \quad \frac{\omega_r}{\omega_n} = \sqrt{1-2\zeta^2}$$

$$\boxed{\omega_r = \omega_n\sqrt{1-2\zeta^2}} \qquad ....(4.14)$$

Form equation 4.13.

$$\theta = -\tan^{-1}\frac{2\xi u_r}{1-u^2}$$

$$= -\tan^{-1}\frac{2\xi\sqrt{1-2\xi^2}}{1-1+2\xi^2}$$

$$\boxed{\theta = -\tan^{-1}\frac{\sqrt{1-2\xi^2}}{\xi}} \qquad ...(4.15)$$

For max.^{M} value of magnitude put $u_r = \sqrt{1-2\xi^2}$ in eqn 4.12.

$$M_r = \frac{1}{\sqrt{[1-(1-2\xi^2)]^2 + 4\xi^2(1-2\xi^2)}}$$

$$\boxed{M_r = \frac{1}{2\xi\sqrt{1-\xi^2}}} \quad ...(4.16)$$

where M_r is known as resonant peak.

For step response of second order system (for $0 \le \zeta \le 1$) maximum overshoot M_p is given by

$$M_p = e^{\frac{\pi\xi}{\sqrt{1-\xi^2}}} \quad ...(4.17)$$

and damped frequency of oscillation is given by

$$\omega_d = \omega_n\sqrt{1-\xi^2} \quad ...(4.18)$$

From equation (4.14) and (4.18)

$$\frac{\omega_r}{\omega_d} = \frac{\omega_n\sqrt{1-2\xi^2}}{\omega_n\sqrt{1-\xi^2}}$$

$$\omega_r = \omega_d\left[\frac{\sqrt{1-2\xi^2}}{\sqrt{1-\xi^2}}\right] \quad ...(4.19)$$

From above expression if the value of ξ is small, the damped frequency ω_d and resonant frequency are nearly same. Hence for large value of ω_r, the time response is faster.

From equation (4.16) and (4.17) it is clear that both M_r and M_p are the function of ζ. As ζ increases both M_r and M_p decreases, but the M_p is limited to 1 and M_r having very large values for $\zeta < 0.4$ as shown in fig (4.42).

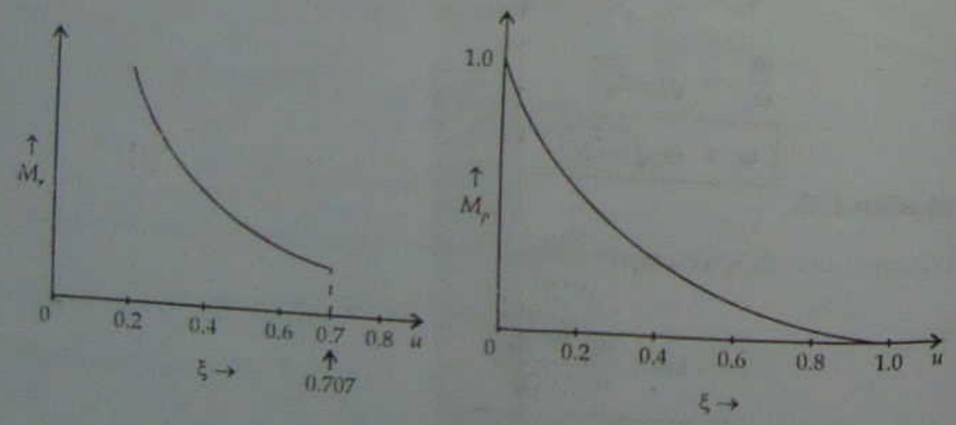


Fig. 4.43.

The bandwidth is also defined as the range of frequencies over which M is equal or greater than $1/\sqrt{2}$.

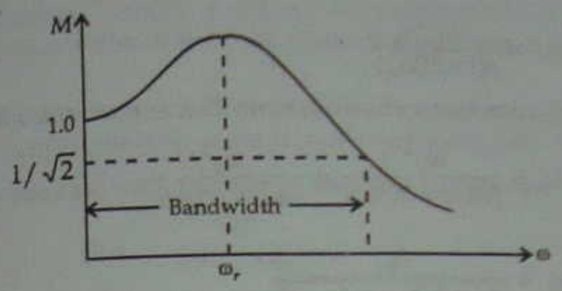


Fig. 4.44.

From equation no. (4.12).

$$M = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}}$$

let u_b = normalized bandwidth = $\frac{\omega_b}{\omega_n}$ then

$$M = \frac{1}{\sqrt{(1-u_b^2)^2 + (2\zeta u_b)^2}} = \frac{1}{\sqrt{2}}$$

Or, $(1-u_b^2)^2 + (2\xi u_b)^2 = 2$

$1 + u_b^4 - 2u_b^2 + 4\xi^2 u_b^2 = 2$

or, $u_b^4 - 2u_b^2 + 4\xi^2 u_b^2 - 1 = 0$

or, $u_b^4 - 2u_b^2(1 - 2\xi^2) - 1 = 0$

Put $u_b^2 = x$

$$x^2 - 2x(1-2\xi^2) - 1 = 0$$

$$\therefore \quad x = \frac{+2(1-2\zeta^2) \pm \sqrt{4(1-2\zeta^2)^2 + 4}}{2}$$

$$= 1 - 2\zeta^2 \pm \sqrt{1 - 4\zeta^2 + 4\zeta^4 + 1}$$

$$= 1 - 2\zeta^2 \pm \sqrt{2 - 4\zeta^2 + 4\zeta^4}$$

consider the positive part

$$u_b^2 = 1 - 2\zeta^2 + \sqrt{2 - 4\zeta^2 + 4\zeta^4}$$

$$\therefore \quad \boxed{u_b = \sqrt{1 - 2\zeta^2 + \sqrt{2 - 4\zeta^2 + 4\zeta^4}}} \quad ...(4.20)$$

But $u_b = \frac{\omega_b}{\omega_n}$

$\therefore \omega_b = u_b\,\omega_n$ = denormalized bandwidth

$$\therefore \quad \boxed{\omega_b = \omega_n\sqrt{1 - 2\zeta^2 + \sqrt{2 - 4\zeta^2 + 4\zeta^4}}} \quad ...(4.21)$$

CALCULATION OF PHASE MARGIN

consider open loop transfer function

$$G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$$

Put $s = j\omega$

$$G(j\omega) = \frac{\omega_n^2}{j\omega(j\omega+2\zeta\omega_n)} = \frac{\omega_n^2}{-\omega^2+j2\zeta\omega\omega_n}$$

$$|G(j\omega)| = \frac{\omega_n^2}{\sqrt{(-\omega^2)^2+(2\zeta\omega\omega_n)^2}}$$

At gain cross over frequency (ω_1) $|G(j\omega)| = 1$

$$\therefore \quad \omega_n^2 = \sqrt{(-\omega_1^2)^2+(2\zeta\omega_1\omega_n)^2}$$

$$\omega_1^4 + 4\xi^2\omega_1^2\omega_n^2 - \omega_n^4 = 0$$

Put $x = \omega_1^2$

$$x^2 + (4\xi^2\omega_n^2)\,x - \omega_n^4 = 0$$

$$x = \frac{-4\zeta^2\omega^2_n \pm \sqrt{(4\zeta^2\omega^2_n)^2+4\omega_n^4}}{2}$$

$$x = -2\zeta^2\omega^2_n \pm \omega^2_n\sqrt{4\zeta^2+1}$$

or, $$\omega_1^2 = -2\zeta^2\omega^2_n \pm \omega^2_n\sqrt{4\zeta^2+1}$$

or, $$\omega_1 = \omega_n\sqrt{\sqrt{4\zeta^4+1}-2\zeta^2}$$

$$\angle G(j\omega_1) = \tan^{-1}\frac{2\zeta\omega_1\omega_n}{-\omega_1^2} = +\tan^{-1}\frac{2\zeta\omega_n}{\omega_1}$$

Put the value of ω_1

$$\angle G(j\omega_1) = +\tan^{-1}\frac{2\zeta\omega_n}{\omega_n\sqrt{\sqrt{4\zeta^4+1}-2\zeta^2}}$$

$$= +\tan^{-1}\frac{2\zeta}{\sqrt{\sqrt{4\zeta^4+1}-2\zeta^2}}$$

∴ Phase margin

$$\phi = -180+\tan^{-1}\frac{2\zeta}{\sqrt{\sqrt{4x^4+1}-2\zeta^2}}$$

or, $$\boxed{\phi = \tan^{-1}\frac{2\zeta}{\sqrt{\sqrt{4\zeta^4+1}-2\zeta^2}}}$$

$\because$ $\tan(180+\phi) = \tan\phi$

4.16. RELATIVE AND ABSOLUTE STABILITY

There are two types of stability namely absolute stability and relative stability. Absolute stability means whether the system is stable or not *i.e.* the system is stable or unstable. If the system is stable then we determine how stable it is *i.e.* we measure the degree of stability, it is known as relative stability.

In time domain the relative stability is measured by maximum overshoot and damping ratio. In frequency domain relative stability is measured by resonant peak (M_r).

Example 4.18. The forward path transfer function of a unity feedback control system is

$$G(s) = \frac{100}{s(s+6.54)}$$

Find the resonance peak M_r, resonant frequency ω_r and bandwidth of the closed loop system.

Solution : Given that

$$G(s) = \frac{100}{s(s+6.54)}$$

$$H(s) = 1$$

$$\therefore \quad \frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)\,H(s)}$$

$$= \frac{100/s(s+6.54)}{1+\dfrac{100}{s(s+6.54)}}$$

$$= \frac{100}{s(s+6.54)+100} = \frac{100}{s^2+6.54s+100}$$

compare with $$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+w_n^2}$$

$$\omega_n^2 = 100 \qquad \boxed{\omega_n = 10 \text{ rad/sec.}}$$

$$2\xi\,\omega_n = 6.54$$

$$2\xi\,10 = 6.54 \qquad \therefore \quad \boxed{\xi = 0.327}$$

$$\omega_r = \omega_n\sqrt{1-2\zeta^2}$$

$$\omega_r = 10\sqrt{1-2(0.327)^2} = 8.86 \text{ rad/sec.}$$

$$M_r = \frac{1}{2\xi\sqrt{1-\xi^2}} = \frac{1}{2(0.327)\sqrt{1-(0.327)^2}}$$

$$\therefore \quad \boxed{M_r = 1.618}$$

Bandwidth $$= \omega_n\sqrt{1-2\xi^2+(2-4\xi^2+4\xi^4)^{1/2}}$$

$$= 10\sqrt{1-2(0.327)^2+(2-4\times0.327^2+4\times0.327^4)^{1/2}}$$

$$= 14.34 \text{ rad/sec.}$$

$$\boxed{\text{Bandwidth} = 14.34 \text{ rad/sec.}}$$

Example 4.19. The specification on a second order unity feedback control system with closed loop transfer function.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega^2_n}$$

are that the maximum overshoot must not exceed 30%. Find the corresponding limiting values of M_r and BW.

Solution : we know that

$$Mp = e^{-\pi\xi/\sqrt{1-\xi^2}} \times 100$$

$$\therefore \quad e^{-\pi\xi/\sqrt{1-\xi^2}} \times 100 = 30$$

or, $$e^{+\pi\xi/\sqrt{1-\xi^2}} = \frac{100}{30}$$

$$\frac{\pi\xi}{\sqrt{1-\xi^2}} = 1.2 \qquad \text{(take the natural log)}$$

or, $$\pi^2\xi^2 = 1.2^2(1-\xi^2) \quad \text{or,} \quad \xi = 0.356$$

$$M_r = \frac{1}{2\xi\sqrt{1-\xi^2}}$$

$$M_r = \frac{1}{2\times 0.356\sqrt{1-0.356^2}}$$

$$\therefore \quad \boxed{M_r = 1.5}$$

Normalized B.W $$= \sqrt{1-2\xi^2+(2-4\xi^2+4\xi^4)^{1/2}}$$

$$= \sqrt{1-2\times 0.356^2+(2-4\times 0.356^2+4\times 0.356^4)^{1/2}} = 1.41 \text{ rad/sec.}$$

Example 4.20. The closed loop frequency response magnitude versus frequency of a second order system is shown in fig. 4.45. Find %M_p (max^m. overshoot) and peak time.

Solution : From fig. $M_r = 1.4$ & $\omega_r = 3$ rad/sec.

since, $$M_r = \frac{1}{2\xi\sqrt{1-\xi^2}}$$

$$\therefore \quad 1.4 = \frac{1}{2\xi\sqrt{1-\xi^2}}$$

$$\xi^2(1-\xi^2) = 0.1275$$

$$\xi^2 - \xi^4 - 0.1275 = 0$$

or $$\xi^4 - \xi^2 + 0.1275 = 0$$

Put $x^2 = \xi$

$$x^2 - x + 0.1275 = 0$$

$$x = 0.8495, 0.1505$$

consider $x = 0.1505$ or $\xi^2 = 0.1505$ $\therefore$ $\boxed{\xi = 0.387}$

$$M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} \times 100 = e^{-\pi 0.387/\sqrt{1-0.387^2}} \times 100 = 26\% \quad \textbf{Ans.}$$

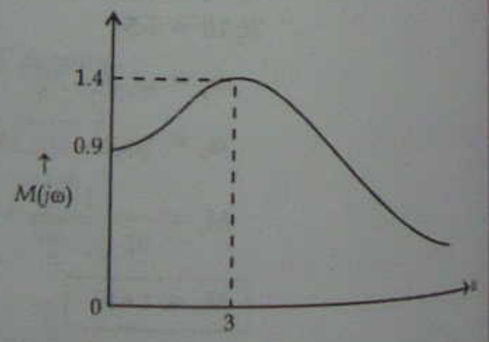


Fig. 4.45.

Peak time $$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$$

$$\omega_r = \omega_n\sqrt{1-2\xi^2}$$

$$3 = \omega_n\sqrt{1-2(0.387)^2}$$

or $$\omega_n = 3.58 \text{ rad/sec.}$$

$\therefore$ $$t_p = \frac{\pi}{3.58\sqrt{1-0.387^2}} = 0.95 \text{ sec.} \qquad \boxed{t_p = 0.95 \text{ sec.}} \quad \textbf{Ans.}$$

$\therefore$

4.17. FREQUENCY RESPONSE OF A CLOSED LOOP SYSTEM

Consider the transfer function

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

For unity feedback $H(s) = 1$

$\therefore$ $$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} \qquad \text{...(4.24)}$$

Put $s = j\omega$

$$\frac{C(j\omega)}{R(j\omega)} = \frac{G(j\omega)}{1+G(j\omega)} \qquad \text{...(4.25)}$$

The polar plot of eqn 4.25 is shown in fig. 4.46.

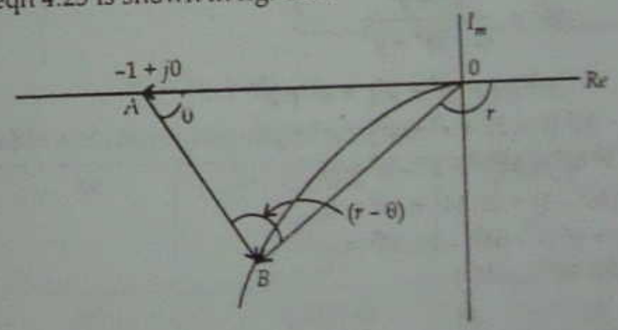


Fig. 4.46.

From Fig.4.45 $$\overrightarrow{OB} = G(j\omega)$$

$$\overrightarrow{OA} = -1$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = G(j\omega) - (-1)$$

$$\overrightarrow{AB} = 1 + G(j\omega)$$

From equation (4.25)

$$\left|\frac{C(j\omega)}{R(j\omega)}\right| = M(\omega) = \frac{\overrightarrow{OB}}{\overrightarrow{AB}}$$

$$\frac{C(j\omega)}{\angle R(j\omega)} = \frac{\angle \overrightarrow{OB}}{\angle \overrightarrow{AB}} = \frac{r}{\theta} = r - \theta$$

$$\therefore \quad \frac{C(j\omega)}{R(j\omega)} = M(\omega)\, e^{j\phi(\omega)}$$

Where $M(j\omega)$ is the magnitude and $\phi(\omega) = r - \theta$

Frequency response consists of two parts (1) magnitude (2) phase angle. Both can be plotted against defferent values of ω.

The constant magnitude loci and constant phase angle can be plotted in complex plane $G(j\omega)$ from these locii we can obtain the frequency response of closed loop system.

4.18. CONSTANT MAGNITUDE CIRCLE (M – CIRCLE)

Let $G(j\omega) = x + jy$

Then from eq^n (4.25)

$$M = M(\omega) = \frac{|C(j\omega)|}{|R(j\omega)|} = \left|\frac{G(j\omega)}{1+G(j\omega)}\right|$$

$$M = \frac{|x+jy|}{|1+x+jy|} = \frac{\sqrt{x^2+y^2}}{\sqrt{(1+x)^2+y^2}} \qquad ...(4.26)$$

Squaring both sides

$$M^2 = \frac{x^2+y^2}{(1+x)^2+y^2}$$

or

$$M^2[(1+x)^2+y^2] = x^2+y^2$$

$$M^2[1+2x+x^2+y^2] = x^2+y^2$$

$$M^2 + 2xM^2 + x^2M^2 + M^2y^2 - x^2 - y^2 = 0$$

$$x^2(M^2-1) + y^2(M^2-1) + 2xM^2 + M^2 = 0$$

or, $\quad x^2(1-M^2) + y^2(1-M^2) - 2xM^2 = M^2$

Divide both the sides by $(1-M^2)$

$$x^2 - y^2 - 2x\frac{M^2}{1-M^2} = \frac{M^2}{1-M^2}$$

$\left(\frac{M^2}{1-M^2}\right)^2$ add in both sides

$$x^2+y^2-2x\frac{M^2}{1-M^2}+\left(\frac{M^2}{1-M^2}\right)^2 = \frac{M^2}{1-M^2}+\left(\frac{M^2}{1-M^2}\right)^2$$

$$\left(x-\frac{M^2}{1-M^2}\right)^2 + (y-0)^2 = \frac{M^2}{1-M^2}+\frac{M^4}{(1-M^2)^2}$$

$$\left(x-\frac{M^2}{1-M^2}\right)^2 + (y-0)^2 = \frac{M^2}{(1-M^2)^2} \qquad ...(4.27)$$

Equation (4) is the equation of the circle with centre $\left(\frac{M^2}{1-M^2}, 0\right)$ and radius $\left(\frac{M}{1-M^2}\right)$

If $M = 1$ Then equation (4.26) becomes

$$(1+x)^2 + y^2 = x^2 + y^2$$

$$x = -\frac{1}{2}$$

or,

This is the equation of the straight line parallel to the y-axis and passing through $-(1/2, 0)$ in the $G(j\omega)$ plane.

The constant M locii for different value of M shown in fig 4.48. From fig. 4.48 it is clear that

(a) The locii are symmetrical with respect to $M = 1$

(b) The M-circles for $M > 1$ are on the left side of the line $M = 1$ and for $M < 1$ the constant M-circles are on right side of the line $M = 1$.

The intersection of $G(j\omega)$ plot (Nyquist plot) & constant M locii gives the value of magnitude M. The M-circle which is tangent to the $G(j\omega)$ plot will give the value of resonance peak (M_r) and resonance frequency ω_r (as shown in fig. 4.47).

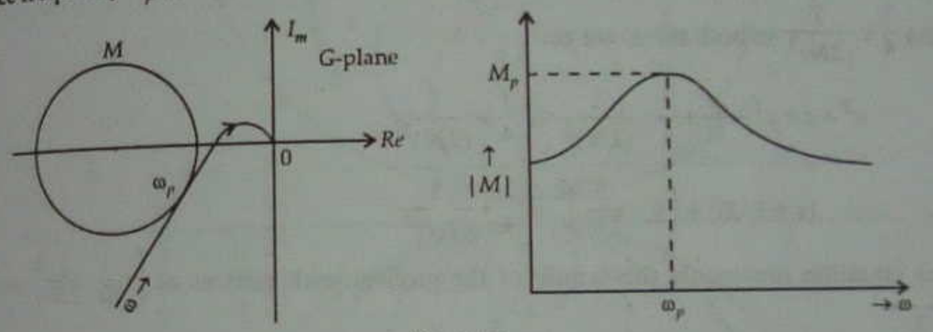


Fig. 4.47.

Table 4.10. For construction of M-circle

S.No.	M	Centre $\frac{M^2}{1-M^2}, 0$	Radius $\frac{M}{1-M^2}$
1.	0.3	(0.098,0)	0.329
2.	0.5	(0.33,0)	0.666
3.	0.67	(0.814, 0)	1.215
4.	0.766	(1.42, 0)	1.854
5.	0.833	(2.27, 0)	2.72
6.	1.2	(-3.27, 0)	-2.72
7.	1.3	(-2.45, 0)	- 1.88
8	1.5	(- 1.8, 0)	- 1.2
9	2.0	(-1.33, 0)	- 0.66
10	3.0	(-1.125, 0)	- 0.375

4.19. CONSTANT N-CIRCLES (PHASE ANGLE LOCII) :

From eq. (4.26), the phase shift can be written as

$$\phi = \arg \text{ of } (x + jy) - \arg (1 + x + jy)$$

$$\phi = \tan^{-1}\frac{y}{x} - \tan^{-1}\frac{y}{1+x}$$

or, $$\tan\phi = \frac{\frac{y}{x} - \frac{y}{1+x}}{1+\left(\frac{y}{x}\right)\left(\frac{y}{1+x}\right)}$$

$$\tan\phi = \frac{y}{x^2+x+y^2}$$

let $\tan\phi = N$

$$\therefore \quad N = \frac{y}{x^2+x+y}$$

or, $N(x^2+x+y^2) = y$

$$x^2+x+y^2-\frac{y}{N} = 0$$

Add $\frac{1}{4}+\frac{1}{(2N)^2}$ to both sides, we get

$$x^2+x+y^2-\frac{y}{N}+\frac{1}{4}+\frac{1}{(2N)^2} = \frac{1}{4}+\frac{1}{(2N)^2}$$

or, $$(x+1/2)^2+\left(y-\frac{1}{2N}\right)^2 = \frac{1}{4}+\frac{1}{(2N)^2}$$

This equation represents the family of the circles, with centres at $\left(-\frac{1}{2},\frac{1}{2N}\right)$ and ra

$$\sqrt{\frac{1}{4}+\frac{1}{4N^2}}$$

centre $(^{-1}/_{2}, ^{1}/_{2N})$

$$\text{Radius} = \sqrt{\frac{N^2+1}{4N^2}}$$

For different values of ϕ, N circles are shown in fig. 4.49
From the fig. 4.49 it is observed that
(a) The centre is lying always at a distance $x = -1/2$ and y depends upon the phase shift.
(b) All the circles passes through - 1 as well as 0.

Table 4.11

ϕ	$N = \tan\phi$	centre	Radius
30°	0.577	(–0.5, 0.866)	1.0
45°	1.0	(–0.5, 0.5)	0.7
60°	1.732	–0.5, 0.288	0.577
15°	0.267	0.5, 1.87	1.9
– 60°	– 1.732	– 0.5, – 0.288	0.577
– 45°	– 1.0	– 0.5, – 0.5	0.7
– 30°	– 0.577	–0.5, – 0.866	1.0

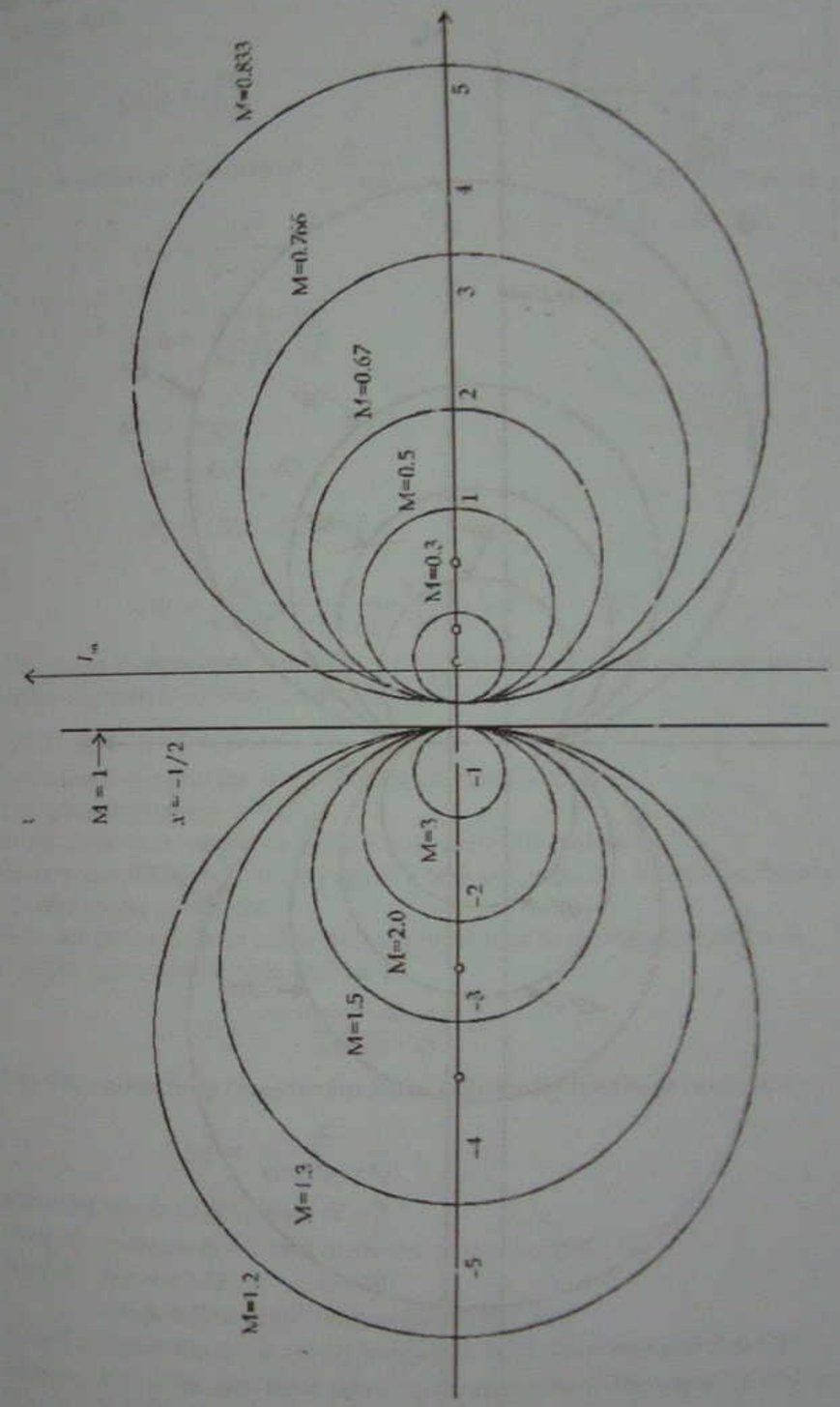


Fig. 4.48. Const. M-Circles

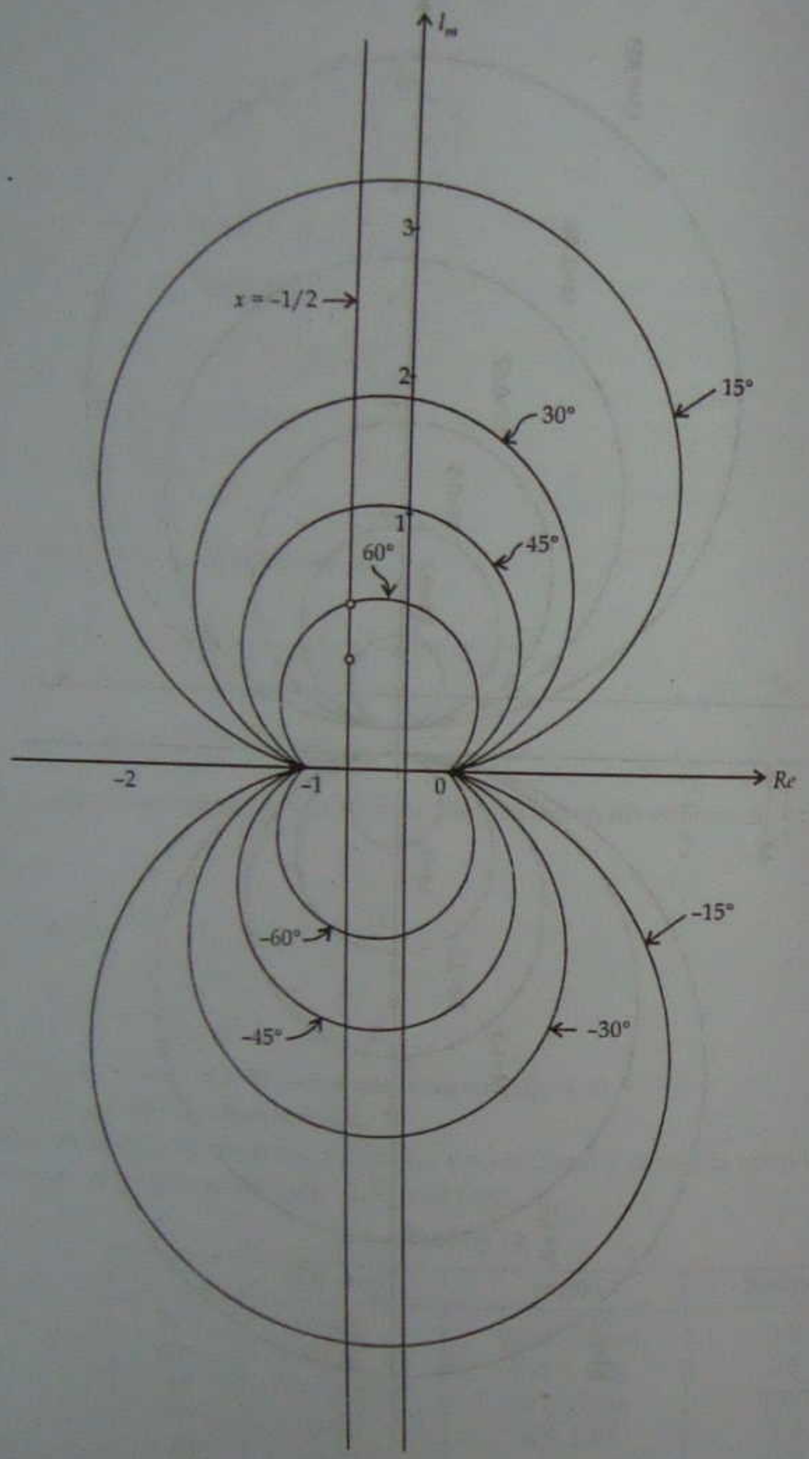


Fig. 4.49. Constt. N-Circles

4.20. GAIN ADJUSTMENT BY M-CIRCLE

Consider the fig. 4.50

From ΔAOC :

$$\sin\theta = \frac{AC}{OA}$$

Where AC = radius of the circle = $\dfrac{M}{1-M^2}$

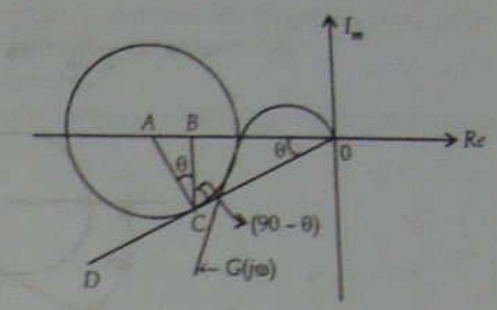


Fig. 4.50.

$$OA = \frac{M^2}{1-M^2}$$

$$\therefore \quad \sin\theta = \frac{M/1-M^2}{M^2/1-M^2}$$

$$\sin\theta = \frac{1}{M}$$

$$OB = OA - AB$$

$$OB = OA - AC\sin\theta \qquad \because \frac{AB}{AC} = Sin\,\theta$$

$$\therefore \quad OB = \frac{M^2}{1-M^2} - \frac{M}{1-M^2}\cdot\frac{1}{M} = -1$$

Thus, The point B represents the point $-1 + j0$ in the GH-plane and gives the factor by which the gain can be adjusted to obtain desired M.

PROCEDURE :

Step 1 : Draw the polar plot for $K = 1$ (or any assumed value)

Step 2 : Calculate θ from $\sin\theta = 1/M$

Step 3 : Draw a line from origin at angle θ with negative real axis

Step 4 : Draw a circle with centre on negative real axis, which is tangent to the line OD and to the polar plot.

Step 5 : From the point C draw a line BC perpendicular to the negative real axis.

Step 6 : Required gain can be obtained as

$$K = \frac{\text{assumed value of K}}{\text{length OB}}$$

Example 4.21. The open loop transfer function with unity feedback is given by

$$G(s) = \frac{K}{s(s+2)(s+4)}$$

Determine the value of K so that $M = 2$

Solution : Step 1 : Assume K = 1 and draw the polar plot (fig 4.51)

Step 2 : $\sin\theta = 1/2 \quad \therefore \theta = 30°$

Draw a line at 30° with negative real axis

Step 3 : Draw the circle that is tangent to both polar plot and line OD.

Step 4 : Draw the line from point c perpendicular to the negative real axis intersects at (– 0.06, 0)

Step 5 : Calculate K

$$K = \frac{1}{0.06} = 16.66 \approx 17$$

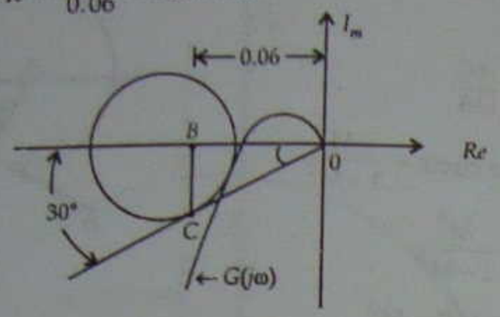


Fig. 4.51.

4.21. CLOSED LOOP FREQUENCY RESPONSE FROM M & N-CIRCLES

Frequency response of closed loop system can be obtained with the help of M and N circles. Conside the following example.

The open loop transfer function with unity feedback system is

$$G(j\omega) = \frac{10}{j\omega(1+j0.2\omega)(1+j0.05\omega)}$$

Step 1 : Draw the polar plot of the given transfer function.

Step 2 : Draw *M*-circles

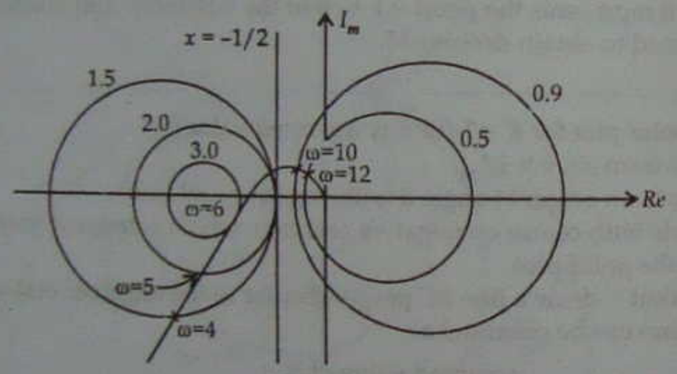


Fig. 4.52.

Step 3 : Draw N-circles

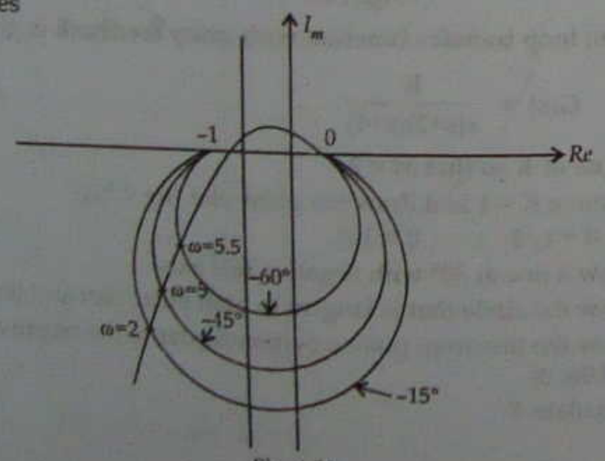


Fig. 4.53.

Step 4 : Point of intersection of M-circle gives the magnitude and read the frequency from polar plot. The circle which is tangent to the polar plot gives the valueof M_p.

Step 5 : Point of intersection of N-circle gives the phase angle.

Step 6 : With the help of above data plot the frequency response of closed loop system. (as shown in fig. 4.54).

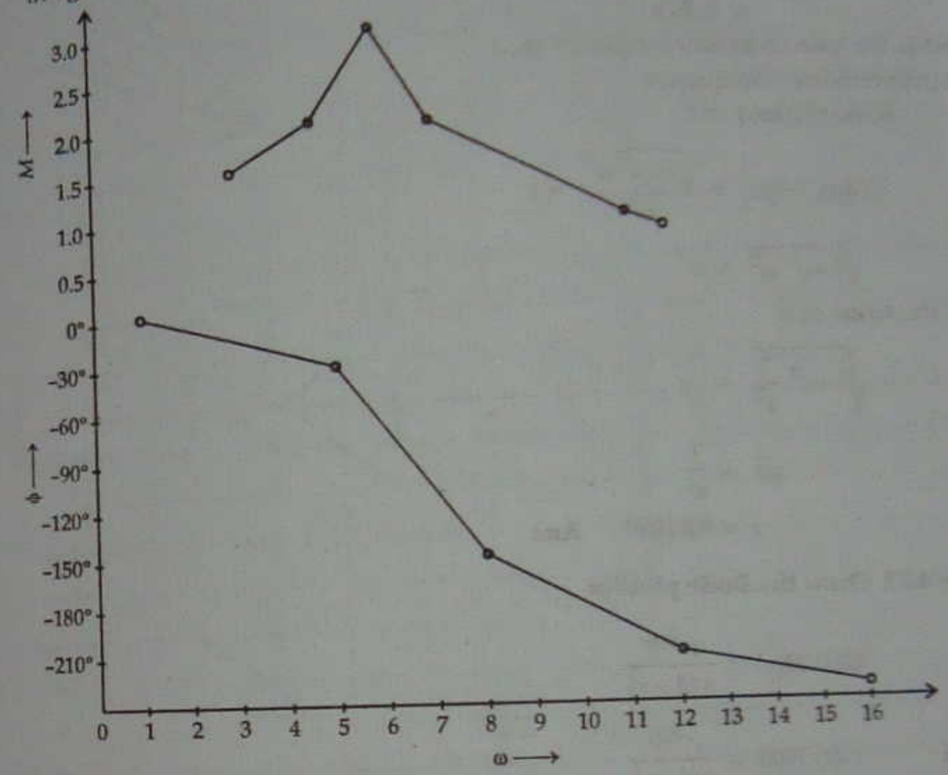


Fig. 4.54.

Example 4.22. Consider the unity feedback control system whose open loop transfer function is

$$G(s) = \frac{a\,s+1}{s^2}.$$

Determine the value of *'a'* so that the phase margin is 45°.

Solution : Given that $G(s)\,H(s) = \dfrac{1+a\,s}{s^2}$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{1+j\,a\omega}{(j\omega)^2}$$

$$\therefore \quad \angle G(j\omega)\,H(j\omega) = -180° + \tan^{-1} a\omega$$

We know that

$$\text{P.M} = 180° + \angle G(j\omega)\,H(j\omega)$$
$$= 180 + (-180° + \tan^{-1} a\,\omega) = \tan^{-1} a\,\omega$$

P.M = 45° (given)

$45° = \tan^{-1} a\omega$

taking tan on both sides

$\tan 45° = \tan(\tan^{-1} a\omega)$

$a\omega = 1$

$\omega = 1/a$

This is the gain crossover frequency ω_{c1}.

At gain crossover frequency

$|G(j\omega)\,H(j\omega)| = 1$

$$|G(j\omega)\,H(j\omega) = \frac{\sqrt{1+a^2\omega^2}}{\omega^2} = 1$$

$$\sqrt{1+a^2\omega^2} = \omega^2$$

Put the value of ω

$$\sqrt{1+a^2.\frac{1}{a^2}} = \frac{1}{a^2}$$

or, $$\sqrt{2} = \frac{1}{a^2}$$

$\therefore$ $a = 0.84089$ **Ans.**

Example 4.23. Draw the Bode plot for

$$G(s)\,H(s) = \frac{e^{-0.3s}}{s(1+s)}.$$

Solution : $$G(s)\,H(s) = \frac{e^{-0.3s}}{s(1+s)}$$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{e^{-j0.3\omega}}{j\omega(1+j\omega)}$$

Since, there is a pole at origin, the initial slope of magnitude plot will -20db/dec. $e^{-j0.3\omega}$ has no effect on magnitude pole.

The corner frequency $\omega = 1$ rad/sec.

From $\omega = 1$, the slope of next line will be $-20 + (-20) = -40$db/dec. as there is a simple pole.

PHASE PLOT : $\angle e^{-j0.3\omega} = -57.3 \times 3\omega = -17.19\omega$

$\angle G(j\omega)\,H(j\omega) = -90° - \tan^{-1}\omega - 17.19\omega$

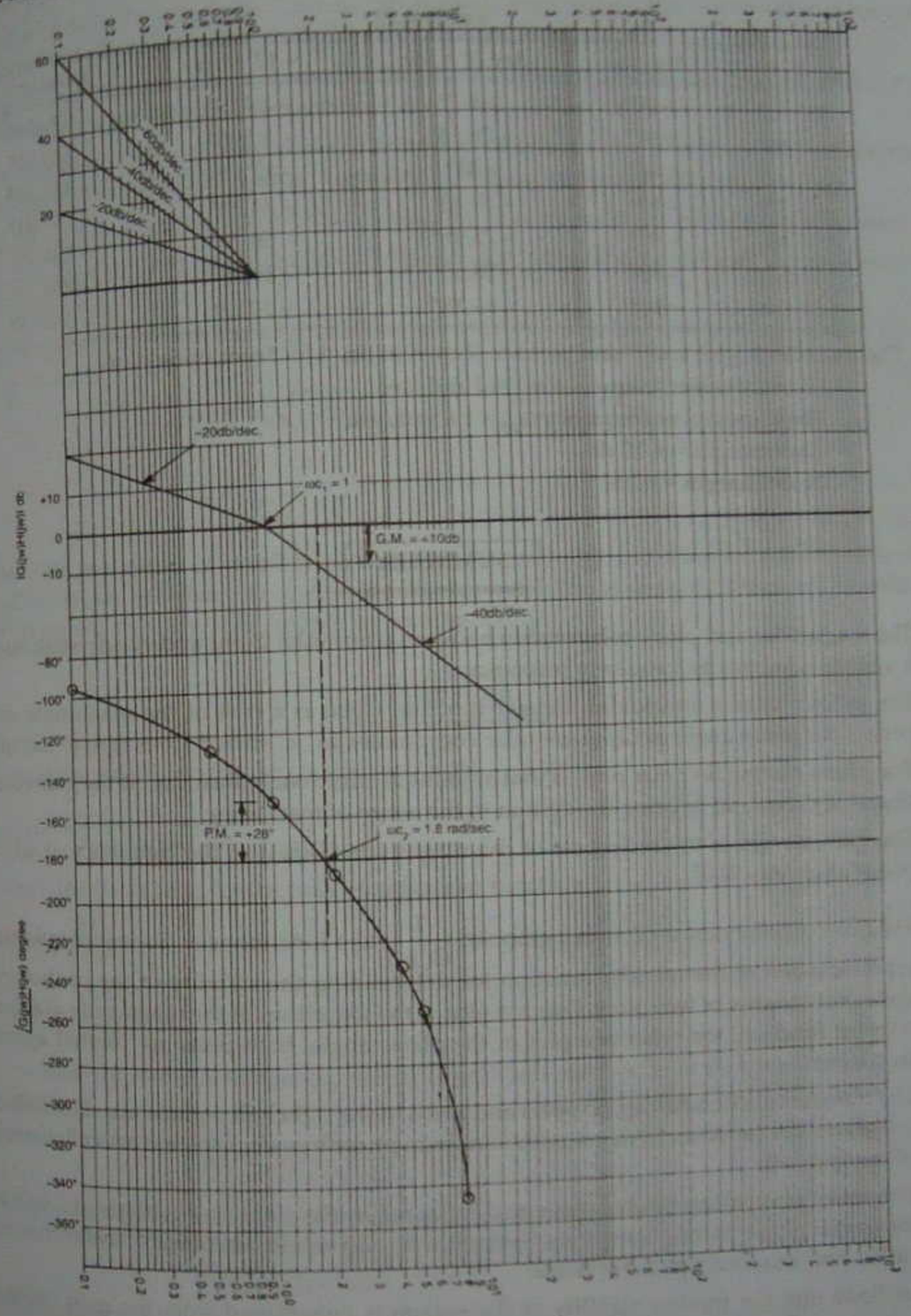


Fig. 4.55

Table 4.12.

ω	$Arg\ (j\omega)$	$Arg\ (1 + j\omega)$	$-17.19\ \omega$	Resultant
0.1	- 90°	- 5.7°	- 1.719°	- 97.419°
0.5	- 90°	- 26.56°	- 8.595°	- 125.15°
1.0	- 90°	- 45°	- 17.19°	- 152.19°
2.0	- 90°	- 63.43°	- 34.38°	- 187.81°
4.0	- 90°	- 75.96°	- 68.76°	- 234.72°
5.0	- 90°	- 78.69°	- 85.95°	- 254.64°
8.0	- 90°	- 84.28°	- 171.9	- 346.18°

The magnitude plot and phase plot are shown in fig. 4.55 From Bode plot:

Gain crossover frequency $\omega_{c1} = 1$ rad/sec.

Phase crossover frequency $\omega_{c2} = 1.8$ rad/sec.

Gain margin = + 10 db

Phase margin = + 28°

SUMMARY

1. The magnitude and phase relationship between sinusoidal input and steady state output of a system is known as frequency response.
2. The polar plot of a sinusoidal transfer function $G(j\omega)$ is a plot of the magnitude of $G(j\omega)$ versus the phase angle of $G(j\omega)$ on polar coordinates as 'ω' varied from zero to infinity.
3. The phase margin (in polar plot) is that amount of additional phase lag at the gain crossover frequency required to bring the system to the verge of instability.
4. The gain margin is the reciprocal of the magnitude $[G(j\omega)]$ at the frequency at which the phase angle as - 180°.
5. The inverse polar plot at $G(j\omega)$ is a graph of $\frac{1}{G(j\omega)}$ as a function of ω. Bode plot is a graphical representation of the transfer function for determining the stability of the control system. Bode plot consists of two plots. One is a plot of the logarithm of the magnitude of sinusoidal transfer function, the other is a plot of the phase angle, both plots are plotted against the frequency.
6. The transfer function having no poles and zeros in the right half s-plane are called minimum phase transfer function. Systems with minimum phase transfer function are called minimum phase systems.
7. The transfer function having poles and/or zeros in the right half s-plane are called non-minimum phase transfer functions. Systems with non-minimum phase transfer function are called non-minimum phase systems.
8. In Bode plot the relative stability of the system is determined from the gain margin and phase margin.
9. If gain cross-over frequency is less than phase cross over frequency then gain margin and phase margin both are positive and system is stable.

10. If gain cross over frequency is greater than the phase crossover frequency than both gain margin and phase margin are negative and the system is unstable.
11. If gain cross over frequency is equal to the phase cross-over frequency the gain margin and phase margin are zero and the system is marginally stable.
12. The maximum value of magnitude is known as resonant peak. The magnitude of resonant peak gives the information about the relative stability of the system.
13. The frequency at which magnitude has maximum value is known as resonant frequency.
14. Bandwidth is defined as the range of frequencies in which the magnitude of close loop does not drop -3db.
15. The frequency at which the magnitude is 3db below its zero frequency value is called cut off frequency.

$$\omega_r = \omega_n\sqrt{1-2\zeta^2}$$

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

16. There are two types of stability namely absolute stability and relative stability. Absolute stability means whether the system is stable or not. If the system is stable then we measure the degree of stability. It is known as relative stability.

EXERCISE

4.1. Sketch the polar plots for the transfer functions

(a) $G(s) = \frac{1}{s(1+s)(1+2s)}$ (b) $G(s) = \frac{10}{s(s+1)(s+4)}$

4.2. Sketch the Bode plots for the following transfer function

(a) $\frac{20s}{(s+1)(s+10)}$ (b) $\frac{1000}{s(1+0.1s)(1+0.005s)}$

(c) $\frac{1500}{s(s+2)(s+30)}$

4.3. Determine the resonant frequency, resonance peak and bandwidth for the system whose transfer function is

$$\frac{C(s)}{R(s)} = \frac{5}{s^2+2s+5}$$

4.4. The forward path transfer function of a unity feedback control system is

$$G(s) = \frac{21.39}{s(s+6.54)}$$

Determine the resonance peak, resonant frequency and normalized bandwidth.

4.5. Repeat the problem 4.19 for maximum overshoot 20%.

SEMI-OBJECTIVE TYPE QUESTIONS

(*i*) Define frequency response.
(*ii*) What is a polar plot.
(*iii*) Write the procedure to sketch the polar plot.
(*iv*) Define minimum phase systems and non-minimum phase systems.
(*v*) What is a bode plot?
(*vi*) Define resonant peak, resonant frequency and bandwidth.
(*vii*) Derive the expression for resonant frequency.
(*viii*) Define relative and absolute stability.
(*ix*) Construct the constant N-circle.
(*x*) Define Phase margin and gain margin.
(*xi*) Write a short note on correlation between time domain and frequency domain specifications.
(*xii*) Write a short note on frequency domain specifications.
(*xiii*) With a suitable diagram define (*a*) Phase crossover frequency, (*b*) gain crossover frequency (*c*) P.M. (*iv*) gain margin.
(*xiv*) What information you get from Bode's plot.
(*xv*) Why logarithmic scale is used while plotting the Bode plots?
(*xvi*) State the advantages of Bode plots.
(*xvii*) State the centre and radius of constant M-circle.
(*xviii*) What is bandwidth?
(*xix*) What is corner frequency?
(*xx*) What are the frequency domain specifications?

□□□

Chapter 5

Stability Theory

5.1. CONCEPT OF STABILITY

The concept of stability is very important to analyse and design the system. A system is said to be stable if its response cannot be made to increase indefinitely by the application of a bounded input excitation. If the output approaches towards infinite value for sufficiently large time, the system is said to be unstable.

A linear time invariant (LTI) system is stable if

1. the system is excited by a bounded input, the output is bounded. (BIBO stability criteria)
2. in the absence of the input, the output tends towards zero (the equilibrium state of the system). This is known as asymptotic stable.

Consider the transfer function

$$\frac{C(s)}{R(s)} = \frac{a_m s^m + a_{m-1}s^{m-1} + \ldots . a_1 s + a_0}{b_n s^n + b_{n-1}s^{n-1} + \ldots . + b_1 s + b_0} \quad \ldots(5.1)$$

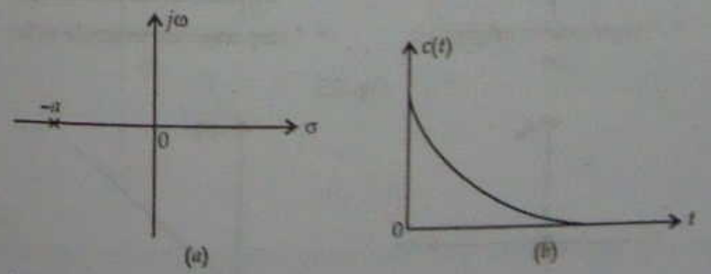
the output is given by

$$C(t) = \int_0^\infty g(\tau) r(t-\tau)\, d\tau \quad \ldots(5.2)$$

where $g(\tau) = \mathcal{L}^{-1} G(s)$ = impulse response of the system. So, a system is said to be stable if the impulse response approaches zero for sufficiently large time. If the impulse response approaches infinity for suffciently large time, the system is said to be unstable. If the impulse response approaches a constant value for sufficiently large time, the system is said to be marginally stable.

5.2. EFFECT OF LOCATION OF POLES ON STABILITY

(*a*) poles on negative real axis



simple pole on negative real axis *response*

Fig. 5.1.

Consider a simple pole at $s = -a$ as shown in fig. 5.1*a*, the corresponding impulse response is given by

$$g(t) = \mathcal{L}^{-1}G(s) = \mathcal{L}^{-1}\frac{K}{s+a} = K.e^{-at}$$

As the time 't' increases, the response approaches zero and the system is stable. The response is shown in fig 5.1(*b*). ...(5.)

(*b*) Pole on positive real axis

Consider a system having simple pole on positive real axis at $s = a$, the corresponding impulse response is given by

$$g(t) = \mathcal{L}^{-1}\frac{K}{s-a} = K.e^{at}$$ ...(5.6)

The response increases exponentially with time, hence the system is unstable. The simple pole and response are shown in fig 5.2 (*a*) and 5.2(*b*).

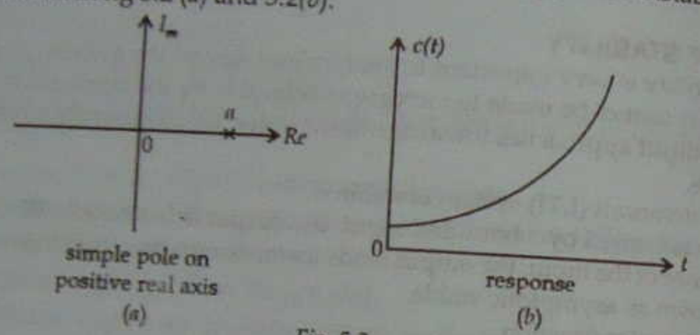


Fig. 5.2.

(*c*) Pole at the origin : Consider a pole at origin

$$\therefore \quad g(t) = \mathcal{L}^{-1}\frac{K}{s} = K$$ ...(5.)

This is constant value, hence the system is marginally stable. If there are two poles at the origin the time response would be

$$g(t) = \mathcal{L}^{-1}\frac{K}{s^2} = Kt$$ ...(5.8)

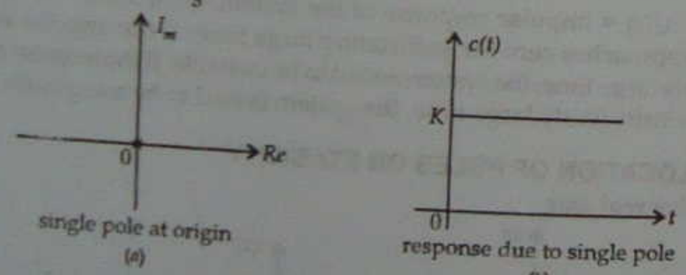


Fig. 5.3.

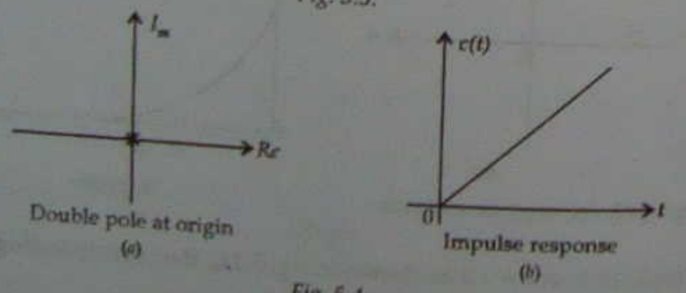


Fig. 5.4.

(*d*) Complex pole in the left half of *s*-plane

Let the transfer function has a complex conjugate poles at $s = -\alpha \pm j\omega$. The time response due to the complex conjugate poles is given by

$$g(t) = \mathcal{L}^{-1}\left[\frac{K}{s+\alpha-j\omega} \cdot \frac{K}{s+\alpha+j\omega}\right]$$

$$= \mathcal{L}^{-1}\left[\frac{2K(s+\alpha)}{(s+\alpha)^2+\omega^2}\right] = 2K\,e^{-\alpha t}\cos\omega t$$ ...(5.7)

When t increases $g(t)$ approaches zero and the system is stable. The complex poles and corresponding time response is shown in fig. [5.5(*a*)] and [5.5(*b*)] respectively.

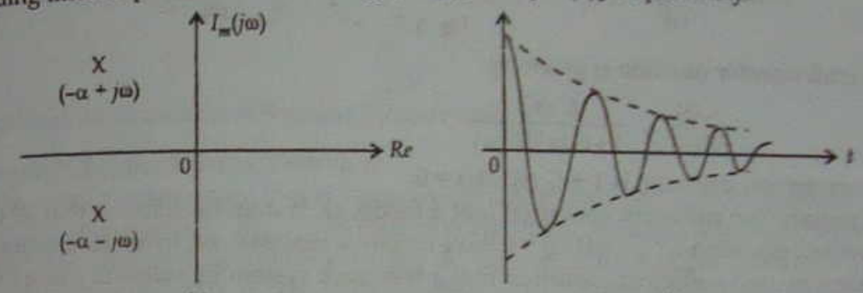


Fig. 5.5.

(*e*) Complex poles in the right half of *s*-plane

Suppose the system has complex conjugate poles at $s = \alpha \pm j\omega$. The time response is given by

$$g(t) = \mathcal{L}^{-1}\left[\frac{A}{s-\alpha-j\omega} + \frac{A}{s-\alpha+j\omega}\right]$$

$$= \mathcal{L}^{-1}\left[\frac{2A(s-\alpha)}{(s-\alpha)^2+\omega^2}\right] = 2A\,e^{\alpha t}\cos\omega t$$ ...(5.8)

Hence, the response increases exponentially sinusoid with time and therefore the response is unstable. The poles and time response shown in fig [5.6(*a*)] and [5.6(*b*)] respectively.

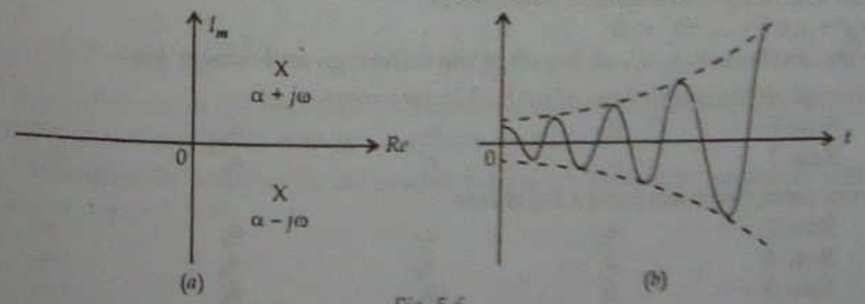


Fig. 5.6.

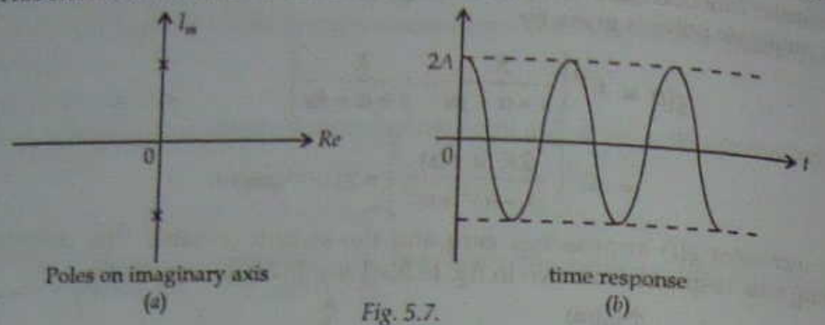
(*f*) Poles on $j\omega$-axis

If the system having the complex poles on $j\omega$-axis the corresponding time response would be

$$g(t) = \mathcal{L}^{-1}\left[\frac{A}{s+j\omega} + \frac{A}{s-j\omega}\right]$$

$$= \mathcal{L}^{-1}\left[\frac{2As}{s^2+\omega^2}\right] = 2A\cos\omega t$$ ...(5.9)

The response is marginally stable. The equation (5.9) shows the sustained oscillations of constant amplitude. This situation will also be considered unstable.



Poles on imaginary axis (a) time response (b)

Fig. 5.7.

The overall transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} \quad ...(5.10)$$

The characteristic equation is $1 + G(s)H(s) = 0$ ...(5.11)

The necessary and sufficient condition that a feedback system be stable is that all the zeros of the characteristic equation $1 + G(s)H(s) = 0$ have negative real part. or, in terms of poles we can say that the necessary and sufficient condition that a feedback system be stable is that all the poles of overall transfer function have negative real part.

5.3. NECESSARY BUT NOT SUFFICIENT CONDITIONS FOR STABILITY

Consider a system with characteristicequation

$$a_0 s^n + a_1 s^{n-1} + ...+ a_n = 0 \quad ...(5.12)$$

(a) All the coefficients of the equation should have same sign,

(b) There should be no missing term.

If above two conditions are not satisfied the system will be unstable. But if all the coefficients have same sign and there is no missing term we have no guarantee that the system will be stable.for stability we use ROUTH-HURWITZ CRITERION.

5.4. THE ROUTH-HURWITZ CRITERION

Consider the following characteristic polynomial

$$a_0 s^n + a_1 s^{n-1} + ...+ a_n = 0 \quad ...(5.13)$$

where the coefficients $a_0, a_1 ....a_n$ are all of the same sign and none is zero.

Step 1 : Arrange all the coefficients of eq^n 5.13 in two rows

Row 1	a_0	a_2	a_4	.....
Row 2	a_1	a_3	a_5	.....

Step 2 : From these two rows form a third row

Row 1	a_0	a_2	a_4	.....
Row 2	a_1	a_3	a_5	.....
Row 3	b_1	b_3	b_5	.....

Where,

$$b_1 = -\frac{1}{a_1}\begin{vmatrix} a_0 & a_2 \\ a_1 & a_3 \end{vmatrix}$$

$$b_3 = -\frac{1}{a_1}\begin{vmatrix} a_0 & a_4 \\ a_1 & a_5 \end{vmatrix}$$

Step 3 : From second and third row, form a fourth row

Row 1	a_0	a_2	a_4	.....
Row 2	a_1	a_3	a_5	.....
Row 3	b_1	b_3	b_5	.....
Row 4	c_1	c_3	c_5	.....

where,

$$c_1 = -\frac{1}{b_1}\begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix}$$

$$c_3 = -\frac{1}{b_1}\begin{vmatrix} a_1 & a_5 \\ b_1 & b_5 \end{vmatrix}$$

Step 4 : Continue this procedure of forming a new rows

5.4.1. Statement of Routh-Hurwitz Criterion

Routh-Hurwitz Criterion states that the system is stable if and only if all the elements in the first column have the same algebaric sign. If all elements are not of the same sign then the number of sign changes of the elements in first column equals the number of roots of the characteristic equation in the right half of the s-plane (or equals to the number of roots with positive real parts)

Example 5.1. Check the stability of the system whose characteristic equation is given by

$$s^4 + 2s^3 + 6s^2 + 4s + 1 = 0$$

Solution : Obtain the array of coefficients as follows

s^4	1	6	1
s^3	2	4	
s^2	4	1	
s^1	3.5		
s^0	1		

$$b_1 = -\frac{1}{2}\begin{vmatrix} 1 & 6 \\ 2 & 4 \end{vmatrix} = 4, \qquad c_1 = -\frac{1}{4}\begin{vmatrix} 2 & 4 \\ 4 & 1 \end{vmatrix} = 3.5$$

$$b_2 = -\frac{1}{2}\begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = 1, \qquad d_1 = -\frac{1}{3.5}\begin{vmatrix} 4 & 1 \\ 3.5 & 0 \end{vmatrix} = 1$$

Since, all the coefficients in the first column are of the same sign (positive), the given equation has no roots with positive real parts. Hence, the system is stable.

Example 5.2. Determine the stability of the system whose characteristic equation is given by

$$2s^4 + 2s^3 + s^2 + 3s + 2 = 0$$

Solution :

s^4	2	1	2
s^3	2	3	
s^2	-2	2	
s^1	5		
s^0	2		

$$b_1 = -\frac{1}{2}\begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} = -2, \qquad c_1 = -\frac{1}{(-2)}\begin{vmatrix} 2 & 3 \\ -2 & 2 \end{vmatrix} = 5$$

$$b_2 = -\frac{1}{2}\begin{vmatrix} 2 & 2 \\ 2 & 0 \end{vmatrix} = 2,$$

$$d_1 = -\frac{1}{5}\begin{vmatrix} -2 & 2 \\ 5 & 0 \end{vmatrix} = 2$$

There are two changes of sign in the first column(from 2 to –2 and from –2 to 5), hence there are two roots in the right half of s-plane. The system is unstable.

Example 5.3. Determine the stability of the system having following characteristic equation

$$2s^4 + 5s^3 + 5s^2 + 2s + 1 = 0$$

Solution :

s^4	2	5	1
s^3	5	2	
s^2	4.2	1	
s^1	0.809		
s^0	1		

From the above Routh table :
No. of sign changes in first column = 0
No. of poles on the right hand side of s-plane = 0
Hence, the system is stable.

Example 5.4. Check the stability of the system having following characteristic equation.

$$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$$

Solution :

s^4	1	3	5
s^3	2	4	
s^2	1	5	
s^1	–6		
s^0	5		

$$b_1 = -\frac{1}{2}\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} = 1, \qquad b_2 = -\frac{1}{2}\begin{vmatrix} 1 & 5 \\ 2 & 0 \end{vmatrix} = 5$$

$$c_1 = -\frac{1}{1}\begin{vmatrix} 2 & 4 \\ 1 & 5 \end{vmatrix} = -6, \qquad d_1 = -\frac{1}{(-6)}\begin{vmatrix} 1 & 5 \\ -6 & 0 \end{vmatrix} = 5$$

From above table :
No. of sign changes in first column = 2
No. of roots in right half of s-plane = 2
Hence, the system is unstable.

Example 5.5. A closed loop control system has the characteristic equation given by

$$s^3 + 4.5s^2 + 3.5s + 1.5 = 0$$

Investigate the stability using Routh Hurwitz criterion. *(R.M.L. University Faizabad 2003)*

Solution :

s^3	1	3.5
s^2	4.5	1.5
s^1	3.17	
s^0	1.5	

No. of sign changes in first column = 0
No. of roots in right half of s-plane = 0
Hence, system is stable.

Example 5.6. Check the stability of the system, having following characteristic equation.

$$s^5 + 6s^4 + 3s^3 + 2s^2 + s + 1 = 0$$

Solution:

s^5	1	3	1
s^4	6	2	1
s^3	2.67	0.83	
s^2	0.135	1	
s^1	–18.95		
s^0	1		

No. of sign change in first column = 2
No. of poles on right half of s-plane = 2
Hence, system is unstable.

SPECIAL CASES

Case 1 : If a first column term in any row is zero, but the remaining terms are not zero or there is no remaining term, then multiply the original equation by a factor $(s + a)$ where 'a' is any positive real number. The simplest value of 'a' is 1 (take $a = 1$). Consider the following ecample.

Example 5.7. Investigate the stability

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

Solution :

s^5	1	2	3
s^4	1	2	5
s^3	0		
s^2			
s^1			
s^0			

Now, multiply the characteristic equation by $(s + 1)$

$(s+1)(s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5) = 0$

or, $s^6 + 2s^5 + 3s^4 + 4s^3 + 5s^2 + 8s + 5 = 0$

s^6	1	3	5	5
s^5	2	4	8	
s^4	1	1	5	
s^3	2	–2		
s^2	2	5		
s^1	–7			
s^0	1			

From the above table :
No. of sign change in the first column = 2
No. of roots in the right half of s-plane = 2
Hence, system is unstable.

Example 5.8. Apply Routh Hurwitz criterion to the following equation and investigate the stability.

$$s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10 = 0$$

Solution :

s^5	1	2	11
s^4	2	4	10
s^3	0		
s^2			
s^1			
s^0			

Since, the third element in first column is zero, so multiply the equation by $(s + 1)$

$$(s+1)(s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10) = 0$$

or, $s^6 + 3s^5 + 4s^4 + 6s^3 + 15s^2 + 21s + 10 = 0$

s^6	1	4	15	10
s^5	3	6	21	
s^4	2	8	10	
s^3	-6	6		
s^2	10	10		
s^1	12			
s^0	1			

From the above table it is clear that there are two changes of sign in first column, therefore there are two roots in the right half of s-plane. The system is unstable.

ALTERNATIVE METHOD

Replace the zero by a small positive quantity ε and continue the procedure. Consider the following example.

Example 5.9. Consider the following equation

$s^3 + s + 2 = 0$

Solution :

s^3	1	1
s^2	0	2
s^1		
s^0		

Replace 0 by ε and continue the procedure.

s^3	1	1
s^2	ε	2
s^1	$-\frac{1}{\varepsilon}(2-\varepsilon)$	-
s^0	2	

No. of sign changes in first column = 2

No. of roots in right of s-plane = 2

Case 2 : When any one row of Routh table is zero.

When any one row of Routh table is zero, it indicates that the equation has at least one pair of roots which lie radially opposite each other and equidistant from origin. The array can be completed by forming the auxiliary polynomial. The polynomial whose coefficients are the element of the row just above the row of zeros in Routh array is called an auxiliary polynomial. Consider the following example

Example 5.10. Consider the equation

$s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50 = 0$

Solution :

s^5	1	24	- 25
s^4	2	48	- 50
s^3	0	0	
s^2			
s^1			
s^0			

From above table it is clear that the third row is zero. Hence form the Auxiliary polynomial $A(s)$

$$A(s) = 2s^4 + 48s^2 - 50$$

$$\frac{dA(s)}{ds} = 8s^3 + 96s$$

Now the Routh array can be written as

s^5	1	24	- 25
s^4	2	48	-50
s^3	8	96	
s^2	24	- 50	
s^1	112.6		
s^0	- 50		

No. of sign changes in first column = 1

No. of roots in right half of s-plane = 1

The roots of the equation formed by the auxiliary polynomial

$$2s^4 + 48s^2 - 50 = 0$$

are also the roots of the original equation.

The given equation can be written in factored form as

$$(s+1)(s-1)(s+j5)(s-j5)(s+2) = 0$$

It is clear that the original equation has one root with positive real part.

The roots of the auxiliary equation are also the roots of the characteristic equation because auxiliary equation is the part of characteristic equation. Suppose, original characteristic equation is of order six i.e., having six roots and auxiliary equation is of order four *i.e.*, having four roots. These four roots are also the roots of the original characteristic equation and remaining (6 – 4 = 2) two roots will always lie to the left half of s-plane and these two roots do not have significant role for stability because they lie far away from imaginary axis. The roots of the auxiliary equations are dominant roots*, the stability can also be determined from the roots of the auxiliary equation.

Consider the following characteristic equation :

$S^6 + 2S^5 + 8S^4 + 12S^3 + 20S^2 + 16S + 16 = 0$

S^6	1	8	20	16
S^5	2	12	16	0
S^4	2	12	16	0
S^3	0	0	0	0
S^2				
S^1				
S^0				

Auxiliary equation $A(s) = 2S^4 + 12S^2 + 16 = 0$, $\frac{dA(s)}{ds} = 8S^3 + 24S$. Since, the order of the characteristic equation is six and order of auxiliary equation is four, the two roots (6 – 4 = 2) lie to the left half of s-plane and four roots of auxiliary equations are dominant roots.

S^6	1	8	20	16
S^5	2	12	16	0
S^4	2	12	16	0
S^3	8	24	0	0
S^2	6	16	0	
S^1	2.67	0		
S^0	16			

No sign change in first column, the system may be stable, since there is a row of zeros it means the roots are lying on imaginary axis. So solve the auxiliary equation $A(s) = 0$.

$$2S^4 + 12S^2 + 16 = 0$$

* dominant roots means the roots are close to the imaginary axis or on the imaginary axis or they are in right half of s-plane

put $S^2 = x$

$\therefore \quad 2x^2 + 12x + 16 = 0$

or, $\quad x^2 + 6x + 8 = 0$

$(x+2)(x+4) = 0$

$x + 2 = 0, \qquad x + 4 = 0$

$x = -2, \qquad x = -4$

$S^2 = -2, \qquad S^2 = -4$

$S = \pm j\sqrt{2}, \qquad S = \pm j2$

Since, the roots are nonrepeated on imaginary axis, hence the system is marginally stable

No. of roots with positive real part = 0

No. of roots with negative real part = 2

No. of roots with zero real part = 4

Important Note : If Sign changes in first column, system is unstable. The number of roots with positive real part of characteristics equation will be equal to the number of sign changes i.e. the roots are located in right half of s-plane. But if no sign changes in first column the system may be stable. In such case stability can be determined by solving auxiliary equation for roots & from the location of roots, stability can be determined.

5.5. APPLICATION OF ROUTH'S STABILITY CRITERION TO CONTROL SYSTEM ANALYSIS

Routh stability criterion is also used for the determination of stability of the linear feedback system. Consider the following example.

Example 5.11. The open loop transfer function of unity feedback system is $\frac{K}{s(1+0.4s)(1+0.25s)}$ Find the restriction of K so that the closed loop system is absolutely stable.

(R.M.L. University Faizabad 200

Solution : Given that

$$G(s) = \frac{K}{s(1+0.4s)(1+0.25s)}$$

$$H(s) = 1$$

The characteristic equation $1 + G(s)H(s) = 0$

$$1 + \frac{K}{s(1+0.4s)(1+0.25s)} = 0$$

or, $\quad s(1+0.4s)(1+0.25s) + K = 0$

or, $\quad s^3 + 6.5s^2 + 10s + 10K = 0$

s^3	1	10
s^2	6.5	10K
s^1	$\frac{65-10K}{6.5}$	
s^0	10K	

For absolute stability, there should be no sign change in the first column i.e no root of the characteristic equation should lie in right half of s-plane. This is possible only when

$K > 0$ and $65 - 10K > 0$

Hence, for closed loop stability

$0 < K < 6.5$

5.6. RELATIVE STABILITY ANALYSIS

Routh stability criterion gives the information about the absolute stability. But we are more interested for the relative stability of the system. Relative stability can be examined by shifting the s-plane and then apply Routh stability criterion. The characteristic equation is modified by shifting the origin of s-plane to $s_1 = -\sigma$ by the substitution

$$s = z - s_1$$

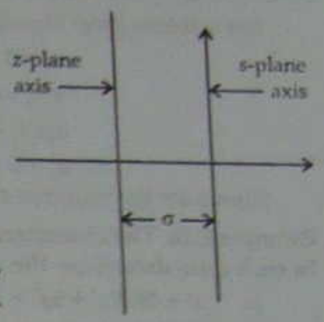


Fig. 5.8.

Now apply the Routh stability criterion, the number of sign change in the first column is equal to the number of roots to the right of the vertical line $s_1 = -\sigma$ or in other words the roots of the original characteristic equation are more negative than $-\sigma_1$.

Example 5.12. Check whether all the roots of the equation $s^3 + 7s^2 + 25s + 39 = 0$ have real parts more negative than – 1.

Solution :

s^3	1	25
s^2	7	39
s^1	19.42	
s^0	39	

Since, there is no sign change in the first column, it means all the roots are in the left half of s plane.

Put $s = z - 1$ in the characteristic equation

$(z-1)^3 + 7(z-1)^2 + 25(z-1) + 39 = 0$

or, $\quad z^3 + 4z^2 + 14z + 20 = 0$

z^3	1	14
z^2	4	20
z^1	9	
z^0	20	

Since, no sign change in the first column, the roots of the characteristic equation lie in the left of z-plane, it means all the roots of the original equation in s-domain lie to the left of $s = -1$.

ILLUSTRATIVE EXAMPLES

Example 5.13. The characteristic equation of a servo system is given by

$$a_0 s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4 = 0$$

Determine the conditions which must be satisfied by the coefficient of the characteristic equation for the system to be stable.

Solution :

s^4	a_0	a_2	a_4
s^3	a_1	a_3	
s^2	$\frac{a_1a_2 - a_0a_3}{a_1}$	$\frac{a_4a_1}{a_1}$	
s^1	$\frac{a_3\left[\frac{a_1a_2 - a_0a_3}{a_1}\right] - \frac{a_1^2a_4}{a_1}}{\frac{a_1a_2 - a_0a_3}{a_1}}$		
s^0	a_4		

For stability there should not be any sign change in first column.

$$\therefore \quad a_1 > 0$$
$$a_1 a_2 - a_0 a_3 > 0$$
$$a_1 a_2 a_3 - a_0 a_3^2 - a_1^2 a_4 > 0$$
$$a_4 > 0$$

Above are the required conditions.

Example 5.14. The characteristic equations for certain feedback control system are given below. In each case, determine the range of values K for the system to be stable.

a. $s^4 + 20Ks^3 + 5s^2 + 10s + 15 = 0$

b. $s^3 + 2Ks^2 + (K + 2)s + 4 = 0$

Solution : a. Routh array is

s^4	1	5	15
s^3	20K	10	
s^2	$\frac{100K - 10}{20K}$	15	
s^1	$10 - \frac{600K^2}{10K - 1}$		
s^0	15		

For stability no root should lie in right half of *s*-plane.

a. $20K > 0 \quad i.e \quad K > 0$

b. $\frac{100K - 10}{20K} > 0$ or, $5 - \frac{1}{2K} > 0 \quad i.e \quad 5 > \frac{1}{2K} \quad$ or $\quad K > \frac{1}{10}$

c. $10 - \frac{600K^2}{10K - 1} > 0$

i.e. $-600K^2 + 100K - 10 > 0$

This gives the complex roots, hence the system is unstable.

(b) $s^3 + 2Ks^2 + (K + 2)s + 4 = 0$

s^3	1	$K + 2$
s^2	$2K$	4
s^1	$\frac{K^2 + 2K - 2}{K}$	
s^0	4	

For stability

$2K > 0 \quad i.e \quad K > 0$

$\frac{K^2 + 2K - 2}{K} > 0 \quad i.e \quad K^2 + 2K - 2 > 0$

$K > 0.73$

Example 5.15. A system oscillates with frequency ω, if it has poles at $s = \pm j\omega$ and no poles in the right half s-plane. Determine the values of *K* and *a* so that the system shown in fig. 5.9 oscillates at a frequency 2 rad/sec.

Solution : The characteristic equation

$$1 + G(s)\,H(s) = 0$$

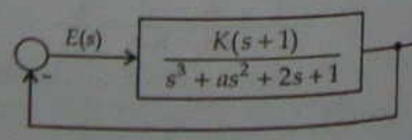


Fig. 5.9.

$$1 + \frac{K(s+1)}{s^3 + as^2 + 2s + 1} \cdot 1 = 0$$

or, $s^3 + as^2 + 2s + 1 + Ks + K = 0$

or, $s^3 + as^2 + (K + 2)s + (K + 1) = 0$

s^3	1	$K + 2$
s^2	a	$K + 1$
s^1	$K + 2 - \frac{(K+1)}{a}$	
s^0	$K + 1$	

The system will have sustained oscillation when

$$K + 2 - \frac{(K+1)}{a} = 0$$

or $\quad a(K + 2) - (K + 1) = 0 \quad$ or $\quad a = \frac{K+1}{K+2}$

The auxiliary equation of row 2

$$as^2 + (K + 1) = 0$$

$$s^2 = -\frac{K+1}{a}$$

Put $\quad s = j\omega$

$$\therefore \quad (j\omega)^2 = -\frac{K+1}{a}$$

$\omega = 2$ rad/sec. (given)

$$(j2)^2 = -\frac{K+1}{a}$$

$$\therefore \quad a = \frac{K+1}{4}$$

Since, $\quad a = \frac{K+1}{K+2}$

$$\therefore \quad \frac{K+1}{4} = \frac{K+1}{K+2}$$

$$\therefore \quad \boxed{K = 2}$$

$$\therefore \quad a = \frac{K+1}{4} = \frac{2+1}{4} = \frac{3}{4} = 0.75$$

$$\therefore \quad \boxed{a = 0.75} \quad \textbf{Ans.}$$

Example 5.16. The open loop transfer function of a unity feedback control system is given by

$$G(s) = \frac{K}{(s+2)(s+4)(s^2 + 6s + 25)}$$

By applying Routh criterion, discuss the stability of the closed loop system as a function of *K*. Determine the value of *K* which will cause sustained oscillations in the closed loop system. What are the corresponding oscillation frequencies ?

Solution : The characteristic equation is given by

$$1 + G(s)\,H(s) = 0$$

$$1 + \frac{K}{(s+2)(s+4)(s^2+6s+25)} \cdot 1 = 0$$

or, $s^4 + 12s^3 + 69s^2 + 198s + (200 + K) = 0$

s^4	1	69	$200 + K$
s^3	12	198	
s^2	52.5	$200 + K$	
s^1	$198 - \frac{12(200+K)}{52.5}$		
s^0	$200 + K$		

system will be stable when

$$200 + K > 0 \quad \text{or} \quad K > -200$$

$$198 - \frac{12(200+K)}{52.5} > 0 \quad \text{or} \quad K < 666.25$$

oscillations will occur when $K = 666.25$

Auxiliary equation : $52.5s^2 + (200 + K) = 0$

$$52.5s^2 + (200 + 666.25) = 0$$

$$52.5s^2 = -866.25$$

$$s^2 = -16.5$$

$$s = \pm j4.06 \text{ or } j\omega = \pm j4.06$$

∴ Frequency of sustained oscillation = 4.06 rad/sec. **Ans.**

Example 5.17. A unity feedback control system is characterized by open loop transfer function

$$G(s) = \frac{K(s+13)}{s(s+3)(s+7)}$$

(a) Using Routh criterion, calculate the range of values of K for the system to be stable.

(b) Check if for $K = 1$, all these roots of the characteristic equation of the abovem system have damping factor greater than 0.5

Solution : The characteristic equation is given by

$$1 + G(s)\,H(s) = 0$$

$$1 + \frac{K(s+13)}{s(s+3)(s+7)} = 0$$

or, $s(s+3)(s+7) + K(s+13) = 0$

or, $s^3 + 10s^2 + (21 + K)s + 13K = 0$

s^3	1	$21 + K$
s^2	10	$13K$
s^1	$21 - \frac{3K}{10}$	
s^0	$13K$	

(a) For stability

$$13K > 0 \quad \text{or} \quad K > 0$$

$$21 - \frac{3K}{10} > 0 \quad \text{or} \quad K < 70$$

∴ Range is $\boxed{0 < K < 70}$

∴ For stability of the system the range of K is $0 < K < 70$

(b) When $K = 1$, the characteristic equation becomes

$$s^3 + 10s^2 + 22s + 13 = 0$$

or, $(s + 7.2)(s^2 + 2.8s + 1.84) = 0$

∴ The roots are $s = -7.2, -1.745, -1.0535$

Since, all the roots are real and negative. Therefore all the roots lie on negative real axis.

$$\text{damping factor} = \cos\theta$$

Since all the roots are lie on negative real axis ∴ $\theta = 180°$

$$\text{damping ratio} = |\cos 180°| = 1$$

∴ Hence for all roots damping ratio is greater than 0.5.

or $s^3 + 10s^2 + 22s + 13 = 0$;

$(s + 1)(s^2 + 9s + 13) = 0$; $\xi = 1.248$

Hence, for all roots ξ is greater than 0.5

Example 5.18. A feedback system has an open loop transfer function

$$G(s)\,H(s) = \frac{Ke^{-s}}{s(s^2+5s+9)}$$

Determine by use of Routh criterion, the maximum value of K for the closed loop system to be stable.

Solution : For low frequencies $e^{-s} = (1 - s)$

$$\therefore \quad G(s)\,H(s) = \frac{K(1-s)}{s(s^2+5s+9)}$$

Characteristic equation is given by $1 + G(s)\,H(s) = 0$

$$1 + \frac{K(1-s)}{s(s^2+5s+9)} = 0$$

$$s(s^2 + 5s + 9) + K - Ks = 0$$

$$s^3 + 5s^2 + 9s + K - Ks = 0$$

$$s^3 + 5s^2 + (9 - K)s + K = 0$$

s^3	1	$9-K$
s^2	5	K
s^1	$9 - \frac{6K}{5}$	
s^0	K	

For stability $K > 0$

$$9 - \frac{6K}{5} > 0 \quad \text{or} \quad K < 7.5$$

Hence, the range of K is $\boxed{0 < K < 7.5}$ **Ans.**

Example 5.19. By means of Routh criterion, determine the stability of the systems represented by the following characteristic equations. For systems found to be unstable, determine the unmber of roots of the characteristic equation in the right half s-plane.

(a) $s^4 + 2s^3 + 8s^2 + 4s + 3 = 0$

(b) $s^4 + 2s^3 + s^2 + 4s + 2 = 0$

(c) $s^5 + 3s^5 + 5s^4 + 9s^3 + 8s^2 + 6s + 4 = 0$

Solution : (a) $s^4 + 2s^3 + 8s^2 + 4s + 3 = 0$

s^4	1	8	3
s^3	2	4	
s^2	6	3	
s^1	3		
s^0	3		

No. of sign change in first column = 0
No. of roots in right half of s-plane = 0
Hence system is stable.

(b) $s^4 + 2s^3 + s^2 + 4s + 2 = 0$

s^4	1	1	2
s^3	2	4	
s^2	-1	2	
s^1	8		
s^0	2		

No. of sign change in first column = 2
No. of roots in right half of s-plane = 2
Hence system is unstable.

(c) $s^6 + 3s^5 + 5s^4 + 9s^3 + 8s^2 + 6s + 4 = 0$

s^6	1	5	8	4
s^5	3	9	6	
s^4	2	6	4	
s^3	0	0		
s^2				
s^1				
s^0				

The elements of fourth row are zero, hence form the auxiliary eqn

$$A(s) = 2s^4 + 6s^2 + 4$$

$$\frac{dA(s)}{ds} = 8s^3 + 12s$$

s^6	1	5	8	4
s^5	3	9	6	
s^4	2	6	4	
s^3	8	12		
s^2	3	4		
s^1	4/3			
s^0	4			

Since, no sign change in the first column, hence system is stable
The roots of auxiliary eq^n are

$$s = \pm j1$$
$$s = \pm j\sqrt{2}$$

∴ Four roots lie on imaginary axis.

Example 5.20. Determine the range of values of K (K > 0) such that the characteristic equation $s^3 + 3(K+1)s^2 + (7K+5)s + (4K+7) = 0$ has roots more negative than $s = -1$

Solution : Put $s = z - 1$ in given characteristic eqn

$$(z-1)^3 + 3(K+1)(z-1)^2 + (7K+5)(z-1) + (4K+7) = 0$$

or, $z^3 + 3Kz^2 + (K+2)z + 4 = 0$

z^3	1	$K+2$
z^2	$3K$	4
z^1	$\frac{3K(K+2)-4}{3K}$	
z^0	4	

For stability

$$\frac{3K(K+2)-4}{3K} > 0 \quad \text{and} \quad 3K > 0 \text{ i.e } K > 0$$

$$3K^2 + 6K - 4 > 0$$

$$K = -2.527, 0.527$$

∴ Hence, the range is

$$\boxed{\infty > K > 0.53} \qquad (\because K > 0)$$

Example 5.21. The characteristic equation of feedback control system is
$s^4 + 20s^3 + 15s^2 + 2s + K = 0$
(a) Determine the range of K for the system to be stable.
(b) Can the system be marginally stable? If so, find the required value of K and the frequency of sustained oscillation. (GATE 1998)

Solution :

s^4	1	15	K
s^3	20	2	
s^2	14.9	K	
s^1	$\frac{29.8-20K}{14.9}$		
s^0	K		

(a) For stability $K > 0$

$$29.8 - 20K > 0$$

or, $K < 29.8/20$

$K < 1.49$

Hence, range $0 < K < 1.49$

(b) For marginally stable $K = 1.49$

Auxiliary eq^n $A(s) = 14.9s^2 + 1.49$

$$14.9s^2 = -1.49$$

$$s^2 = -0.1$$

$$s = \pm j\,0.316$$

∴ $\omega = 0.316$ rad/sec.

∴ Frequency of sustained oscillation = 0.316 rad/sec.

Example 5.22. The characteristic equation for certain feedback control system is given below. Determine the range of value of K for the system to be stable.
$s^4 + 4s^3 + 13s^2 + 36s + K = 0$

Solution :

s^4	1	13	K
s^3	4	36	
s^2	4	K	
s^1	$36-K$		
s^0	K		

For stability $K > 0$

$36 - K > 0$ or $K < 36$

Hence, range is $36 > K > 0$ **Ans.**

5.7. ROOT LOCUS

Root locus is a graphical method in which roots of the characteristic equation are plotted in s-plane for the different values of prameter. The locus of the roots of the characteristic equation when gain is varied from zero to infinity is called root locus.

Consider a unity feedback system as shown in fig. 5.10.

The characteristic equation $1 + G(s)\,H(s) = 0$

$$G(s) = \frac{K}{s(s+2)},\ H(s) = 1$$

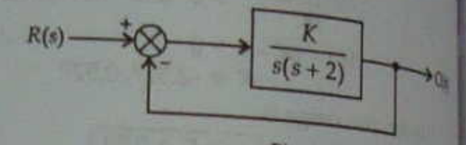


Fig. 5.10.

$$1 + \frac{K}{s(s+2)} \cdot 1 = 0 \text{ or,}$$

$$s^2 + 2s + K = 0$$

The roots of *eq*^{n} (5.14) are ...(5.14)

$$s_1 = -1 + \sqrt{1-K},\quad s_2 = -1 - \sqrt{1-K}$$

As 'K' is varied, the two roots give the locii in s-plane. For various values of K, the location of the roots are

1. when $0 < K < 1$, the roots are real and distinct
2. when $K = 0$, the two roots are $s_1 = 0$ and $s_2 = -2$. These are also the open loop poles.
3. when $K = 1$, both roots are real and equal.
4. when $K > 1$, the roots are complex conjugate with real part = – 1

when 'K' is varying the root locus is shown in the fig. 5.11.

(*a*) when $K = 0$, two branches of root locus starts from $s = 0$ & $s = -2$.

(*b*) when $K = 1$, both roots meet at $s = -1$

(*c*) when $K > 1$, the roots breakaway from the real axis and become complex conjugate having negative real part equal to – 1.

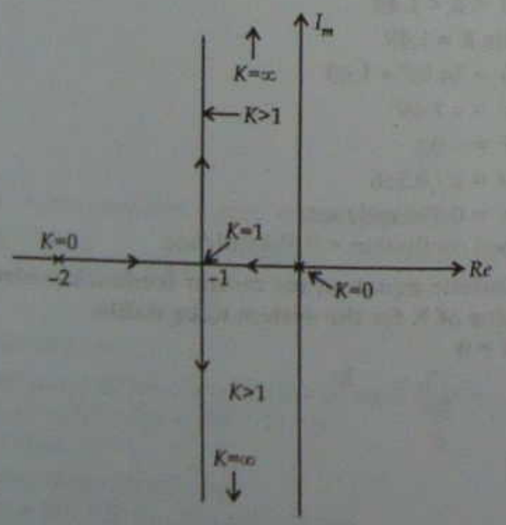


Fig. 5.11.

Consider a closed loop system shown in fig 5.12. The overall transfer function is

$$\frac{c(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

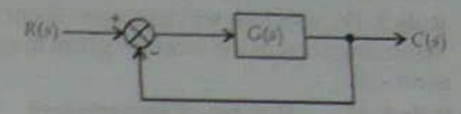


Fig. 5.11 (*a*)

The characteristic equation is $1 + G(s)\,H(s) = 0$

$$G(s)\,H(s) = -1$$

or, $$|G(s)\,H(s)| = -1 \qquad ...(5.15)$$

or, $$\angle G(s)H(s) = \pm 180(2K+1)$$

$$K = 0, 1, 2, 3, \ldots \qquad ...(5.16)$$

Equations (5.15) and (5.16) are the magnitude and angle conditions. The roots of the characteristic equations must satisfied the above two conditions or in other words the values of 's' which satisfies the above said conditions are the roots of the characteristic equation or poles of the closed loop systems. These two conditions are not independent. A plot of the points in 's' plane satisfying the second condition (*eq*. 5.16 known as angle criterion) is the root locus. The value of gain corresponding to a root can be determined from the first condition (*eqn* 5.15 known as magnitude criterion).

Consider a system with $G(s)\,H(s) = \dfrac{K}{s(s+4)(s+5)}$. Find whether $s = -1$ is on root locus or not using angle condition.

The angle condition is

$$\angle G(s)\,H(s) = \pm 180°\,(2K+1).$$

$$\angle G(s)\,H(s)\big|_{s=-1} = \frac{\angle K + jo}{(-1+jo)(3+jo)(4+jo)}$$

$$= \frac{0°}{180°.0°.0°} = -180°$$

Since $\angle G(s)\,H(s) = -180°$ at $s = -1$, this satisfies the angle condition, hence the point at $s = -1$ is on the root locus.

Consider above example for magnitude condition. Now we are interested to find the value of K at which $s = -1$ is one of the roots of $1 + G(s)\,H(s) = 0$.

$$|G(s)\,H(s)|_{s=-1} = 1$$

$$\frac{|K|}{|-1||-1+4||-1+5} = 1$$

$$\therefore \quad K = 12.$$

Therefore for $K = 12$, one of the three roots is located at $s = -1$. Remaining two roots can also be easily calculated.

5.8. RULES FOR CONSTRUCTION OF ROOT LOCII

Following are the rules to sketch the root locus plot.

Rule 1. The root locus is symmetrical about the real axis.

Rule 2. The root locii starts from an open loop pole with $K = 0$ *e.g* For the system having

$$G(s)H(s) = \frac{K(s+3)}{(s+2)} \qquad ...(5.17)$$

Find the starting point of the root locii.

Solution : According to the rule the root locii starts from $s = -2$

Rule 3. The root locii will terminate either on an open loop zeros or on infinity with $K = \infty$ *e.g.* Find the ending point of the root locii given in eqn. 5.17. According to the rule the root locii will terminate at $s = -3$

Rule 4 . If N = No. of separate locii
P = No. of finite poles
Z = No. of finite zeros then

number of root locii will be equal to the no. of poles if number of poles are more than number of zeros *i.e.* $P > Z$

$$N = P \text{ if } P > Z$$

If $Z > P$, then number of root locii will be equal to the number of zeros.

If $P = Z$, then No. of root locii = Poles = Zeros.

e.g. Find the number of separate root locii for the system given by the *eq*^{n} 5.17

Solution : $P = 1$
$Z = 1$
$\therefore$ $N = 1$

Rule 5. ROOT LOCII ON THE REAL AXIS.

Any point on the real axis is a part of the root locus if and only if the number of poles and zeros to its right is odd.

Rule 6. ASYMPTOTES

The branches of root locus tend to infinity along a set of straight line called asymptotes. These asymptotes making an angle with real axis and is given by

$$\phi = \frac{(2K+1)180^\circ}{P-Z} \quad \text{where } K = 0, 1, 2, \ldots$$

The total number of asymptotes = P–Z

e.g. If $G(s)H(s) = \dfrac{K}{s(s^2+6s+10)}$ ...(5.18)

$P = 3$
$z = 0$

No. of asymptotes = $P - Z = 3 - 0 = 3$

$K = 0$ $\quad \phi_1 = \dfrac{(2\times0+1)180^\circ}{3} = 60^\circ$

$K = 1$ $\quad \phi_2 = \dfrac{(2\times1+1)180^\circ}{3} = 180^\circ$

$K = 2$ $\quad \phi_2 = \dfrac{(2\times2+1)180^\circ}{3} = 300^\circ$

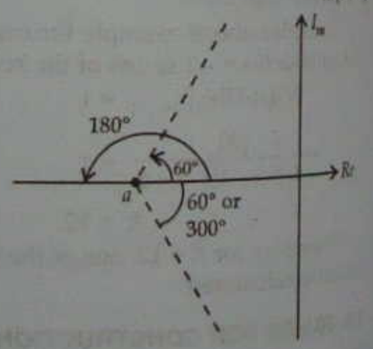


Fig. 5.12.

Rule 7. CENTROID OF ASYMPTOTES

The point of intersection of asymptotes with real axis is called centroid of asymptotes (σ_A) and is given by

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z}$$

e.g. Find the centroid of asymptotes of the system given by eqn. 5.18

Solution : There are three poles at $s_1 = 0, s_2 = -3 + j1, s_3 = -3 - j1$

No. of zeros = 0

$\therefore$ centroid $\quad \sigma_A = \dfrac{0-3+j1-3-j1-0}{3} = -2$

The centroid is shown in fig. 5.12 by the point '*a*'

Rule 8. ANGLE OF DEPARTURE & ANGLE OF ARRIVAL OF THE ROOT LOCII

The angle of departure of the root locus from a complex pole is given by

$\phi_d = 180^\circ$ - sum of angles of vectors drawn to this pole from other poles + sum of angles of vectors drawn to this pole from the zeros.

The angle of arrival at a complex zero is given by

$\phi_a = 180^\circ$-sum of angles of vectors drawn to this zero from other zeros + sum of angles of vectors drawn to this zero from poles.

e.g. For

$$G(s)\,H(s) = \frac{K}{s(s+6)(s^2+4s+13)}$$

Determine the angle of departure from complex poles.

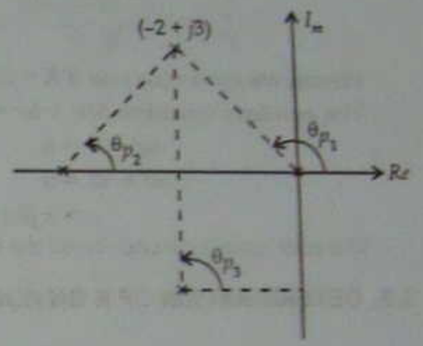


Fig. 5.13.

$$\phi_d = 180^\circ - (\theta_{P_1} + \theta_{P_2} + \theta_{P_3})$$
$$= 180^\circ - (123^\circ + 37^\circ + 90^\circ) = -70^\circ$$

$\therefore$ angle of departure at $(-2 + j3) = -70^\circ$

$\therefore$ angle of departure at $(-2 - j3) = 70^\circ$

Rule No. 9. BREAKAWAY POINT ON REAL AXIS

If the root locus lies between two adjacent open loop poles on the real axis then there will be at least one breakaway point, because the roots move towards each other as K is increased and meet at a point. At this point K is maximum. If we increase the value of K between two poles the root locus breaks in two parts.

Similarly if root locus lies between two adjacent zeros on real axis, then there will be at least one break in point. If the root locus lies between an open loop pole and zero, then then there will be no breakaway or breakin point or may be both occur.

The breakaway or break in points can be determined from the roots of

$$\frac{dK}{ds} = 0 \qquad \text{...(5.19)}$$

e.g. If $G(s)H(s) = \dfrac{K}{s(s^2+6s+10)}$ Determine the breakaway point

$$1 + G(s)\,H(s) = 1 + \frac{K}{s(s^2+6s+10)}$$

or, $s(s^2 + 6s + 10) + K = 0$

$$K = -s^3 - 6s^2 - 10s$$

$$\frac{dK}{ds} = -3s^2 - 12s - 10 = 0$$

or, $3s^2 + 12s + 10 = 0$

$s_1 = -1.1835$ & $s_2 = -2.815$ are the breakaway points.

Rule No. 10. The intersection of root locus branches with jω-axis can be determined through Routh Hurwitz criterion.

e.g. If $G(s)\,H(s) = \dfrac{K}{s(s^2+6s+10)}$. Find the intersection of the root locii with the imaginary axis.

Solution : The characteristic *eq*^{n} $s^3 + 6s^2 + 10s + K = 0$

s^3	1	10
s^2	6	K
s^1	$\dfrac{60-K}{6}$	
s^0	K	

Hence, we get a zero row if $K = 60$

The auxiliary equation $A(s) = 6s^2 + K$

$$6s^2 + K = 0$$
$$6s^2 + 60 = 0$$
$$\therefore \quad s = \pm j3.16$$

The root locus branches cross the imaginary axis at $s = \pm j3.16$ for $K = 60$.

5.9. DETERMINATION OF K ON ROOT LOCII

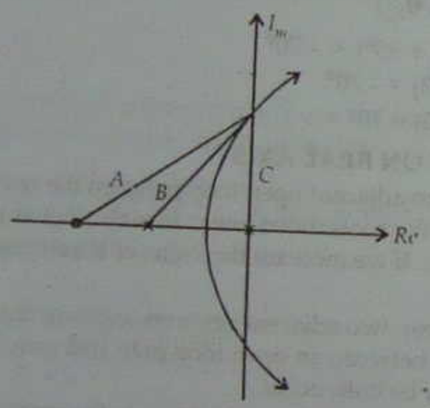


Fig. 5.14.

The value of K can be determined by

$$K = \frac{\text{Product of all vector lengths drawn from the poles of } G(s)\,H(s) \text{ to the point}}{\text{Product of all vector lengths drawn from the zeros of } G(s)\,H(s) \text{ to the point}}$$

e.g. in fig 5.14 determine the value of K at the point of intersection of root locus branch with imaginary axis.

$$K = \frac{B.C}{A}$$

Example 5.23. The forward path transfer function of a unity feedback system is given by $G(s) = \dfrac{K}{s(s+4)(s+5)}$ Sketch the root locus as K varies from zero to infinity.

Solution : Step 1 : Plot the poles and zeros

Poles are at $s = 0, -4$ & -5

No. of poles $P = 3$

No. of zeros $z = 0$

Step 2 : The root locus exists between $s = 0$ & $s = -4$ and to the left of -5. Mark the root locus on real axis.

Step 3 : Number of root locii

$$P = 3$$
$$Z = 0$$

$\therefore$ Number of root locii $(N) = 3$

Step 4 : Centroid of asymptotes

$$\sigma A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z} = \frac{(0-4-5)-(0)}{3-0} = -3$$

Step 5 : Angle of asymptotes

$$\phi = \left(\frac{2K+1}{P-Z}\right)180^\circ$$

$K = 0, \quad \phi_1 = 60^\circ$

$K = 1, \quad \phi_2 = 180^\circ$

$K = 2, \quad \phi_3 = 300^\circ$

Step 6 : Caluculation of breakaway point

The characteristic equation $1 + G(s)\,H(s) = 0$

$$1 + \frac{K}{s(s+4)(s+5)} = 0$$
$$s^3 + 9s^2 + 20s + K = 0$$
$$\therefore \quad K = -s^3 - 9s^2 - 20s$$
$$\frac{dk}{ds} = -3s^2 - 18s - 20 = 0$$
$$3s^2 + 18s + 20 = 0$$
$$s_1 = -1.4725, \; s_2 = -4.5275$$

Since, -4 to -5 is not the segment of root locus. Therefore we consider -1.4725 as a breakaway point.

Step 7 : Determination of point of intersection of branches of root locus with imaginary axis. by Routh Hurwitz

The characteristic *eq*^{n} is

$$s^3 + 9s^2 + 20s + K = 0$$

s^3	1	20
s^2	9	K
s^1	$\dfrac{180-K}{9}$	
s^0	K	

For $K = 180$, the auxiliary *eq*^{n} $A(s) = 9s^2 + K$

$$9s^2 + K = 0$$
$$9s^2 + 180 = 0$$
$$s = \pm j\,4.47$$

The complete root locus is shown in fig. 5.15.

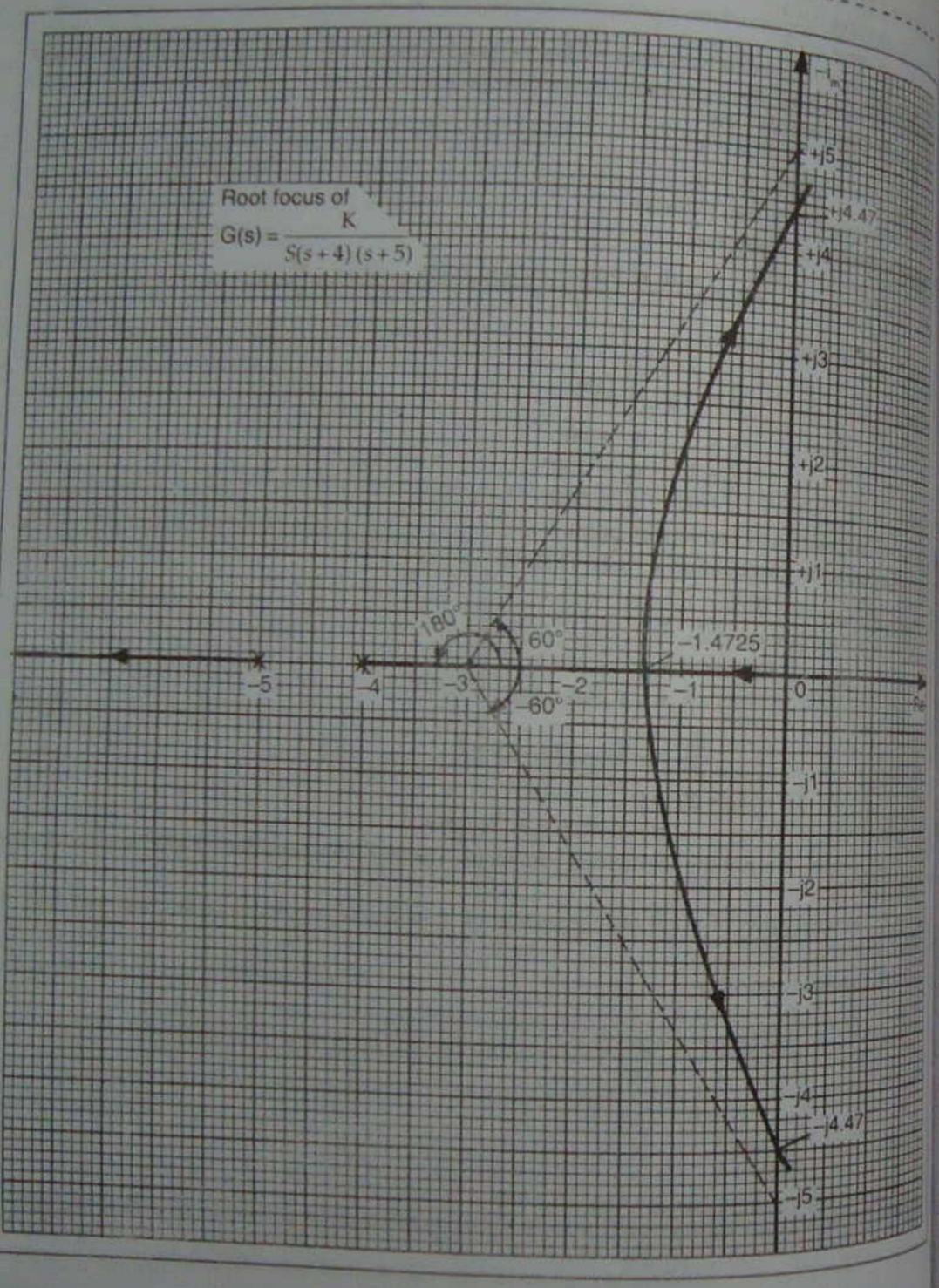


Fig. 5.15

Example 5.24. For a unity feedback system the open loop transfer function is given by

$$G(s) = \frac{K}{s(s+2)(s^2+6s+25)}$$

(a) Sketch the root locus for $0 \le K \le \infty$

(b) At what value of 'K' the system becomes unstable.

(c) At this point of instability determine the frequency of oscillation of the system.

Solution : Step 1 : Plot the poles and zeros

Poles are at $s_1 = 0, s_2 = -2, s_3 = -3 + j4, s_4 = -3 - j4$

$$s^2 + 6s + 25 = 0$$

$$s_3 = -3 + j4$$

$$s_4 = -3 - j4$$

Step 2 : The segment on the real axis between $s = 0$ & $s = -2$ is the part of the root locus.

Step 3 : Number of root locii

Number of Poles $P = 4$

Number of zeros $Z = 0$

No of root locii $N = P = 4$

Step 4 : centroid of the asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z} = \frac{(0-2-3+j4-3-j4)-(0)}{4-0} = -2$$

Step 5 : Angle of asymptotes .

$$\phi = \frac{2K+1}{P-Z}180°$$

$K = 0$ $\quad \phi_1 = \frac{2\times 0+1}{4}180° = 45°$

$K = 1$ $\quad \phi_2 = \frac{2\times 1+1}{4}180° = 135°$

$K = 2$ $\quad \phi_3 = \frac{2\times 2+1}{4}180° = 225°$

$K = 3$ $\quad \phi_4 = \frac{2\times 3+1}{4}180° = 315°$

Step 6 : Breakaway point

The characteristic *eq*^{n} $1 + G(s)\,H(s) = 0$

$$1 + \frac{K}{s(s+2)(s^2+6s+25)} = 0$$

or, $K + s^4 + 8s^3 + 37s^2 + 50s = 0$

$$K = -(s^4 + 8s^3 + 37s^2 + 50s)$$

$$\therefore \quad \frac{dk}{ds} = -(4s^3 + 24s^2 + 74s + 50) = 0$$

or, $4s^3 + 24s^2 + 74s + 50 = 0$

By trial & error method $s = -0.8981$*

* put the values of 's' between 0 & – 2 because the portion between 0 & – 2 is the part of root locus, if there is any breakaway point then it will lie between these points.

Step 7 : Determination of jω crossover (by Routh Hurwitz)

ch. eq^n $s^4 + 8s^3 + 37s^2 + 50s + K = 0$

s^4	1	37	K
s^3	8	50	
s^2	30.75	K	
s^1	$\frac{1537.5 - 8K}{30.75}$		
s^0	K		

$$\frac{1537.5 - 8K}{30.75} = 0 \qquad \therefore K = 192.18$$

For $K = 192.18$ Auxiliary eq^n

$$30.75\, s^2 + K = 0$$

$$30.75\, s^2 = -192.18$$

$$s = \pm j2.5$$

Step 8 : Angle of departure from upper compex pole

$$\phi_d = 180° - (104° + 90° + 127°)$$

$$\phi_d = -141°$$

b. The range of values for stability is $0 < K < 192.18$

The closed loop system becomes unstable for $K < 0$ & $K > 192.18$

c. At this point of instability the gain is $K = 192.18$

$$30.75\, s^2 + 192.18 = 0$$

Put $s = j\omega$

$$-30.75\, \omega^2 + 192.18 = 0$$

$$\therefore \quad \omega = 2.5 \text{ rad/sec.}$$

∴ The frequency of oscillation at the point of instability = 2.5 rad/sec.

The root locus plot is shown in fig 5.16.

Example 5.25. Consider a unity feedback control system with the following feedforward transfer function.

$$G(s) = \frac{K}{s(s^2 + 4s + 8)}$$

Plot the root locii for the system.

Solution : Step 1 : Plot the poles & zeros

$$s^2 + 4s + 8 = 0$$

$$s_2, s_3 = \frac{-4 \pm \sqrt{16 - 32}}{2} = -2 \pm j2$$

Three poles are at $s_1 = 0$, $s_2 = -2 + j2$ & $s_3 = -2 - j2$

Step 2 : Since there is only one pole at $s = 0$, the entire left half of the real axis is the part of the root locus.

Step 3 : Number of root locii

No. of poles $P = 3$

No. of zeros $Z = 0$

∴ No. of root locii $N = P = 3$

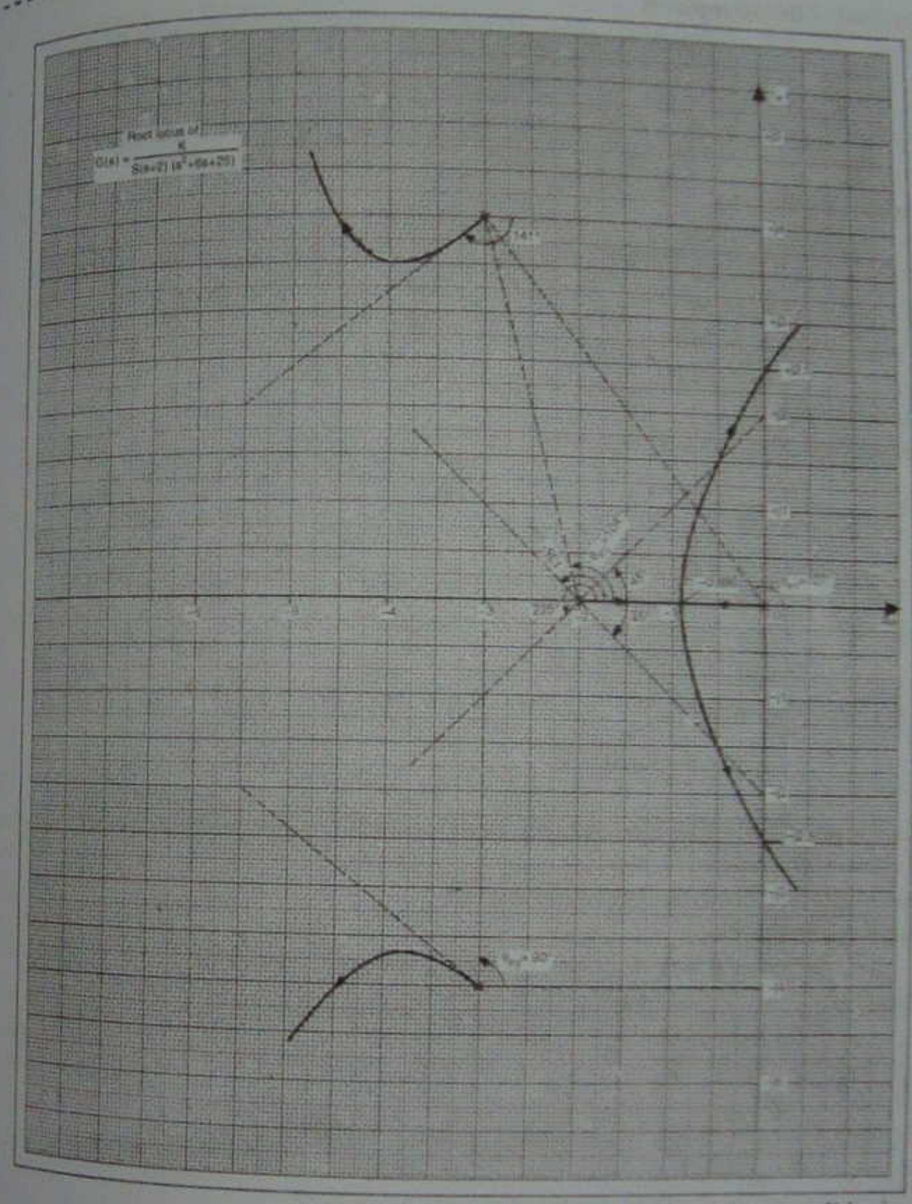


Fig. 5.16

Step 4 : Centroid of the asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z}$$

$$= \frac{(0 - 2 + j2 - 2 - j2) - (0)}{3} = -4/3 = -1.33$$

Step 5 : Angle of asymptotes

$$\phi = \frac{2K+1}{P-Z}180°$$

$K = 0$ $\quad \phi_1 = 60°$

$K = 1$ $\quad \phi_2 = 180°$

$K = 2$ $\quad \phi_3 = 300°$

Step 6 : Breakaway point

The characteristic eqn $1 + G(s)\,H(s) = 0$

$$1 + \frac{K}{s(s^2 + 4s + 8)} = 0$$

$$s^3 + 4s^2 + 8s + K = 0$$

or, $$K = -(s^3 + 4s^2 + 8s)$$

$$\frac{dK}{ds} = -(3s^2 + 8s + 8)$$

$$3s^2 + 8s + 8 = 0$$

$$\therefore \quad s = \frac{-8 \pm \sqrt{64 - 96}}{6} = -1.33 \pm j\,0.943$$

Since, at the point $s = -1.33 \pm j\,0.943$, the angle condition is not satisfied. Hence there is no breakaway point on the real axis.

Step 7 : Point of intersection with imaginary axis.

The characteristic equation

$$s^3 + 4s^2 + 8s + K = 0$$

Routh array is

s^3	1	8
s^2	4	K
s^1	$\frac{32 - K}{4}$	
s^0	K	

For sustained oscillation $32 - K = 0$ or $K = 32$

The auxiliary equation $A(s) = 4s^2 + K$

$$4s^2 + 32 = 0$$

$$\therefore \quad s = \pm j\,2.83$$

Step 8 : The angle of departure from the upper complex pole is

$$\phi_d = 180° - (135° + 90°) = -45°$$

The root locus is shown in fig. 5.17.

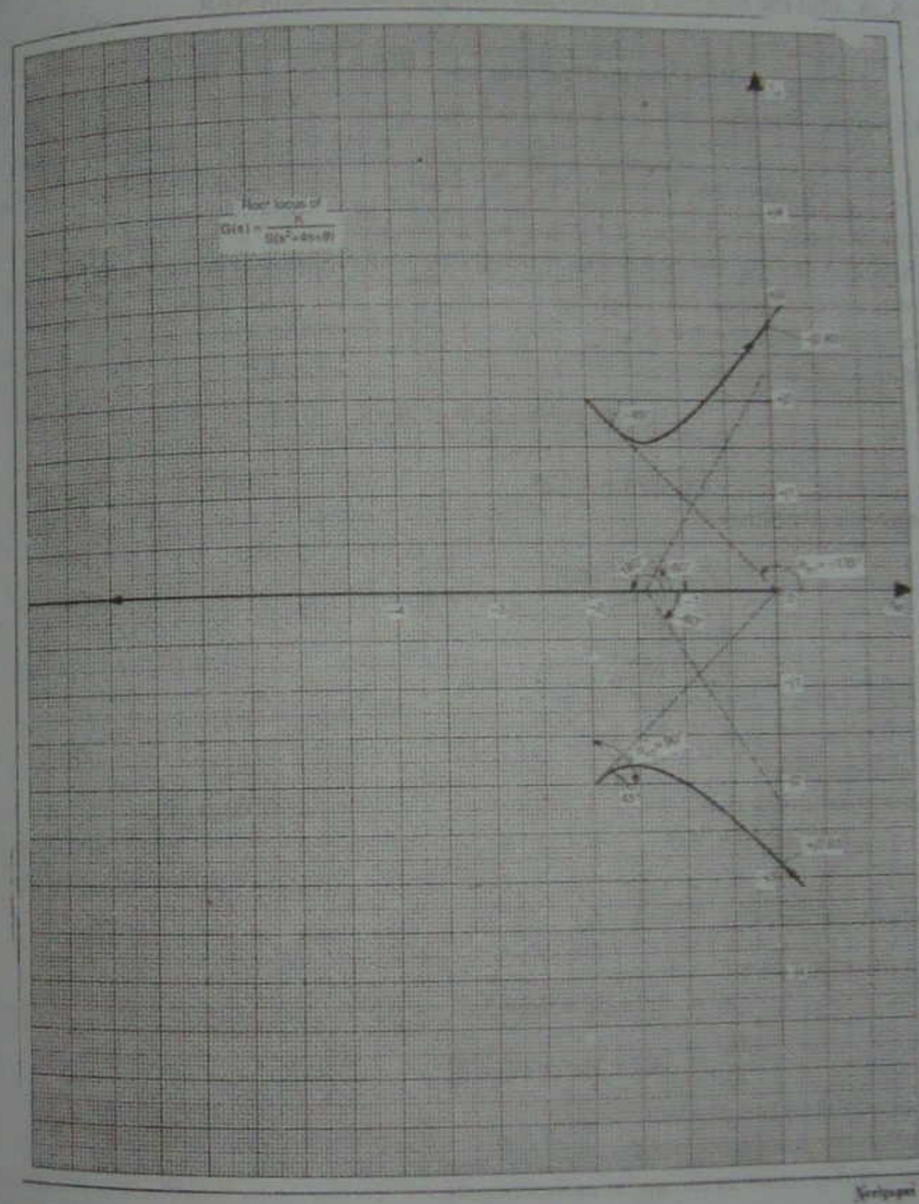


Fig. 5.17.

Example 5.26. Plot the root locii for the closed loop control system with

$$G(s) = \frac{K}{s(s+1)(s^2+4s+5)}, \quad H(s) = 1.$$

Solution : Step 1 : Plot the poles and zeros $s^2 + 4s + 5$

$$s = \frac{-4 \pm \sqrt{16-20}}{2} = -2 \pm j1$$

Poles are at $s_1 = 0$, $s_2 = -1$, $s_3 = -2 + j1$ & $s_4 = -2 - j1$

Step 2 : The segment between $s = 0$ & $s = -1$ is the part of the root locus on real axis.

Step 3 : Number of root locii

Number of root locii $N = P = 4$

Step 4 : Centroid of the asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z} = \frac{0-1-2+j1-2-j1-0}{4}$$

$$= -5/4 = -1.25$$

Step 5 : Angle of asymptotes

$$\phi = \frac{2K+1}{P-Z}180°$$

$K = 0$ $\quad \phi_1 = 45°$

$K = 1$ $\quad \phi_2 = 135°$

$K = 2$ $\quad \phi_3 = 225°$

$K = 3$ $\quad \phi_4 = 315°$

Step 6 : Breakaway point

The characteristic equation

$$1 + G(s)\,H(s) = 0$$

$$1 + \frac{K}{s(s+1)(s^2+4s+5)} = 0$$

$$s^4 + 5s^3 + 9s^2 + 5s + K = 0$$

$$K = -(s^4 + 5s^3 + 9s^2 + 5s)$$

$$\frac{dK}{ds} = -[4s^3 + 15s^2 + 18s + 5]$$

$$4s^3 + 15s^2 + 18s + 5 = 0$$

$\therefore$ Breakaway point is $s = -0.4$

Step 7 : Angle of departure at the upper complex pole

$$\phi_d = 180° - (154° + 136° + 90°) = -200°$$

Step 8 : Point of intersection on $j\omega$ axis

$$s^4 + 5s^3 + 9s^2 + 5s + K = 0$$

s^4	1	9	K
s^3	5	5	
s^2	8	K	
s^1	$\frac{40-5K}{8}$		
s^0	K		

At $40 - 5K = 0$ $\quad K = 8$ (for sustained oscillation)

the auxiliary equation $A(s) = 8s^2 + K$

$$8s^2 + 8 = 0$$

$$s = \pm j1$$

The root locus plot is shown in fig 5.18.

Example 5.27. Sketch the root locii for

$$G(s) = \frac{K(s+1)}{s^2(s+3.6)}$$

$$H(s) = 1$$

Solution : Step 1 : Plot the poles and zeros

Poles are at $\quad s = 0, 0, -3.6$

zeros at $\quad s = -1$

Step 2 : The segment between $s = -1$ & -3.6 is the part of the root locus

Step 3 : Centroid of the asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z}$$

$$= \frac{0+0-3.6+1}{3-1} = -1.3$$

Step 4 : Angle of asymptotes

$$\phi = \frac{2K+1}{P-Z}180°$$

$K = 0$ $\quad \phi_1 = 90°$

$K = 1$ $\quad \phi_2 = 270°$

Step 5 : Breakaway point

characteristic equation $1 + G(s)\,H(s) = 0$

$$1 + \frac{K(s+1)}{s^2(s+3.6)} = 0$$

$$s^2(s+3.6) + K(s+1) = 0$$

$$K = -\frac{s^3 + 3.6s^2}{s+1}$$

$$\frac{dK}{ds} = \frac{(s+1)(3s^2+7.2s) - (s^3+3.6s^2)}{(s+1)^2} = 0$$

or, $\quad s^3 + 3.3s^2 + 3.6s = 0$

$$s(s^2 + 3.3s + 3.6) = 0$$

$$s = 0 \text{ and } s = \frac{-3.3 \pm \sqrt{3.3^2 - 4 \times 1 \times 3.6}}{2}$$

$$s = 0, s = -1.65 \pm j\,0.936$$

Point $s = 0$, is the actual breakaway point

$s = -1.65 \pm j\,0.936$ is a complex quantity, this will not be breakaway point

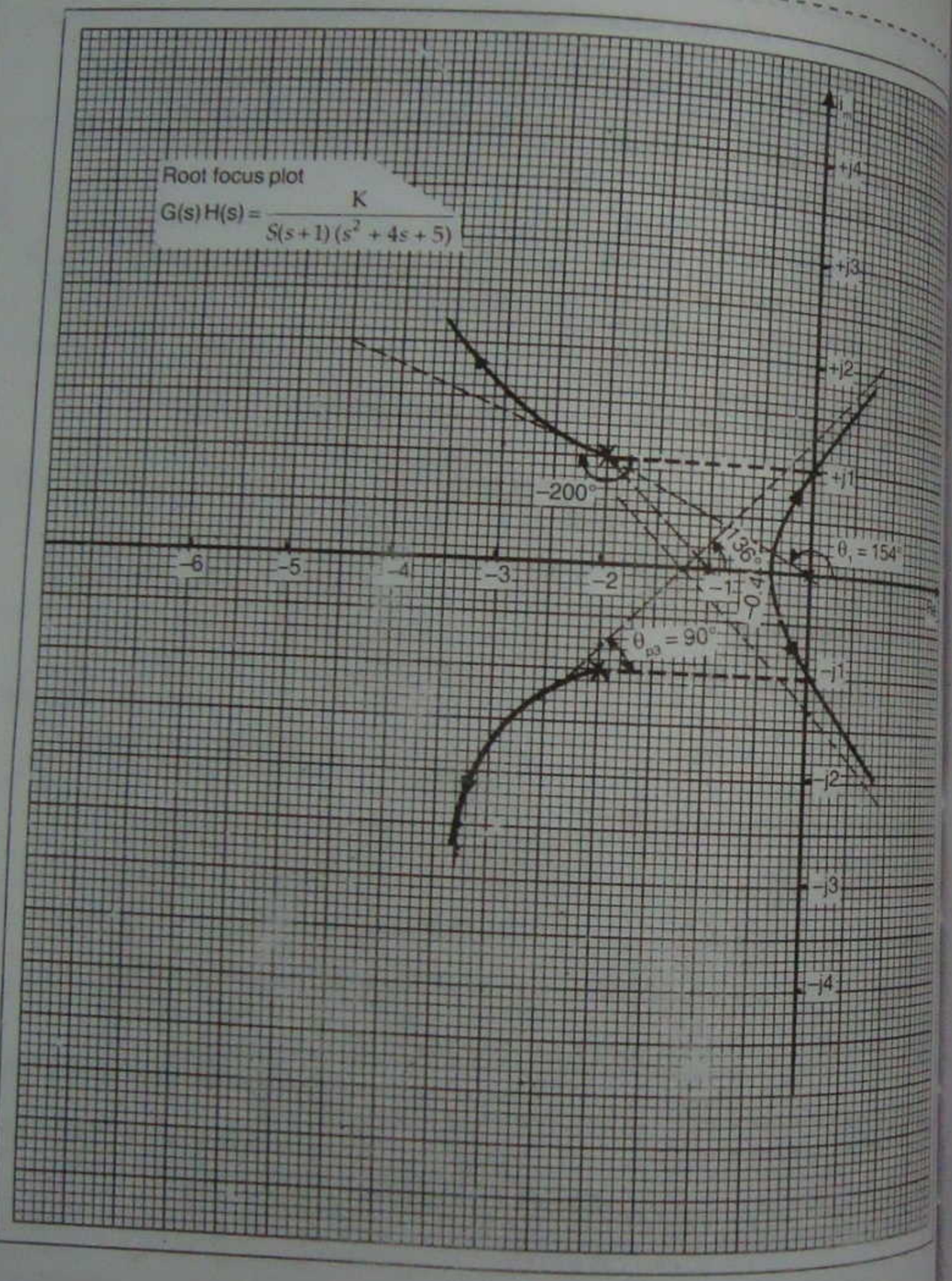


Fig. 5.18.

Step 6 : Point of intersection of root locii on $j\omega$ axis.

The ch. eq^n $s^3 + 3.6s^2 + Ks + K = 0$

s^3	1	K
s^2	3.6	K
s^1	$0.72K$	
s^0	K	

For sustained oscillation $K = 0$

$$A(s) = 3.6\,s^2 + K$$
$$3.6\,s^2 + 0 = 0$$
$$s^2 = 0$$

$\therefore$ Root locus branches do not cross the $j\omega$ axis.

The root locus is shown in fig. 5.19.

Example 5.28. A unity feedback control system has an open loop transfer function

$$G(s) = \frac{K}{s\,(s^2 + 4s + 13)}$$

Sketch the root locus plot of the system by determining the following

(*a*) Centroid, number and angle of asymptotes.

(*b*) Angle of departure of root locii from the poles

(*c*) Breakaway point if any

(*d*) The value of K and the frequency at which the root locii cross $j\omega$ axis.

Solution : Step 1 : Plot the poles & zeros

$$s = 0$$

$$s = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm j3$$

Step 2 : From $s = 0$ to the left is the part of the root locus

Step 3 : Number of root locii $N = P = 3$

Step 4 : Centroid of the asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z}$$

$$= \frac{0 - 2 + j3 - 2 - j3 - 0}{3 - 0} = -4/3 = -1.33$$

Step 5 : Angle of asymptotes

$$\phi = \frac{2K + 1}{P - Z} 180°$$

$K = 0$	$\phi_1 = 60°$
$K = 1$	$\phi_2 = 180°$
$K = 2$	$\phi_3 = 300°$

Step 6 : Breakaway point

The characteristic equation $1 + G(s)\,H(s) = 0$

$$1 + \frac{K}{s\,(s^2 + 4s + 13)} = 0$$

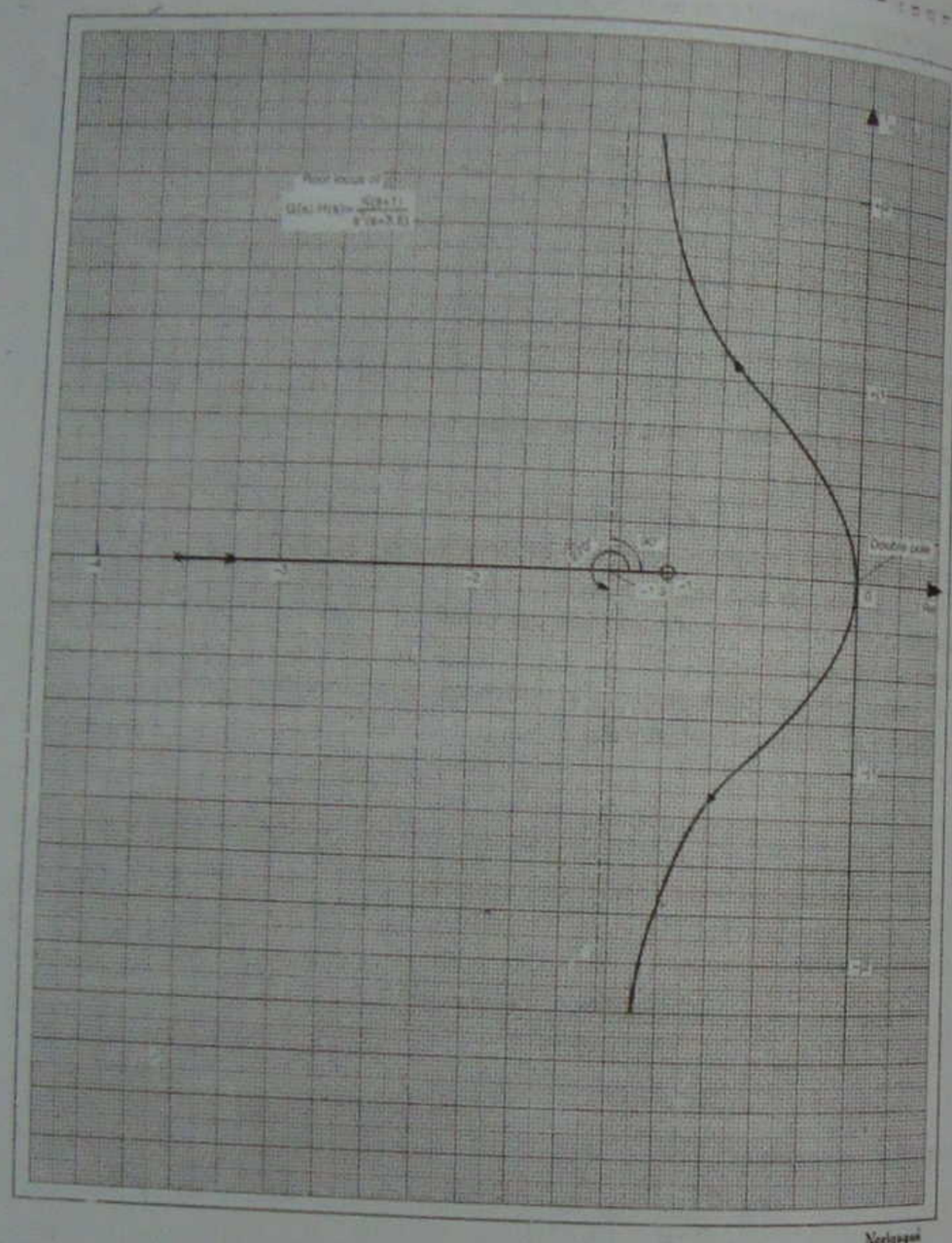


Fig. 5.19.

$$K = -[s^3 + 4s^2 + 13s]$$

$$\frac{dK}{ds} = -[3s^2 + 8s + 13] = 0$$

$$s = \frac{-8 \pm \sqrt{64 - 156}}{6} = \frac{-8 \pm j\,9.59}{6} = -1.33 \pm j\,1.59$$

∴ Since, it is a complex root, there will be no breakaway point on real axis.

Step 7 : Point of intersection of root locii on imaginary axis characteristic eqn

$$s^3 + 4s^2 + 13s + K = 0$$

s^3	1	13
s^2	4	K
s^1	$\frac{52 - K}{4}$	
s^0	K	

For sustained oscillation $52 - K = 0$ or $K = 52$

Auxiliary equation $A(s) = 4s^2 + K$

$$4s^2 + 52 = 0$$

$$s^2 = -13$$

$$s = \pm j\,3.6$$

∴ The value of K at the point of intersection on imaginary axis = 52

The frequency at this point = 3.6 rad/sec.

Step 8 : angle of departure at the upper complex pole

$$\phi_d = 180° - (124° + 90°) = -34°$$

The root locus is shown in fig 5.20.

Example 5.29. Sketch the root locus for

$$G(s)\,H(s) = \frac{K}{s(s+2)(s+4)}$$

and evaluate the value of K at the point where the root locii crosses the imaginary axis. Also determine the frequency. Also, Determine the value of 'K' so that the dominant pair of complex poles of the system has a damping ratio of 0.5.

Solution : Step 1 : Plot the poles and zeros.

Poles are at $s_1 = 0, s_2 = -2, s_3 = -4$

Step 2 : The segment between $s = 0$ & -2 and from $s = -4$ to the left are the parts of root locus.

Step 3 : Number of root locii $N = P = 3$

Step 4 : Centroid of the asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z}$$

$$= \frac{0 - 2 - 4 - 0}{3 - 0} = -2$$

Step 5 : Angle of asymptotes

$$\phi = \frac{2K + 1}{P - Z} 180°$$

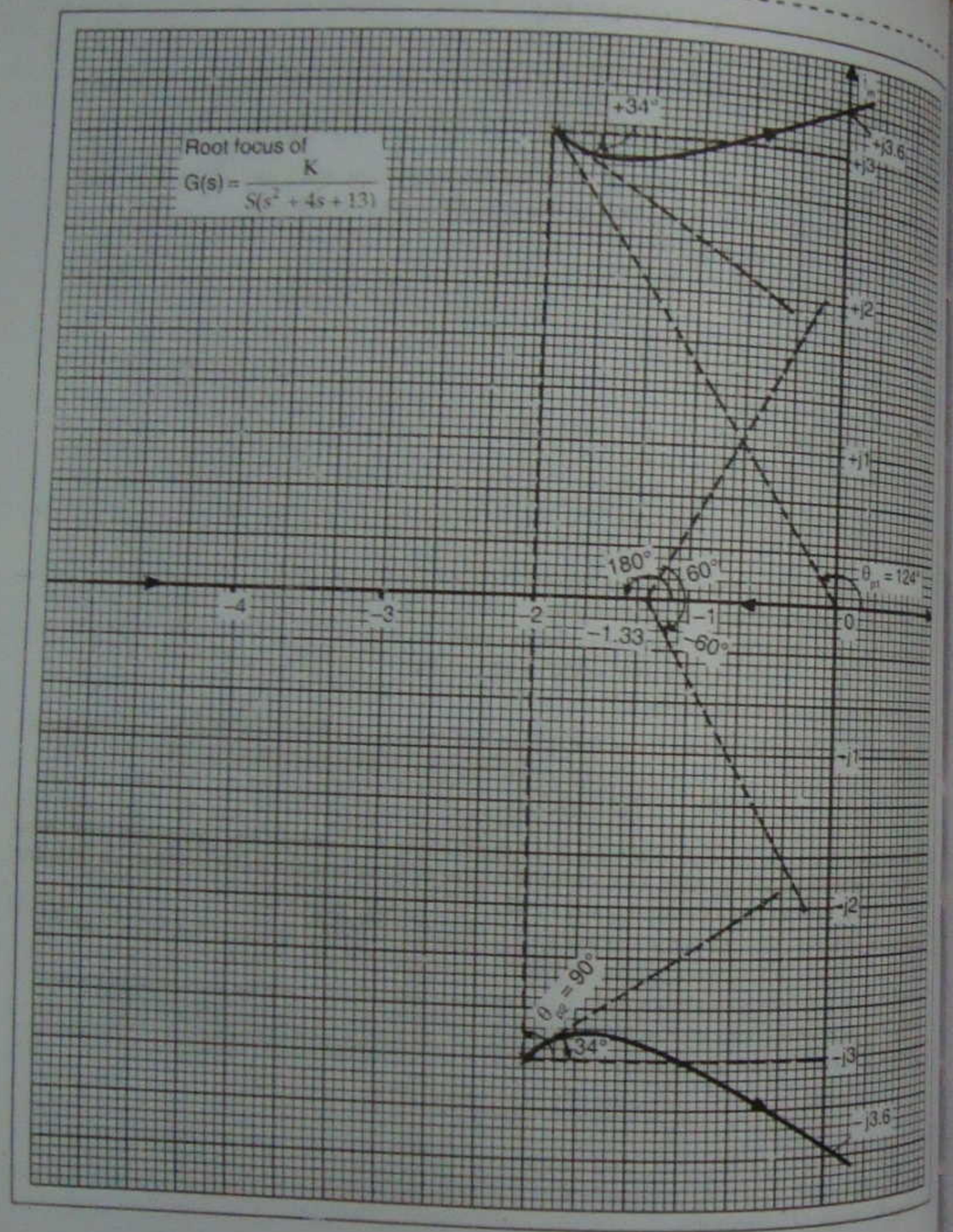


Fig. 5.20.

$K = 0$ $\quad \phi_1 = 60°$

$K = 1$ $\quad \phi_2 = 180°$

$K = 2$ $\quad \phi_3 = 300°$

Step 6 : Breakaway point

The characteristic equation

$$s(s+2)(s+4) + K = 0$$

$$K = -(s^3 + 6s^2 + 8s)$$

$$\frac{dK}{ds} = -(3s^2 + 12s + 8) = 0$$

$$3s^2 + 12s + 8 = 0$$

$$s = -0.85 \text{ \& } -3.15$$

Since, – 3.15 is not the part of root locus therefore breakaway point is – 0.85.

Step 7 : Point of intersection of root locii on imaginary axis

characteristic equation

$$s^3 + 6s^2 + 8s + K = 0$$

s^3	1	8
s^2	6	K
s^1	$\frac{48-K}{6}$	
s^0	K	

For sustained oscillation $K = 48$

Auxiliary eq^n $\quad A(s) = 6s^2 + K$

$$6s^2 + 48 = 0$$

$$s = \pm j\,2.8$$

Put $\quad s = j\omega$

$$\therefore \quad \omega = 2.8$$

$\therefore$ frequency of oscillation = 2.8 rad/sec.

The value of K at the point of intersection of root locii with the imaginary axis = 48

This value of K can also be determined from the graph. Draw the lines from all poles to the point of intersection on imaginary axis (c).

$$K = AC \times BC \times OC$$

from graph

$AC = 9.8$ cm. $= 4.9$

$BC = 6.9$ cm. $= 3.45$

$OC = 5.8$ cm. $= 2.9$

$\therefore \quad K = 4.9 \times 3.45 \times 2.9 = 49$

In second quadrant draw a ζ-line at $\theta = \cos^{-1} 0.5 = 60°$ with negative real axis. This line intersect the root locii at point D. The value of 'K' at this point can be obtained as :

Connect all the poles to this point and measure all the distance of all poles to this point

$\therefore$

$OD = 2.6$ c m. $= 1.3$

$BD = 3.6$ c m. $= 1.8$

$AD = 7.2$ c.m. $= 3.6$

$\therefore \quad K = 1.3 \times 1.8 \times 3.6 = 8.424$

The root locus is shown in the fig. 5.21.

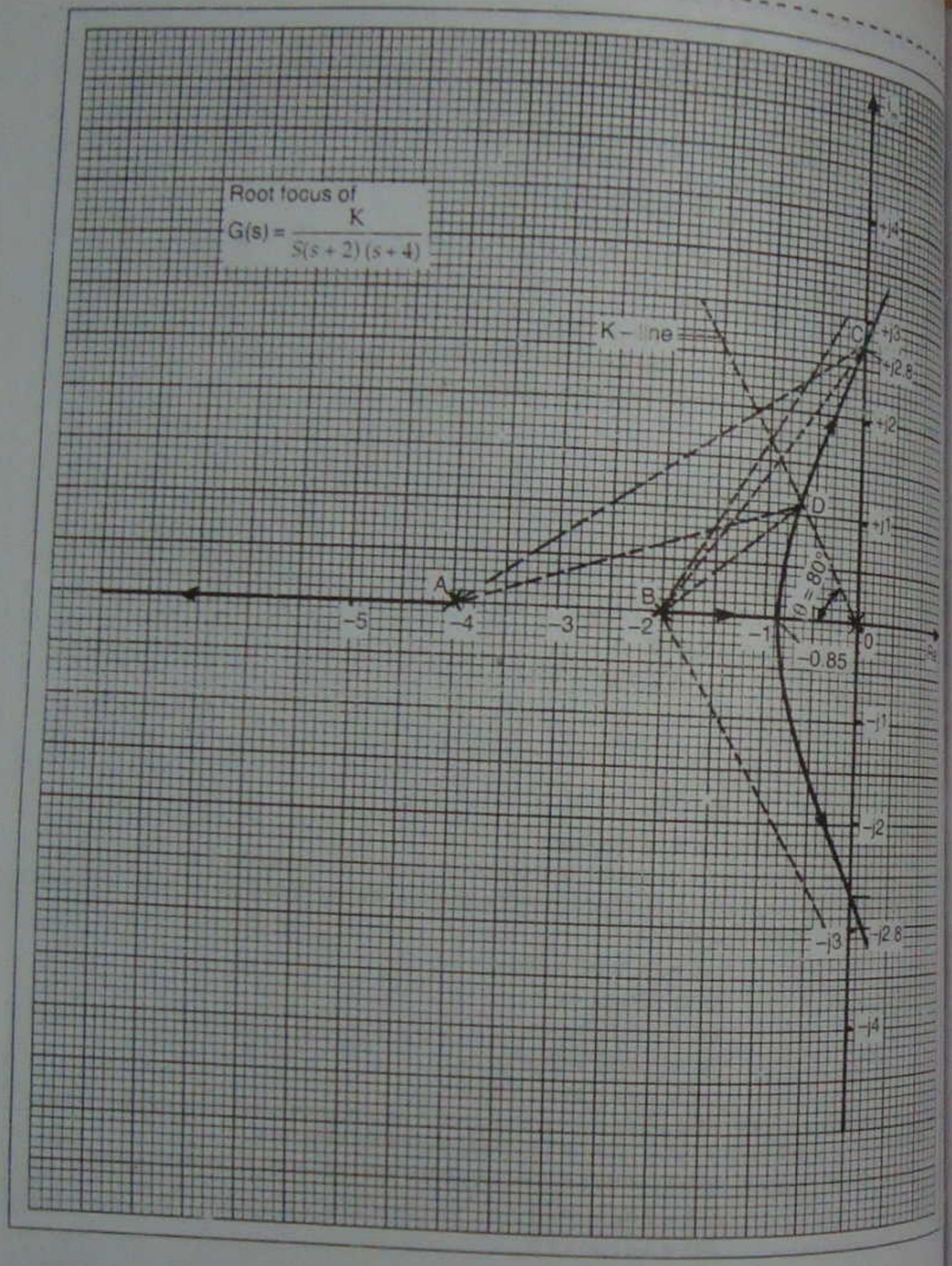


Fig. 5.21.

Example 5.30. A unity feedback system has an open loop transfer function

$$G(s) = \frac{K(s+1)}{s(s-1)}$$

Sketch the root locus plot with 'K' as variable parameter and show that the loci of complex roots are part of a circle with (-1, 0) as centre and radius = $\sqrt{2}$.

Solution : Step 1 : Plot the poles & zeros

Poles. $s_1 = 0, s_2 = 1$

zeros. $s_3 = -1$

Step 2 : Centroid of asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z} = \frac{0+1-(-1)}{2-1} = 2$$

Step 3 : Angle of asymptotes

$$\phi = \frac{2K+1}{P-Z}180°$$

$K = 0$, $\quad \phi = \frac{2\times 0+1}{2-1}\times 180° = 180°$

Step 4 : Breakaway point

The characteristic equation

$$1 + \frac{K(s+1)}{s(s-1)} = 0$$

$$K = -\frac{s(s-1)}{s+1}$$

$$\frac{dK}{ds} = s^2 + 2s - 1 = 0$$

$$s = -2.414 \text{ and } 0.414$$

Since, both values are the part of the root locus, hence there will be a breakaway point and other is re-entry or breaking point.

As $s = 0.414$ lies between two poles $s = 0$ and $s = 1$, therefore it is breakaway point and $s = -2.414$ lies between zero & infinity therefore it is a breakin or re-entry point.

Step 5 : Point of intersection on imaginary axis.

characteristic equation

$$s^2 + (K-1)s + 1 = 0$$

s^2	1	1
s	$K-1$	0
s^0	1	

For stability $\quad K - 1 > 0$

$$K > 1$$

Auxiliary eq^n $\quad A(s) = s^2 + 1$

$$s = \pm j1$$

The root locus is shown in fig 5.22.

From the root locus it is clear that centre at (- 1, 0) & radius = $\sqrt{2}$

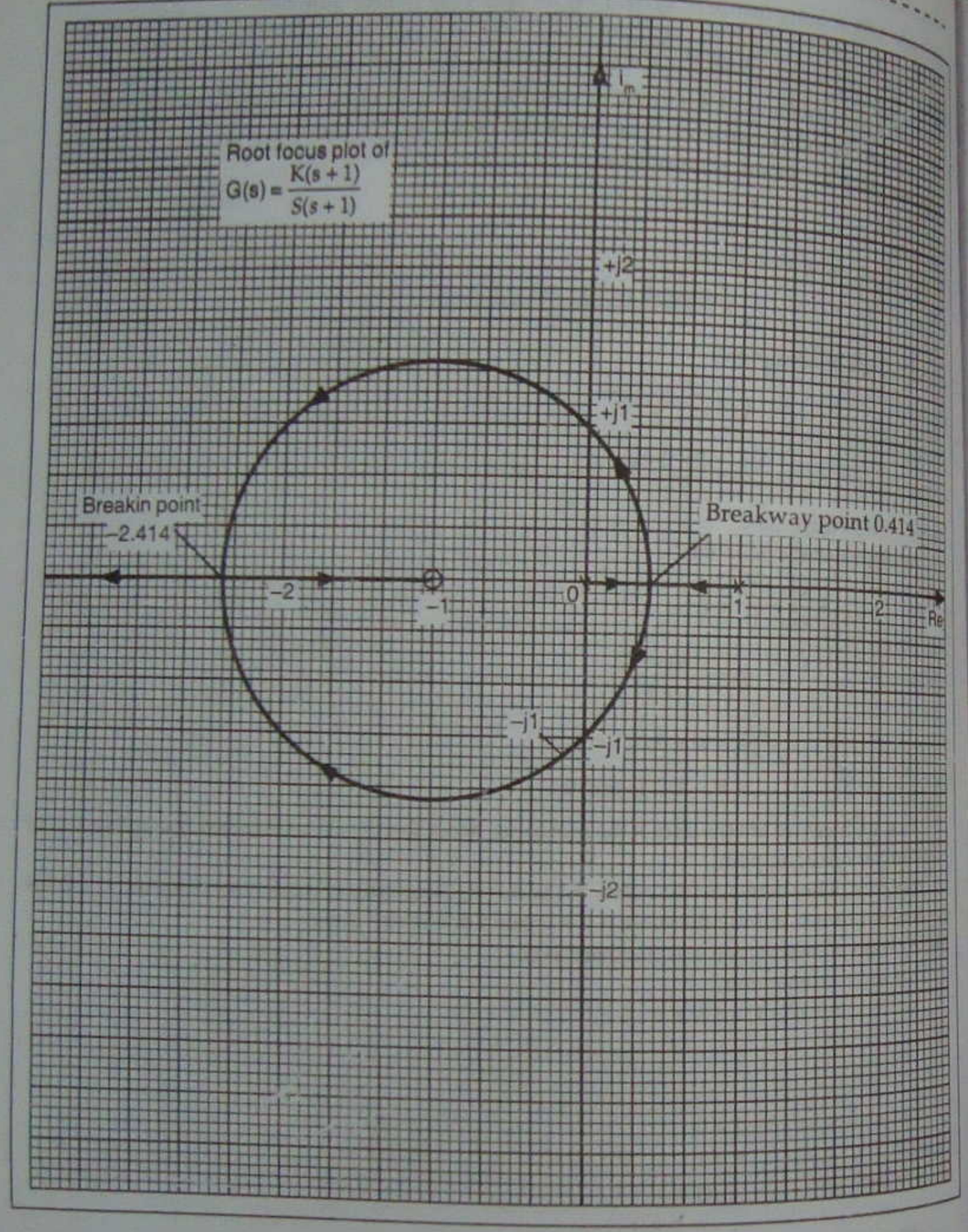


Fig. 5.22.

Example 5.31. In example 5.30 Is the system stable for all values of K? If not determine the range of K for stable system operation. Find also the marginal value of K which causes sustained oscillations and the frequency of these oscillations.

Solution : In example 5.30 step 5 it is clear that the system is not stable for all values of K. The range of the K for stable operation is

$$1 < K < \infty$$

Marginal value of 'K' for sustained oscillations = 1.

Frequency of sustained oscillations = 1 rad/sec.

Example 5.32. The open loop transfer function of a control system is given by

$$G(s)\,H(s) = \frac{K}{s\,(s+6)\,(s^2+4s+13)}$$

sketch the root locus and determine the breakaway point, the angle of departure from complex poles and the stability condition. *(R.M.L. University Faizabad 2002, LST)*

Solution : Step 1 : Plot poles and zeros.

Poles are at $s_1 = 0, s_2 = -6$

$$s^2 + 4s + 13 = 0$$

$$s = \frac{-4 \pm \sqrt{16-52}}{2} = -2 \pm j3$$

$$s_3 = -2 + j3, s_4 = -2 - j3$$

Step 2 : The segment between $s = 0$ and $s = -6$ is the part of the root locus.

Step 3 : Number of root locii $N = P = 4$

Step 4 : Centroid of the asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z} = \frac{0-6-2+j3-2+j3-0}{4} = -2.5$$

Step 5 : Angle of asymptotes

$$\phi = \frac{2K+1}{P-Z}180°$$

$eK = 0$ $\quad \phi_1 = 45°$

$K = 1$ $\quad \phi_2 = 135°$

$K = 2$ $\quad \phi_3 = 225°$

$K = 3$ $\quad \phi_4 = 315°$

Step 6 : Break away point

The characteristic equation $1 + G(s)\,H(s) = 0$

$$1 + \frac{K}{s\,(s+6)\,(s^2+4s+13)}$$

or, $s\,(s+6)\,(s^2+4s+13) + K = 0$

or, $K = -(s^4 + 10s^3 + 37s^2 + 7s) = 0$

$$\frac{dK}{ds} = -(4s^3 + 30s^2 + 74s + 78) = 0$$

$$4s^3 + 30s^2 + 74s + 78 = 0$$

By trial and error method $s = -4.2$

Breakaway point = -4.2

Step 7 : Point of intersection of root locii on imaginary axis.

characteristic equation is

$$s^4 + 10s^3 + 37s^2 + 78s + K = 0$$

Routh array

s^4	1	37	K
s^3	10	78	
s^2	29.2	K	
s^1	$78 - 0.342K$		
s^0	K		

For sustained oscillations $78 - 0.34\,K = 0 \therefore K = 229.41$

The auxiliary eq^n $\quad A(s) = 29.2\,s^2 + K$

$$29.2\,s^2 + 229.41 = 0$$

$$s = \pm j\,2.8$$

Step 8 : From Routh table For stability

$$K > 0$$

$78 - 0.342\,K > 0$ or $K < 229.41$

Step 9 : The angle of departure from upper complex pole is

$$\phi_d = 180° - (124° + 90° + 37°)$$

$$\phi_d = -71°$$

similarly, the angle of departure from lower complex pole is

$$\phi_d = +71°$$

The root locus plot is shown in the fig 5.23.

Example 5.33. Draw the root locus for a system whose open loop transfer function is given by

$$G(s)\,H(s) = \frac{K}{s\,(s+4)\,(s^2+4s+20)}$$

show all the salient points on the locus. *(Control system, R.M.L. University Faizabad 2002)*

Solution : Step 1. Plot poles and zeros

Poles are at $\quad s = 0, s = -4$

$$s^2 + 4s + 20 = 0$$

$$s = -2 \pm j4$$

Step 2 : The segment between $s = 0$ and $s = -4$ is the part of the root locus.

Step 3 : Number of root locii $N = P = 4$

Step 4 : Centroid of asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z} = \frac{0 - 4 - 2 + j4 - 2 - j4 - 0}{4} = -2$$

Step 5 : Angle of asymptotes

$$\phi = \frac{2K+1}{P-Z}180°$$

$K = 0$	$\phi_1 = 45°$
$K = 1$	$\phi_2 = 135°$
$K = 2$	$\phi_3 = 225°$
$K = 3$	$\phi_4 = 315°$

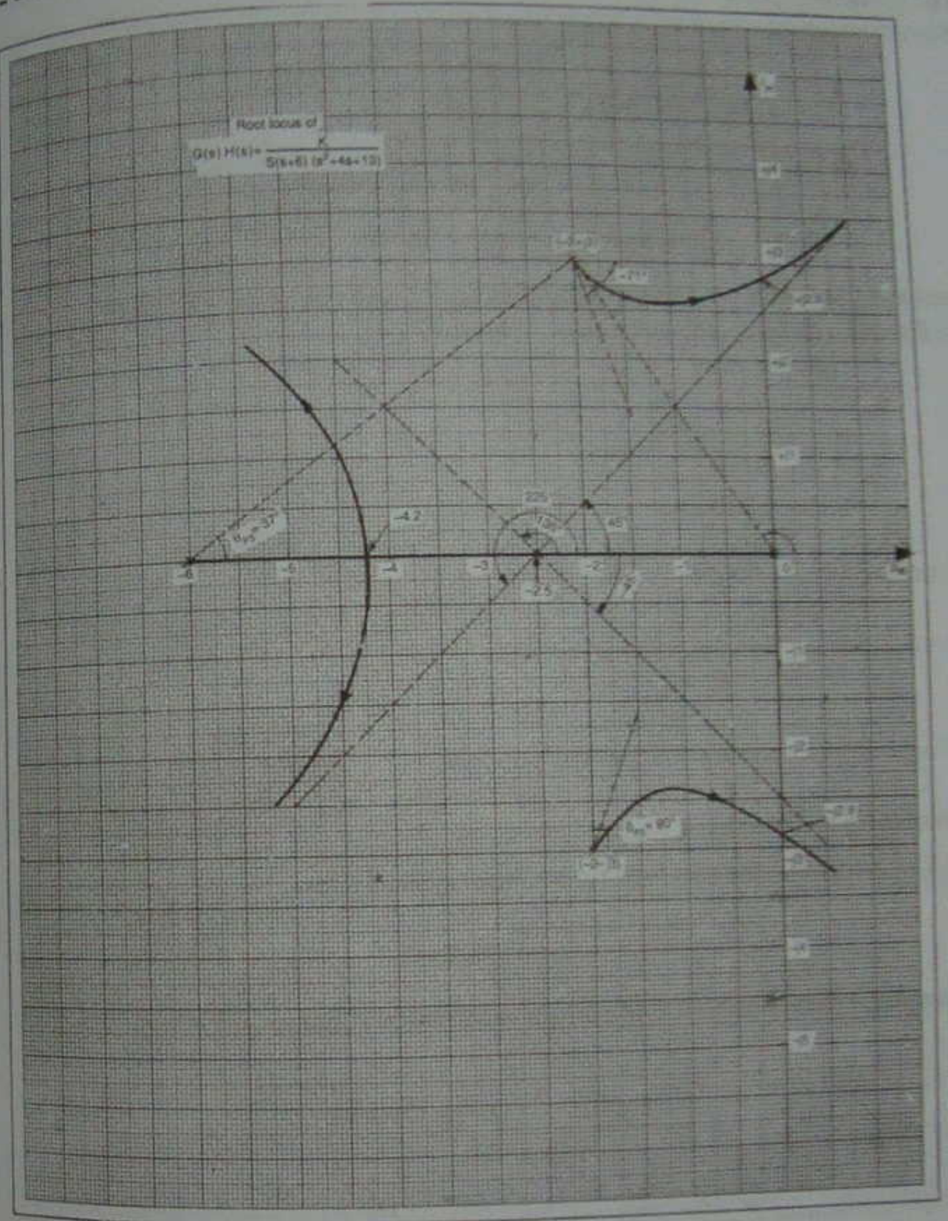


Fig. 5.23.

Step 6 : Breakaway point

The characteristic equation . $1 + G(s)\,H(s) = 0$

$$1+\frac{K}{s(s+4)(s^2+4s+20)} = 0$$

$$s^4 + 8s^3 + 36\,s^2 + 80\,s + K = 0$$

$$K = -(s^4 + 8s^3 + 36s^2 + 80s)$$

$$\frac{dK}{ds} = -(4s^3 + 24s^2 + 72s + 80) = 0$$

Breakaway point is $s = -2$ and two complex breakaway points are $-2 \pm j\,2.45$

Step 7 : Point of intersection of root locii on imaginary axis

Routh array

s^4	1	36	K
s^3	8	80	
s^2	26	K	
s^1	$80 - 0.307K$		
s^0	K		

For stability $K > 0$

$80 - 0.307K > 0$ or $K < 260$

at $K = 260$, the auxiliary eq^n $A(s) = 26s^2 + K$

$$26s^2 + 260 = 0$$

$$s = \pm j\,3.16$$

Step 8 : The angle of departure at upper complex pole

$$\phi_d = 180° - (117° + 90° + 63°) = -90°$$

The root locus is shown in fig 5.24.

Example 5.34. Show that the root locii for a control system with

$$G(s) = \frac{K(s^2+6s+10)}{s^2+2s+10}, \quad H(s) = 1$$

are the arcs of the circle centered at the origin with radius equal to $\sqrt{10}$.

Solution : $G(s)\,H(s) = \dfrac{K(s^2+6s+10)}{s^2+2s+10} = \dfrac{K(s+3-j1)(s+3+j1)}{(s+1-j3)(s+1+j3)}$

Put $s = \sigma + j\omega$

$$G(s)H(s)\big|_{s=\sigma+j\omega} = \frac{K[s+j\omega+3-j1][\sigma+j\omega+3+j1]}{[\sigma+j\omega+1-j3][\sigma+j\omega+1+j3]}$$

$$= \frac{K[(\sigma+3)+j(\omega-1)][(\sigma+3)+j(\omega+1)]}{[(\sigma+1)+j(\omega-3)][(\sigma+1)+j(\omega+3)]}$$

$$\angle G(j\omega)H(j\omega) = \left[\tan^{-1}\frac{\omega-1}{\sigma+3} + \tan^{-1}\frac{\omega+1}{\sigma+3}\right] - \left[\tan^{-1}\frac{\omega-3}{\sigma+1} + \tan^{-1}\frac{\omega+3}{\sigma+1}\right]$$

To satisfy the angle criterion equate the above eq^n to 180°

$$\left[\tan^{-1}\frac{\omega-1}{\sigma+3} + \tan^{-1}\frac{\omega+1}{\sigma+3}\right] - \left[\tan^{-1}\frac{\omega-3}{\sigma+1} + \tan^{-1}\frac{\omega+3}{\sigma+1}\right] = 180$$

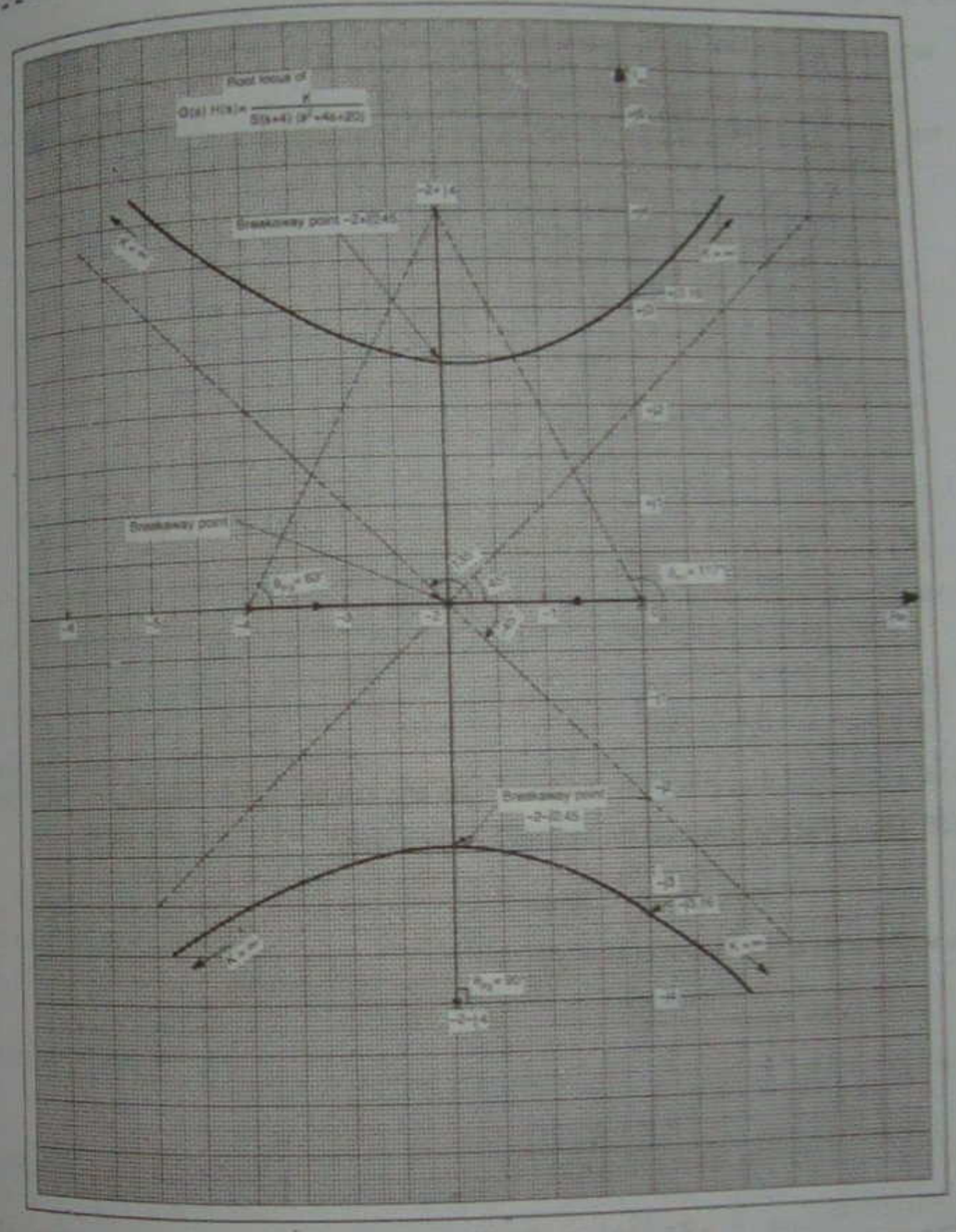


Fig. 5.24

Taking tangent of both sides

$$\tan\left[\tan^{-1}\frac{\omega-1}{\sigma+3}+\tan^{-1}\frac{\omega+1}{\sigma+3}\right]-\tan\left[\tan^{-1}\frac{\omega-3}{\sigma+1}+\tan^{-1}\frac{\omega+3}{\sigma+1}\right]=\tan 180°$$

$$\tan\left[\tan^{-1}\frac{\omega-1}{\sigma+3}+\tan^{-1}\frac{\omega+1}{\sigma+3}\right]=\tan\left[\tan^{-1}\frac{\omega-3}{\sigma+1}+\tan^{-1}\frac{\omega+3}{\sigma+1}\right]$$

$$\text{or,}\quad \frac{\frac{\omega-1}{\sigma+3}+\frac{\omega+1}{\sigma+3}}{1-\frac{(\omega-1)(\omega+1)}{(\sigma+3)^2}}=\frac{\frac{\omega-3}{\sigma+1}+\frac{\omega+3}{\sigma+1}}{1-\frac{(\omega-3)(\omega+3)}{(\sigma+1)^2}}$$

$$\text{or,}\quad \frac{\frac{2\omega}{\sigma+3}}{\frac{(\sigma+3)^2-(\omega^2-1)}{(\sigma+3)^2}}=\frac{\frac{2\omega}{\sigma+1}}{\frac{(\sigma+1)^2-(\omega^2-9)}{(\sigma+1)^2}}$$

$$\text{or,}\quad \frac{\sigma+3}{(\sigma+3)^2-(\omega^2-1)}=\frac{\sigma+1}{(\sigma+1)^2-(\omega^2-9)}$$

$$\text{or, } (\sigma+1)\left[(\sigma+3)^2-(\omega^2-1)\right]=(\sigma+3)\left[(\sigma+1)^2-(\omega^2-9)\right]$$

$$\text{or,}\quad 2\sigma^2+2\omega^2-20=0$$

$$\text{or,}\quad \sigma^2+\omega^2=10$$

$$\text{or}\quad (\sigma-0)2+(\omega-0)^2=\left(\sqrt{10}\right)^2$$

This is the equation of the circle with centre at origin and radius = $\sqrt{10}$ *Proued.*

Example 5.35. Show that the root locii for a control system with

$$G(s)=\frac{K(s+1)}{s(s-1)},\ H(s)=1$$

is the circle with centre at (-1, 0) and radius $\sqrt{2}$

Solution : $$G(s)\,H(s)=\frac{K(s+1)}{s(s-1)}$$

Put $s=\sigma+j\omega$

$$G(s)H(s)\big|_{s=\sigma+j\omega}=\frac{K(\sigma+j\omega+1)}{(\sigma+j\omega)(\sigma+j\omega-1)}$$

$$\angle G(s)H(s)\big|_{s=\sigma+j\omega}=\angle\sigma+j\omega+1-\angle\sigma+j\omega-\angle\sigma+j\omega-1$$

$$=\tan^{-1}\frac{\omega}{\sigma+1}-\tan^{-1}\frac{\omega}{\sigma}-\tan^{-1}\frac{\omega}{\sigma-1}$$

To satisfy the angle criterion

$$\tan^{-1}\frac{\omega}{\sigma+1}-\tan^{-1}\frac{\omega}{\sigma}-\tan^{-1}\frac{\omega}{\sigma-1}=180°$$

Taking tangent on both sides

$$\tan\left[\tan^{-1}\frac{\omega}{\sigma+1}-\tan^{-1}\frac{\omega}{\sigma-1}\right]-\tan\left[\tan^{-1}\frac{\omega}{\sigma}\right]=\tan 180°$$

$$\text{or,}\quad \frac{\frac{\omega}{\sigma+1}-\frac{\omega}{\sigma-1}}{1+\frac{\omega}{\sigma+1}\cdot\frac{\omega}{\sigma-1}}=\frac{\omega}{\sigma}$$

$$\text{or,}\quad \frac{-2}{\sigma^2-1+\omega^2}=\frac{1}{\sigma}$$

$$-2\sigma=\sigma^2-1+\omega^2$$

$$\text{or,}\quad \omega^2+\sigma^2-1+2\sigma=0$$

$$\text{or,}\quad \omega^2+\sigma^2+2\sigma-1+2-2=0$$

$$\text{or,}\quad \omega^2+(\sigma+1)^2=2$$

$$\text{or,}\quad (\sigma+1)^2+(\omega)^2=\left(\sqrt{2}\right)^2$$

$$(\sigma+1)^2+(\omega-0)^2=\left(\sqrt{2}\right)^2$$

This is the equation of the circle with centre at (-1, 0) and radius $\sqrt{2}$ **(Proved).**

Example 5.36. The open loop transfer function of a system is given by

$$G(s)\,H(s)=\frac{K(s+12)}{s^2(s+20)}$$

sketch the root locus for the system. *(Control system, R.M.L. University Faizabad 2001)*

Solution : Step 1 : Plot the poles and zero

Poles are at $s_1=0, s_2=0, s_3=-20$

zero is at $s_4=-12$

Step 2 : The segment between $s=-20$ and $s=-12$ is the part of the root locus.

Step 3 : Centroid of asymptotes

$$\sigma_A=\frac{\text{sum of poles}-\text{sum of zeros}}{P-Z}=\frac{0+0-20+12}{3-1}=-4$$

Step 4 : Angle of asymptotes

$$\phi=\frac{2K+1}{P-Z}180°$$

$K=0$ $\quad\phi_1=90°$

$K=1$ $\quad\phi_2=270°$

Step 5 : Breakaway point The characteristic eqn

$$1+\frac{K(s+12)}{s^2(s+20)}=0$$

$$\text{or,}\quad K=-\frac{(s^3+20s^2)}{s+12}$$

$$\frac{dK}{ds}=-\left[\frac{(s+12)(3s^2+40s)-(s^3+20s^2)}{(s+12)^2}\right]=0$$

or, $s^3 + 28s^2 + 240\,s = 0$

$s(s^2 + 28s + 240) = 0$

we get $s = 0, -14 \pm j\,6.63$

Breakaway point $s = 0$, points $-14 \pm j\,6.63$ are neither breakaway point nor breakin point, because the corresponding gain values K becomes complex quantities.

Step 6 : Point of intersection of rod locii with imaginary axis.

The characteristic eq^n

$$s^3 + 20\,s^2 + Ks + 12\,K = 0$$

Put $s = j\omega$

$$(j\omega)^3 + 20\,(j\omega)^2 + K\,(j\omega) + 12\,K = 0$$

$$(12\,K - 20\omega^2) + j\omega\,(K - \omega^2) = 0$$

if $\omega = 0, \quad K = 0$

because of double pole at the origin, the root locus is tangent to the imaginary axis at $\omega = 0$

The root locus is shown in fig. 5.25.

Example 5.37. Sketch the root locus for

$$G(s) = \frac{K}{s(s^2 + 6s + 12)} \; ; H(s) = 1$$

Solution : Step 1 : Draw the pole zero plot

poles at $s = 0$

$s^2 + 6s + 12 = 0$

$s = -3 \pm j\,1.73$

Step 2 : The left portion of the negative real axis from origin is the part of the root locus.

Step 3 : Centroid of asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z} = \frac{0 - 3 + j1.73 - 3 - j1.73}{3} = -2$$

Step 4 : Angle of asymptotes

$$\phi = \frac{2K + 1}{P - Z} 180°$$

$K = 0 \qquad \phi_1 = 60°$

$K = 1 \qquad \phi_2 = 180°$

$K = 2 \qquad \phi_3 = 300°$

Step 5 : Breakaway point

The characteristic equation

$$1 + \frac{K}{s(s^2 + 6s + 12)} = 0$$

$$K = -(s^3 + 6s^2 + 12)$$

$$\frac{dK}{ds} = -(3s^2 + 12\,s + 12) = 0$$

$$3s^2 + 12\,s + 12 = 0$$

$s = -2$ is the breakaway point

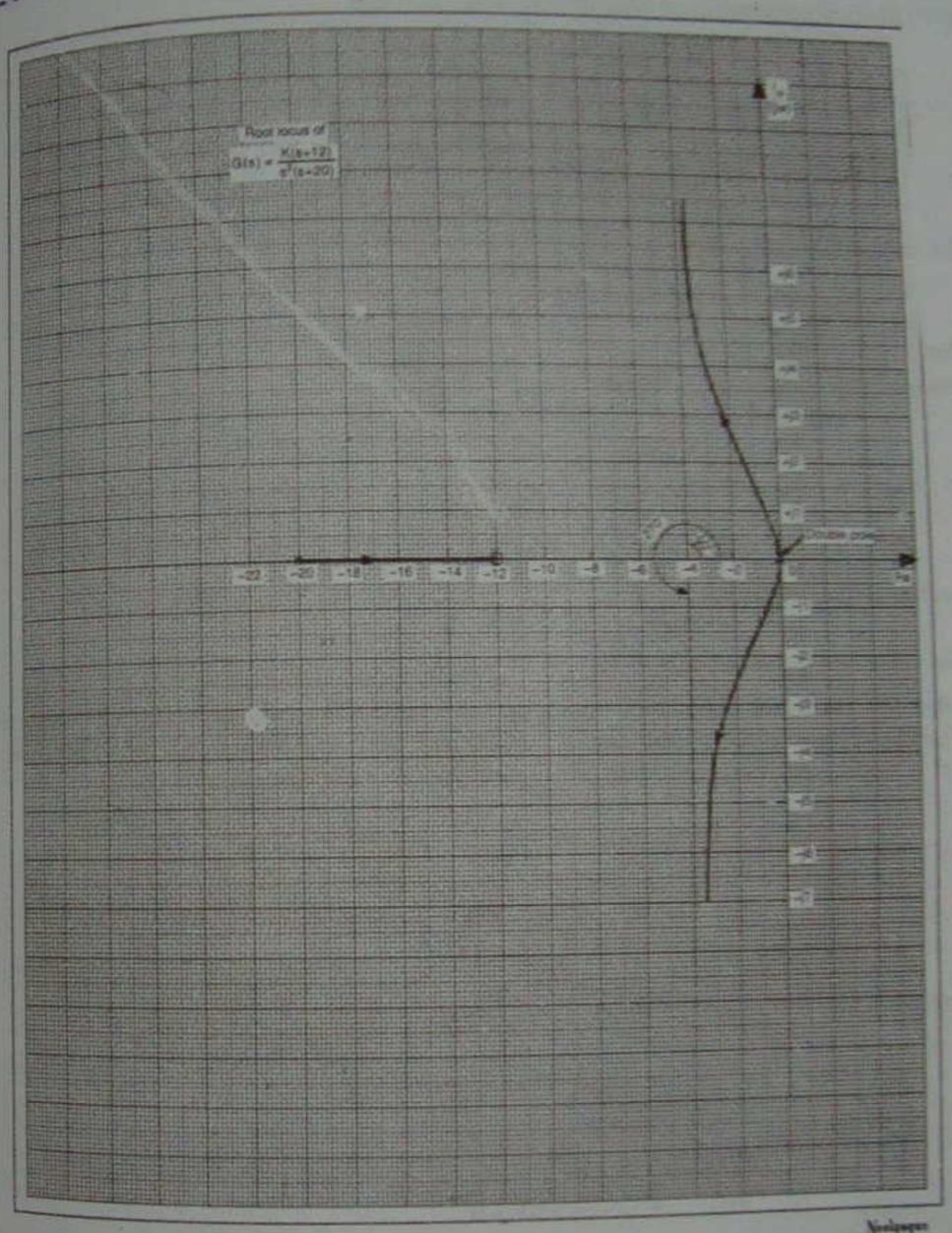


Fig. 5.25.

Step 6 : Point of intersection of root locii on imaginary axis

Routh array

s^3	1	12
s^2	6	K
s^1	$\frac{72-K}{6}$	
s^0	K	

for stability $K > 0$

$$\frac{72-K}{6} > 0 \quad \text{or } K < 72$$

For $K = 72$

the auxiliary equation $A(s) = 6s^2 + K$

$$6s^2 + 72 = 0$$

$$s = \pm j3.46$$

Step 7 : Angle of departure from upper complex pole

$$\phi_d = 180° - (150° + 90°)$$

$$\phi_d = -60°$$

similarly from lower complex pole

$$\phi_d = +60°$$

The root locus is shown in fig. 5.26.

Example 5.38. The block diagram of a position control system is shown in fig. 5.27. Draw the root locus for the system as the parameter 'α' varies from zero to infinity and hence determine the value of 'α' so that the damping ratio of the dominat poles is 0.6.

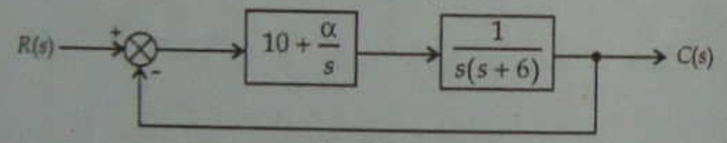


Fig. 5.27.

Solution : $G(s) = \dfrac{10s+\alpha}{s^2(s+6)}$, $H(s) = 1$.

$$\therefore \quad \frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$\frac{C(s)}{R(s)} = \frac{10s+\alpha}{s^2(s+6)+10s+\alpha}$$

The characteristic equation. $s^2(s+6) + 10s + \alpha = 0$

or, $$1 + \frac{\alpha}{s^2(s+6)+10s} = 0$$

let $$G'(s) = \frac{\alpha}{s^2(s+6)+10s} = \frac{\alpha}{s(s^2+6s+10)}$$

Step 1 : Poles are at $s_1 = 0$, $s_2 = -3 + j1$, $s_3 = -3 - j1$

Step 2 : The entire left half of real axis is the part of root locus.

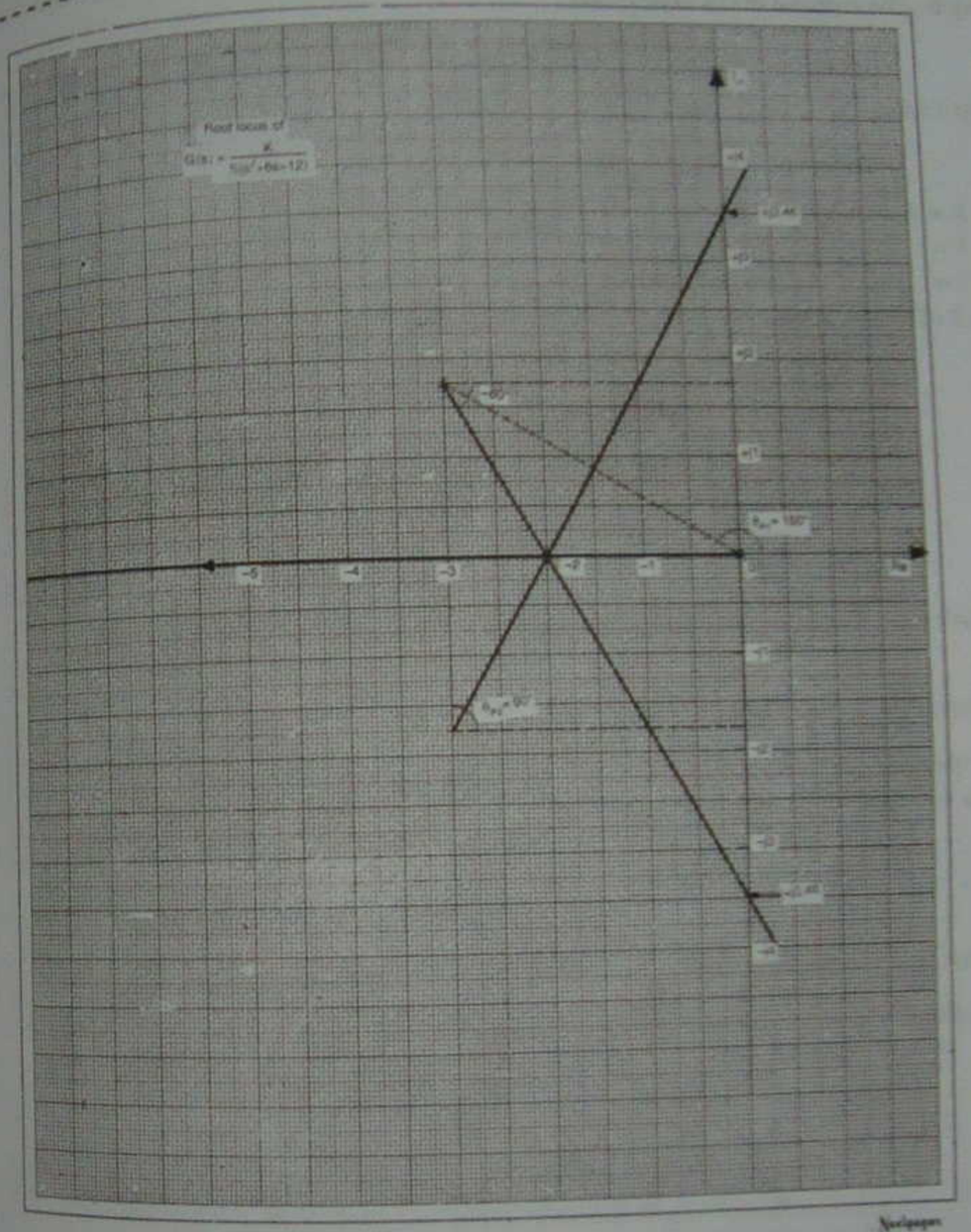


Fig. 5.26.

Step 3 : The centroid of the asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P - Z} = \frac{0 - 3 + j1 - 3 - j1}{3} = -2$$

Step 4 : Angle of asymptotes

$$\phi = \frac{2K+1}{P-Z}180°$$

$K = 0$ $\quad \phi_1 = 60°$

$K = 1$ $\quad \phi_2 = 180°$

$K = 2$ $\quad \phi_3 = 300°$

Step 5 : The characteristic equation

$$\alpha = -(s^3 + 6s^2 + 10s)$$

$$\frac{d\alpha}{ds} = -3s^2 - 12s - 10$$

$$s = -1.18, -2.816$$

$$\frac{d^2\alpha}{ds^2} = -6s - 12$$

Put $s = -1.18$ $\quad \frac{d^2\alpha}{ds^2} = +7.08 - 12 = -4.92$

Since, $\frac{d^2\alpha}{ds^2}$ is negative for $s = -1.18$

$\therefore$ Breakaway point $= -1.18$

Put $s = -2.816$

$$\frac{d^2\alpha}{ds^2} = +4.896$$

Since $\frac{d^2\alpha}{ds^2}$ is positive for $s = -2.816$, therefore $s = -2.816$ is the re rentry point.

Step 6 : Point of intersection of root locii on $j\omega$ axis

$$s^3 + 6s^2 + 10s + \alpha = 0$$

s^3	1	10
s^2	6	α
s^1	$\frac{60-K}{6}$	
s^0	α	

At $K = 60$, auxiliary equation $A(s) = 6s^2 + \alpha$

$$6s^2 + 60 = 0$$

$\therefore$ $\quad s = \pm j3.16$

Step 7 : Angle of departure from upper complex pole.

$$\theta_{P1} = 162°$$

$$\theta_{P2} = 90°$$

$$\phi_d = 180° - (\theta_{P1} + \theta_{P2}) = 180° - (162° + 90°) = -72°$$

The root locus plot is shown in the fig 5.28.

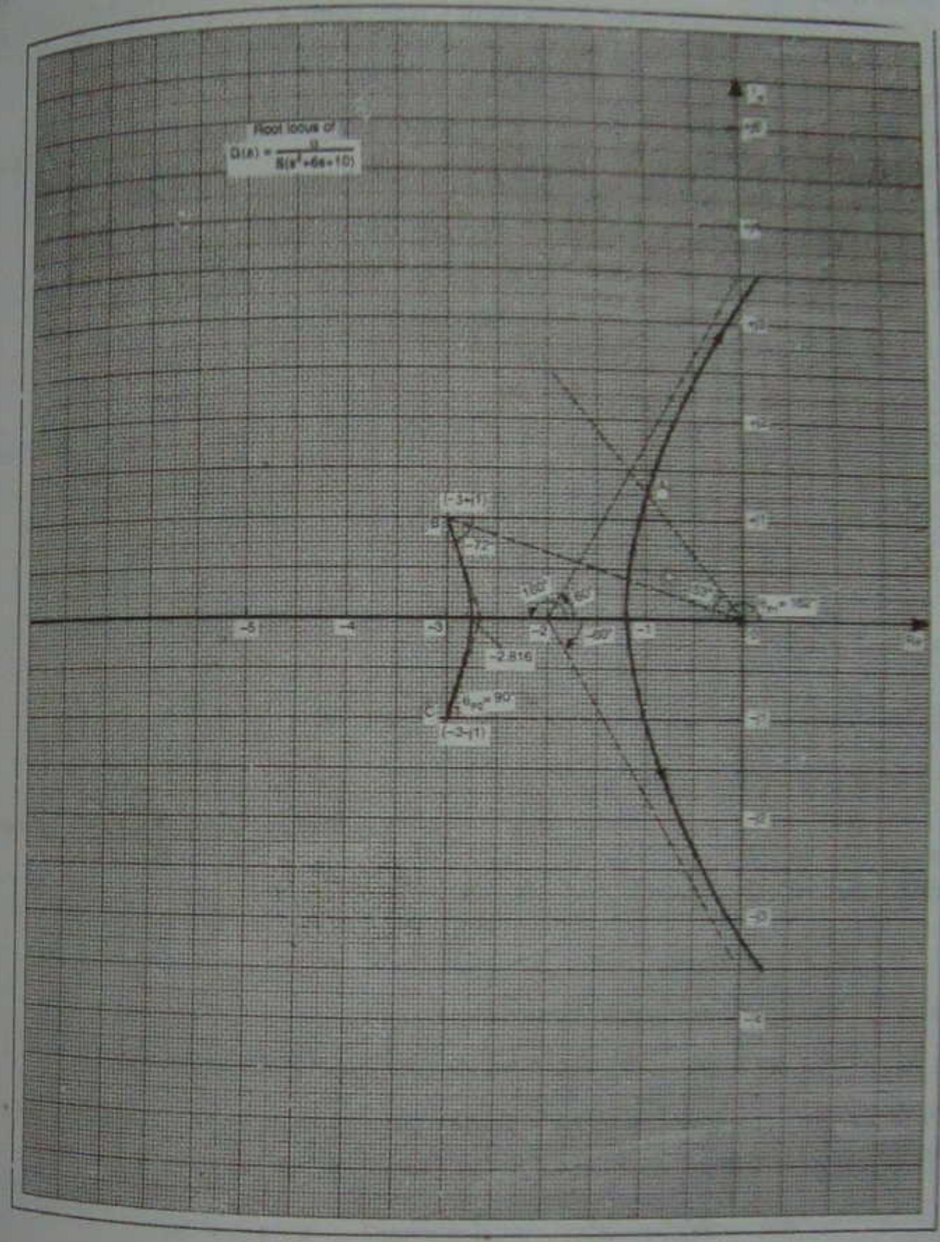


Fig. 5.28.

Step 8 : Draw ζ-line at $\cos^{-1} 0.6 = 53.1°$ from negative real axis. This line intersects the root locus at point A. Measure the distance of point B and point C from point A.

$$OA = 1.55,\ AB = 2.1,\ AC = 3.05$$

$\therefore$ required value of $\alpha = 1.55 \times 2.1 \times 3.05 = 9.927$

Example 5.39. For the example 5.29 determine the phase margin and gain margin for $K = 12$. Also find the roots at the point where $K = 12$.

Solution. The root locus is shown in Fig. 5.29.

GAIN MARGIN. *(a)* The desired value of $K = 12$

For marginal stability the value of $K = 48$ (from example 5.29)

$$\text{Gain margin} = \frac{\text{Value of K for marginal stability}}{\text{Desired value of K}} = \frac{48}{12} = 4$$

Gain margin in $db = 20\log_{10} 4 = 12.04\ db$.

(b) For the root at $K = 12$, locate the point 'x' by trial and error method (the value of K at this point should be 12). In this case

$$Ox = 1.55$$
$$Ax = 3.8$$
$$Bx = 2.05$$

At the point x : $\dfrac{K}{0x \| Ax \| Bx} = 1$

$$\frac{K}{1.55 \times 3.8 \times 2.05} = 1 \qquad \therefore \ K = 12.07$$

The roots at this point $= -0.5 \pm j\,1.5$

(c) For $P.M$ at $K = 12$, locate a point on imaginary axis such that $|G(j\omega)\,H(j\omega)| = 1$ (by trial and error method)

In present case it is 1.2 rad/sec.

$$\frac{12}{j1.2(2+j1.2)(4+j1.2)} = 1.02$$

Hence $\omega_1 = 1.2$ (rad/sec). is the gain crossover frequency

$$P.M. = 180° - \left(\theta_{P_1} + \theta_{P_2} + \theta_{P_3}\right) = 180° - (90° + 32° + 17°) = 41°$$

Example 5.40. For the system shown in fig 5.30 determine the value of α so that the poles of the closed loop system transfer function have a damping ratio of 0.6.

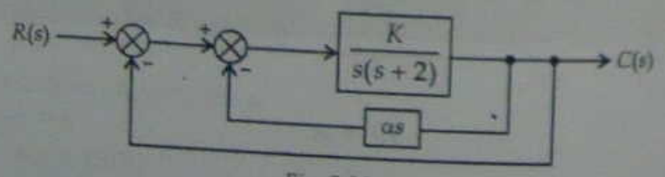


Fig. 5.30.

Solution : For the above fig. $\dfrac{C(s)}{R(s)} = \dfrac{K}{s^2 + 2s + 25\alpha s + 25}$

Characteristic equation.

$$s^2 + 2s + 25\alpha s + 25 = 0$$

or, $\quad s^2 + 2s + 25 + 25\,\alpha s = 0$

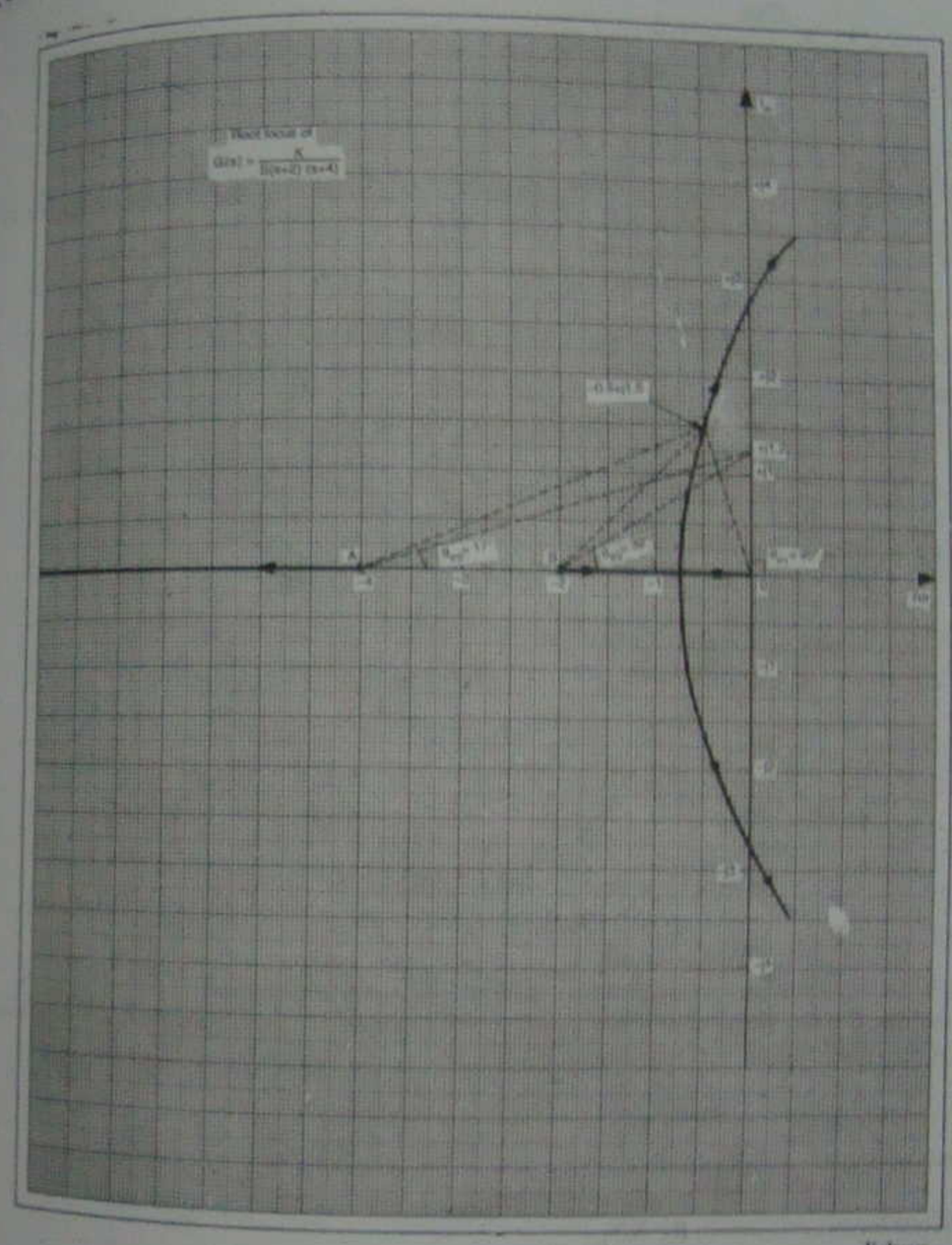


Fig. 5.29

or, $$1+\frac{25\alpha s}{s^2+2s+25}=0$$

Put $$K = 25\,\alpha$$

$$1+\frac{Ks}{s^2+2s+25}=0$$

The given system has a zero at s= 0 and two poles at – 1 ± j4.89, there is a possibility of circular root locus. Since, the angle condition is

$$\angle\frac{Ks}{s^2+2s+25} = \pm 180^\circ (2K+1).$$

We have $\angle s - \angle s+1+j4.89 - \angle s+1-j4.89 = \pm 180^\circ(2K+1)$

Put $s = \sigma + j\omega$

$$\angle\sigma+1+j(4.89+\omega) + \angle\sigma+1+j(\omega-4.89) = \angle\sigma+j\omega+180^\circ$$

$$\tan^{-1}\frac{\omega+4.89}{\sigma+1} + \tan^{-1}\frac{\omega-4.89}{\sigma+1} = \tan^{-1}\frac{\omega}{\sigma}$$

Taking tangent on both sides we get

$$\frac{\frac{\omega+4.89}{\sigma+1}+\frac{\omega-4.89}{6+1}}{1-\left(\frac{\omega+4.89}{\sigma+1}\right)\left(\frac{\omega-4.89}{\sigma+1}\right)} = \frac{\omega}{\sigma}$$

or $$\frac{\frac{2\omega}{\sigma+1}}{1-\frac{(\omega^2-23.9121)}{(\sigma+1)^2}} = \frac{\omega}{\sigma}$$

or, $$2\sigma(\sigma+1) = (\sigma+1)^2 - \omega^2 + 23.9121$$

or, $$\sigma^2+\omega^2 = 24.9121$$

or, $$\sigma^2+\omega^2 = 25$$

This is the equation of the circle* with center at origin and radius is 5.

We require $\zeta = 0.6$, draw. a line at an angle $\cos^{-1}(0.6) = 53.13^\circ$ with negative real axis. This line intersects the root locus at 'x'. The value of K at 'x' is

$$K = \frac{D_1D_3}{D_2} = \frac{2.2\times 9.1}{5} = 4.004$$

Since, $25\alpha = K$

$25\alpha = 4.00\ \alpha = 0.16016$

The root locus is shown in fig. 5.31.

The root at the point of intersection is $-3 + j\,3.95$.

5.10. EFFECT OF ADDITION OF POLES AND ZEROS ON ROOT LOCUS

Consider $G(s)\,H(s) = \frac{K}{s(s+4)}$

* For $K > 0$, the portion of the circle lies to the left of the complex poles. For $K < 0$ the portion of the circle lies to the right of the complex poles. Hence, for present case, it is not the root locus, where $K > 0$.

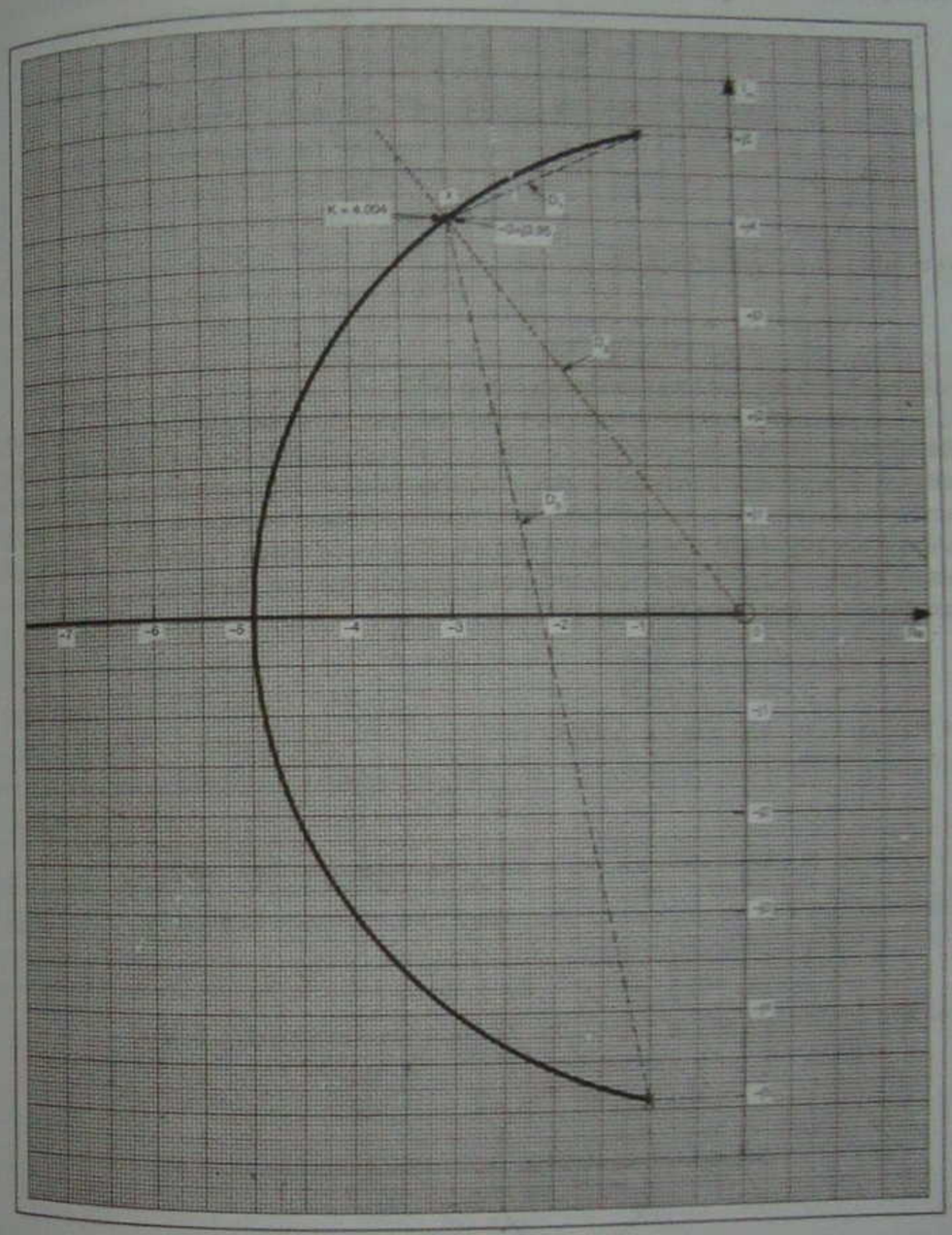


Fig. 5.31.

The root locus is shown in fig. 5.32(*a*)

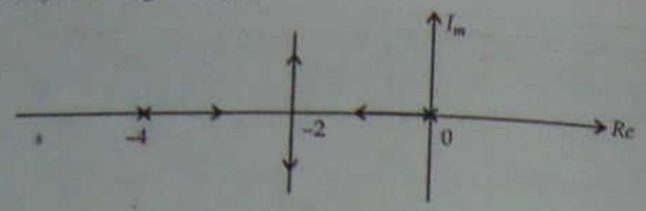


Fig. 5.32.(a)

Now add one pole at $s = -5$, then

$$G(s)\,H(s) = \frac{K}{s(s+4)(s+5)}$$

the corresponding root locus is shown in fig. 5.15.

So, before addition of pole for any value of K the system is stable but after addition of pole to the left half, the two branches of root locus moves to the right half for some value of K. Therefore the system will be stable for this value of K, after this value of K the system becomes unstable. So the stability of the system gets restricted. For further addition of poles to left half, the breaking point moves towards right. Therefore we can say that by addition of poles to left half, the root locus shifted towards right hand side and stability of the system decreases.

Now consider open loop transfer function

$$G(s)\,H(s) = \frac{K}{s(s+2)}$$

The root locus is shown in fig. 5.32(*b*)

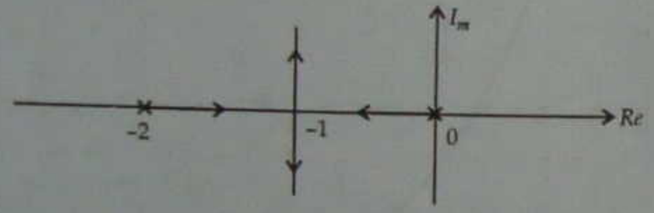


Fig. 5.32(b)

Now add zero to left half at s = – 3, the root locus is shown in fig. 5.32(*c*)

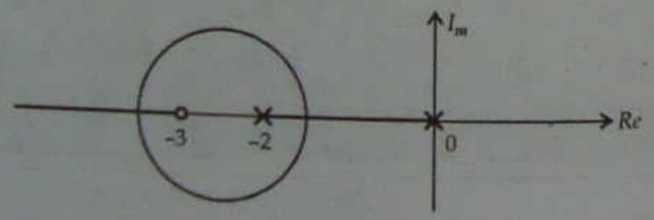


Fig. 5.32(c)

$$G(s)\,H(s) = \frac{K(s+3)}{s(s+2)}$$

It can be seen that by addition of zero towards left, the root locus shifts towards left half. Since root locus shifts towards left half, the relative stability increases.

In conclusion we can say that by addition of poles, the root locus shifts towards imaginary axis and system stability decreases, while by addition of zeros towards left half, the root locus moves away from the imaginary axis & system stability increases.

5.11. INTRODUCTION

Consider the closed loop transfer function

$$\frac{C(s)}{R(s)} = T(s) = \frac{G(s)}{1+G(s)H(s)}$$

The characteristic equation can be obtained by equating the denominator polynomial to zero.

∴ The characteristic equation is $1 + G(s)\,H(s) = 0$

$G(s)\,H(s)$ is the loop transfer function, $L(s)$

where, Loop transfer function zeros : zeros of $L(s)$

Loop transfer function poles : Poles of $L(s)$

Closed loop transfer function poles : zeros of $1 + G(s)\,H(s)$

Zeros of $1 + G(s)\,H(s)$: Roots of the characteristic equation.

For the stability of the system we have already studied the Routh Hurwitz and root locus technique. For stability of the system, the roots of the characteristic equation should not lie in the right half of the s-plane or on the imaginary axis. In other words, we can say that a system is said to be closed loop stable if the poles of the close loop transfer function or zeros of $1 + G(s)\,H(s)$ are all in the left half of s-plane. Similarly, for open loop stability, a system is said to be open loop stable if the poles of the loop transfer function are all in the left half of s-plane.

Now we are going to study the Nyquist stability criterion. The Nyquist stability criterion is based on the principle of argument. The principle of argument is related with the theory of mapping. So first we will study the mapping technique and principle of argument then Nyquist stability criterion.

5.12. MAPPING

Consider a function $D(s) = s^2 + 1$

Any point in the s-plane can be mapped by locating the values of u and v for the given value of s.

let $s_1 = 2 + j\,4$

$$D(s_1) = D(2 + j4) = (2 + j4)^2 + 1 = -11 + j16$$

The mapping is shown in the fig. 5.33.

Therefore, we can say that every point in the s-plane maps into one and only one point in the D(s) plane. Any close contour in the s-plane maps into the closed contour in the $D(s)$ plane. These are the properties of the mapping.

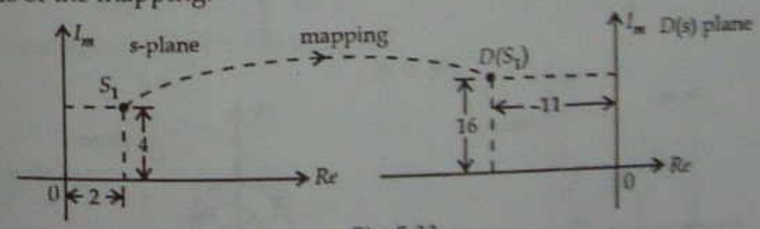


Fig. 5.33.

5.13. MAPPING OF CLOSE CONTOUR AND PRINCIPLE OF ARGUMENT

Consider the characteristic equation

$$D(s) = 1 + G(s)\,H(s) = \frac{K(s-a_1)(s-a_2)\ldots}{(s-b_1)(s-b_2)\ldots} \qquad \ldots(5.20)$$

where $a_1, a_2 \ldots$ are zeros and $b_1, b_2 \ldots$ are poles. let 's' be a complex variable $s = 6 + j\omega$, then, $D(s)$ will also be complex

$$D(s) = U(s) + jV(s) \qquad \ldots(5.21)$$

Now choose a closed path 'C' arbitrarily. The contour 'C' in s-plane is mapped onto the $D(s)$ plane as contour Γ.

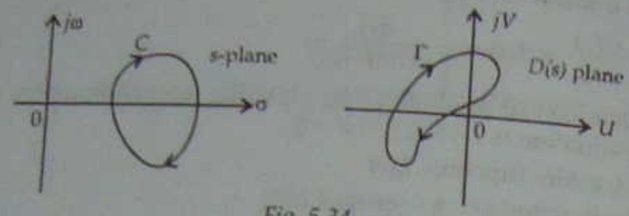


Fig. 5.34.

Now, consider a contour 'C' in s-plane. This contour encircles neither zeros nor poles, then the contour 'Γ' in $D(s)$ plane cannot encircle the origin or the point at infinity. The inside region of the contour 'C' does not have any pole or zero. If we trace the contour 'C' in clockwise direction then the inside region will be on right of the contour. The corresponding region in $D(s)$ plane which contains neither origin nor the point at infinity should also be to the right of the contour Γ.

If the contour 'C' encircles a zero but not poles then the point $D_1(s)$ is given by

$$|D_1(s)| = \frac{s-a_1||s-a_2|\ldots}{s-b_1||s-b_2|\ldots}$$

$$\angle D_1(s) = \angle s-a_1 + \angle s-a_1 \ldots - \angle s-b_1 - \angle s-b_2$$

i.e. $$D_1(s) = r\angle\phi$$

It means that the tip of $D_1(s)$ forms a close contour about the origin in clockwise direction. Similarly, if the contour 'C' encircles 'z' zeros in clockwise direction then the contour in $D(s)$ plane encircles the origin of $D(s)$ plane 'z' times in clockwise direc on. In fig. 5.36(a) the contour 'C' encircles two zeros, the corresponding contour in $D(s)$ plane is shown in fig 5.36(b).

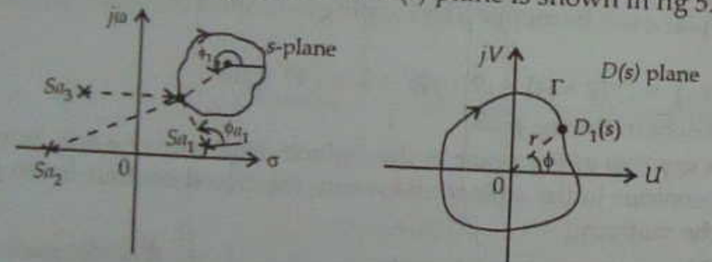


Fig. 5.35.

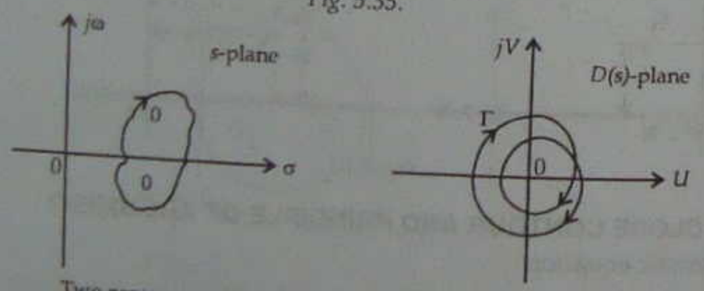


Fig. 5.36.

If any point traces the curve 'C' in clockwise direction and return to the starting point then that phasor will generate an angle -2π.

Consider another case when a pole is encircled by the contour 'c' in clockwise direction then the angle will be $\angle -\phi$. The phasor $s-b_1$ will also generate an angle $-2\pi^*$ but due to in denominator the net phase change will be $+2\pi$. Therefore it means that if the contour 'C' encircles 'P' number of poles in clockwise direction then the corresponding contour Γ will encircle the origin P times in counter clockwise direction.

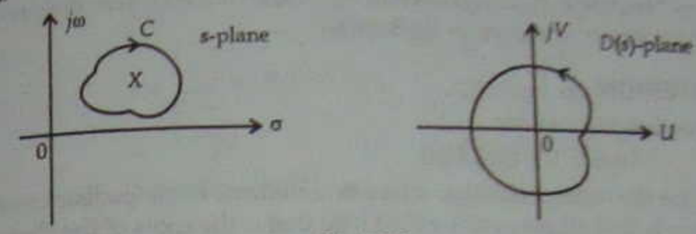


Fig. 5.37.

Now, if contour 'C' encircles both zeros and poles in clockwise direction then the corresponding contour 'Γ' encircles the origin of $D(s)$ plane $Z-P = N$ times in clockwise direction.

This relation between enclosuer of poles and zeros by the contour in s-plane and encirclement of origin by the contour 'Γ' in $D(s)$ plane is known as principle of argument.

Conclusion : $N = Z-P$

(*a*) If $N > 0$ *i.e.* $Z > P$, then N is positive integer. In this case Γ will encircle the origin N times in the same direction as that of contour 'C'.

(*b*) If $N = 0$, *i.e.*, $Z = P$, the contour 'C' will not encircle the origin.

(*c*) If $N < 0$ *i.e.* $Z < P$, then N is negative integer, in this case Γ will encircle the origin N times in opposite direction as that of 'C'.

For example. The contour 'C' encircles three poles and one zero in clockwise then the number of encirclement $N = 1-3 = -2$. Then the contour 'Γ' will encircle the origin two times in counterclockwise direction.

5.14. NYQUIST PATH OR NYQUIST CONTOUR

The overall transfer of a system is given by

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} \qquad \ldots(5.22)$$

The characteristic equation is $1 + G(s)\,H(s) = 0$ ...(5.23)

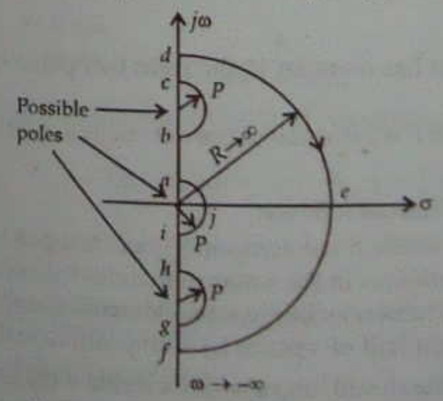


Fig. 5.38(a)

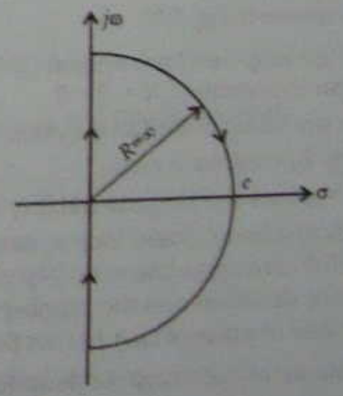


Fig. 5.38(b)

The main purpose in study the stability of the closed loop system is to determine whether the characteristic equation has any root in the right half of s-plane *i.e* whether $C(s)/R(s)$ has any pole in right half of s-plane.

For this purpose, we use a contour in s-plane which encloses the entire right half plane. This contour having the encirclement in clockwise direction and radius 'R' approaches infinity. Thus path or contour is known as Nyquist contour (shown in fig. 5.38*a*). If the system does not have any pole or zero at origin then the contour is shown in fig 5.38(*b*).

5.15. NYQUIST CRITERION

The characteristic equation is given by

$$D(s) = 1 + G(s)\,H(s) \qquad ...(5.24)$$

The zeros of $D(s)$ are the roots of the characteristic equation. For a feedback system the necessary and sufficient condition is that all zeros of $1 + G(s)\,H(s)$ that is the roots of the characteristic equation must have negative real part *i.e.*, they must lie in the left half of s-plane. In order to determine the presence of zeros in right half of s-plane we choose a contour as shown in fig 5.38 called Nyquist contour. Let there are 'Z' zeros and 'P' poles in the right half of s-plane . If this contour is mapped in $D(s)$ plane as Γ_D then Γ_D enclosesthe origin N times (where $N = Z\text{-}P$) in clockwise. Hence the system is unstable because the clockwise encirclement is possible only when there are zeros of $D(s)$ in right half of s-plane.

A feedback system (close loop system) is stable if and only if there is no zeros of $D(s)$ in the right half of s-plane. *i.e.* $Z = 0$

$$\therefore \qquad N = -P$$

Therefore, for a closed loop system to be stable, the number of counter-clockwise encirclement of the origin of $D(s)$ plane by Γ_D should equal the number of right half s-plane poles of $D(s)$ which are the poles of open loop transfer function $G(s)\,H(s)$.

Since $\quad D(s) = 1 + G(s)\,H(s)$

or, $\quad G(s)\,H(s) = D(s) - 1$

The contour Γ_D in $D(s)$ plane can be mapped in $G(s)\,H(s)$ plane. Γ_{GH} by shifting horizontally to the left by one unit. Thus the encirclement of the origin by the contour Γ_D is equivalent to the encirclement of the point $(-1+j0)$ by the contour Γ_{GH} as shown in fig. 5.39.

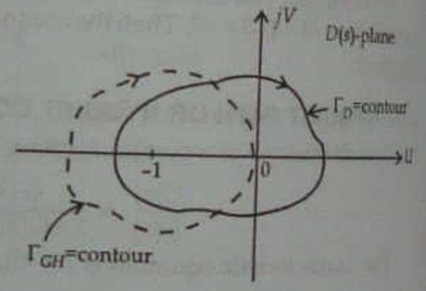


Fig. 5.39.

In most single loop feedback system $G(s)\,H(s)$ has no poles in the right half plane *i.e*, $P = 0$ then closed loop system is stable if $N = P = 0$

So, we can say that A closed loop system with $P = 0$ is stable if the net encirclement of the origin of $D(s)$ plane by Γ_D contour is zero.

Now, we can state the Nyquist stability criterion as follows:

A feedback system or closed loop system is stable if the contour Γ_{GH} of the open loop transfer function $G(s)\,H(s)$ corresponding to the Nyquist contour in the s-plane encircles the point $(-1 + j0)$ in counterclockwise direction and the number of counterclockwise encirclements about the $(-1 + j0)$ equals the number of poles of $G(s)\,H(s)$ in the right half of s-plane *i.e* with positive real parts.

In common case of open loop stable system, the closed loop system is stable if the contour Γ_{GH} of $G(s)\,H(s)$ does not pass through or does not encircle $(-1 + j0)$ point, *i.e* net encirclement is zero.

5.16. GENERAL CONSTRUCTION RULES OF THE NYQUIST PATH

Consider the fig. 5.37

Table 5.1.

Path *ab*	$s = j\omega$	$0 < \omega < \omega_1$	...(5.25)
Path *bc*	$s = \lim_{P\to 0} (j\omega_1 + Pe^{j\theta})$	$-90° \le \theta \le 90°$	...(5.26)
Path *c d*	$s = j\omega$	$\omega_1 \le \omega \le \infty$	...(5.27)
Path *def*	$s = \lim_{R\to\infty} Re^{j\theta}$	$-90° \le \theta \le 90°$	...(5.28)
Path *fg*	$s = j\omega$	$-\infty < \omega < -\omega'$	...(5.29)
Path *gh*	$s = \lim_{P\to 0} (j\omega_1 + Pe^{j\theta})$	$-90° \le \theta \le 90°$	...(5.30)
Path *hi*	$s = j\omega$	$-\omega_1 \le \omega \le 0$	...(5.31)
Path *ija*	$s = \lim_{P\to 0} Pe^{j\theta}$	$-90° \le \theta \le 90°$	...(5.32)

Step 1 : Check $G(s)$ for poles on $j\omega$ axis and at the origin

Step 2 : Using *eq*^{n} 5.25 to *eq*^{n} 5.27 sketch the image of the path $a - d$ in the $G(s)$-plane. If there are no poles on $j\omega$ axis equation (5.26) need not be employed.

Step 3 : Draw the mirror image about the real axis of the sketch resulting from step 2.

Step 4 : Use *eq*^{n} (5.28) plot the image of path def. This path at infinity usually plot into a point in the $G(s)$ -plane.

Step 5 : Use *eq*^{n} (5.32) plot the image of path *ija* (pole at origin)

Step 6 : Connect all curves drawn into the previous steps.

Example 5.41. Determine the closed loop stability of a control system whose open loop transfer function is

$$G(s)\,H(s) = \frac{K}{s(1+sT)} \qquad \text{(type '1' system)}$$

Solution : Given that

$$G(s)\,H(s) = \frac{K}{s(1+sT)}$$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{K}{j\omega(1+j\omega T)} \qquad ...(5.33)$$

Rationalizing the equation (5.33) and separating into real and imaginary parts.

$$G(j\omega)\,H(j\omega) = -\frac{KT}{1+\omega^2T^2} - j\frac{K}{\omega(1+\omega^2T^2)} \qquad ...(5.34)$$

$$\lim_{\omega\to 0} |G(j\omega)H(j\omega) = \infty$$

$$\lim_{\omega\to 0} \angle G(j\omega)H(j\omega) = -90°$$

$$\lim_{\omega\to\infty} |G(j\omega)H(j\omega) = 0$$

$$\lim_{\omega\to\infty} \angle G(j\omega)H(j\omega) = -180°$$

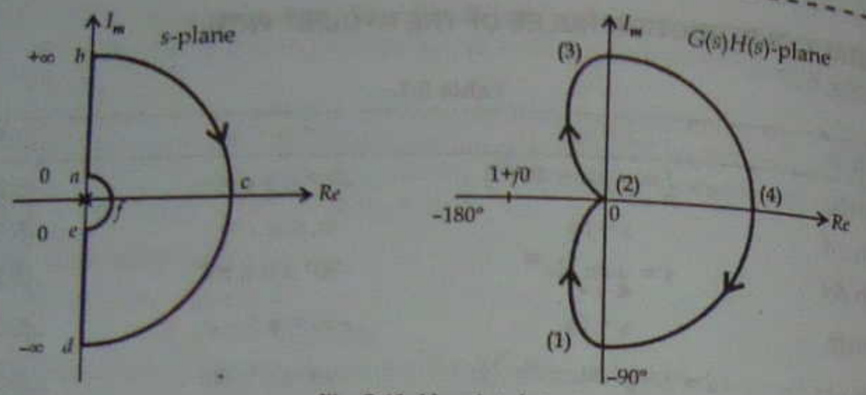


Fig. 5.40. Nyquist plot

The polar plot will lie in third quadrant.

The Nyquist plot is shown in fig 5.40. The part for $0 < \omega < +\infty$ is drawn (1) (2) and for $-\infty < \omega < 0$ is shown by the point (2) (3) which is the mirror image of (1) (2). The semicircular detour around the origin in *s*-plane is mapped into a semicircular path of infinite radius representing a change of phase from $+\pi/2$. to $-\pi/2$.

As the point $(-1 + j0)$ is not encircled by the plot, $N = 0$

$$N = 0 \qquad P = 0$$

$$\because \quad N = Z\text{-}P \qquad \therefore \quad Z = 0$$

The number of zeros or roots of the characteristic equation with positive real part is nil and hence the closed loop system is stable.

Example 5.42. Sketch the Nyquist plot and determine the stability of a unity feedback control system.

$$G(s) = \frac{K}{(1+sT_1)(1+sT_2)} \qquad \text{(Type O system)}$$

Solution : Given that :

$$G(s)\,H(s) = \frac{K}{(1+sT_1)(1+sT_2)}$$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{K}{(1+j\omega T_1)(1+j\omega T_2)} \qquad ...(5.35)$$

$$|G(j\omega)\,H(j\omega)| = \frac{K}{\sqrt{1+\omega^2T^2{}_1}\,\sqrt{1+\omega^2T_2{}^2}} \qquad ...(5.36)$$

$$\angle G(j\omega)H(j\omega) = -\tan^{-1}\omega T_1 - \tan^{-1}\omega T_2 \qquad ...(5.37)$$

$$\lim_{\omega\to 0} |G(j(\omega)H(j\omega) = K$$

$$\lim_{\omega\to 0} \angle G(j\omega)H(j\omega) = 0$$

$$\lim_{\omega\to \infty} |G(j\omega)H(j\omega) = 0$$

$$\lim_{\omega\to \infty} \angle G(j\omega)H(j\omega) = -180°$$

Rationalize the equation (5.35) and separate the real and imaginary parts.

$$\frac{K}{(1+j\omega T_1)(1+j\omega T_2)} = \frac{K(1-\omega^2T_1T_2)}{(1+\omega^2T_1{}^2)(1+\omega^2T_2{}^2)} - j\,\frac{\omega(T_1+T_2)K}{(1+\omega^2T_1{}^2)(1+\omega^2T_2{}^2)} \qquad ...(5.38)$$

Equate the real part to zero, we get

$$\omega = \frac{1}{\sqrt{T_1T_2}}$$

$$G(j\omega)H(j\omega)\Big|_{\omega=\frac{1}{\sqrt{T_1T_2}}} = \frac{KT_1T_2}{T_1+T_2}$$

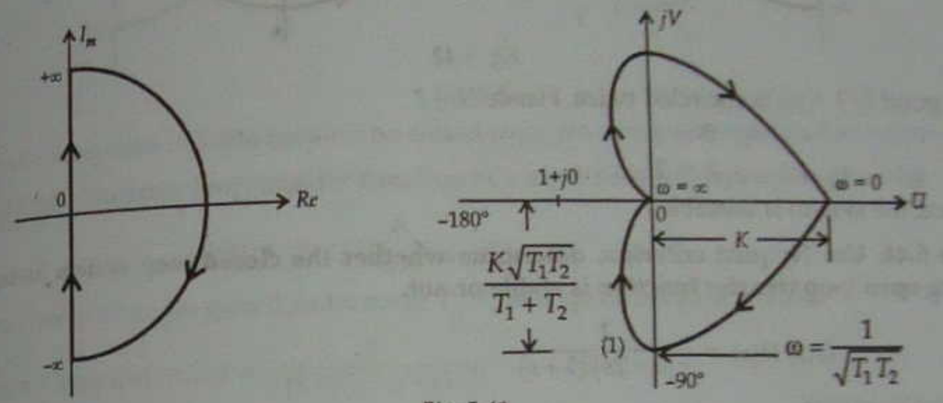


Fig. 5.41.

The plot of $G(j\omega)\,H(j\omega)$ is shown in fig. 5.41. The infinite semicircular arc of the Nyquist contour maps into origin. As the point $(-1 + j0)$ is not encircled by the plot

$$\therefore \quad N = 0$$

$$P = 0$$

$$\therefore \quad Z = 0$$

Hence, the system is stable.

Example 5.43. Using Nyquist criterion, determine the stability of the feedback system which has the following open loop transfer function.

$$G(s)\,H(s) = \frac{K}{s^2(1+sT)} \qquad \text{(Type '2' system)}$$

Solution : Given that

$$G(s)\,H(s) = \frac{K}{s^2(1+sT)}$$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{K}{(j\omega)^2(1+j\omega T)} \qquad ...(5.39)$$

Rationalizing the equation 5.39 and separating the real and imaginary part

$$G(j\omega)\,H(j\omega) = \frac{K}{-\omega^2(1+\omega^2T^2)} + j\,\frac{K}{\omega(1+\omega^2T^2)} \qquad ...(5.40)$$

The Nyquist diagram is shown in the fig (5.42). Because of the double pole at $s = 0$, a small semicircular detour at the origin should be made.

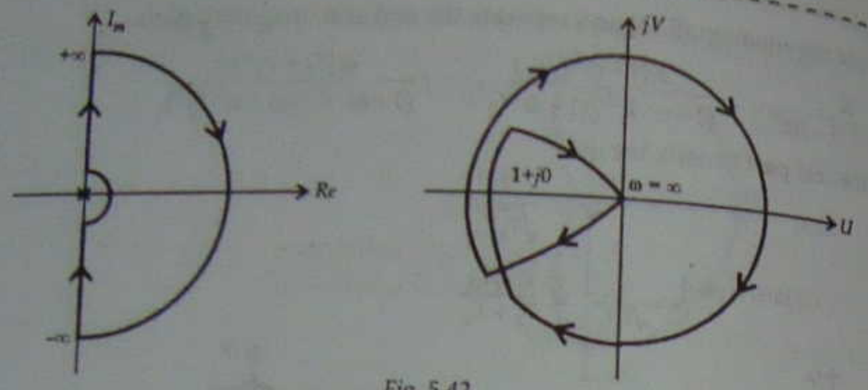


Fig. 5.42.

The point $(-1 + j0)$ is encircled twice. Hence $N = 2$

$$P = 0$$

$$\therefore \quad Z = 2$$

Hence, the system is unstable.

Example 5.44. Use Nyquist criterion, determine whether the closed loop system having the following open loop transfer function is stable or not.

$$G(s)\,H(s) = \frac{1}{s(1+2s)(1+s)}$$

Solution : Given that

$$G(s)\,H(s) = \frac{1}{s(1+2s)(1+s)}$$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{1}{j\omega(1+j2\omega)(1+j\omega)} \qquad ...(5.41)$$

Rationalizing the equation 5.41 and separate the real and imaginary part.

$$G(j\omega)\,H(j\omega) = \frac{-3}{(1+4\omega^2)(1+\omega^2)} - j\frac{1-2\omega^2}{\omega(1+4\omega^2)(1+\omega^2)} \qquad ...(5.42)$$

Equate the imaginary part to zero, we get the point of intersection on real axis

$$1 - 2\omega^2 = 0$$

$$\therefore \quad \omega = 0.707$$

$$\left|G(j\omega)H(j\omega)\right|_{j\omega = 0.707} = 0.66$$

The Nyquist plot is shown in fig. 5.43.

Since, OA = 0.66, the point $(-1 + j0)$ is not encircled

$$\therefore \quad N = 0$$

$$P = 0$$

$$\therefore \quad Z = 0$$

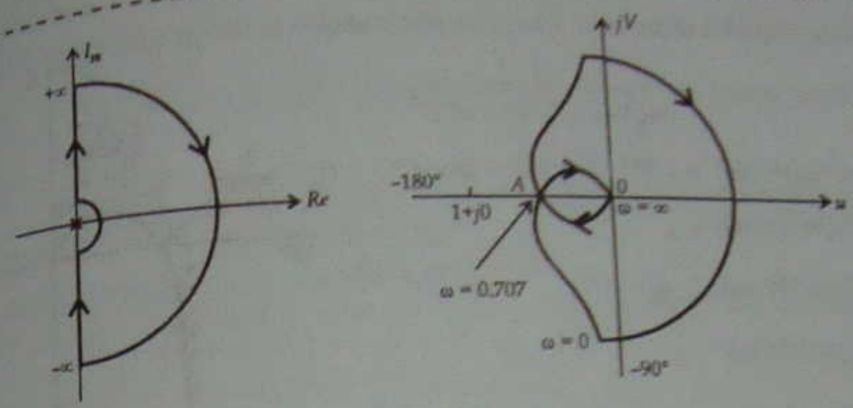


Fig. 5.43.

Hence, the system is stable because no closed loop poles lie in the right half of s-plane.

Example 5.45. The open loop transfer function of a unity feedback system is given by

$$G(s) = \frac{K}{s(1+sT_1)(1+sT_2)}$$

Derive an expression for gain K in terms of T_1, T_2 and specified gain margin.

Solution : Given that $G(s) = \dfrac{K}{s(1+sT_1)(1+sT_2)}$

Put $s = j\omega$

$$G(j\omega) = \frac{K}{j\omega(1+j\omega T_1)(1+j\omega T_2)} \qquad ...(5.43)$$

Rationalizing the eqn 5.43 and separate the real and imaginary parts

$$G(j\omega)\,H(j\omega) = \frac{-K(T_1+T_2)}{(1+\omega^2T_1^2)(1+\omega^2T_2^2)} - j\frac{K(1-\omega^2T_1T_2)}{\omega(1+\omega^2T_1^2)(1+\omega^2T_2^2)} \qquad ...(5.44)$$

Equating the imaginary part to zero of eqn 5.44

$$K(1 - \omega^2 T_1T_2) = 0$$

$$\therefore \quad \omega = \frac{1}{\sqrt{T_1T_2}}$$

$$G(j\omega)H(j\omega)\Big|_{\omega = \frac{1}{\sqrt{T_1T_2}}} = -\frac{KT_1T_2}{T_1+T_2} \qquad ...(5.45)$$

Gain margin $\quad Gm = \dfrac{1}{a}$

Here, $\quad a = \dfrac{KT_1T_2}{T_1+T_2}$

$$\therefore \quad Gm = \frac{T_1+T_2}{K(T_1T_2)}$$

$$K = \frac{1}{Gm}\left[\frac{1}{T_1} + \frac{1}{T_2}\right]$$

This is the required expression. The polar plot is shown in the fig. 5.44

$$|G(j\omega)\,H(j\omega)| = \frac{1}{\omega\sqrt{1+\omega^2T_1^2}\sqrt{1+\omega^2T_2^2}}$$

$$\angle G(j\omega)H(j\omega) = -90^\circ - \tan^{-1}\omega T_1 - \tan^{-1}\omega T_2$$

$$\lim_{\omega\to 0}|G(j\omega)H(j\omega) = \infty$$

$$\lim_{\omega\to 0}\angle G(j\omega)H(j\omega) = -90^\circ$$

$$\lim_{\omega\to\infty}|G(j\omega)H(j\omega) = 0$$

$$\lim_{\omega\to\infty}\angle G(j\omega)H(j\omega) = -270^\circ$$

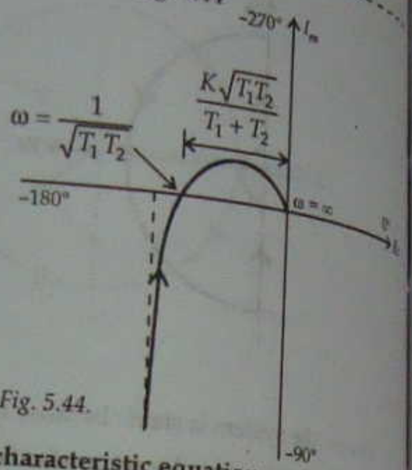


Fig. 5.44.

Example 5.46. Consider a feedback system having the characteristic equation

$$1+\frac{K}{(s+1)(s+1.5)(s+2)} = 0$$

It is desired that all the roots of the characteristic equation have real parts less than -1. Extend the Nyquist stability criterion to find the largest value of K, satisfying this condition.

Solution : Given that

$$G(s)\,H(s) = \frac{K}{(s+1)(s+1.5)(s+2)}$$

Put $s = -1 + j\omega$

$$G(-1+j\omega)\,H(-1+j\omega) = \frac{K}{j\omega(o.5+j\omega)(1+j\omega)} \quad ...(5.46)$$

Rationalizing the equation 5.46 and separate the real and imaginary parts.

$$G(-1+j\omega)\,H(-1+j\omega) = \frac{-1.5K}{(0.25\omega^2)(1+\omega^2)} - j\frac{K(0.5-\omega^2)}{\omega(0.25+\omega^2)(1+\omega^2)}$$

Equate the imaginary term to zero

$$0.5-\omega^2 = 0$$

$$\therefore \quad \omega = 0.707 \text{ rad/sec.}$$

$$\left|G(-1+j\omega)H(-1+j\omega)\right|_{\omega=0.707} = 1.33K$$

As per condition given

$$1.33K < 1 \quad \text{or,} \quad K < 0.7518$$

$\therefore$ Required largest value of K is 0.7518 **Ans.**

Example 5.47. Use Nyquist criterion, determine whether the closed loop system having the following open loop transfer function is stable or not. If not how many closed loop poles lie in the right half s-plane?

$$G(s)\,H(s) = \frac{1+4s}{s^2(1+s)(1+2s)}$$

(R.M.L. University Faizabad 2002, UST)

Solution : Given that

$$G(s)\,H(s) = \frac{1+4s}{s^2(1+s)(1+2s)}$$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{1+j4\omega}{(j\omega)^2(1+j\omega)(1+j2\omega)}$$

The mapping of Nyquist contour is as follows:

1. Along the jω-axis, for various values of 'ω', $G(j\omega)\,H(j\omega)$ is plotted.

ω	0	∞	0.1	1.0	2.0	4.0	10.0	20.0
$G(j\omega)\,H(j\omega)$	$\infty\angle -180^\circ$	$0\angle -270^\circ$	105	1.3	0.21	0.03	0.0019	0.00024
			$\angle -175.22^\circ$	-212.47°	-236.5°	-252.4°	-262.8°	-266.4°

$$M = |G(j\omega)\,H(j\omega)| = \frac{\sqrt{1+16\omega^2}}{\omega^2\sqrt{1+\omega^2}\sqrt{1+4\omega^2}}$$

$$\angle G(j\omega)H(j\omega) = \tan^{-1}4\omega - 180^\circ - \tan^{-1}\omega - \tan^{-1}2\omega$$

The point of intersection of $G(j\omega)\,H(j\omega)$ with real axis is calculated as

$$\angle G(j\omega)H(j\omega) = -180^\circ$$

$$+\tan^{-1}4\omega - 180^\circ - \tan^{-1}\omega - \tan^{-1}2\omega = -180^\circ$$

or, $\quad \tan^{-1}4\omega = \tan^{-1}\omega + \tan^{-1}2\omega$

taking tangent on both sides

$$4\omega = \frac{\omega+2\omega}{1-2\omega^2}$$

$$\therefore \quad \omega = 0.354 \text{ rad/sec.}$$

$$\left|G(j\omega)H(j\omega)\right|_{\omega=0.354} = \frac{1+j4\times 0.354}{(j0.354)^2(1+j0.354)(1+j2\times 0.354)}$$

$$= -10.64$$

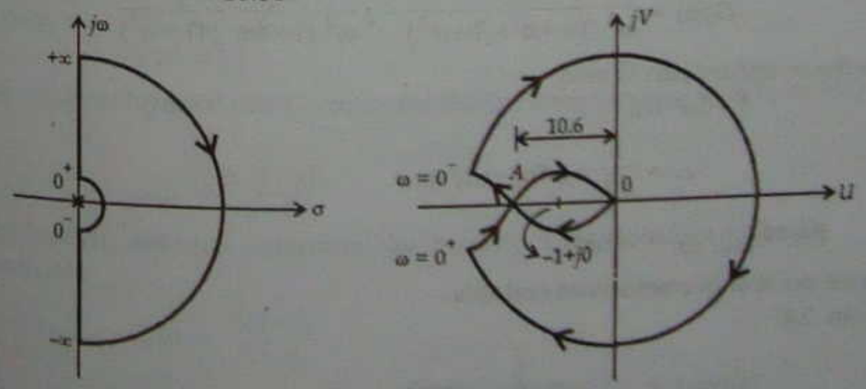


Fig. 5.45.

2. Mapping of infinite semicircle

$$\lim_{R\to\infty}\frac{1+Re^{j\theta}\cdot 4}{(R^2e^{j2\theta})(1+Re^{j\theta})(1+2Re^{j\theta})}=0\angle -270^\circ$$

3. Mapping of $j\omega$-axis from $-\infty$ to $\bar{0}$, which is the mirror image of $j\omega$-axis from $\overset{+}{0}$ to $+\infty$

4. Mapping small semicircle at origin

$$\lim_{P\to 0}\left[\frac{1+4Re^{j\theta}}{(P^2e^{j2\theta})(1+Pe^{j\theta})(1+2Pe^{j\theta}}\right]=\infty(\angle 180^\circ\to\angle 0^\circ\to\angle -180^\circ)$$

The complete Nyquist plot is shown in the fig. 5.45.

Since, $OA = -10.64$, hence $(-1+j0)$ encircles two times in clockwise direction.

$$\therefore \quad N = 2$$
$$P = 0$$
$$N = Z - P$$
$$2 = Z - 0$$
$$\therefore \quad Z = 2$$

Hence, two roots of the characteristic equation lies in right half of s-plane. The closed loop system is unstable.

Example 5.48. A unity feedback system has open loop transfer function.

$$G(s) = \frac{1}{s(1+2s)(1+s)}$$

Sketch Nyquist plot for the system and therefore obtain the gain margin and the phase margin. (GATE 1992)

Solution : Given that $G(s) = \dfrac{1}{s(1+2s)(1+s)}$

Put $s = j\omega$

$$G(j\omega) = \frac{1}{j\omega(1+j2\omega)(1+j\omega)} \qquad ...(5.47)$$

Rationalizing the equation (5.47) and separate the real and imaginary parts.

$$G(j\omega) = \frac{-3\omega^2}{\omega^2(1+4\omega^2)(1+\omega^2)} - j\frac{\omega(1-2\omega^2)}{\omega^2(1+4\omega^2)(1+\omega^2)}$$

Equate the imaginary part to zero

$$1-2\omega^2 = 0$$

$$\therefore \quad \omega = \frac{1}{\sqrt{2}} = 0.707 \text{ rad/sec.}$$

$$|G(j\omega)|_{\omega=0.707} = -0.667$$

This is the point of intersection on real axis.

From eqn. 5.47

$$|G(j\omega)| = \frac{1}{\omega^2\sqrt{1+4\omega^2}\sqrt{1+\omega^2}}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}2\omega - \tan^{-1}\omega$$

$$\lim_{\omega\to 0}|G(j\omega) = \infty, \qquad \lim_{\omega\to 0}\angle G(j\omega) = -90^\circ$$

$$\lim_{\omega\to\infty}|G(j\omega) = 0, \qquad \lim_{\omega\to\infty}\angle G(j\omega) = -270^\circ$$

The Nyquist contour and Nyquist plot are shown in fig. 5.46

The mapping is as follows:

1. For different values of ω between 0+ and $+\infty$

2. Mapping of infinite semicircle

$$\lim_{R\to\infty}\frac{1}{Re^{j\theta}(1+2Re^{j\theta})(1+Re^{j\theta})} = 0\ \angle -270^\circ$$

3. Mapping of $j\omega$ axis from $\bar{0}$ to $-\infty$, which is the mirror image of $j\omega$ axis from 0+ to $+\infty$.

4. Mapping of semicircle at origin

$$\lim_{P\to\infty}\left[\frac{1}{Pe^{j\theta}(1+2Pe^{j\theta})(1+Pe^{j\theta})}\right] = \infty\angle -90^\circ$$

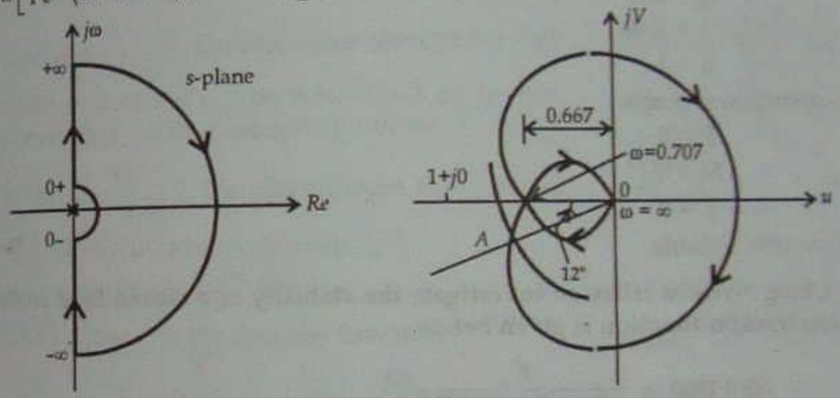


Fig. 5.46.

$$\text{Gain margin} = 20\log\frac{1}{0.667}$$
$$= 3.51$$

For phase margin draw the unit circle which intersect the plot at A. Join the point 0 to A and measure the angle.

$$\therefore \quad \text{P.M} = 12$$

Example 5.49. Draw the Nyquist plot for unity feedback control system, with open loop transfer function

$$G(s) = \frac{K(1-s)}{s+1}$$

Using Nyquist stability criterion, determine the stability of the closed loop system.

Solution : Given that

$$G(s) = \frac{K(1-s)}{(1+s)}$$

Put $s = j\omega$

$$G(j\omega) = \frac{K(1-j\omega)}{(1+j\omega)} \quad |G(j\omega)| = \frac{K\sqrt{1+\omega^2}}{\sqrt{1+\omega^2}}$$

$\angle G(j\omega) = -2\tan^{-1}\omega$

$\omega = 0 \quad |G(j\omega)| = K$

$\angle G(j\omega) = 0°$

$\omega = 1 \quad |G(j\omega)| = K$

$\angle G(j\omega) = -90°$

$\omega = \infty \quad |G(j\omega)| = K$

$\angle G(j\omega) = -180°$

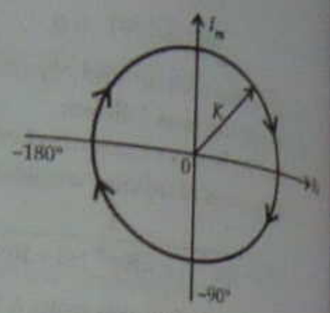


Fig. 5.47.

The Nyquist plot is shown in fig. 5.47.

If $K > 1$: No open loop pole on right half of s-plane $P = 0$.

Number of encirclement of point $(-1 + j0)$ $N = 1$

$\therefore \quad N = Z - P$

$1 = Z - 0$

$\therefore \quad Z = 1$

Hence, the system is unstable

If $K < 1$: $\quad P = 0$

$N = 0$

$\therefore \quad Z = 0$

Hence, the system is stable.

Example 5.50. Using Nyquist criterion investigate the stability of a closed loop control system whose open loop transfer function is given below

$$G(s)\,H(s) = \frac{K}{s(1+sT_1)(1+sT_2)}$$

(R.M.L. University Faizabad, 2002, control system)

Solution : Given that

$$G(s)\,H(s) = \frac{K}{s(1+sT_1)(1+sT_2)}$$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{K}{j\omega(1+j\omega T_1)(1+j\omega T_2)} \qquad ...(5.48)$$

Rationalizing the equation 5.48 and separate the real and imaginary part

$$G(j\omega)\,H(j\omega) = \frac{-K(T_1+T_2)}{(1+\omega^2T_1^2)(1+\omega^2T_2^2)} - j\frac{K(1-\omega^2T_1T_2)}{\omega(1+\omega^2T_1^2)(1+\omega^2T_2^2)} \qquad ...(5.49)$$

The value of 'ω' at the point of intersection of plot with real axis can be obtained by equating the imaginary part to zero.

$$\frac{K(1-\omega^2T_1T_2)}{\omega(1+\omega^2T_1^2)(1+\omega^2T_2^2)} = 0$$

$$1 - \omega^2T_1T_2 = 0$$

or,

$$\omega = \frac{1}{\sqrt{T_1T_2}}$$

or,

$$|G(j\omega)H(j\omega)|_{\omega=\frac{1}{\sqrt{T_1T_2}}} = -\frac{KT_1T_2}{T_1+T_2}$$

For stable system

$$\frac{KT_1T_2}{T_1+T_2} < 1$$

or,

$$K < \frac{T_1+T_2}{T_1T_2}$$

(a) When the gain K is less than $\frac{T_1+T_2}{T_1T_2}$, the point $(-1 + j0)$ is not encircled by $G(j\omega)\,H(j\omega)$ plot and the system is stable

(b) When $K = \frac{T_1+T_2}{T_1T_2}$, the plot passes through $(-1 + j0)$ which indicates the system has roots on $j\omega$-axis. Hence, the system is marginally stable.

(c) When $K > \frac{T_1+T_2}{T_1T_2}$, the plot encircles the point $(-1 + j0)$. Hence the system is unstable.

The plot is shown in fig. 5.48.

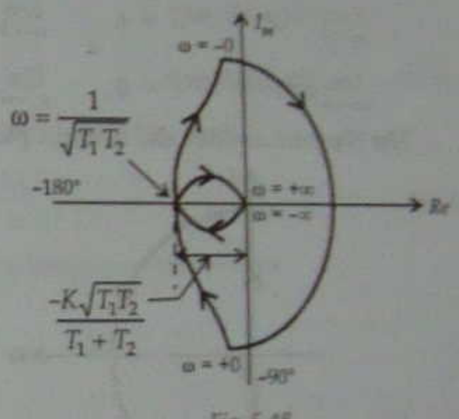


Fig. 5.48.

Example 5.51. Consider the transfer function

$$G(s)\,H(s) = \frac{60}{(s+1)(s+2)(s+5)}$$

Using Nyquist stability criterion determine whether the closed loop system is stable or not.

(KNIT, Sultanpur 1998-99)

Solution : Given that

$$G(s)\,H(s) = \frac{60}{(s+1)(s+2)(s+5)}$$

Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{60}{(1+j\omega)(2+j\omega)(5+j\omega)} \qquad ...(5.50)$$

Rationalizing the equation 5.50 and separate the real and imaginary parts.

$$G(j\omega)\,H(j\omega) = \frac{60(10-8\omega^2)}{(1+\omega^2)(4+\omega^2)(25+\omega^2)} - j\frac{60(17\omega-\omega^3)}{(1+\omega^2)(4+\omega^2)(25+\omega^2)}$$

Equate the real part to zero

$$\frac{60(10-8\omega^2)}{(1+\omega^2)(4+\omega^2)(25+\omega^2)} = 0$$

$$\omega = 1.25 \text{ rad/sec.}$$

Point of intersection on imaginary axis

$$|G(j\omega)H(j\omega)|_{\omega=1.25} = -3.057$$

Equate the imaginary part to zero

$$\frac{60(17\omega-\omega^3)}{(1+\omega^2)(4+\omega^2)(25+\omega^2)} = 0$$

$$\therefore \quad \omega = 4.123 \text{ rad/sec.}$$

Point of intersection of plot on real axis

$$|G(j\omega)H(j\omega)|_{\omega=4.123} = -0.47$$

From equation 5.50

$$\lim_{\omega\to 0}|G(j\omega)H(j\omega)| = 6, \qquad \lim_{\omega\to 0}\angle G(j\omega)H(j\omega) = 0°$$

$$\lim_{\omega\to \infty}|G(j\omega)H(j\omega)| = 0, \qquad \lim_{\omega\to \infty}\angle G(j\omega)H(j\omega) = -270°$$

The Nyquist contour and Nyquist plot are shown in fig. 5.49

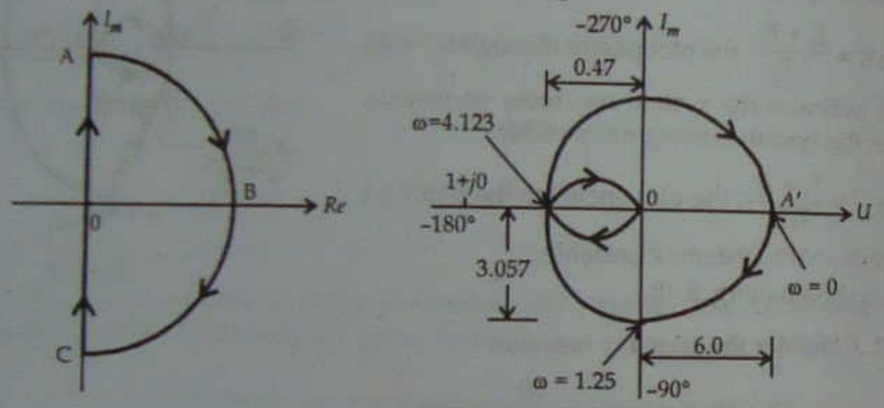


Fig. 5.49.

The mapping is as follows:

1. Part OA in the s-plane is the $j\omega$-axis from $\omega = 0$ to $\omega = \infty$ and maps into the polar plot $A'O$ in GH-plane.
2. The infinite semicircle ABC maps into origin in GH-plane.
3. Part CO which is the negative part of $j\omega$ axis, maps into OA' From Nyquist plot, the point $(-1 + j0)$ is not encircled by the plot

$$\therefore \quad N = 0$$

No pole in the right half

$$\therefore \quad P = 0$$

$$N = Z - P$$

$$\therefore \quad Z = 0$$

Therefore, no zero in right half and hence the system is stable.

Example 5.52. The open loop transfer function of a control system is given by

$$G(s) = \frac{K}{s(s+2)(s+10)}$$

Determine the value of K so that the system may be stable with

a. **gain margin equal to 6 *db***

b. **P.M equal to 45°**

Solution :

$$G.M = 20\log_{10}\left(\frac{1}{a}\right)$$

Where 'a' is the point of intersection of Nyquist plot on negative real axis.

$$6 = 20\log_{10}{}^{1}/_{a}$$

$$\therefore \quad a = 0.501$$

$$\therefore \quad G(s) = \frac{K}{s(s+2)(s+10)}$$

Since,

Put $s = j\omega$

$$G(j\omega) = \frac{K}{j\omega(2+j\omega)(10+j\omega)} \qquad ...(5.51)$$

Rationalizing the equation 5.51 and separate the real and imaginary parts.

$$G(j\omega) = \frac{-12K\omega^2}{\omega^2(4+\omega^2)(100+\omega^2)} - j\frac{K(20\omega-\omega^3)}{\omega^2(4+\omega^2)} \qquad ...(5.52)$$

To get the point of intersection on real axis, equate the imaginary part to zero.

$$\frac{K(20\omega-\omega^3)}{\omega^2(4+\omega^2)(100+\omega^2)} = 0$$

$$\therefore \quad \omega = 4.47 \text{ rad/sec.}$$

$$|G(j\omega)|_{\omega=4.47} = -0.0041\,K^*$$

Put $\omega = 4.47$ rad/sec. in eq^n no. 5.52

$$0.0041K = 0.501$$

$$\therefore \quad a = 0.501$$

$$\therefore \quad \boxed{K = 122}$$

(ii) From equation 5.51

$$P.M = \angle G(j\omega)H(j\omega) + 180°$$

$$45° = -90° - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{10} + 180°$$

$$\tan^{-1}\frac{\omega}{2} + \tan^{-1}\frac{\omega}{10} = 45°$$

Taking tangent on both sides

$$\frac{\frac{\omega}{2}+\frac{\omega}{10}}{1-\frac{\omega}{2}\cdot\frac{\omega}{10}} = 1$$

$$\therefore \quad \omega^2 + 12\omega - 20 = 0$$

$$\therefore \quad \omega = -13.48, 1.48$$

Consider the positive value of ω *i.e.* $\omega = 1.48$ rad/sec.

$$|G(j\omega| = 1$$

$$\frac{K}{j\omega(2+j\omega)(10+j\omega)} = 1$$

$$\frac{K}{j1.48(2+j1.48)(10+j1.48)} = 1$$

$$\boxed{K = 37.22}$$

5.17. ANALYSIS OF STABILITY BY LYAPUNOV'S DIRECT METHOD

5.17.1 Introduction

This method is useful for determing the stability of the non-linear system. This method is also applicable for linear systems. In this method the stability of the system can be determined without actually solving the differential equation, that's why it is called direct method.

The basic concept of this, is that the energy stored in a stable system cannot increase with time. If we obtain a function $V(x)$ which is suitable function of energy and examine it as a function of time, then for stability the function of this type approaches zero as the time approaches infinity.

5.17.2 The Concept of Definiteness

Let $V(x)$ is a real scalar function.

The scalar function $V(s)$ is said to be positive definite if the function $V(x)$ has always positive sign in the given region about the origin, except only at the origin where it is zero. or in other words the scalar function $V(x)$ is positive definite in the given region if $V(x) > 0$ for all non zero states x in the region and $V(0) = 0$.

The scalar function $V(x)$ is said to be negative definite if the function $V(x)$ has always negative sign in the given region about the origin, except only at the origin where it is zero or a scalar function $V(x)$ is said to be negative definite if $-V(x)$ is positive definite.

For example :

$v(x) = x_1^2 + x_2^2 + \ldots + x_n^2$ is positive definite

$v(x) = -x_1^2 - (x_1 + x_2)^2$ is negative definite

$v(x) = -x_1^2 - (3x_1 + 2x_2)^2$ is negative definite

5.17.3. Lyapunov Stability Theorem.

THEOREM 5.17.4. suppose a system is described by

$$\dot{x} = f(x)$$

If there exists a scalar function $V(x)$ which is real, continuous and has continuous first partial derivatives with

$$V(x) > 0 \text{ for } x \neq 0$$

$$V(0) = 0$$

$$\dot{V}(x) < 0 \text{ for all } x \neq 0$$

then the system is asymptotically stable.

$V(x)$ is the Lyapunov function.

Example 5.53. The system is given by

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 - x_2^3$$

Investigate the system by Lyapunov's method using

$$v = x_1^2 + x_2^2$$

Solution :

$$\dot{V}(x) = \frac{\partial V}{\partial x_1}\dot{x}_1 + \frac{\partial V}{\partial x_2}\dot{x}_2$$

$$\dot{V}(x) = 2x_1\dot{x}_1 + 2x_2\dot{x}_2$$

Put the values of $\dot{x}_1$ and $\dot{x}_2$

$$\dot{V}(x) = 2x_1x_2 + 2x_2(-x_1 - x_2^3)$$

$$= -2x_2^4$$

Hence, $\dot{V}(x) < 0$ for all non zeros values of x_2 *i.e.* negative definite, hence the system is asymptotically stable.

Example 5.54. Investigate the system described by

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Solution :

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 - x_2$$

let $V(x) = x_1^2 + x_2^2$

$$\dot{V}(x) = \frac{\partial V}{\partial x_1}\dot{x}_1 + \frac{\partial V}{\partial x_2}\dot{x}$$

$$\dot{V}(x) = 2x_1\dot{x}_1 + 2x_2\dot{x}_2$$

Now, substitute the values of $\dot{x}_1$ and $\dot{x}_2$

$$\dot{V}(x) = 2x_1x_2 + 2x_2(-x_1 - x_2)$$

$$\dot{V}(x) = -2x_2^2$$

$\dot{V}(x)$ = negative definite

Hence, the system is asymptotically stable

THEOREM 5.17.5. Suppose a system is described by

$$\dot{x} = f(x)$$

If there exists a scalar function $V(x)$ which is real, continuous and has continuous first partial derivatives with

$$V(x) \geq 0 \text{ for } x \neq 0$$

$$V(0) = 0$$

$$\dot{V}(x) > 0 \text{ for all } x \neq 0$$

then the system is unstable.

5.17.6. The Direct Method of Lyapunov and Linear System

Let the system is given by

$$\dot{x} = Ax \quad \ldots(5.53)$$

Select the Lyapunov function as

$$V(x) = x^T Px \quad \ldots(5.54)$$

$$\dot{V}(x) = \dot{x}^T Px + x^T P\dot{x} \quad \ldots(5.55)$$

Substiture the value of x from 5.53

$$\dot{V}(x) = (Ax)^T Px + x^T PAx = x^T A^T Px + x^T PAx$$
$$= x^T[A^T P + PA]x$$
$$\dot{V}(x) = x^T Qx$$

where, $-Q = [A^T P + PA]$ ...(5.56)

where Q is positive definite matrix. Select a positive definite Q and findout P from equation (5.57) if P is positive definite then the system will be stable. ...(5.57)

PROCEDURE. Step1 : Select Q as positive definite

Step 2 : Obtain P from the equation 5.57 for this we will have to solve $\frac{n(n+1)}{2}$ number of equations where 'n' is the order of matrix A.

Step 3 : Using Sylvester theorem (Sylvester theorem is given at the end of this chapter) determine the definitness of P. If P is positive definite the system is stable otherwise unstable. Generally Q can be taken as identity matrix.

Example 5.55. Determine the stability of the system

$$\dot{x} = Ax$$

$$A = \begin{bmatrix} -1 & -2 \\ +1 & -4 \end{bmatrix}$$

Solution : Given that

$$A = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}$$

$$\therefore \quad A^T = \begin{bmatrix} -1 & 1 \\ -2 & 4 \end{bmatrix}$$

Select $$Q = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

we know that $Q = A^T P + PA$

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}\begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} \quad ...(5.58)$$

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -2p_{11} + 2p_{12} & -2p_{11} - 5p_{12} + p_{22} \\ -2p_{11} - 5p_{12} + p_{22} & -4p_{12} - 8p_{22} \end{bmatrix} \quad ...(5.59)$$

Take $p_{12} = p_{21}$, this is because solution matrix P is known to be a positive definite real symmetric matrix for a stable system

$$-2p_{11} + 2p_{12} = -1 \quad ...(5.60)$$
$$-2p_{11} - 5p_{12} + p_{22} = 0 \quad ...(5.61)$$
$$-4p_{12} - 8p_{22} = -1 \quad ...(5.62)$$

Solving the equation 5.60 and 5.61 we get

$$7p_{12} - p_{22} = -1 \quad ...(5.63)$$

Now solving (5.62) and (5.63) we get

$$p_{12} = -7/60 \therefore p_{21} = -7/60$$

From (5.60) we get

$$p_{11} = 23/60$$

From (5.62)

$$p_{22} = 11/60$$

$$\therefore \quad P = \begin{bmatrix} 23/60 & -7/60 \\ -7/60 & 11/60 \end{bmatrix} = \frac{1}{60}\begin{bmatrix} 23 & -7 \\ -7 & 11 \end{bmatrix}$$

Use sylvestor's theorem.

(a) $23 = r_{11} > 0$

(b) $23 \times 11 - 49 = 204 > 0$

Hence, P is positive definite.

Therefore, the origin of the system under consideration is asymptotically stable in-the-large.

Example 5.56. Consider a non-linear system described by the equations

$$\dot{x}_1 = -3x_1 + x_2$$
$$\dot{x}_2 = -x_1 - x_2 - x_2^3$$

Investigate the sbtability of the equilibrium state.

Solution : Let $V(x) = x_1^2 + x_2^2$

$$\dot{V}(x) = \frac{\partial V}{\partial x_1}\dot{x}_1 + \frac{\partial V}{\partial x_2}\dot{x}_2$$

$$\dot{V}(x) = 2x_1\ \dot{x}_1 + 2x_2\ \dot{x}_2$$

Substitute the values of $\dot{x}_1$ and $\dot{x}_2$

$$\dot{V}(x) = 2x_1(-3x_1 + x_2) + 2x_2(-x_1 - x_2 - x_2^3)$$
$$\dot{V}(x) = -6x_1^2 - 2x_2^2 - 2x_2^4 = \text{negative definite}$$

Hence, the system is asymptotically stable in-the-large.

Example 5.57. Check the stability of the system described by

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -x_1 - x_1^2 x_2$$

Solution : Let $V(x) = x_1^2 + x_2^2$

$$\dot{V}(x) = \frac{\partial V}{\partial x_1}\dot{x}_1 + \frac{\partial V}{\partial x_2}\dot{x}_2$$

$$\dot{V}(x) = 2x_1\ \dot{x}_1 + 2x_2\ \dot{x}_2$$

Substitute the values of $\dot{x}_1$ and $\dot{x}_2$

$$\dot{V}(x) = 2x_1x_2 + 2x_2(-x_1 - x_1^2x_2)$$
$$\dot{V}(x) = -2x_1^2x_2^2 = \text{negative definite}$$

Hence, the system is asymptotically stable.

Example 5.58. Examine the stability of the system described by the equation

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -6x_1 - 5x_2$$

Solution : Let $V(x) = x_1^2 + x_2^2$

$$\dot{V}(x) = \frac{\partial V}{\partial x_1}\dot{x}_1 + \frac{\partial V}{\partial x_2}\dot{x}_2$$

$$\dot{V}(x) = 2x_1\dot{x}_1 + 2x_2\dot{x}_2$$

Put the values of $\dot{x}_1$ and $\dot{x}_2$

$$\dot{V}(x) = 2x_1(x_2) + 2x_2(-6x_1 - 5x_2)$$

$$= -10x_1x_2 - 10x_2^2 = \text{negative definite}$$

Hence, the system is asymptotically stable.

THEOREM 5.17.7. If theorem 5.16.4 is satisfied and in addition

$$V(x) \to \infty$$

$$\text{as } ||x|| \to \infty$$

then the system is asymptotically stable-in-the large at origin.

5.18. SYLVESTER'S THEOREM

This theorem states that the necessary and sufficient conditions that the quadratic form $V(x)$ be positive definite are that all the svccessive principal minors of P be positive *i.e.*

$$p_{11} > 0$$

$$\begin{vmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{vmatrix} > 0$$

$$\begin{vmatrix} P_{11} \cdots \\ p_{1n} \cdots \end{vmatrix} > 0$$

(*a*) $V(x)$ is negative definite if $-V(x)$ is positive definite.

(*b*) $V(x) = x^T Px$ is positive semidefinite if P is singular and all the principal minors are non negative

Example 5.59. Determine whether the following quadratic form is negative definite.

$$-Q = x_1^2 + 3x_2^2 + 11x_3^2 - 2x_1x_2 + 4x_2x_3 + 2x_1x_3$$

Solution :
$$[x_1\ x_2\ x_3]\begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 11 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Note. The middle matrix is found as the coeff of

$$\begin{bmatrix} x_1^2 & \frac{x_1x_2}{2} & \frac{x_1x_3}{2} \\ \frac{x_2x_1}{2} & x_2^2 & \frac{x_2x_3}{2} \\ \frac{x_1x_3}{2} & \frac{x_2x_3}{2} & x_3^2 \end{bmatrix}$$

Now, $p_{11} = 1 > 0$

$$\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix} = 2 > 0; \quad \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 11 \end{bmatrix} = 11 > 0$$

Hence $-Q$ is positive definite therefore quadratic form is negative definite.

5.19. HOW TO OBTAIN OPEN LOOP TRANSFER FUNCTION $G(s)\ H(s)$ FROM CHARACTERISTIC EQUATION

The following steps are required to obtain the open loop transfer function $G(s)\ H(s)$.

Step 1 : Write all the terms of 's' together without K.

Step 2 : Write all the terms of K together

Step 3 : Divide the equation by polynomial containing the terms of s only without K.

Step 4 : Compare with $1 + G(s)\ H(s) = 0$

For example consider the characteristic equation

$$S^3 + 10\,S^2 + 21\,S + KS + 13\,K = 0$$

Step 1 :
$$(S^3 + 10\,S^2 + 21\,S) + K(S + 13) = 0$$

Step 2 : Divide the equation by $S^3 + 10\,S^2 + 21\,S$, we get

$$1 + \frac{K(s+13)}{S^3 + 10s^2 + 21s} = 0$$

Step 3 : Compare with $1 + G(s)\ H(s) = 0$

$$\therefore \quad G(s)\ H(s) = \frac{K(s+13)}{S^3 + 10\,s^2 + 21\,s}$$

$$\therefore \quad G(s)\ H(s) = \frac{K(s+13)}{s(s+3)\,(s+7)}$$

SUMMARY

A linear time invariant system is stable if (*a*) the system is excited by a bounded input, the output is bounded (*b*) in the absence of the input the output tends towards zero.

A linear time invariant system is unstable if (*a*) the system is excited by a bounded input, it produces unbounded output (*b*) in the absence of the input, output may not tends towards zero.

If one or more pairs of simple non-repeated roots are located on the imaginary axis of s-plane but there are no roots in right half of s-plane the response will be undamped sinusoidal oscillations of constant amplitude and frequency as shown in fig 5.7.

A linear time invariant system is said to be conditionally stable if for certain condition of a particular parameter of the system, its output is bounded otherwise for other conditions the output becomes unbounded and system becomes unstable. Such system is called conditionally stable.

Routh-Hurwitz method is used to determine the location of poles of a characteristic equation in left and right half of s-plane without actually solving the equation.

Routh-Hurwitz criterion states that the system is stable is stable if and only if all the elements in the first column have the same algebraic sign. If all elements are not of the same sign then the number of sign changes of the elements in first column equals the number of roots of the characteristic equation in the right half of s-plane.

Advantages of Routh-Hurwitz criterion:

1. Stability of the system can be determine without solving the equation.
2. Relative stability can be easily determined.
3. By this method we can determine the range of K for stability.

4. By this method we can determine the point of intersection of root locus with imaginary axis.
5. For unstable system it gives the number of roots of characteristic equation having positive real part.

Limitations of Routh-Harwitz criterion:

1. This is applicable only for linear systems.
2. It does not provide the exact location of closed loop poles in left or right half of s-plane.
3. It is valid only for real coefficients of the characteristic equation.

The locus of the roots of the characteristic equation when gain is varied from zero to infinity is called root locus.

The root locii starts form open loop poles and terminates on open loop zeros or on infinity.

The asymptotes are the guide for the branches approaching to infinity.

The point of intersection of asymptotes with real axis is called centroid.

Angle conditions used to test whether a point in s-plane is on the root locus or not.

Magnitude conditions gives the value of K corresponding to any point which is on the root locus.

Nyquist stability criterion states that a closed loop system is stable if the contour GH of the open loop transfer function $G(s)\,H(s)$ corresponding to the Nyquist contour in s-plane encircles the point $(-1 + j0)$ in counterclockwise direction and the number of counterclockwise encirclement about $(-1 + j0)$ equals the number of poles of $G(s)\,H(s)$ in the right half of s-plane.

The closed loop system is stable if the contour GH of $G(s)\,H(s)$ does not pass through or does not encircles $(-1 + j0)$ point *i.e.*, net encirclement is zero.

Lyapunou stability Theorem

If a system is described by

$$\dot{x} = f(x)$$

If there exists a scalar function $V(x)$ which is real, continuous and has continuous first partial derivatives with

$$V(x) > 0 \quad \text{for} \quad x \neq 0$$
$$V(0) = 0$$
$$\dot{V}(x) < 0 \quad \text{for all} \quad x \neq 0$$

then the system is asymptotically stable.

EXERCISE

5.1. Sketch the root locus for open loop transfer function

$$\frac{K(s+1)}{s(s^2+2s+2)}$$

5.2. Consider the closed loop feedback control system shown in fig. 5.50. Using routh Hurwitz criterion determine the range of K for which the system is stable. Find also the number of roots of the characteristic equation that are in the right half of s-plane for $K = 0.5$. *(Calir 96)*

5.3. Sketch the root locus for the given system (fig. 5.51). Determine the value of K at which the system becomes unstable.

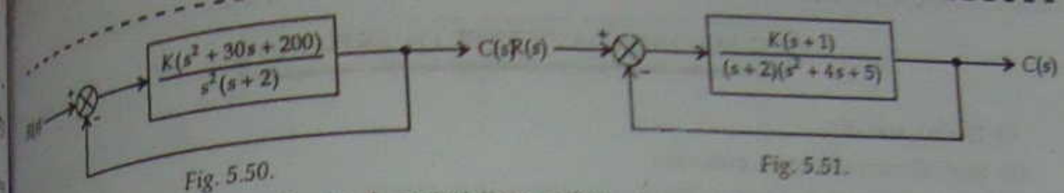


Fig. 5.50.

Fig. 5.51.

5.4. Sketch the root locus plot for the characteristic equation

$$(s^2 + 8s + 25) + K\,S(s + 4) = 0$$

5.5. The characterisic equation of a system is given by

$$s^4 + 20s^3 + 15s^2 + 2s + K = 0$$

(i) Determine range of K, for system to stable.
(ii) Can system be marginally stable ? If so, find the required value of K and the frequency of sustained oscillation.

5.6. Apply Routh's criterion, determine the number of roots that lie in the right half-plane of the following characteristic equation

(i) $s^4 + 6s^5 + 10s^4 + 12s^3 + 13s^2 - 18s - 24 = 0$
(ii) $s^3 + 2s^2 + 4s + 8 = 0$

5.7. The characteristic equation for a control system is

$$s(s^2 + 5s + 20) + K(s + 2) = 0$$

Use Routh's criterion to determine the range of value of K such that the system is stable.

5.8. Sketch the Nyquist plot for $0 < \omega < \infty$ and find the phase crossover frequency for

$$G(s)\,H(s) = \frac{200K}{s(s+2)(s+10)}$$

5.9. For open transfer function $G(s)\,H(s) = \dfrac{K(s+2)}{(s+1)(s-3)}$, sketch the Nyquist plot for determine whether the system is stable or not.

5.10. Point out the limitations of Routh Hurwitz stability criterion. The open loop transfer function of a unity feedback control system is given by $G(s) = \dfrac{K}{s(s^2 + 8s + T)}$. Using Routh-Hurwitz criterion determine the values of K & T which corresponds to a stable system.

Also determine the new values of K & T if it is required that all the roots of the system characteristic equation lie in the region to the left of the line $s = -1$. *(UPSC 1974)*

(*Hint* : For second part modified the characterisitc equation by putting $s = s + 1$ in the characteristic equation & then apply routh Hurwitz criterion).

ANSWERS

5.2. Stable for $0 < K < 4.67$,
For $K = 0.5$, two roots in right half
5.5. $0 < K < 1.49$, $\omega = 0.316$ rad/sec.
5.8. Phase cross over frequency $= 4.472$ rad/sec.
5.9. Stable if $K > 2$
5.10. $K > 0$ and $T > K/B$
For second part $200 + 10T - K > 0$
$9 + T + K > 0$.

SEMI-OBJECTIVE TYPE QUESTIONS

(i) Define stability (Article 5.1)
(ii) State Routh Hunwitz criterion (Article 5.4)
(iii) State necessary but not sufficient conditions for stability. (Article 5.3)
(iv) Define marginally stable system. (Article 5.3)
(v) What are the two special cases, which occur in Routh's table? How are those cases handled? (Article 5.5)
(vi) Define relative stability (Article 5.4)
(vii) What is root locus? (Article 5.6)
(viii) What it breakaway point? (Article 5.7)
(ix) Write short note on Nyquist criterion. (Article 5.7)
(x) Write short note on mapping. (Article 5.15)
(xi) State Sylvester's Theorem. (Article 5.12)
(xii) State advantages of Routh Hurwitz criterion. (Article 5.18)
(xiii) State limitations of Routh Hurwitz criterion. (Summary)
(xiv) State Lyapunov stability theorem. (Summary)
(xv) Write a note on angle and magnitude condition of root locus. (Article 5.17.3)
(xvi) Explain the term (a) asymptotes (b) centroid. (Article 5.7)
(xvii) Write a short note on effects of adding pole and zero on the root locus. (Article 5.8)
(xviii) State the rule for obtaining the breakaway point in the root locii. (Article 5.10)
(xix) How to obtain the open gain at any point on the root locii?
(xx) How location of roots of the characteristic equation are related to stability?
(xxi) How will you obtain the angle of departure and angle of arrival of root locii?
(xxii) How will you determine the absolute value of K an any point S_1 on the root locii?

□□□

Chapter 6

Compensation Techniques

6.1. INTRODUCTION

Sometimes it is necessary to compensate an unstable system to make it stable or it may be necessary to improve the existing system to satisfy or to meet the required specifications. The system performance is given by time response and frequency response.

For the design of system in s-domain, introduction of poles and zeros at suitable places will give satisfactory performance. Main requirement of the control system is accuracy and stability. For greater accuracy of a system steady state error should be small, but to reduce the steady state error the gain of the amplifier must be increased, overshoot will also increase and stability will decrease. But we are interested in both accuracy and stability. This can be done by connecting a circuit between error detector and plant is known as compensation.

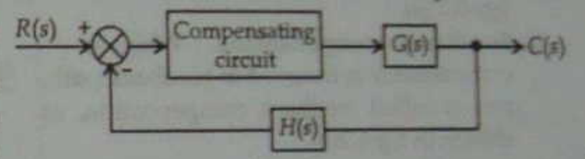


Fig. 6.1.

Now consider a transfer function

$$G(s) = \frac{K}{s(s+1)(s+2)} \quad ...(6.1)$$

It is required that velocity constant Kv corresponding to steady state error should be 10 sec^{-1}.

We know that

$$K_v = \lim_{s \to 0} sG(s)$$

$$K_v = \lim_{s \to 0} s.\frac{K}{s(s+1)(s+2)} = K/2$$

$$10 = K/2$$

$$\therefore \quad K = 20$$

Therefore for accuracy K should be equal to 20.

Now equation 6.1 becomes

$$G(s) = \frac{20}{s(s+1)(s+2)} \quad ...(6.2)$$

The characteristic equation of 6.1 will be

$$1 + G(s)\,H(s) = 0$$

$$1 + \frac{K}{s(s+1)(s+2)} = 0$$

$$s^3 + 3s^2 + 2s + K = 0$$

Apply	Routh Hurwitz	criteria
s^3	1	2
s^2	3	K
s^1	$\frac{6-K}{3}$	0
s^0	K	

For stable system

$$\frac{6-K}{3} > 0$$

or $K < 6$

Hence, sometimes accuracy and stability can not be met at same time, then a compensating system is used.

6.2. TYPES OF COMPENSATION

Depending upon the location of compensating network, following are the types of compen sation.

1. **Series compensation :** When a compensating network is inserted in the forward path, this is called series or cascade compensation, as shown in fig. 6.2 (*a*)

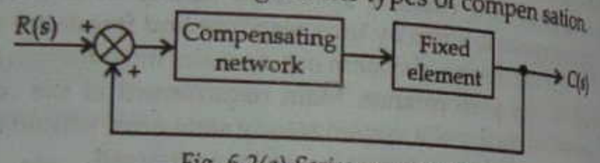


Fig. 6.2(a) Series compensation

2. **Feedback compensation :** When a compensator is inserted in feedback path, this is called feedback compensation, as shown in fig 6.2(*b*)

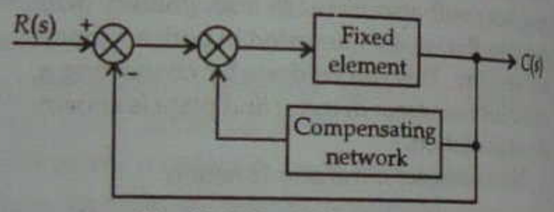


Fig. 6.2(b) Feedback compensation

3. **Load compensation :** A combination of series and feedback compensation is called load compensation or combined cascade and feedback compensation as shown in fig. 6.2*c*.

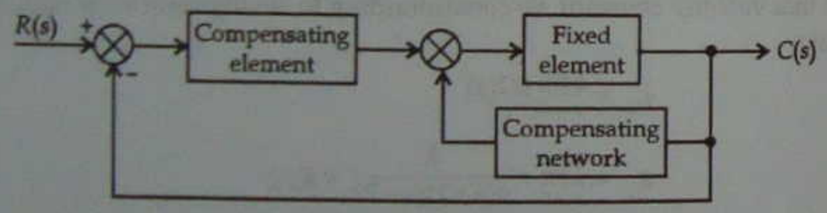


Fig. 6.2(c) Load compensation

The compensators may be electrical, mechanical, pneumatic or hydraulic type. Mostly electrical networks are used. It is easy to design R.C. filters. For the design of compensation network mainly transfer function approach is used. The simplest networks are lead, lag and lag-lead networks.

6.3. DESIGN OF COMPENSATION USING BODE'S PLOT

Bode diagram approach is commonly used because of its simplicity and performance specifications are given in frequency domain.

6.3.1. Phase Lead Network

From circuit diag.

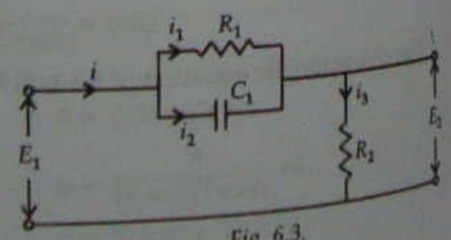


Fig. 6.3.

$$i = i_1 + i_2$$

$$i_1 = \frac{E_1 - E_2}{R_1}$$

$$i_2 = C\frac{d}{dt}(E_1 - E_2)$$

$$i = \frac{E_1 - E_2}{R_1} + C\frac{d}{dt}(E_1 - E_2) \quad \text{...(6.3)}$$

$$i_3 = i = \frac{E_2}{R_2} \quad \text{...(6.4)}$$

Put the value of i from eq^n. 6.4 to (6.3)

$$\frac{E_2}{R_2} = \frac{E_1 - E_2}{R_1} + C\frac{d}{dt}(E_1 - E_2) \quad \text{...(6.5)}$$

Laplace transform of equation 6.5

$$\frac{1}{R_2}E_2(s) = \frac{1}{R_1}[E_1(s) - E_2(s)] + sc[E_1(s) - E_2(s)]$$

$$\frac{E_2(s)}{E_1(s)} = \frac{R_2 + R_1R_2Cs}{R_1 + R_2 + R_1R_2Cs} \quad \text{...(6.6)}$$

Equation (6.6) can be written as

$$\frac{E_2(s)}{E_1(s)} = \frac{R_2}{R_1 + R_2}\left[\frac{1 + R_1Cs}{1 + \frac{R_1R_2Cs}{R_1 + R_2}}\right] \quad \text{...(6.7)}$$

Put $$a = \frac{R_1 + R_2}{R_2} \qquad a > 1$$

$$T = \frac{R_1R_2}{R_1 + R_2} \cdot C$$

From equation 6.7

$$\frac{E_2(s)}{E_1(s)} = \frac{R_2}{R_1 + R_2}\left[\frac{1 + \left(\frac{R_1 + R_2}{R_2}\right)\left(\frac{R_1R_2C}{R_1 + R_2}\right)s}{1 + \frac{R_1R_2Cs}{R_1 + R_2}}\right]$$

$$\therefore \quad \frac{E_2(s)}{E_1(s)} = \frac{1}{a}\left[\frac{1 + aTs}{1 + Ts}\right] \quad \text{...(6.8)}$$

6.3.1.1. Pole-Zero Plot of Equation 6.8

From the pole-zero plot of equation 6.8 shown in fig 6.3*a*, it is clear that zero is nearer to the imaginary axis as compared to the pole. The zero being more dominant than pole it gives a positive phase shift (phase shift increased) and therefore is called phase lead network. Phase lead network allow to pass high frequencies and low frequncies are attenuated.

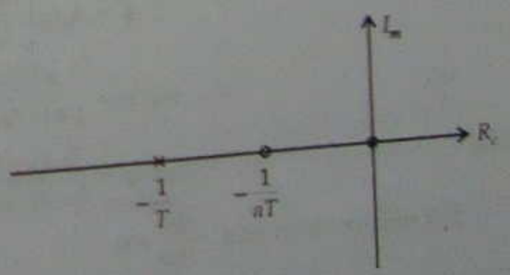


Fig. 6.3(a).

Polar Plot of Equation 6.8.

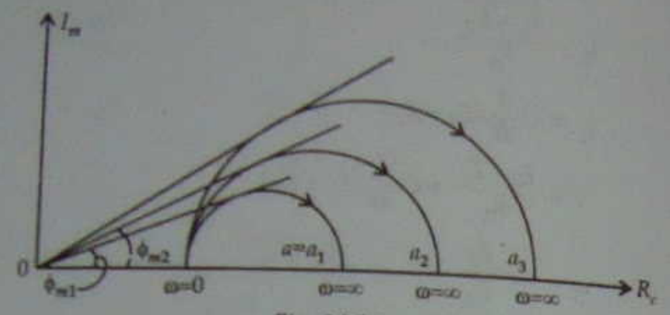


Fig. 6.3(b)

6.3.1.2. Bode Plot for Phase Lead Network

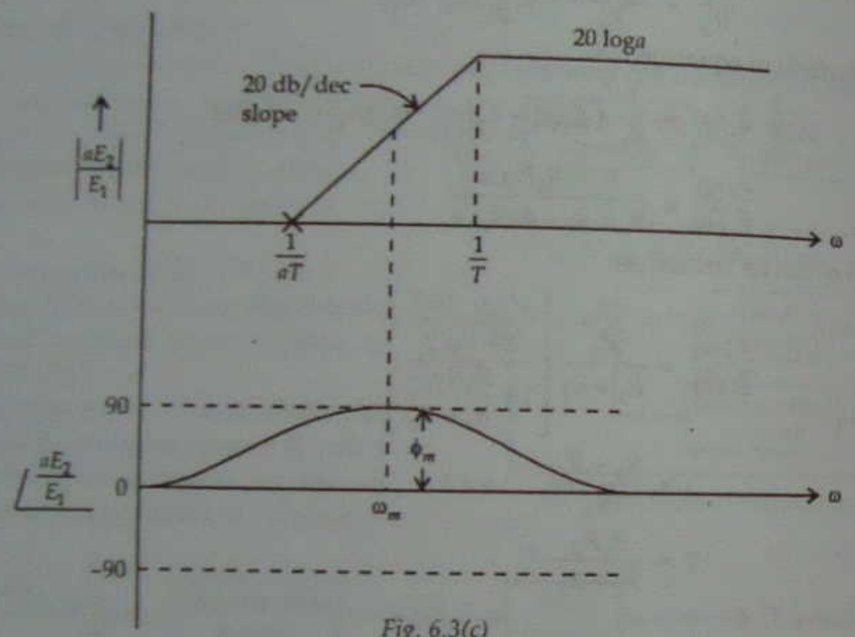


Fig. 6.3(c)

6.3.1.3. Calculation of ϕ_m and ω_m

From equation 6.8

$$\frac{aE_2(s)}{E_1(s)} = \frac{1+aTs}{1+Ts}$$

$$\phi = \text{Arg}\left[\frac{aE_2(s)}{E_1(s)}\right] = \tan^{-1} aT\omega - \tan^{-1}\omega T$$

or,
$$\tan\phi = \frac{aT\omega - T\omega}{1+(aT\omega)(T\omega)}$$

$$\tan\phi = \frac{T\omega(a-1)}{1+aT^2\omega^2} \quad ...(6.9)$$

For maxm. Value of ϕ : $\frac{d\phi}{d\omega} = 0$ ...(6.10)

We get
$$\omega_m = \frac{1}{T\sqrt{a}}$$

Alternative : Since ω_m is the geometric mean of two corner frequencies we can write.

$$\log_{10}\omega_m = \frac{1}{2}\left[\log_{10}\frac{1}{aT} + \log_{10}\frac{1}{T}\right] = \left[\log\frac{1}{aT}\cdot\frac{1}{T}\right]^{1/2} = \left[\log\frac{1}{aT^2}\right]^{1/2}$$

$$\omega_m = \left[\frac{1}{aT^2}\right]^{1/2} = \frac{1}{\sqrt{a}\,T}$$

Therefore maximum phase lead occurs at a frequency of $\omega_m = \frac{1}{\sqrt{a}\,T}$. Put the value of ω_m in equation (6.9)

$$\tan\phi_m = \frac{\frac{1}{T\sqrt{a}}\cdot T(a-1)}{1+aT^2\frac{1}{aT^2}}$$

$$\tan\phi_m = \frac{a-1}{2\sqrt{a}} \quad ...(6.11)$$

$$\sin\phi_m = \frac{a-1}{a+1} \quad ...(6.12)$$

From eq^n 6.12 we can calculate the value of 'a'

$$a = \frac{1+\sin\phi_m}{1-\sin\phi_m} \quad ...(6.13)$$

DESIGN PROCEDURE FOR PHASE LEAD COMPENSATION

Step 1 : The magnitude and phase Vs frequency curves are plotted for G(s) of the uncompensated system, with gain constant K set according to steady state error requirement.

Step 2 : From the Bode's plot determine the phase margin and gain margin.

Let ϕ = phase margin of uncompensated system

ϕ_s = specified phase margin

ε = margin of safety

then, $\phi_m = \phi_s - \phi + \varepsilon$

If $\phi_m > 60°$, two identical network each contributing maximum lead of $\phi_m/2$ are used.

Step 3 : Use equation 6.12 calculate the value of 'a'.

Step 4 : Calculate ω_m at which uncompensated system will have a gain equal to -10log a. (the phase lead network causes an attenuation of $-\frac{1}{2}$ (20log a) = -10loga db at frequency ω_m)

Step 5 : Once 'a' is determined, calculate the value of 'T' from $\omega_m = \frac{1}{\sqrt{a}\,T}$

Step 6 : Transfer function of phase lead network is determined from the values of 'a' & 'T'.

Step 7 : Draw the Bode's plot of the compensated system and check that all performance specifications are met or not. If not, a new value of ϕ_m must be estimated.

6.3.2. Effect of Phase Lead Network

1. The velocity constant is usually increased.
2. The slope of the magnitude curve is reduced at the gain cross over frequency, with the result relative stability improve.

3. Phase margin increased.
4. The bandwidth increased.
5. The response is faster.

Example 6.1. Design a cascade compensation for a system whose transfer function is

$$G(s) = \frac{K}{s(1+0.1s)(1+0.001s)}$$

It will fullfill the following specifications
Phase margin ≥ 45°
Velocity constant K_v = 1000 sec^{-1} *(R.M.L., University, Faizabad, 2003)*

Solution : Step 1 :

$$K_v = \lim_{s\to 0} s.G(s) = \lim_{s\to 0} s \cdot \frac{K}{s(1+0.1s)(1+0.001s)}$$

$$K_v = K$$

$$\therefore \quad K = 1000$$

$$\therefore \quad G(s) = \frac{1000}{s(1+0.1s)(1+0.001s)} \qquad ...(6.14)$$

Step 2 : Draw the Bode's plot for the transfer function given by the eq^n. 6.14
Two corner frequencies are 1/0.1 = 10 rad/sec. and 1/0.001 = 1000 rad/sec.

Table 6.1.

ω	Arg(1000) ϕ_1	- Arg(0 + jω) ϕ_2	- Arg(1 + 0.1jω) ϕ_3	- Arg(1 + j0.001ω) ϕ_4	Resultant $\phi_1+\phi_2+\phi_3+\phi_4$
1	0	- 90°	- 5.7°	- 0.06	- 95.7°
5	0	- 90°	- 26.5°	- 0.28	- 116.5°
10	0	- 90°	- 45°	- 0.57	- 135.63°
50	0	- 90°	- 78.6°	- 2.86°	- 171.46°
100	0	- 90°	- 84.2°	- 5.71°	- 179.9°
150	0	- 90°	- 86.2°	- 8.5°	- 184°
200	0	- 90°	- 87.13°	- 11.3°	- 188.43°
500	0	- 90°	- 88.85°	- 26.56°	- 205.41°

Step 3 : From Bode plot :
Phase margin avilable ϕ = 0°
Specified phase margin ϕ_s = 45°
Margin of safety ε = 5°

$$\therefore \quad \phi_m = 45° - 0 + 5° = 50°$$

Step 4 : Calculation of 'a'

$$\sin\phi_m = \frac{a-1}{a+1}$$

$$\sin 50 = \frac{a-1}{a+1}$$

$$\therefore \quad a = 7.51$$

Step 5 : Calculation of ω_m
Zero frequency attenuation = - 10log a
= - 10log 7.51 = - 8.75 db

At the gain of - 8.75 *db* draw a line on magnitude curve, this will gives ω_m (new gain cross over frequency).

$\therefore \omega_m$ = 170 rad/sec. (from Bode plot)

Step 6 : Calculation of 'T'

$$\omega_m = \frac{1}{T\sqrt{a}}$$

$$a = 7.51$$

$$\omega_m = 170$$

$$T = 0.00214$$

$$\therefore$$

Step 7 : Transfer Function of Compensator

$$\boxed{G_c(s) = \frac{1}{7.51}\left(\frac{1+0.016s}{1+0.00214s}\right)}$$

The amplification necessary to cancel the lead network attenuation of 7.51

$$\therefore \quad G_c(s) = \frac{1+0.016s}{1+0.00214s}$$

Step 8 : Overall Transfer Function

$$G(s) = G(s) \,.\, Gc\,(s)$$

$$= \frac{1000(1+0.016s)}{s(1+0.1s)(1+0.001s)(1+0.00214s)}$$

Step 9 : Draw the Bode plot of overall transfer function & check
The corner frequencies are
ω_1 = 10 rad/sec.
ω_2 = 62.5 rad/sec.
ω_3 = 476.5 rad/sec.
ω_4 = 1000 rad/sec.

Table 6.2.

ω	Arg 1000 ϕ_1	-Arg (jω) ϕ_2	-Arg (1+j0.1ω) ϕ_3	-Arg (1+j0.001ω) ϕ_4	-Arg (1+j0.002ω) ϕ_5	-Arg (1+j0.016ω) ϕ_6	Resultant $\phi_1+\phi_2+\phi_3+\phi_4+\phi_5+\phi_6$
1	0	-90°	-5.7°	0.05	0.11	0.91	-95°
10	0	-90°	-45°	0.57	1.14	9	-126°
50	0	-90°	-78.6°	-2.86°	-6°	38.65°	-138.8°
100	0	-90°	-84.2°	-5.71°	-11.85°	57.99°	-133.77°
150	0	-90°	-86.1°	-8.5°	-17.4°	63.38°	-134.6°
800	0	-90°	-89.2°	-38.65°	-59.23°	85.5°	-191.58°
200	0	-90°	-87.13°	-11.3°	-22.78°	72.6°	-138.61°

From Bode's plot of copensated system P.M. = 45°

6.3.3. Limitations of Phase Lead Networks

For getting large phase margin we required large value of 'a' which increases the bandwidth, this may increase the transmission of noise. In practice the value of 'a' should not be greater than 15.

If the large phase lead is required two or more caseade compensation should be used.

If the phase shift decreases rapidly near the gain cross over frequency, phase lead compensation becomes ineffective because the additional phase lead at new gain crossover frequency is added to a much smaller phase angle than that at the old gain cross over frequency. The desired phase margin can be achieved only with large value of 'a' which is not desireable.

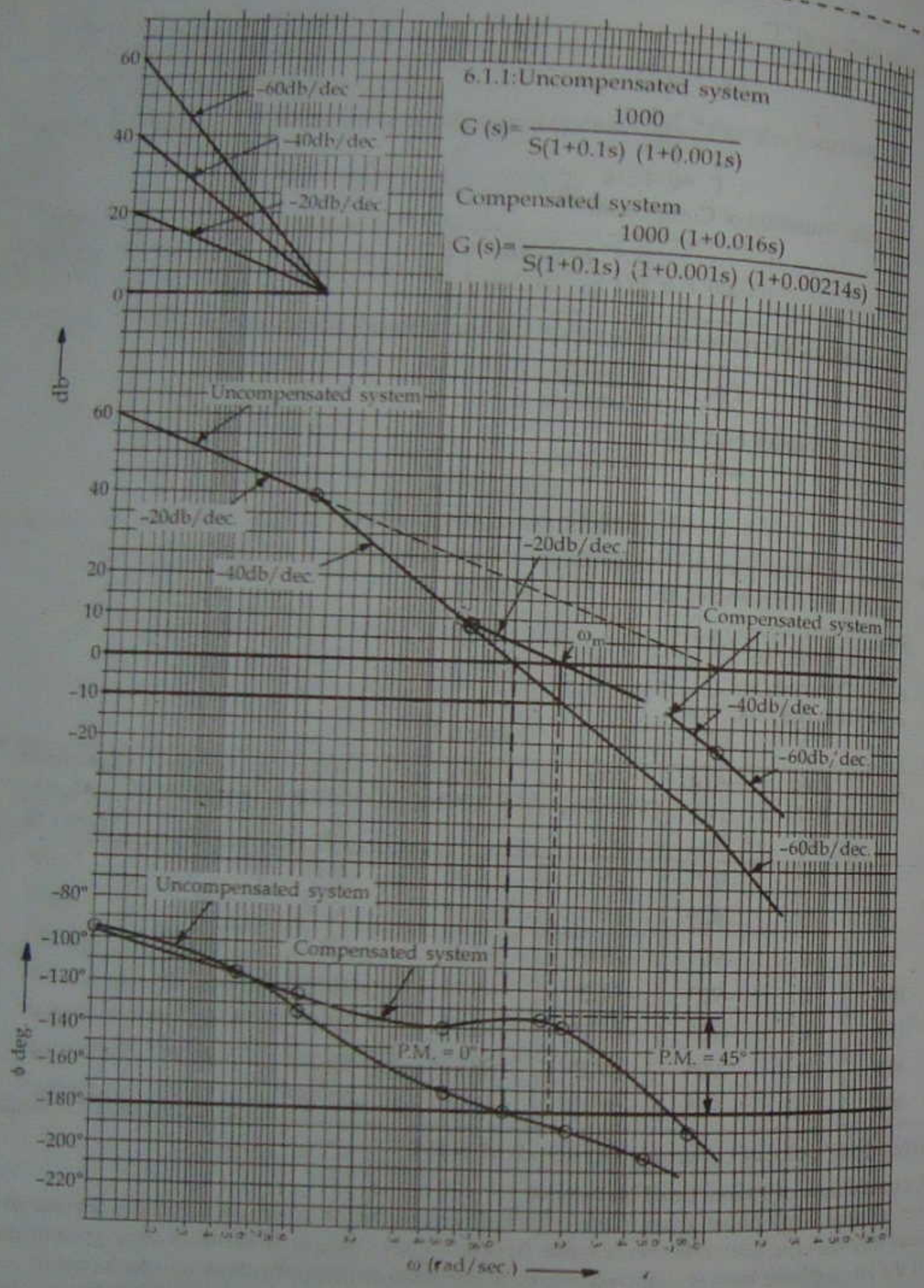


Fig. 6.3(d).

6.3.4. Phase Lag Network

A simple and commonly used network is shown in fig. 6.4.

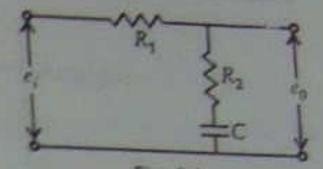


Fig. 6.4.

Transfer Function

Apply *KVL* in both mesh

$$e_i = R_1 i + R_2 i + \frac{1}{c}\int i\,dt \qquad ...(6.15)$$

$$e_0 = R_2 i + \frac{1}{c}\int i\,dt \qquad ...(6.16)$$

Take Laplace transform of eqn 6.15 & 6.16

$$E_i(s) = R_1 I(s) + R_2 I(s) + \frac{1}{cs} I(s) \qquad ...(6.17)$$

$$E_0(s) = R_2 I(s) + \frac{1}{Cs} I(s) \qquad ...(6.18)$$

From eq^n. (6.17) & (6.18)

$$\frac{E_0(s)}{E_i(s)} = \frac{1 + R_2 Cs}{(R_1 + R_2)Cs + 1} \qquad ...(6.19)$$

Equation 6.19 is the transfer function of lag network

Put $aT = R_2 C$

$$a = \frac{R_2}{R_1 + R_2}$$

then $T = (R_1 + R_2)\,C$

$$\boxed{\frac{E_0(s)}{E_i(s)} = \frac{1 + aTs}{1 + Ts}} \qquad a < 1 \qquad ...(6.20)$$

Pole-Zero Plot

The lag network has a pole at s = – 1/T and zero at s = – 1/aT. From pole zero plot shown in fig 6.4a, the pole is nearer to the imaginary axis as compared to zero, hence the effect of pole is dominant, therefore introduction of phase lag network in forward path reduces the phase shift.

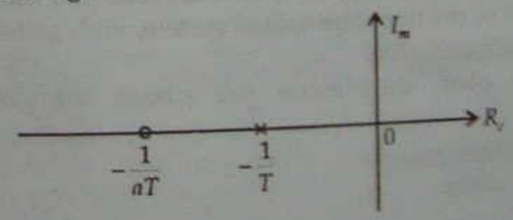


Fig. 6.4(a)

Bode Plot for Phse Lag Network

From Bode plot, the phase lag network provides an attenuation of $20\log_{10} a$ at high frequencies. The lag network allows to pass low frequencies and high frequencies are attenuated.

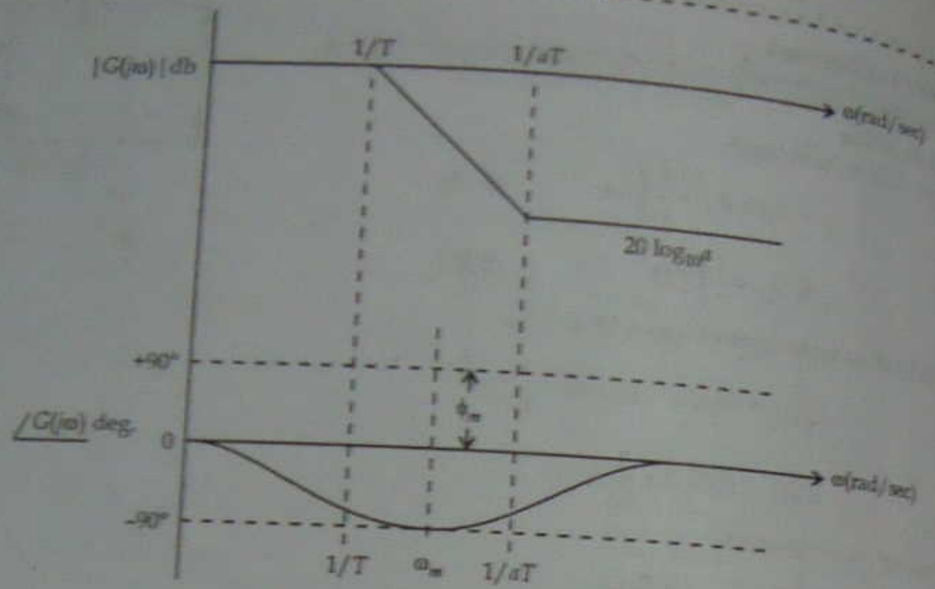


Fig. 6.4(b)

Polar Plot of Phase Lag Network

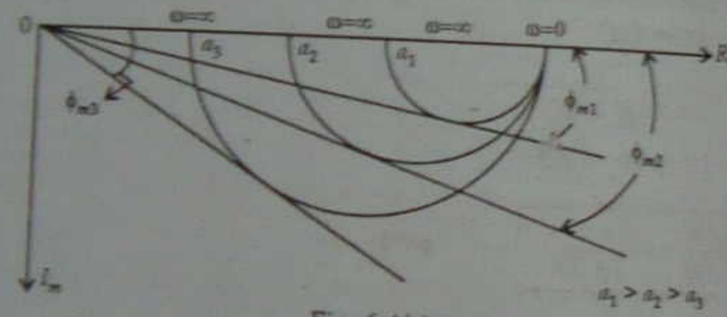


Fig. 6.4(c)

DESIGN PROCEDURE FOR PHASE LAG NETWORK

Step 1 : The magnitude and phase *Vs* frequency curves (Bode plot using asymptotic approximation are plotted for $G(s)$ of the uncompensated system, with gainconstant K set according to steady state error requirement.

Step 2 : From the Bode plot, determine the phase margin of the uncompensated system.

Step 3 : If ϕ_s = specified phase margin
ε = marign of safety
$\phi = \phi_s + \varepsilon$.

Step 4 : Determine the frequency corresponding to the required phase margin from the phase curve. This frequency is new gain cross over frequency (ω'_m)

Step 5 : The magnitude curve is brought down to 0 db at the new gain cross-over frequency where the phase margin is satisfied, the phase lag network must provide the amount of attenuation equal to the value of magnitude curve at ω'_m

$$|G(j\omega'_m)| = -20 \log a \qquad a < 1$$

or, $$a = 10^{-|G(j\omega'_m)|/20} \qquad a < 1$$

calculate 'a' from above expression.

Step 6 : Calculate 'T' from

$$\frac{1}{aT} = \frac{\omega'_m}{10}$$

usually the upper corner frequency $(1/aT)$ is placed at a frequency about one decade below the new gain-cross over frequency.

Step 7 : Draw the Bode's plot for compensated network & investigate to see if the required phase margin is met or not, if not, adjust the value of 'a' & 'T'.

6.3.5. Effect of Phase Lag Network

1. For a given relative stability, the velocity constant is increased.
2. There is decrease in the gain cross over frequency thus decreasing the bandwidth.
3. Phase margin increase.
4. Response will be slower.
5. Due to the decrease in bandwidth, the rise time and the settling time become large.

Example 6.2. Design a sutable lag compensating network for

$$G(s) = \frac{K}{s(s+2)(s+20)}$$

to meet the following specifications

$K_v = 20 \text{ sec}^{-1}$
P.M $\geq 35°$

Solution : Step 1 :

$$K_v = \lim_{s \to 0} s.G(s) = \lim_{s \to 0} s.\frac{K}{s(s+2)(s+20)}$$

$$20 = \frac{K}{40}$$

$$\therefore \quad K = 800$$

$$\therefore \quad G(s) = \frac{800}{s(s+2)(s+20)} = \frac{20}{s(1+0.5s)(1+0.05s)} \quad \text{...(6.21)}$$

Step 2 : Draw the Bode plot for eq^n 6.21

There are two corner frequencies

$\omega_1 = 1/0.5 = 2$ rad/sec.
$\omega_2 = 1/0.05 = 20$ rad/sec.

Table 6.3.

ω	$-\text{Arg}(j\omega)$ ϕ_1	$-\text{Arg}(1 + j\,0.5\omega)$ ϕ_2	$-\text{Arg}(1 + j\,0.05\omega)$ ϕ_3	Resultant $\phi = \phi_1 + \phi_2 + \phi_3$
0.1	- 90°	- 2.86°	- 0.286°	- 93.14°
0.2	- 90°	- 5.71°	- 0.57°	- 96.28°
0.5	- 90°	- 14.04°	- 1.43°	- 105.47°
1	- 90°	- 26.56°	- 2.86°	- 119.42°
2	- 90°	- 45°	- 5.71°	- 140.71°
5	- 90°	- 68.2°	- 14.04°	- 172.24°
6	- 90°	- 71.56°	- 16.7°	- 178.26°
7	- 90°	- 74.05°	- 19.29°	- 183.34°
8	- 90°	- 75.96°	- 21.8°	- 187.76°
10	- 90°	- 78.69	- 26.57°	- 195.26°

Step 3 : *P.M* avilable = 0°

Specified phase margin = 35°

Margin of safety ε = 5°

$$\therefore \quad \phi = 35° + 5° = 40°$$

For desired *P.M.* the gain crossover frequency must be shifted to a lower value and this is possible by using phase lag compensation.

Step 4 : Required phase margin is avilable at ω = 2 rad/sec.

This is new gain cross over frequency. Draw a line at – 140° the point where it cuts the phase curve project it to the magnitude curve.

From Bode plot gain avilable at ω'_m (= 2 rad/sec) = 20 db

Step 5 :
$$|G(j\omega'_m)| = -20\log a$$
$$20 = -20\log a$$
$$\therefore \quad a = 0.1$$

Step 6 : Calculation of *T* :

$$\frac{1}{aT} = \frac{\omega'_m}{10}$$

$$\frac{1}{0.1T} = \frac{2}{10}$$

$$T = 50$$

Step 7 : Calculation of *Gc*

$$Gc(s) = \frac{1+aTs}{1+Ts}$$

$$Gc(s) = \frac{1+5s}{1+50s}$$

Step 8 : Overall transfer function after compensation

$$G(s) = \frac{20(1+5s)}{s(1+0.5s)(1+.05s)(1+50s)} \qquad ...(6.22)$$

Step 9 : Draw the Bode plot for *eqn* 6.22.

Corner frequencies are

ω_1 = 0.02 rad/sec.

ω_2 = 0.2 rad/sec.

ω_3 = 2 rad/sec.

ω_4 = 20 rad/sec.

Table 6.4.

ω	– Arg(jω)	– Arg (1+ j 50ω)	+Arg (1 + j 5ω)	– Arg (1 + 0.5ω)	– Arg (1+ 0.05ω)	Resultant
0.1	90°	– 78.69°	26.56°	– 2.86°	– 0.286°	– 145.28°
0.2	– 90°	– 84.29°	45°	– 5.71°	– 0.573°	– 135.57°
0.5	– 90°	– 87.70°	68.2°	– 14.04°	– 1.43°	– 124.57°
1.0	– 90°	– 88.85°	78.69°	– 26.56°	– 2.86°	– 129.58°
2.0	– 90°	– 89.43°	84.29°	– 45°	-5.71°	– 145.85°
5.0	– 90°	– 89.77°	87.71°	– 68.2°	– 14.04°	– 174.8°
6.0	– 90°	–89 80°	88.09°	– 71.56°	– 16.70°	– 179.9°
7.0	– 90°	– 89.84°	88.36°	– 74.05°	– 19.29°	– 184.29°
10.0	– 90°	– 89.88°	88.85°	– 78.69°	– 26.57°	– 196.29°

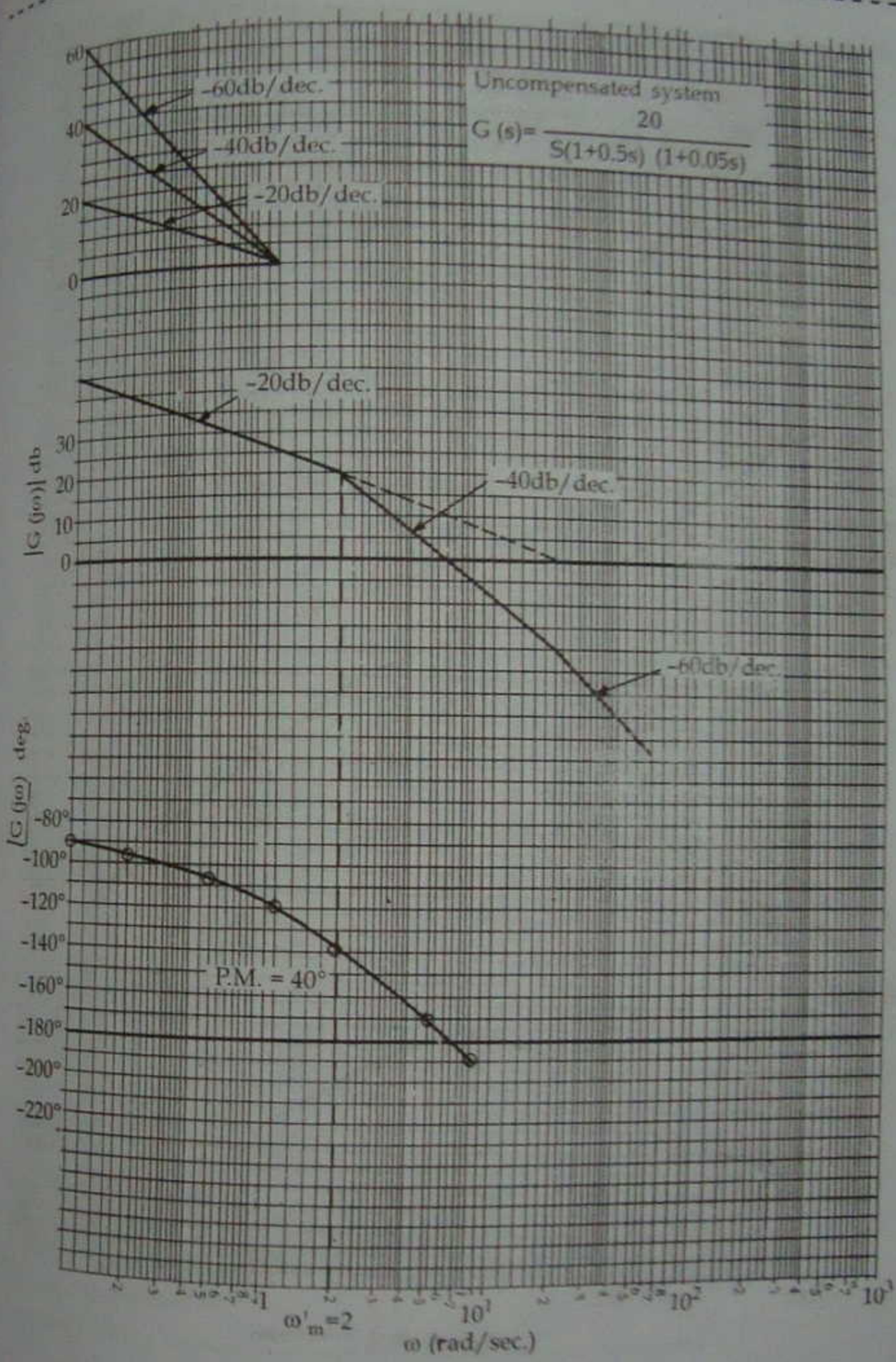


Fig. 6.4(d).

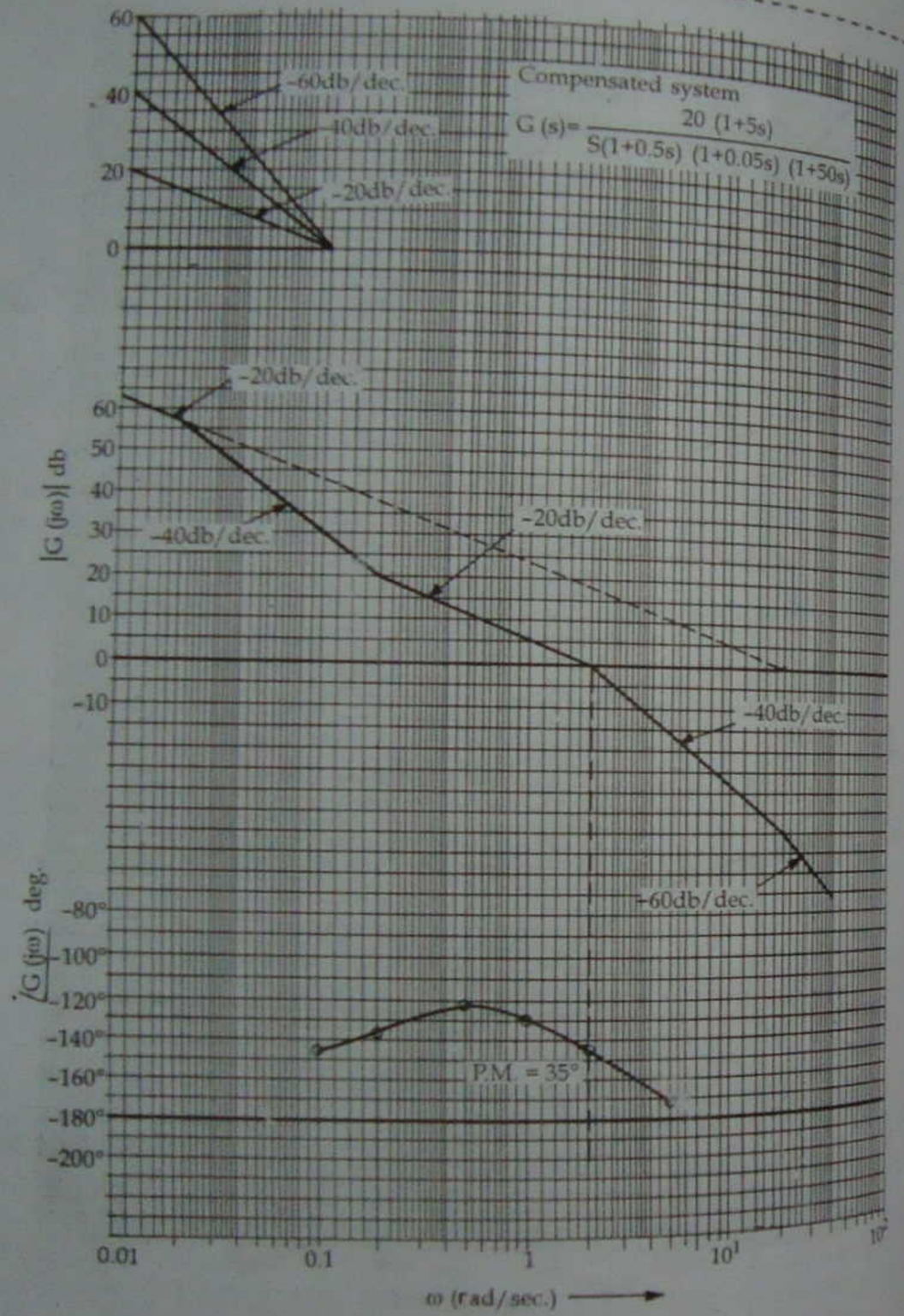


Fig. 6.4(e).

6.16 Lead-Lag Compensation

In phase lead compensation, the gain cross-over frequency shifts to a higher value and therefore the bandwidth increases, speed of response becomes fast but the steady state error does not show much improvement.

In phase lag compensation, the gain cross-over frequency shifts to a lower value and therefore the bandwidth decrease, speed of response reduced but the steady state error improve.

The speed of response and steady state error can be simultaneously improved if both phase lead and phase lag compensation networks are used.

The transfer function of lead-lag network can be written as

$$G(s) = \underbrace{\left(\frac{1+aT_1s}{1+T_1s}\right)}_{lead}\underbrace{\left(\frac{1+bT_2s}{1+T_2s}\right)}_{lag} \quad \text{...(6.23)}$$

The circuit of lead-lag compensation is shown in fig. 6.5.

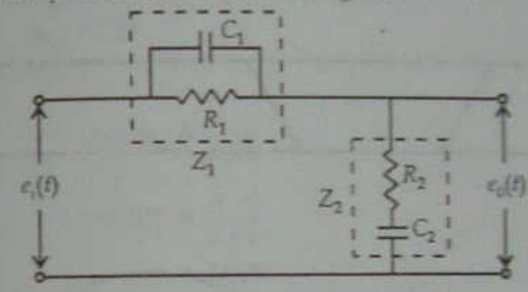


Fig. 6.5.

$$Z_1 = \frac{R_1}{1+R_1C_1s} \quad \& \quad Z_2 = \frac{1+R_2C_2s}{C_2s}$$

$$\frac{E_0(s)}{E_i(s)} = \frac{Z_2(s)}{Z_1(s)+Z_2(s)} = \frac{(1+R_2C_2s)/C_2s}{\frac{1+R_2C_2s}{C_2s}+\frac{R_1}{1+R_1C_1s}} \quad \text{...(6.24)}$$

$$\frac{E_0(s)}{E_i(s)} = \frac{(1+R_1C_1s)(1+R_2C_2s)}{1+(R_1C_1+R_2C_2+R_1C_2)s+R_1R_2C_1C_2s^2} \quad \text{...(6.25)}$$

compare the equation 6.23 with 6.25

$$aT_1 = R_1C_1 \quad \text{...(6.26)}$$

$$bT_2 = R_2C_2 \quad \text{...(6.27)}$$

$$T_1T_2 = R_1R_2C_1C_2 \quad \text{...(6.28)}$$

Multiplying the equation (6.26) & (6.27)

$$abT_1T_2 = R_1R_2C_1C_2 \quad \text{...(6.29)}$$

From eq^n 6.28 and 6.29

$$\boxed{ab = 1}$$

$a > 1$ lead compensation

$b < 1$ lag compensation

Pole Zero Plot

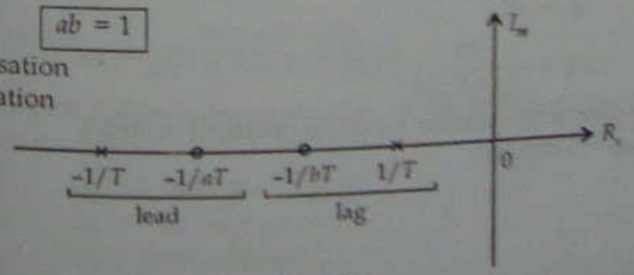


Fig. 6.5(a)

Polar Plot of Lead-Lag Network

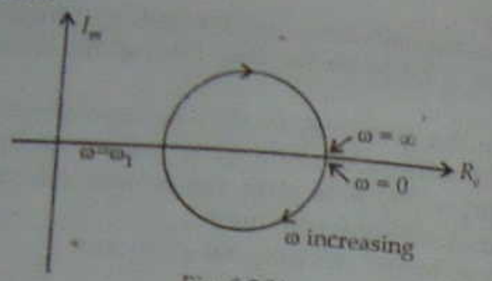


Fig. 6.5(b)

Bode Plot for Lead-Lag Network

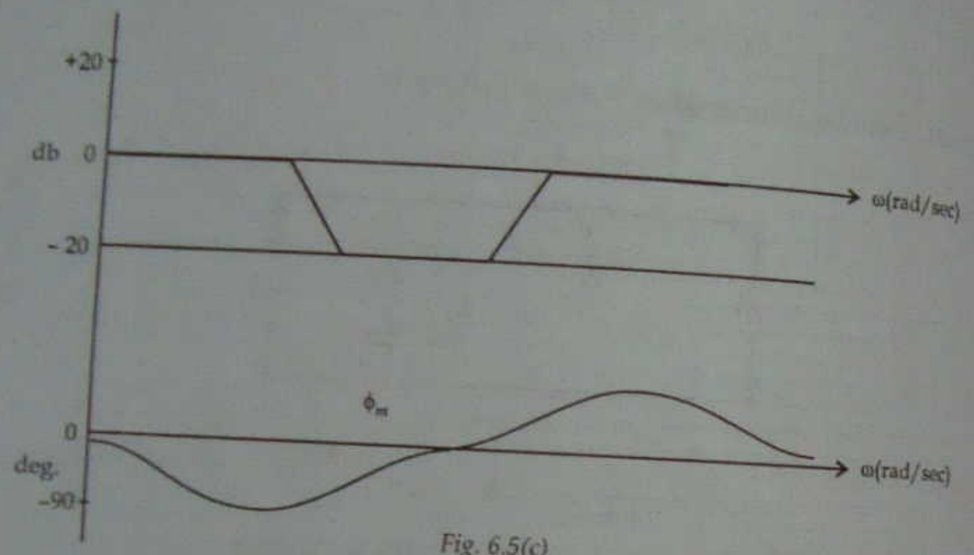


Fig. 6.5(c)

Example 6.3. The open loop transfer function of a unity feedback is

$$G(s) = \frac{K}{s(1+0.2s)}$$

It is required that the velocity error constant should be at least 20 and the phase margin should be 44°. Does the system meet the required specifications. If not, design the compensating network to satisfy the required specifications. *(R.M.L. Univ. Faizabad 2001)*

Solution : Step 1 :

$$K_v = \lim_{s \to 0} s.G(s)$$

$$K_v = \lim_{s \to 0} s.\frac{K}{s(1+0.2s)}$$

$$K_v = K$$

$$\therefore \quad K = 20$$

$$\therefore \quad G(s) = \frac{20}{s(1+0.2s)}$$

Step 2 : Draw the Bode plot of the transfer function obtained in step 1.
corner frequency = 5 rad/sec.

Table 6.5.

S. No.	ω	Arg(jω)	Arg(1 + j0.2ω)	Resultant
1.	1	- 90°	- 11.3°	- 101.3°
2.	2	- 90°	- 21.8°	- 111.8°
3.	4	- 90°	- 38.65°	- 128.65°
4.	6	- 90°	- 50°	- 140°
5.	8	- 90°	- 58°	- 148°
6.	10	- 90°	- 63.4°	- 153.4°
7.	20	- 90°	- 76°	- 166°

Step 3 : Phase margin avilable = 28°
Specified phase margin = 44°

$$\phi_m = 44 - 28 + 4$$

$$\phi_m = 20°$$

Step 4 : Calculation of 'a' and 'T' :

$$\sin \phi_m = \frac{a-1}{a+1}$$

$$\sin 20° = \frac{a-1}{a+1}$$

$$\therefore \quad a = 2.039$$

zero frequency attenuation = $-10\log a = -10\log 2.039 = -3.1$db

Draw a horizontal line at -3.1 db to cut the magnitude curve, measure the frequency at the point of intersection, this will ω_m.

From Bode plot $w_m = 12$ rad/sec.

since, $$\omega_m = \frac{1}{T\sqrt{a}}$$

$$\therefore \quad T = \frac{1}{\omega_m \sqrt{a}}$$

$$T = \frac{1}{12\sqrt{2.039}} = 0.058$$

$\therefore$ The transfer function of compensating network

$$Gc(s) = \frac{1}{2.039} \cdot \frac{1+0.118s}{1+0.058s}$$

The overall transfer function of the system with A = 2.039

$$G'(s) = G(s).\, Gc(s)$$

$$G'(s) = \frac{20(1+0.118s)}{s(1+0.2s)(1+0.058s)}$$

Step 5 : Draw the Bode plot and investigate whether all the specifications met or not.
Corner frequencies are

$\omega_1 = 5$ rad/sec.
$\omega_2 = 8.47$ rad/sec.
$\omega_3 = 17.24$ rad/sec.

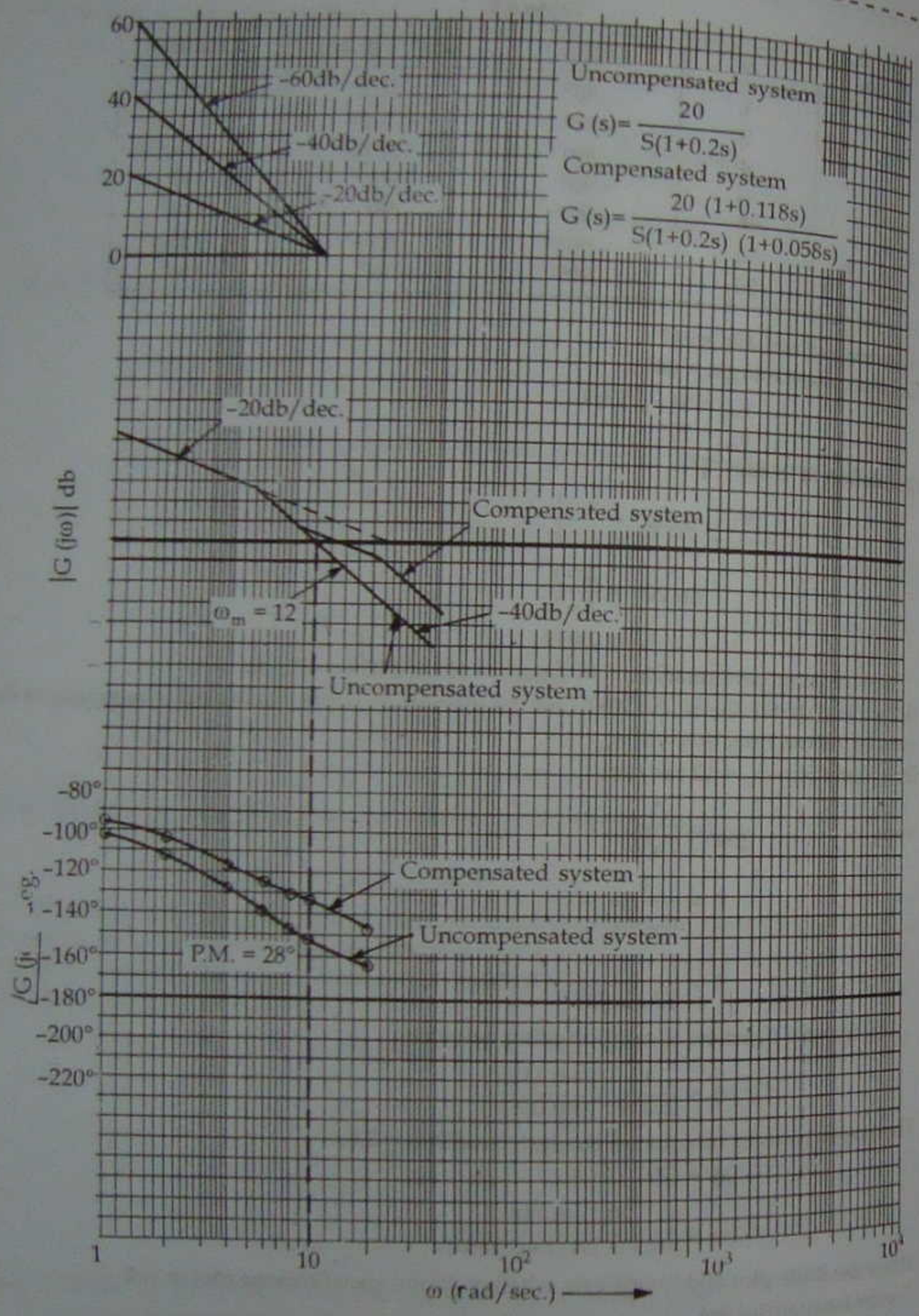


Fig. 6.6.

Table 6.6.

ω	- Arg (jω)	- Arg (1 + j0.2ω)	- Arg(1 + j0.058ω)	Arg(1 + j0.118 ω)	Resultant
1	- 90°	- 11.3°	- 3.31°	6.73°	- 97.88°
2	- 90°	- 21.8°	- 6.61°	13.27°	- 105.14°
4	- 90°	- 38.65°	- 13°	25.26°	- 116.4°
6	- 90°	- 50°	- 19.2°	35.29°	- 124°
8	- 90°	- 58°	- 24.89°	43.35°	- 130°
10	- 90°	- 63.4°	- 30°	49.72°	- 133.68°
20	- 90°	- 76°	- 49°	67.03°	- 148°

Example 6.4. Design a suitable phase lag compensating network for

$$G(s) = \frac{K}{s(1+0.1s)(1+0.2s)}$$

to meet the following specifications

K_v = 30 sec-1

P.M. ≥40°

Solution : Step 1 :

$$K_v = \lim_{s\to 0} s.G(s) = \lim_{s\to 0} s.\frac{K}{s(1+0.1s)(1+0.2s)} = K$$

$$\therefore \quad K = 30$$

$$G(s) = \frac{30}{s(1+0.1s)(1+0.2s)}$$

Step 2: Draw the Bode plot of the transfer function obtained in step 1.' corner frequencies are

ω_1 = 5 rad/sec.

ω_2 = 10 rad/sec.

Table 6.7.

ω	Arg (jω)	Arg (1 + j 0.1ω)	Arg (1 + j0.2ω)	Resultant
1	- 90°	- 5.71°	- 11.31°	- 107.02°
2	- 90°	- 11.31°	- 21.8°	- 123.11°
4	- 90°	- 21.8°	- 38.65°	- 150.45°
6	- 90°	- 31°	- 50.19°	- 171.19°
8	- 90°	- 38.65°	- 58°	- 186.65°
10	- 90°	- 45°	- 63.4°	- 198.4°
20	- 90°	- 63.4°	- 76°	- 229.4°
30	- 90°	- 71.56°	- 80.5°	- 242.06°
40	- 90°	- 76°	- 82.87°	- 248.87°
50	- 90°	- 78.6°	- 84.28°	- 252.88°
100	- 90°	- 84.28°	- 87.13°	- 261.41°

Step 3 : From Bode Plot

Phase margin avilable = - 22°

Required phase margin 40° is avilable at ω = 3 rad/sec.

Draw a line at - 140° the point of intersection of this line with phase curve is '*x*'. From the point '*x*' draw a line on the magnitude curve.

∴ The gain avilable at ω'_m = 3 rad/sec. is 20 db (from Bode plot). ω'_m is the new cross-over frequency.

Step 4 : Calculation of 'a' and 'T'

$$|G(j\omega'_m)| = -20\log a$$

$$-20\log a = 20$$

$$\therefore a = 0.1$$

$$\frac{1}{aT} = \frac{\omega'_m}{10}$$

$$\frac{1}{0.1T} = \frac{2.8}{10} \qquad \therefore \quad T = 35.7$$

Note : It will be better $\omega'_m = 2.8$ rad/sec. instead of 3 rad/sec.

Step 5 : Calculation of transfer function of compensator

$$Gc(s) = \frac{1+aTs}{1+Ts}$$

$$Gc(s) = \frac{1+3.57s}{1+35.7s}$$

Step 6 : Calculation of overall transfer function of the system.

$$G(s) = \frac{30(1+3.57s)}{s(1+0.1s)(1+0.2s)(1+35.7s)}$$

Step 7 : Draw the Bode plot for the transfer function obtained in step 6.

Corner frequencie are

$\omega_1 = 0.028$ rad/sec.

$\omega_2 = 0.28$ rad/sec.

$\omega_3 = 10$ rad/sec.

$\omega_4 = 5$ rad/sec.

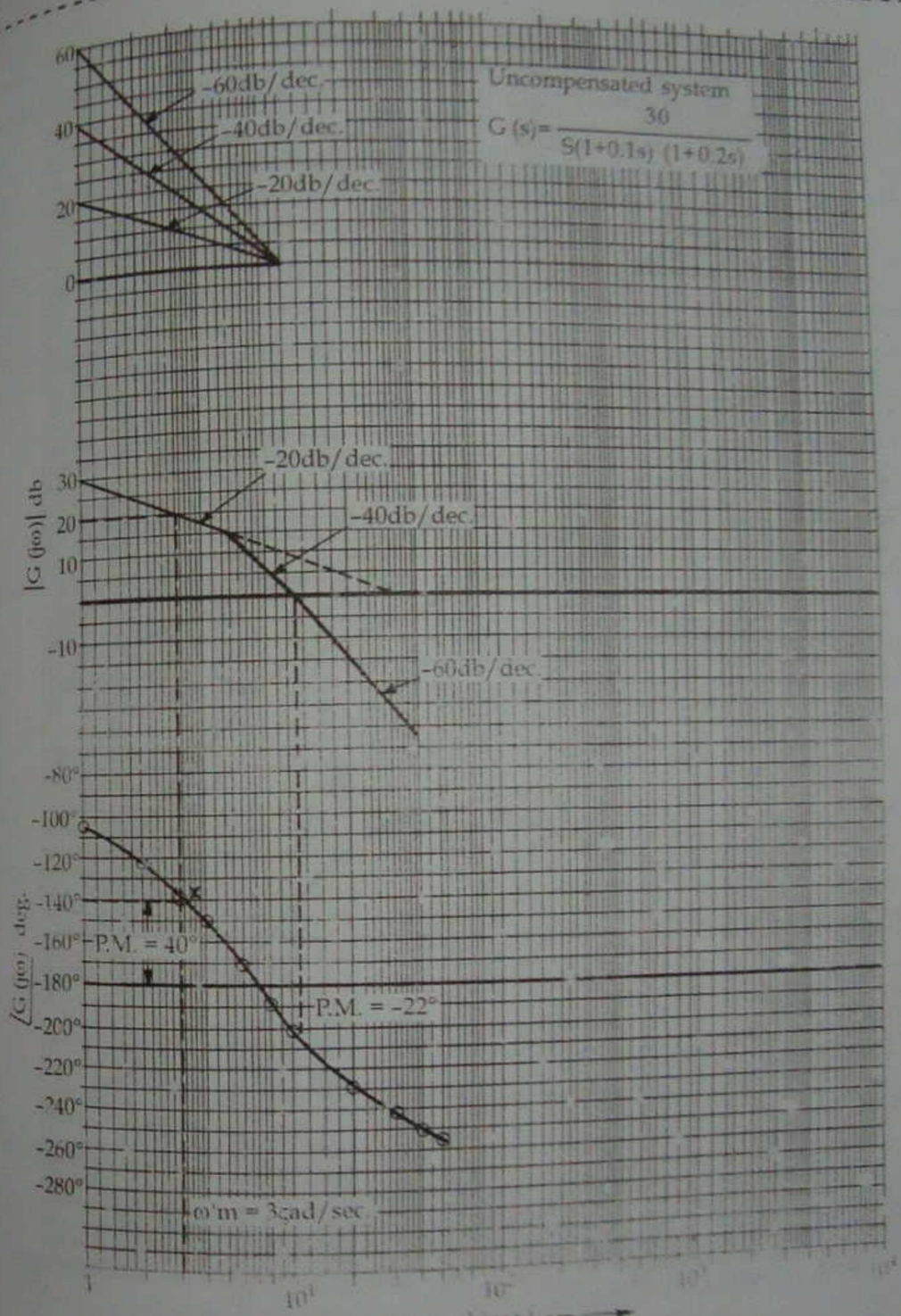
Table 6.8.

ω	$-\text{Arg}(j\omega)$	$-\text{Arg}$ $(1+j\,0.1\,\omega)$	$-\text{Arg}$ $(1+j\,0.2\omega)$	$-\text{Arg}$ $(1+35.7\,\omega)$	$+\text{Arg}$ $(1+j\,3.57\,\omega)$	Resultant
0.1	- 90°	- 0.57°	-1.14°	- 74.35°	+ 19.65°	- 146.41
0.3	- 90°	- 1.71°	- 3.4°	- 84.66°	+ 46.96°	- 132.81
0.5	- 90°	- 2.8°	- 5.7°	- 86.79°	+ 60.74°	- 124.55
0.8	- 90°	- 4.5°	- 9.1°	- 87.99°	+ 70.70°	- 120.89°
1.0	- 90°	- 5.7°	- 11.3°	- 88.39°	+ 74.23°	- 121.04°
2.0	- 90°	- 11.3°	-21.8°	- 89.19°	+ 82.03°	- 130.26°
5.0	- 90°	- 26.5°	- 45°	- 89.67°	+ 86.79°	- 164.38°
10.0	- 90°	- 45°	- 63.43°	- 89.83°	+ 88.39°	- 199.71°
20.0	- 90°	- 63.4°	- 76°	- 89.91°	+ 89.19°	- 230.12°
40.0	- 90°	- 76°	- 82.8°	- 89.59°	+ 89.59°	- 249.16°
100.0	- 90°	- 84°	- 87.13°	- 89.98°	+ 89.84°	- 261.27

Example 6.5. Consider a laglead network defined by

$$Gc(s) = K\frac{\left(s+\frac{1}{T_1}\right)\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{\beta}{T_1}\right)\left(s+\frac{1}{\beta T_2}\right)}$$

Show that at frequency ω_1, where $\omega_1 = \frac{1}{\sqrt{T_1 T_2}}$, the phase angle of $Gc(j\omega)$ becomes zero.



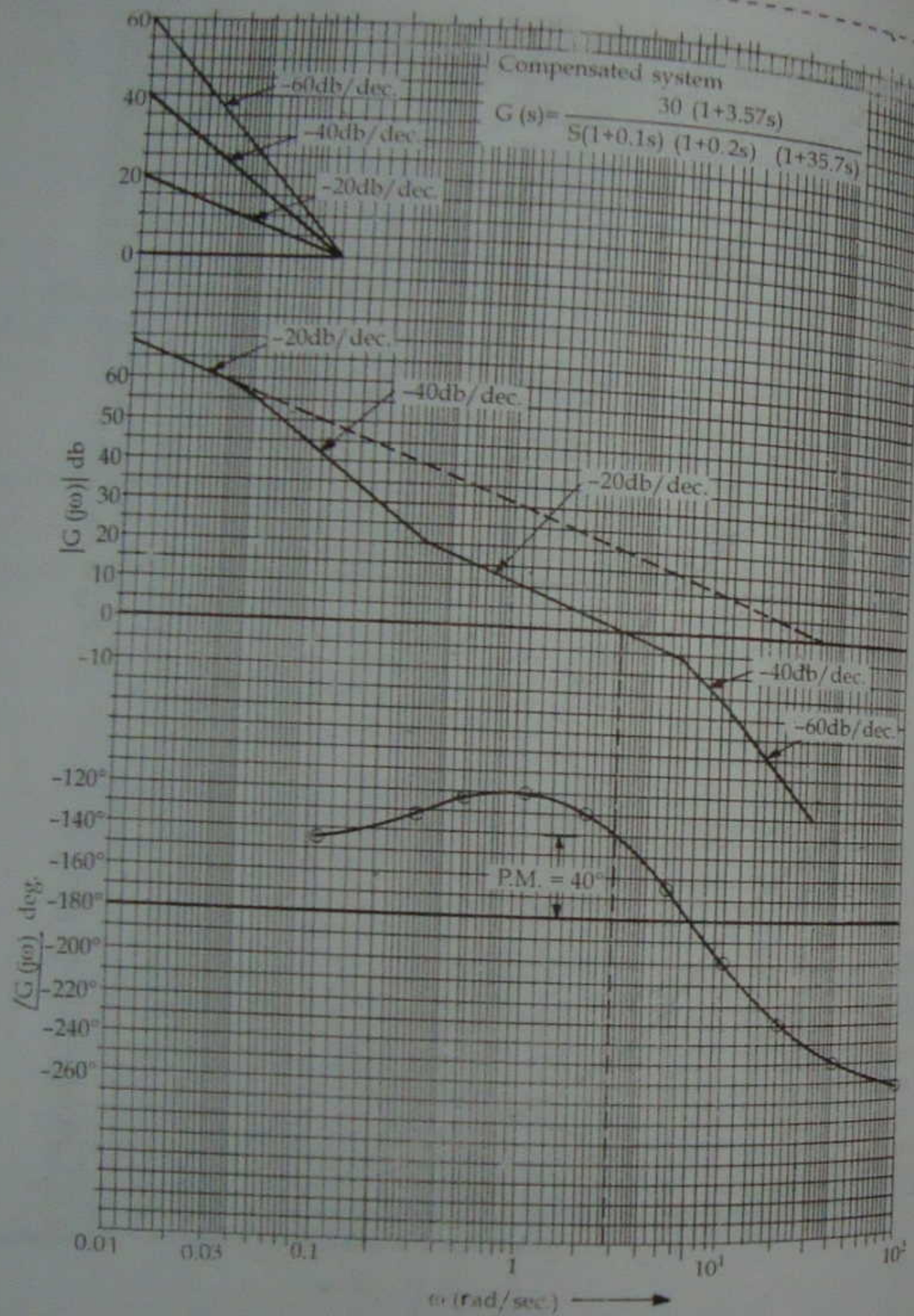


Fig. 6.7(b).

Solution : From the given transfer function put $s = j\omega$

$$Gc(j\omega) = K\frac{\left(j\omega + 1/T_1\right)\left(j\omega + 1/T_2\right)}{\left(j\omega + \beta/T_1\right)\left(j\omega + 1/\beta T_2\right)}$$

$$\angle Gc(j\omega) = \tan^{-1}\omega T_1 + \tan^{-1}\omega T_2 - \tan^{-1}\frac{\omega T_1}{\beta} - \tan^{-1}\omega T_2\beta \quad \text{...(6.30)}$$

Put $\omega = \omega_1 = \dfrac{1}{\sqrt{T_1T_2}}$ in equation 6.30

$$\angle Gc(j\omega) = \tan^{-1}\frac{T_1}{\sqrt{T_1T_2}} + \tan^{-1}\frac{T_2}{\sqrt{T_1T_2}} - \tan^{-1}\frac{T_1}{\beta}\cdot\frac{1}{\sqrt{T_1T_2}} - \tan^{-1}\frac{T_2}{\sqrt{T_1T_2}}\beta$$

$$\angle Gc(j\omega) = \tan^{-1}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\sqrt{\frac{T_2}{T_1}} - \tan^{-1}\frac{1}{\beta}\sqrt{\frac{T_1}{T_2}} - \tan^{-1}\beta\sqrt{\frac{T_2}{T_1}} \quad \text{...(6.31)}$$

From equation 6.31 take first two terms

$$\tan\left[\tan^{-1}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\sqrt{\frac{T_2}{T_1}}\right] = \frac{\sqrt{\frac{T_1}{T_2}} + \sqrt{\frac{T_2}{T_1}}}{1 - \sqrt{\frac{T_1}{T_2}}\sqrt{\frac{T_2}{T_1}}} \cdots \infty$$

$$\therefore \quad \tan^{-1}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\sqrt{\frac{T_2}{T_1}} = \pi/2 \quad \text{...(6.32)}$$

similarly,

$$\tan\left[\tan^{-1}\frac{1}{\beta}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\beta\sqrt{\frac{T_2}{T_1}}\right] = \frac{\frac{1}{\beta}\sqrt{\frac{T_1}{T_2}} + \beta\sqrt{\frac{T_2}{T_1}}}{1 - \frac{1}{\beta}\sqrt{\frac{T_1}{T_2}}\cdot\beta\sqrt{\frac{T_2}{T_1}}} = \infty$$

$$\therefore \quad \tan^{-1}\frac{1}{\beta}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\beta\sqrt{\frac{T_2}{T_1}} = \pi/2 \quad \text{...(6.33)}$$

From eq^n (6.31)

$$\angle Gc(j\omega) = \tan^{-1}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\sqrt{\frac{T_2}{T_1}} - \left(\tan^{-1}\frac{1}{\beta}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\beta\sqrt{\frac{T_2}{T_1}}\right) \quad \text{...(6.34)}$$

Put the values from 6.32 & 6.33 in eq^n 6.34

$$\angle Gc(j\omega) = \pi/2 - \pi/2 = 0$$

Therefore, the angle of $Gc(j\omega)$ becomes zero at $\omega_1 = 1/\sqrt{T_1T_2}$ **Proved.**

Example 6.6. The open loop transfer function of type two system with unity feedback is given by

$$G(s) = \frac{K}{s^2(1 + 0.25s)}$$

Design a lead compensator to meet the following specifications
1. acceleration constant $Ka = 10$
2. P.M = 35°

Solution : Step 1 :

$$Ka = \lim_{s\to 0} s^2 C(s) = \lim_{s\to 0} s^2 \frac{K}{s^2(1+0.25s)} = K$$

$$\therefore \quad K = Ka = 10$$

$$\therefore \quad G(s) = \frac{10}{s^2(1+0.25s)}$$

Step 2 : Draw the Bode plot for uncompensated system.

Table 6.9.

ω	- Arg $(j\omega)^2$	- Arg $(1+ j0.25\omega)$	Resultant
0.1	- 180°	- 1.43°	- 181.43°
0.2	- 180°	- 2.86°	- 182.86°
0.4	- 180°	- 5.71°	- 185.71°
0.8	- 180°	- 11.3°	- 191.3°
1.0	- 180°	- 14.03°	- 194.03°
2.0	- 180°	- 26.56°	- 206.56°
3.0	- 180°	- 36.87°	- 216.87°
5.0	- 180°	- 51.34°	- 231.34°
10.0	- 180°	- 68.19°	- 248.19°
20.0	- 180°	- 78.69°	- 258.69°
30.0	- 180°	- 82.4°	- 262.4°
50.0	- 180°	- 85.43°	- 265.42°

Step 3 : From Bodes plot :

specified $P.M\ \phi_s = 35$

$P.M$ avilable = - 38

$$\epsilon = 11$$

$$\therefore \quad \phi_m = 35 - (-38) + 11 = 84°$$

Since $\phi_m > 60°$, two identical network each contributing a maximum lead of $\phi_m/2$ are used

$$\therefore \quad \phi_m/2 = 42°$$

$$\sin \phi_m = \frac{a-1}{a+1}$$

$$\therefore \quad \sin 42 = \frac{a-1}{a+1}$$

$$\boxed{\therefore \quad a = 5.04}$$

Step 4 : Zero freq. attenuation = $\angle$ [-10loga]

= - 14.04 db.

Draw a line at - 14.04 db on magnitude curve. From graph $\omega_m = 6$ rad/sec.

Step 5 : Calculation of 'T'

$$\omega_m = \frac{1}{\sqrt{a}\ T}$$

$$6 = \frac{1}{\sqrt{5.04}\ T}$$

$$\therefore \quad \boxed{T = 0.0742}$$

Step 6 : Calculation of $Gc\ (s)$

$$Gc(s) = \frac{1}{a}\left(\frac{1+aTs}{1+Ts}\right) = \frac{(1+0.374s)^2}{(1+0.074s)^2}$$

Step 7 : Transfer function of compensated system

$$G'(s) = Gc(s)\ G(s)$$

$$G'(s) = \frac{10(1+0.374s)^2}{s^2(1+0.25s)\ (1+0.074s)^2}$$

Step 8 : Draw the Bode plot for $G'(s)$

Corner frequencies are $\omega_1 = 2.67$ rad/sec.

$\omega_2 = 4$ rad/sec.

$\omega_3 = 13.5$ rad/sec.

From the graph P.M = 28° instead of 35°, this is because of the excessive lag of the fixed part of the system at the new cross over frequency. If P.M = 28° is not acceptable, redesigned the system with greater value of ϵ.

Table 6.10.

ω	-Arg$(j\omega)^2$	-Arg $(1+j0.25\omega)$	-Arg $(1+j0.074\omega)^2$	+Arg $(1+0.374\omega)^2$	Resultant
0.1	-180°	-1.43°	-0.85°	+ 4.28°	-178°
0.2	-180°	-2.86°	-1.69°	+ 8.55°	-176°
0.4	-180°	-5.71°	-3.39°	+ 17.02°	- 172.08°
0.8	-180°	-11.3°	-6.76°	+ 33.31°	- 164.75°
1.0	-180°	- 14.03°	- 8.46°	+ 41.01°	- 161.48°
2.0	-180°	- 26.56°	- 16.83°	+ 73.59°	- 149.8°
3.0	-180°	- 36.87°	- 25.0°	+ 96.58°	- 145.29°
5.0	-180°	- 51.34°	- 40.6°	+ 123.73°	- 148.21°
10.0	-180°	- 68.19°	- 73°	+ 150°	- 171.19°
20.0	-180°	- 78.69°	- 111.9°	+ 164.77°	- 205.82°
30.0	-180°	- 82.4°	- 131.5°	+ 169.81°	- 224°
50.0	-180°	- 85.43°	- 149.75°	+ 173.87°	- 241.31°

6.4. COMPENSATION USING ROOT LOCUS

Root locus method can also be used to design a compensator. When an improvement is required in steady state performance, a lag compensator is designed. The poles and zeros of lag compensator are placed in such a way that it does not affect the root loci carrying the dominant closed loop poles. First plot the root locus for uncompensated system and determine the value of 'K' for desired damping ratio and obtain the dominant poles. If steady state specifications are not met design a lag compensator. The pole and zero of the compensator should be closed to the origin so that root locus remains the same.

When an improvement in transient performance is required lead compensator should be designed. The poles and zeros of the lead compensator should be so placed that the resulting closed loop transfer function has a pair of complex poles in the left half of s-plane with the required damping ratio and natural frequency ω_n, and also all other poles are either near a zero or much

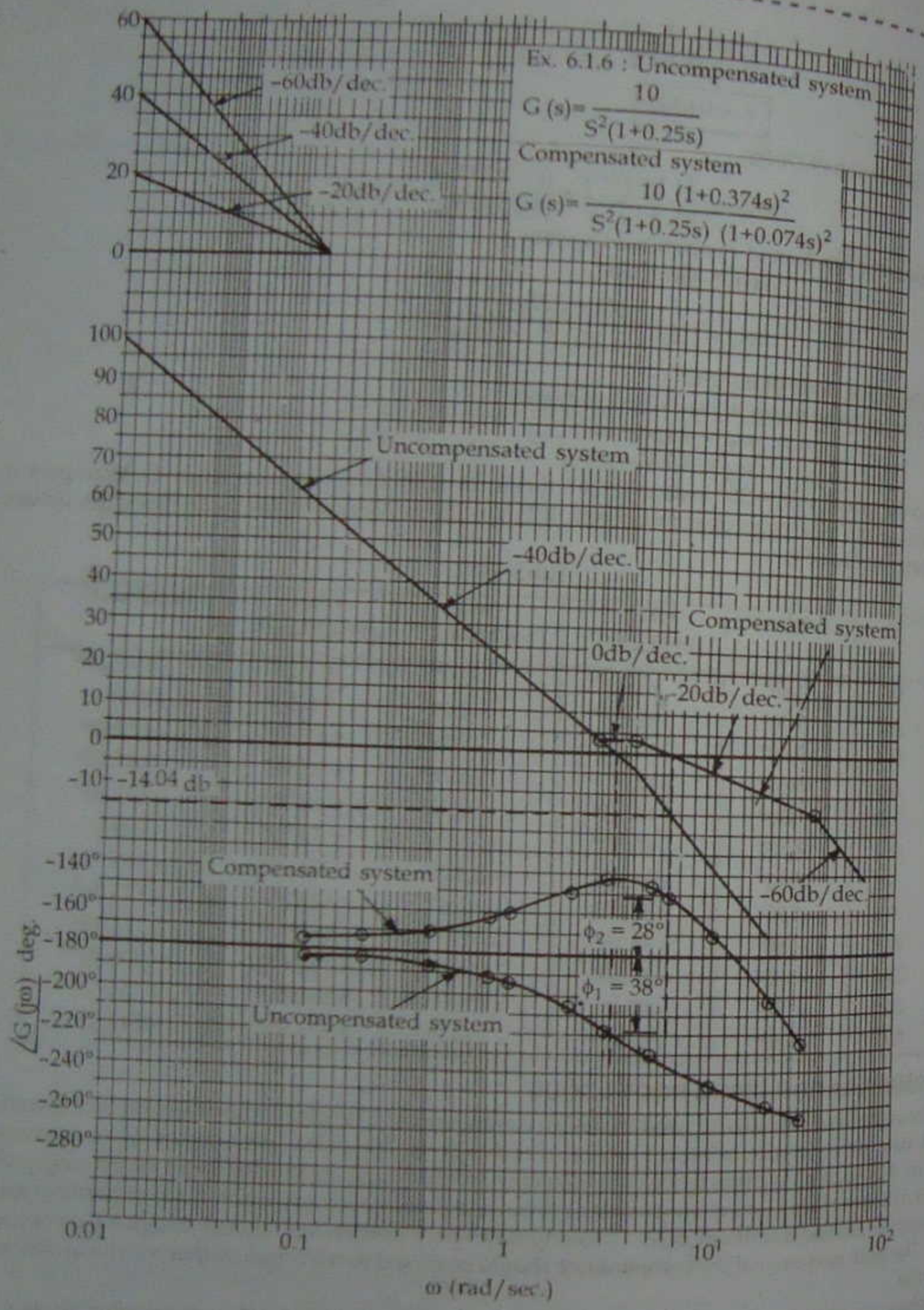


Fig. 6.8.

away from the origin. In such cases the poles (other than complex poles) cannot be contributed to the system transient.

6.4.1. Lag Compensation

The compensator having the transfer function

$$Gc(s) = \frac{1+aTs}{1+Ts} \qquad a < 1$$

is known as lag compensator.

The lag compensator inproves the steady state performance of the system.

Example 6.7. Design a lag compensator for a system whose open loop transfer function is

$$G(s) = \frac{K}{s(s+1)(s+4)}$$

to meet the following specifications.

Damping ratio = 0.5

Settling time t_s = 10 sec.

Velocity error constant $K_v \geq 5$

Solution: Step 1 : Draw the root locus for uncompensated system.

(a) Number of Poles $P = 3$

$s_1 = 0, s_2 = -1, s_3 = -4$

$z = 0$

Plot the pole and zero

(b) The segment between $s = 0$ and $s = -1$, from $s = -4$ to onwords are the parts of the root locus.

(c) Centroid of the asymptotes

$$\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{P-Z} = \frac{0-1-4}{3} = -5/3$$

(d) Angles of asymptotes

$$\phi = \frac{2K+1}{P-Z}180°$$

$K = 0$ $\quad \phi_1 = 60°$

$K = 1$ $\quad \phi_2 = 180°$

$K = 2$ $\quad \phi_3 = 300°$

(e) Breakaway point

The characteristic equation $1 + G(s)\,H(s) = 0$

or, $s(s+1)(s+4) + K = 0$

or, $s^3 + 5s^2 + 4s + K = 0$

or, $K = -(s^3 + 5s^2 + 4s)$

$$\frac{dK}{ds} = 0$$

$$\frac{dK}{ds} = -(3s^2 + 10s + 4) = 0$$

$3s^2 + 10s + 4 = 0$

$s = -2.86, -0.46$

$s = -0.46$ is the breakaway point

(f) point of intersection of root locus with $j\omega$ axis

characteristic eq^n. $s^3 + 5s^2 + 4s + K = 0$

Routh array

s^3	1	4
s^2	5	K
s^1	$\frac{20-K}{5}$	
s^0	K	

When $K = 20$, the auxiliary equation

$$A(s) = 5s^2 + K = 0$$
$$5s^2 + 20 = 0$$
$$5s^2 = -20$$
$$s^2 = -4$$
$$s = \pm j2$$

The complete root locus is shown in the fig 6.9

Step 2: Draw a line for $\zeta = 0.5$

$$\cos^{-1} 0.5 = 60^\circ$$

Draw a line at angle of 60° with negative real axis.

Step 3 : Find the value of K at the point A

$$OA = 0.8$$
$$AB = 0.9$$
$$AC = 3.7$$
$$\therefore \quad K = OA \times AB \times AC$$
$$= 0.8 \times 0.9 \times 3.7 = 2.664$$

Step 4 : The velocity constant for the cascade system should satisfy

$$K_v = \lim_{s \to 0} s \cdot \frac{s + \frac{1}{aT}}{s + \frac{1}{T}} \cdot \frac{2.664}{s(s+1)(s+4)}$$

$$5 = \frac{1}{a} \cdot \frac{2.664}{4}$$

$$\therefore \quad a = 0.13$$

The lag compensator should be so placed that the root locus is not disturbed by putting the compensator pole and zero near the origin. Generally, 10% of distance between the origin and the first pole on the negative real axis is taken for the location of compensator zero

$\therefore$ The zero at $-\frac{1}{aT}$ is at -0.1 (10% of 1)

$$-\frac{1}{aT} = -0.1$$
$$a = 0.13$$
$$\frac{1}{T} = 0.013$$

Hence, the transfer function of compensator $Gc = \frac{s+0.1}{s+0.013}$

The overall transfer function = $G(s)\,G(c)$

$$G'(s) = \frac{2.664(s+0.1)}{s(s+1)(s+4)(s+0.013)}$$

Example 6.8. For example (5.23) design a lag compensator for

$$\mathbf{K}v \geq 5$$
$$\zeta = 0.707$$
$$\omega_n = 2 \text{ rad/sec.}$$

Solution : The complete root locus is shown in fig 6.10. (For calculations readers please see the example 5.23)

The value of K at the point A :

$$K = OA \times AB \times AC = 1.8 \times 3.2 \times 4.1 = 23.6$$

The velocity constant for cascade system should satisfy

$$K_v = \lim_{s \to 0} s \cdot \frac{s + \frac{1}{aT}}{s + \frac{1}{T}} \frac{23.6}{s(s+4)(s+5)}$$

$$5 = \frac{1}{a} \frac{23.6}{20} \qquad \therefore \quad a = 0.236$$

Take the distance 10% of the first pole from the origin for compensator zero i.e. $\frac{-1}{aT}$ will be placed at -0.4 from origin.

$$\frac{1}{aT} = 0.4 \qquad \therefore \frac{1}{T} = 0.0944$$

$\therefore$ Transfer function of the compensator $Gc = \frac{s+0.4}{s+0.0944}$

The overall transfer function after compensation $G'(s) = G(s)\,Gc(s)$

$$G'(s) = \frac{23.6(s+0.4)}{s(s+4)(s+5)(s+0.0944)}$$

6.4.2. Lead Compensation

The following steps are involved for designing a lead compensator.

1. Draw the root locus for the given open-loop transfer function.
2. From the transient response specification determine the desired location for the dominant closed-loop poles by

$$S_d = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2} \qquad \ldots(6.35)$$

Where, ξ is the damping ratio and ω_n is the undamped natural frequency. The damping ratio determines the angular location of the poles and distance of the pole from origin is determined by undamped natural frequency ω_n.

3. Connect all the open-loop poles and zeros to one of the dominant closed loop poles (at desired location) and measure the angles & then find the sum of angles. Now calculate the angle deficiency ϕ to be added, so that sum of angle is equal to $\pm 180\,(2k+1)$ at that point.

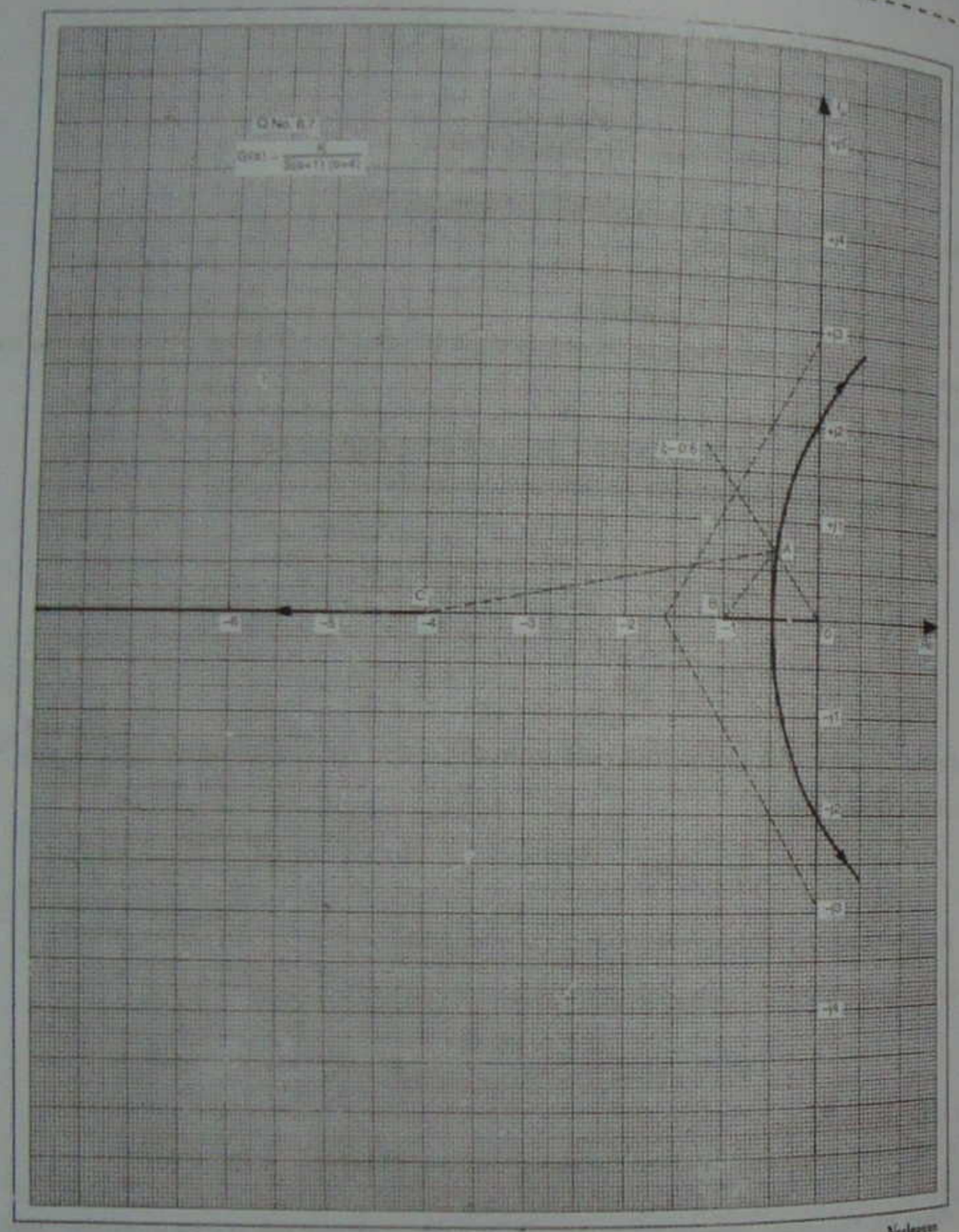


Fig. 6.9.

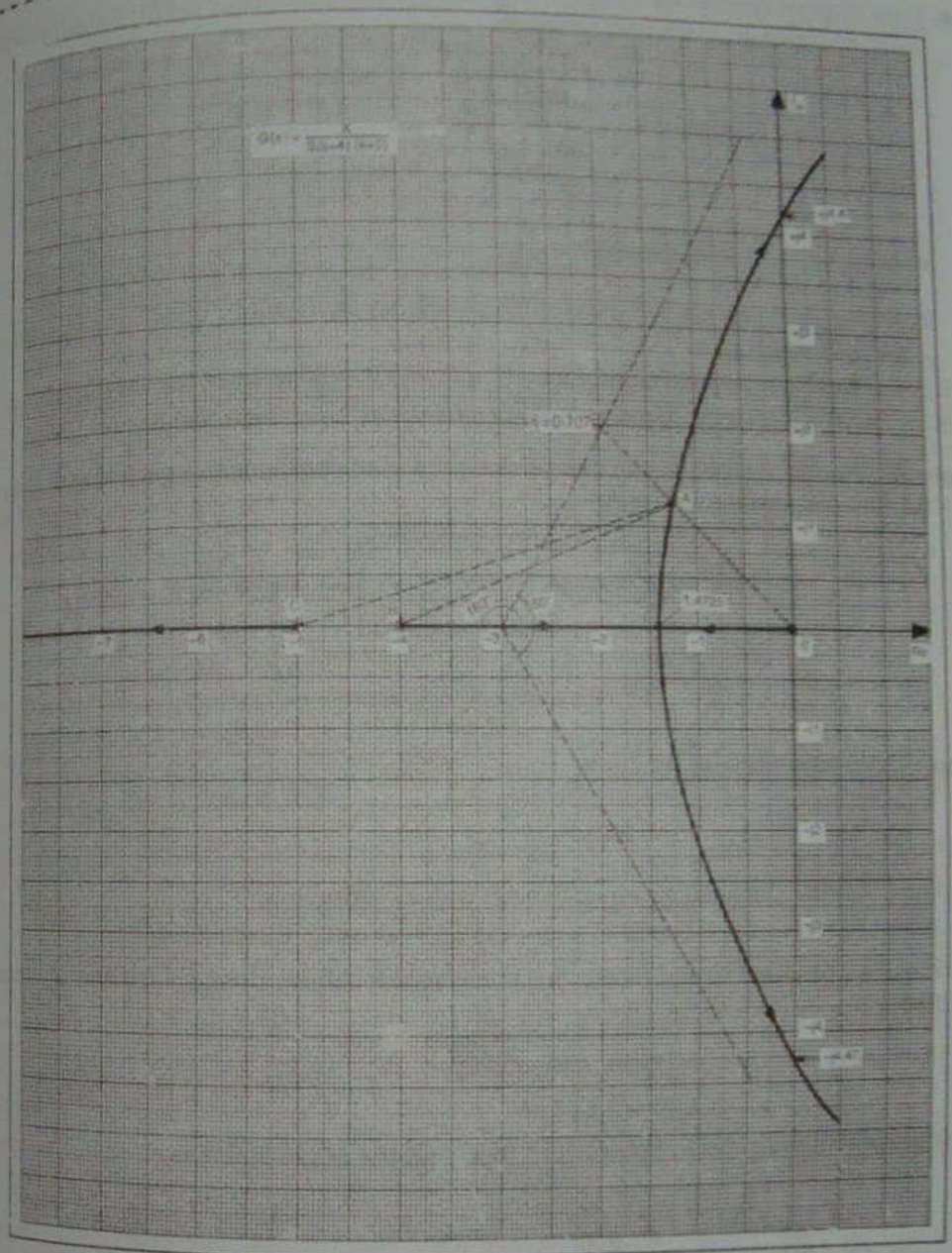


Fig. 6.10

4. If the original system has the open-loop transfer function $G(s)$, then the compensated system will have open-loop transfer function.

$$G_c(s)\,G(s) = \left(k_c \frac{s+\frac{1}{T}}{s+\frac{1}{\alpha T}} \right) G(s) \quad \text{...(6.36)}$$

where, $$G_c(s) = k_c \alpha \frac{1+sT}{1+\alpha T s} \quad \text{...(6.37)}$$

α and T can be determined from angle deficiency. K_c is determined from the requirement of the open-loop gain.

5. Determine the open-loop gain of the compensated system from the magnitude condition.
6. Now check whether all performance specifications have been met or not.

Example 6.9. For the system shown above, design a lead compensator such that $\omega_n = 4$ rad/sec. and $\xi = 0.5$ for compensated system. *(UPTU 2002-2003)*

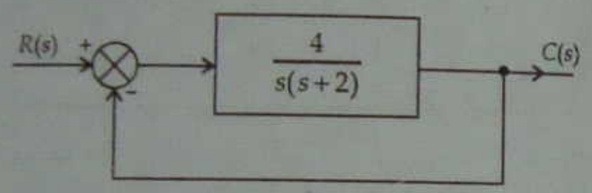


Fig. 6.11.

Solution : Open-loop transfer function

$$G(s)\,H(s) = \frac{4}{s(s+2)} \quad \text{...(6.38)}$$

Step 1 : Draw the root locus of given open loop transfer function.

Note : Students are advised to draw the root-locus by themselves.

Step 2 : Plot the dominant closed loop poles at desired location. Here, $\xi = 0.5$ and $\omega_n = 4$rad/sec. given

$$\therefore \quad s_d = \xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$$

$$\therefore \quad s_d = -2 \pm j\,3.46 \quad \text{...(6.39)}$$

Step 3 : Connect all the poles to point P and then find the angle deficiency.

$$\angle \left.\frac{4}{s(s+2)}\right|_{s=-2\pm j3.46} = -210°$$

[Alternative $180 - (90 + 120) = -30°$]

Thus lead compensator must contribute $\phi = 30°$ at this point

Step 4 : Determine location of zeros and pole of the lead compensator as follows.

Draw a horizontal line passing through point P to A connect the point P to the origin. Measure the angle APO and bisect it. PB is the bisector. Draw two lines PC & PD that makes angle $\pm\,\phi/2 = 30/2 = 15°$. The intersection of PC & PD with negative real axis gives the location for zero and pole of lead network.

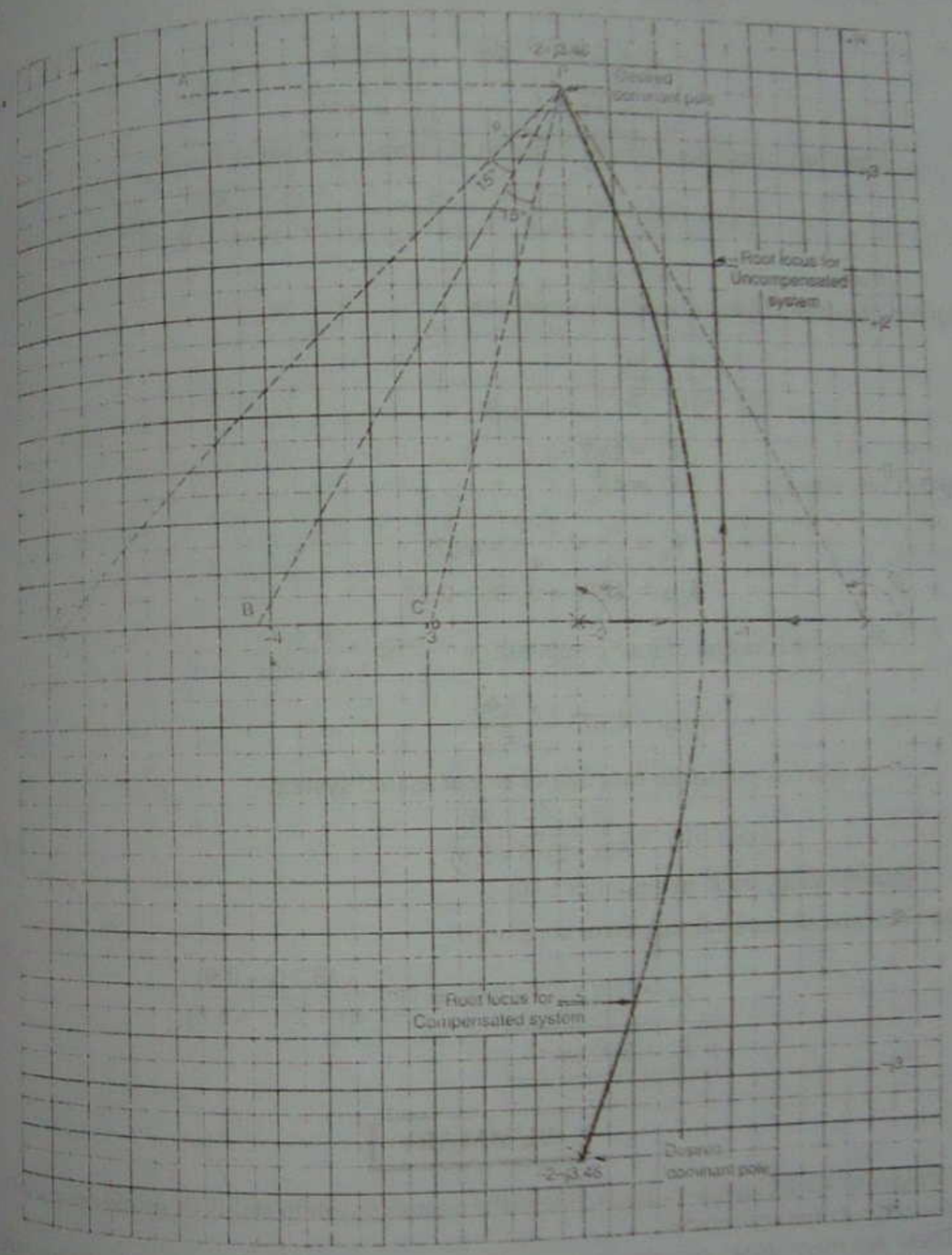


Fig. 6.12

Step 5 : From the plot

zero at $s = -2.96$

Pole at $s = -5.5$

$$\therefore \quad \frac{1}{T} = 2.96 \quad \& \quad \frac{1}{\alpha T} = 5.5$$

$$\boxed{T = 0.337} \qquad \boxed{\alpha = 0.538}$$

Therefore open-loop transfer function of compensated system is

$$G_c(s)\,G(s) = K_c \frac{s+2.96}{s+5.5} \cdot \frac{4}{s(s+2)} = \frac{K'(s+2.96)}{s(s+2)(s+5.5)} \quad \ldots(6.40)$$

Where $K' = 4K_c$

Step 6 : Calculate K' from magnitude condition

$$\left|\frac{K'(s+2.96)}{s(s+2)(s+5.5)}\right|_{s_d=-2\pm j3.46} = 1$$

$$K' = 18.7$$

Step 7 : Calculate K_c $\qquad K' = 4K_c$

$$\therefore \quad K_c = \frac{K'}{4} = \frac{18.7}{4} = 4.675$$

$$\therefore \quad K_c\alpha = 4.675 \times 0.538 = 2.52$$

$$\therefore \text{ Transfer function of lead compensator} = 2.52\frac{1+0.337s}{1+0.182s}$$

or, $$G_c(s) = 4.675\,\frac{s+2.96}{s+5.5} \quad \ldots(6.41)$$

Also, open loop transfer function of compensated system

$$G_c(s)\,G(s) = \frac{18.7\,(S+2.96)}{S(S+2)\,(S+5.5)} \quad \ldots(6.42)$$

(put the value of K' in equation 6.40)

Step 8 : We can calculate velocity error constant K_v

$$K_v = \lim_{s\to 0} s\,G_c(s)\,G(s) = \lim_{s\to 0} \frac{s\,18.7(s+2.9)}{s(s+2)(s+5.5)}$$

$$\therefore \quad \boxed{K_v = 5.02\ \text{sec}^{-1}}$$

SUMMARY

Compensation technique is used to make the unstable system stable by introducing the poles and/or zeros at suitable places.

There are many types of compensation depending upon the location of the compensating network. Such as series compensation, feedback compensation and load compensation.

Transfer function of phase lead network is given by

$$\frac{E_2(s)}{E_1(s)} = \frac{1}{a}\left[\frac{1+aTs}{1+Ts}\right]$$

In phase lead network zero is nearer to the imaginary axis as compared to pole, this gives the positive phase shift.

In phase lead network phase margin increased, bandwidth increased and response is faster.

Transfer function of lag network is given by

$$\frac{E_2(s)}{E_1(s)} = \frac{1+aTs}{1+Ts}$$

In lag network pole is near to the imaginary axis. Lag network reduces the phase shift. In lag network there is decrease in gain cross over frequency, bandwidth decrease, rise time and setting time is large. Response is slower.

In lead-lag network speed of the response and steady state error can be simultaneously increased.

SEMI-OBJECTIVE TYPE QUESTIONS

(i) Write short note on different type of compensations. (Article 6.2)
(ii) Draw the Bode plot for phase lead network. (Article 6.3)
(iii) Write the procedure for phase lead compensation. (Article 6.3)
(iv) Write the effect of phase lead network. (Article 6.3)
(v) Write short note on phase lag network. (Article 6.3.4)
(vi) Draw the Bode plot for phase lag network. (Article 6.3.4)
(vii) What is the effect of phase lag network. (Article 6.3.5)
(viii) Write short note on lead-lag compensation. (Article 6.3.6)

□□□

Chapter 7

Non-Linear System Analysis

7.1. INTRODUCTION

In a linear system the form of output does not depend on the magnitude of the input, when the input increases the output will also increases but the form remains the same. In non-linear system form may change with change in magnitude of the input. For example in a linear system, if the input is sinusoidal the output of the linear system will also be sinusoidal but in non linear the output will be non-sinusoidal. The non linear systems does not possess the homogeneity and superposition properties*. The stability of the non-linear systems depends on the input and the initial state of the system. If a non-linear system, the input and outputs are dercribed by the non linear differential equation then it is called dynamic non-linearity and if the input and output are not describe by the differential equations then it is known as static non linearity. Sometime to improve the characteristic of the system some non-linearities are intentionally added to the system. Such type of non-linearities are called intentional non linearities. If the non-linearties are inherently present in the system (*e.g.* saturation, dead zone, friction etc.) then these non linearities are known as incidental non-linearities.

7.2 SOME COMMON TYPES OF NON-LINEARITIES

Some basic common non-linearities are discuss in brief.

7.2.1. Saturation

The saturation characteristic is shown in fig. 7.1. The common example of saturation is the relationship between magnetic flux and current in an iron cored coil. An iron core coil has linear inductance for low values of current but if the current increases, the core becomes saturate and inductance becomes non linear. Another example is resistor, a resistor is linear for low values of current but becomes non-linear for higher values of current because of the temperature. Transistors, operational amplifiers and magnetic components are having this type of nonlinearity.

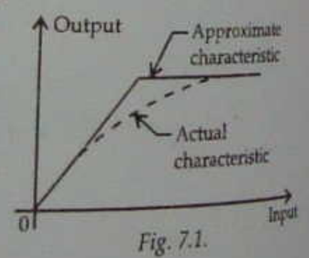


Fig. 7.1.

*If the magnitude of the input increase '*K*' times, the output magnitude also increased '*K*' times then this property is called homogeneity. *e.g.*, if the input is '*x*' and output is '*y*' and the input becomes *Kx* then the output magnitude should be *Ky*.

If the two inputs are applied simultaneously, then output will be the sum of two outputs. This property is called superposition. *eg.* if one input is x_1 and output y_1 and other input is x_2 and output y_2 then $x_1 + x_2 \rightarrow y_1 + y_2$.

7.2.2. Dead Zone

Sometimes the output is zero when the input crosses the certain value. Such type of non-linearities are shown in motors, actuators and in *d.c.* servomotors. This characteristic is shown in fig. 7.2.

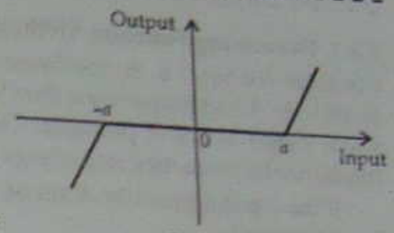


Fig. 7.2.

7.2.3. Friction

Frictional force opposes the motion of a body. The friction is of two types (*a*) static friction (*b*) Dynamic friction. Static friction is the friction when the body tends to move and the dynamic friction is experienced by a body when in motion. The dynamic friction can further classified as sliding friction and rolling friction. The sliding friction experienced by a body when it slides over another body. The rolling friction experienced by a body when it rolls over the other. In mechanical system there are two types of friction (*i*) viscous friction (*ii*) static friction. Viscous friction is proportional to the relative velocity between the moving surfaces. The force-velocity relationship is linear. This characteristic is shown in fig 7.3.

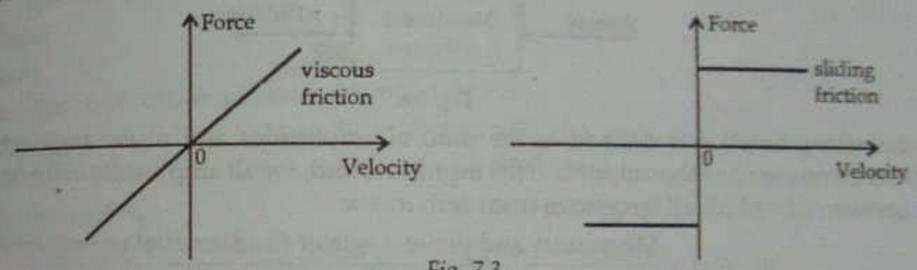


Fig. 7.3.

7.2.4. Relays

A relay is a non-linear power amplifier which provides a large power amplifications, therefore they are used in control system. The characteristic of ideal relay is shown in fig. 7.4*a*. The actual relay having a dead zone as shown in fig. 7.4*b*.

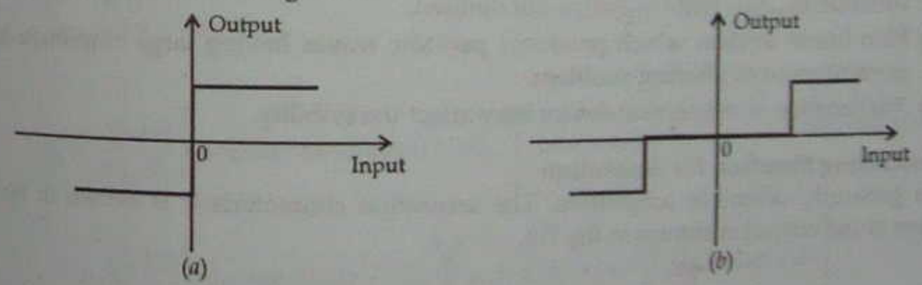


Fig. 7.4.

7.2.5. Backlash

Backlash is the play between the teeth of the drive gear and driven gear. The characteristic is shown in fig. 7.5.

7.3. STUDY OF NON-LINEAR SYSTEMS

For the study of non-linear systems we shall consider the following methods :

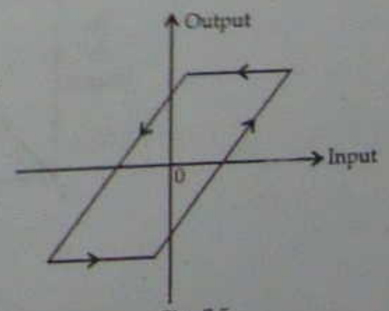


Fig. 7.5.

7.3.1. Describing Function Method

Consider the fig. 7.6. A non-linear device is represented by a block. If we apply a sine wave of amplitude A and frequency ω then the actual output of a non-linear system does not have sinusoidal wave shape but it is periodic. The Fourier series of non-sinusoidal wave form contains the fundamental frequency and a large number of harmonics.

If the input is given by A sin ωt, then the steady state output may be expressed in the following Fourier series

$$y(t) = \frac{A_o}{2} + A_1 \cos\omega t + B_1 \sin\omega t + A_2 \cos 2\omega t + B_2 \sin 2\omega t + \ldots \quad \ldots(7.1)$$

The first two components of equation (7.1) are the fundamental components and others are harmonics. For writing the equation (7.1) it is assumed that non-linearity does not generate subharmonics and if non-linearity is symmetrical then the average value of $y(t)$ will be zero, so the output is given by

$$y(t) = A_1 \cos\omega t + B_1 \sin\omega t + A_2 \cos 2\omega t + B_2 \sin 2\omega t + \ldots \quad \ldots(7.2)$$

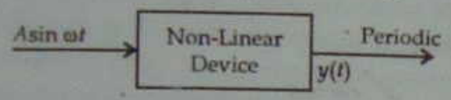


Fig. 7.6.

The describing function is defined as the ratio of amplitudes and phase angle between the fundamental components of the output and the input sinusoid, for all amplitudes of the input which are to be considered and for all freqencies from zero to + ∞

$$\therefore\ G_D(j\omega) = \frac{\text{Magnitude and phase angle of fundamental component of output}}{\text{Magnitude and phase angle of input}}$$

7.3.2. Limitations

(*a*) If the system having one nonlinear part can be analysed by this method. If the system having two non-linear part, this method is not suitable because the product of two describing function $G_{D1}(j\omega)$ and $G_{D2}(j\omega)$ is not defined.

(*b*) Non-linear devices which produces periodic waves having large amplitude harmonics present more of filtering problem.

(*c*) The location of non-linear device may affect the stability.

7.3.3. Describing Function for Saturation

This generally occurs in amplifiers. The saturation characteristic is shown in fig. 7.7. The waveform of the output is shown in fig. 7.8.

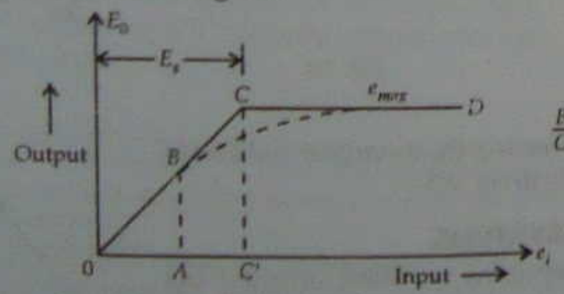


Fig. 7.7.

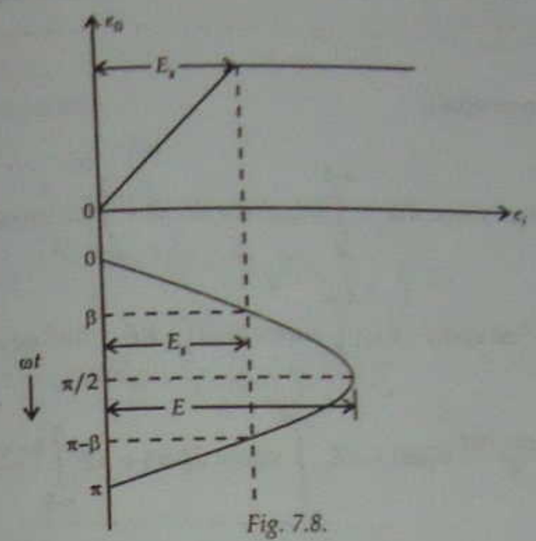


Fig. 7.8.

E_s = Value of input required to saturate the system.

let input $\quad e_i = E \sin\omega t$ $\quad \ldots(7.3)$

output $\quad e_o = Ke_i \quad 0 \le \omega t \le \beta$

The output will remain constant from β to (π − β)

$$e_o = KE_s \quad \beta \le \omega t \le (\pi - \beta)$$

$$e_o = Ke_i \quad (\pi - \beta) \le \omega t \le \pi$$

Calculation of A_1 :

$$A_1 = \frac{2}{\pi}\int_o^{\pi} y(t)\cos\omega t\, dt$$

$$A_1 = \frac{2}{\pi}\left[\int_o^{\beta} KE\sin\omega t.\cos\omega t\, d(\omega t) + \int_{\beta}^{\pi-\beta} KE_s\cos\omega t\, d(\omega t) + \int_{\pi-\beta}^{\pi} KE\sin\omega t\cos\omega t\, d(\omega t)\right] \quad \ldots(7.4)$$

$$A_1 = \frac{2}{\pi}\left[\frac{KE}{2}\int_o^{\beta} 2\sin\omega t\cos\omega t\, d(\omega t) + KE_s\int_{\beta}^{\pi-\beta}\cos\omega t\, d(\omega t) + \frac{KE}{2}\int_{\pi-\beta}^{\pi} 2\sin\omega t\cos\omega t\, d(\omega t)\right]$$

$$= \frac{2}{\pi}\left[\frac{KE}{2}\int_o^{\beta}\sin 2\omega t\, d(\omega t) + KE_s\int_{\beta}^{\pi-\beta}\cos\omega t\, d(\omega t) + \frac{KE}{2}\int_{\pi-\beta}^{\pi}\sin 2\omega t\, d\omega t\right]$$

$$= \frac{2}{\pi}\left[\frac{-KE}{2}\cdot\frac{1}{2}\{\cos 2\omega t\}_o^{\beta} + KE_s\{\sin\omega t\}_{\beta}^{\pi-\beta} - \frac{KE}{2}\cdot\frac{1}{2}\{\cos 2\omega t\}_{\pi-\beta}^{\pi}\right]$$

$$= \frac{2}{\pi}\left[-\frac{KE}{4}\cos 2\beta + \frac{KE}{4} + KE_s\{\sin\beta - \sin\beta\} - \frac{KE}{4}\{1-\cos 2\beta\}\right]$$

$$= 0$$

$$\therefore\ \boxed{A_1 = 0}$$

CALCULATION OF B_1 :

$$B_1 = \frac{2}{\pi}\int_0^{\pi} y(t)\sin\omega t\, d(\omega t)$$

$$B_1 = \frac{2}{\pi}\left[\int_0^{\beta} KE\sin\omega t.\sin\omega t\, d\omega t + \int_{\beta}^{\pi-\beta} KE_s\sin\omega t\, d(\omega t) + \int_{\pi-\beta}^{\pi} KE\sin\omega t.\sin\omega t\, d(\omega t)\right]$$

$$= \frac{2}{\pi}\left[\int_0^{\beta} KE\sin^2\omega t\, d(\omega t) + KE_s\int_{\beta}^{\pi-\beta}\sin\omega t\, d(\omega t) + KE\int_{\pi-\beta}^{\pi}\sin^2\omega t\, d(\omega t)\right]$$

$$= \frac{2}{\pi}\left[KE\int_0^{\beta}\frac{1-\cos 2\omega t}{2}d(\omega t) + KE_s\int_{\beta}^{\pi-\beta}\sin\omega t\, d(\omega t) + KE\int_{\pi-\beta}^{\pi}\frac{1-\cos 2\omega t}{2}d(\omega t)\right]$$

$$= \frac{2}{\pi}\left[\frac{KE}{2}\left\{\int_0^{\beta} d(\omega t) - \int_0^{\beta}\cos 2\omega t\, d\omega t\right\} - KE_s|\cos\omega t|_{\beta}^{\pi-\beta} + \frac{KE}{2}\left\{\int_{\pi-\beta}^{\pi} d\omega t - \int_{\pi-\beta}^{\pi}\cos 2\omega t\, d\omega t\right\}\right]$$

$$B_1 = \frac{2}{\pi}\left[\frac{KE}{2}\left\{\beta - \frac{1}{2}|\sin 2\omega t|_0^{\beta}\right\} - KE_s|\cos(\pi-\beta) - \cos\beta| + \frac{KE}{2}\left\{(\pi-\pi+\beta) - \frac{1}{2}|\sin 2\omega t|_{\pi-\beta}^{\pi}\right\}\right]$$

$$= \frac{2}{\pi}\left[KE\beta + 2KE_s\cos\beta - \frac{KE}{2}\sin 2\beta\right] = \frac{2}{\pi}\left[KE\beta + 2KE_s\cos\beta - \frac{KE}{2}\cdot 2\sin\beta\cos\beta\right]$$

$$= \frac{2}{\pi}[KE\beta + 2KE_s\cos\beta - KE\sin\beta\cos\beta] = \frac{2KE}{\pi}\left[\beta + 2\frac{E_s}{E}\cos\beta - \sin\beta\cos\beta\right]$$

Since, $e_i = E\sin\omega t$

Put $e_i = E_s$ and $\omega t = \beta$

$\therefore$ $E_s = E\sin\beta$

$$\sin\beta = \frac{E_s}{E}$$

$\therefore$ $$\cos\beta = \sqrt{1-\left(\frac{E_s}{E}\right)^2}$$

$\therefore$ $$B_1 = \frac{2KE}{\pi}\left[\sin^{-1}\frac{E_s}{E} + 2\cdot\frac{E_s}{E}\sqrt{1-\left(\frac{E_s}{E}\right)^2} - \frac{E_s}{E}\sqrt{1-\left(\frac{E_s}{E}\right)^2}\right]$$

$$= \frac{2KE}{\pi}\left[\sin^{-1}\frac{E_s}{E} + \frac{E_s}{E}\sqrt{1-\left(\frac{E_s}{E}\right)^2}\right]$$

$\therefore$ $$G_D(j\omega) = \frac{\sqrt{A_1^2+B_1^2}}{E}\angle\tan^{-1}\frac{A_1}{B_1}$$

*The output in fig 7.8 has odd symmetry, its Fourier series will have only sine terms.

Put the values of A_1 and B_1

$\therefore$ Required describing function

$$G_D(j\omega) = \frac{B_1}{E}\angle\tan^{-1}\frac{A_1}{B_1}$$

$$G_D(j\omega) = \frac{1}{E}\cdot\frac{2KE}{\pi}\left[\sin^{-1}\frac{E_s}{E} + \frac{E_s}{E}\sqrt{1-\left(\frac{E_s}{E}\right)^2}\right]$$

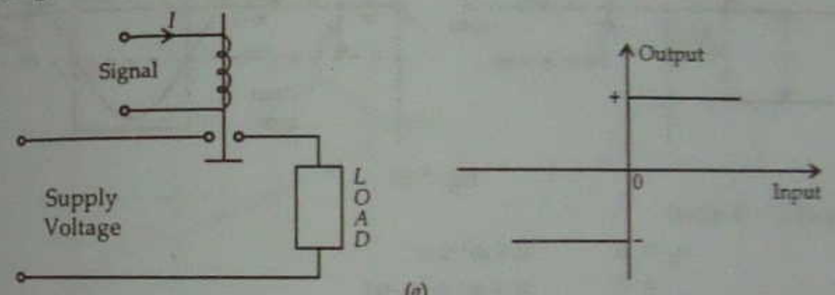
$$= \frac{2K}{\pi}\left[\sin^{-1}\frac{E_s}{E} + \frac{E_s}{E}\sqrt{1-\left(\frac{E_s}{E}\right)^2}\right]$$

If $\frac{E_s}{E} \triangleq R$, then

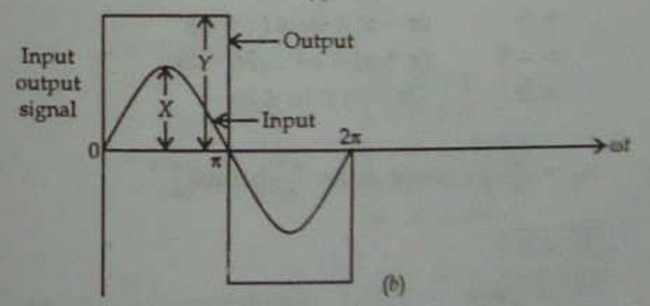
$\therefore$ $$\boxed{G_D(j\omega) = \frac{2K}{\pi}\left[\sin^{-1}R + R\sqrt{1-R^2}\right]\angle 0^{\circ}} \quad \ldots(7.5)$$

7.14. Describing Function for Ideal Relay

(a)



(a)



(b)

Fig. 7.9.

Response of the ideal relay

Let input to the relay $x = X\sin\omega t$

output $e_0 = Y \quad 0 \le \omega t \le \pi$

$= -Y \quad \pi \le \omega t \le 2\pi$

$$A_1 = \frac{2}{\pi}\int_0^{\pi} e_0\cos\omega t\, d\omega t = \frac{2}{\pi}\int_0^{\pi} Y\cos\omega t\, d(\omega t) = \frac{2}{\pi}Y|\sin\omega t|_0^{\pi}$$

$= 0$

$\therefore$ $$\boxed{A_1 = 0}$$

$$B_1 = \frac{2}{\pi}\int_0^{\pi} e_o \sin\omega t\, d(\omega t) = \frac{2}{\pi}\int_0^{\pi} Y \sin\omega t\, d(\omega t)$$

$$B_1 = \frac{2Y}{\pi}\int_0^{\pi} \sin\omega t\, d(\omega t) = \frac{4Y}{\pi}$$

$$\therefore \quad G_D(j\omega) = \frac{\sqrt{A_1^2 + B_1^2}}{X} \angle \tan^{-1} A_1/B_1$$

$$\boxed{G_D(j\omega) = \frac{4Y}{\pi X}\angle 0^\circ}$$

(b) RELAY WITH DEAD ZONE (PRACTICAL RELAY)

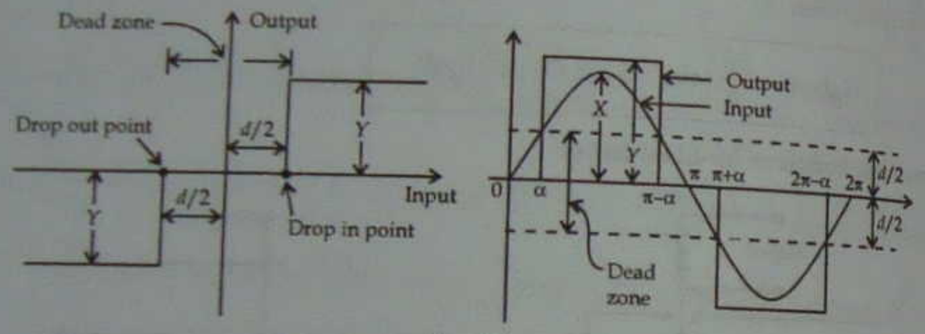


Fig. 7.10.

Let the input $x = X\sin\omega t$

output

$$\begin{aligned} e_0 &= 0 && 0 \le \omega t \le \alpha \\ &= Y && \alpha \le \omega t \le (\pi-\alpha) \\ &= 0 && (\pi-\alpha) \le \omega t \le (\pi+\alpha) \\ &= -Y && (\pi+\alpha) \le \omega t \le (2\pi-\alpha) \\ &= 0 && (2\pi-\alpha) \le \omega t \le 2\pi \end{aligned}$$

$$A_1 = \frac{2}{\pi}\int_{\alpha}^{\pi-\alpha} Y\cos\omega t\, d\omega t = \frac{2Y}{\pi}\left|\sin\omega t\right|_{\alpha}^{\pi-\alpha}$$

$$\therefore \quad \boxed{A_1 = 0}$$

$$B_1 = \frac{2}{\pi}\int_{\alpha}^{\pi-\alpha} Y\sin\omega t\, d\omega t = \frac{2Y}{\pi}\int_{\alpha}^{\pi-\alpha}\sin\omega t\, d(\omega t) = \frac{4Y}{\pi}\cos\alpha$$

when $\omega t = \alpha$

$$x = d/2 = X\sin\alpha$$

$$\sin\alpha = \frac{d}{2X}$$

$$\cos\alpha = \left[1-\left(\frac{d}{2X}\right)^2\right]^{1/2}$$

$$\therefore \quad B_1 = \frac{4Y}{\pi}\left[1-\frac{d^2}{4X^2}\right]^{1/2}$$

$$\therefore \quad G_D(j\omega) = \frac{\sqrt{A_1^2 + B_1^2}}{X} \angle \tan^{-1} A_1/B_1$$

$$= \frac{B_1}{X}\angle 0^\circ = \frac{4Y}{\pi X}\left[1-\frac{d^2}{4X^2}\right]^{1/2} \angle 0^\circ$$

$$\boxed{G_D(j\omega) = \frac{4Y}{\pi X}\left[1-\frac{d^2}{4X^2}\right]^{1/2} \angle 0^\circ}$$

7.35. Describing Function for Combination of Dead Zone and Saturation

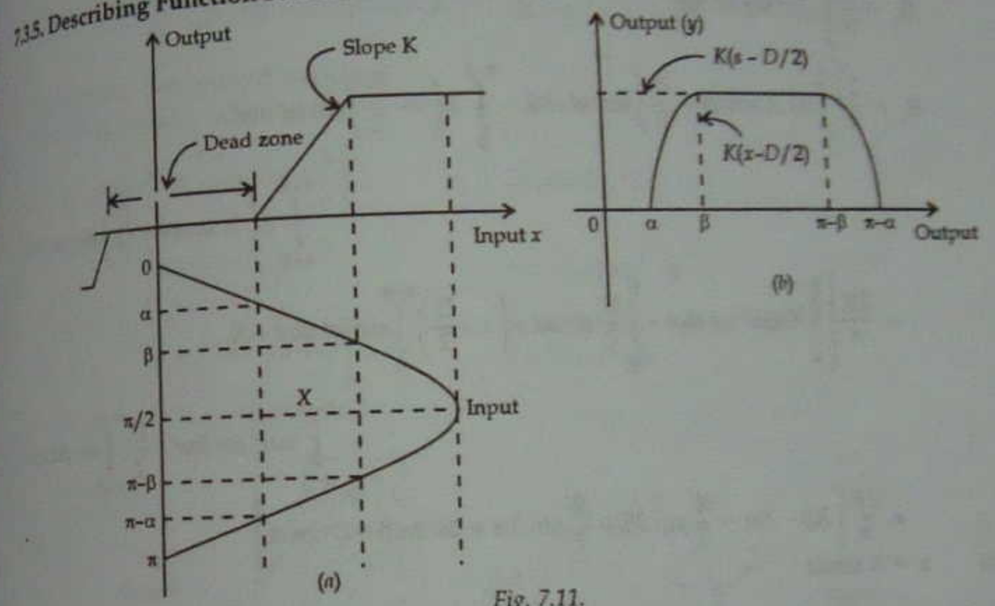


Fig. 7.11.

Let input $x = X\sin\omega t$

output

$$\begin{aligned} y &= 0 && 0 \le \omega t \le \alpha \\ &= K(x - D/2) && \alpha \le \omega t \le \beta \\ &= K(s - D/2) && \beta \le \omega t \le (\pi-\beta) \\ &= K(x - D/2) && (\pi-\beta) \le \omega t \le (\pi-\alpha) \\ &= 0 && (\pi-\alpha) \le \omega t \le \pi \end{aligned}$$

$$A_1 = \frac{2}{\pi}\left[\int_{\alpha}^{\beta} K\left(X\sin\omega t - \frac{D}{2}\right)\cos\omega t\, d(\omega t) + \int_{\beta}^{\pi-\beta} K\left(s - \frac{D}{2}\right)\cos\omega t\, d(\omega t) + \int_{\pi-\beta}^{\pi-\alpha} K\left(X\sin\omega t - \frac{D}{2}\right)\cos\omega t\, d\omega t\right]$$

$$A_1 = \frac{2K}{\pi}\left[\frac{X}{2}\int_{\alpha}^{\beta} 2\sin\omega t\cos\omega t\, d\omega t - \frac{D}{2}\int_{\alpha}^{\beta}\cos\omega t\, d\omega t + \left(s - \frac{D}{2}\right)\int_{\beta}^{\pi-\beta}\cos\omega t\, d\omega t + \frac{X}{2}\int_{\pi-\beta}^{\pi-\alpha} 2\sin\omega t\cos\omega t\, d\omega t - \frac{D}{2}\int_{\pi-\beta}^{\pi-\alpha}\cos\omega t\, d\omega t\right]$$

$$A_1 = \frac{2K}{\pi}\left[\frac{X}{2}\int_{\alpha}^{\beta}\sin 2\omega t\, d\omega t - \frac{D}{2}\int_{\alpha}^{\beta}\cos\omega t\, d\omega t + \left(s-\frac{D}{2}\right)\int_{\beta}^{\pi-\alpha}\cos\omega t\, d\omega t + \frac{X}{2}\int_{\pi-\beta}^{\pi-\alpha}\sin 2\omega t\, d\omega t - \frac{D}{2}\int_{\pi-\beta}^{\pi-\alpha}\cos\omega t\, d\omega t\right]$$

$= 0$

$$\boxed{A_1 = 0}$$

$$B_1 = \frac{2}{\pi}\int_{0}^{\pi} y(t)\sin\omega t\, d\omega t$$

$$B_1 = \frac{2}{\pi}\left[\int_{\alpha}^{\beta} K\left(X\sin\omega t - \frac{D}{2}\right)\sin\omega t\, d\omega t + \int_{\beta}^{\pi-\beta} K\left(s-\frac{D}{2}\right)\sin\omega t\, d\omega t + \int_{\pi-\beta}^{\pi-\alpha} K\left(X\sin\omega t - \frac{D}{2}\right)\sin\omega t\, d\omega t\right]$$

$$= \frac{2K}{\pi}\left[\int_{\alpha}^{\beta} X\sin^2\omega t\, d\omega t - \int_{\alpha}^{\beta}\frac{D}{2}\sin\omega t + \left(s-\frac{D}{2}\right)\int_{\beta}^{\pi-\beta}\sin\omega t\, d\omega t + X\int_{\pi-\beta}^{\pi-\alpha}\sin^2\omega t\, d\omega t - \frac{D}{2}\int_{\pi-\beta}^{\pi-\alpha}\sin\omega t\, d\omega t\right]$$

$$= \frac{2K}{\pi}\left[X\beta - X\alpha - \frac{X}{2}\sin 2\beta + \frac{X}{2}\sin 2\alpha + 2s\cos\beta - D\cos\alpha\right] \quad ...(7.9)$$

Input $x = X\sin\omega t$

$$\omega t = \alpha,\ \frac{D}{2} = X\sin\alpha \text{ or } D = 2X\sin\alpha$$

$$\sin\alpha = \frac{D}{2X}$$

b. $\omega t = \beta$, $s = X\sin\beta$

$$\therefore \quad \sin\beta = \frac{s}{X}$$

put these values in eq^n (7.9)

$$B_1 = \frac{2K}{\pi}\left[X\beta - X\alpha - \frac{X}{2}\sin 2\beta + \frac{X}{2}\sin 2\alpha + 2X\sin\beta\cos\beta - 2X\sin\alpha\cos\alpha\right]$$

$$= \frac{2K}{\pi}\left[X(\beta-\alpha) - \frac{1}{2}X\sin 2\beta + \frac{1}{2}X\sin 2\alpha + X\sin 2\beta - X\sin 2\alpha\right]$$

$$= \frac{2KX}{\pi}\left[(\beta-\alpha) + \frac{1}{2}\sin 2\beta - \frac{1}{2}\sin 2\alpha\right] = \frac{KX}{\pi}\left[2(\beta-\alpha) + \sin 2\beta - \sin 2\alpha\right]$$

$$= \frac{KX}{\pi}\left[2(\beta-\alpha) + \sin 2\beta - \sin 2\alpha\right]$$

$$\therefore \quad G_D(j\omega) = \frac{\sqrt{A_1^2 + B_1^2}}{X}\angle\tan^{-1}A_1/B_1 = \frac{B_1}{X}\angle 0^\circ$$

$$\therefore \quad \boxed{G_D(j\omega) = \frac{K}{\pi}\left[2(\beta-\alpha) + \sin 2\beta - \sin 2\alpha\right]\angle 0^\circ} \quad ...(7.10)$$

The Describing function is given by

$$G_D(j\omega) = \begin{cases} 0\ ;\ x < D/2,\ \alpha = \beta = \pi/2 \\ \left[1 - \frac{2}{\pi}(\alpha + \sin\alpha\cos\alpha)\right]K \quad ;\ D/2 < X < s \\ \frac{K}{\pi}\left[2(\beta-\alpha) + \sin 2\beta - \sin 2\alpha\right] \quad ;\ X > s \end{cases}$$

7.3.6. Describing Function of Backlash

Input $i = I_m\sin\omega t$

output $y(t)$

$$\begin{cases} = K\left[I\sin\omega t - \frac{b}{2}\right] & 0 \le \omega t \le \pi/2 \\ = K\left[I - \frac{b}{2}\right] & \pi/2 \le \omega t \le \pi - \beta \\ = K\left[I\sin\omega t + \frac{b}{2}\right] & \pi - \beta \le \omega t \le \pi \end{cases}$$

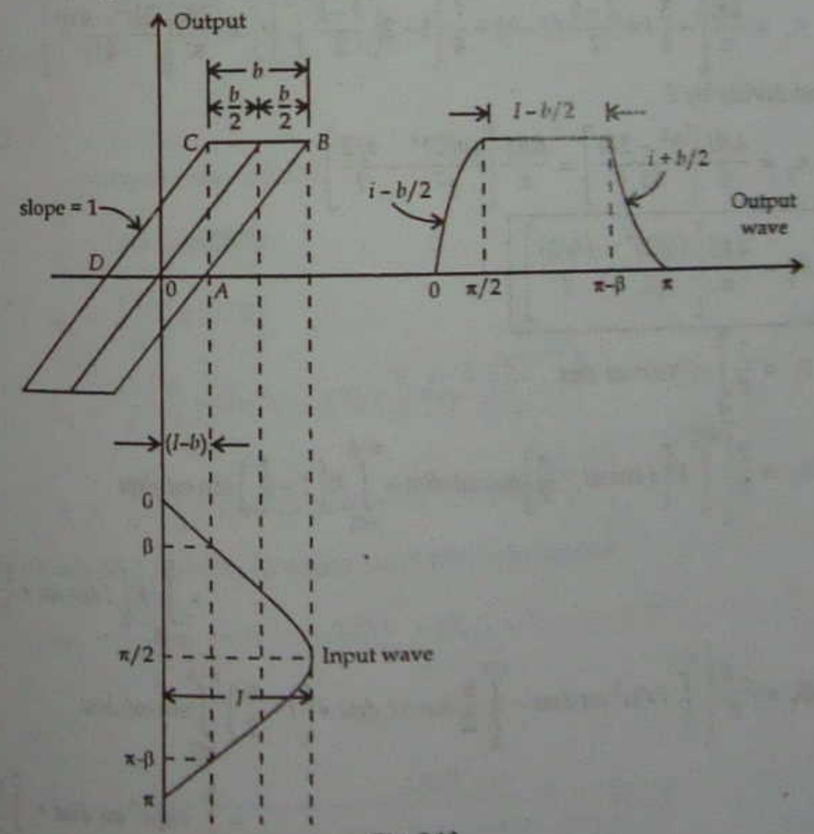


Fig. 7.12.

$$A_1 = \frac{2}{\pi}\left[\int y(t)\cos\omega t\, d(\omega t)\right]$$

$$A_1 = \frac{2K}{\pi}\left[\int_0^{\pi/2}\left(I\sin\omega t - \frac{b}{2}\right)\cos\omega t\, d\omega t + \int_{\pi/2}^{\pi-\beta}\left(I - \frac{b}{2}\right)\cos\omega t\, d\omega t + \int_{\pi-\beta}^{\pi}\left(I\sin\omega t + \frac{b}{2}\right)\cos\omega t\, d\omega t\right]$$

$$A_1 = \frac{2K}{\pi}\left[\int_0^{\pi/2} I\sin\omega t\cos\omega t\, d\omega t - \frac{b}{2}\int_0^{\pi/2}\cos\omega t\, d\omega t + \left(I - \frac{b}{2}\right)\int_{\pi/2}^{\pi-\beta}\cos\omega t\, d\omega t + \int_{\pi-\beta}^{\pi} I\sin\omega t\cos\omega t\, d\omega t + \frac{b}{2}\int_{\pi-\beta}^{\pi}\cos\omega t\, d\omega t\right]$$

$$A_1 = \frac{2K}{\pi}\left[-\frac{3}{4}I + I\sin\beta - b\sin\beta + \frac{I}{4}\cos 2\beta\right]$$

$$\sin\beta = \frac{I-b}{I}$$

$$A_1 = \frac{2K}{\pi}\left[-\frac{3}{4}I + \sin\beta(I-b) + \frac{I}{4}\left\{1 - 2\sin^2\beta\right\}\right]$$

$$A_1 = \frac{2K}{\pi}\left[-\frac{3}{4}I + \frac{I-b}{I}(I-b) + \frac{I}{4}\left\{1 - 2\left(\frac{I-b}{I}\right)^2\right\}\right] = \frac{2K}{\pi}\left[\frac{2b^2 - 4Ib}{4I}\right]$$

Multiplying and divide by I

$$A_1 = \frac{4KI}{\pi}\left[\frac{b^2 - 2Ib}{4I.I}\right] = \frac{4KI}{\pi}\left[\frac{(b/2)^2}{I^2} - \frac{b/2}{I}\right]$$

$$\boxed{A_1 = \frac{4KI}{\pi}\left[\frac{(b/2)^2}{I^2} - \frac{(b/2)}{I}\right]}$$

$$B_1 = \frac{2}{\pi}\int_0^{\pi} y(t)\sin\omega t\, d\omega t$$

$$B_1 = \frac{2}{\pi}\left[\int_0^{\pi/2} K\left[I\sin\omega t - \frac{b}{2}\right]\sin\omega t\, d\omega t + \int_{\pi/2}^{\pi-\beta} K\left(I - \frac{b}{2}\right)\sin\omega t\, d\omega t + \int_{\pi-\beta}^{\pi} K\left(I\sin\omega t + \frac{b}{2}\right)\sin\omega t\, d\omega t\right]$$

$$B_1 = \frac{2K}{\pi}\left[\int_0^{\pi/2} I\sin^2\omega t\, d\omega t - \int_0^{\pi/2}\frac{b}{2}\sin\omega t\, d\omega t + \left(I - \frac{b}{2}\right)\int_{\pi/2}^{\pi-\beta}\sin\omega t\, d\omega t + \int_{\pi-\beta}^{\pi} I\sin^2\omega t\, d\omega t + \int_{\pi-\beta}^{\pi}\frac{b}{2}\sin\omega t\, d\omega t\right]$$

$$B_1 = \frac{2K}{\pi}\left[\frac{I}{2}\int_0^{\pi/2}(1-\cos 2\omega t)\, d\omega t - \frac{b}{2}\int_0^{\pi/2}\sin\omega t\, d\omega t + \left(I - \frac{b}{2}\right)\int_{\pi/2}^{\pi-\beta}\sin\omega t\, d\omega t + \frac{I}{2}\int_{\pi-\beta}^{\pi}(1-\cos 2\omega t)\, d\omega t + \frac{b}{2}\int_{\pi-\beta}^{\pi}\sin\omega t\, d\omega t\right]$$

$$B_1 = \frac{2K}{\pi}\left[\frac{\pi I}{4} + \cos\beta\,(I-b) + \frac{I}{2}\beta - \frac{I}{2}\sin\beta\cos\beta\right]$$

$$\sin\beta = \frac{I-b}{I} \qquad \therefore \cos\beta = \frac{\sqrt{2Ib - b^2}}{I}$$

$$B_1 = \frac{2K}{\pi}\left[\frac{\pi I}{4} + \frac{\sqrt{2Ib-b^2}}{I}\cdot(I-b) + \frac{I}{2}\beta - \frac{I}{2}\cdot\frac{I-b}{I}\cdot\frac{\sqrt{2Ib-b^2}}{I}\right]$$

$$= \frac{KI}{\pi}\left[\frac{\pi}{2} + \beta + \frac{b(I-b)}{I^2}\sqrt{\frac{2I}{b} - 1}\right]$$

$$\boxed{B_1 = \frac{KI}{\pi}\left[\frac{\pi}{2} + \beta + \frac{b(I-b)}{I^2}\sqrt{\frac{2I}{b} - 1}\right]}$$

$$\therefore G_D(j\omega) = \frac{\sqrt{A_1^2 + B_1^2}}{I}\angle\tan^{-1}A_1/B_1 \qquad \text{...(7.11)}$$

Let $R \triangleq \frac{b/2}{I}$

where I = input (maximum)

$\frac{b}{2}$ = half backlash

$$A_1 = \frac{4KI}{\pi}\left[R^2 - R\right]$$

$$B_1 = \frac{KI}{\pi}\left[\frac{\pi}{2} + \sin^{-1}(1-2R) + \frac{b}{I}\cdot\frac{I-b}{I}\sqrt{\frac{2I}{b} - 1}\right]$$

$$B_1 = \frac{KI}{\pi}\left[\frac{\pi}{2} + \sin^{-1}(1-2R) + 2R\,(1-2R)\sqrt{\frac{1-R}{R}}\right]$$

Put the values of A_1 and B_1 in equation (7.11) and simplify

$$G_D(j\omega) = \frac{K}{\pi}\left[\left\{\frac{\pi}{2} + \sin^{-1}(1-2R)\right\}^2 + 4R\,(1-R) + \{\pi + 2\sin^{-1}(1-2R)\}\, 2R(1-2R)\sqrt{\frac{1-R}{R}}\right]^{1/2}$$

$$\angle G_D(j\omega) = \angle\tan^{-1}\frac{4R(R-1)}{\pi/2 + \sin^{-1}(1-2R) + 2R\,(1-2R)\sqrt{\frac{1-R}{R}}}$$

Example 7.1. Determine the describing function of the following non-linearity.

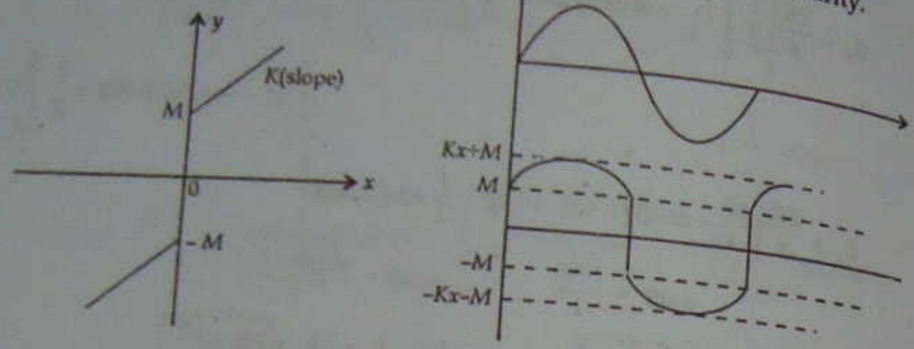


Fig. 7.13.

Solution : Input $x = X \sin\omega t$

Output can be expressed as Fourier series

$$y(t) = A_0 + \sum_{n=1}^{\infty}(A_n \cos n\omega t + B_n \sin n\omega t)$$

For symmetrical non-linearity $A_0 = 0$

$$y(t) = M + Kx \sin\omega t$$

Since, $y(t)$ is an odd function, its Fourier series has only sine terms

$\therefore$ Fundamental harmonic components is

$$y(t) = B_1 \sin\omega t$$

$$B_1 = \frac{1}{\pi}\int_0^{2\pi} y(t)\sin\omega t\, d\omega t = \frac{4}{\pi}\int_0^{\pi/2}(M + Kx\sin\omega t)\sin\omega t\, d\omega t$$

$$= \frac{4}{\pi}\int_0^{\pi/2}(M\sin\omega t + Kx\sin^2\omega t)\, d\omega t = \frac{4}{\pi}\int_0^{\pi/2}\left[\left(M\sin\omega t + \frac{Kx}{2}(1-\cos 2\omega t\right)d\omega t\right]$$

$$B_1 = \frac{4M}{\pi} + Kx$$

$$\therefore \quad G_D(j\omega) = \frac{\sqrt{A_1^2 + B_1^2}}{x}$$

$$\therefore \quad \boxed{G_D(j\omega) = \frac{4M}{\pi x} + K} \quad \textbf{Ans.}$$

Example 7.2. Determine the describing function of the following non-linearity.

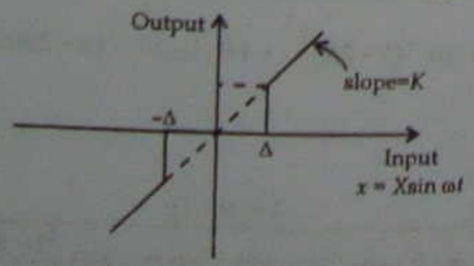


Fig. 7.14.(a)

Solution :

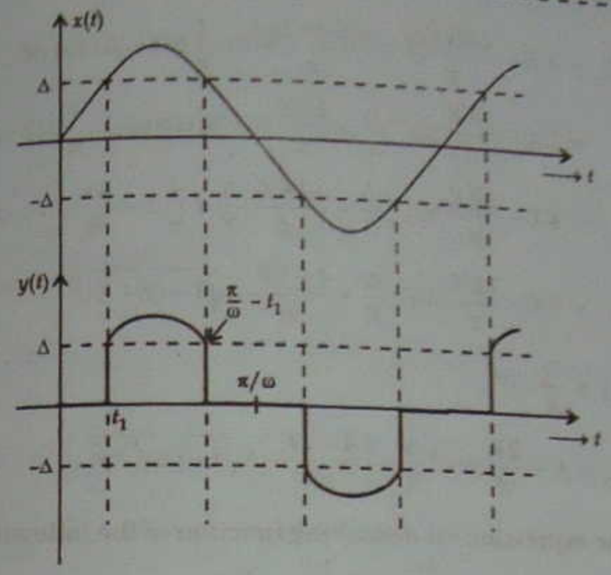


Fig. 7.14(b)

output $y(t) = 0 \qquad 0 < t < t_1$

$= K(X\sin\omega t - \Delta) + \Delta \qquad t_1 < t < \frac{\pi}{\omega} - t_1$

$= 0 \qquad \frac{\pi}{\omega} - t_1 < t < \frac{\pi}{\omega}$

For symmetrical non-linearity $A_0 = 0$

$y(t)$ is an odd function $\quad A_n = 0$

$\therefore \quad y_1(t) = B_1 \sin\omega t$

where $\quad B_1 = \frac{1}{\pi}\int_0^{2\pi} y(t)\sin\omega t\, d\omega t = \frac{4}{\pi}\int_0^{\pi/2} y(t)\sin\omega t\, d\omega t$

$$B_1 = \frac{4}{\pi}\int_{\omega t_1}^{\pi/2}\{K(X\sin\omega t - \Delta) + \Delta\}\sin\omega t\, d(\omega t)$$

$$= \frac{4}{\pi}\int_{\omega t_1}^{\pi/2}\{KX\sin^2\omega t + \Delta(1-K)\sin\omega t\}\, d(\omega t)$$

$$= \frac{4}{\pi}\left[\frac{KX}{2}\omega t - \frac{KX}{4}\sin 2\omega t + \Delta(1-K)(-\cos\omega t)\right]_{\omega t_1}^{\pi/2}$$

$$= \frac{4}{\pi}\left[\frac{KX}{2}\cdot\frac{\pi}{2} - \frac{KX\omega t_1}{2} + \frac{KX}{4}\sin 2\omega t_1 + \Delta(1-K)\cos\omega t_1\right]$$

since

$$\Delta = X\sin\omega t_1$$

$$\sin\omega t_1 = \Delta/X$$

$$\cos\omega t_1 = \sqrt{1-(\Delta/X)^2}$$

$$\therefore \quad B_1 = KX - \frac{2KX\omega t_1}{\pi} + \frac{KX\sin 2\omega t_1}{\pi} - \frac{4}{\pi}\Delta(1-K)\cos\omega t_1$$

$$= KX - \frac{2KX}{\pi}\sin^{-1}\frac{\Delta}{x} + \frac{2KX}{\pi}\sin\omega t_1\cos\omega t_1 + \frac{4}{\pi}\Delta(1-K)\cos\omega t_1$$

$$= KX - \frac{2KX}{\pi}\sin^{-1}\frac{\Delta}{X} + \left\{\frac{2KX}{\pi}\cdot\frac{\Delta}{X} + \frac{4}{\pi}\Delta - \frac{4K}{\pi}\Delta\right\}\cos\omega t_1$$

$$= KX - \frac{2KX}{\pi}\sin^{-1}\frac{\Delta}{X} + \frac{4-2K}{\pi}\Delta\sqrt{1-(\Delta/X)^2}$$

$$\therefore \quad G_D(j\omega) = \frac{B_1}{X}\angle 0^\circ$$

$$\therefore \quad G_D(j\omega) = K - \frac{2K}{\pi}\sin^{-1}\frac{\Delta}{X} + \left(\frac{4-2K}{\pi X}\right)\Delta\sqrt{1-(\Delta/X)^2}$$

Example 7.3. Derive the expression for describing function of the following non-linearity.

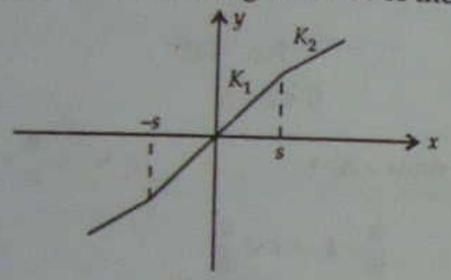


Fig. 7.15(a)

Solution :

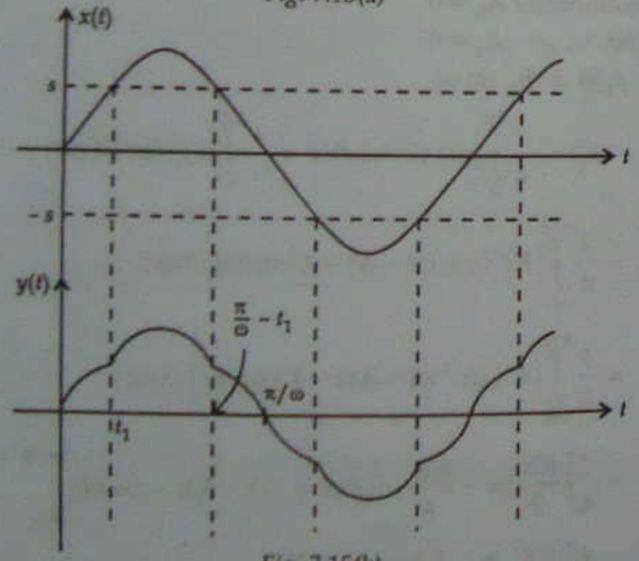


Fig. 7.15(b)

$$y(t) = K_1 X\sin\omega t \qquad 0 < t < t_1$$

$$= K_2 X\sin\omega t + (K_1 - K_2)\,s \qquad t_1 < t < \frac{\pi}{\omega} - t_1$$

$$= K_1 X\sin\omega t \qquad \frac{\pi}{\omega} - t_1 < t < \pi/\omega$$

For symmetrical non-linearity $A_0 = 0$

$y(t)$ is an odd function $A_n = 0$

$$B_1 = \frac{1}{\pi}\int_0^{2\pi} y(t)\sin\omega t\, d(\omega t)$$

$$= \frac{4}{\pi}\int_0^{\pi/2} y(t)\sin\omega t\, d(\omega t)$$

$$B_1 = \frac{4}{\pi}\int_0^{\omega t_1} K_1 X\sin^2\omega t\, d(\omega t) + \frac{4}{\pi}\int_{\omega t_1}^{\pi/2} K_2 X\sin^2\omega t + (K_1 - K_2)\,s\sin\omega t\, d(\omega t)$$

$$= \frac{4}{\pi}\left[\frac{K_1 X\omega t}{2} - \frac{K_1 X}{4}\sin 2\omega t\right]_0^{\omega t_1} + \frac{4}{\pi}\left[\frac{K_2 X\omega t}{2} - \frac{K_2 X\sin^2\omega t}{4} - (K_1 - K_2)\,s\cos\omega t\right]_{\omega t_1}^{\pi/2}$$

$$= \frac{4}{\pi}\left[\frac{K_1 X\omega t_1}{2} - \frac{K_1 X\sin 2\omega t_1}{4}\right] + \frac{4}{\pi}\left[\frac{K_2 X\pi}{4} + (K_1 - K_2)\,s\cos\omega t_1 + \frac{K_2 X\sin 2\omega t_1}{4} - \frac{K_2 X\omega t}{2}\right]$$

$$= \frac{4}{\pi}\left[\frac{K_2 X\pi}{4} + \frac{K_1 - K_2}{2}X\sin^{-1}\frac{s}{x} + (K_1 - K_2)\,s\cos\omega t_1 - \frac{(K_1 - K_2)}{4}X2\sin\omega t_1\cos\omega t_1\right]$$

$$= \frac{4}{\pi}\left[\frac{K_2 X\pi}{4} + \frac{K_1 - K_2}{2}X\sin^{-1}\frac{s}{X} + \frac{K_1 - K_2}{2}s\cos\omega t_1\right]$$

$$= K_2 X + \frac{2}{\pi}(K_1 - K_2)\,X\sin^{-1}\frac{s}{X} + \frac{2}{\pi}(K_1 - K_2)\,s\sqrt{1-\frac{s^2}{X^2}}$$

$$= K_2 X + \frac{2}{\pi}(K_1 - K_2)\,X\left[\sin^{-1}\frac{s}{X} + \frac{s}{X}\sqrt{1-\frac{s^2}{X^2}}\right]$$

$$\therefore \quad G_D(j\omega) = \frac{B_1}{X}\angle 0$$

$$\therefore \quad \boxed{G_D(j\omega) = K_2 + \frac{2}{\pi}(K_1 - K_2)\left[\sin^{-1}\frac{s}{X} + \frac{s}{X}\sqrt{1-\frac{s^2}{X^2}}\right]}$$

7.4. STABILITY ANALYSIS WITH DESCRIBING FUNCTION

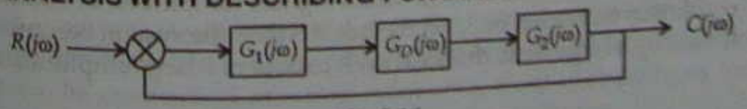


Fig. 7.16.

Consider the fig. 7.16, where $G_1(j\omega)$, $G_2(j\omega)$ are the linear part of the system and non-linearity is replaced by the describing function $G_D(j\omega)$. For above fig 7.16.

$$\frac{C(j\omega)}{R(j\omega)} = \frac{G_1(j\omega)G_D(X,j\omega)G_2(j\omega)}{1+G_1(j\omega)G_D(j\omega)G_2(j\omega)}$$

The characteristic equation is

$$1 + G_1(j\omega)\, G_D(X,j\omega)\, G_2(j\omega) \quad \text{...(7.12)}$$

According to Nyquist stability criterion, the system will exhibit sustained oscillations or limit cycle* when ...(7.13)

$$G_1(j\omega)\, G_D(X,j\omega)\, G_2(j\omega) = -1$$

This implies that the plot of $G_1(j\omega)\, G_D(j\omega)\, G_2(j\omega)$ passes through the critical point. Generally ...(7.14)
$G_D(j\omega)$ is amplitude dependent, rearrange the equation 7.14

$$G_1(j\omega)\, G_2(j\omega) = -\frac{1}{G_D(X,j\omega)} \quad \text{...(7.15)}$$

Let $G_1(j\omega)\, G_2(j\omega)$ plot be superimposed on $-\frac{1}{G_D(X,j\omega)}$. When $G_D(j\omega)$ is a function of amplitude only, the resulting plot is a single curve but if it is a function of frequency and amplitude then there is a family of curves. Then three situations arises.

(*a*) When there is no intersection between $G_1(j\omega)\, G_2(j\omega)$ and $-\frac{1}{G_D(X,j\omega)}$ and the plot of $-\frac{1}{G_D(X,j\omega)}$ is completely enclosed by $G_1(j\omega)\, G_2(j\omega)$ as shown in fig. 7.17 *a*, then the system is absolutely unstable.

(*b*) When there is no intersection between $G_1(j\omega)\, G_2(j\omega)$ and $-\frac{1}{G_D(X,j\omega)}$ and the plot of $-\frac{1}{G_D(X,j\omega)}$ does not enclose by the plot of $G_1(j\omega)\, G_2(j\omega)$ as shown in fig. 7.17*b* then the system is absolutely stable.

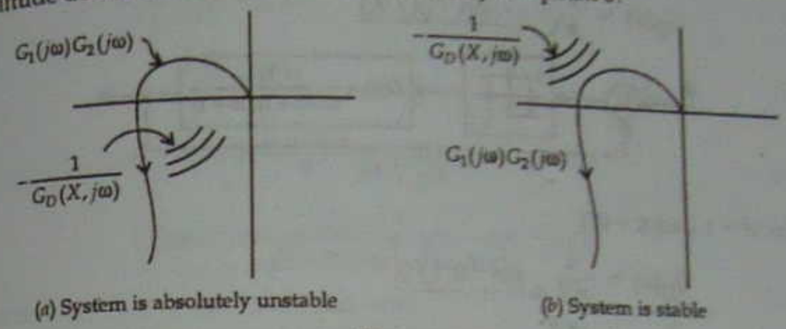
(*c*) When there is no intersection between $G_1(j\omega)\, G_2(j\omega)$ and $-\frac{1}{G_D(X,j\omega)}$ and few curves are enclosed and few curves of $-\frac{1}{G_D(X,j\omega)}$ does not enclose by $G_1(j\omega)\, G_2(j\omega)$ as shown in fig. 7.17(*c*) then the system is unstable for ω_1, ω_2 and stable for ω_3, ω_4.

Now consider another case *i.e.* when there is a point of intersection of $G_1(j\omega)\, G_2(j\omega)$ with $-\frac{1}{G_D(X,j\omega)}$ (as shown in fig. 7.18*a*)

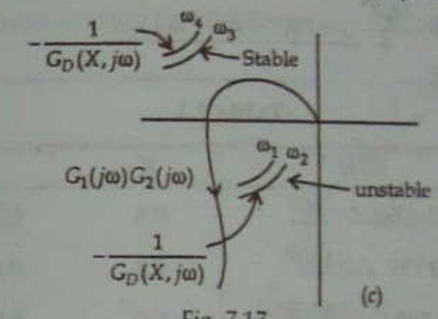
Suppose a system is actuated by a small disturbance, so a signal of small amplitude is obtained so that the point on $-\frac{1}{G_D(X,j\omega)}$ locus corresponding to this small signal will be inside the $G_1(j\omega)$ $G_2(j\omega)$ locus. Hence the system is unstable for small signals (amplitude of oscillation increases). If the amplitude of oscillation increased beyond the point 5 then the system becomes stable. Similarly, if the system is actuated by a large disturbance then it produces a large amplitude signal so the point

* detail of limit cycle is given at the end of the chapter.

on $-\frac{1}{G_D(X,j\omega)}$ locus corresponding to this signal will be outside of $G_1(j\omega)\, G_2(j\omega)$. Hence the system is stable (amplitude decreases and the value is defined by the point 5.

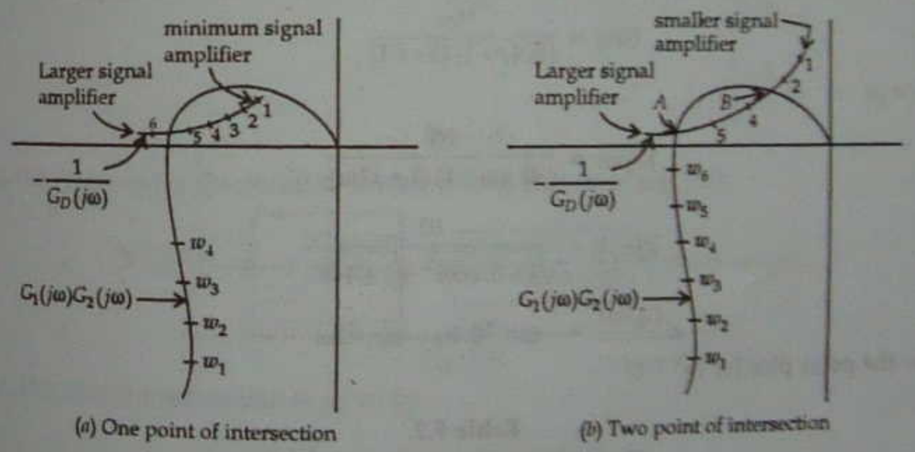


(*a*) System is absolutely unstable (*b*) System is stable



(*c*)

Fig. 7.17.



(*a*) One point of intersection (*b*) Two point of intersection

Fig. 7.18.

Now consider when there are two points of intersection, for small disturbance signal, the system will be stable between the point B and the point of small signal (because amplitude of oscillation decreases and it will reach to its steady state equilibrium). If we start from point 4 *i.e.*, between the point B and A, the amplitude starts increasing, then the system is unstable. still for large signal the system is stable and the oscillation decreases in amplitude to the value defined at point A.

The stability can be determined by using polar plot, Bode plot and root locus. Here we will study the stability using polar plot. Consider the following examples.

Example 7.4. For the non-linear system shown in the fig. 7.19 determine the amplitude and frequency of limit cycle. The describing function of non-linearity is given by

$$G_D(x) = \frac{4M}{\pi X} \angle -\sin^{-1}(n/X)$$

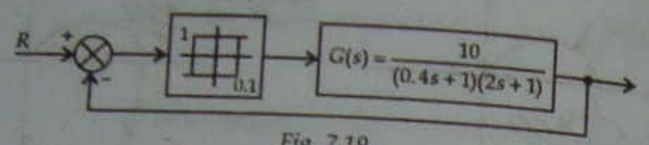


Fig. 7.19.

Solution : Here $M = 1$ and $n = 0.1$

$$\therefore \quad G_D(x) = \frac{4}{\pi X} \angle \sin^{-1} 0.1/X$$

$$-\frac{1}{G_D(x)} = -\frac{\pi X}{4} \angle -180° + \sin^{-1} 0.1/X \qquad ...(7.16)$$

Plot the curve for *eq*^{n} (7.16)

Table 7.1.

X	-1/N	X	-1/N
0.1	$0.0785\angle -90°$	0.5	$0.393\angle -168.5°$
0.2	$0.157\angle -150°$	0.8	$0.628\angle -172.8°$
0.3	$0.235\angle -160.5°$	1.0	$0.785\angle -174.3°$

$$G(s) = \frac{10}{(0.4s+1)(2s+1)} \qquad ...(7.17)$$

put $s = j\omega$

$$G(j\omega) = \frac{10}{(j0.4\omega+1)(1+j2\omega)}$$

$$|G(j\omega)| = \frac{10}{\sqrt{1+0.16\omega^2}\sqrt{1+4\omega^2}}$$

$$\angle G(j\omega) = -\tan^{-1}0.4\omega - \tan^{-1}2\omega$$

Draw the polar plot for *eq*^{n} 7.17

Table 7.2.

ω	G(jω)
3	$1.052\angle -130.7°$
4	$0.057\angle -140.8°$
6	$0.319\angle -152.6°$
7	$0.239\angle -156.6°$
8	$0.186\angle -159°$
10	$0.121\angle -163.1°$
20	$0.031\angle -171.4°$

The curves of $-\frac{1}{G_D(X)}$ and $G(j\omega)$ intersect at the point A which corresponds to stable limit cycle and magnitude at this point is 0.21. from fig. 7.20

$$-\frac{1}{G_D(X)} = 0.21$$

and

$$\frac{\pi X}{4} = 0.21$$

$$\therefore X = \frac{0.21\times 4}{\pi} = 0.267$$

$$|G(j\omega)| = \frac{10}{\sqrt{1+0.16\omega^2}\sqrt{1+4\omega^2}} = 0.21$$

on simplification $\omega = 7.92$ rad/sec.

Thus, stable limit cycle corresponds to X = 0.267 and ω = 7.92 rad/sec.

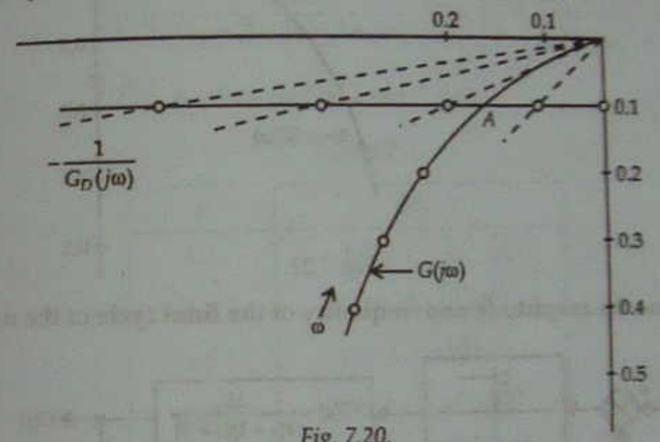


Fig. 7.20.

Example 7.5. Determine the stability of the system shown in fig. 7.21.

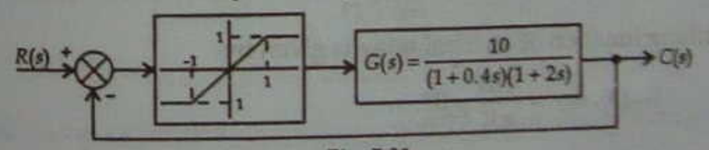


Fig. 7.21.

Solution. The describing function is given by

$$G_D(X) = \frac{2K}{\pi}\left[\sin^{-1}\frac{s}{X} + \frac{s}{X}\sqrt{1-\left(\frac{s}{X}\right)^2}\right] \angle 0°$$

Here, $s = 1,\ K = 1$

$$G_D(X) = \frac{2}{\pi}\left[\sin^{-1}\frac{1}{X} + \frac{1}{X}\sqrt{1-\left(\frac{1}{X}\right)^2}\right]$$

$$G(s) = \frac{10}{(1+0.4s)(1+2s)}$$

put $s = j\omega$

$$G(j\omega) = \frac{10}{(1+j0.4\omega)(1+j2\omega)}$$

$$|G(j\omega)| = \frac{10}{\sqrt{1+0.16\omega^2}\sqrt{1+4\omega^2}}$$

$$\angle G(j\omega) = -\tan^{-1} 0.4\,\omega - \tan^{-1} 2\omega$$

Draw the plot for $-\frac{1}{G_D(X)}$ and $G(j\omega)$ (shown in fig. 7.22) From the graph we find that $-\frac{1}{G_D(X)}$ curve lies outside the $G(j\omega)$ plot. Hence, the system is always stable.

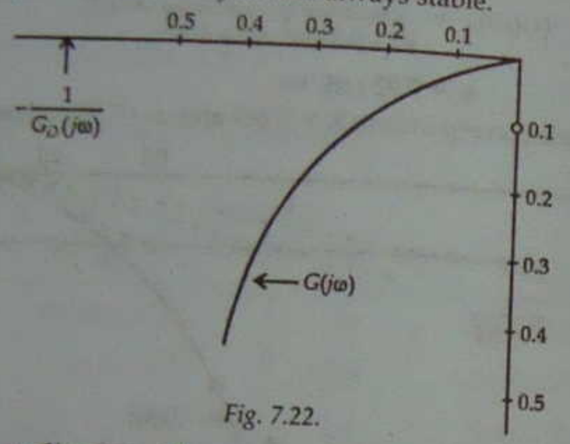


Fig. 7.22.

Example 7.6. Determine the amplitude and frequency of the limit cycle of the non-linearity shown in fig. 7.23.

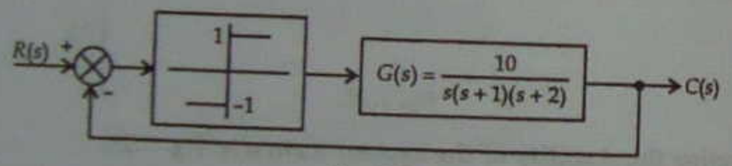


Fig. 7.23.

Solution : The describing function of an ideal relay is given by

$$G_D(X, j\omega) = \frac{4M}{\pi X}\angle 0° \qquad ...(7.18)$$

$$-\frac{1}{G_D(X, j\omega)} = \frac{\pi X}{4}\angle -180°$$

$\because$ $M = 1$ and angle $= 0°$

Table 7.3.

X	$-1/G_D(X)$
0.1	$0.0785\angle -180°$
0.2	$0.157\angle -180°$
0.3	$0.237\angle -180°$
0.5	$0.393\angle -180°$
0.8	$0.628\angle -180°$
1.0	$0.785\angle -180°$
2.0	$1.57\angle -180°$
2.5	$1.97\angle -180°$

$$G(j\omega) = \frac{10}{j\omega(1+j\omega)(2+j\omega)} \quad |G(j\omega)| = \frac{10}{\omega\sqrt{1+\omega^2}\sqrt{4+\omega^2}}$$

$$\angle G(j\omega) = -90° - \tan^{-1}\omega - \tan^{-1}\omega/2$$

Table 7.4.

ω	$G(j\omega)$
1.2	$2.287\angle -171.2°$
1.4	$1.701\angle -179.45°$
1.5	$1.479\angle -183.2°$
1.8	$1.003\angle -192.93°$
2.0	$0.79\angle -198.43°$
5.0	$0.073\angle -236.9°$
10	$0.0098\angle -253°$

Intersection of curves $G(j\omega)$ and $-\frac{1}{G_D(X, j\omega)}$ gives a point which corresponds to stable limit cycle.

Here, $\frac{\pi X}{4} = 1.68$

$$X = 2.14$$

Also, intersection is at $-180°$

$$\Rightarrow -180° = -90° - \tan^{-1}\omega - \tan^{-1} 0.5\omega$$

$$\therefore \quad \omega = 1.41 \text{ rad/sec.}$$

Thus, the stability limit cycle corresponds to

$$X = 2.14 \text{ and } \omega = 1.41 \text{ rad/sec.}$$

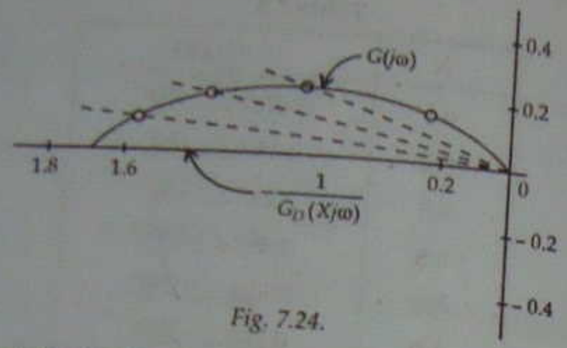


Fig. 7.24.

7.5. THE PHASE PLANE TECHNIQUE

The phase plane method is a graphical method for solving the second order non-linear differential equations. This method also gives the information about the types of transient response for various initial conditions.

Consider the equation

$$\frac{d^2x}{dt^2} + A\frac{dx}{dt} + Bx = 0 \qquad ...(7.19)$$

If, $\frac{dx}{dt} = \dot{x}$, $\frac{d^2x}{dt^2} = \frac{d\dot{x}}{dt}$ then equation 7.19 becomes

$$\frac{d\dot{x}}{dt} + A\dot{x} + Bx = 0$$

or, $$\frac{d\dot{x}}{dt} = -(A\dot{x} + Bx) \qquad ...(7.20)$$

Divide both sides by $\frac{dx}{dt}$ we get

$$\frac{d\dot{x}}{dx} = \frac{-(A\dot{x} + Bx)}{\dot{x}}$$

$$\frac{d\dot{x}}{dt} = f(x, \dot{x}) \qquad ...(7.21)$$

The coordinate plane corresponding to x and $\dot{x}$, is called phase plane. The curve described in phase plane with respect to time is called phase trajectory. The phase trajectory can be plotted by graphical or analytical method. The family of trajectory for different initial conditions is known as phase-portrait. The phase portrait gives the information about the stability and limit cycle (if exists).

The equation 7.21 can be plotted directly as there are three variables $\frac{d\dot{x}}{dt}$, $\dot{x}$ and x. To get the relationship between $\dot{x}$ and x there are several methods.

(i) For differential equation : get the solution and obtain x and then differentiate. This is to get $\dot{x}$ and eliminate independent variables then draw a curve.

(ii) Put the equation in general form, integrate and plot the result.

(iii) Equation 7.21 is the general equation for phase plane, if a constant value K is selected for $\frac{d\dot{x}}{dt}$ the equation 7.21 becomes

$$K = f(\dot{x}, x) \qquad ...(7.22)$$

then curve defined by the equation 7.22 is the locus of constant slope for phase trajectories of isoclines. Two methods are employed for phase-plane plot (*a*) method of isoclines (*b*) delta method.

7.6. PHASE PORTRAITS

Consider the second order system shown in fig. 7.25.

The equation of the system

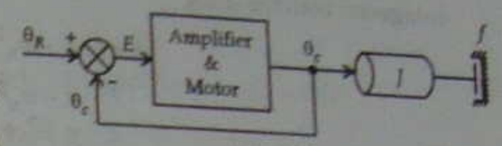


Fig. 7.25.

$$J\frac{d^2\theta_c}{dt^2} + f\frac{d\theta_c}{dt} = KE \qquad ...(7.23)$$

$$E = \theta_R - \theta_c \qquad ...(7.24)$$

$$\theta_c = \theta_R - E$$

or, $$\dot{\theta}_c = \dot{\theta}_r - \dot{E} \qquad ...(7.25)$$

$$\ddot{\theta}_c = \ddot{\theta}_r - \ddot{E} \qquad ...(7.26)$$

The equation 7.23 can be written as

$$J\ddot{\theta}_c + f\dot{\theta}_c = KE \qquad ...(7.27)$$

Put the values form 7.25 and 7.26 in *eq*^{n} 7.27

$$J(\ddot{\theta}_r - \ddot{E}) + f(\dot{\theta}_r - \dot{E}) = KE$$

or, $$J\ddot{E} + f\dot{E} + KE = J\ddot{\theta}_r + f\dot{\theta}_r$$

$$\ddot{E} + \frac{f}{J}\dot{E} + \frac{K}{J}E = \ddot{\theta}_r + \frac{f}{J}\dot{\theta}_r$$

Assume, $$\zeta = \frac{f}{2\sqrt{KJ}}, \omega_n^2 = \frac{K}{J}$$

$$f = 2\zeta\sqrt{KJ}$$

$$\therefore \ddot{E} + \frac{2\zeta\sqrt{KJ}}{J}\dot{E} + \frac{K}{J}E = \ddot{\theta}_r + 2\zeta\frac{\sqrt{KJ}}{J}\dot{\theta}_r$$

or, $$\ddot{E} + 2\zeta\omega_n\dot{E} + \omega^2_n E = \ddot{\theta}_r + 2\zeta\omega\dot{\theta}_r$$

ζ = damping constant

ω_n = undamped natural oscillation

For undamped system, step displacement input

$$f = 0$$

$$\ddot{\theta}_R = \dot{\theta}_R = 0$$

$$\therefore \ddot{E} + \omega^2_n E = 0$$

or $$\frac{d\dot{E}}{dt} + \omega^2_n E = 0 \qquad ...(7.24)$$

Divide the equation (7.24) by $\frac{dE}{dt}$

$$\frac{d\dot{E}}{dt} + \omega_n^2\frac{E}{\dot{E}} = 0$$

$$\frac{d\dot{E}}{dt} = -\omega_n^2 \frac{E}{\dot{E}}$$

$$\dot{E}\, d\dot{E} = -\omega_n^2 E\, dE$$

Integerate both the sides

$$\int \dot{E}\, d\dot{E} = -\omega_n^2 \int E\, dE$$

$$\dot{E}^2 = -\omega_n^2 E^2 + C^2$$

or, $$C^2 = \dot{E}^2 + \omega_n^2 E^2$$

or, $$\left(\frac{\dot{E}}{\omega_n}\right)^2 + E^2 = \left(\frac{c}{\omega_n}\right)^2 \qquad ...(7.25)$$

Equation (7.25) is the equation of the circle with radius $= \frac{c}{\omega_n}$

The trajectory is shown in the fig. 7.26. From Fig. 7.26 the system having sustained oscillations.

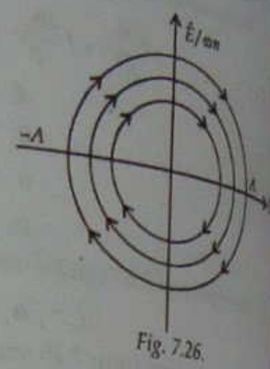


Fig. 7.26.

7.7. PHASE TRAJECTORY OF A SECOND ORDER SYSTEM USING METHOD OF ISOCLINES

Second order system is given by

$$\ddot{E} + 2\zeta\omega_n \dot{E} + \omega^2_n E = 0$$

or, $$\frac{d\dot{E}}{dt} + 2\zeta\omega_n \frac{dE}{dt} + \omega_n^2 E = 0 \qquad ...(7.27)$$

Divide the equation 7.27 by $\frac{dE}{dt} = \dot{E}$, we get

$$\frac{d\dot{E}}{dt} + 2\zeta\omega_n + \omega_n^2 \frac{E}{\dot{E}} = 0 \qquad ...(7.28)$$

put $\frac{d\dot{E}}{dt} = N = \tan\phi$ = slope of the trajectory

$$N + 2\zeta\omega_n + \omega^2_n \frac{E}{\dot{E}} = 0$$

$$N = -2\zeta\omega_n - \omega_n^2 \frac{E}{\dot{E}}$$

or, $$\dot{E}(N + 2\zeta\omega_n) + \omega_n^2 E = 0$$

$$\frac{\dot{E}}{E} = -\frac{\omega_n^2}{N + 2\zeta\omega_n} \qquad ...(7.29)$$

$$\frac{\dot{E}}{E} = \text{slope of isocline} = \tan\theta \qquad ...(7.30)$$

Consider the following example.

Example 7.7. Draw the trajectory when $\zeta = 0.5$, $\omega_n = 1$ rad/sec. for unit step input.

Solution : From equation 7.28

$$\frac{\dot{E}}{E} = -\frac{1}{N+1} = \tan\theta$$

Starting point of the trajectory.

$$E = \theta_r - \theta_c = 1 - 0 = 1$$

$$\dot{E} = 0$$

Step 1: Assume suitable scale, let 15cm = 1 unit, mark the initial point as shown in fig. 7.27.

Step 2: Select θ at equidistance (equal angle) say is 15° each

Step 3: Calculate N and then ϕ

Step 4: Draw a line parallel to first slope marker from (1,0) and from the same point draw a line parallel to second slope marker, middle of these will give second point of the trajectory i.e point on isocline 2.

Step 5: From point 2 draw a line parallel to second slope marker and third slope marker, their middle gives the third point and so on.

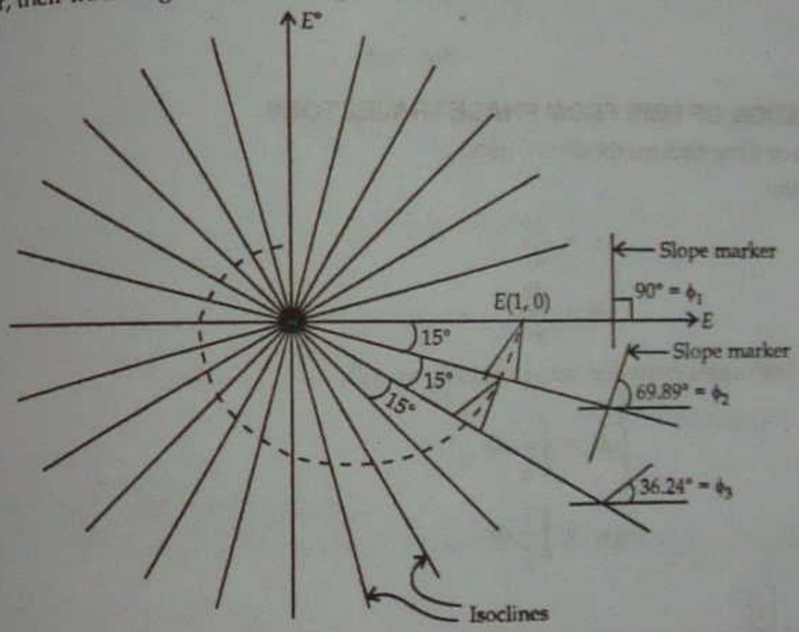


Fig. 7.27.

Table 7.5

$\theta = \tan^{-1}\frac{1}{N+1}$	$\phi = \tan^{-1}N$	N
0°	90°	∞
-15°	69.89°	2.73
:	:	:
:	:	:

SECOND METHOD

Draw a line parallel to first slope marker of first isocline, let intersect at a_1, find the mid point of $\dot{E}_1$. Let this be b_1, from b_1 draw a line parallel to second slope marker. Let it cut at a_2, the point

where second slope cuts the second isoclines gives the second point of trajectory and so on. (shown in fig. 7.28)

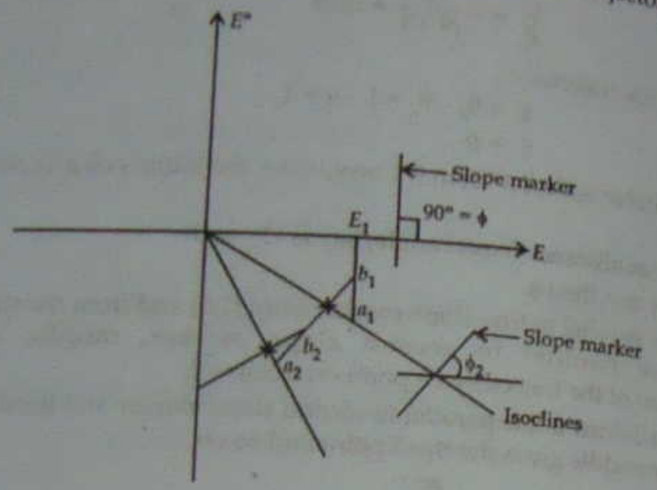


Fig. 7.28.

7.8. CALCULATION OF TIME FROM PHASE TRAJECTORY

For evaluation of time two methods are used

(*a*) We define

$$\dot{x} = \frac{dx}{dt}$$

or, $$dt = \frac{dx}{\dot{x}}$$

Integerate both sides between two points P_1 and P_2

$$\int_{P_1}^{P_2} dt = \int \frac{1}{\dot{x}}\, dx$$

$$t_{P_1P_2} = \int \frac{1}{\dot{x}}\, dx \qquad ...(7.30)$$

If a curve of $\left|\frac{1}{\dot{x}}\right|$ v_s x is plotted, the area between two limits P_1,P_2 will represent the time.

(*b*) Mark the points on phase trajectory. The points should be so choosen that the curve between two consecutive points could be approximated as a straight line, and read the average speed or velocity at the center between the two points and the corresponding change in x *i.e* Δx measured. Do the calculation for each segment (between two points) and calculate the total time upto any point.

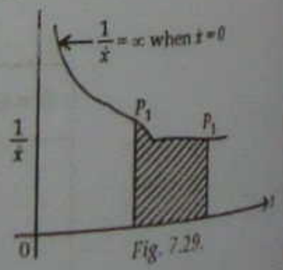


Fig. 7.29.

Hence,

$$\Delta t = \frac{\Delta x}{\dot{x}_{average}}$$

and $$T = \Sigma \Delta t$$

Consider the fig. 7.30.

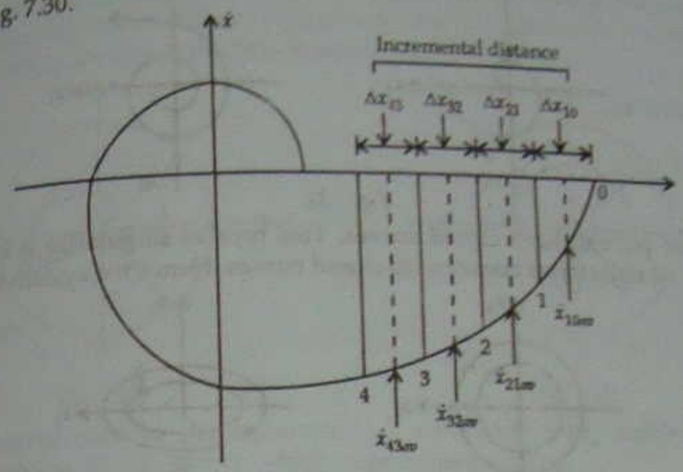


Fig. 7.30.

$$t_{01} = \frac{\Delta x_{10}}{\dot{x}_{10av}}, \qquad t_{01} = \frac{\Delta x_{10}}{\dot{x}_{10av}}$$

$$t_{12} = \frac{\Delta x_{21}}{\dot{x}_{21av.}}, \qquad t_{02} = t_{01} + t_{12} = \frac{\Delta x_{10}}{\dot{x}_{10av}} + \frac{\Delta x_{21}}{\dot{x}_{21av}}$$

$$t_{23} = \frac{\Delta x_{32}}{\dot{x}_{32av}} \qquad t_{03} = t_{01} + t_{12} + t_{23}$$

$$t_{03} = \frac{\Delta x_{10}}{\dot{x}_{10av}} + \frac{\Delta x_{21}}{\dot{x}_{21av}} + \frac{\Delta x_{32}}{\dot{x}_{32av.}} \text{ and so on.}$$

7.9. STABILITY FROM THE PHASE PLANE

The stability can be determined from the phase- plane trajectory near the point of equilibrium called singular points.

Consider the second order system

$$\frac{dx}{dt} = P(x,y) \qquad ...(7.31)$$

$$\frac{dy}{dt} = Q(x,y) \qquad ...(7.32)$$

$$\frac{dy}{dx} = \frac{Q(x,y)}{P(x,y)} \qquad ...(7.33)$$

The singular points are those values of x and y satisfying

$$P(x,y) = 0$$

$$Q(x,y) = 0$$

In fig. 7.31 (*a*) the trajectory approaches the origin. It will appear as a spiral and this type of singularity is known as Focus. Hence it is a stable focus. If the trajectory diverge from the focus then this will be unstable focus (fig. 7.31*b*).

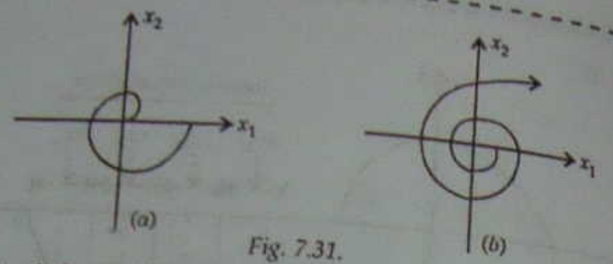


Fig. 7.31.

In fig. 7.31*c*, the portrait have closed curves. This type of singularity is known as center or vortex. In this case all trajectories converge to closed curves. Hence the system is limitedly stable.

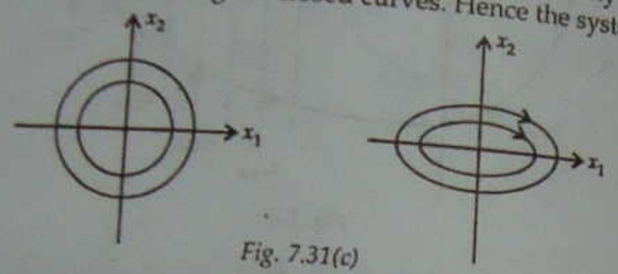


Fig. 7.31(c)

In fig. 7.31*d*, the trajectories are converged to the origin. This is called nodal point, hence the nodal point is stable. In fig. 7.31*e* the nodal point is unstable because trajectories passes through this point but do not stop here.

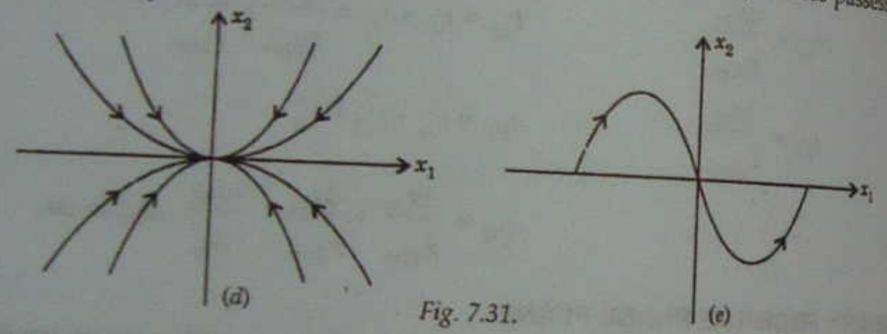


Fig. 7.31.

7.10. AUTONOMOUS SYSTEM

Consider a non linear system represented by the equation 7.34

$$\dot{x} = f(x,u) \qquad ...(7.34)$$

where $u(t)$ is the input vector and 'f' is the non-linear function of 'x' and 'u'.

If $u(t)$ is identically zero, the system is known as an autonomovs system and represented by equation (7.35)

$$\dot{x} = f(x) \qquad ...(7.35)$$

The autonomous system is also known as unforced system.

The points at which the derivatives of all state variables are zero, called the points of equilibrium (also called singular points).

$$i.e.\ \dot{x} = 0 \qquad ...(7.36)$$

7.11. ASYMPTOTIC STABILITY

Consider a hypersphere of finite radius surrounding the point of equilibrium is described by the equation (7.37)

$$x_1^2 + x_2^2 ..... + x_n^2 = R^2 \qquad ...(7.37)$$

The system is said to be asymptotically stable if there exists $\delta > 0$ such that trajectory starts from any point $x(0)$ wihtin s (δ) does not leave s (R) at any time and returns to the origin. (shown in fig 7.32) Fig 7.33 represents unstable equilibrium state.

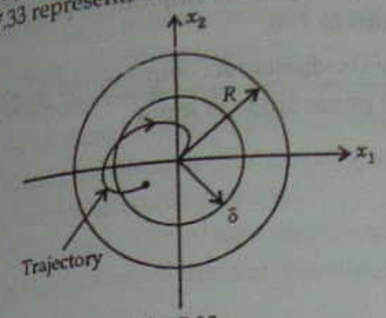


Fig. 7.32.

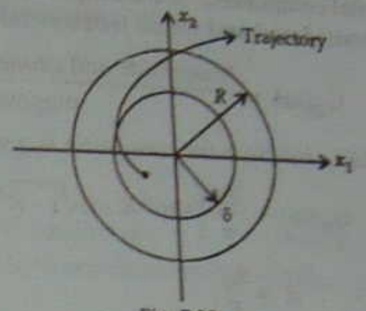


Fig. 7.33.

The system represented by the equation 7.35 is asymptotically stable in the large if it is asymptotically stable for all states from which trajectories originate.

7.12. LIMIT CYCLES

The definition of stability in the sense of Lyapunov includes the possibility of the state of a perturbed non-linear system following a closed trajectory within the tolerance limits specified by the region S(R). This behaviour is called limit cycle. The limit cycle describes the oscillations of non-linear systems. The limit cycle corresponds to an oscillation of fixed amplitude and period but not necessarily sinusoidal.

Consider Vander Pol's differential equation

$$\frac{d^2x}{dt^2} - \mu(1-x^2)\frac{dx}{dt} + x = 0 \qquad ...(7.38)$$

Compare the equation 7.38 with linear differential equation

$$\frac{d^2x}{dt^2} + 2\zeta\frac{dx}{dt} + x = 0 \qquad ...(7.39)$$

$$2\zeta = -\mu(1 - x^2)$$

$$\zeta = \frac{\mu}{2}(1-x^2) \qquad ...(7.40)$$

If $|x| >> 1$, then damping factor has positive value. Therefore the system behaves like overdamped system with decrease of amplitude of x. (shown by the outer trajectory fig. 7.34).

If $|x| << 1$, then the damping factor has negative value with increase of amplitude of x till the system state again enters the limit cycle. (shown by the inner trajectory fig. 7.34).

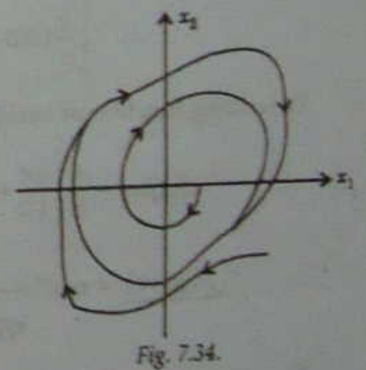


Fig. 7.34.

SUMMARY

A non-linear system does not posses the homogeneity and superposition properties.

If the non-linearities are inherently present in the system then these are known as incidental non-linearities. If some non-linearities are intentionally added to the system, then such types of non-linearities are called intentional non-linearities.

Some common type of non-linearities are saturation, dead zone, friction etc.

The describing function is defined as the ratio of amplitudes and phase angle between the fundamental components of the output and input sinusoid, for all amplitudes of the input which are to be considered and for all frequencies from zero to + ∞.

$$G_D(j\omega) = \frac{\text{Magnitude and phase angle of fundamental component of output}}{\text{magnitude and phase angle of input}}$$

Describing function for saturation is given by

$$G_D(j\omega) = \frac{2K}{\pi}\left[\sin^{-1} R + R\sqrt{1-R^2}\right]\angle 0°$$

where $R = \frac{E_S}{E}$

Describing function for ideal relay is

$$G_D(j\omega) = \frac{4Y}{\pi x}\angle 0°$$

Describing function for practical relay

$$G_D(j\omega) = \frac{4Y}{\pi x}\left[1-\frac{d^2}{4x^2}\right]^{\frac{1}{2}}\angle 0°$$

Describing function for the combination of dead zone and saturation

$$G_D(j\omega) = \begin{cases} 0 & ; x < \frac{D}{2}, \alpha = \beta = \pi/2 \\ \left[1-\frac{2}{\pi}(\alpha + \sin\alpha\cos\alpha\right]k & ; \frac{D}{2} < x < s \\ \frac{k}{\pi}[2(\beta-\alpha)+\sin 2\beta - \sin 2\alpha] & ; x > s \end{cases}$$

Describing function of backlash is given by

$$G_D(j\omega) = \frac{K}{\pi}\left[\left\{\frac{\pi}{2} + \sin^{-1}(1-2R)\right\}^2 + 4R(1-R) + \{\pi + 2\sin^{-1}(1-2R)\}\,2R(1-2R)\sqrt{\frac{1-R}{R}}\right]^{\frac{1}{2}}$$

$$\angle G_D(j\omega) = \tan^{-1}\frac{4R(R-1)}{\pi/2 + \sin^{-1}(1-2R) + 2R(1-2R)\sqrt{\frac{1-R}{R}}}$$

The phase plane method is a graphical method for solving the second order non-linear differential equations. This method gives the information about the types of transient response for various initial conditions.

SEMI-OBJECTIVE TYPE QUESTIONS

(i) Define non-linear system. (Article 7.1)
(ii) Write short note on common types of non-linearities. (Article 7.2)
(iii) Define Describing function. (Article 7.3)
(iv) Write short note on phase plane technique. (Article 7.5)
(v) Write short note on phase portraits. (Article 7.6)
(vi) Write the procedure for drawing a trajectory. (Ex. 7.7)
(vii) Define autonomous system. (Article 7.10)
(viii) Write short note on asymptotic stability and limit cycles. (Articles 7.11 and 7.12)
(ix) Is the concept of transfer function applicable to non-linear system also?

□□□

Chapter 8

State Space Analysis of Control System

8.1. ANALYSIS OF SYSTEMS

The procedure for determining the state of a system is called state variable analysis. The state of a dynamic system is the smallest set of variables such that the knowledge of these variables at $t = t_0$ with the knowledge of the input for $t \geq t_0$ completely determines the behaviour of the system for any time $t \geq t_0$. This set of variables is called state variables.

In earlier chapters we studied the linear system by transfer function, block diagram etc. The transfer function has some drawbacks *e.g* transfer function is only defined under zero initial conditions and also it is applicable to linear time invariant systems.[1] Therefore due to these limitation state variable approach is developed. This technique can be used for analysis and design of linear and non linear, time invariant or time variant and multi input multi output systems.[2] The state space analysis involves the description of the system in terms of I order differential equations by selecting suitable state variables, the first order derivatives are arranged on left hand side and on right hand side the terms are free from derivatives. The state space techniques have many advantages (Given in next article *i.e.* 8.2).

8.2. ADVANTAGES OF STATE SPACE TECHNIQUES

This technique has the following advantages.

1. This approach can be applied to linear or nonlinear, time variant or time invariant systems.
2. It is easier to apply where the laplace transform cannot be applied
3. n^{th} order differential equations can be expressed as 'n' equation of first order whose solutions are easier.
4. It is a time domain approach.
5. This method is suitable for digital computer computation because this is a time domain approach.
6. The system can be designed for optimal conditions with respect to given performance indices.

8.3. SOME IMPORTANT DEFINITIONS

State : The state of a system at any time 't_0' is the minimum set of numbers $x_1, x_2, \ldots, x_n$ which along with the input to the system for time $t \geq t_0$ is sufficient to determine the behaviour of the system for all $t \geq t_0$. In other words, the state of a system represents the minimum amount of information that we need to know about a system at "t_0" such that its future behaviour can be determined without reference to the input before 't_0'. The state can also be defined as the state of a system at time t_0 is the

1. If the characteristic of a system does not change with time, then the system is said to be time invariant.
2. A system is said to be a single variable system if and only if it has only one input terminal and only one output terminal. A system is said to be multi variable system if and only if it has more than one input terminal or more than one output terminal.

amount of information at t_0, that, together with input $u(t_0, \infty)$ determines the unique behaviour of the system for all $t \geq t_0$. By the behaviour of the system, we mean all responses, including the state of the system. If the system is a network we mean the voltage and current of every branch of the network.

Consider the network shown in fig. 8.1 if the initial current through the inductor and initial voltage across the capacitor are known, then for any driving voltage the behaviour of the netwrok can be determined. Hence, the inductor current and capacitor voltage can be considered as the state of the network.

State Variables : The definition is given in article 8.1

State Vector : If we need n variables to completely describe the behaviour of a given system, then these n state variables may be considered as n component of a vector x. Such a vector is called state vector, A state vector is thus a vector which determines uniquely the system state $x(t)$ for any time $t \geq t_0$, once the state at $t = t_0$ is given and the input $u(t)$ for $t \geq t_0$ is specified.

State space : The n-dimensional space whose coordinate axes consists of the x_1 axis, x_2 axis ....., x_n axis is called state space. Any state can be represented by a point in the state space.

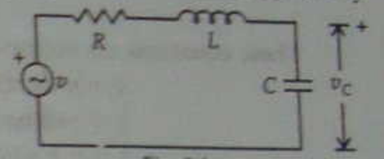


Fig. 8.1.

8.4. STATE SPACE REPRESENTATION

8.4.1. State Space Representation for Electrical Network (Physical Variable Form)

Consider an *RLC* network shown in fig. 8.2. Let, the current at time $t = 0$ be $i_L(0)$ and capacitor voltage at time $t = 0$ be $V_c(0)$. Thus, the state of the network at time $t = 0$ is specified by the inductor current and capacitor voltage. Hence, the pair $i_L(0)$, $V_c(0)$ is called the initial state of the network. Similarly at time 't', the pair $i_L(t)$, $V_c(t)$ is called the state of the network at 't'. The variable i_L and V_c are called state variables of the network.

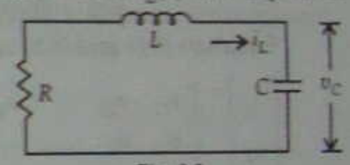


Fig. 8.2.

Apply *KVL*

$$Ri_L + L\frac{di_L}{dt} + V_c = 0 \quad \ldots(8.1)$$

Also,

$$i_c = i_L = C\frac{dV_c}{dt} \quad \ldots(8.2)$$

from (8.1)

$$\frac{di_L}{dt} = -\frac{R}{L}i_L - \frac{1}{L}V_c \quad \ldots(8.3)$$

$$\frac{dv_c}{dt} = \frac{1}{C}i_L \quad \ldots(8.4)$$

Equations of this form are called state equations. In such equations all the variables present are state variables.

Equations 8.3 and 8.4 can be written in matrix form as

$$\begin{bmatrix} \frac{di_L}{dt} \\ \frac{dv_c}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} i_L \\ V_C \end{bmatrix} \quad \ldots(8.5)$$

let $x(t) \Delta \begin{bmatrix} i_L \\ V_C \end{bmatrix}$ and $A = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix}$

then equation 8.5 can be written as

$$\frac{d}{dt} x(t) = A\, x(t)$$

or, $$\dot{x}(t) = A\, x(t)$$

In the linear time-invariant systems, the general form of state equations are

$$\dot{x}(t) = A\, x(t) + B\, u(t) \quad \text{...(8.6)}$$

$$y(t) = C\, x(t) + D\, u(t) \quad \text{...(8.7)}$$

These equations are vector differential equations where 'x' is the n-dimensional state vector

y = n-dimensional output vector

u = r-dimensional control vector or input vector

A = $n \times n$ system matrix

B = $n \times r$ control matrix

C = $n \times n$ output matrix

In some cases there is no direct connection between input and output so $D\, u(t)$ will not be there.

$$y(t) = C\, x(t)$$

Equation (8.6) and (8.8) can be expressed as ...(8.8)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ \vdots & & & \\ a_{n1} & & & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1r} \\ \vdots & & & \\ \vdots & & & \\ b_{n1} & & & b_{nr} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ \vdots \\ u_n \end{bmatrix} \quad \text{...(8.9)}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ \vdots & & & \\ \vdots & & & \\ C_{m1} & \cdots & \cdots & C_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{...(8.10)}$$

8.4.2. State Space Representation of nth Order Differential Equations

Consider the following examples

8.4.2 (a) For nth order differential equation

Example 8.1. A system is described by the differential equation

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 10y = 8\, u(t)$$

where y is the output and u is the input to the system. Obtain state space representation of the system.

(R.M.L., University, Faizabad, 2010)

Solution : Select the state variables as

$x_1 = y$, $x_2 = \dot{y}$ and $x_3 = \ddot{y}$ then

$\dot{x}_1 = x_2$

$\dot{x}_2 = x_3$

$\dot{x}_3 = 8\, u(t) - 10\, x_1 - 11 x_2 - 6x_3$

The last equation is obtained from the given equation.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -10 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 8 \end{bmatrix} u(t) \quad \text{...(8.11)}$$

Compare equation (8.11) with equation (8.6) we get

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -10 & -11 & -6 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 8 \end{bmatrix}, \quad x(t) = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

8.4.2 (b) : State space representation of n^{th} order Linear system with r Forcing function

Consider the following example

Example 8.2. A system is described by the following differential equation. Represent the system in state space.

$$\frac{d^3x}{dt^3} + 3\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = u_1(t) + 3u_2(t) + 4\, u_3(t)$$

and outputs are

$$y_1 = 4\frac{dx}{dt} + 3u_1$$

$$y_2 = \frac{d^2x}{dt^2} + 4u_2 + u_3$$

Solution : Select the state variables as

$x_1 = x$

$\dot{x}_1 = \dot{x} = x_2$

$\dot{x}_2 = x_3$

$\dot{x}_3 = u_1(t) + 3u_2(t) + 4u_3(t) - 3x_3 - 4x_2 - 4x_1$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

outputs :

$y_1 = 4x_2 + 3u_1$

$y_2 = x_3 + 4u_2 + u_3$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

8.4.3. State Space Representation for Transfer Function

Consider the following example

Example 8.3. For the given transfer function, obtain the state model.

$$G(s) = \frac{y(s)}{u(s)} = \frac{K}{s^3 + a_3 s^2 + a_2 s + a_1}$$

Solution : This transfer function has no zeros.

$$(s^3 + a_3 s^2 + a_2 s + a_1)\, y(s) = Ku(s)$$

or, $$s^3 y(s) + a_3 s^2 y(s) + a_2 s\, y(s) + a_1 y(s) = Ku(s)$$

Taking inverse laplace

$$\dddot{y}(t) + a_3 \ddot{y}(t) + a_2 \dot{y}(t) + a_1 y(t) = K u(t)$$

or $$\dddot{y}(t) = Ku(t) - a_3 \ddot{y}(t) - a_2 \dot{y}(t) - a_1 y(t)$$

Select the state variables as, first state variable as output

$$y(t) = x_1$$
$$\dot{y}(t) = \dot{x}_1 = x_2$$
$$\ddot{y}(t) = \dot{x}_2 = x_3$$
$$\dddot{y}(t) = \dot{x}_3$$
$$\therefore \quad \dddot{y}(t) = -a_3 x_3 - a_2 x_2 - a_1 x_1 + Ku(t)$$

Rewriting the equations

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = -a_3 x_3 - a_2 x_2 - a_1 x_1 + Ku(t)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_1 & -a_2 & -a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ K \end{bmatrix} u(t)$$

$$y(t) = [1\ 0\ 0]\, x_1(t)$$

BLOCK DIAGRAM :

The block diagram of the given transfer function is shown in fig. 8.3.

Now consider another case when the transfer function has zeros.

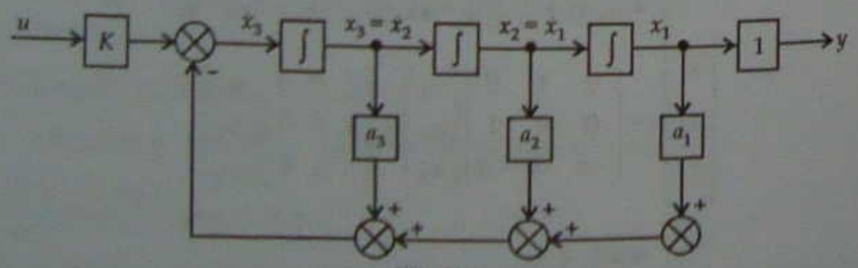


Fig. 8.3.

Example 8.4. Obtain the state model for the given transfer function.

$$G(s) = \frac{y(s)}{u(s)} = \frac{K(C_2 s + C_1)}{s^3 + a_3 s^2 + a_2 s + a_1}$$

Solution : Break the transfer function in two parts

$$\frac{y(s)}{u(s)} = \frac{x_1(s)}{u(s)} \cdot \frac{y(s)}{x_1(s)} = \frac{K}{s^3 + a_3 s^2 + a_2 s + a_1} (C_2 s + C_1)$$

Now consider $$\frac{x_1(s)}{u(s)} = \frac{K}{s^3 + a_3 s^2 + a_2 s + a_1}$$

$$[s^3 + a_3 s^2 + a_2 s + a_1]\, x_1(s) = Ku(s)$$

Taking inverse laplace

$$\dddot{x}_1(t) + a_3 \ddot{x}_1(t) + a_2 \dot{x}_1(t) + a_1 x_1(t) = Ku(t)$$
$$\dddot{x}_1(t) = -a_3 \ddot{x}_1(t) - a_2 \dot{x}_1(t) - a_1 x_1(t) + Ku(t)$$

Select the state variables as

$$\dot{x}_1 = x_2$$
$$\ddot{x}_1 = \dot{x}_2 = x_3$$
$$\dddot{x}_1 = \dot{x}_3$$

Rewrite

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = -a_3 x_3 - a_2 x_2 - a_1 x_1 + Ku$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_1 & -a_2 & -a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ K \end{bmatrix} u(t)$$

Consider, $$\frac{y(s)}{x_1(s)} = C_2 s + C_1$$

$$y(s) = x_1(s)\,[C_2 s + C_1]$$

Take inverse laplace

$$y(t) = C_1 x_1 + C_2 \dot{x}_2$$
$$y(t) = [C_1\ C_2\ 0]\, x(t)$$
$$= [C_1\ C_2\ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The block diagram is shown in fig. 8.4.

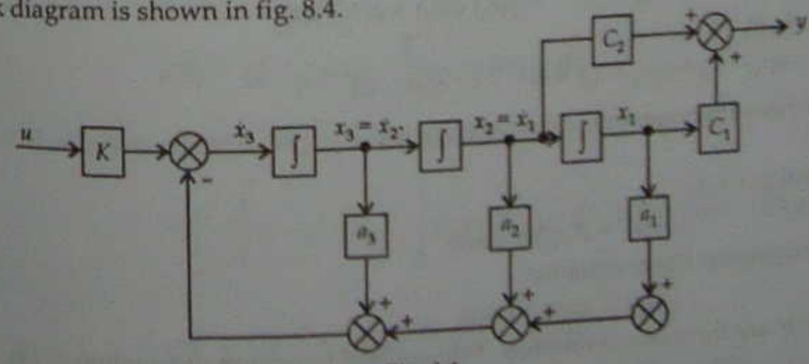


Fig. 8.4.

8.5. SOLUTION OF THE TIME-INVARIANT STATE EQUATION

8.5.1. Solution of Homogeneous State Equation : Laplace Transform Method

We know that

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$u(t) = 0$ for unforced response

then

$$\dot{x}(t) = Ax(t) \quad ...(8.12)$$

let us consider the analogous scalar equation

$$\dot{x}(t) = ax(t) \quad ...(8.13)$$

Take the laplace transform of equation 8.14

$$sX(s) - x(0) = aX(s) \quad ...(8.14)$$

$$(s - a)\, X(s) = x(0)$$

or,
$$X(s) = (s - a)^{-1} x(0)$$

Take the inverse laplace of equation 8.15

$$x(t) = e^{at} x(0) \quad ...(8.15)$$

If equation (8.16) is the solution of equation (8.14) then the solution of equation (8.13)

$$x(t) = e^{At} x(0) \quad ...(8.16)$$

$e^{At} = \phi(t)$ = State Transition Matrix (STM)

$$= I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + ... = \sum_{i=0}^{n} \frac{A^i t^i}{i!} \quad ...(8.17)$$

$$\phi(t) = \mathcal{L}^{-1}\phi(s) = \mathcal{L}^{-1}[sI - A]^{-1}$$

where $\phi(s)$ = Resolvant matrix

8.5.2. Properties of State Transition Matrices

For time invariant system

$$\dot{x} = Ax \text{ and}$$

$$\phi(t) = e^{At} = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + ...$$

(i) $\phi(0) = e^{A(0)} = I$

(ii) $\phi(t) = e^{At} = (e^{-At})^{-1} = [\phi(-t)]^{-1}$

(iii) $\phi^{-1}(t) = \phi(-t)$

(iv) $\phi(t_1 + t_2) = e^{A(t_1 + t_2)} = e^{At_1}\, e^{At_2} = \phi(t_1)\, \phi(t_2) = \phi(t_2)\, \phi(t_1)$

(v) $[\phi(t)]^n = \phi(nt)$

(vi) $\phi(t_2 - t_1)\, \phi(t_1 - t_0) = \phi(t_2 - t_0) = \phi(t_1 - t_0)\, \phi(t_2 - t_1)$

8.5.3. Solution of Non-Homogeneous State Equations

Here $u(t) \neq 0$

a. Initial condition $t = 0$

$$\dot{x}(t) = Ax(t) + Bu(t) \quad ...(8.18)$$

Consider an analogous scalar equation

$$\dot{x}(t) = ax(t) + bu(t) \quad ...(8.19)$$

where '*a*' and '*b*' are the scalar quantities. Take laplace transform of equation (8.19).

$$sX(s) - x(0) = aX(s) + bu(s)$$

$$(s - a)\, X(s) = x(0) + bu(s)$$

or,
Multiplying both the sides by $(s - a)^{-1}$ we get

$$X(s) = (s - a)^{-1} x(0) + (s - a)^{-1} bu(s) \quad ...(8.20)$$

Take inverse laplace of equation (8.20)

$$x(t) = e^{at} x(0) + \int_0^t e^{a(t-\tau)}\, bu(\tau)\, d\tau \quad ...(8.21)$$

Equation (8.21) is the solution of equation (8.19), then the solution of equation (8.18) will be

$$x(t) = e^{At} x(0) + \int_0^t e^{A(t-\tau)}\, bu(\tau)\, d\tau \quad ...(8.22)$$

or,
$$x(t) = \phi(t)\, x(0) + \int_0^t \phi(t-\tau)\, bu(\tau)\, d\tau \quad ...(8.23)$$

b. When initial time is given by t_0 *i.e* $t = t_0$

$$x(t) = \phi(t)\, x(0) + \int_0^t \phi(t-\tau)\, Bu(\tau)\, d\tau$$

at
$$t = t_0$$

$$x(t_0) = \phi(t_0)\, x(0) + \int_0^{t_0} \phi(t_0 - \tau)\, Bu(\tau)\, d\tau \quad ...(8.24)$$

Multiply both the sides by $\phi^{-1}(t_0)$

$$\phi^{-1}(t_0)\, x(t_0) = x(0) + \int_0^{t_0} \phi(-t_0)\, \phi(t_0 - \tau)\, Bu(\tau)\, d\tau$$

$$= x(0) + \int_0^{t_0} \phi(-t_0 + t_0 - \tau)\, Bu(\tau)\, d\tau$$

$$\phi^{-1}(t_0)\, x(t_0) = x(0) + \int_0^{t_0} \phi(-\tau)\, Bu(\tau)\, d\tau$$

or
$$x(0) = \phi(-t_0)\, x(t_0) - \int_0^{t_0} \phi(-\tau)\, Bu(\tau)\, d\tau \quad ...(8.25)$$

Put the value of $x(0)$ from equation (8.25) in equation (8.23)

$$x(t) = \left[\phi(-t_0)\, x(t_0) - \int_0^{t_0} \phi(-\tau)\, Bu(\tau)\, d\tau\right] \phi(t) + \int_0^t \phi(t-\tau)\, Bu(\tau)\, d\tau$$

or,
$$x(t) = \phi(t - t_0) x(t_0) + \int_{t_0}^t \phi(t-\tau)\, Bu(\tau)\, d\tau \quad ...(8.26)$$

Equation (8.26) can also be written as

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}\, Bu(\tau)\, d\tau \qquad \text{...(8.27)}$$

Example 8.5. Find the time response of the system described by the equation

$$\dot{x}(t) = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$x(0) = \begin{bmatrix} -1 \\ 0 \end{bmatrix},\ u(t) = 1,\ t > 0$$

Solution :

$$x(t) = \phi(t)\, x(0) + \int_0^t \phi(t-\tau)\, Bu(\tau)\, d\tau \qquad \text{...(8.28)}$$

Step 1 : Calculation of $\phi(t)$:

$$\phi(t) = \mathcal{L}^{-1}\phi(s) = \mathcal{L}^{-1}(sI - A)^{-1}$$

$$\phi(s) = [sI - A]^{-1}$$

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} s+1 & -1 \\ 0 & s+2 \end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix} \dfrac{s+2}{s^2+3s+2} & \dfrac{1}{s^2+3s+2} \\ 0 & \dfrac{s+1}{s^2+3s+2} \end{bmatrix}$$

$$\therefore \quad \phi(t) = \mathcal{L}^{-1}\,\phi(s) = \begin{bmatrix} \mathcal{L}^{-1}\dfrac{s+2}{s^2+3s+2} & \mathcal{L}^{-1}\dfrac{1}{s^2+3s+2} \\ 0 & \mathcal{L}^{-1}\dfrac{s+1}{s^2+3s+2} \end{bmatrix}$$

$$\therefore \quad \phi(t) = \begin{bmatrix} e^{-t} & e^{-t} - e^{-2t} \\ 0 & e^{-2t} \end{bmatrix}$$

Put all the values in equation (8.28)

$$x(t) = \begin{bmatrix} e^{-t} & e^{-t} - e^{-2t} \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \int_0^t \begin{bmatrix} e^{-(t-\tau)} & e^{-(t-\tau)} - e^{-2(t-\tau)} \\ 0 & e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} 1\, dt$$

$$= \begin{bmatrix} -e^{-t} \\ 0 \end{bmatrix} + \int_0^t \begin{bmatrix} e^{-(t-\tau)} - e^{-2(t-\tau)} \\ e^{-2(t-\tau)} \end{bmatrix} d\tau$$

$$x_1(t) = -e^{-t} + \int_0^t e^{-(t-\tau)} - e^{-2(t-\tau)}\, d\tau$$

$$x_2(t) = \int_0^t e^{-2(t-\tau)}\, d\tau$$

$$x_1(t) = \frac{1}{2} - 2e^{-t} + \frac{1}{2}e^{-2t} \qquad \textbf{Ans.} \qquad \text{...(8.29)}$$

or,

$$x_2(t) = \frac{1}{2} - \frac{1}{2}e^{-2t}$$

$$y = Cx \qquad C = [1\ 0]$$

$$y = [1\ 0] \begin{bmatrix} \dfrac{1}{2} - 2e^{-t} + \dfrac{1}{2}e^{-2t} \\ \dfrac{1}{2} - \dfrac{1}{2}e^{-2t} \end{bmatrix} = \frac{1}{2} - 2e^{-t} + \frac{1}{2}e^{-2t} \qquad \text{...(8.30)}$$

Equations 8.29 and 8.30 are the required answers.

8.8 TRANSFER MATRIX

We know that

$$\dot{x}(t) = Ax(t) + Bu(t) \qquad \text{...(8.31)}$$

$$y(t) = Cx(t) \qquad \text{...(8.32)}$$

Take the laplace transform, assuming initial conditions zero*

$$sX(s) = AX(s) + Bu(s) \qquad \text{...(8.33)}$$

$$Y(s) = CX(s) \qquad \text{...(8.34)}$$

Consider the equation 8.33, I is the identity martix

$$sIX(s) = AX(s) + Bu(s)$$

or,

$$X(s)\,[sI - A] = Bu(s)$$

Multiplying both the side by $[sI - A]^{-1}$ we get

$$X(s) = u(s)[sI - A]^{-1}B \qquad \text{...(8.35)}$$

Put the values of $X(s)$ from equation (8.35) in equation (8.34)

$$Y(s) = C[sI - A]^{-1}\, Bu(s) \qquad \text{...(8.36)}$$

We know that

$$Y(s) = G(s)\, u(s) \qquad \text{...(8.37)}$$

Compare the equation 8.36 and equation 8.37 we get

$$G(s) = C[sI - A]^{-1}\, B \qquad \text{...(8.38)}$$

where $G(s)$ = Transfer matrix

The transfer matrix given by the equation 8.38 is for MIMO system. For single input single output system, the transfer matrix is given by

$$G(s) = C^T[sI - A]^{-1}\, b \qquad \text{...(8.39)}$$

Also, $[sI - A]^{-1} = \phi(s)$ = Resolvant matrix, Then equation (8.39) becomes

$G(s) = C\phi(s)\, B$ ...(8.40) For MIMO system

$G(s) = C^T\phi(s)\, b$ ...(8.41) For SISO system

Example 8.6. The system equations are given by

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [1\ \ 0]\, x(t)$$

Find the transfer function of the system.

Solution : Given that

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix},\ B = \begin{bmatrix} 0 \\ 1 \end{bmatrix},\ C^T = [1\ \ 0]$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix} \quad \therefore [sI - A]^{-1} = \begin{bmatrix} \frac{s+3}{s^2+3s+2} & \frac{1}{s^2+3s+2} \\ \frac{-2}{s^2+3s+2} & \frac{s}{s^2+3s+2} \end{bmatrix}$$

$$C^T[sI - A]^{-1}.\, b = [1\ \ 0] \begin{bmatrix} \frac{s+3}{s^2+3s+2} & \frac{1}{s^2+3s+2} \\ \frac{-2}{s^2+3s+2} & \frac{s}{s^2+3s+2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{s^2+3s+2}$$

Hence, required transfer function = $\frac{1}{s^2+3s+2}$ **Ans.**

8.7. COMPUTATION OF STATE TRANISITION MATRIX : e^{At} :

8.7.1. Laplace Transform Method

We know that

$$\dot{x}(t) = Ax + Bu(t)$$

For unforced system $u(t) = 0$

$$\dot{x}(t) = Ax$$

Take the laplace transform

$$s\,X(s) - x(0) = A\,X(s)$$

or $$[sI - A]\,X(s) = x(0)$$

Multiply both the sides by $[sI - A]^{-1}$

$$X(s) = [sI - A]^{-1}\,x(0)$$

$[sI - A]^{-1} = \phi(s)$ = Resolvant matrix

$$\phi(t) = e^{At} = \mathcal{L}^{-1}\,\phi(s) = \mathcal{L}^{-1}[sI - A]^{-1}$$

Where, $\phi(t) = e^{At}$ = state transition matrix

Example 8.7. Compute the STM when

$$A = \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}$$

Solution :

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} s+1 & -1 \\ 0 & s-2 \end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix} \frac{s-2}{s^2+3s+2} & \frac{1}{s^2+3s+2} \\ 0 & \frac{s+1}{s^2+3s+2} \end{bmatrix}$$

* By the definition of transfer function.

$$\phi(t) = \mathcal{L}^{-1}[sI - A]^{-1} = \begin{bmatrix} \mathcal{L}^{-1}\frac{s-2}{s^2+3s+2} & \mathcal{L}^{-1}\frac{1}{s^2+3s+2} \\ 0 & \mathcal{L}^{-1}\frac{s+1}{s^2+3s+2} \end{bmatrix}$$

$$\phi(t) = \begin{bmatrix} e^{-t} & e^{-t} - e^{-2t} \\ 0 & e^{-2t} \end{bmatrix}$$

8.7.2. Series Summation Method

$$\phi(t) = e^{At} = I + At + \frac{A^2t^2}{2!} + \ldots\ldots$$

Example. 8.8. Evaluate the STM by series summation method

$$\mathbf{A} = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$$

Solution : Given that

$$A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & 3 & 1 \\ 4 & 2 & 3 \\ 4 & -2 & 7 \end{bmatrix}, A^3 = \begin{bmatrix} 14 & -4 & 17 \\ 13 & 3 & 11 \\ 13 & 11 & 3 \end{bmatrix} \text{ and so on.}$$

$$e^{At} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 2t & -2t & 3t \\ t & t & t \\ t & 3t & -t \end{bmatrix} + \frac{t^2}{2}\begin{bmatrix} 5 & 3 & 1 \\ 4 & 2 & 3 \\ 4 & -2 & 7 \end{bmatrix} + \frac{t^3}{6}\begin{bmatrix} 14 & -4 & 17 \\ 13 & 3 & 11 \\ 13 & 11 & 3 \end{bmatrix} + \ldots$$

$$= \begin{bmatrix} 1+2t+\frac{5}{2}t^2+\frac{14}{6}t^3+\ldots & -2t+\frac{3t^2}{2}-\frac{2t^2}{3}+\ldots & 3t+\frac{t^2}{2}+\frac{17}{6}t^3+\ldots \\ t+2t^2+\frac{13}{6}t^3+\ldots & 1+t+t^2+\ldots & t+\frac{3t^2}{2}+\frac{11t^3}{6}+\ldots \\ t+2t^2+\frac{13}{6}t^3+\ldots & 3t-t^2+\frac{11t^3}{6}+\ldots & 1-t+\frac{7t^2}{2}+\frac{3t^3}{6}+\ldots \end{bmatrix}$$

8.7.3 Eigen Values

The roots of the characteristic equation are known as eigen values of matrix.

8.7.4. Eigen Vectors

Any non zero vector P_i that satisfies the matrix equation $(\lambda_i I - A)\,P_i = 0$ where λ_i is i^{th} eigen value of A is known as eigen vector of A associated with the eigen value λ_i.

8.7.5. Use of Diagonal

$$\text{let } A = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \ldots$$

$$e^{At} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} + \begin{bmatrix}\lambda_1 t & 0 & 0\\ 0 & \lambda_2 t & 0\\ 0 & 0 & \lambda_3 t\end{bmatrix} + \ldots = \begin{bmatrix}e^{\lambda_1 t} & 0 & 0\\ 0 & e^{\lambda_2 t} & 0\\ 0 & 0 & e^{\lambda_3 t}\end{bmatrix}$$

let λ_i is the eigen value of A and P_i is the eigen vector, then for each of the n eigen values of λ

$$AP_1 = \lambda_1 P_1$$
$$AP_2 = \lambda_2 P_2$$
$$\vdots$$
$$AP_n = \lambda_n P_n$$

or $$AM = M\wedge$$

Where $\wedge$ = diagonal matrix with eigen values on its main diagonal
M = modal matrix —(8.42)

$$\wedge = M^{-1}AM$$

or $$e^{\wedge t} = M^{-1}e^{At}M$$

or $$e^{At} = M\,e^{\wedge t}\,M^{-1}$$

Example 8.9. Compute e^{At} when

$$\mathbf{A} = \begin{bmatrix}0 & 1\\ -3 & -4\end{bmatrix}$$

Solution :

$$\lambda I - A = \lambda\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} - \begin{bmatrix}0 & 1\\ -3 & -4\end{bmatrix} = \begin{bmatrix}\lambda & -1\\ 3 & \lambda+4\end{bmatrix} \quad —(8.43)$$

$$|\lambda I - A| = \lambda^2 + 4\lambda + 3 = 0$$

or, $$\lambda_1 = -1$$
$$\lambda_2 = -3$$

let the eigen vector

$$P_1 = \begin{bmatrix}p_{11}\\ p_{21}\end{bmatrix} \text{ and } P_2 = \begin{bmatrix}p_{12}\\ p_{22}\end{bmatrix}$$

Put $\lambda = -1$ in equation (8.43)

$$\begin{bmatrix}-1 & -1\\ 3 & 3\end{bmatrix}\begin{bmatrix}p_{11}\\ p_{21}\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$$-p_{11} - p_{21} = 0$$
$$3p_{11} + 3p_{21} = 0$$

let $$p_{11} = 1 \quad \text{then} \quad p_{21} = -1$$

$\therefore$ $$P_1 = \begin{bmatrix}1\\ -1\end{bmatrix}$$

Put $$\lambda = -3$$

$$\begin{bmatrix}-3 & -1\\ 3 & 1\end{bmatrix}\begin{bmatrix}p_{12}\\ p_{22}\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$$-3p_{12} - p_{22} = 0$$
$$3p_{12} + p_{22} = 0$$

let $p_{12} = 1$ then $p_{22} = -3$

$$P_2 = \begin{bmatrix}1\\ -3\end{bmatrix}$$

$$M \doteq [P_1\ P_2] = \begin{bmatrix}1 & 1\\ -1 & -3\end{bmatrix}$$

$$M^{-1} = \begin{bmatrix}1.5 & 0.5\\ -0.5 & -0.5\end{bmatrix}$$

$$\wedge = \begin{bmatrix}-1 & 0\\ 0 & -3\end{bmatrix}$$

$$e^{At} = M\,e^{\wedge t}\,M^{-1} = \begin{bmatrix}1 & 1\\ -1 & -3\end{bmatrix}\begin{bmatrix}e^{-t} & 0\\ 0 & e^{-3t}\end{bmatrix}\begin{bmatrix}1.5 & 0.5\\ -0.5 & -0.5\end{bmatrix}$$

$$e^{At} = \begin{bmatrix}1.5\,e^{-t} - 0.5\,e^{-3t} & 0.5e^{-t} - 0.5e^{-3t}\\ 1.5e^{-3t} - 1.5e^{-t} & 1.5e^{-3t} - 0.5e^{-t}\end{bmatrix} \quad \textbf{Ans.}$$

8.8. BLOCK DIAGRAM OF A LINEAR SYSTEM IN STATE VARIABLE FORM

8.8.1 MIMO System

The system is represented by the following equations

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

The block diagram is shown in the fig. 8.5.

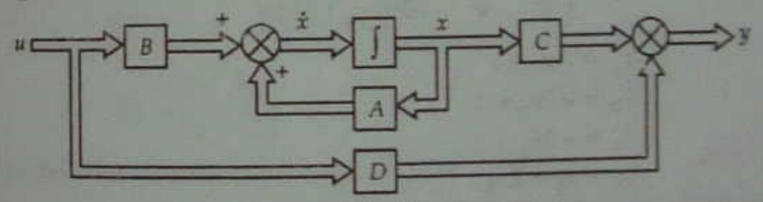


Fig. 8.5.

8.8.2 Single Input Single Output System

$$\dot{x}(t) = Ax + bu$$
$$y(t) = C^T x + du$$

The block diagram is shown in fig 8.6.

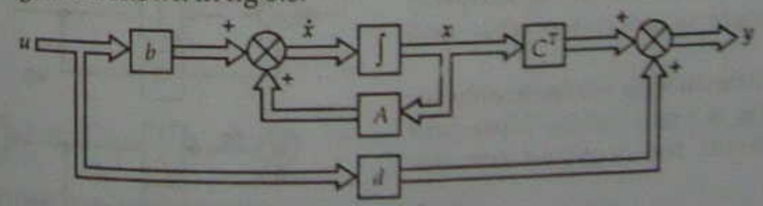


Fig. 8.6.

8.9. CONTROLLABILITY AND OBSERVABILITY

A system is said to be controllable if any initial state $x(t_0)$ or x_0 can be transferred to any final state $x(t_f)$ in a finite time interval $(t_f - t_0)$, $t \geq 0$ by some control u.

A system is said to be observable if every state x_0 can be exactly determined from the measurement of the output 'y' over a finite interval of time $0 \leq t \leq t_f$

The concept of controllability and observability were introduced by Kalman and they play an important role in the design of control system in state space.

There are four possible states for the system.

(*i*) The system is controllable and observable

(*ii*) The system is controllable and unobservable

(*iii*) The system is uncontrollable and observable

(*iv*) The system is uncontrollable and unobservable

Consider the following example to determine the controllability and observability.

8.9.1. With the Help of Block Diagram

Example 8.10. Determine the controllability and observability of the system described by the state equation. Find out the transfer function and draw the block diagram.

$$\dot{x}(t) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 & +2 \end{bmatrix} x(t)$$

Solution :

$$\dot{x}(t) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u$$

$$\dot{x}_1 = -x_1 + u$$

$$\dot{x}_2 = -2x_2 + u$$

$$\dot{x}_3 = -3x_3$$

$$y = x_1 + 2x_3$$

The block diagram is shown in fig. 8.7.

From the block diagram, the first state is controllable and observable, second state is controllable and unobservable (*scu*), the third state is uncontrollable and observable (*suo*). Since, all the states are not controllable and obserbvable the whole system is uncontrollable and unobservable.

The transfer function is available only for the state which is both controllable and observable. Hence, the required transfer function is

$$G(s) = \frac{1/s}{1+1/s} = \frac{1}{s+1}$$

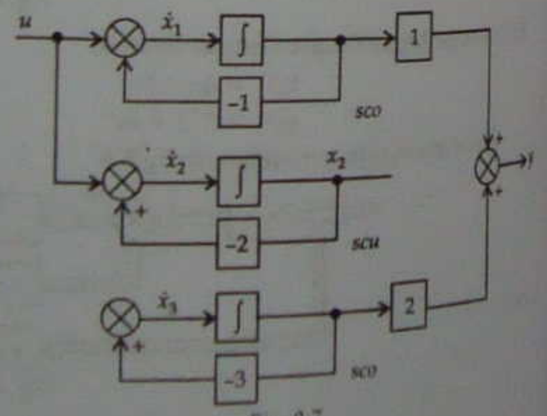


Fig. 8.7.

8.9.2. Kalman's Test

A linear time invariant continuous system described by the state equation

$$\dot{x} = Ax + Bu \qquad \text{...(8.44)}$$

$$y = Cx$$

is completely controllable if and only if the rank of the controllability matrix is defined as

$$Q = [B : AB : A^2 B : \ldots A^{n-1} B]$$

is equal to rank 'n'.

The system represented by the equation (8.44) is observable if and only if, the $n \times n_m$ composite matrix is obtained as

$$Q = [C^T : A^T C^T : (A^T)^2 C^T : \ldots A^{T(n-1)} C^T]$$

Example 8.11. Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -0.5 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Test for controllability and observability.

Solution :

$$Q = [B : AB]$$

Given that

$$A = \begin{bmatrix} -0.5 & 0 \\ 0 & -2 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -0.5 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

$$\therefore \quad Q = \begin{bmatrix} 0 & 0 \\ 1 & -2 \end{bmatrix}$$

Rank of 'Q' is one. Hence system is uncontrollable.*

$$C = \begin{bmatrix} 0 & 1 \end{bmatrix} \therefore C^T = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} -0.5 & 0 \\ 0 & -2 \end{bmatrix} A^T C^T = \begin{bmatrix} -0.5 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

$$Q' = [C^T : A^T C^T]$$

$$Q' = \begin{bmatrix} 0 & 0 \\ 1 & -2 \end{bmatrix}$$

Rank of 'Q' is one. Hence the system is unobservable. For controllability and observability the rank should be two.

The given system is uncontrollable and unobservable

8.10. TIME VARYING SYSTEM

Consider the n-dimensional linear time-varying dynamical equation

$$\dot{x}(t) = A(t)\,x(t) + B(t)\,u(t)$$
$$y(t) = C(t)\,x(t) + D(t)\,u(t) \qquad ...(8.45)$$

Where,

$x(t)$ is an n column vector, $A(t)$ and $B(t)$ are respectively $n \times n$ matrix and $n \times m$ matrix whose elements are function of time, $u(t)$ is m column vector of forcing function.

The solution of equation (8.45) with $x(t_0) = x_0$ is given by

$$x(t) = \phi(t, t_0)\,x_0 + \int_{t_0}^{t} \phi(t, \tau)\,B(\tau)\,u(\tau)\,d\tau \qquad ...(8.46)$$

$$x(t) = \phi(t, t_0)\left[x_0 + \int_{t_0}^{t} \phi(t_0, \tau)\,B(\tau)\,u(\tau)\,d\tau\right] \qquad ...(8.47)$$

Where $\phi(t, \tau)$, is the state transition matrix of $\dot{x} = A(t)\,x$, or, equivalently, the unique solution of

$$\frac{\partial}{\partial t}\phi(t, \tau) = A(t)\,\phi(t, \tau) \qquad \phi(\tau, \tau) = I \qquad ...(8.48)$$

Equation (8.47) is obtained from the equation (8.46) by using

$$\phi(t, \tau) = \phi(t, t_0)\,\phi(t_0, \tau) \qquad ...(8.49)$$

$\phi(t, t_0)$ is given by the series

$$\phi(t, t_0) = I + \int_0^t A(\tau_1)\,d\tau_1 + \int_{t_0}^{t} A(\tau_1)\left[\int_{t_0}^{\tau_1} A(\tau_2)\,d\tau_2\right]d\tau_1 + \ldots$$

Example 8.12. Determine the state transition matrix $\phi(t, 0)$ for the system

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ t & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution : $t_0 = 0$

$$\int_0^t A(\tau)\,d\tau = \int_0^t \begin{bmatrix} 0 & 1 \\ \tau & 0 \end{bmatrix} d\tau = \begin{bmatrix} 0 & t \\ t^2/2 & 0 \end{bmatrix}$$

$$\int_0^t A(\tau_1)\left[\int_0^{\tau_1} A(\tau_2)\,d\tau_2\right]d\tau_1 = \int_0^t \begin{bmatrix} 0 & 1 \\ \tau_1 & 0 \end{bmatrix}\begin{bmatrix} 0 & \tau_1 \\ \tau_1^2/2 & 0 \end{bmatrix} d\tau_1 = \begin{bmatrix} t^3/6 & 0 \\ 0 & t^3/3 \end{bmatrix}$$

$$\int_0^t \begin{bmatrix} 0 & 1 \\ \tau_1 & 0 \end{bmatrix}\begin{bmatrix} \tau_1^3/6 & 0 \\ 0 & \tau_1^3/3 \end{bmatrix} d\tau_1 = \begin{bmatrix} 0 & t^4/12 \\ t^5/30 & 0 \end{bmatrix}$$

$$\therefore \quad \phi(t, 0) = \begin{bmatrix} 1 + \frac{t^3}{6} + \ldots & t + \frac{t^4}{12} + \ldots \\ \frac{t^2}{2} + \frac{t^5}{30} + \ldots & 1 + \frac{t^3}{3} + \ldots \end{bmatrix}$$

*For controllability and observability the rank should be two.

8.11. CONTROLLABILITY : AND OBSERVABILITY : (Time Variant Systems)

Consider n-dimensional state equation

$$\dot{x} = A(t)\,x(t) + B(t)\,u(t)$$

Assume that the matrices A and B in the above state equation are $n-1$ times continuously differentiable. Then the state equation is controllable at t_0 if there exists a finite $t_1 > t_0$ such that

$$P[M_0(t_1) : M_1(t_1) : \ldots\ldots : M_{n-1}(t_1)] = n \qquad ...(8.50)$$

$$M_{K+1}(t) = -A(t)\,M_K(t) + \frac{d}{dt}M_K(t) \qquad ...(8.51)$$

$$K = 0, 1, 2 \ldots . n - 1$$

$$M_0(t) = B(t) \qquad ...(8.52)$$

Example 8.13. Consider

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} t & 1 & 0 \\ 0 & t & 0 \\ 0 & 0 & t^2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u$$

From *eq*^{n} 8.52

$$M_0(t) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$M_1(t) = -A(t)\,M_0(t) + \frac{d}{dt}M_0(t) = -\begin{bmatrix} t & 1 & 0 \\ 0 & t & 0 \\ 0 & 0 & t^2 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \frac{d}{dt}\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -t \\ -t^2 \end{bmatrix}$$

$$M_2(t) = -A(t)\,M_1(t) + \frac{d}{dt}M_1(t) = -\begin{bmatrix} t & 1 & 0 \\ 0 & t & 0 \\ 0 & 0 & t^2 \end{bmatrix}\begin{bmatrix} -1 \\ -t \\ -t^2 \end{bmatrix} + \frac{d}{dt}\begin{bmatrix} -1 \\ -t \\ -t^2 \end{bmatrix}$$

$$M_2(t) = \begin{bmatrix} 2t \\ t^2 - 1 \\ t^4 - 2t \end{bmatrix}$$

$$[M_0(t) : M_1(t) : M_2(t) = \begin{bmatrix} 0 & -1 & 2t \\ 1 & -t & t^2 - 1 \\ 1 & -t^2 & t^4 - 1 \end{bmatrix}$$

The above matrix has rank 3 for all $t \neq 0$, hence the system is controllable at every time 't'.

OBSERVABILITY : Consider the state equation

$$\dot{x} = A(t)\,x(t) + B(t)\,u(t)$$
$$\dot{y} = C(t)\,x(t) + D(t)\,u(t)$$

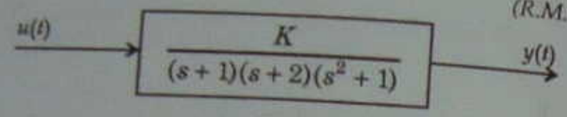
assume that the matrices A and C in the above state equation are $n-1$ times continuously differentiable. The state equation is observable at t_0 if there exists a finite $t_1 > 0$ such that

$$P\begin{bmatrix} N_0(t_1) \\ N_1(t_1) \\ \vdots \\ N_{n-1}(t_1) \end{bmatrix} = n \qquad ...(8.53)$$

where
$$N_{K+1}(t) = N_K(t)\,A(t) + \frac{d}{dt}N_K(t)$$
$$K = 0, 1, 2, \ldots, n-1 \quad \text{...(8.54)}$$
$$N_0(t) = C(t) \quad \text{...(8.55)}$$

ILLUSTRATIVE EXAMPLES

Example 8.14. A system is given below, find the state and output equation of the system. *(R.M.L. University Faizabad, 2001)*



Solution :

$$\frac{y(s)}{u(s)} = G(s) = \frac{K}{(s+1)(s+2)(s^2+1)}$$

$$\frac{y(s)}{u(s)} = \frac{K}{s^4+3s^3+3s^2+3s+2}$$

$$y(s)[s^4 + 3s^3 + 3s^2 + 3s + 2] = Ku(s)$$

$$s^4y(s) + 3s^3y(s) + 3s^2y(s) + 3sy(s) + 2y(s) = Ku(s)$$

Taking inverse laplace

$$\dddot{y}(t) + 3\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = Ku(t)$$

$$y(t) = x_1$$
$$\dot{y}(t) = \dot{x}_1 = x_2$$
$$\ddot{y}(t) = \dot{x}_2 = x_3$$
$$\dddot{y}(t) = \dot{x}_3 = x_4$$
$$\ddddot{y}(t) = \dot{x}_4$$

$$\therefore \ddddot{y}(t) = \dot{x}_4 = Ku(t) - 3x_4 - 3x_3 - 3x_2 - 2x_1$$

Rewriting

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = x_4$$
$$\dot{x}_4 = Ku(t) - 3x_4 - 3x_3 - 3x_2 - 2x_1$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\\ \dot{x}_4\end{bmatrix} = \begin{bmatrix}0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\\ -2 & -3 & -3 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 0\\ K\end{bmatrix}u(t)$$

$$y(t) = x_1$$
$$y(t) = \begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix}x(t)$$

Block diagram :

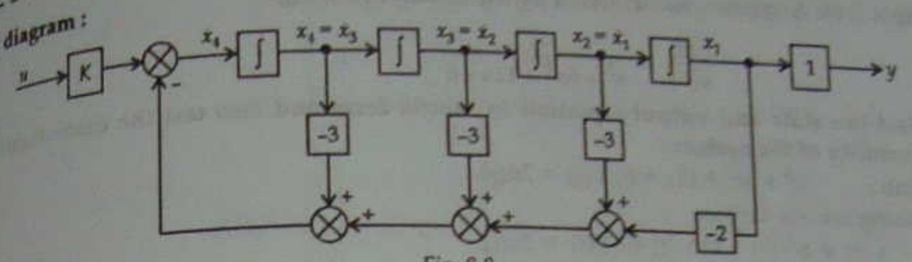


Fig. 8.8.

Example 8.15. A system is described by the following transfer function

$$G(s) = \frac{20\,(10s+1)}{s^3+3s^2+2s+1}$$

Find the state and output equation of the system. *(A.M.U., Aligarh 1989-90)*

Solution :
$$G(s) = \frac{y(s)}{u(s)} = \frac{20\,(10s+1)}{s^3+3s^2+2s+1}$$

$$\frac{y(s)}{u(s)} = \frac{X_1(s)}{u(s)} \cdot \frac{y(s)}{X_1(s)} = \frac{20}{s^3+3s^2+2s+1} \cdot (10s+1)$$

Consider
$$\frac{X_1(s)}{u(s)} = \frac{20}{s^3+3s^2+2s+1}$$

$$X_1(s)\,[s^3 + 3s^2 + 2s + 1] = 20\,u(s)$$

Taking inverse laplace

$$\dddot{X}_1(t) + 3\ddot{X}_1(t) + 2\dot{X}_1(t) + X_1(t) = 20\,u(t)$$

Select
$$\dot{X}_1 = X_2$$
$$\ddot{X}_1 = \dot{X}_2 = X_3$$
$$\dddot{X}_1 = \dot{X}_3 = X_4$$
$$\ddot{X}_1 = 20u - 3\ddot{X}_1 - 2\dot{X}_1 - X_1 = 20u - 3X_3 - 2X_2 - X_1$$

Rewriting
$$\dot{X}_1 = X_2$$
$$\dot{X}_2 = X_3$$
$$\dot{X}_3 = 20u - 3X_3 - 2X_2 - X_1$$

$$\begin{bmatrix}\dot{X}_1\\ \dot{X}_2\\ \dot{X}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -2 & -3\end{bmatrix}\begin{bmatrix}X_1\\ X_2\\ X_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 20\end{bmatrix}u$$

$$\frac{y(s)}{X_1(s)} = 10s + 1$$

or,
$$y(s) = (10s + 1)\,X_1(s)$$
$$y(t) = X_1 + 10\,X_2$$
$$y(t) = \begin{bmatrix}1 & 10 & 0\end{bmatrix}X(t)$$

Example 8.16. A system characterised by the transfer function

$$\frac{Y(s)}{u(s)} = \frac{2}{s^3+6s^2+11s+6}$$

Find the state and output equation in matrix form and also test the controllability and observability of the system.

Solution : $(s^3 + 6s^2 + 11s + 6)\ Y(s) = 2u(s)$.

Taking inverse laplace

$$\dddot{Y}(t) + 6\ddot{Y}(t) + 11\dot{Y}(t) + 6Y(t) = 2u(t).$$

let

$$Y(t) = x_1$$
$$\dot{Y}(t) = \dot{x}_1 = x_2$$
$$\ddot{Y}(t) = \dot{x}_2 = x_3$$
$$\dddot{Y}(t) = \dot{x}_3 = x_4$$

$\therefore$
$$\dddot{Y}(t) = x_4 = \dot{x}_3 = 2u(t) - 6x_3 - 11x_2 - 6x_1$$

Rewriting

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = 2u(t) - 6x_3 - 11x_2 - 6x_1$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 2\end{bmatrix}u(t)$$

$$y(t) = \begin{bmatrix}1 & 0 & 0\end{bmatrix}x(t)$$

For controllability

$$Q = [b : Ab : A^2 b]$$

$$A = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}, b = \begin{bmatrix}0\\ 0\\ 2\end{bmatrix}$$

$$Ab = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}\begin{bmatrix}0\\ 0\\ 2\end{bmatrix} = \begin{bmatrix}0\\ 2\\ -12\end{bmatrix}$$

$$A^2 b = \begin{bmatrix}2\\ -12\\ 50\end{bmatrix}$$

$$Q = \begin{bmatrix}0 & 0 & 2\\ 0 & 2 & -12\\ 2 & -12 & 50\end{bmatrix}$$

$|Q| \neq 0$

It is full of rank, hence the system is controllable

For observability :

$$Q' = \left[C^T : A^T C^T : A^{T^2} C^T\right]$$

$$C^T = \begin{bmatrix}1\\ 0\\ 0\end{bmatrix}, A^T = \begin{bmatrix}0 & 0 & -6\\ 1 & 0 & -11\\ 0 & 1 & -6\end{bmatrix}$$

$$A^T C^T = \begin{bmatrix}0\\ 1\\ 0\end{bmatrix}, \quad A^{T^2} C^T = \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}$$

$\therefore$
$$Q' = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

Rank of the matrix = 3, Hence the system observable.

Example 8.17. A system is described by the following equations

$$\dot{x}(t) = \begin{bmatrix}-1 & 1\\ 0 & -2\end{bmatrix}x(t) + \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix}u(t)$$

$$y(t) = \begin{bmatrix}1 & 2\\ 1 & 0\\ 1 & 1\end{bmatrix}x(t)$$

Find the transfer function of the system. *(B.E., A.M.U., Aligarh 89-90) (R.M.L. University, Faizabad, 2003)*

Solution : The given system is MIMO system.

$\therefore$
$$G(s) = C[sI - A]^{-1} B$$

$$B = \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix}, C = \begin{bmatrix}1 & 2\\ 1 & 0\\ 1 & 1\end{bmatrix}$$

$$sI - A = \begin{bmatrix}s & 0\\ 0 & s\end{bmatrix} - \begin{bmatrix}-1 & 1\\ 0 & -2\end{bmatrix} = \begin{bmatrix}s+1 & -1\\ 0 & s+2\end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix}s+2 & 1\\ 0 & s+1\end{bmatrix}\frac{1}{s^2+3s+2}$$

$$C[sI - A]^{-1} = \begin{bmatrix}1 & 2\\ 1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}s+2 & 1\\ 0 & s+1\end{bmatrix}\frac{1}{s^2+3s+2} = \begin{bmatrix}s+2 & 2s+3\\ s+2 & 1\\ s+2 & s+2\end{bmatrix}\frac{1}{s^2+3s+2}$$

$$C[sI - A]^{-1} B = \frac{1}{s^2+3s+2}\begin{bmatrix}s+2 & 2s+3\\ s+2 & 1\\ s+2 & s+2\end{bmatrix}\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix} = \frac{s^3-3s^2-9s+2}{s^2+3s+2}$$

$\therefore$
$$G(s) = \frac{s^3-3s^2-9s+2}{s^2+3s+2}$$ is the required transfer function.

Example 8.18. A system is characterised by the equation

$$\frac{y(s)}{u(s)} = \frac{20\,(4s+2)}{s^3+5s^2+8s+2}$$

Find its state and output equation and express in matrix form.

(B.T.E., A.M.U., Aligarh 86-87)

Solution :

$$\frac{y(s)}{u(s)} = \frac{X_1(s)}{u(s)} \cdot \frac{y(s)}{X_1(s)} = \frac{20}{s^3+5s^2+8s+2} \cdot (4s+2)$$

$$\frac{X_1(s)}{u(s)} = \frac{20}{s^3+5s^2+8s+2}$$

$$(s^3+5s^2+8s+2)\,X_1(s) = 20\,u(s)$$

Taking inverse laplace

$$\dddot{x}_1 + 5\ddot{x}_1 + 8\dot{x}_1 + 2x_1 = 20u$$

Select

$$x_1 = x_2$$
$$\dot{x}_1 = \dot{x}_2 = x_3$$
$$\ddot{x}_1 = \dot{x}_3 = x_4$$
$$\dddot{x}_1 = \dot{x}_3 = -5x_3 - 8x_2 - 2x_1 + 20u$$

Rewriting

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$x_3 = -5x_3 - 8x_2 - 2x_1 + 20u$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -2 & -8 & -5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 20\end{bmatrix}u$$

$$\frac{y(s)}{X_1(s)} = 4s+2$$

$$y(s) = (4s+2)\,X_1(s)$$
$$y(t) = 4\dot{x}_1 + 2x_1 = 4x_2 + 2x_1$$
$$y(t) = \begin{bmatrix}2 & 4 & 0\end{bmatrix}x(t)$$

Example 8.19. A system is described by the following transfer function

$$G(s) = \frac{s+2}{s^3+3s^2+2s+10}$$

Find the values of A, b and C^T

Solution :

$$\frac{Y(s)}{u(s)} = \frac{s+2}{s^3+3s^2+2s+10}$$

$$\frac{Y(s)}{u(s)} = \frac{X_1(s)}{u(s)} \cdot \frac{y(s)}{X_1(s)} = \frac{1}{s^3+3s^2+2s+10}(s+2)$$

$$\frac{X_1(s)}{u(s)} = \frac{1}{s^3+3s^2+2s+10}$$

or, $(s^3+3s^2+2s+10)\,X_1(s) = u(s)$

Taking inverse laplace

$$\dddot{X}_1 + 3\ddot{X}_1 + 2\dot{X}_1 + 10X_1 = u$$
$$\dddot{X}_1 = u - 3\ddot{X}_1 - 2\dot{X}_1 - 10X_1$$

Select

$$\dot{X}_1 = x_2$$
$$\ddot{X}_1 = \dot{X}_2 = x_3$$
$$\dddot{X}_1 = \dot{X}_3$$

Rewriting

$$\dot{X}_1 = x_2$$
$$\dot{X}_2 = x_3$$
$$\dot{X}_3 = -3x_3 - 2x_2 - 10x_1 + u$$

$$\begin{bmatrix}\dot{X}_1\\ \dot{X}_2\\ \dot{X}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -10 & -2 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u$$

$$A = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -10 & -2 & -3\end{bmatrix} \text{ and } b = \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}$$

$$\frac{y(s)}{X_1(s)} = s+2$$

$$y(s) = (s+2)\,X_1(s)$$
$$= \dot{X}_1 + 2X_1$$
$$y(t) = x_2 + 2x_1$$
$$y(t) = \begin{bmatrix}2 & 1 & 0\end{bmatrix}x(t) \;\therefore\; C^T = \begin{bmatrix}2 & 1 & 0\end{bmatrix}$$

Example 8.20. A single input single output system is given as

$$\dot{x}(t) = \begin{bmatrix}-1 & 0 & 0\\ 0 & -2 & 0\\ 0 & 0 & -3\end{bmatrix}x(t) + \begin{bmatrix}1\\ 1\\ 0\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 0 & 2\end{bmatrix}x(t)$$

Test for controllability and observability. (R.M.L. University Faizabad 2001)

Solution :

$$A = \begin{bmatrix}-1 & 0 & 0\\ 0 & -2 & 0\\ 0 & 0 & -3\end{bmatrix},\; B = \begin{bmatrix}1\\ 1\\ 0\end{bmatrix},\; C = \begin{bmatrix}1 & 0 & 2\end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$$

$$A^2B = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}$$

$$Q = [B : AB : A^2B] = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of the matrix = 2

Hence, the system is uncontrollable

For observability

$$Q' = \left[C^T : A^T C^T : A^{T^2} C^T\right]$$

$$A^T C^T = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ -6 \end{bmatrix}$$

$$A^{T^2} C^T = \begin{bmatrix} 1 \\ 0 \\ 18 \end{bmatrix}$$

$$Q' = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 2 & -6 & 18 \end{bmatrix}$$

Rank of matrix = 2

Hence, system is unobservable

∴ The given system is uncontrollable and unobservable.

Example 8.21. Consider the system shown in fig. 8.9 and investigate whether it is observable or not.

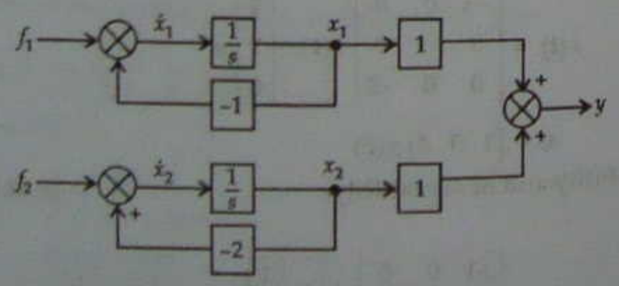


Fig. 8.9.

Solution :

$$\dot{X}_1 = -X_1 + f_1$$
$$\dot{X}_2 = -2X_2 + f_2$$

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$$

$$y = [1 \ 1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

For observability

$$Q' = [C^T : A^T C^T]$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \therefore A^T = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$$

$$C = [1 \ 1] \quad C^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A^T C^T = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

$$Q' = \begin{bmatrix} 1 & -1 \\ 1 & -2 \end{bmatrix}$$

Rank of the matrix = 2

The system is observable.

Example 8.22. A system is described by the matrices

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, C = [1 \ 2 \ 0]$$

Determine the transfer function.

Solution :

$$sI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0 & 2 & s+3 \end{bmatrix}$$

$$[sI - A]^{-1} = \frac{1}{s^3 + 3s^2 + 2s} \begin{bmatrix} s^3 + 3s + 2 & s+3 & 1 \\ 0 & s^2 + 3s & s \\ 0 & -2s & s^2 \end{bmatrix}$$

$$C^T[sI - A]^{-1} = [1 \ 2 \ 0] \begin{bmatrix} s^3 + 3s + 2 & s+3 & 1 \\ 0 & s^2 + 3s & s \\ 0 & -2s & s^2 \end{bmatrix} \frac{1}{s^3 + 3s^2 + 2s}$$

$$C^T[SI - A]^{-1} b = \begin{bmatrix} s^2 + 3s + 2 & 2s^2 + 7s + 3 & 2s+1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \frac{1}{s^3 + 3s^2 + 2s} = \frac{2s+1}{s^3 + 3s^2 + 2s}$$

Hence required transfer function = $\frac{2s+1}{s^3+3s^2+2s}$

Example 8.23. Obtain the transfer function of the system.

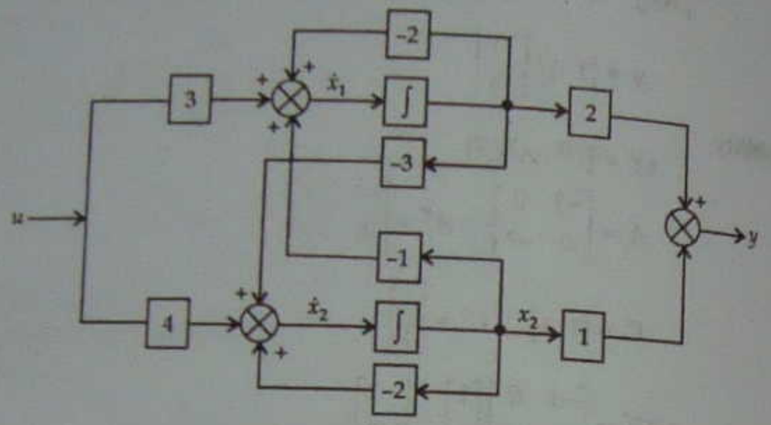


Fig. 8.10.

Solution : From fig. 8.10.

$$\dot{x}_1 = -2x_1 - x_2 + 3u$$
$$\dot{x}_2 = -3x_1 - 2x_2 + 4u$$

Output equation:

$$y = 2x_1 + x_2$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-2 & -1\\ -3 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}3\\ 4\end{bmatrix}u$$

$$y = [2 \;\; 1]\, x(t)$$

$$[sI - A]^{-1} = \frac{1}{s^2+4s+1}\begin{bmatrix}s+2 & -1\\ -3 & s+2\end{bmatrix}$$

$$C^T[sI - A]^{-1} b = \frac{7s+3}{s^2+4s+1}$$

Hence, required transfer function = $\frac{7s+3}{s^2+4s+1}$

Example 8.24. Consider the following matrix

$$\dot{x} = \begin{bmatrix}0 & 1\\ -6 & -5\end{bmatrix} \quad x(0) = \begin{bmatrix}1\\ 0\end{bmatrix}$$

Find the state transition matrix, also determine $x(t)$

Solution :

Given $$A = \begin{bmatrix}0 & 1\\ -6 & -5\end{bmatrix}$$

$$\phi(s) = [sI - A]^{-1} = \frac{1}{s^2+5s+6}\begin{bmatrix}s+5 & 1\\ -6 & s\end{bmatrix}$$

$$\phi(t) = \mathcal{L}^{-1}\phi(s) = \mathcal{L}^{-1}\begin{bmatrix}\frac{s+5}{s^2+5s+6} & \frac{1}{s^2+5s+6}\\ \frac{-6}{s^2+5s+6} & \frac{s}{s^2+5s+6}\end{bmatrix}$$

$$\phi(t) = \begin{bmatrix}3e^{-2t} - 2e^{-3t} & e^{-2t} - e^{-3t}\\ -6(e^{-2t} - e^{-3t}) & 3e^{-3t} - e^{2t}\end{bmatrix}$$

We know that $x(t) = \phi(t)\, X(0)$

$$X(t) = \begin{bmatrix}3e^{-2t} - 2e^{-3t} & e^{-2t} - e^{-3t}\\ -6(e^{-2t} - e^{-3t}) & 3e^{-3t} - e^{-2t}\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix}$$

$$= \begin{bmatrix}3e^{-2t} - 2e^{-3t}\\ -6(e^{-2t} - e^{-3t})\end{bmatrix}$$ **Ans.**

Example 8.25. Write the state equation for the circuit shown in fig 8.11.

Solution : Select the state variables as i_{L_1}, i_{L_2} and V_c

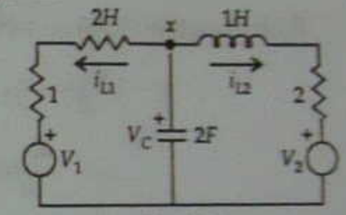


Fig. 8.11.

KVL in mesh (1) :

$$\frac{-2di_{L_1}}{dt} - 1.i_{L_1} - V_1 + V_c = 0$$

$$\frac{di_{L_1}}{dt} = -\frac{1}{2}i_{L_1} - \frac{1}{2}V_1 + \frac{1}{2}V_c$$

KVL in mesh(2) :

$$-1\frac{di_{L_2}}{dt} - 2i_{L_2} - V_2 + V_c = 0$$

$$\frac{di_{L_2}}{dt} = -2i_{L_2} - V_2 + V_c$$

KCL at node x :

$$i_{L_1} + i_{L_2} + 2\frac{dV_c}{dt} = 0$$

$$\frac{dV_c}{dt} = -\frac{1}{2}i_{L_1} - \frac{1}{2}i_{L_2}$$

$$\begin{bmatrix}\dot{i}_{L_1}\\ \dot{i}_{L_2}\\ \dot{V}_c\end{bmatrix} = \begin{bmatrix}-1/2 & 0 & 1/2\\ 0 & -2 & 1\\ -1/2 & -1/2 & 0\end{bmatrix}\begin{bmatrix}i_{L_1}\\ i_{L_2}\\ V_c\end{bmatrix} + \begin{bmatrix}-1/2 & 0\\ 0 & -1\\ 0 & 0\end{bmatrix}\begin{bmatrix}V_1\\ V_2\end{bmatrix}$$

Example 8.26. Write the state equation for the circuit shown in fig. 8.12.

Solution : Select the state variable as V_{c1}, V_{c2} and i_L

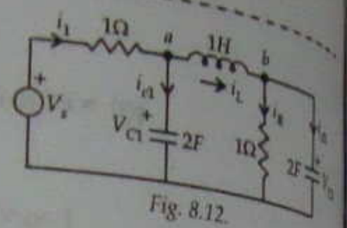


Fig. 8.12.

KVL in mesh (1) :

$$V_s - 1.i_1 - V_{c1} = 0 \qquad ...(8.56)$$

KVL in mesh (2) :

$$V_{C_1} - 1 \cdot \frac{di_L}{dt} - 1.i_R = 0 \qquad ...(8.57)$$

KVL in mesh (3) :

$$1.\, i_R - V_{C2} = 0$$

KCL at node (*a*) :-

$$i_1 = i_{C1} + i_L \qquad ...(8.58)$$

or

$$i_1 = 2\frac{dVc_1}{dt} + i_L \qquad ...(8.59)$$

KCL at node *b* :

$$i_L = i_R + i_{C2}$$

$$i_R = i_L - i_{C2} = i_L - 2\frac{dv_{c2}}{dt} \qquad ...(8.60)$$

From equation (8.58) and (8.60)

$$i_L - 2\frac{dv_{c2}}{dt} - V_{c2} = 0$$

$$\frac{dv_{c_2}}{dt} = \frac{1}{2}i_L - \frac{1}{2}V_{c_2} \qquad ...(8.61)$$

$$i_R = \frac{V_{C_2} - 0}{1} = V_{C_2}$$

From eqn(8.57)

$$V_{C_1} - \frac{di_L}{dt} - V_{C_2} = 0$$

$$\frac{di_L}{dt} = V_{C_1} - V_{C_2} \qquad ...(8.62)$$

Put the value of i_1 form 8.59 in *eq*^{n} 8.56

$$-V_{C_1} - i_L - 2\frac{dV_{C_1}}{dt} + V_s = 0$$

$$\frac{dV_{C_1}}{dt} = \frac{1}{2}V_s - \frac{1}{2}i_L - \frac{1}{2}V_{C_1}$$

$$\begin{bmatrix} \dot{V}_{C_1} \\ \dot{V}_{C_2} \\ \dot{i}_L \end{bmatrix} = \begin{bmatrix} -1/2 & 0 & -1/2 \\ 0 & -1/2 & 1/2 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} V_{C_1} \\ V_{C_2} \\ i_L \end{bmatrix} + \begin{bmatrix} 1/2 \\ 0 \\ 0 \end{bmatrix} [V_s]$$

Example 8.27. Write the state equation for the circuit shown in fig. 8.13.

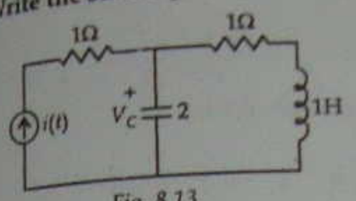


Fig. 8.13.

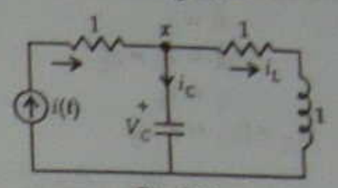


Fig. 8.14.

Solution : Select the state variables as V_C and i_L.

KVL in mesh (1) : $i(t) \cdot 1 - V_C = 0$

KVL in mesh (2) : $-2i_L - 1\frac{di_L}{dt} + V_C = 0$

$$\frac{di_L}{dt} = -2i_L + V_C$$

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KCL at node *x* :

$$i(t) = 2\frac{dV_C}{dt} + i_L$$

$$\frac{dV_C}{dt} = \frac{1}{2}i(t) - \frac{1}{2}i_L$$

$$\begin{bmatrix} \dot{V}_C \\ \dot{i}_L \end{bmatrix} = \begin{bmatrix} 0 & -1/2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} V_C \\ i_L \end{bmatrix} + \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} i(t) \qquad \textbf{Ans.}$$

Example 8.28. Express the system performance by a suitable state space representation of the system shown in fig 8.15.

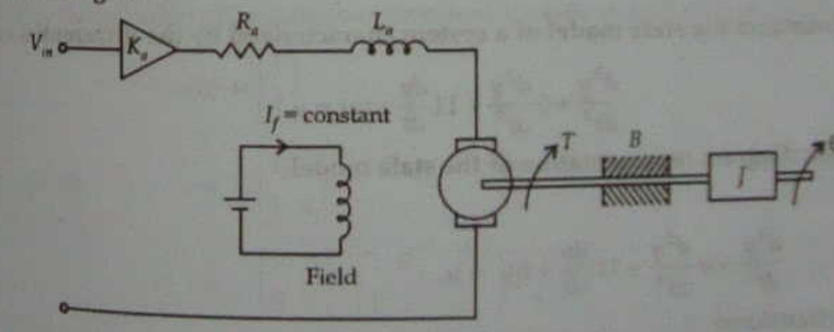


Fig. 8.15. Field control d.c. servometer

Solution : System equations

$$J\ddot{\theta} + \beta\dot{\theta} = K_T i_a \qquad ...(8.63)$$

$$L_a i_a + R_a i_a = V_a + K_V\dot{\theta} \qquad ...(8.64)$$

Where β = Coefficient of viscous friction

K_T = Torque constant of the motor

K_V = Back *e.m.f* constant of the motor

$V_a = K_a V_{in}$

$V_{in} = u$

$V_a = K_a u$

Select state variables as

$$\theta = x_1$$
$$\dot{\theta} = \dot{x}_1 = x_2$$
$$i_a = x_3$$
$$J\dot{x}_2 + \beta x_2 = K_T x_3$$

or,
$$\dot{x}_2 = -\frac{\beta}{J}x_2 + \frac{K_T}{J}x_3 \quad \text{--(8.65)}$$

$$L_a\dot{x}_3 + R_a x_3 = V_a + K_V x_2 \quad \text{--(8.66)}$$
$$\dot{x}_3 = -\frac{R_a}{L_a}x_3 + \frac{K_V}{L_a}x_2 + \frac{K_a}{L_a}u$$

Rewriting

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -\frac{\beta}{J}x_2 + \frac{K_T}{J}x_3$$
$$\dot{x}_3 = -\frac{R_a}{L_a}x_3 + \frac{K_V}{L_a}x_2 + \frac{K_a}{L_a}u$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & -\beta/j & K_T/J\\ 0 & K_V/L_a & -R_a/L_a\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ K_a/L_a\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 0 & 0\end{bmatrix}x$$

Example 8.29. Construct the state model of a system characterized by the differential equation.

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y = u$$

Give the block diagram representation of the state model.

Solution :,

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y = u$$

Take laplace transform

$$s^3Y(s) + 6s^2Y(s) + 11sY(s) + 6Y(s) = u(s)$$
$$(s^3 + 6s^2 + 11s + 6)Y(s) = u(s)$$

Taking inverse laplace transform

$$\dddot{Y}(t) + 6\ddot{Y}(t) + 11\dot{Y}(t) + 6Y(t) = u(t)$$

let
$$Y(t) = x_1$$
$$\dot{Y}(t) = \dot{x}_1 = x_2$$
$$\ddot{Y}(t) = \dot{x}_2 = x_3$$
$$\dddot{Y}(t) = \dot{x}_3$$
$$\dddot{Y}(t) = \dot{x}_3 = 2u(t) - 6x_3 - 11x_2 - 6x_1$$

Rewriting

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = 2u(t) - 6x_1 - 11x_2 - 6x_3$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u(t)$$

$$y(t) = \begin{bmatrix}1 & 0 & 0\end{bmatrix}x(t)$$

Block diag. shown in fig 8.16.

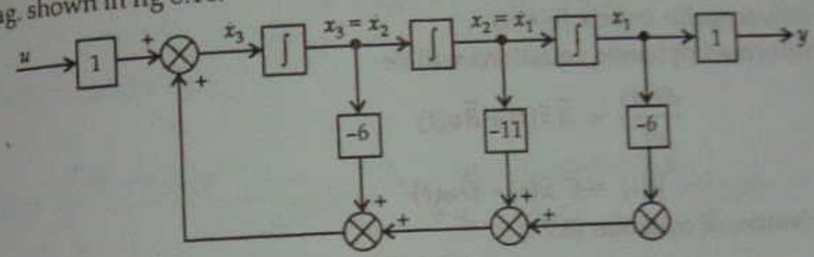


Fig. 8.16.

Example 8.30. A linear time invariant system is characterised by the homogeneous state equation

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Compute the solution of homogeneous equation, assuming the initial state vector

$$x(0) = \begin{bmatrix}1\\ 0\end{bmatrix}$$

Solution : Given
$$A = \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$$

$$sI - A = \begin{bmatrix}s & 0\\ 0 & s\end{bmatrix} - \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix} = \begin{bmatrix}s-1 & 0\\ -1 & s-1\end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix}\frac{1}{s-1} & 0\\ \frac{1}{(s-1)^2} & \frac{1}{s-1}\end{bmatrix}$$

$$\phi(t) = \mathcal{L}^{-1}[sI - A]^{-1} = \begin{bmatrix}e^t & 0\\ te^t & e^t\end{bmatrix}$$

$$x(t) = \phi(t)\,x(0) = \begin{bmatrix}e^t & 0\\ te^t & e^t\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix} = \begin{bmatrix}e^t\\ te^t\end{bmatrix}$$ **Ans.**

8.12. SIMILARITY TRANSFORMATION

Similarity transformation means transformation of one set of dynamic equations to another set of dynamic equations. Consider the dynamic equation of a single input single output system

$$\frac{dx(t)}{dt} = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t) \quad \text{...(8.67)}$$

Where $x(t)$ = state vector $(n \times 1)$ ...(8.68)

$u(t)$ = input vector

$y(t)$ = output vector

Let the dynamic equations (8.67) & (8.68) be transformed into another set of equations.

let $x(t) = P\,\bar{x}(t)$

or, $\bar{x}(t) = P^{-1}x(t)$ ...(8.69)

Where, P = non-singular matrix $(n \times n)$ ...(8.70)

then the transformed dynamic equations will be

$$\frac{d\bar{x}(t)}{dt} = \bar{A}\bar{x}(t) + \bar{B}u(t) \quad \text{...(8.71)}$$

$$\bar{y}(t) = \bar{C}\,\bar{x}(t) + \bar{D}u(t) \quad \text{...(8.72)}$$

Take the derivative of equation (8.70)

$$\frac{d\bar{x}(t)}{dt} = P^{-1}\frac{dx(t)}{dt} \quad \text{...(8.73)}$$

Put the value of $\frac{dx(t)}{dt}$ from equation (8.67) in equation (8.73)

$$\frac{d\bar{x}(t)}{dt} = P^{-1}Ax(t) + P^{-1}B\,u(t) \quad \text{...(8.74)}$$

Put the value of $x(t)$ from (8.69) in equation (8.74)

$$\frac{d\bar{x}(t)}{dt} = P^{-1}A\,Px(t) + P^{-1}B\,u(t) \quad \text{...(8.75)}$$

Compare equation (8.71) with (8.75)

$$\boxed{\bar{A} = P^{-1}AP} \quad \text{and} \quad \boxed{\bar{B} = P^{-1}B}$$

Put the value of $x(t)$ from equation (8.69) in equation (8.68).

$$y(t) = CP\bar{x}(t) + Du(t) \quad \text{...(8.76)}$$

compare equation (8.76) with equation (8.72)

$$\boxed{\bar{C} = CP} \quad \text{and} \quad \boxed{\bar{D} = D}$$

By transformation, there is no change in characteristic equation, eigen values and transfer function.

8.13. STATE SPACE REPRESENTATION OF TRANSFER FUNCTION SYSTEMS :

8 13.0. State Space Representation in Canonical Form

Consider the transfer function

$$\frac{Y(s)}{u(s)} = \frac{b_o s^n + b_1 s^{n-1} + \ldots\ldots + b_{n-1}s + bn}{s^n + a_1 s^{n-1} + \ldots\ldots + a_{n-1}s + a_n} \quad \text{... (8.77)}$$

8.13.0.1. Controllable Canonical Form

Equation (8.77) can be written as

$$\frac{Y(s)}{u(s)} = b_0 + \frac{(b_1 - a_1b_0)s^{n-1} + \ldots\ldots + (b_{n-1} - a_{n-1}b_0)s + (b_n - a_nb_0)}{s^n + a_1s^{n-1} + \ldots\ldots + a_{n-1}s + a_n} \quad \text{...(8.78)}$$

Multiply both sides by $u(s)$

$$Y(s) = b_o u(s) + \left[\frac{(b_1 - a_1b_o)s^{n-1} + \ldots\ldots + (b_{n-1} - a_{n-1}b_o)s + (b_n - a_nb_o)}{s^n + a_1s^{n-1} + \ldots\ldots a_{n-1}s + a_n}\right]u(s) \quad \text{...(8.79)}$$

or, $Y(s) = b_o u(s) + \dot{Y}(s)$ ...(8.80)

where

$$\dot{Y}(s) = \left[\frac{(b_1 - a_1b_0)s^{n-1} + \ldots\ldots + (b_{n-1} - a_{n-1}b_0)s + (b_n - a_nb_o)}{s^n + a_1s^{n-1} + \ldots\ldots + a_{n-1}\,s + a_n}\right]u(s) \quad \text{...(8.81)}$$

Equation (8.81) can be written as

$$= \frac{\dot{Y}(s)}{(b_1 - a_1b_0)s^{n-1} + \ldots\ldots + (b_{n-1} - a_{n-1}b_0) + (b_n - a_nb_0)}$$

$$= \frac{u(s)}{s^n + a_1s^{n-1} + \ldots\ldots + a_{n-1}s + a_n} = Q(s) \quad \text{...(8.82)}$$

Where $Q(s)$ is a dummy variable

Equate L.H.S of equation (8.82) to $Q(s)$

$$\dot{Y}(s) = Q(s)\,[(b_1 - a_1b_0)\,s^{n-1} + \text{- - - - +}(b_{n-1} - a_{n-1}\,b_0) + (b_n - a_nb_0)]$$

or $\dot{Y}(s) = (b_1 - a_1b_0)\,S^{n-1}\,Q(s) + \text{- - - - +}(b_{n-1} - a_{n-1}\,b_0)\,Q(s) + (b_n - a_nb_o)\,Q(s)$ ...(8.83)

Equate R.H.S of equation (8.82) to $Q(s)$

$$u(s) = Q(s)[s^n + a_1s^{n-1} + \text{- - - -} + a_{n-1}\,s + a_n]$$

or, $u(s) = s^n\,Q(s) + a_1s^{n-1}\,Q(s) + \text{- - - - +}a_{n-1}\,s\,Q(s) + a_n\,Q(s))$

or, $S^nQ(s) = u(s) + [-a_1s^{n-1}\,Q(s) + \text{- - - - -}\,a_{n-1}\,s\,Q(s) - a_n\,Q(s)]$ ...(8.84)

Select the state variables

$$\left.\begin{aligned} X_1(s) &= Q(s) \\ X_2(s) &= sQ(s) \\ X_3(s) &= s^2Q(s) \\ &\vdots \\ X_{n-1}(s) &= s^{n-2}Q(s) \\ X_n(s) &= s^{n-1}Q(s) \end{aligned}\right\} \quad \text{...(8.85)}$$

Multiplying both sides of first equation of equation (8.85) by s

$$s\,X_1(s) = sQ(s) = X_2(s)$$

i.e $sX_1(s) = X_2(s)$

similarly $s^2X_1(s) = s^2Q(s) = X_3(s)$

i.e.

$$\left.\begin{aligned} sX_1(s) &= X_2(s) \\ sX_2(s) &= X_3(s) \\ &\vdots \quad \vdots \\ sX_{n-1}(s) &= x_n(s) \end{aligned}\right] \quad \text{...(8.86)}$$

Equation (8.86) can be written as

$$\left.\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \end{aligned}\right] \quad \text{...(8.87)}$$

Also, $S^n\,Q(s) = s\,X_n(s)$

Equation (8.84) can be written as (with the help of equation 8.85)

$$s\,X_n(s) = u(s) + [-a_1\,X_n(s) \ldots\ldots\ldots - a_{n-1}\,X_2(s) - a_n\,X_1(s)]$$

or, $\dot{x}_n = u(s) + [-a_1x_1 \ldots\ldots - a_{n-1}\,x_2 \ldots\ldots - a_1\,x_n]$

$$\dot{x}_n = -a_nx_1 \ldots\ldots - a_{n-1}\,x_2 \ldots\ldots a_1x_n + 4 \quad \text{...(8.88)}$$

Equation (8.87) & (8.88) can be written in matrix form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u \quad \text{...(8.89)}$$

From equation (8.80) and (8.83)

$$Y(s) = b_o\,U(s) + (b_1 - a_1b_o)\,s^{n-1}\,Q(s) + \ldots. + (b_{n-1} - a_{n-1}\,b_o)\,sQ(s) + (b_n - a_nb_o)\,Q(s)$$

from equation (8.85)

$$Y(s) = b_o\,u(s) + (b_1 - a_1b_o)\,x_n + \ldots + (b_{n-1} - a_{n-1}\,b_o)\,X_2(s) + (b_n - a_nb_o)\,X_1(s)$$

Taking inverse laplace

$$y = b_ou + (b_1 - a_1b_o)x_n + \ldots. + (b_{n-1} - a_{n-1}\,b_o)\,x_2 + (b_n - a_nb_o)\,x_1$$

or,

$$y = \begin{bmatrix} (b_n - a_nb_o), (b_{n-1} - a_{n-1}b_o), \ldots., (b_1 - a_1b_o) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_ou \quad \text{...(8.90)}$$

Equation (8.89) and (8.90) are said to be in the controllable canonical form (C

Now compare the equation (8.89) with (8.71)

$$\overline{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} = P^{-1}AP \quad \text{...(8.91)}$$

$$\overline{B} = P^{-1}B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \quad \text{...(8.92)}$$

where $P = SM$ ...(8.93)

and $S = \left[B\;AB\;A^2B \ldots A^{n-1}B\right]$ ...(8.94)

$$M = \begin{bmatrix} a_{n-1} & a_{n-2} & \cdots & a_1 & 1 \\ a_{n-2} & \vdots & & 1 & \vdots \\ \vdots & \vdots & & \vdots & \vdots \\ a_1 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix} \quad \text{...(8.95)}$$

∵ The characteristic equation is

$$|sI - A| = s^n + a_1s^{n-1} + \ldots. + a_{n-1}\,s + a_n$$

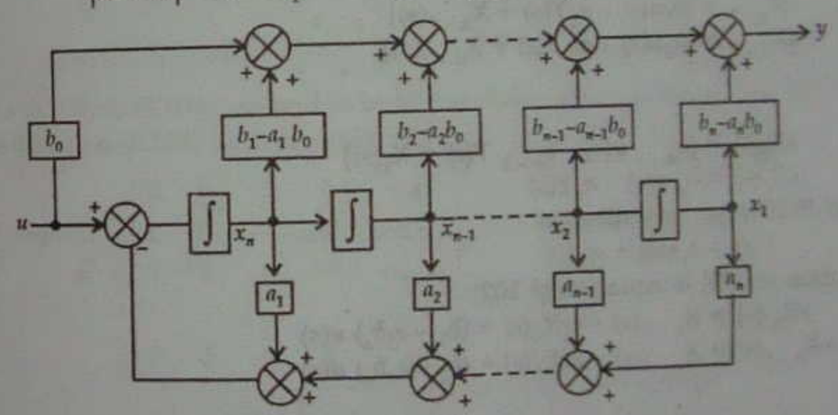


Fig. 8.17.

8.13.0.2. Observable Canonical Form

Consider the transfer function given in equation (8.77) Equation (8.77) can be written as

$$Y(s)\,[s^n + a_1s^{n-1} + \ldots. a_2s^{n-2} + \ldots. a_{n-1}\,s + a_n]$$
$$= u(s)\,[b_os^n + b_1s^{n-1} + b_2s^{n-2} + \ldots. + b_{n-1}\,s + b_n] \quad \text{...(8.96)}$$

or, $s^n\,Y(s) + a_1s^{n-1}\,Y(s) + a_2s^{n-2}\,Y(s) + \ldots. a_{n-1}\,sY(s) + a_nY(s)$

$$= b_os^nu(s) + b_1s^{n-1}\,u(s) + b_2s^{n-2}\,u(s) + \ldots b_{n-1}\,su(s) + b_nu(s) \quad \text{...(8.97)}$$

Rearrange equation (8.97) by transferring RHS to LHS

$$s^nY(s) - b_os^nu(s) + a_1s^{n-1}\,Y(s) - b_1s^{n-1}\,u(s) + a_2s^{n-2}\,Y(s) - b_2s^{n-2}\,u(s) + \ldots a_{n-1}\,sY(s) - b_{n-1}\,su(s) + a_nY(s) - b_nu(s) = 0$$

or, $s^n [Y(s) - b_o u(s)] + s^{n-1} [a_1 Y(s) - b_1 u(s)] + s^{n-2} [a_2 Y(s) - b_2 u(s)] + \ldots + s[a_{n-1} Y(s) - b_{n-1} u(s)] + a_n Y(s) - b_n u(s) = 0$

Divide the equation (8.98) by s_n

$$[Y(s) - b_o u(s)] + s^{-1} [a_1 Y(s) - b_1 u(s)] + s^{-2} [a_2 Y(s) - b_2 u(s)] + \ldots + s^{1-n} [a_{n-1} Y(s) - b_{n-1} u(s)] + a_n s^{-n} Y(s) - b_n s^{-n} u(s) = 0 \quad \ldots(8.99)$$

Rearrange equation (8.99)

$$Y(s) = b_o u(s) + \frac{1}{s}[b_1 u(s) - a_1 Y(s)] + \frac{1}{s^2}[b_2 u(s) - a_2 Y(s)] + \ldots\ldots + \frac{1}{s^{n-1}}[b_{n-1} Y(s) - a_{n-1} Y(s)] + \frac{1}{s^n}[b_n u(s) - a_n Y(s)] = 0 \quad \ldots(8.100)$$

Select the state variable as

$$\left.\begin{aligned} X_n(s) &= \frac{1}{s}[b_1 u(s) - a_1 Y(s) + X_{n-1}(s)] \\ X_{n-1}(s) &= \frac{1}{s}[b_2 u(s) - a_2 Y(s) + X_{n-2}(s)] \\ &\vdots \\ X_2(s) &= \frac{1}{s}[b_{n-1} u(s) - a_{n-1} Y(s) + X_1(s)] \\ X_1(s) &= \frac{1}{s}[b_n u(s) - a_n Y(s)] \end{aligned}\right\} \quad \ldots(8.101)$$

Equation (8.101) can be written as

$$sX_{n-1} = [b_1 u(s) - a_1 Y(s) + X_{n-1}(s)]$$
$$sX_{n-2} = [b_2 u(s) - a_2 Y(s) + X_{n-2}(s)]$$
$$\vdots$$
$$sX_2(s) = [b_{n-1} u(s) - a_{n-1} Y(s) + X_1(s)]$$
$$sX_1(s) = b_n u(s) - a_n Y(s)$$

Equation (8.100) can be written as

$$Y(s) = b_o u(s) + X_n(s) \quad \ldots(8.103)$$

Put equation (8.103) in equation (8.102)

$$sX_n(s) = X_{n-1}(s) - a_1 X_n(s) + (b_1 - a_1 b_o) u(s)$$
$$sX_{n-1}(s) = X_{n-2}(s) - a_2 X_n(s) + (b_2 - a_2 b_o) u(s)$$
$$\vdots$$
$$sX_2 = X_1(s) - a_{n-1} X_n(s) + (b_{n-1} - a_{n-1} b_o) u(s)$$
$$sX_1 = -a_n X_n + (b_n - a_n b_o) u(s)$$

Taking inverse laplace of equation (8.104)

$$\dot{x}_n = x_{n-1} - a_1 x_n + (b_1 - a_1 b_0) u$$
$$\dot{x}_{n-1} = x_{n-2} - a_2 x_n + (b_2 - a_2 b_0) u$$
$$\vdots \quad \ldots(8.105)$$
$$\dot{x}_2 = x_1 - a_{n-1} x_n + (b_{n-1} - a_{n-1} b_o) u$$
$$\dot{x}_1 = -a_n x_n + (b_n - a_n b_o) u$$

Rearrange the equation (8.105)

$$\dot{x}_1 = -a_n x_n + (b_n - a_n b_o) u$$
$$\dot{x}_2 = x_1 - a_{n-1} x_n + (b_{n-1} - a_{n-1} b_o) u$$
$$\vdots$$
$$\dot{x}_{n-1} = x_{n-2} a_2 x_n + (b_2 - a_2 b_o) u \quad \ldots(8.106)$$
$$\dot{x}_n = x_{n-1} - a_1 x_n + (b_1 - a_1 b_o) u$$

Inverse laplace of equation (8.103)

$$y = b_o u(s) + x_n$$
$$y = x_n + b_o u(s) \quad \ldots(8.107)$$

equation (8.106) and (8.107) can be written in matrix form as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ 0 & 1 & \cdots & 0 & \vdots \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_n - a_n b_o \\ b_{n-1} - a_{n-1} b_o \\ \vdots \\ b_1 - a_1 b_0 \end{bmatrix} u \quad \ldots(8.108)$$

$$y = [0\ 0\ \cdots\ 0\ 1] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + b_o u \quad \ldots(8.109)$$

Equations (8.108) & (8.109) are said to be in the observable canonical form (OCF)

Compare equation (8.108) with equation (8.71)

$$\overline{A} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ 0 & 1 & \cdots & \cdots & -a_{n-2} \\ \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 1 & a_n \end{bmatrix} \quad \ldots(8.110)$$

$$\overline{B} = \begin{bmatrix} b_n - a_n b_o \\ b_{n-1} - a_{n-1} b_0 \\ \vdots \\ b_1 - a_1 b_o \end{bmatrix} \quad \ldots(8.111)$$

$$\overline{C} = [0\ 0 \ldots 0\ 1] \quad \ldots(8.112)$$

Here, we use Q instead of P as used in controllable canonical form.

$$\overline{A} = Q^{-1} A Q \quad \ldots(8.113)$$

$$\overline{C} = CQ \quad \ldots(8.114)$$

where $Q = [MV]^{-1}$ $\ldots(8.115)$

The value of M wil be same as given in equation (8.95) & V is

$$V = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \end{bmatrix}$$

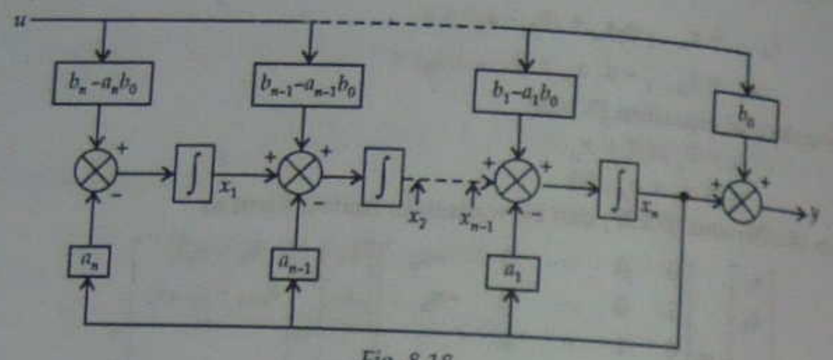


Fig. 8.18.

8.13.0.3. Diagonal Canonical Form

Let the equation (8.77) is written as

$$\frac{Y(s)}{u(s)} = b_0 + \frac{d_1}{s+\lambda_1} + \frac{d_2}{s+\lambda_2} + \ldots + \frac{d_n}{s+\lambda_n} \quad \ldots(8.117.0)$$

or,

$$Y(s) = b_0 u(s) + \frac{d_1}{s+\lambda_1} u(s) + \frac{d_2}{s+\lambda_2} u(s) + \ldots + \frac{d_n}{s+\lambda_n} u(s) \quad \ldots(8.117.1)$$

Define the state variable as

$$X_1(s) = \frac{1}{s+\lambda_1} u(s)$$

$$X_2(s) = \frac{1}{s+\lambda_2} u(s) \quad \ldots(8.117.2)$$

$$\vdots$$

$$X_n(s) = \frac{1}{s+\lambda_n} u(s)$$

Above equations can be written as

$$sX_1(s) = -\lambda_1 X_1(s) + u(s) \qquad e.g.\ X_1(s)[s+\lambda_1] = u(s)$$
$$sX_2(s) = -\lambda_2 X_2(s) + u(s) \qquad sX_1(s) + X_1(s)\lambda_1 = u(s)$$
$$\vdots \qquad \text{or}\quad sX_1(s) = -\lambda_1 X_1(s) + u(s)$$
$$sX_n(s) = -\lambda_n X_n(s) + u(s) \quad \ldots(8.117.3)$$

The inverse laplace of (8.117.3)

$$\dot{x}_1 = -\lambda_1 x_1 + u$$
$$\dot{x}_2 = -\lambda_2 x_2 + u \quad \ldots(8.117.4)$$
$$\vdots$$
$$\dot{x}_n = -\lambda_n x_n + u$$

put the equation (8.117.2) in (8.117.1)

$$Y(s) = b_0 u(s) + d_1 X_1(s) + d_2 X_2(s) + \ldots + d_n X_n(s) \quad \ldots(8.117.5)$$

Inverse laplace of equation (8.117.5)

$$y = b_0 u + d_1 x_1 + d_2 x_2 + \ldots + d_n x_n \quad \ldots(8.117.6)$$

equation (8.117.6) is the output equation.

The equations (8.117.4) & (8.117.6) can be written in matrix form as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -\lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & -\lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & -\lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \\ 0 & 0 & 0 & & -\lambda_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u \quad \ldots(8.117.7)$$

$$y = \begin{bmatrix} d_1 & d_2 & d_3 & \ldots d_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + b_0 u \quad \ldots(8.117.8)$$

Equations (8.117.7) & (8.177.8) are said to be in diagonal canonical form.

[*Note*: $\lambda_1, \lambda_2, \ldots$ roots of characteristic equation or eigen value of matrix A.

If = only 1, 1, 1, ... no other values (means it is controllable)

If any of $d_1, d_2, \ldots$ is absent then system is unobservable]

We can also select the state variables as

$$X_1(s) = \frac{d_1}{s+\lambda_1} u(s)$$

$$X_2(s) = \frac{d_2}{s+\lambda_2} u(s) \quad \ldots(8.117.9)$$

$$\vdots$$

$$Xn(s) = \frac{d_n}{s+\lambda_n} u(s)$$

equation (8.117.9) can be written as

$$sX_1(s) = -\lambda_1 X_1(s) + d_1 u(s)$$
$$sX_2(s) = -\lambda_2 X_2(s) + d_2 u(s) \quad \ldots(8.117.10)$$
$$\vdots$$
$$sX_n(s) = -\lambda_n X_n(s) + d_n u(s)$$

Laplace transform of (8.117.10)

$$\dot{x}_1 = -\lambda x_1 + d_1 u$$
$$\dot{x}_2 = -\lambda_2 x_2 + d_2 u \quad \ldots(8.117.11)$$
$$\vdots$$
$$\dot{x}_n = -\lambda_n x_n + d_n u$$

from equation (8.117.1)

$$Y(s) = b_0 u(s) + X_1(s) + X_2(s) + X_3(s) + \ldots X_n \quad \ldots(8.177.12)$$

Inverse laplace of (8.117.12)

$$y = b_o u + x_1 + x_2 + x_3 + ... x_n$$

Equation (8.117.12) & (8.117.13) can be written in matrix form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -\lambda_1 & \cdots & 0 & \cdots & 0 \\ 0 & -\lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & -\lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & & -\lambda_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{bmatrix} u \quad ...(8.117.13)$$

...(8.117.14)

$$y = [1\ 1\ 1\ \ldots\ 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + b_o u \quad ...(8.117.15)$$

Equation (8.117.14) & (8.117.15) are also called diagonal canonical form. This is an alternative approach.

let $x(t) = T\bar{x}(t)$

or, $\bar{x}(t) = T^{-1}x(t)$ ...(8.177.16)

derivative of equation (8.117.17) ...(8.117.17)

$$\frac{d\bar{x}(t)}{dt} = T^{-1}\frac{dx(t)}{dt} \quad ...(8.117.18)$$

Put the value of $\frac{dx(t)}{dt}$ from equation (8.67)

$$\frac{d\bar{x}(t)}{dt} = T^{-1}Ax(t) + T^{-1}B\,u(t)$$

or, $$\frac{d\bar{x}(t)}{dt} = T^{-1}AT\,\bar{x}(t) + T^{-1}B\,u(t) \quad ...(8.117.19)$$

compare equation (8.177.19) with (8.71)

$$\boxed{\bar{A} = T^{-1}AT}, \quad \boxed{\bar{B} = T^{-1}B} \quad ...(8.177.20)$$

from equation (8.68) and (8.72)

$$y(t) = CT\,\bar{x}(t) + Du(t) \quad ...(8.117.21)$$

Compare equation (8.117.21) with (8.72)

$$\boxed{\bar{C} = CT} \text{ and } \boxed{\bar{D} = D}$$

T matrix can be formed by the use of eigen vectors of A.

$$T = [p_1 \quad p_2 \quad p_3 \quad \cdots \quad p_n]$$

If the matrix A is of CCF and A has distinct eigen values, the DCF transformation matrix is the vandermonde matrix, then

$$T = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_n^2 \\ \vdots & & & \\ \lambda^{n-1} & \lambda_2^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix} \quad ...(8.177.22)$$

where, $\lambda_1, \lambda_2, \lambda_3 \ldots \lambda_n$ are the eigen values of A.

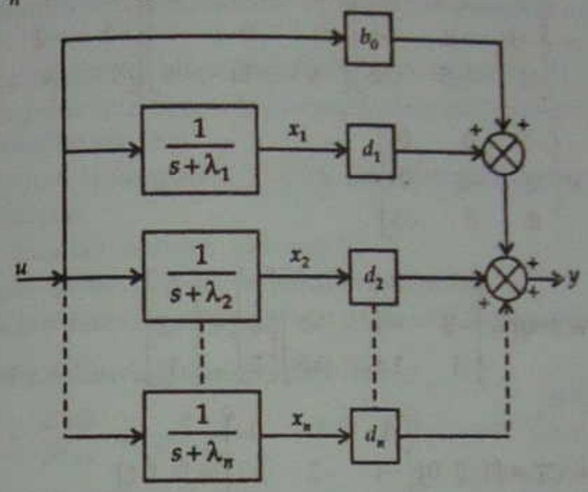


Fig. 8.19.

Example 8.31. Consider the state space representation given in example 8.16, change it into diagonal canonical form (DCF).

Solution : Compare state equation (8.67) & (8.68)

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \quad C = [1\ 0\ 0]$$

First calculate eigen vector of A.

$$|\lambda I - A| = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} = \lambda^3 + 6\lambda^2 + 11\lambda + 6$$

or $$|\lambda I - A| = \lambda^3 + 6\lambda^2 + 11\lambda + 6 = 0$$

$$(\lambda + 1)(\lambda + 2)(\lambda + 3) = 0$$

$$\lambda_1 = -1, \lambda_2 = -2, \lambda_3 = -3$$

Now, three eigenvalues are distinct, then we calculate $\bar{A}, \bar{B}$ by the use of T, where

$$T = \begin{bmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$T^{-1} = \begin{bmatrix} 3 & 2.5 & 0.5 \\ -3 & -4 & -1 \\ 1 & 1.5 & 0.5 \end{bmatrix}$$

Note : students are advised to calculate T^{-1} themselves

$$\overline{A} = T^{-1}AT$$

$$= \begin{bmatrix} 3 & 2.5 & 0.5 \\ -3 & -4 & -1 \\ 1 & 1.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$\overline{B} = T^{-1}B = \begin{bmatrix} 3 & 2.5 & 0.5 \\ -3 & -4 & -1 \\ 1 & 1.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$\overline{C} = CT = [1\ 0\ 0] \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix} = [1\ 1\ 1]$$

Now the state equation can be written in transform form (DCF) as with the help of equation (8.117.19) & (8.117.21)

$$\begin{bmatrix} \dot{\overline{x}}_1 \\ \dot{\overline{x}}_2 \\ \dot{\overline{x}}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} \overline{x}_1 \\ \overline{x}_2 \\ \overline{x}_3 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} u \qquad \ldots(8.117.23)$$

$$y = [1\ 1\ 1] \begin{bmatrix} \overline{x}_1 \\ \overline{x}_2 \\ \overline{x}_3 \end{bmatrix} \qquad \ldots(8.117.24)$$

From equations (8.117.23) & (8.117.24) it is clear that transformation matrix T modifies the coefficient matrix into diagonal matrix. The diagonal elements of equation (8.117.23) are same as eigenvalues of A.

8.14. DECOMPOSITION TRANSFER FUNCTION

The process by which transfer function changes to state diagram or state equations is called decomposition of the transfer function. The transfer function can be decmposed by three methods known as direct decomposition, cascade decomposition and parallel decomposition.

8.14.1. Direct Decomposition

8.14.1.1 Direct decomposition to controllable canonical form (CCF)

Direct decomposition method is used when the transfer function of a system is not in factored form

let the transfer function is

$$\frac{C(s)}{R(s)} = \frac{b_o s^n + b_1 s^{n-1} + \ldots + b_{n-1}s + b_n}{s^n + a_1 s^{n-1} + \ldots + a_{n-1}s + a_n} \qquad \ldots(8.118)$$

The following steps are involved

Step 1 : Express the numerator & denominator in the form of rational polynomials.

Step 2 : Divide the numerator and denominator by the higest power of s in the denominator or the numerator.

Step 3 : Multipiy the numerator and denominator by $Q(s)$, where $Q(s)$ is a dummy variable.

Step 4 : Express the numerator and denominator separately and use the denominator equation to write an expression for $Q(s)$.

Step 5 : Draw the state variable diagram (SVD) for $Q(s)$ obtained in step 4.

Step 6 : Draw SVD for output.

Step 7 : Combine SVD obtained in step 5 and step 6.

Step 8 : Write expressions for the derivatives of state variables.

Step 9 : Feedback paths are determined by system poles and forward paths by system zero.

Example 8.32. Consider the following transfer function

$$\frac{C(s)}{R(s)} = \frac{s+6}{s^2+5s+6}$$

obtain the state space representation.

Solution :

$$\frac{C(s)}{R(s)} = \frac{s+6}{s^2+5s+6} \qquad \ldots(8.119(a))$$

Step 1 : Divide numerator and denominator by s^2

$$\frac{C(s)}{R(s)} = \frac{s^{-1}+6s^{-2}}{1+5s^{-1}+6s^{-2}}$$

Step 2 : Multiply numerator and denominator by $Q(s)$

$$\frac{C(s)}{R(s)} = \frac{s^{-1}+6s^{-2}}{1+5s^{-1}+6s^{-2}} \cdot \frac{Q(s)}{Q(s)} \qquad \ldots(8.119(b))$$

Step 3 : Express numerator and denominator separately.

$$C(s) = Q(s)\ [s^{-1} + 6s^{-2}] \qquad \ldots(8.119(c))$$

$$R(s) = Q(s)\ [1 + 5s^{-1} + 6s^{-2}] \qquad \ldots(8.119(c))$$

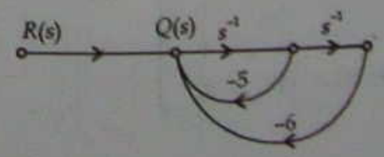
Equation (8.32.3) can be written as

$$R(s) = Q(s) + Q(s)\ [5s^{-1} + 6s^{-2}] \qquad \ldots(8.119(d))$$

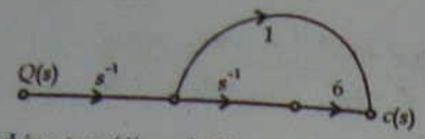
or,

$$Q(s) = R(s) - Q(s)\ [5s^{-1} + 6s^{-2}] \qquad \ldots(8.119(e))$$

Step 4 : Draw SVD for equation (8.119(e))



Step 5 : Draw SVD for equation (8.119(*c*)).



Step 6 : Combine SVD obtained in step (4) and (5)

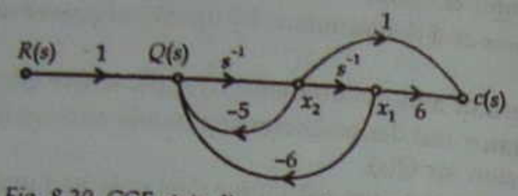


Fig. 8.20. CCF state diagram of transfer function

From diagram

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = r(t) - 6x_1 - 5x_2$$

Therefore

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -6 & -5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix} r(t) \qquad ...(8.119(f))$$

$$c(t) = 6x_1 + x_2$$

$$c(t) = [6\ \ 1]\begin{bmatrix}x_1\\ x_2\end{bmatrix} \qquad ...(8.119(g))$$

Equations (8.119(*f*)) & (8.119(*g*)) are required state space representation.
The above equations are controllable canonical form (CCF).

8.14.1.2. Direct Decomposition to Observable Canonical Form (OCF)
Multiply numerator and denominator by s^{-n} of equation (8.77)

$$\frac{Y(s)}{U(s)} = \frac{b_o + b_1 s^{-1} + ... + b_{n-1}s^{1-n} + b_n s^{-n}}{1 + a_1 s^{-1} + ... + a_{n-1}s^{1-n} + a_n s^{-n}}$$

$$(1 + a_1 s^{-1} + ... + a_{n-1}\, s^{1-n} + a_n s^{-n})\ Y(s) = (b_o + b_1 s^{-1} + ... + b_{n-1}\, s^{1-n} + b_n s^{-n})\ u(s)$$

or,
$$Y(s) = (b_o + b_1 s^{-1} + ... b_{n-1} s^{1-n} + b_n s^{-n})\ u(s) - Y(s)\ [a_1 s^{-1} + ... + a_{n-1}\, s^{1-n} + a_n s^{-n}] \qquad ...(8.120)$$

The output of integrators as designated as the state variables. The state variables are assigned in descending order from right to left. Apply SFG gain formula to state variable diagram and write the dynamic equation.

$$\overline{A} = \begin{bmatrix}0 & 0 & \cdots & 0 & -a_n\\ 1 & 0 & \cdots & 0 & -a_{n-1}\\ 0 & 1 & & \vdots & \vdots\\ \vdots & \vdots & & \vdots & \vdots\\ 0 & 0 & \cdots & 1 & a_n\end{bmatrix}$$

$$\overline{B} = \begin{bmatrix}b_n - a_n b_0\\ b_{n-1} - a_{n-1} b_0 -\\ \vdots\\ b_1 - a_1 b_0\end{bmatrix}, \quad \overline{C} = [0\ \ 0\ \ 0\ \ ...\ \ 1]$$

Consider the following example

Example 8.33. Obtain the state model of the system whose transfer function is given by

$$T(s) = \frac{s^2 + 3s + 3}{s^3 + 2s^2 + 3s + 1} \qquad ...(8.121)$$

(UPTU 2003-2004)

Solution :
$$\frac{Y(s)}{U(s)} = \frac{s^2 + 3s + 3}{s^3 + 2s^2 + 3s + 1}$$

divide numerator and denominator by $s^n = s^3$

$$\frac{Y(s)}{U(s)} = \frac{s^{-1} + 3s^{-2} + 3s^{-3}}{1 + 2s^{-1} + 3s^{-2} + s^{-3}}$$

or, $(s^{-1} + 3s^{-2} + 3s^{-3})\ u(s) = (1 + 2s^{-1} + 3s^{-2} + s^{-3})\ Y(s)$

or, $Y(s) = (s^{-1} + 3s^{-2} + 3s^{-3})\ u(s) - (2s^{-1} + 3s^{-2} + s^{-3})\ Y(s) \qquad ...(8.122)$

with the help of equation (8.122) draw state variable diagram (SVD)

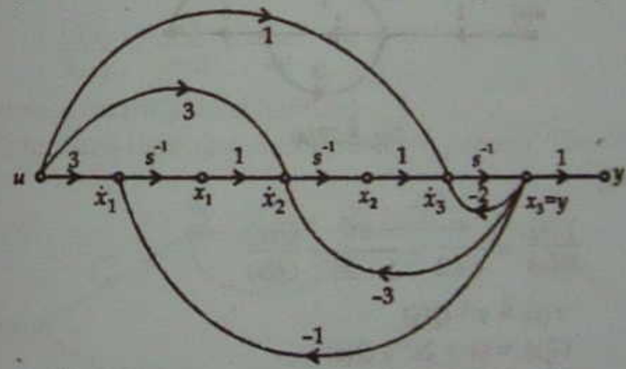


Fig. 8.21. OCF state diagram of the transfer function.

From OCF diagram shown in fig 8.22.

$$y = x_3$$

$$\dot{x}_3 = u + x_2 - 2x_3 = Ox_1 + x_2 - 2x_3 + u$$

$$\dot{x}_2 = 3u - 3x_3 + x_1 = 1x_1 + Ox_2 - 3x_3 + 3u$$

$$\dot{x}_1 = 3u + x_3 = Ox_1 + Ox_2 + x_3 + 3u$$

state equations are

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 0 & 1\\ 1 & 0 & -3\\ 0 & 1 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}3\\ 3\\ 1\end{bmatrix} u \qquad y = [0\ \ 0\ \ 1]\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

The above equations are required answer.

8.14.2. Cascade Decomposition

When the transfer function is in factored form then cascade decomposition is used. First make the transfer function in factored form, then draw the transfer function of each part and combined all part by unity transmittances inserted between each SVD. Consider the following example.

Example 8.34. Consider the transfer function given in example 8.32 draw SVD and obtain state space model by series (cascade) decomposition.

Solution : The given transfer function can be written in factored form

$$\frac{Y(s)}{U(s)} = \frac{s+6}{(s+3)(s+2)}$$

let, $G_1(s) = \frac{s+6}{s+3}$ and $G_2 = \frac{1}{s+2}$

First draw SVD for G_1

$$\frac{Y(s)}{U(s)} = \frac{1+6s^{-1}}{1+3s^{-1}} \cdot \frac{Q(s)}{Q(s)}$$

$$Y(s) = (1 + 6s^{-1})\, Q(s)$$
$$U(s) = (1 + 3s^{-1})\, Q(s)$$

or, $$Q(s) = u(s) - 3s^{-1}\, Q(s)$$

SVD for G_1

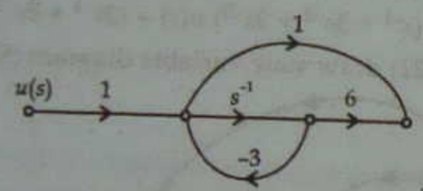


Fig. 8.22(a)

Now draw SVD for G_2

$$\frac{Y(s)}{U(s)} = \frac{1}{s+2} = \frac{s^{-1}}{1+2s^{-1}} \frac{Q(s)}{Q(s)}$$

$$Y(s) = s^{-1}\, Q(s)$$
$$U(s) = (1 + 2s^{-1})\, Q(s)$$

or, $$Q(s) = U(s) - 2s^{-1}\, Q(s)$$

Draw SVD for G_2

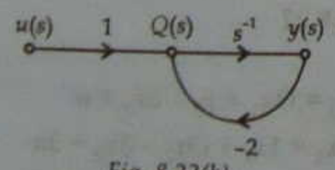


Fig. 8.22(b).

Combine SVD for G_1 & G_2 with unity transmittance, we get complete SVD

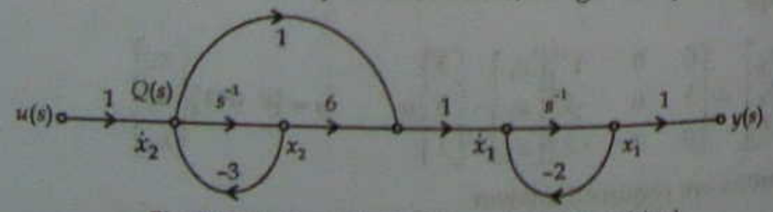


Fig. 8.22(c). Complete SVD cascode decomposition)

From SVD :

$$\dot{x}_1 = -2x_1 + \dot{x}_2 + 6x_2 \quad \text{...(8.124)}$$

$$\dot{x}_2 = -3x_2 + u \quad \text{...(8.125)}$$

Put the value of $\dot{x}_2$ from (8.125) in (8.124), and simplified

$$\dot{x}_1 = -2x_1 + 3x_2 + u$$

$$\dot{x}_2 = -3x_2 + u$$

In matrix form $$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-2 & 3\\ 0 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 1\end{bmatrix}u \qquad y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

The above equations are the required state space equations.

8.14.3. Parallel Composition

This method can be used when denominator of transfer function is in factrored form and numerator may or may not be factored form. Transfer function is expanded by partial fraction and draw SVD for each part, connect them in parallel.

Example 8.35. Obtain the state space representation of the transfer function given in example 8.32.

Solution :

$$\frac{C(s)}{R(s)} = \frac{s+6}{(s+3)(s+2)} \quad \text{...(8.126)}$$

After partial fraction equation (8.126) can be written as

$$\frac{C(s)}{R(s)} = \frac{-3}{s+3} + \frac{4}{s+2} \quad \text{...(8.127)}$$

(students are advised do the partial fraction)

Now draw SVD for each term and then connect in parallel. The resultant SVD is shown in figure 8.23.

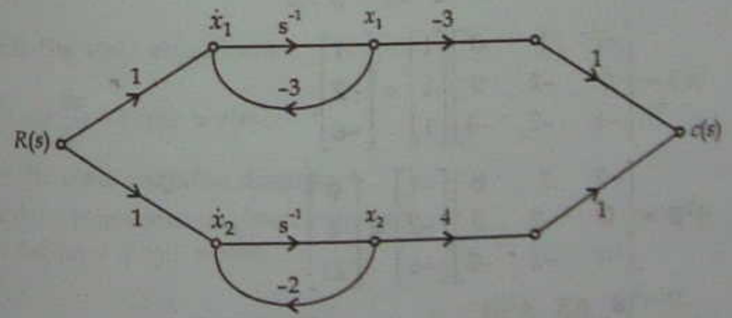


Fig. 8.23. Complete SVD parallel decomposition)

The set of state equation

$$\dot{x}_1 = -3x_1 + r$$

$$\dot{x}_2 = -2x_2 + r$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-3 & 0\\ 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 1\end{bmatrix}r$$

output equation

$$c(t) = -3x_1 + 4x_2$$

$$c(t) = \begin{bmatrix} -3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Example 8.36. Given the dynamic equation

$$\frac{dx(t)}{dt} = Ax(t) + Bu(t)$$

$$y(t) = Cx(t)$$

where,

$$A = \begin{bmatrix} -2 & +1 & 0 \\ 0 & -2 & 0 \\ -1 & -2 & -3 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, C = [1\ 0\ 0]$$

Find the transformation matrix $x(t) = P\bar{x}(t)$.

Solution : First find out the characteristic equation

$$SI - A = s\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 0 \\ -1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} s+2 & -1 & 0 \\ 0 & s+2 & 0 \\ 1 & 2 & s+3 \end{bmatrix}$$

$$|SI - A| = s^3 + 7s^2 + 16s + 12 = 0$$

Here, $a_0 = 12, a_1 = 16, a_2 = 7, a_3 = 1$

$$M = \begin{bmatrix} a_1 & a_2 & 1 \\ a_2 & a_3 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 16 & 7 & 1 \\ 7 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 0 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ -6 \end{bmatrix}$$

$$A^2B = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 0 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \\ -6 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 23 \end{bmatrix}$$

$$S = [B \quad AB \quad A^2B]$$

$$\therefore \quad S = \begin{bmatrix} 1 & -1 & 0 \\ 1 & -2 & 4 \\ 1 & -6 & 23 \end{bmatrix}$$

$\therefore$ Transformation matrix $P = SM$

$$P = \begin{bmatrix} 1 & -1 & 0 \\ 1 & -2 & 4 \\ 1 & -6 & 23 \end{bmatrix} \begin{bmatrix} 16 & 7 & 1 \\ 7 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 9 & 6 & 1 \\ 6 & 5 & 1 \\ -3 & 1 & 1 \end{bmatrix}$$

Ans.

Example 8.37. Draw state variable diagram (SVD) f or the given transfer function

$$\frac{C(s)}{R(s)} = \frac{5s}{3s^2 + 3s + 1}$$

Solution : Multiply numerator and denominator by $\frac{1}{s^2}$

$$\frac{C(s)}{R(s)} = \frac{5s}{3 + 3s^{-1} + s^{-2}} \cdot \frac{Q(s)}{Q(s)}$$

$$C(s) = 5s^{-1}\, Q(s)$$

$$R(s) = (3 + 3s^{-1} + s^{-2})\, Q(s)$$

$$R(s) = 3Q(s) + 3s^{-1}\, Q(s) + s^{-2}\, Q(s)$$

$$3Q(s) = R(s) - 3s^{-1}\, Q(s) - s^{-2}\, Q(s)$$

$$Q(s) = \frac{1}{3}R(s) - s^{-1}Q(s) - \frac{1}{3}s^{-2}Q(s)$$

$$Q(s) = \frac{1}{3}R(s) - Q(s)\left[s^{-1} + \frac{1}{3}s^{-2}\right]$$

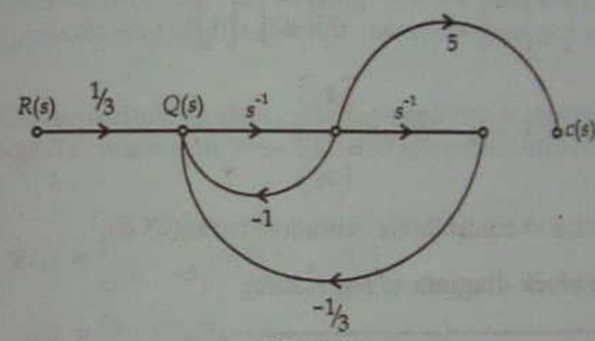


Fig. 8.24.

Example 8.38. Write the state equations

$$\frac{d^3c}{dt^3} + 6\frac{dc}{dt} + 5c(t) = r(t)$$

Also draw the state variable diagram

Solution : Take laplace transform of given equation

$$s^3C(s) + 6sC(s) + 5C(s) = R(s)$$

$$\frac{C(s)}{R(s)} = \frac{1}{s^3 + 6s + 5}$$

Divide numerator and denominator by s^3

$$\frac{C(s)}{R(s)} = \frac{s^{-3}}{1 + 6s^{-2} + 5s^{-3}}$$

$$\frac{C(s)}{R(s)} = \frac{s^{-3}}{1 + 6s^{-2} + 5s^{-3}} \cdot \frac{Q(s)}{Q(s)}$$

$$C(s) = s^{-3}\, Q(s)$$

$$R(s) = Q(s)\,[1 + 6s^{-2} + 5s^{-3}]$$

$$R(s) = Q(s) + Q(s)\,[6s^{-2} + 5s^{-3}]$$
$$Q(s) = R(s) - Q(s)\,[6s^{-2} + 5s^{-3}]$$

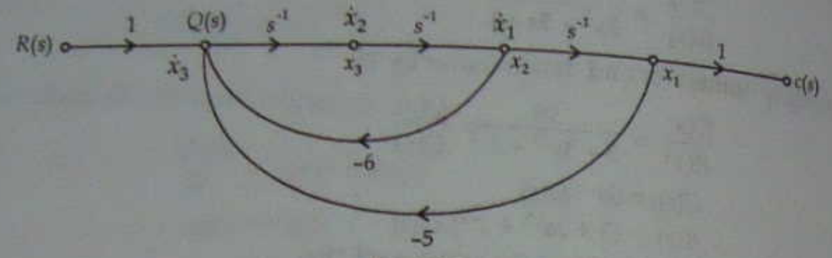


Fig. 8.25.

From SVD $C(s) = x_1$ $\dot{x}_1 = x_2$

$\dot{x}_2 = x_3$ $\dot{x}_3 = -6x_2 - 5x_1 + r$

Above equation can be written in matrix form

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -5 & -6 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix} r$$

$$C(s) = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

The above equations are in controllable canonical form (CCF).

Example 8.39. Draw the block diagram of fig. 8.22(c)

Solution :

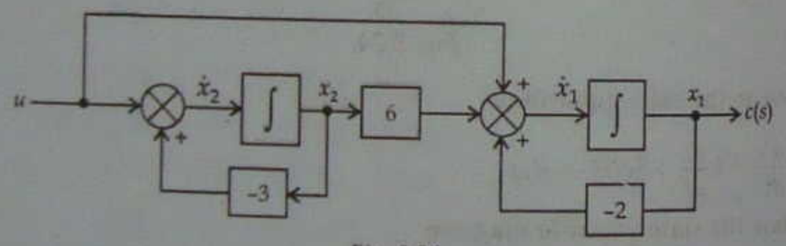


Fig. 8.26.

Example 8.40. Draw the block diagram of example 8.35.

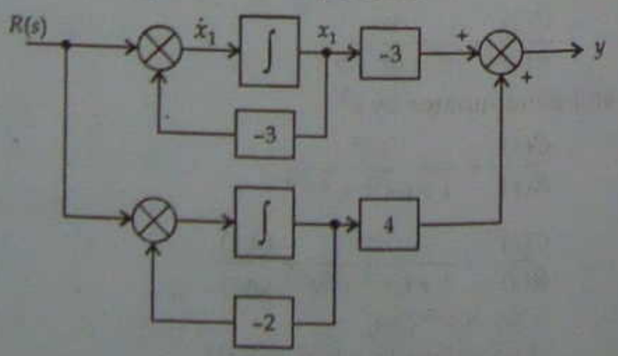


Fig. 8.27.

8.15. EFFECT OF POLE-ZERO CANCELLATION IN TRANSFER FUNCTION

Consider the transfer function

$$\frac{Y(s)}{u(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \ldots + b_{n-1}s + b_n}{s^n + a_1 s^{n-1} + \ldots + a_{n-1}s + a_n}$$

The above transfer function can be written in factored form

$$\frac{Y(s)}{u(s)} = \frac{k(s-z_1)(s-z_2)\ldots(s-z_n)}{(s-p_1)(s-p_2)\ldots(s-p_n)}$$

$$= \sum_{k=1}^{k} \frac{d_k}{(s-p_k)}, \text{ where } d_k \text{ are residues of poles at } s = p_k$$

Suppose, transfer function has identical pole and zero at $z = p$ thus $d_i = 0$. The effect of this cancellation on controllability and observability depends on how the state variables are defined. For example if the state variables are defined by equation (8.117.7) & (8.117.8) the d_i will be zero in output vector c and if the state variables are defined by equation (8.117.14 & 8.117.15) then d_i will appear in control vector B. Thus, if pole zero cancellation occurs in transfer function, then the controllability and ovservability vary, depending how the state variables are choosen. So, for completely state controllable and observable the transfer function must not have any pole zero cancellation.

Example 8.41. Consider the following state equations and check controllability and observability of the given systems and explain why there is a difference in controllability and observability of the same system.

(i) $$\dot{X}(t) = \begin{bmatrix}0 & 1\\ -6 & -5\end{bmatrix}x(t) + \begin{bmatrix}0\\ 1\end{bmatrix}r(t)$$

$$c(t) = \begin{bmatrix}2 & 1\end{bmatrix}x(t)$$

(ii) $$\dot{X}(t) = \begin{bmatrix}0 & -6\\ 1 & -5\end{bmatrix}x(t) + \begin{bmatrix}2\\ 1\end{bmatrix}r(t)$$

$$c(t) = \begin{bmatrix}0 & 1\end{bmatrix}x(t).$$

Solution : (i) $$A = \begin{bmatrix}0 & 1\\ -6 & -5\end{bmatrix},\ B = \begin{bmatrix}0\\ 1\end{bmatrix},\ C = [2 \quad 1]$$

For controllability $Q = [B \quad AB]$

$$AB = \begin{bmatrix}0 & 1\\ 1 & -5\end{bmatrix};\ Q = \begin{bmatrix}0 & 1\\ 1 & -5\end{bmatrix}$$

$Q \neq 0$, Rank = 2, system is controllable

$$C^T = \begin{bmatrix}2\\ 1\end{bmatrix},\ A^T = \begin{bmatrix}0 & -6\\ 1 & -5\end{bmatrix}$$

$$Q' = [C^T \quad A^T C^T] = \begin{bmatrix}2 & -6\\ 1 & -3\end{bmatrix}$$

Rank = 1, system is unobservable.

(ii) $$A = \begin{bmatrix} 0 & -6 \\ 1 & -5 \end{bmatrix},\quad B = \begin{bmatrix} 2 \\ 1 \end{bmatrix},\quad C = [0 \quad 1]$$

For controllability $Q = [B \quad AB]$; $AB = \begin{bmatrix} -6 \\ -3 \end{bmatrix}$

$$Q = \begin{bmatrix} 2 & -6 \\ 1 & -3 \end{bmatrix}$$

Rank = 1, the system in uncontrollable.

For observability $Q' = [C^T \quad A^T C^T]$

$$A^T C^T = \begin{bmatrix} 1 \\ -5 \end{bmatrix} \; ; Q' = \begin{bmatrix} 0 & 1 \\ 1 & -5 \end{bmatrix}$$

Rank = 2, the system is observable.

Now find the transfer function

$$SI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & -6 \\ 1 & -5 \end{bmatrix} = \begin{bmatrix} s & 6 \\ -1 & s+5 \end{bmatrix}$$

$$|SI - A| = s^2 + 5s + 6 = (s+2)(s+3)$$

$$[SI - A]^{-1} = \begin{bmatrix} \frac{s+5}{s^2+5s+6} & \frac{-6}{s^2+5s+6} \\ \frac{1}{s^2+5s+6} & \frac{s}{s^2+5s+6} \end{bmatrix}$$

Transfer function is given by

$$C^T [SI - A]^{-1} B = [0 \; 1] \begin{bmatrix} \frac{s+5}{s^2+5s+6} & \frac{-6}{s^2+5s+6} \\ \frac{1}{s^2+5s+6} & \frac{s}{s^2+5s+6} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\text{Transfer function} = \frac{s+2}{(s+2)(s+3)}$$

From the transfer function it is clear that there is a pole at $s = -2$ and a zero at $s = -2$, there is a pole zero cancellation. Therefore for completely controllable and observable the transfer function must not have any pole zero cancellation.

SUMMARY

Advantages of State Space Techniques

1. This method can be applied to linear or non-linear, time variant or time invariant system.
2. It is easy to apply where laplace transform cannot applied.
3. It is time domain approach.
4. This method is suitable for digital computer computation.

5. This method can be designed for optimal conditions.

A state vector is a vector which determines uniquely the system state $x(t)$ for any time $t \geq t_0$.

$$f(t) = \text{State transition matrix} = e^{AT}$$

$$= I + At + \frac{A^2 t^2}{21} + \ldots.$$

Properties of state transition matrix (STM)

1. $\phi(0) = e^{A(0)} = I$
2. $\phi(t) = [\phi(-t)]^{-1}$
3. $\phi^{-1}(t) = \phi(-t)$
4. $[\phi(t)]^n = \phi(nt)$
5. $\phi(t_1 + t_2) = \phi(t_2)\,\phi(t_1)$

$$\phi(t) = \pounds^{-1}\,\phi(s) = \pounds^{-1}\,(SI - A)^{-1}$$

$\phi(s)$ = resolvent matrix

Transfer matrix

$G(s) = C\phi(s)\,B$ for MIMO system

$G(s) = C^T \phi(s)\,b$ for SISO system

The roots of the characteristic equation are known as eigen values of matrix.

Any non-zero vector P_i that satisfies the matrix equation $(\lambda_i I - A)\,P_i = 0$.

λ_i is the eigen value of A

A system is said to be controllable if any initial state $x(t_0)$ or x_0 can be transferred to any final state $x(t_f)$ in a finite time interval $(t_f - t_0)$, $t_0 \geq 0$ by control U.

A system is said to be observable if every state x_0 can be exactly determined from the measurement of the output y over a finite interval of time $0 < t < t_f$

There are four types of states :

1. The system is controllable and observable. (SCO)
2. The system is controllable and unobservable. (SCU)
3. The system is uncontrollable and observable (SUO)
4. The system is uncontrollable and unobservable (SUU)

KALMAN'S TEST

A linear time invariant continuous system is completely controllable if and only if the rank of the controllability matrix is defined as

$$Q = [B : AB : A^2 B ; \ldots . A^{n-1} B]$$

is equal to rank 'n'.

EXERCISE

Q1. The state equation of a linear time invariant system is given by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

Find the state transition matrix and the characteristic equation.

8.2. Find the state transition matrix for

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -6x_1 - 5x_2$$

8.3. The state space of a system is represented by the following equations

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad ; t > 0$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find the transfer function of the system.

8.4. Obtain A, B, C and D matrices of the circuit

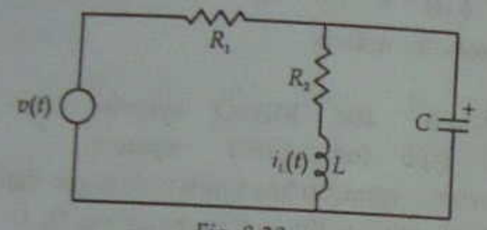


Fig. 8.28.

8.5. The state space representation of a system is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 & 2 & 1 \\ 0 & 0 & 1 \\ -1 & -4 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

output : $y = x_1$

Is the system controllable ?

8.6. Obtain a state space representation for the system

$$\frac{d^3y}{dt^3} + 5\frac{dy^2}{dt^2} + 6\frac{dy}{dt} + 3y = 6u(t)$$

8.7. For question no. (8.35) determine the transfer function.

8.8. Obtain the transfer function representation for a system represented by matrices

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -10 & -2 & -3 \end{bmatrix}; \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}; \quad C = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix}$$

8.9. The transfer function of a system is $G(s) = \dfrac{2}{(s+1)(s+2)}$. Obtain a state variable representation.

8.10. Obtain the state space representation of the following control system. *(UPSC 1979)*

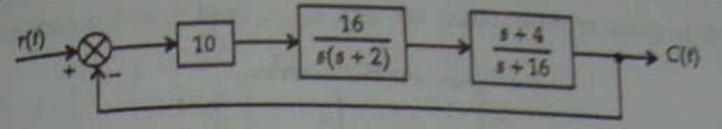


Fig. 8.29.

8.11. Determine the controllability of the equation

$$\dot{x} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} x + \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} x$$

8.12. Check the contrllability & observability of the equation

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -4 & -3 \end{bmatrix} x + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix} x$$

8.13. Determine the state model for the electrical circuit snown in fig. select i_{L1} & i_{L2} as state variables.

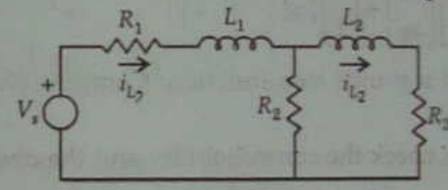


Fig. 8.30.

8.14. Obtained the state model for the given electrical circuit. Select the state variables as V_c, i_{L1} & i_{L2}

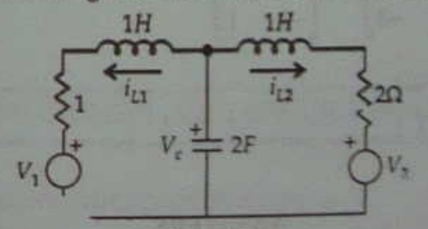


Fig. 8.31

8.15. In question no. 8.52 check the observability.

8.16. The state equation of linear time invariant system *i.e.*

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

Find the stable transition matrix *(UPTU Question Bank)*

8.17. Give the advantage of state space technique & obtain the state space representation of the system

$$\frac{d^3y}{dt^3} + 5\frac{d^2y}{dt^2} + 10\frac{dy}{dt} + 20y = 8u(t)$$

(UPTU Question Bank)

8.18. Write the state space represe.itation taking i, i_{L_1}, i_{L_2} and V_c as the variables.

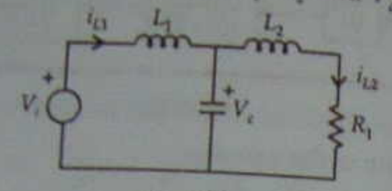


Fig. 8.32.

8.19. (*a*) Wite the properties of transition matrix.

(*b*) If $A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}$; $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$; $C = [0\ 1]$

Determine $\phi(t)$.

8.20. (*a*) For the given transfer function, obtain the state model *(UPTU Question Bank)*

$$G(s) = \frac{k}{s^3 + a_3 s^2 + a_2 s + a_1}$$

OR

consider a control system with the state model given by

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} [u]$$

$X_1(0)$ and $X_2(0) = 0$ and u = unit step function. Compute the state transition matrix and hence find state response. *(UPTU Question Bank)*

(*b*) For the given system check the controllability and the observability of the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -4 & -3 \end{bmatrix} x + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} u$$

$$Y = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix} x$$

(UPTU Question Bank)

ANSWERS

8.1. $\begin{bmatrix} te^{-t} + e^{-t} & te^{-t} \\ -te^{-t} & -te^{-t} + e^{-t} \end{bmatrix}$, characteristic equation $= s^2 + 2s + 1 = 0$

8.2. $\begin{bmatrix} 3e^{-2t} + 2e^{-3t} & e^{-2t} - e^{-3t} \\ -6e^{-2t} + 6e^{-3t} & -2e^{-2t} + 3e^{-3t} \end{bmatrix}$

8.3. $\frac{1}{(s+1)(s+2)}$

8.4. $A = \begin{bmatrix} -\frac{1}{R_1 C} & -\frac{1}{C} \\ \frac{1}{L} & -\frac{R_2}{L} \end{bmatrix}$, $B = \begin{bmatrix} \frac{1}{R_1 C} \\ 0 \end{bmatrix}$, $C = [1\ 0]$, $D = [0]$

8.5. $\frac{s+2}{s^3 + 3s^2 + 2s + 10}$

8.6. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$

and $y(t) = [1\ \ 0]\, x(t)$

8.7. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ -640 & -172 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 610 & 160 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$

8.8. not contrabble

8.9. $\begin{bmatrix} \dot{i}_{L1} \\ \dot{i}_{L2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{L_1}(R_1 + R_2) & \frac{R_2}{L_1} \\ \frac{R_2}{L_2} & -\frac{1}{L_2}(R_2 + R_3) \end{bmatrix} \begin{bmatrix} i_{L_1} \\ i_{L_2} \end{bmatrix} + \begin{bmatrix} \frac{1}{L_1} \\ 0 \end{bmatrix} [V_s]$

8.10. $\begin{bmatrix} \dot{V}_c \\ \dot{i}_{L1} \\ \dot{i}_{L2} \end{bmatrix} = \begin{bmatrix} 0 & -0.5 & -0.5 \\ 0.5 & -0.5 & 0 \\ 1 & 0 & -2 \end{bmatrix} \begin{bmatrix} V_c \\ i_{L1} \\ i_{L2} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -0.5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$

SEMI-OBJECTIVE TYPE QUESTIONS

(i) What are the advantages of state space techniques. (Article 8.2)
(ii) Define state, state variables, state vector and state space. (Article 8.3)
(iii) What is state transition matrix? (Article 8.6)
(iv) Define Eigen values and Eigen vectors. (Articles 8.7.3 and 8.7.4.)
(v) What are the properties of STM. (Article 8.5.2)
(vi) Define controllability and observability. (Article 8.9)
(vii) Write a short note on Kalman's Test. (Article 8.9.2)
(viii) Write a short note limitations of state variable approach.

□□□

Chapter 9

Control System Components

9.1. INTRODUCTION

In control system, the devices which are used to convert the process variables in one form to another from is known as transducers. Transducers can also be defined as a device which transforms the energy from one form to another, for example a thermocouple converts the heat energy into electrical voltage. In control system the following devices are used as a transducer.

1. Potentiometers
2. D.C. servomotors
3. A.C. servomotors
4. Synchros
5. Stepper motors
6. Magnetic amplifier
7. Tachogenerators
8. Gyroscope
9. Differential transformer

9.2. POTENTIOMETERS

A potentiometer is a simple device which is used for mechanical displacement either linear or angular. Thus, a potentiometer is an electromechanical transducer which converts the mechanical energy into electrical energy. The input to the device is in the form of linear mechanical displacement or rotational mechanical displacement, when the voltage is applied across the fixed terminals the output voltage is proportional to the displacement.

Consider the fig. 9.1 (translational potentiometer).

Let E_i = input voltage

E_0 = output voltage

x_1 = displacement from zero position

x_t = total length of translational potentiometer

R = Total resistance of potentiometer

Under ideal condition the output voltage E_0 is given by

$$E_0 = \frac{x_1}{x_t} E_i$$

Equation (9.1) shows a linear relationship shown in fig. 9.2.

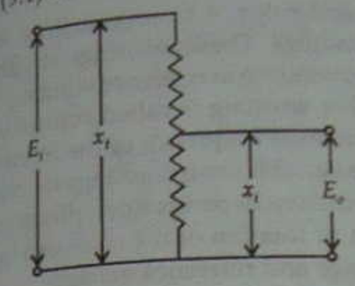


Fig. 9.1.

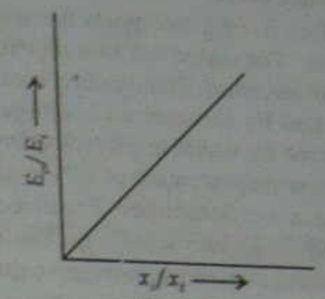


Fig. 9.2

Similarly for rotational motion, the output voltage E_0 is given by

$$E_0 = E_i \cdot \frac{\theta_i}{\theta_t} \quad \text{...(9.2)}$$

where, θ_i = input angular displacement (degree or radians)

θ_t = total travel of wiper (degree or radians)

Fig. 9.3(*a*) Shows an arrangement of error sensing transducer. In this arrangement two potentiometers are connected in parallel. The output voltage taken across the variable terminals of the two potentiometer. The output voltage E_0 is given by

$$E_o = K(\theta_1 - \theta_2) \quad \text{...(9.3)}$$

E_i is the voltage applied, θ_1 and θ_2 are the angular displacement of the wiper, K is constant and is known as sensitivity. The block diagram is shown in fig. 9.3(*b*)

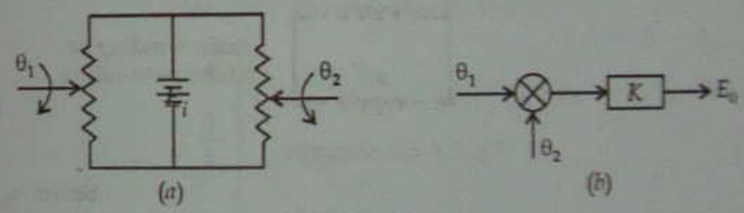


Fig. 9.3.

9.3. SERVOMOTORS

Servomotors are used in feedback control systems. Servomotors have low rotor inertia and high speed of response. The servomotors are also know as control motors. The servomotors which are used in feedback control system should have linear relationship between electrical control signal and rotor speed, torque speed characteristic should be linear, the response of the servomotor should be fast and inertia should be low.

9.4. TYPES OF SERVOMOTORS

The servomotors are classified as

(i) A.C. servomotors

(ii) D.C. servomotors

(iii) Special servomotors

D.C. servomotors are further classified as armature controlled d.c. servomotors and field control d.c. servomotors.

9.4.1. A.C. Servomotors

These motors having two parts namely stator and rotor. A.C. servomotors are two phase induction motor. The stator has two distributed windings. These windings are displaced from each other by 90° electrical. One winding is called main winding or reference winding. The reference winding is excited by constant a.c. voltage. The other winding is called control winding. This winding is excited by variable control voltage of the same frequency as the reference winding but having a phase displacement of 90° electrical. The variable control voltage for control winding is obtained from a servoamplifier. The direction of rotation depends upon phase relationship of voltages applied to the two windings. The direction of rotation of the rotor can be reversed by reversing the phase difference between control voltage and reference voltage.

The rotor of a.c. servomotors are of two types (*a*) squirrel cage rotor (*b*) drag cup type rotor. The squirrel cage rotor having large length and small diameter, so its resistance is very high. The air gap of squirrel cage is kept small. In drag cup type there are two air gaps. For the rotor a cup of non-magnetic conducting material is used. A stationary iron core is placed between the conducting cup to complete the magnetic circuit. The resistance of drag cup type is high and therefore having high starting torque. Generally aluminium is used for cup. Fig. 9.4. Shows the schematic diagram of two phase a.c. servomotor and 9.5(*a*), 9.5 (*b*) shows the two types of rotor.

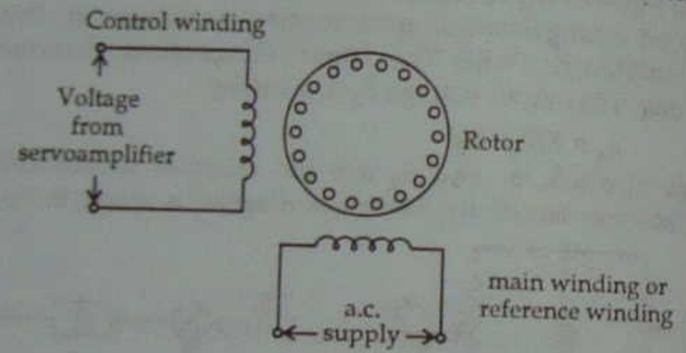


Fig. 9.4. a.c. servomotor

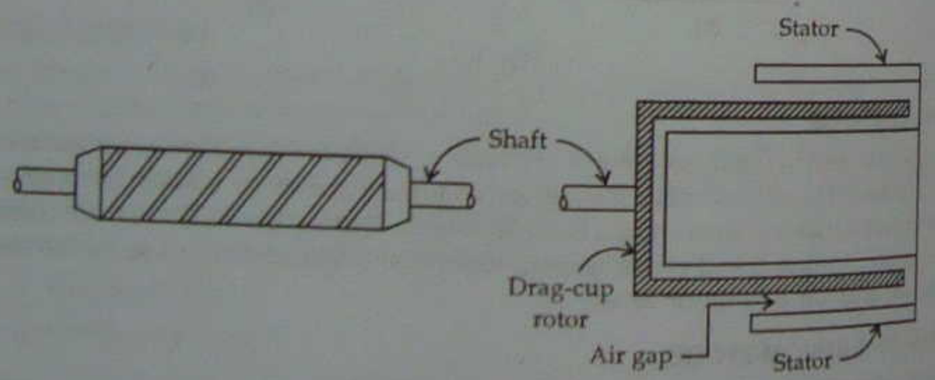


Fig. 9.5(a) Squirrel cage rotor

Fig. 9.5(b) Drag cup type rotor

9.4.2. Torque-speed Characteristic

The torque speed characteristic of two phase induction motor depends upon the ratio of reactance to resistance. For high resistance and low reactance, the characteristic is linear and for large ratio of X to R it becomes non-linear as shown in Fig. 9.6(*a*). The torque-speed characteristics for various control voltages are almost linear as shown in fig. 9.6(*b*).

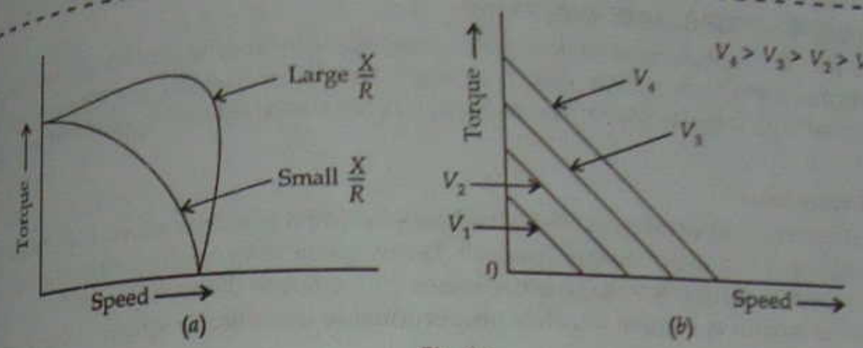
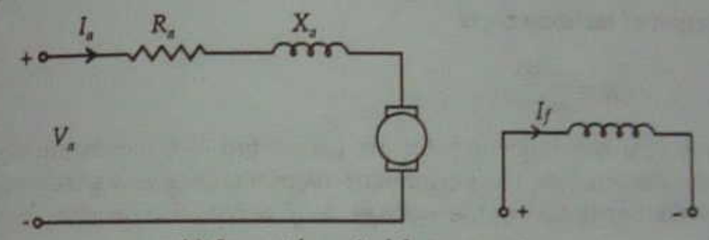


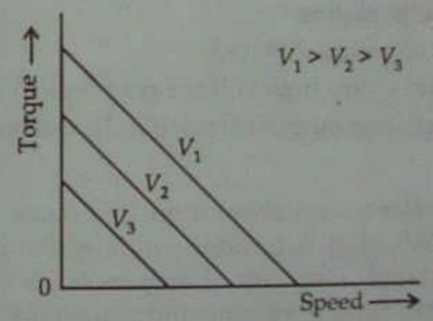
Fig. 9.6

9.4.3. D.C. Servomotors

D.C. servomotors are separately excited or permanent magnet d.c. servomotors. The armature of d.c. servomotor has a large resistance, therefore torque speed characteristic is linear. The torque speed characteristic shows in fig. 9.7 (*b*). Fig 9.7 (*a*) shows the schematic diagram of separately excited d.c. servomotor.



(a) Separately excited d.c. servometer



(b) Torque speed characteristic

Fig. 9.7.

The d.c. servomotors can be controlled from armature side or from field. In field controlled d.c. servomotors the ratio of L/R is large *i.e.*, the time constant for field circuit is large. Due to large time constant, the response is slow and therefore they are not commonly used. Transfer function of field controlled d.c. servomotor is given in chapter 1. The speed of the motor can be controlled by adjusting the voltage applied to the armature. In armature controlled d.c. servomotor the time constant is small and hence the response is fast. The efficiency is better than the field controlled motor. The transfer function of armature controlled d.c. servomotor is derived in chapter 1.

9.4.4. Application of Servomotors

Servomotors are widely used in radars, electromechanical actuators, computers, machine tools, tracking and guidance system, process controllers and robots.

9.5. TACHOGENERATORS (TACHOMETERS)

Tachometers are electromechanical devices, which transforms the mechanical energy into electrical energy. In tachometers its magnetic flux is constant and induced e.m.f is proportional to the speed of the shaft i.e., angular speed. The tachometers are classified as d.c. tachometers and ac tachometers.

9.5.1. D.C. Tachometer

In control systems most common type of tachometers are d.c. tachometers. D.C. tachometers contains an iron core rotor and permanent magnet. The magnetic field is provided by the permanent magnet and no external supply voltage is necessary. The input to the tachometer is the speed of the shaft and the output is voltage which is proportional to the angular speed of the shaft.

$$\therefore \qquad e = K\omega(t) \qquad \ldots(9.4)$$

where, e = tachometer generator voltage

K = tachometer sensitivity

ω = angular speed of the shaft.

Laplace transform of equation 9.4

$$E(S) = K\omega(S)$$

∴ Transfer function of tachometer is

$$K = \frac{E(S)}{\omega(S)} \qquad \ldots(9.5)$$

$$\ldots(9.6)$$

In d.c. tachometers the windings on rotor are connected to the commutator and the output voltage is taken across the brushes. The permanent magnet tachometers are compact and reliable but having high inertia. For reducing the voltage drop across the brushes, metal brushes with silver tips are used.

9.5.2. Advantages of D.C. Tachometers

1. At zero speed there is no residual voltage.
2. It is possible to generate a very high voltage gradients in small size.
3. It can be used with high pass output filters to reduce servo velocity lags.

9.5.3. A.C. Tachometers

A.C. tachometers are similar to two phase induction motor. The schematic diagram of a.c. tachometer is shown in fig. 9.8. Here two stator windings are placed in quadrature with each other and rotor is short circuited. The rotor of tachometer is a thin aluminium cup. The two stator windings are known as primary winding and secondary winding. Primary winding also known as reference winding.

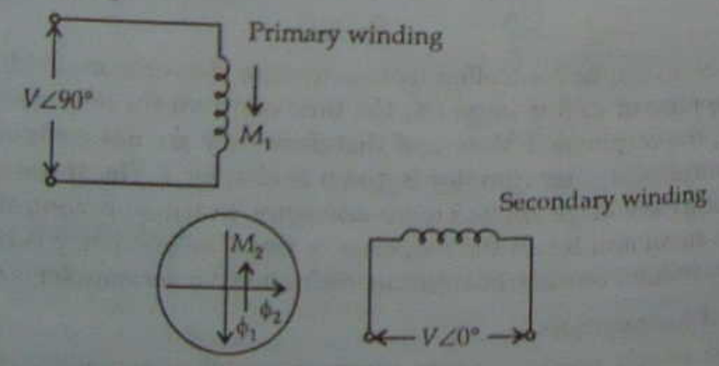


Fig. 9.8. A.C. tachometer

The primary winding carries a voltage $V\angle 90°$. This voltage is fixed in magnitude at fixed frequency known as carrier frequency. The coil axis of primary winding is called direct axis. When the current in primary winding flows a pulsating field M_1 is produced along direct axis. When the rotor is stationary an e.m.f. is induced in it. Since the motor is short circuited eddy current flows in it. Due to this eddy current a pulsating flux M_2 is produced opposite to the M_1. Therefore a pulsating flux ϕ_1 is produced due to the vector sum of M_1 and M_2. Since the secondary winding is in quadrature no e.m. f. will produced in it due to ϕ_1. Now if the rotor rotates it cuts the flux and an e.m.f is induced in it, due to this induced e.m.f a current will flow in the rotor and a pulsating flux will appear along the coil axis of secondary winding. This flux ϕ_2 will induce the voltage in secondary winding. The magnitude of the output voltage will be proportional to the rotor speed.

9.6. STEPPER MOTORS

In stepper motors, the movement of the rotor is in discrete steps. A stepper motor is electromechanical device. There are three types of stepper motors:

(i) Variable reluctance motors
(ii) Permanent magnet motors
(iii) Hybrid type

9.6.1. Variable Reluctance Stepper Motors

Variable reluctance stepper motors are of two types namely single-stack variable reluctance motor and multi stack variable reluctance motor.

9.6.1.1. Single-stack Variable Reluctance Motor

A 4-phase, 4-stator pole, 2-tooth rotor variable reluctance stepper motor is shown schematrically in fig. 9.9.

In fig. 9.9. four phases are connected to d.c. source through switches S_1, S_2, S_3 and S_4. When phase 1 is excited, the rotor aligns with the axis of phase 1 as shown in fig. 9.9. If now switch S_1 off and S_2 switched on, the rotor moves through 90° in clockwise to align with the resultant air gap field which lies along the axis of phase 2. Now, phase 3 is excited and phase 2 is disconnected, the rotor aligns with the resultant air gap field which now lies along phase 3 and so on. The windings of the stator are energized in sequence 1, 2, 3, 4, 1.

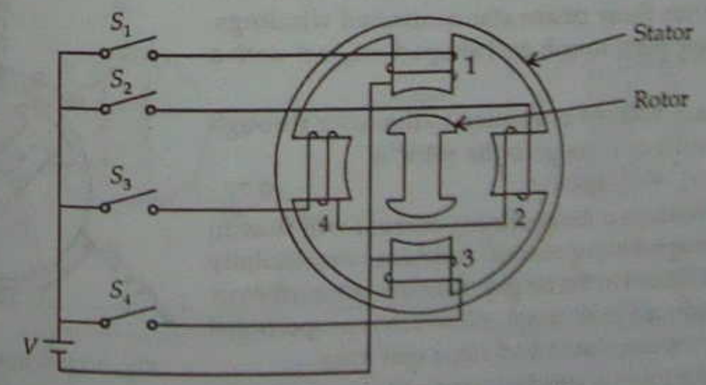


Fig. 9.9. Variable reluctance stepper motor

As the phases are energized as 1, 2, 3, 4, 1, the rotor moves through a step of 90° in clockwise direction. Rotor can be made to rotate in anticlockwise direction by reversing the switching sequence.

9.6.1.2. Multistack Variable Reluctance Stepper Motor

A multistack variable reluctance stepper motor is shown in Fig. 9.10. In this type of stepper motor, rotors have a common shaft and stator have a common frame. The stators and rotors have same number of poles with same teeth size. The stators are pulse excited while rotors are unexcited. The windings of all stator poles are excited simultaneously. Since, stators and rotors have same number poles, therefore pole pitch is also same. If T_r is the number of rotor teeth and n is the number of stacks or phases, then tooth pitch is given by $360°/T_r$ and angular displacement or step angle is given by $360°/nTr$. For exmaple 12 pole rotor, the pitch is $360/12 = 30$ and the step angle will be $360/3 \times 12 = 10°$ *i.e.*, rotor poles are displaced from each other by 10°.

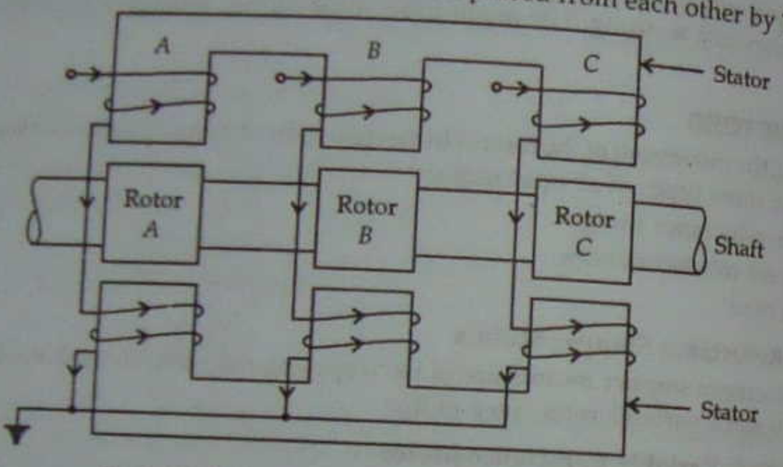


Fig. 9.10. Multi stack variable reluctance stepper motor

9.7. SYNCHROS

A synchro is an electromagnetic transducer which converts the angular position of a shaft into an electric signal. Synchros are used as detectors and encoders.

9.7.1. Synchro Transmitter

The construction of synchro transmitter is very similar to that of a three phase alternator. The stator is made of laminated silicon steel and carries three phase star connected windings. The rotor is a rotating part, dumb-bell shaped magnet with a single winding.

A single phase a.c. voltage is applied to the rotor through slip rings. Let applied a.c. voltage to the rotor is

$$e_r = E_r \sin \omega_0 t \qquad ...(9.7)$$

due to this applied voltage a magnetizing current will flow in the rotor coil. This magnetizing current produces sinusoidally varying flux and distributed in the air gap. Because of transformer action voltages get induced in all stator coil which is proportional to cosine of angle between stator and rotor coil axes.

Now, consider the rotor of synchro transmitter is at an angle θ, then voltages ineach stator coil with respect to neutral are

$$E_{an} = KE_r \sin\omega_0 t \cos\theta \qquad ...(9.8)$$

$$E_{bn} = KE_r \sin\omega_0 t \cos(\theta + 120°) \qquad ...(9.9)$$

$$E_{cn} = KE_r \sin\omega_0 t \cos(\theta + 240°) \qquad ...(9.10)$$

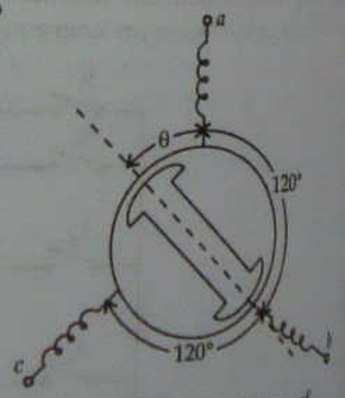


Fig. 9.11. Schematic diagram of synchro transmitter

Magnitudes of stator terminal voltages are

$$E_{cb} = E_{cn} - E_{bn}$$
$$= KE_r \sin\omega_0 t [\cos(\theta + 240°) - \cos(\theta + 120°)]$$
$$= KE_r \sin\omega_0 t \left[\sqrt{3}\sin\theta\right]$$

$$\therefore \quad E_{cb} = \sqrt{3}\, KE_r \sin\omega_0 t \sin\theta \qquad ...(9.11)$$

Similarly, $$E_{ac} = \sqrt{3}\, KE_r \sin\omega_0 t \sin(\theta + 120°) \qquad ...(9.12)$$

$$E_{ba} = \sqrt{3}\, KE_r \sin\omega_0 t \sin(\theta + 240°) \qquad ...(9.13)$$

When $\theta = 0$, the maximum induced voltage will be E_{an} and E_{cb} will be zero. This position of the rotor is defined as electrical zero of the transmitter and is used as the reference for indicating the angular position of the rotor.

Thus, the input to the synchro transmitter is the angular position of the rotor shaft and the output are the three single phase voltages which are the function of the shaft position.

9.7.2. Synchro Control Transformer

Principle of operation of synchro control transformer is same as that of synchro transmitter. Rotor of snchro control transformer is cylindrical type. Synchro control transformer is an electromechanical device. The combination of synchro transmitter and synchro control transformer is used as an error detector. The function of error detector is to convert the difference of two shaft positions into an electrical signal. The fig. 9.12, shows schematic diagram of synchro error detector.

The output of synchro transmitter is connected to the stator winding of the synchro control transformer. Therefore the same current will flow in the stator windings of synchro control transformer but in opposite direction. The voltage across the rotor terminals of control transformer is

$$e(t) = K_1\, V_r \cos\phi \sin\omega_0 t \qquad ...(9.14)$$

where ϕ = angular displacement between the two rotors. When the two rotors are at an angle 90°, the voltage induced in control trnsformer is zero. This position is known as electrical zero position control transformer.

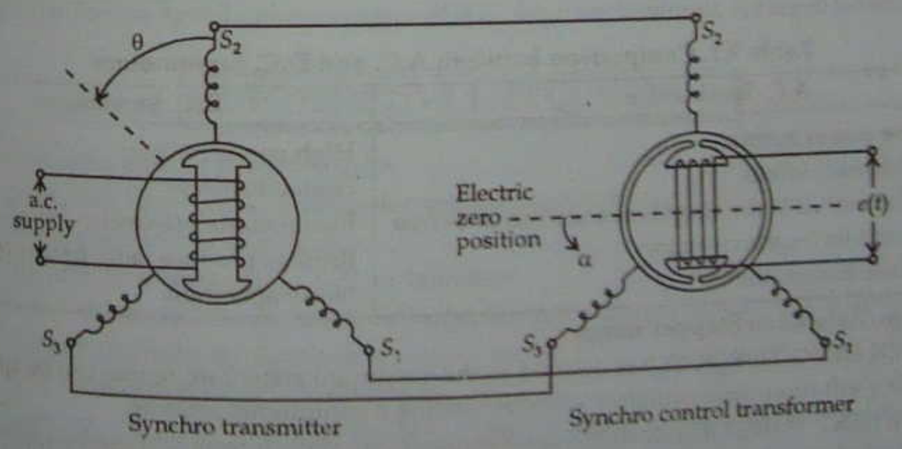


Fig. 9.12. Synchro error detector

Let the transmitter rotate through an angle 'θ' in the direction indicated and let control transformer rotor rotates in the same direction through an angle 'α'. Then

$$\phi = (90° - \theta + \alpha)$$

put the value of ϕ in equation 9.14, we get

$$e(t) = K_1 V_r \sin(\theta - \alpha) \sin \omega_0 t \quad \text{...(9.15)}$$

From eqn. (9.16) it is clear that when two rotor shafts are not in alignment, the rotor voltage ...(9.16) of control transformer is approximately a sine function of the difference between the two shaft angles.

For small angular displacement between two rotor position

$$e(t) = K_1 V_r (\theta - \alpha) \sin\omega_0 t \quad \text{...(9.17)}$$

SUMMARY

For feedback control system, servomotor should have following properties

(*i*) It should be reversible.

(*ii*) Response should be fast.

(*iii*) Relationship between electrical control system and rotor speed should be linear.

(*iv*) It should have linear torque-speed characteristic.

A.C. Servomotor has following features.

(*i*) light in weight.

(*ii*) Reliable and stable.

(*iii*) Maintenance free.

(*iv*) large *R* to *X* ratio.

(*v*) Smooth & noise free.

Field controlled D.C. motor has following features.

(*i*) It has large time constant.

(*ii*) It is open loop system.

(*iii*) Control circuit is simple.

Table 9.1. Comparison between A.C. and D.C. servomotors

S.No.	A.C. Servomotor	D.C. Servomotor
1.	low power output	High power output
2.	Efficiency is less	High efficiency
3.	No brushes & slip rings hence maintenance free	Frequent maintenance required
4.	No radio frequency noise	Brushes produce radio frequency noise
5.	Smooth operation	Noisy operation

Definitions Related to Stepper motor

(*i*) **HOLDING TORQUE:** It is defined as the maximum static torque that can be applied to the shaft of an excited motor without causing a continuous rotation.

(*ii*) **DETENT TORQUE:** It is defined as the maximum static torque that can be applied to the shaft of an unexcited motor without causing a continuous rotation.

(*iii*) **STEP ANGLE:** It is defined as the angular displacement of the rotor in response to each pulse.

(*iv*) **CRITICAL TORQUE:** It is defined as the maximum load torque at which rotor does not move when an exciting winding is energized. This is also known as pullout torque.

(*v*) **LIMITING TORQUE:** It is defined for a given pulsing rate or stepping rate measured in pulses per second, as the maximum load torque at which motor follows the control pulses without missing any step. This is also called as pull in torque.

(*vi*) **SYNCHRONOUS STEPPING RATE:** It is defined as the maximum rate at which the motor can step without missing steps. The motor can start, stop or reverse at this rate.

(*vii*) **SLEWING RATE :** It is defined as the maximum rate at which the motor can step unidirectionally. The slewing rate is much higher than the synchronous stepping rate. Motor will not be able to stop or reverse without missing steps at this rate.

Stepper motors are used in computer peripheral systems such as printers, tape drives etc., process control systems, robotic control systems, watches etc.

Tachnometers or Tachogenerator is an electro-mechanical unit which generated an electrical output proportional to the speed of the shaft. Tachometers are classified as D.C. Tachometers & A.C. Tachometers.

Potentiometer is an electromechanical transducer that converts mechanical energy into electrical energy. The input to the device is in the form of a mechanical displacement either linear or rotational.

Different type of potentiometers are single turn & multiturn potentiometer, linear motion potentiometers and wire wound potentiometers.

EXERCISE

1. Write short note on stepper motor.
2. Explain the working of Tachometers.
3. State the applications of a.c. servomotor.
4. Compare A.C. servomotors with D.C. servomotors.
5. Write short note on potentiometer.
6. Explain the principle of operation of synchro error detector.
7. Sketch the Torque-speed characteristics of A.C. servomotor.

SEMI-OBJECTIVE TYPE QUESTIONS

(*i*) Write short note on potentiometers. (Article 9.2)

(*ii*) Distinguish between a.c. servomotor and D.C. servomotor. (Summary)

(*iii*) State applications of servomotor. (Article 9.4.4)

(*iv*) What are the advantages of D.C. tachometers. (Article 9.5.2)

(*v*) Write a short note on multistack variable reluctance stepper motor. (Article 9.6.1.2)

(*vi*) What are the major applications of stepper motor. (Summary)

(*vii*) Write short note on synchro transmitter. (Article 9.7.1)

(*viii*) Draw the torque-slip characteristic of D.C. servomotor. (Article 9.4.3)

(*ix*) Draw the torque-slip characteristic of A.C. servomoter. (Article 9.4.2)

(*x*) Write short note on synchro control transformer. (Article 9.7.2)

(*xi*) What is a synchro transmitter?

(*xii*) Draw the torque-speed characteristics of a two phase servomotor.

(xiii) What are the two varieties of stepper motors?
(xiv) Explain the use of synchros as transmission of position over long distances.
(xv) Distinguish between d.c motor and d.c. servomotors.
(xvi) What is the function of tachogenerator?
(xvii) Gives the applications of A.C. tachogenerator.
(xviii) Explain the use of synchro pair as an error detector.
(xix) What are the applications of synchro?
(xx) Give a trade name of synchro.

□□□

Chapter 10

The Laplace Transforms

10.1. INTRODUCTION

In laplace transform method, the differential equation in the time domain is transformed into s-plane. The complex frequency variable is given by $S = \sigma + j\omega$, where σ and ω are both real quantities. σ is the real part and ω is the imaginary part. The advantage of laplace transform is that this method gives total solution. For example when we solve the differential equation by laplace transform, both transient and steady state component of the solution can be obtained simultaneously.

10.2. DEFINITION OF LAPLACE TRANSFORM

One-sided Laplace transform of a functio.. of time $f(t)$ is given by

$$\mathcal{L}[f(t)] = \int_0^\infty f(t) \cdot e^{-st} dt$$

$$\mathcal{L}[f(t)] = F(s) = \int_0^\infty f(t) \cdot e^{-st} dt \qquad \text{...(10.1)}$$

where $f(t)$ = is time function defined in the interval $0 \le t \le \infty$
s = complex variable
$F(s)$ = Laplace transform of $f(t)$

10.3. LAPLACE TRANSFORMS OF SOME TIME FUNCTIONS

10.3.1. Exponential Function

Consider the exponential function

$$f(t) = 0 \quad \text{for} \quad t < 0$$
$$= Ke^{-at} \quad \text{for} \quad t \ge 0$$

where 'K' and 'a' are constants.

$$\mathcal{L}[K e^{-at}] = \int_0^\infty K e^{-at} \cdot e^{-st} dt = K \int_0^\infty e^{-(a+s)t} dt$$

$$= K\left[-\frac{1}{a+s} e^{-(a+s)t}\right]_0^\infty = -\frac{K}{s+a}[e^{-\infty} - e^{0}]$$

$$\mathcal{L}[K e^{-at}] = \frac{K}{s+a}$$

10.3.2. Step Function

Consider the step function

$f(t) = 0 \quad \text{for} \quad t < 0$

$f(t) = A \quad \text{for} \quad t > 0$

where 'A' is a constant

$$\mathcal{L} f(t) = \int_0^\infty A e^{-st}\,dt = A\int_0^\infty e^{-st}\,dt = A\left[\frac{e^{-st}}{-s}\right]_0^\infty$$

$$\mathcal{L} f(t) = \frac{A}{s}$$

10.3.3. Unit Step Function

Consider the unit step function

$f(t) = 0 \quad \text{for} \quad t < 0$

$1 \quad \text{for} \quad t > 0$

$$\mathcal{L} f(t) = \int_0^\infty 1.e^{-st}\,dt = \left[\frac{e^{-st}}{-s}\right]_0^\infty$$

$$\mathcal{L} f(t) = F(s) = \frac{1}{s}$$

10.3.4. Ramp Function

Consider the ramp function

$f(t) = 0 \quad \text{for} \quad t < 0$

$= At \quad \text{for} \quad t \geq 0$

$$\mathcal{L}[At] = \int_0^\infty At.e^{-st}\,dt = \frac{A}{s^2}$$

10.3.5. Sinusoidal Function

Consider the sinusoidal function

$f(t) = 0 \quad \text{for} \quad t < 0$

$= A \sin \omega t \quad \text{for} \quad t \geq 0$

$\mathcal{L} f(t) = \mathcal{L}(A \sin \omega t) = A\,\mathcal{L}(\sin \omega t)$

$$= A\,\mathcal{L}\frac{e^{j\omega t} - e^{-j\omega t}}{2j} = \frac{A}{2j}\left[\mathcal{L} e^{j\omega t} - \mathcal{L} e^{-j\omega t}\right]$$

$$= \frac{A}{2j}\left[\frac{1}{s - j\omega} - \frac{1}{s + j\omega}\right] = \frac{A\omega}{s^2 + \omega^2}$$

$$\mathcal{L}(A \sin \omega t) = \frac{A\omega}{s^2 + \omega^2}$$

For $A = 1$

$$\mathcal{L} \sin \omega t = \frac{\omega}{s^2 + \omega^2}$$

10.3.6. Laplace Transform of t^n

$$\mathcal{L} t^n = \int_0^\infty t^n e^{-st}\,dt$$

let $st = x$

$$dt = \frac{1}{s}dx$$

$$t^n = \left(\frac{x}{s}\right)^n$$

When $t = 0, x = 0$

$t = \infty, x = \infty$

$$\mathcal{L} t^n = \int_0^\infty \left(\frac{x}{s}\right)^n e^{-x}.\frac{1}{s}dx = \frac{1}{s^{n+1}}\int_0^\infty x^n e^{-x}dx$$

Since $\overline{n+1} \triangleq \int_0^\infty x^n e^{-x}dx$

$$\overline{n+1} = \overline{n} = \underline{n}$$

$$\mathcal{L} t^n = \frac{1}{s^{n+1}}\overline{n+1} = \frac{\lfloor n}{s^{n+1}}$$

$$\boxed{\mathcal{L} t^n = \frac{\lfloor n}{s^{n+1}}}$$

10.3.7. Hyperbolic Sine and Cosine Functions

$f(t) = \cosh \beta t$

$\mathcal{L} f(t) = \mathcal{L} \cosh \beta t$

$$= \mathcal{L}\left[\frac{e^{\beta t} + e^{-\beta t}}{2}\right] \qquad \because \cosh \beta t = \frac{e^{\beta t} + e^{-\beta t}}{2}$$

$$= \frac{1}{2}\left[\mathcal{L} e^{\beta t} + \mathcal{L} e^{-\beta t}\right]$$

$$= \frac{1}{2}\left[\frac{1}{s - \beta} + \frac{1}{s + \beta}\right] \qquad \because \mathcal{L} e^{\beta t} = \frac{1}{s - \beta}\ \mathcal{L} e^{-\beta t} = \frac{1}{s + \beta}$$

$$\mathcal{L} \cosh \beta t = \frac{s}{s^2 - \beta^2}$$

$$\mathcal{L} \cosh \beta t = \frac{s}{s^2 - \beta^2}$$

Similarly, $\mathcal{L} \sinh \alpha t = \mathcal{L}\frac{e^{\alpha t} - e^{-\alpha t}}{2} = \frac{1}{2}[\mathcal{L} e^{\alpha t} - \mathcal{L} e^{-\alpha t}]$

$$= \frac{1}{2}\left[\frac{1}{s-\alpha} - \frac{1}{s+\alpha}\right] = \frac{\alpha}{s^2-\alpha^2}$$

$$\boxed{\mathcal{L} \sinh \alpha t = \frac{\alpha}{s^2-\alpha^2}}$$

10.3.8. Damped Sine and Cosine Functions

$$f(t) = e^{-\alpha t} \sin \beta t$$

$$f(t) = e^{-\alpha t}\left[\frac{e^{j\beta t} - e^{-j\beta t}}{2j}\right] = \frac{1}{2j}\left[\mathcal{L}\, e^{j\beta t}\, e^{-\alpha t} - \mathcal{L}\, e^{-j\beta t} e^{-\alpha t}\right]$$

$$= \frac{1}{2j}\left[\int_0^\infty e^{j\beta t} e^{-\alpha t} e^{-st}\, dt - \int_0^\infty e^{-j\beta} e^{-\alpha t}\, e^{-st}\, dt\right]$$

$$= \frac{1}{2j}\left[\int_0^\infty e^{-t(s+\alpha-j\beta)} dt - \int_0^\infty e^{-t(s+\alpha+j\beta)}\right]$$

$$= \frac{1}{2j}\left[\frac{1}{s+\alpha-j\beta} - \frac{1}{s+\alpha+j\beta}\right] = \frac{\beta}{(s+\alpha)^2+\beta^2}$$

$$\boxed{\mathcal{L}\, e^{-\alpha t} \sin \beta t = \frac{\beta}{(s+\alpha)^2+\beta^2}}$$

$$f(t) = e^{-\alpha t} \cos \beta t$$

$$\mathcal{L} f(t) = F(s) = \mathcal{L}\, e^{-\alpha t} \cos \beta t = \mathcal{L}\, e^{-\alpha t}\, \frac{e^{j\beta t} + e^{-j\beta t}}{2}$$

$$= \frac{1}{2}\left[\int_0^\infty e^{-t(s+\alpha-j\beta)}\, dt + \int_0^\infty e^{-t(s+\alpha+j\beta)}\, dt\right]$$

$$= \frac{1}{2}\left[\frac{1}{s+\alpha-j\beta} + \frac{1}{s+\alpha+j\beta}\right] = \frac{s+\alpha}{(s+\alpha)^2+\beta^2}$$

$$\boxed{\mathcal{L}\, e^{-\alpha t} \cos \beta t = \frac{s+\alpha}{(s+\alpha)^2+\beta^2}}$$

Example 10.1. Find the Laplace transform of

$$f(t) = e^{at} \sin \omega t$$

Solution : $\mathcal{L}\, e^{at} \sin \omega t = \frac{1}{2j}[\mathcal{L}\, e^{j\omega t} e^{-at} - \mathcal{L}\, e^{-j\omega t} e^{at}]$

$\because$ $\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$

$$= \frac{1}{2j}\left[\int_0^\infty e^{j\omega t} e^{at} e^{-st} dt - \int_0^\infty e^{-j\omega t} e^{-at} e^{-st} dt\right]$$

$$= \frac{1}{2j}\left[\int_0^\infty e^{-t(s-a-j\omega)} dt + \int_0^\infty e^{-t(s-a+j\omega)}\, dt\right]$$

$$= \frac{1}{2j}\left[\frac{1}{s-a-j\omega} - \frac{1}{s-a+j\omega}\right] = \frac{\omega}{(s-a)^2+\omega^2}$$

$$\boxed{\mathcal{L}\, e^{at} \sin \omega t = \frac{\omega}{(s-a)^2+\omega^2}}$$

10.4. PROPERTIES OF LAPLACE TRANSFORMS

1. If 'a' is a constant and $F(s)$ is the Laplace transform of $f(t)$ then
$$\mathcal{L}\, af(t) = a\,\mathcal{L} f(t) = aF(s)$$
2. If $f_1(t)$ and $f_2(t)$ are the time functions and $F_1(s)$, $F_2(s)$ are their Laplace transform then
$$\mathcal{L}[f_1(t) \pm f_2(t)] = \mathcal{L} f_1(t) \pm \mathcal{L} f_2(t) = F_1(s) + F_2(s)$$
3. Time shift property : If 'a' is a positive real number and $F(s)$ is the Laplace transform of $f(t)$ then
$$\mathcal{L}[f(t-a)] = e^{-as} F(s)$$
Thus, if the original time function is shifted to the right by 'a' on time axis, its laplace transform is equal to the original laplace transform multiplied by e^{-as}. This is the time shifting theorem.
4. If $F(s)$ is the Laplace transform of $f(t)$ then
$$\mathcal{L}\, e^{at} f(t) = F(s-a)$$
$$\mathcal{L}\, e^{-at} f(t) = F(s+a)$$
e.g., $\mathcal{L}\, e^{-at} \sin \omega t$

since $\mathcal{L} \sin \omega t = \frac{\omega}{s^2+\omega^2}$

$\therefore$ $\mathcal{L}\, e^{-at} \sin\omega t = F(s+a) = \frac{\omega}{(s+a)^2+\omega^2}$

5. Real differentiation theorem:
The Laplace transform of the derivative of a function $f(t)$ is given by
$$\mathcal{L}\left[\frac{d}{dt} f(t)\right] = sF(s) - f(o)$$
where $f(o)$ is the initial value of $f(t)$ at $t = 0$
6. Laplace transform of higher derivative
$$\mathcal{L}\left[\frac{d^n f(t)}{dt^n}\right] = S^n F(s) - s^{n-1} f(o) \ldots\ldots - \frac{d^{n-1} f(o)}{dt^{n-1}}$$
where $f(o)$ is the value of $f(t)$ at $t = 0$
$\frac{d^{n-1} f(o)}{dt^{n-1}}$ is the value of $\frac{d^{n-1} f(t)}{dt^{n-1}}$ at $t = 0$ and so on.
7. Laplace transform of integral of $f(t)$
$$\mathcal{L}\left[\int f(t)\, dt\right] = \frac{F(s)}{s} + \frac{f^{-1}(o)}{s}$$
where $F(s)$ is the laplace transform of $f(t)$ and

$$f^{-1}(o) = \int f(t)\,dt \quad at\ t = 0$$

8. Complex differentiation theorem :

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} F(s)$$

e.g., Find the Laplace transform of t sinh at

$$\mathcal{L} \sinh at = \frac{a}{s^2 - a^2}$$

$$\therefore \quad \mathcal{L}[t \sin h\, at] = -\frac{d}{ds}\left[\frac{a}{s^2 - a^2}\right]$$

$$\mathcal{L}[t \sinh at] = \frac{2as}{(s^2 - a^2)^2}$$

9. Complex integration :
If $F(s)$ is the laplace transform of $f(t)$ then

$$\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_0^\infty F(s)\,ds$$

e.g., find the Laplace transform of $\frac{\sin 2t}{t}$

$$\mathcal{L} \sin 2t = \frac{2}{s^2 + 4}$$

$$\mathcal{L}\frac{\sin 2t}{t} = \int_s^\infty \frac{2}{s^2 + 4}\,ds = 2 \cdot \frac{1}{2}\left[\tan^{-1}\frac{s}{2}\right]_s^\infty$$

$$= \left[\tan^{-1}\infty - \tan^{-1}\frac{s}{2}\right] \qquad \because \int \frac{1}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$$

$$= \frac{\pi}{2} - \tan^{-1}\frac{s}{2}$$

10. Initial value theorem:
If initial value of the function $f(t)$ is required, then it can be calculated by initial value theorem which states

$$\lim_{t \to 0} f(t) = \lim_{s \to \infty} sF(s)$$

e.g., Find the initial value of e^{-at}

$$f(t) = e^{-at}$$

$$\therefore \quad F(s) = \mathcal{L} f(t) = \frac{1}{s + a}$$

$$\lim_{t \to 0} f(t) = \lim_{s \to \infty} s \cdot \frac{1}{s + a} = \lim_{s \to \infty} \frac{1}{1 + \frac{a}{s}} = 1$$

11. Final value theorem :
If final value of $f(t)$ is required then it can be calculated by final value theorem, which states that

$$\lim_{s \to \infty} f(t) = \lim_{s \to 0} sF(s)$$

e.g., find the final value of e^{-at}

$$f(t) = e^{-at}$$

$$F(s) = \mathcal{L} f(t) = \frac{1}{s + a}$$

$$\lim_{t \to \infty} f(t) = \lim_{s \to 0} s \cdot \frac{1}{s + a} = 0$$

10.5. INVERSE LAPLACE TRANSFORM

If $F(s)$ is the Laplace transform of $f(t)$ then

$$\mathcal{L}^{-1} F(s) = f(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(s)\, e^{st} ds \qquad ...(10.2)$$

But this method is not generally used. In practice if $F(s)$ is a rational function obtained the partial fraction and take the inverse Laplace with the help of standard table of transform pairs.

Example 10.2. Find the inverse Laplace transform of $\frac{2}{s(s+2)}$.

Solution : Given: $F(s) = \frac{2}{s(s+2)}$ ...(10.3)

The partial fraction of $F(s)$ is

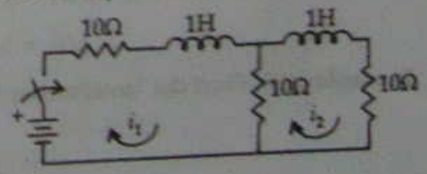
$$F(s) = \frac{1}{s} - \frac{1}{s+2}$$

$$f(t) = \mathcal{L}^{-1} F(s) = \mathcal{L}^{-1}\frac{1}{s} - \mathcal{L}^{-1}\frac{1}{s+2} = 1 - e^{-2t} \qquad \textbf{Ans.}$$

10.6. APPLICATION OF LAPLACE TRANSFORM IN CONTROL SYSTEM

In control systems the system is described by the integrodifferential equations. It is very difficult to solve the such equations in time domain. The laplace transforms of such equations converts them into simple algebaric equations. Then by using the inverse laplace time domain function can be obtained.

Example 10.3. Find i_2 by using Laplace transformation.



Solution : For mesh 1:

$$\frac{di_1}{dt} + 20i_1 - 10i_2 = 100\,u(t) \qquad ...(10.4)$$

For mesh 2 :

$$\frac{di_2}{dt} + 20i_2 - 10i_1 = 0 \qquad ...(10.5)$$

Laplace transform of (10.4) & (10.5)

$$sI_1(s) + 20I_1(s) - 10\,I_2(s) = \frac{100}{s}$$

$$-10\,I_1(s) + sI_2(s) + 20\,I_2(s) = 0$$

equation (10.6) and (10.7) can be written as ...(10.6) ...(10.7)

$$I_1(s)\,[s + 20] - 10\,I_2(s) = \frac{100}{S}$$

$$-I_1(s)\,10 + (s + 20)\,I_2(s) = 0$$ ...(10.8) ...(10.9)

Solve (10.8) and (10.9)

$$I_2(s) = \frac{1000}{s(s^2+40s+300)}$$ ...(10.10)

or, $$I_2(s) = \frac{1000}{s(s^2+40s+300)}$$

$$= \frac{3.33}{s} - \frac{5}{s+10} + \frac{1.67}{s+30}$$ ...(10.11)

Inverse laplace of 10.11

$$\mathcal{L}^{-1}\,I_2(s) = i_2(t) = \mathcal{L}^{-1}\frac{3.33}{s} - \mathcal{L}^{-1}\frac{5}{s+10} + \mathcal{L}^{-1}\frac{1.67}{s+30}$$

$$= 3.33 - 5\,e^{-10t} + 1.67e^{-30t}$$ **Ans.**

Example 10.4. Obtian the solution of differential equation given below

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 5$$ **assuming zero initial conditions.**

Solution : $s^2\,y(s) + 3sy(s) + 2y(s) = \frac{5}{s}$

$$y(s) = \frac{5}{s(s^2+3s+2)}$$

$$y(s) = \frac{5}{s(s+1)(s+2)} = \frac{2.5}{s} - \frac{5}{s+1} - \frac{2.5}{s+2}$$

$$\mathcal{L}^{-1}\,y(s) = y(t) = \mathcal{L}^{-1}\frac{2.5}{s} - \mathcal{L}^{-1}\frac{5}{s+1} - \mathcal{L}^{-1}\frac{2.5}{s+2}$$

$$y(t) = 2.5\,u(t) - 5\,e^{-t} - 2.5\,e^{-2t}$$ **Ans.**

Example 10.5. Find the inverse Laplace transform of

$$\frac{s+4}{s(s-1)(s^2+4)}$$

Solution : Resolve the given function into partial fraction

$$\frac{s+4}{s(s-1)(s^2+4)} = \frac{A}{s} + \frac{B}{s-1} + \frac{Cs+D}{s^2+4}$$

Put $s = 0, \quad A = -1$

$s = 1, \quad B = 1$

$C = 0, \quad D = -1$

Similarly,

$$\frac{s+4}{s(s-1)(s^2+4)} = \frac{-1}{s} + \frac{1}{s-1} - \frac{1}{s^2+4}$$

$$\mathcal{L}^{-1}\left[\frac{s+4}{s(s-1)(s^2+4)}\right] = \mathcal{L}^{-1}(-1/s) + \mathcal{L}^{-1}\left(\frac{1}{s-1}\right) - \mathcal{L}^{-1}\frac{1}{s^2+4}$$

$$= -1 + e^t - \frac{1}{2}\sin 2t$$ **Ans.**

Example 10.6. What is Laplace transform of the function $f(t)$ shown in fig. 10.1.

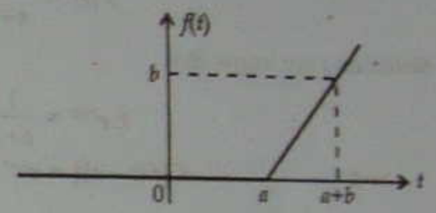


Fig. 10.1.

Solution : The given waveform is a straight line, so, first calculate $f(t)$ using the equation.

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$ ...(10.12)

From fig. 10.1 : $x_1 = a, \quad x_2 = a + b$

$y_1 = 0, \quad y_2 = b$

Put all these values in eq. 10.12, we get

$$f(t) = t - a$$ ...(10.13)

Taking laplace transform of equation 10.13, we get

$$F(s) = \frac{1}{s^2} - \frac{a}{s} = \frac{1-as}{s^2}$$ **Ans.**

Example 10.7. Obtain the Laplace transform of the waveform shown in fig. 10.3.

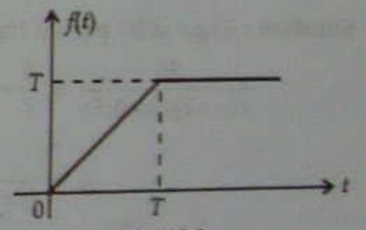


Fig. 10.2.

Solution: The given waveform can be divided into two parts

$$f(t) = f_1(t) + f_2(t)$$

$f_1(t)$ can be calculated with the help of equation 10.12

$\therefore$

$$f_1(t) = t \qquad 0 < t < T$$

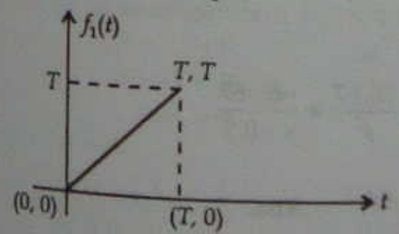


Fig. 10.2(a)

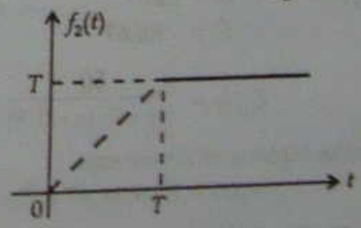


Fig. 10.2(b)

$f_2(t)$ is shifted unit step function given by

$$f_2(t) = T\,u(t - T) \qquad T < t < \infty$$

$$f(t) = \begin{cases} t & 0 < t < T \\ Tu(t-T) & t < t < \infty \end{cases}$$ ...(10.4)

∴ Laplace transform of 10.14 will be

$$F_1(s) = \int_0^T t\, e^{-st} dt = \frac{-1}{s} T e^{-sT} + \frac{1}{s^2}(1 - e^{-sT})$$

$$F_2^*(s) = \mathcal{L}\{T\, u(t-T)\} = e^{-sT} \cdot \frac{1}{s} \cdot T$$

$$(s) = F_1(s) + F_2(s) = \frac{1}{s^2}(1 - e^{-sT}) \quad \textbf{Ans.}$$

Example 10.8. Obtain the inverse Laplace transform of the following function.

$$F(s) = \frac{5e^{-s}}{s+1}$$

Solution : we know that

$$\mathcal{L}\, e^{-at} = \frac{1}{s+a}$$

and $$\mathcal{L}[f(t-a)] = e^{-as} F(s)$$

$$\therefore \quad F(s) = 5e^{-s} \frac{1}{S+1}$$

Take the inverse laplace transform

$$f(t) = 5\, e^{-(t-1)} \quad \textbf{Ans.}$$

Example 10.9. Find the Laplace transform of the given function

$$E_0(s) = \frac{50}{s(s+2)(s+0.5)}$$

Solution : Expand by partial fraction

$$\frac{50}{s(s+2)(s+0.5)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+0.5}$$

$$A = \left.\frac{50}{(s+2)(s+0.5)}\right|_{s=0} \quad \therefore \quad A = 50$$

Similarly, $B = 16.67$

$C = -66.67$

$$E_0(s) = \frac{50}{s(s+2)(s+0.5)} = \frac{50}{s} + \frac{16.67}{s^2} + \frac{-66.67}{s+0.5}$$

The inverse laplace of above eqn

$$e_0(t) = 50 + 16.67\, e^{-2t} - 66.67\, e^{-0.5t} \quad \textbf{Ans.}$$

Example 10.10. Find the inverse Laplace transform of

(*a*) $$F_1(s) = \frac{6S+3}{s^2}$$

(*b*) $$F_2(s) = \frac{5s+2}{(s+1)(s+2)^2}$$

*Using time shift property from art. 10.4

Solution : $$F_1(s) = \frac{6s+3}{s^2} = \frac{6}{s} + \frac{3}{s^2}$$

Taking inverse Laplace on both sides we have

$$f_1(t) = 6u(t) + 3t \quad \text{for} \quad t \geq 0 \quad \textbf{Ans.}$$

(*b*) $$F_2(s) = \frac{5s+2}{(s+1)(s+2)^2}$$

Solution : Expand by partial fraction

$$F_2(s) = \frac{5s+2}{(s+1)(s+2)^2} = \frac{A_1}{(s+2)} + \frac{A_2}{(s+2)^2} + \frac{B}{s+1}$$

$$B = (s+1)\, F_2(s)\big|_{s=-1} = -3$$

$$A_2 = (s+2)^2\, F_2(s)\big|_{s=-2} = 8$$

$$A_1 = \frac{d}{ds}\left.\frac{5s+2}{s+1}\right|_{s=-2} = 3$$

$$\therefore \quad F_2(s) = \frac{3}{s+2} + \frac{8}{(s+2)^2} + \frac{-3}{s+1}$$

Inverse Laplace of above equation

$$f_2(t) = 3\, e^{-2t} + 8\, t\, e^{-2t} - 3\, e^{-t} \quad t \geq 0 \quad \textbf{Ans.}$$

Example 10.11. Find the inverse laplace transform of

$$F(s) = \frac{1}{s^2(s^2+\omega^2)}$$

Solution : Expand $F(s)$ by partial fraction

$$F(s) = \frac{1}{s^2(s^2+\omega^2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+\omega^2}$$

$$\therefore \quad 1 = A\, s\,(s^2+\omega^2) + B\,(s^2+\omega^2) + (Cs+D)\, s^2$$

Comparing the coefficients of s^3, s^2, s, s^0 we have

$A + C = 0$, $\omega^2 B = 1$

$B + D = 0$

$A\omega^2 = 0$

Thus, $A = 0$

$B = 1/\omega^2$

$C = 0$

$D = -1/\omega^2$

$$\therefore \quad F(s) = \frac{1}{\omega^2 s^2} - \frac{1}{\omega^2(s^2+\omega^2)}$$

$$= \frac{1}{\omega^2 s^2} - \frac{\omega}{\omega^3} \cdot \frac{1}{s^2+\omega^2}$$

Inverse Laplace transform of above equation

$$f(t) = \frac{t}{\omega^2} - \frac{1}{\omega^3}\sin\omega t \qquad t \geq 0$$

$$f(t) = \frac{1}{\omega^2}\left(t - \frac{\sin\omega t}{\omega}\right) \qquad \textbf{Ans.}$$

Example 10.12. Obtain the inverse Laplace transform of the given function

$$F(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

Solution: Expand the given function F(s) by partial fraction.

$$\frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n{}^2 = A(s^2 + 2\zeta\omega_n s + \omega_n{}^2) + (Bs + C)s$$

Comparing the coefficient, we have

$$A + B = 0$$
$$2\zeta\omega_n A + C = 0$$
$$\omega_n{}^2 A = \omega_n{}^2$$
$$\therefore \quad A = 1, B = -1, C = -2\zeta\omega_n$$

$$\therefore \quad F(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s} - \frac{s + \zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} - \frac{\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2 - \zeta^2\omega_n^2} - \frac{\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2 - \zeta^2\omega_n^2}$$

$$= \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} - \frac{\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

Since, $1 - \zeta^2$ is a positive quantity for $0 < \zeta < 1$

therefore $\sqrt{1-\zeta^2}$ is also a positive quantity.

$$\therefore \quad F(S) = \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta w_n)^2 + \left(\omega_n\sqrt{1-\zeta^2}\right)^2} - \frac{\zeta\omega_n\sqrt{1-\zeta^2}}{\left[(s + \zeta\omega_n)^2 + \left(w_n\sqrt{1-\zeta^2}\right)^2\right]\sqrt{1-\zeta^2}}$$

Taking the inverse laplace transform

$$f(t) = 1 - e^{-\zeta\omega_n t}\cos\left(\omega_n\sqrt{1-\zeta^2}\,t\right) - \frac{\zeta e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_n\sqrt{1-\zeta^2}\right)t$$

Example 10.13. Obtain the solution of the differential equation

$$\dot{x} + ax = A\sin\omega t,\ x(o) = b$$

Solution : The given differential equation

$$\dot{x} + ax = A\sin\omega t$$

$$sX(s) - x(0) + aX(s) = \frac{A\omega}{s^2 + \omega^2}$$

given that $x(0) = b$

$$sX(s) - b + aX(s) = \frac{A\omega}{s^2 + \omega^2}$$

$$X(s) = \frac{A\omega}{(s^2 + \omega^2)(s + a)} - \frac{(a - b)}{(s + a)} \qquad \ldots(10.15)$$

$$\therefore$$

let $$\frac{A\omega}{(s^2+\omega^2)(s+a)} = \frac{K_1 s + K_2}{s^2 + \omega^2} + \frac{K_3}{s + a}$$

$$\frac{A\omega}{(s^2+\omega^2)(s+a)} = \frac{(K_1 s + K_2)(s + a) + K_3(s^2 + \omega^2)}{(s^2 + \omega^2)(s + a)}$$

Comparing Numerator on both sides and hence coefficients of s^2, s and s^0 we have

$$K_1 + K_3 = 0$$
$$aK_1 + K_2 = 0$$
$$aK_2 + K_3\omega^2 = A\omega$$

$$\therefore \quad K_1 = -\frac{A\omega}{a^2 + \omega^2},\ K_2 = \frac{a\omega A}{a^2 + \omega^2} \text{ and } K_3 = \frac{A\omega}{a^2 + \omega^2}$$

Now put these values in equation (10.15) we get

$$X(s) = -\frac{A\omega}{a^2 + \omega^2}\cdot\frac{s}{s^2 + \omega^2} + \frac{aA}{a^2 + \omega^2}\cdot\frac{\omega}{s^2 + \omega^2} + \left(\frac{A\omega}{a^2 + \omega^2} + b - a\right)\cdot\frac{1}{s + a}$$

Now apply inverse laplace transform on both sides

$$x(t) = -\frac{A\omega}{a^2 + \omega^2}\cos\omega t + \frac{aA}{a^2 + \omega^2}\sin\omega t + \left(\frac{A\omega}{a^2 + \omega^2} + b - a\right)e^{-at}$$

$$x(t) = A\left[\frac{a\sin\omega t}{a^2 + \omega^2} - \frac{\omega\cos\omega t}{a^2 + \omega^2}\right] + \left[\frac{A\omega}{a^2 + \omega^2} + b - a\right]e^{-at} \quad \textbf{Ans.}$$

Example 10.14. Solve the differential equation

$$\frac{d^2x(t)}{dt^2} + 2\frac{d}{dt}x(t) + 10\,x(t) = e^{-t}$$

when $$x(0) = 0$$

$$\frac{d}{dt}x(0) = 0$$

The forcing function e^{-t} is given at $t = 0$ when the system is at rest.

Solution : The given equation is

$$\frac{d^2x(t)}{dt^2} + 2\frac{d}{dt}x(t) + 10\,x(t) = e^{-t} \qquad \ldots(10.16)$$

Taking laplace transform on both sides of equation (10.16)

$$[s^2X(s) - sx'(0_+) - x(0+)] + 2[sX(s) - x(0_+)] + 10X(s) = \frac{1}{s+1}$$

Put the initial conditions

$$s^2X(s) + 2sX(s) + 10\,X(s) = \frac{1}{s+1}$$

$$\therefore \quad X(s) = \frac{1}{(s+1)(s^2+2s+10)}$$

Expand the equation 10.17 by partial fraction

$$\frac{1}{(s+1)(s^2+2s+10)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+2s+10}$$

$$1 = A(s^2 + 2s + 10) + (Bs + C)(s + 1)$$

Comparing the coefficients of s^2, s and s^0 we have

$$A + B = 0$$
$$2A + B + C = 0$$
$$10A + C = 1$$

$$\therefore \quad A = 1/9,\ B = -1/9 \text{ and } C = -1/9$$

put all these values in equation 10.18 we get

$$X(s) = \frac{1}{9(s+1)} - \frac{s+1}{9[(s+1)^2+9]}$$

Using, $\quad \mathcal{L}\,e^{-at} = \frac{1}{s+a}$

$$\mathcal{L}[e^{-at} f(t)] = F(s+a)$$

$$\mathcal{L}[e^{-at}\cos\omega t] = \frac{s+a}{(s+a)^2+\omega^2}$$

Taking inverse laplace of equation (10.19)

$$x(t) = \frac{1}{9}e^{-t} - \frac{1}{9}e^{-t}\cos 3t \quad \textbf{Ans.}$$

Example 10.15. For the circuit shown in fig. 10.3, find the current at $t > 0$. The switch is closed at $t = 0$. Assume no initial charge on the capacitor.

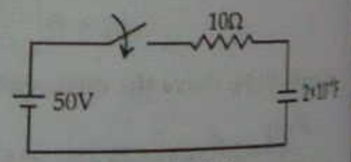


Fig. 10.3.

Solution : From the given fig. $Ri + \frac{1}{C}\int_{-\infty}^{t} i\,dt = 50$

$$10i + \frac{1}{2\times10^{-4}}\int_{0}^{t} i\,dt = 50 \qquad \text{...(10.20)}$$

(because for $t < 0$ there is no charge on capacitor)

laplace transform of eqn. (10.20)

$$10I(s) + \frac{1}{s(2\times10^{-4})}I(s) = \frac{50}{s}$$

$$I(s)\left[\frac{s(2\times10^{-3})+1}{s(2\times10^{-4})}\right] = \frac{50}{s}$$

$$I(s) = \frac{5}{s+500} \qquad \text{...(10.21)}$$

Inverse laplace transform of equation 10.21

$$i(t) = 5\,e^{-500t} \quad \textbf{Ans.}$$

Example 10.16. In RL circuit the input is given by $Vs = 10t$. Determine the response of the circuit *i.e.*, $i(t)$ when $R = 50\ \Omega$ and $L = 5H$.

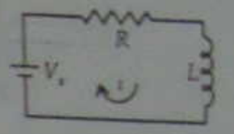


Fig. 10.4.

Solution: The differential equation of the given circuit

$$Ri + L\frac{di}{dt} = V_s$$

$$50i + 5\frac{di}{dt} = 10\,t \qquad \text{...(10.22)}$$

laplace transform of eqn. 10.22

$$50\,I(s) + 5\,I(s) = \frac{10}{s^2}$$

$$I(s) = \frac{2}{s^2(s+10)} \qquad \text{...(10.23)}$$

Expand the equation 10.23 by partial fraction

$$I(s) = \frac{2}{s^2(s+10)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+10} \qquad \text{...(10.24)}$$

$$2 = A\,s(s+10) + B(s+10) + Cs^2$$

compare the coefficients of s^3, s^2, s and s^0

$$10A + B = 0$$
$$A + C = 0$$
$$10B = 2$$
$$\therefore \quad B = 0.2$$
$$\therefore \quad A = -0.02$$
$$C = 0.02$$

equation 10.24 becomes

$$I(s) = \frac{-0.02}{s} + \frac{0.2}{s^2} + \frac{0.02}{s+10} \qquad \text{...(10.25)}$$

Inverse Laplace transform of 10.15

$$i(t) = -0.02 + 0.2t + 0.02\,e^{-10t}$$
$$i(t) = 0.02\,e^{-10t} + 0.2t - 0.02 \quad \textbf{Ans.}$$

Example 10.17. Express the pulse shown in fig. 10.5 as a combination of step functions and obtain its laplace transform.

Solution :

$$f(t) = f_1(t) + f_2(t)$$
$$f_1(t) = Au_{-1}(t - t_o) \quad \text{(shifted step function)}$$
$$f_2(t) = -Au_{-1}[t - (t_o + T)]$$
$$f(t) = f_1(t) - f_2(t)$$
$$= Au_{-1}(t - t_o) - A\,u_{-1}[t - (t_o + T)]$$
$$\mathcal{L}f(t) = \mathcal{L}\,Au_{-1}(t - t_o) - \mathcal{L}Au_{-1}[t - (t_o + T)]$$

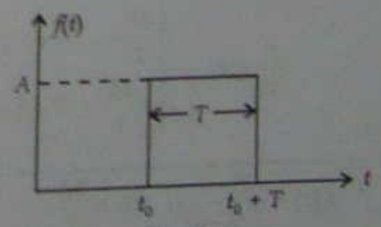


Fig. 10.5.

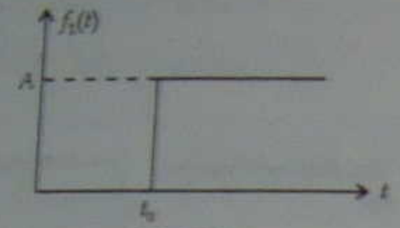


Fig. 10.5(a)

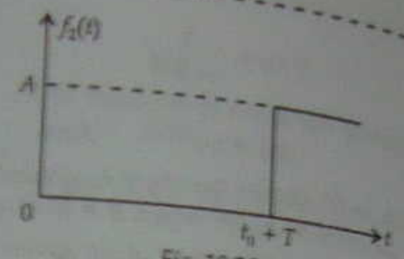


Fig. 10.5(b)

$$F(s) = A\cdot\frac{1}{s}e^{-st_0} - A\frac{1}{s}e^{-s(t_0+T)}$$

$$F(s) = \frac{A}{s}\left[e^{-st_0} - e^{-s(t_0+T)}\right] \quad \textbf{Ans.}$$

10.7. LAPLACE TRANSFORM OF PERIODIC FUNCTIONS

Theorem : The laplace transform of a periodic function with time period T is equal to the product of $\frac{1}{1-e^{-Ts}}$ and laplace transform of the first cycle.

Example 10.18. Find the laplace transform of the periodic triangular wave in fig. 10.6.

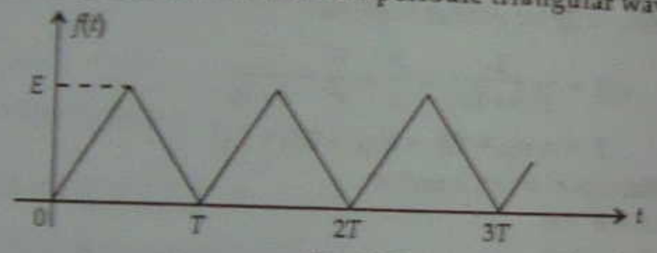


Fig. 10.6.

Solution : To find laplace transform of a periodic wave, first evaluate $f(t)$ for one cycle i.e. finite waveform shown in fig. (10.6 *a*).

$$g(t) = \begin{cases} \frac{2E}{T}t^* & 0<t<T/2 \\ -\frac{2E}{T}t+2E^* & T/2<t<T \\ 0 & \text{otherwise} \end{cases}$$

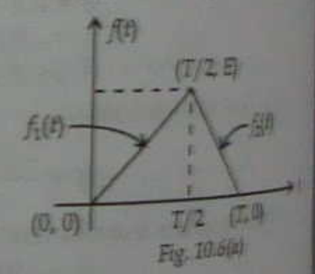


Fig. 10.6(a)

$$\frac{d}{dt}g(t) = \begin{cases} \frac{2E}{T} & 0<t<T/2 \\ -\frac{2E}{T} & T/2<t<T \end{cases} \quad ...(10.26)$$

* $f_1(t) = \frac{E-0}{T/2-0}(t-0) = \frac{2E}{T}t$

$f_2(t) = \frac{E-0}{T/2-0}(t-T) = -\frac{2E}{T}t + 2E$

The waveform of equation 10.26 is shown in fig. 10.6(*b*).

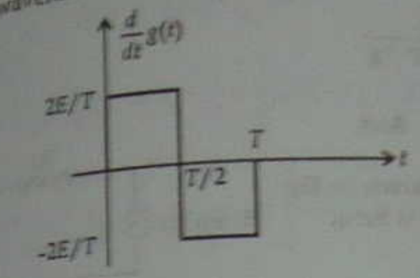


Fig. 10.6(b)

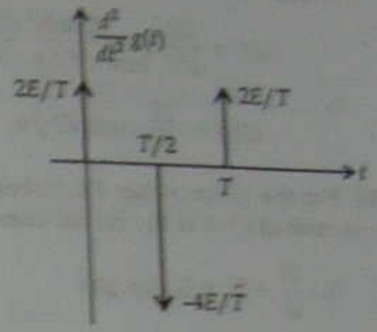


Fig. 10.6(c)

From Fig. 10.6 (*c*)

$$\frac{d^2}{dt^2}g(t) = \frac{2E}{T}\delta(t) - \frac{4E}{T}\delta(t-T/2) + \frac{2E}{T}\delta(t-T) \quad ...(10.27)$$

Take Laplace transform of eqn 10.27

$$s^2 G(s) = \frac{2E}{T} - \frac{4E}{T}e^{-sT/2} + \frac{2E}{T}e^{-sT}.1$$

$$G(s) = \frac{2E}{Ts^2}\left(1-2e^{-sT/2}+e^{-sT}\right) \quad ...(10.28)$$

Therefore, Laplace transform of periodic function will be

$$G(s) = G_1(s)\cdot\frac{1}{1-e^{-sT}}$$

$$G(s) = \frac{2E}{s^2T(1-e^{-sT})}(1-2e^{-sT/2}+e^{-sT}) = \frac{2E}{s^2T}\left(\frac{1-2e^{-sT/2}+e^{-sT}}{1-e^{-sT}}\right)$$

$$= \frac{2E}{s^2T}\frac{e^{-sT/2}(e^{sT/2}-2+e^{-sT/2})}{e^{-sT/2}(e^{sT/2}-e^{-sT/2})}$$

$$= \frac{2E}{s^2T}\left[\frac{e^{sT/2}+e^{-sT/2}-2}{e^{sT/2}-e^{-sT/2}}\right] \quad ...(10.29)$$

Since $\frac{e^x+e^{-x}}{2} = \cosh x$

$\therefore\ e^x + e^{-x} = 2\cosh x$

then the eq^n 10.29 can be written as

$$G(s) = \frac{2E}{s^2T}\left[\frac{e^{sT/2}+e^{-sT/2}-2}{\frac{e^{sT/2}-e^{-sT/2}}{2}\cdot 2}\right] = \frac{2E}{s^2T}\left[\frac{2\cosh sT/2-1}{2\sinh sT/2}\right]$$

$$= \frac{2E}{s^2T}\left[\frac{\cosh sT/2-1}{\sinh sT/2}\right] \quad ...(10.30)$$

Also, $\cosh x - 1 = 2\sin^2 h\, x/2$

$$\therefore \quad G(s) = \frac{2E}{s^2T}\,\frac{2\sinh^2 sT/4}{2\sinh T/4\cosh sT/4}$$

$$G(s) = \frac{2E}{s^2T}\tanh sT/4 \qquad \textbf{Ans.}$$

Example 10.19. For the given series *RL* circuit shown in fig. 10.7 find the current at $t > 0$ if the initial current is zero.

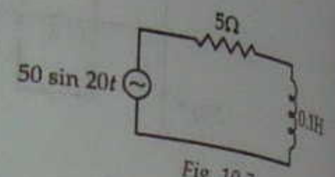


Fig. 10.7.

Solution : $0.1\frac{di}{dt} + 5i = 50\sin 20\,t$

or, $\frac{di}{dt} + 50i = 500\sin 20\,t$

Take laplace transform of equation (10.31) ...(10.31)

$$sI(s) + 50\,I(s) = 500\cdot\frac{20}{s^2+400} \qquad ...(10.32)$$

$$I(s) = \frac{10000}{(s+50)(s^2+400)} \qquad ...(10.35)$$

$$\frac{10000}{(s+50)(s^2+400)} = \frac{A}{s+50} + \frac{Bs+C}{s^2+400} \qquad ...(10.34)$$

from 10.34 we get

$A = 3.4$

$B = -3.4$

$C = 172.4$

then the equation 10.34 becomes

$$\frac{10000}{(s+50)(s^2+400)} = \frac{3.4}{s+50} - 3.4\frac{s}{s^2+400} + \frac{172.4}{s^2+400}$$

$$= \frac{3.4}{s+50} - 3.4\frac{s}{s^2+400} + 8.62\frac{20}{s^2+400}$$

Take inverse laplace transform of above eqn.

$$i(t) = 3.4\,e^{-50t} - 3.4\cos 20\,t + 8.62\sin 20\,t \qquad \textbf{Ans.}$$

10.8. CONVOLUTION THEOREM

If $\mathcal{L} f_1(t) = F_1(s)$ and $\mathcal{L} f_2(t) = F_2(s)$ then

$$\mathcal{L}\left[\int_0^t f_1(t-\tau)\,f_2(\tau)\,d\tau\right] = \mathcal{L}\left[\int_0^t f_1(\tau)\,f_2(t-\tau)\,d\tau\right] = F_1(s)\,F_2(s)$$

Example 10.20. Find $\mathcal{L}^{-1}\frac{1}{(s^2+a^2)^2}$

Solution: $F_1(s) = F_2(s) = \frac{1}{s^2+a^2}$

$$f_1(t) = f_2(t) = \frac{1}{a}\sin at$$

$$\mathcal{L}^{-1}\frac{1}{(s^2+a^2)^2} = f_1 * f_2 = \int_0^t f_1(t-\tau)\,f_2(\tau)\,d\tau = \int_0^t\left[\frac{1}{a}\sin a(t-\tau)\right]\left[\frac{1}{a}\sin a\tau\right]d\tau$$

$$\mathcal{L}^{-1}\frac{1}{(s^2+a^2)^2} = \frac{1}{a^2}\left[\sin at\int_0^t \cos a\tau\sin a\tau\,d\tau - \cos at\int_0^t \sin^2 a\tau\,d\tau\right]$$

$$= \frac{1}{a^3}(\sin at - at\cos at) \qquad \textbf{Ans.}$$

Example 10.21. Find i_2 by using laplace transformation.

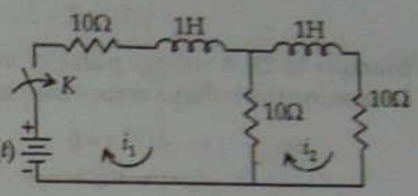


Fig. 10.8.

Solution : For mesh (1) :

$$\frac{di_1}{dt} + 20i_1 - 10i_2 = 100\,u(t) \qquad ...(10.35)$$

For mesh (2)

$$\frac{di_2}{dt} + 20i_2 - 10i_1 = 0 \qquad ...(10.36)$$

The initial values of i_1 and i_2 are zero. The laplace transform of eqn 10.35 and 10.36.

$$(s+20)\,I_1(s) - 10I_2(s) = \frac{100}{s} \qquad ...(10.37)$$

$$-10\,I_1(s) + (s+20)\,I_2(s) = 0 \qquad ...(10.38)$$

Solve 10.37 and 10.38 for $I_2(s)$

$$I_2(s) = \frac{1000}{s(s^2+40s+300)} \qquad ...(10.39)$$

Expand (10.39) by partial fraction

$$I_2(s) = \frac{3.33}{s} - \frac{5}{s+10} + \frac{1.67}{s+30} \qquad ...(10.40)$$

Taking inverse laplace transform of eqn 10.40

$$\mathcal{L}^{-1}\,I_2(s) = 3.33\,\mathcal{L}^{-1}(1/s) - 5\,\mathcal{L}^{-1}\frac{1}{s+10} + 1.67\,\mathcal{L}^{-1}\frac{1}{s+30}$$

$$i_2(t) = 3.33 - 5\,e^{-10t} + 1.67\,e^{-30t} \qquad \textbf{Ans.}$$

Example 10.22. Solve the differential equation

$$\ddot{y}(t) + 4\dot{y}(t) + 3y(t) = 6$$

if

$$y(0) = \dot{y}(0) = 0$$

Solution : The given equation is

$$\ddot{y}(t) + 4\dot{y}(t) + 3y(t) = 6 \qquad ...(10.41)$$

laplace transform of eqn. 10.41

$$[s^2Y(s) - y(0) - \dot{y}(0)] + 4\,[sY(s) - y(0)] + 3Y(s) = \frac{6}{s}$$

Put the initial values

$$Y(s)(s^2 + 4s + 3) = \frac{6}{S}$$

$$Y(s) = \frac{6}{s(s^2+4s+3)}$$

Expand the equation (10.42) by partial fraction

$$Y(s) = \frac{6}{s(s^2+4s+3)} = \frac{2}{s} - \frac{3}{s+1} + \frac{1}{s+3}$$

Take inverse laplace of eqn (10.43)

$$y(t) = 2 - 3e^{-t} + e^{-3t}$$ Ans.

Example 10.23. A voltage pulse of unit height and width T is applied to the RC circuit at $t = 0$. Determine the voltage across the capacitance C as a function of time.

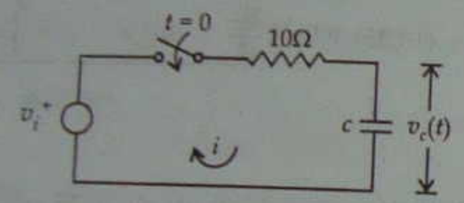


Fig. 10.9(a)

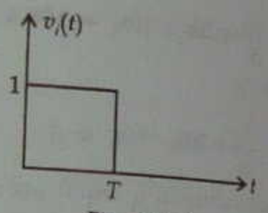


Fig. 10.9(b)

Solution : From fig 10.9 *(b)*

$$^*v_i(t) = u(t) - u(t - T)$$

Now at $t = 0$, switch will be closed, the circuit is shown in fig 10.9 (c)

Applying KVL in the circuit, we have

$$v_i(t) = Ri(t) + \frac{1}{C}\int_{-\infty}^{t} i(t)\,dt \quad ...(10.45)$$

$$\frac{1}{C}\int_{-\infty}^{t} i\,dt = \frac{q}{C}$$

$$i = \frac{dq}{dt} = C\frac{dv_c}{dt}$$

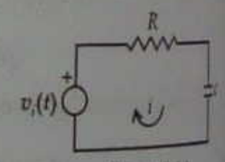


Fig. 10.9 (c)

$$\therefore \quad v_i(t) = RC\frac{d}{dt}v_c(t) + v_c(t) \quad ...(10.46)$$

$$u(t) - u(t - T) = RC\frac{d}{dt}v_c(t) + v_c(t) \quad ...(10.47)$$

Taking Laplace transform on both sides

$$\frac{1}{s} - \frac{e^{-sT}}{s} = RC[sV_c(s) - v_c(0_+)] + V_C(s)$$

$$\frac{1}{s} - \frac{e^{-sT}}{s} = (1 + sCR)\,V_c(s)$$

* See the example 10.17 for $v_i(t)$, Put $t_0 = 0$ such form is called as rectangular function or gate function.

$$V_c(s) = \frac{1}{s(1+sCR)} - \frac{e^{-sT}}{s(1+sCR)} \quad ...(10.48)$$

$$\therefore \quad V_{c1} = \frac{1}{s(1+sCR)} \quad \text{and} \quad V_{c2} = \frac{e^{-sT}}{s(1+sCR)}$$

let

Expand V_{c1} and V_{c2} by partial fraction

$$V_{c1} = \frac{1}{s} - \frac{1}{s+\frac{1}{RC}} \text{ and } V_{c2} = e^{-sT}\left[\frac{1}{s} - \frac{1}{s+\frac{1}{RC}}\right]$$

$$V_c(s) = \frac{1}{s} - \frac{1}{s+\frac{1}{RC}} - e^{-sT}\cdot\frac{1}{s} + e^{-sT}\cdot\frac{1}{s+\frac{1}{RC}} \quad ...(10.49)$$

Taking inverse laplace of equation (10.49)

$$v_c(t) = u(t) - e^{-t/RC}\,u(t) - u(t - T) + e^{-(t-T)/RC}\cdot u(t - T)$$

$$v_c(t) = (1 - e^{-t/RC})\,u(t) - [1 - e^{-(t-T)/RC}]\,u(t - T) \quad ...(10.50)$$

The graph of eqn 10.50 is shown in fig. 10.9*(d)*

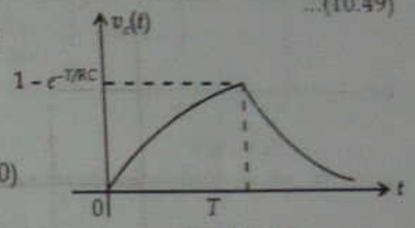


Fig. 10.9 (d).

Example 10.24. The circuit shown in fig. 10.10 is initially under steady-state condition. The switch is moved from position 1 to position 2 at $t = 0$. Find the current after switching.

Solution: Consider the circuit when switch is at position 1 from fig 10.10*(a)*.

$$i_L(0_-) = \frac{V}{R_1} \quad ...(10.51)$$

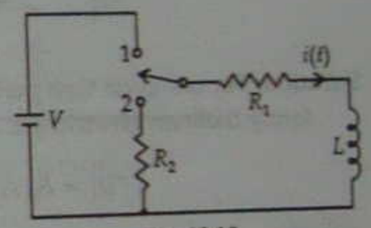


Fig. 10.10.

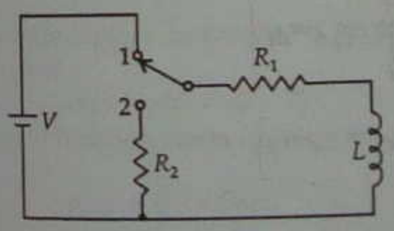


Fig. 10.10(a)

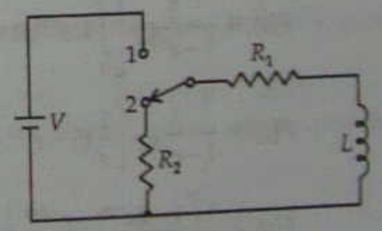


Fig. 10.10(b)

Now, when the switch at position 2, the equivalent circuit will be

Apply KVL in fig. 10.10*(b)*

$$L\frac{di}{dt} + (R_1 + R_2)\,i = 0 \quad ...(10.52)$$

Taking the inverse laplace of eqn (10.52)

$$LsI(s) - i_L(O_+)\,(R_1 + R_2)\,I(s) = 0$$

$$I(s) = \frac{Li_L(O_+)}{sL + R_1 + R_2}$$

$$I(s) = \frac{i_L(O_+)}{s + \frac{R_1 + R_2}{L}}$$

Taking inverse laplace of eqn 10.53 ...(10.53)

$$i(t) = i_L(O_+)\, e^{-\left(\frac{R_1+R_2}{L}\right)t}$$

Put the value of $i_L(O_+)$ from eqn (10.51) Hence required value of $i(t)$ will be ...(10.54)

$$i(t) = \frac{V}{R_1} e^{-\left(\frac{R_1+R_2}{L}\right)t} \quad \textbf{Ans.}$$

Example 10.25. Find the Laplace transform of the wave shown in fig. 10.11.

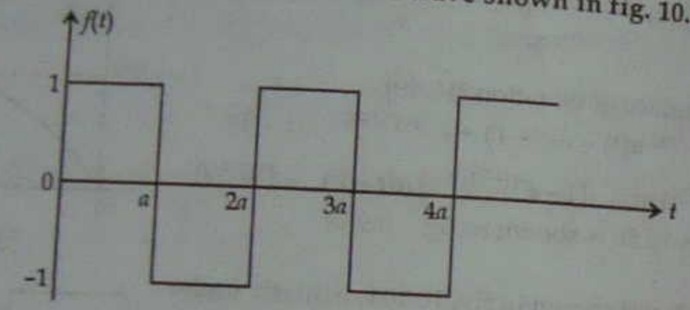


Fig. 10.11.

Solution: For one cycle time period = $2T$

Apply theorem given in art. 10.7

$$\mathcal{L} f(t) = F_1(s) \cdot \frac{1}{1 - e^{-Ts}}$$

$$\therefore \quad F(s) = \frac{1}{1 - e^{-2as}} \left[\int_0^a 1.e^{-st}\, dt + \int_a^{2a} (-1)\, e^{-st}\, dt \right] \qquad ...(10.55)$$

$$F(s) = \frac{1}{1 - e^{-2as}} \left[\frac{1}{s} (1 - e^{-as} + e^{-2as} - e^{-as}) \right]$$

$$F(s) = \frac{1}{s} \left[\frac{1 - 2e^{-as} + e^{-2as}}{1 - e^{-2as}} \right] \qquad ...(10.56)$$

$$F(s) = \frac{1}{s} \frac{e^{-as}(e^{as} - 2 + e^{-as})}{e^{-as}(e^{as} - e^{-as})} = \frac{1}{s} \left[\frac{e^{as} + e^{-as} - 2}{\frac{e^{as} - e^{-as}}{2} \times 2} \right]$$

$$F(s)4 = \frac{1}{s} \left[\frac{2\cosh as - 2}{\sinh as \times 2} \right] \qquad \because \quad \frac{e^x + e^{-x}}{2} = \cosh ax$$

$$= \frac{1}{s} \left[\frac{2\sin^2 h \frac{as}{2}}{2\sinh \frac{as}{2} . \cosh \frac{as}{2}} \right] \qquad \because \quad \cosh x - 1 = 2\sin^2 h\, x/2$$

$$F(s) = \frac{1}{s} \tanh \frac{as}{2} \quad \textbf{Ans.}$$

10.9. GATE FUNCTIONS

A rectangular pulse starting at $t = t_1$ and duration of pulse is T having unit height is represented as

$$G(T) = u(t - t_1) - u(t - t_1 - T) \qquad ...(10.57)$$

is known as gate function.

Any function multiplied by a gate function will have zero value outside the duration of the gate $(t_1 + T) < t < t_1$ and the value of the function will be unaffected within the duration of the gate $t_1 < t < (t_1 + T)$. The gate function is also known as rectangular function.

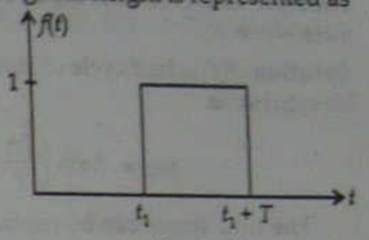


Fig. 10.12.

Exmaple 10.26. Determine the laplace transform of saw-tooth waveform shown in fig. 10.13(a).

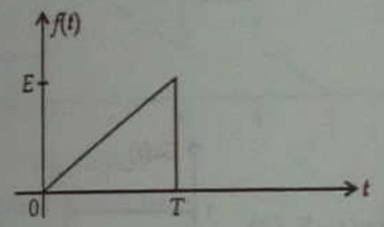


Fig. 10.13(a)

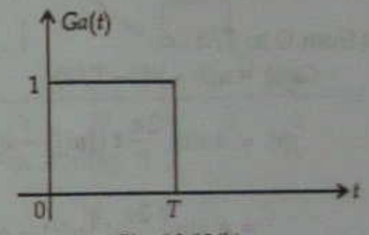


Fig. 10.13(b)

Solution : $f(t) = \frac{E}{T} t \quad 0 < t < T$ ...(10.58)

We can evaluate laplace transform of $f(t)$ by replacing the time limits by gate function as shown in fig 10.13(*b*)

$$Ga(t) = u(t) - u(t - T) \qquad \because \ t_1 = 0 \quad \text{From eqn. (10.57)}$$

Hence, the gate function starts from $t = 0$ to $t = T$ and $f(t)$ can be represented as

$$f(t) = \frac{E}{T} t \times Ga(t)$$

$$f(t) = \frac{E}{T} t\,[u(t) - u(t - T)] = \frac{E}{T} t\, u(t) - \frac{E}{T} t\, u(t - T)^*$$

$$= \frac{E}{T} t\, u(t) - \frac{E}{T} (t - T + T)\, u(t - T)$$

$$= \frac{E}{T} t\, u(t) - \frac{E}{T} (t - T)\, u(t - T) - \frac{E}{T} T\, u(t - T)$$

* In the second term we cannot apply time shifting property as the whole function $tu(t - T)$ is not of form $f(t - a)$

$$f(t) = \frac{E}{T} t\, u(t)^{**} - \frac{E}{T}(t-T)\, u(t-T) - E u(t-T)$$

Taking Laplace transform of eqn (10.59) we get ...(10.59)

$$F(s) = \frac{E}{T}\frac{1}{s^2} - \frac{E}{T} e^{-sT}\frac{1}{s^2} - E\, e^{-sT}\frac{1}{s}$$

$$F(s) = \frac{E}{s^2 T}[1 - e^{-sT} - sT\, e^{-sT}] \quad \textbf{Ans.}$$

Example 10.27. Find the laplace transform of following half cycle sine wave.

Solution : $f(t)$ is half cycle of sine wave (whose period was T) can be express as

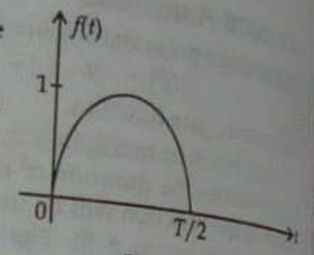


Fig. 10.14.

$$f(t) = A\sin\left(\frac{2\pi}{T}t\right) \qquad 0 < t < T/2 \qquad ...(10.60)$$

The time limits can be replaced by the use of gate function.

$$f(t) = A\sin\left(\frac{2\pi}{T}t\right) \times Ga(t)$$

$Ga(t)$ is from O to $T/2$ i.e.,

$$Ga(t) = u(t) - u(t - T/2)$$

$$\therefore \quad f(t) = A\sin\left(\frac{2\pi}{T}t\right)[u(t) - u(t - T/2)] \qquad ...(10.61)$$

$$= A\sin\left(\frac{2\pi}{T}t\right)u(t) - A\sin\left(\frac{2\pi}{T}t\right)u(t - T/2)$$

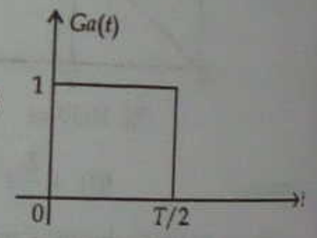


Fig. 10.14(a).

Now,

$$= A\sin\left(\frac{2\pi}{T}t\right)u(t - T/2)$$

$$= A\sin\left[\frac{2\pi}{T}\left(t - \frac{T}{2}\right) + \frac{T}{2}\right]u\left(t - \frac{T}{2}\right)$$

$$= A\left[\sin\frac{2\pi}{T}\left(t - \frac{T}{2}\right)\cos\left(\frac{2\pi}{T}\cdot\frac{T}{2}\right) + \cos\frac{2\pi}{T}\left(t - \frac{T}{2}\right)\cdot\sin\frac{2\pi}{T}\cdot\frac{T}{2}\right]u\left(t - \frac{T}{2}\right)$$

$$= A\sin\left[\frac{2\pi}{T}\left(t - \frac{T}{2}\right)\right](-1)u\left(t - \frac{T}{2}\right) = -A\sin\left[\frac{2\pi}{T}\left(t - \frac{T}{2}\right)\right]u\left(t - \frac{T}{2}\right)$$

$$\therefore \quad f(t) = A\sin\left(\frac{2\pi}{T}t\right)u(t) + A\sin\left[\frac{2\pi}{T}(t - T/2)\right]u(t - T/2) \qquad ...(10.62)$$

** Here, multiplying $f(t)$ by $u(t)$ makes $\mathcal{L}f(t)\,u(t) = \int_0^\infty f(t).1.e^{-st}\,dt = \mathcal{L}f(t)$. Here $f(t) = t$, therefore multiplication by $u(t)$ does affects its laplace transform.

$$\mathcal{L}[t\,u(t)] = \int_0^\infty t.1.e^{-st}dt = \mathcal{L}[t]$$

Take the laplace transform of 10.62.

$$F(s) = \frac{A\cdot(2\pi/T)}{s^2 + (2\pi/T)^2} + \frac{e^{-sT/2}\cdot A(2\pi/T)}{s^2 + (2\pi/T)^2}$$

$$F(s) = \frac{2\pi A}{T\left[s^2 + \left(\frac{2\pi}{T}\right)^2\right]}(1 + e^{-sT/2}) \quad \textbf{Ans.}$$

Example 10.28. Resolve $f(t)$ in terms of step, impulse and ramp functions. Also find the laplace transform of $f(t)$. *(I.E.S 2002, Elec. & Tele. Comm.)*

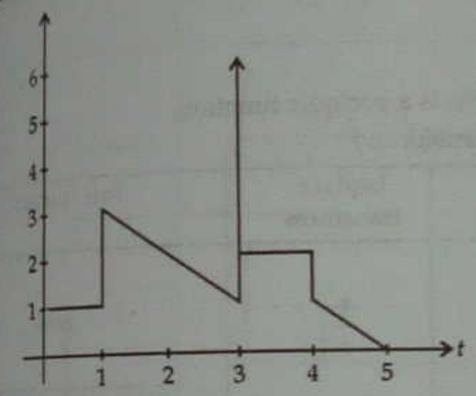


Fig. 10.15.

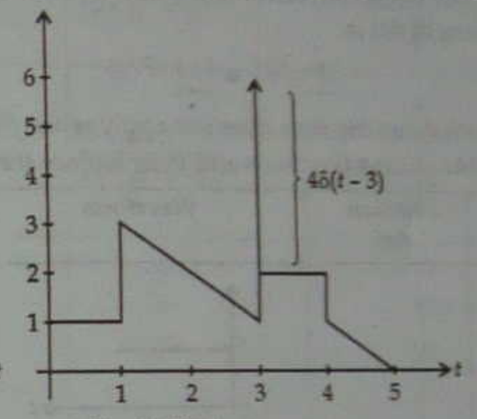


Fig. 10.15(a)

Solution : From fig. 10.15(a)

$$f(t) = \begin{cases} 1 & 0 \le t < 1 \\ (4-t) & 1 \le t < 3 \\ 6 & t = 3 \\ 2 & 3 < t \le 4 \\ (5-t) & 4 \le t \le 5 \end{cases} \qquad ...(10.63)$$

$$f(t) = [u(t) - u(t-1)] + (4-t)\,[u(t-1) - u(t-3] + 4\,\delta(t-3) + 2[u(t-3) - u(t-4)] + (5-t)\,[u(t-4) - u(t-5)]$$

$$f(t) = u(t) - u(t-1) - (t-4)\,[u(t-1) - u(t-3)] + 4\delta(t-3) + 2u(t-3)\,2u(t-4) - (t-5)\,u(t-4) + (t-5)\,u(t-5)$$

$$= u(t) - u(t-1) - (t-1)\,u(t-1) + 3\,u(t-1) - (t-3)\,u(t-3) - u(t-3) + 4\delta(t-3) + 2u(t-3) - 2u(t-4) - (t-4)\,u(t-4) + u(t-4) + (t-5)\,u(t-5)$$

$$f(t) = u(t) + 2u(t-1) - (t-1)\,u(t-1) + (t-3)\,u(t-3) + u(t-3) + 4\delta(t-3) - (t-4)\,u(t-4) - u(t-4) + (t-5)\,u(t-5) \qquad ...(10.64)$$

Taking the laplace transform of eqn 10.64

$$F(s) = \frac{1}{s} + 2\frac{e^{-s}}{s} - \frac{1}{s^2}e^{-s} + \frac{1}{s^2}e^{-3s} + \frac{e^{-3s}}{s} + 4e^{-3s} - \frac{1}{s^2}e^{-4s} - \frac{e^{-4s}}{s} + \frac{e^{-5s}}{s^2}$$

$$F(s) = 4e^{-3s} + \frac{1}{s}\left[1 + 2e^{-s} + e^{-3s} - e^{-4s}\right] + \frac{1}{s^2}\left[-e^{-s} + e^{-3s} - e^{-4s} + e^{-5s}\right] \quad \textbf{Ans.}$$

SUMMARY

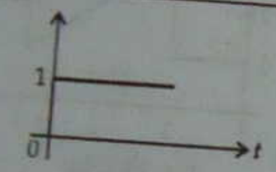
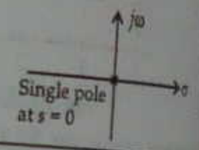
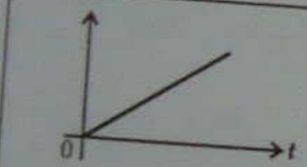
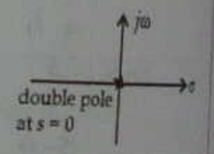
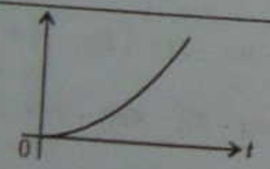
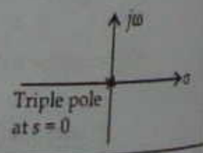
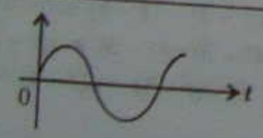
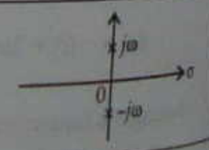
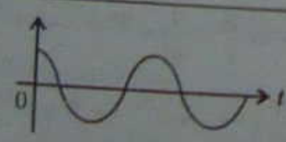
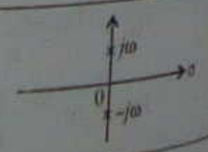
1. Advantages of Laplace transforms
   (a) It simplifies the operations
   (b) By Laplace transform, the complete solution comes out.
   (c) Linear differential equations can be solved easily.
2. Initial value theorem. If $f(t)$ and its first derivative are laplace transform able, then initial value of $f(t)$ is

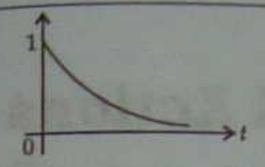
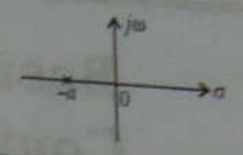
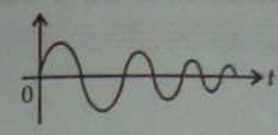
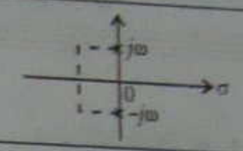
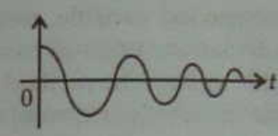
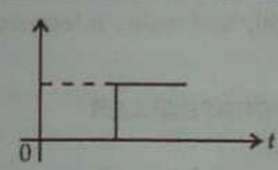
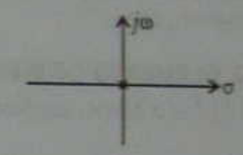
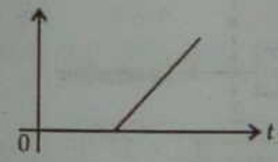
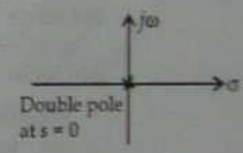
$$f(O_+) = \lim_{t \to O_+} f(t) = \lim_{s \to \infty} sF(s)$$

3. Final value theorem : If $f(t)$ and its first derivatives are laplace transformable then final value of $f(t)$ is

$$\lim_{t \to \infty} f(t) = \lim_{s \to 0} sF(s)$$

4. Final value theorem does not apply when $f(t)$ is a periodic function.
5. Table of time functions and their laplace transform.

S.No.	Function $f(t)$	Waveform	Laplace transform	Pole location
1.	$u(t)$		$\frac{1}{s}$	
2.	t		$\frac{1}{s^2}$	
3.	t^2		$\frac{2}{s^3}$	
4.	$\sin \omega t$		$\frac{\omega}{s^2+\omega^2}$	
5.	$\cos \omega t$		$\frac{s}{s^2+\omega^2}$	

S.No.	Function $f(t)$	Waveform	Laplace	Pole location
6.	e^{-at}		$\frac{1}{s+a}$	
7.	$e^{-at} \sin \omega t$		$\frac{\omega}{(s+a)^2+\omega^2}$	
8.	$e^{-at} \cos \omega t$		$\frac{s+a}{(s+a)^2+\omega^2}$	
9.	$u(t-T)$		$\frac{e^{-sT}}{s}$	
10.	$t-T$		$\frac{e^{-sT}}{s^2}$	

□□□

Chapter 11

Basic Control Actions and Controller Characteristics

11.1. INTRODUCTION

The automatic controller determines the value of controlled variable, compare the actual value to the desired value (reference input), determines the deviation and produces a control signal that will reduce the deviation to zero or to a smallest possible value. The method by which the automatic controller produces the control signal is called mode of control or control action.

The control action may operate through either mechanical, hydraulic, pneumatic or electro-mechanical means *i.e.*, controllers can be electrical, hydraulic, pneumatic, electromechanical or electronics types.

11.2. ELEMENTS OF INDUSTRIAL AUTOMATIC CONTROLLER

Fig. 11.1 is a block diagram of an industrial controller.

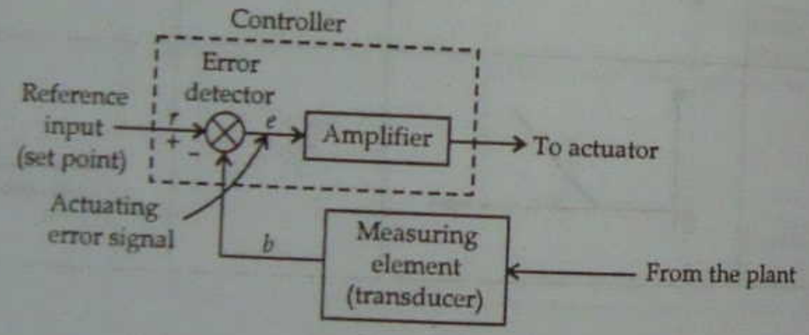


Fig. 11.1.

The controller consists of an error detector and amplifier. The measuring element is a device which converts the output variable to another suitable variable such as displacement, pressure or electrical signals, which can be used for comparing the output to the reference input signal. The set point of a controller must be converted to reference input of the same units as feedback signal from the measuring element. From Fig. 11.1.

$$e = r - b \qquad \ldots(11.1)$$

i.e., deviation is the difference between controlled variable and set point (reference input)

11.3. CLASSIFICATION OF CONTROLLERS

Controllers are classified depending upon the type of controlling action used. Therefore, they can be classified as

(*i*) Two-position or ON-OFF controllers.
(*ii*) Proportional controllers

(*iii*) Integral controllers
(*iv*) Proportional-plus-integral controllers
(*v*) Proportional-plus-derivative controllers
(*vi*) Proportional-plus-integral-plus derivative controllers

They can also be classified according to the power source used for actuating mechanisms, such as electrical, electronics pneumatic and hydraulic controllers. Hydraulic controllers are used for controlling heavy loads, pneumatic controllers are suitable for shop floor applications.

The selection of a particular type of controller depends on the nature of plant, operating conditions such as safety, cost, availability, accuracy, weight and size.

11.4. TWO POSITION CONTROL

This is also known as ON-OFF control or bang-bang control. This type of controllers are simple and inexpensive and are generally employed on home heating systems, domestic water heaters and industrial control systems.

In this type of control the output of the controller is quickly changed to either a maximum or minimum value depending upon whether the controlled variable (b) is greater or less than the set point or in other words depends upon the actuating error signal (e) The minimum value is usually zero.

Let, m = output of the controller
M_1 = Maximum value of output of the controller
M_2 = Minimum value of output of the controller
e = actuating error signal or deviation.

The equations for two-position control will be

$m = M_1$ when $e > 0$ ...(11.2)
$m = M_2$ when $e < 0$ ...(11.3)

The minimum value M_2 is usually either zero or $-M_1$

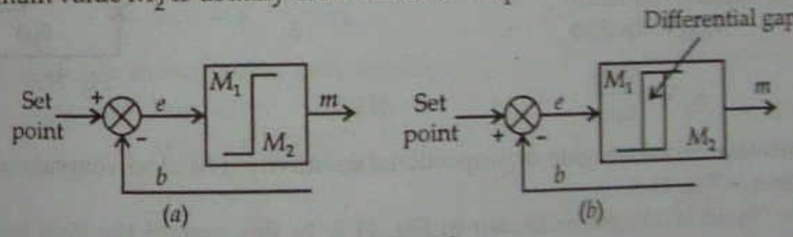


Fig. 11.2.

The block diagram of two position controllers are shown in Fig. 11.2. In ON-OFF controller there is an overlap as the error increases through zero or decreases through zero. This overlap creats a span of error. During this span of error, there is no change in the controller output. This span of error is known as dead zone or dead band. Dead band is shown in Fig. 11.3.

From Fig. 11.3, it is clear that till the error changes by Δe there is no change in the controller output. Similarly while decreasing the error must decrease beyond Δe below 0 to change the controller output. Hence during $2\Delta e$ there is no change in the controller output. This zone is known as differential gap. The differential gap

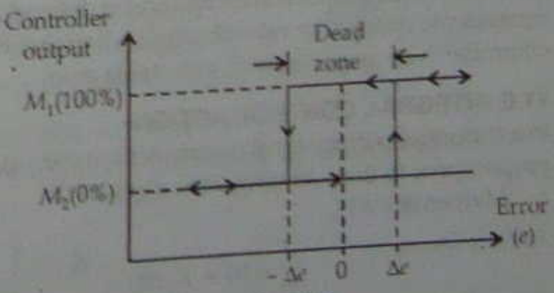


Fig. 11.3.

can also be defined as the range through which the actuating error signal must move before the switching occurs.

In this type of controller, the control variable always oscillates with a frequency which increases with decreasing width of the dead zone (differential gap). The decrease in dead zone, the number of ON-OFF switching of controller increases, hence therefore the useful life of the component decreases. Hence dead band should be designed to prevent the oscillations in ON-OFF controllers.

Two position control mode are used in room air conditioners, heaters, liquid level control in large volume tank.

The example of this type of controller is a room heater. Fig. 11.4 shows the controlling of temperature. If the temperature drops below the set point, an error signal is produced by the error detector, this error signal energised the relay and the heater turned ON. Similarly, if the temperature increase above a set point, the heater is turned OFF.

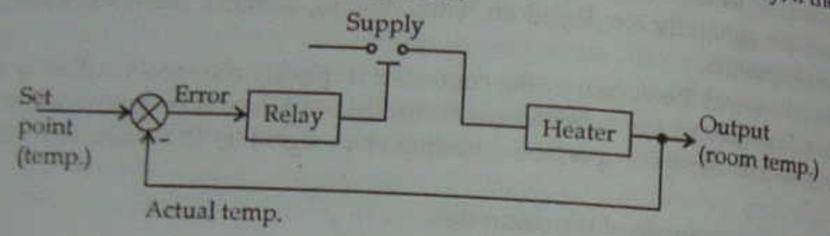


Fig. 11.4.

11.5. PROPORTIONAL CONTROL ACTION

In a controller with proportional control action, there is a continuous linear relation between the output of the controller m (manipulated variable) and actuating error signal e(deviation).

Mathematically

$$m(t) = K_p\, e(t) \qquad \text{...(11.4)}$$

or in terms of Laplace transform

$$M(s) = Kp\, E(s)$$

$$K_p = \frac{M(s)}{E(s)} \qquad \text{...(11.5)}$$

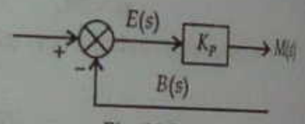


Fig. 11.5.

where K_p is known as proportional gain or proportional sensitivity. The block diagram of proportional controller is shown in Fig. 11.5.

Consider the liquid level system shown in Fig. 11.6. In this system the float lever is directly connected to the control value. When the level of the liquid rise, the valve close in proportionate amount this reduces the inflow to the vessel and vice versa. The inverse of proportional gain or proportional sensitivity is the proportional band and is defined as the change in level necessary to operate the valve through full stroke. Basically proportional controller is an amplifier with adjustable gain.

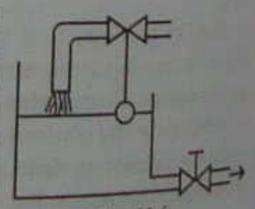


Fig. 11.6.

11.6. INTEGRAL CONTROL ACTION

In a controller with integral control action, the output of the controller is changed at a rate which is proportional to the actuating error signal $e(t)$.

Mathematically,

$$\frac{d}{dt} m(t) = K_i\, e(t) \qquad \text{...(11.6)}$$

where K_i is a constant.

Equation 11.6 can also be written as

$$m(t) = K_i \int e(t) + m(0) \qquad \text{...(11.7)}$$

where $m(0)$ = control output at $t = 0$.

Laplace transform of Eqn. 11.6

$$SM(s) = K_i\, E(s)$$

$$\frac{M(s)}{E(s)} = \frac{K_i}{s} \qquad \text{...(11.8)}$$

or

Equation 11.8 is the transfer function of integral controller. The block diagram of integral controller is shown in Fig. 11.7.

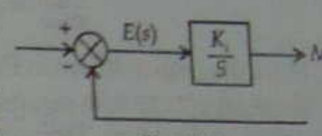


Fig. 11.7.

The inverse of K_i is called integral time T_i and defined as the time of change of output caused by a unit change of actuating error signal. The step response of integral controller is shown in Fig. 11.8.

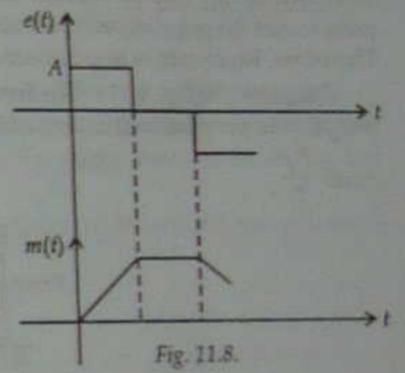


Fig. 11.8.

From the Fig. 11.8 it is clear that for positive error, the output of the controller is ramp (positive), for zero error, there is no change in the output of the controller and for negative error the output of the controller is negative ramp. The integral control action is also known as "Re-set control".

11.7. DERIVATIVE CONTROL ACTION

In a controller with derivative control action the output of the controller depends on the rate of change of actuating error signal $e(t)$.

Mathematically,

$$m(t) = K_d \frac{d}{dt} e(t) \qquad \text{...(11.9)}$$

where K_d is known as derivative gain constant

Laplace transform of Eqn. 11.9

$$M(s) = K_d\, SE(s)$$

$$\frac{M(s)}{E(s)} = sK_d \qquad \text{...(11.10)}$$

or

Equation 11.10 is the transfer function of the controller. The block diagram is shown in Fig. 11.9.

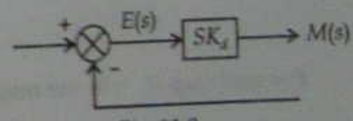


Fig. 11.9.

From equation 11.9 it is clear that when the error is zero or constant, the output of the controller will be zero. Therefore this type of controller cannot be used alone. For this type of controller the gain should be small. The derivative control action also known as rate control.

11.8. PROPORTIONAL-PLUS-INTEGRAL CONTROL ACTION

This is the combination of proportional and integral control action. Mathematically it can be represented by equation 11.11.

$$m(t) = K_p\, e(t) + K_p K_i \int_0^t e(t)\, dt \qquad \text{...(11.11)}$$

or $$m(t) = K_p\, e(t) + K_p \frac{1}{T_i} \int_0^t e(t)\, dt$$

Laplace transform of Eqn. 11.12 ...(11.12)

$$M(s) = K_p\, E(s) + \frac{K_p}{ST_i} E(s) = E(s) \left[1 + \frac{1}{ST_i}\right] Kp$$

$$\frac{M(s)}{E(s)} = K_p \left[1 + \frac{1}{ST_i}\right] \quad ...(11.13)$$

Block diagram is shown in Fig. 11.10.

In equation 11.13 both parameters K_p and T_i are adjustable. T_i is called integral time. The inverse of integral time is called reset rate. Reset rate is defined as the number of times per minute that the proportional part of the response is duplicated. Therefore, Reset rate is also known as "repeats per minute."

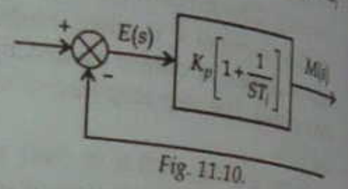


Fig. 11.10.

Consider the Fig. 11.11. The error varies at $t = t_1$. The output of the controller suddenly changes to m_p due to proportional action, after that controller output changes linearly with respect to time at a rate $\frac{K_p}{T_i}$.

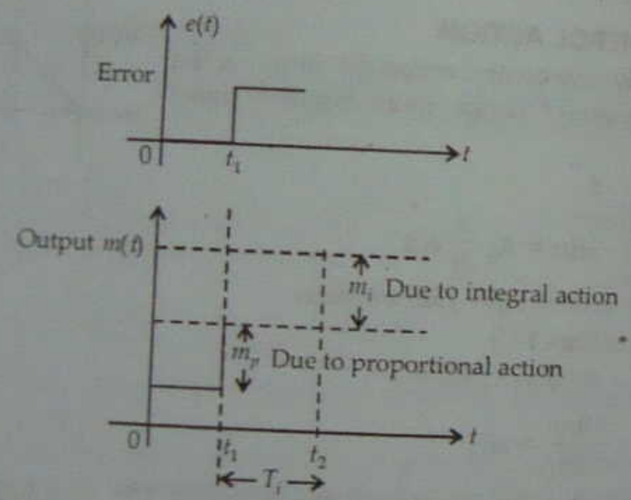


Fig. 11.11.

For unit step ($t_1 = 0$) the response is shown in Fig. 11.12.

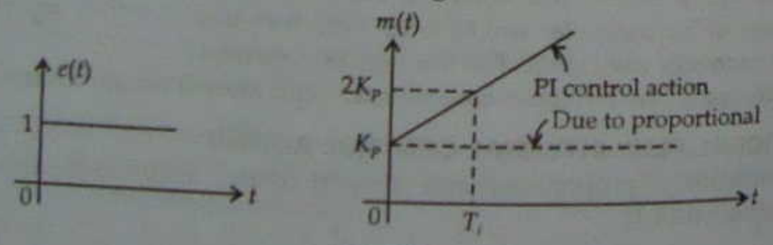


Fig. 11.12.

From the equation 11.12, it is clear that the proportional sensitivity K_p affects both the proportional and integral parts of the action.

11.8. PROPORTIONAL-PLUS-DERIVATIVE CONTROL ACTION

When a derivative control action is added in series to proportional control action, then this combination is termed as proportional-derivative control action. Mathematically it can be defined as

$$m(t) = K_p\, e(t) + K_p\, T_d \frac{d}{dt} e(t) \quad ...(11.14)$$

Laplace transform of Eqn. 11.14

$$M(s) = K_p\, E(s) + K_p\, T_d\, SE(s) = E(s)\, [1 + sT_d]\, K_p$$

$$\frac{M(s)}{E(s)} = K_p\, (1 + sT_d) \quad ...(11.15)$$

This is the transfer function.

The block diagram is shown in Fig. 11.13.

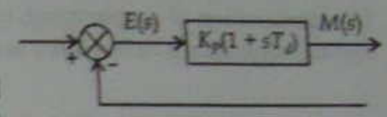


Fig. 11.13.

T_d is known as derivative time. Derivative time is defined as the time interval by which the rate action advances the effect of the proportional control action or in other words derivative time is defined as the amount of lead, expressed in units of time, that the control action is given.

If the actuating error signal $e(t)$ is a ramp function at $t = t_1$. The derivative mode causes a step m_d at t_1 and proportional mode causes a rise of m_p equal to m_d at t_2. This is for direct action PD control (Fig. 11.14)

PD control action reduces the rise time, faster response, improves the bandwidth and improves the damping etc.

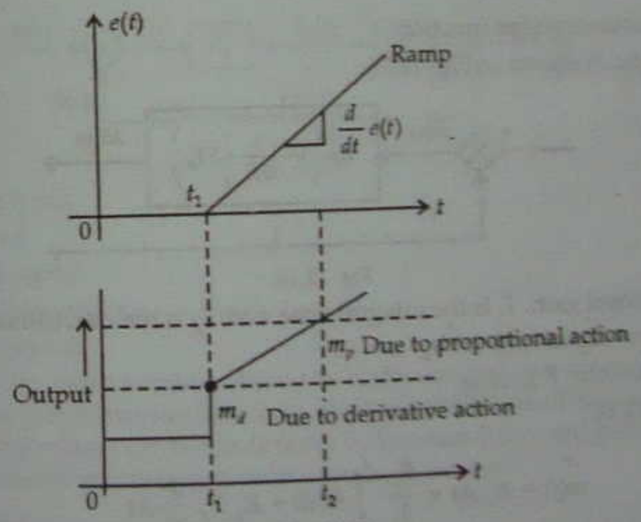


Fig. 11.14.

For unit ramp input.

If the actuating error signal is unit ramp then output of the controller is shown in Fig. 11.15.

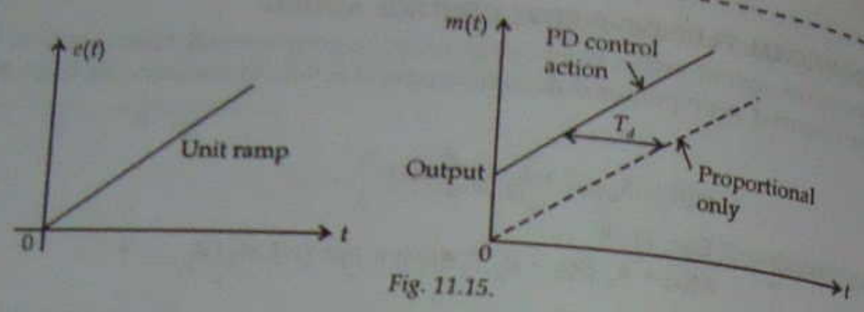


Fig. 11.15.

11.10. PROPORTIONAL-PLUS-INTEGRAL-PLUS-DERIVATIVE CONTROL ACTION

The combination of proportional, integral and derivative control action is called PID control action and the controller is called three action controller. Mathematically

$$m(t) = K_p\, e(t) + K_p \frac{1}{T_i} \int_0^t e(t)\, dt + K_p\, T_d \frac{d}{dt} e(t) \qquad ...(11.16)$$

Laplace transform

$$M(s) = K_p\, E(s) + \frac{K_p}{sT_i}\, E(s) + K_p\, T_d\, SE(s)$$

$$\frac{M(s)}{E(s)} = K_p \left(1 + \frac{1}{sT_i} + sT_d\right) \qquad ...(11.17)$$

Equation 11.17 is the transfer function.

The block diagram is shown in Fig. 11.16.

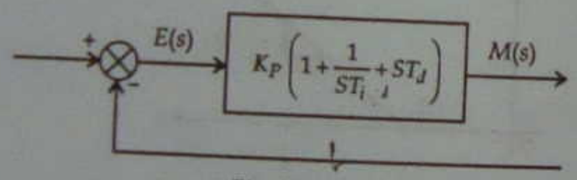


Fig. 11.16..

K_p is the proportional gain, T_i is the integral time and T_d is the derivative time.

Let actuating error signal is given by $e = At$

where 'A' is a constant and 't' is time.

Put in equation 11.16

$$m(t) = K_p\, At + \frac{K_p}{T_i} \int_0^t At\, dt + K_p\, T_d \frac{d}{dt} At$$

$$m(t) = K_p\, A \left[t + \frac{t^2}{2T_i} + T_d\right] \qquad ...(11.18)$$

From the above equation, the proportional part of the control action repeats the change of error (lower straight line) Fig. 11.17. The derivative part of the control action adds an increment of output so that proportional plus derivative action is shifted ahead in time (middle straight line). The integral part adds a further increment of output proportional to the area under the deviation line. The combination of proportional, integral and derivative action may be made in any sequence.

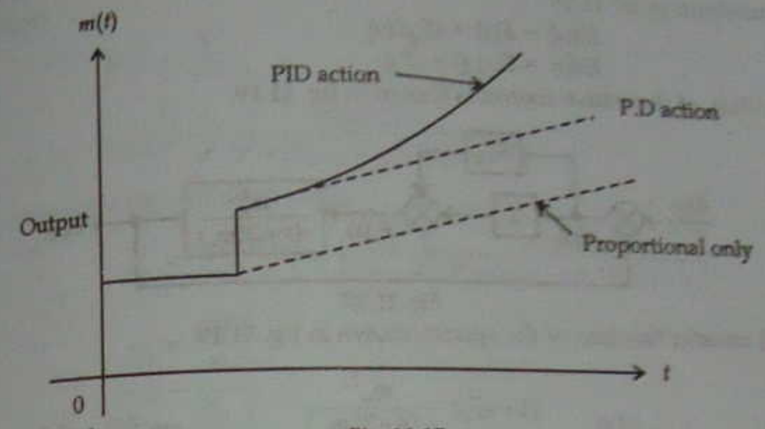


Fig. 11.17.

11.11. RESPONSE WITH *P*, *PI* AND PID CONTROLLERS

11.11.1. Proportional Control

The error signal is the difference of reference input and feedback signal. If $R(s)$ is the input signal and $B(s)$ is the feedback signal then the error signal is given by

$$E(s) = R(s) - B(s)$$

In proportional control the actuating signal is proportional to the error signal $E(s)$. Therefore it is known as proportional control system. The proportional control action is shwon by the block diag. (fig. 12.18)

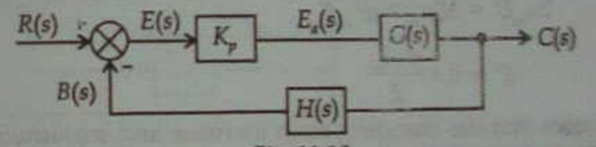


Fig. 11.18.

$R(s)$ = Reference input

$E(s)$ = Error signal

$Ea(s)$ = Actuating signal

$C(s)$ = Output of the system

$B(s)$ = Feedback signal

For quick response, the control system should be underdamped. During the transient period in output the underdamped system has exponentially decaying oscillations. The sluggish (slow moving) overdamped response of a system can be made faster by increasing forward path gain of the system, the steady state error reduced but the maximum overshoot increase :

$$\frac{Ea(s)}{E(s)} = K_P$$

Where, K_P is known as proportional gain.

11.11.2 PD Controller

In derivative control the actuating signal consists of proportional error signal and derivative of the error signal.

$$\therefore \quad ea(t) = e(t) + T_d \frac{d}{dt} e(t) \qquad ...(11.19)$$

laplace transform of eq^n 11.19

$$Ea(s) = E(s) + sT_d E(s)$$
$$Ea(s) = E(s)\,[1 + sT_d]$$

The block diag. of derivative control is shown in fig. 11.19

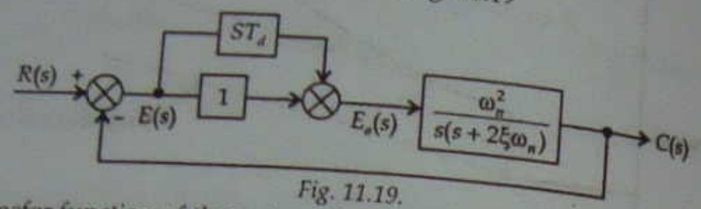


Fig. 11.19.

The overall transfer function of the system shown in fig. 11.19

$$\frac{c(s)}{R(s)} = \frac{(1+sT_d)\cdot\dfrac{\omega_n^{\,2}}{s(s+2\zeta\omega_n)}}{1+(1+sT_d)\dfrac{w_n^{\,2}}{s(s+2\zeta\omega_n)}} = \frac{\omega_n^{\,2}(1+sT_d)}{s(s+2\zeta\omega_n)+(1+sT_d)\omega^2_n}$$

$$= \frac{\omega_n^{\,2}(1+sT_d)}{s^2+(2\zeta\omega_n+\omega_n^{\,2}T_d)\,s+\omega_n^{\,2}}$$

The characteristic equation of 11.21 is ...(11.21)

$$s^2 + (2\zeta\omega_n + \omega^2_n\, T_d)\, s + \omega^2_n = 0$$

Compare the equation 11.22 with $s^2 + 2\zeta'\omega_n\, s + \omega^2_n = 0$ ...(11.22)

$$2\omega_n\zeta' = 2\zeta\omega_n + \omega^2_n\, T_d$$

$$\therefore \quad \zeta' = \xi + \frac{\omega_n T_d}{2} \qquad \text{...(11.23)}$$

Equation (11.23) shows that the damping ratio increase and maximum overshoot is reduced. Now, the overall transfer function in eq^n 11.21 can be written as

$$\frac{c(s)}{R(s)} = \omega_n^{\,2} T_d \cdot \frac{\left(s+\dfrac{1}{T_d}\right)}{s^2+2\zeta'\omega_n s+\omega_n^{\,2}} \qquad \text{...(11.24)}$$

The error function is given by.

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

From fig. 11.19

$$G(s) = (1+sT_d)\cdot\frac{\omega_n^{\,2}}{s(s+2\zeta\omega_n)}$$

$$H(s) = 1$$

$$\therefore \quad \frac{E(s)}{R(s)} = \frac{1}{1+\dfrac{\omega_n^{\,2}(1+sT_d)}{s(s+2\zeta\omega_n)}} = \frac{s(s+2\zeta\omega_n)}{s(s+2\zeta\omega_n)+\omega^2_n(1+sT_d)}$$

$$\frac{E(s)}{R(s)} = \frac{s(s+2\zeta\omega_n)}{s^2+(2\xi\omega_n+\omega_n^2T_d)\,s+\omega_n^2} \qquad \text{...(11.25)}$$

For ramp input

$$R(s) = \frac{1}{s^2}$$

$$\therefore \quad E(s) = \frac{1}{s^2}\,\frac{s(s+2\zeta\omega_n)}{s^2+(2\zeta\omega_n+\omega^2{}_nT_d)\,s+\omega^2{}_n}$$

The steady state error for unit ramp input

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\cdot\frac{1}{s^2}\,\frac{s(s+2\zeta\omega_n)}{s^2+(2\zeta\omega_n+\omega_n^{\,2}T_d)\,s+\omega_n^{\,2}}$$

$$e_{ss} = \frac{2\zeta\omega_n}{\omega_n^{\,2}}$$

$$\boxed{e_{ss} = \frac{2\zeta}{\omega_n}} \qquad \text{...(11.26)}$$

Hence, the steady state error is same in derivative control action. From eqn (11.26) a zero is added and hence rise time (t_r) reduced.

11.11.3. Derivative Feedback Control or Rate Feedback Controller

In derivative feedback control, the actuating signal is the difference of proportional error signal and the derivative of the output. The derivative feecback control is also known as tachometer feedback control. The block diag. is shown in fig. 11.20*a*.

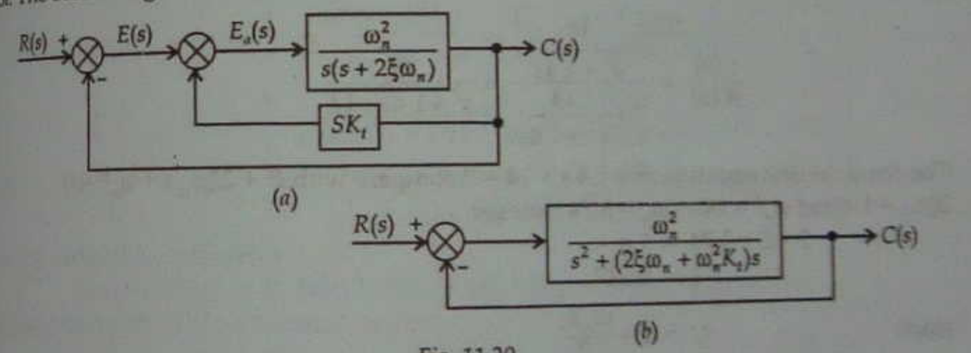


Fig. 11.20

The transfer function of the system shown in fig. 11.20

$$\frac{c(s)}{R(s)} = \frac{\omega_n^{\,2}}{s^2+(2\zeta\omega_n+\omega_n^{\,2}K_t)\,s+\omega_n^{\,2}} \qquad \text{...(11.27)}$$

The characteristic equation

$$s^2 + (2\zeta\omega_n + \omega_n^{\,2}K_t)\, s + \omega_n^{\,2} = 0$$

compare with $s^2 + 2\zeta'\omega_n s + \omega_n^{\,2} = 0$

$$2\omega_n\zeta' = 2\xi\omega_n + \omega_n^{\,2}\,Kt$$

$$\therefore \quad \zeta' = \zeta + \frac{\omega_n K_t}{2} \qquad \text{...(11.28)}$$

Hence, by using the derivative feedback control, the damping ratio is increased, maximum overshoot reduced.

$$\frac{E(s)}{R(s)} = \frac{1}{G(s)H(s)+1} = \frac{1}{1+\dfrac{\omega_n^{\,2}}{s^2+(2\zeta\omega_n+\omega_n^{\,2}K_t)s}} = \frac{s^2+(2\zeta\omega_n+\omega_n^{\,2}K_t)\,s}{s^2+(2\zeta\omega_n+\omega^2{}_nK_t)\,s+\omega_n^{\,2}}$$

For ramp input

$$R(s) = \frac{1}{s^2}$$

$$\therefore \quad E(s) = \frac{1}{s^2} \cdot \frac{s^2 + (2\zeta\omega_n + \omega_n{}^2 K_t)\, s}{s^2 + (2\zeta\omega_n + \omega^2{}_n K_t)\, s + \omega_n{}^2}$$

Steady state error

$$e_{ss} = \lim_{s \to 0} s.E(s) = s \cdot \frac{1}{s^2} \cdot \frac{s^2 + (2\zeta\omega_n + \omega^2{}_n K_t)s}{s^2 + (2\zeta\omega_n + \omega_n{}^2 K_t)\, s + \omega^2{}_n}$$

$$= s \cdot \frac{1}{s^2} \cdot \frac{s[s + (2\zeta\omega_n + \omega_n{}^2 K_t)\, s}{s^2 + (2\zeta\omega_n + \omega_n{}^2 K_t)\, s + \omega_n{}^2} = \frac{2\zeta\omega_n + \omega_n{}^2 K_t}{\omega_n{}^2}$$

$$e_{ss} = \frac{2\zeta}{\omega_n} + K_t$$

Hence, by using the derivative feedback control the steady state error is increased. —(11.2)

Example 11.1. A closed loop control system with unity feedback is shown in fig. 11.21. By using derivative control the damping ratio is to be made 0.7. Determine the value of T_d, also determine the rise time, peak time and maximum overshoot without derivative control and with derivative control. The input to the system is unit step.

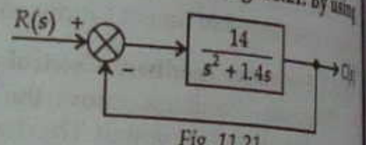


Fig. 11.21.

Solution :

$$\frac{c(s)}{R(s)} = \frac{\dfrac{14}{s^2 + 1.4s}}{1 + \dfrac{14}{s^2 + 1.4s}} = \frac{14}{s^2 + 1.4s + 14}$$

The characteristic equation $s^2 + 1.4\,s + 14 = 0$ compare with $s^2 + 2\zeta\omega_n s + \omega_n{}^2 = 0$

$2\zeta\omega_n = 1.4$ and $\omega_n{}^2 = 14 \therefore \omega_n = 3.74$ rad/sec.

$$2 \times \zeta \times 3.74 = 1.4$$

$$\therefore \quad \zeta = 0.187$$

since,

$$\zeta' = \zeta + \frac{\omega_n T_d}{2}$$

$$\therefore \quad \zeta' = 0.187 + \frac{3.74 T_d}{2}$$

$$0.7 = 0.187 + \frac{3.74 T_d}{2} \qquad \zeta' = 0.7 \text{ given}$$

$$\therefore \quad T_d = 0.274$$

CALCULATION OF RISE TIME, PEAK TIME AND MAXIMUM OVERSHOOT WITHOUT DERIVATIVE CONTROL

$$\text{Rise time } t_r = \frac{\pi - \tan^{-1}\dfrac{\sqrt{1-\zeta^2}}{\zeta^2}}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi - \tan^{-1}\dfrac{\sqrt{1-(0.187)^2}}{0.187}}{3.74\sqrt{1-(0.187)^2}} = \frac{1.757}{3.67} = 0.478 \text{ sec.}$$

$$\text{Peak time } t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{3.74\sqrt{1-(0.187)^2}} = 0.854 \text{ sec.}$$

$$\text{Max}^M. \text{ overshoot } M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = e^{\frac{-\pi(0.187)}{\sqrt{1-(0.187)^2}}} \times 100 = 55\%$$

The overall transfer function with derivative control

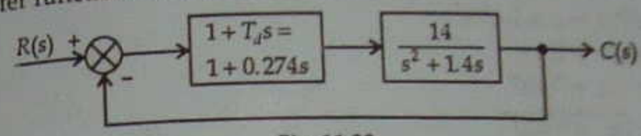


Fig. 11.22.

$$\frac{c(s)}{R(s)} = \frac{\dfrac{14(1+0.274s)}{s^2+1.4s}}{1 + \dfrac{14(1+0.274s)}{s^2+1.4s}} = \frac{14(1+0.274s)}{s^2+1.4s+14+3.836s}$$

$$\frac{c(s)}{R(s)} = \frac{14(1+0.274s)}{s^2+5.236s+14}$$

Input

$$R(s) = \frac{1}{s}$$

$$\therefore \quad c(s) = \frac{1}{s} \cdot \frac{14(1+0.274s)}{s^2+5.236s+14} = \frac{1}{s}\left[\frac{14+3.836s}{s^2+5.236s+14}\right]$$

$$= \frac{1}{s} \cdot \frac{14}{s^2+5.236s+14} + \frac{1}{s} \cdot \frac{3.836s}{s^2+5.236s+14}$$

$$= \frac{K_1}{s} + \frac{K_2 s + K_3}{s^2+5.236s+14} + \frac{3.836}{s^2+5.236s+14}$$

Now calculate the coefficients

$$14\,(1+0.274s) = K_1\,(s^2 + 5.236\,s + 14) + (K_2\,s + K_3)\,s + 3.836\,s$$

Equating the coeff. of s^2, s & constt. terms

$$K_1 + K_2 = 0$$

$$5.236\,K_1 + K_3 + 3.836 = 3.836$$

$$14\,K_1 = 14$$

$$\therefore \quad K_1 = 1,\ K_2 = -1 \text{ and } K_3 = -5.236$$

$$\therefore \quad C(s) = \frac{1}{s} - \frac{s+5.236}{s^2+5.236s+14} + \frac{3.836}{s^2+5.236s+14}$$

$$= \frac{1}{s} - \frac{s+5.236}{(s+2.618)^2 + (14-2.618^2)} + \frac{3.836}{(s+2.618)^2 + (14-2.618^2)}$$

$$= \frac{1}{s} - \frac{s+5.236}{(s+2.618)^2 + (2.673)^2} + \frac{3.836}{(s+2.618)^2 + (2.673)^2}$$

$$C(s) = \frac{1}{s} - \frac{s+2.618-2.618}{(s+2.618)^2 + (2.673)^2} + \frac{3.836}{(s+2.618)^2 + (2.673)^2}$$

$$= \frac{1}{s} - \frac{s+2.618}{(s+2.618)^2 + (2.673)^2} + \frac{1.218}{(s+2.618)^2 + (2.673)^2}$$

$$C(s) = \frac{1}{s} - \frac{s+2.618}{(s+2.618)^2+(2.673)^2} + \frac{2.673}{(s+2.618)^2+(2.673)^2}\cdot\frac{1.218}{2.673}$$

Take inverse laplace

$$C(t) = 1 - e^{-2.618t}\cos 2.673t + 0.455\, e^{-2.618t}\sin 2.673t$$

For rise time put $C(t) = 1$

$$1 = 1 - e^{-2.618t}\cos 2.673t + 0.455\, e^{-2.618t}\sin 2.673t \quad ...(12.30)$$

$$0.455 \sin 2.673t = \cos 2.673t$$

$$\tan 2.673t = 2.19$$

$$\therefore \quad t = 0.427$$

$$\therefore \quad \boxed{t_r = 0.427}$$

For peak time differentiate the output equation w.r.t 't' and equate to zero

$$\frac{d}{dt}C(t) = 0$$

$$C(t) = 1 - e^{-2.618t}[\cos 2.673t - 0.455 \sin 2.673\, t].$$

$$\frac{dC(t)}{dt} = -[e^{-2.618t}[-\sin 2.673\, t \times 2.673 - 0.455 \times 2.673 \cos 2.673t] + [\cos 2.673t - 0.455 \sin 2.673t]\, e^{-2.618t}(-2.618)]$$

$$= -[e^{-2.618t}[-2.673 \sin 2.673t - 1.22 \cos 2.673\, t + (\cos 2.673t - 0.455 \sin 2.673t)(-2.618)]$$

$$= -[e^{-2.618t}[-2.673 \sin 2.673t - 1.22 \cos 2.673t - 2.618 \cos 2.673t + 1.19 \sin 2.673t]]$$

$$= -e^{-2.618t}[-1.483 \sin 2.673t - 3.838 \cos 2.673t]$$

$$\tan 2.673t = -2.58$$

$$2.673\, t_p = \tan^{-1}(-2.58)$$

$$t_p = \frac{\pi - \tan^{-1}(2.58)}{2.673} = 0.72 \text{ sec.}$$

$$C(t)_{max} = 1 - e^{-2.618 \times 0.72}[\cos 2.673 \times 0.72 - 0.455 \sin 2.673 \times 0.72]$$

$$= 1 - 0.15[-0.346 - 0.94] = 1.1929$$

$$M_p = \frac{C(t)_{max-1}}{1} \times 100 = \frac{1.1929-1}{1} \times 100$$

$$M_p = 19.29\%$$

Example 11.2. In example 11.1 it is desired the damping ratio be 0.7. Determine the derivative rate feedback K_t. Also determine the peak time, rise time, steady state error and max. ^{m}overshoot with and without feedback for unit ramp input.

Solution : The characteristic eq^n. $s^2 + 1.4s + 14 = 0$

$$\therefore \quad \zeta = 0.187 \;\&\; \omega_n = 3.74 \text{ rad/sec.}$$

$$\zeta' = \zeta + \frac{\omega_n K_t}{2}$$

$$0.7 = 0.187 + \frac{3.74 K_t}{2} \qquad \boxed{\therefore K_t = 0.274}$$

The calculation for rise time, peak time and max.^{m} overshoot without feedback is same as in previous solution. The steady state error without feedback.

$$e_{ss} = \frac{2\zeta}{\omega_n} = \frac{2 \times 0.187}{3.74} = 0.1$$

The overall transfer function using derivative feedback control is

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + (2\zeta\omega_n + \omega_n^2 K_t)s + \omega_n^2}$$

Given that $\zeta = 0.187$, $\omega_n = 3.74$ rad/sec, $K_t = 0.274$

$$\therefore \quad \frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 5.23s + 14}$$

characteristic eqn $s^2 + 5.23s + 14 = 0$

$$2\zeta'\omega_n = 5.23, \; \omega_n = 3.74$$

$$\zeta' = 0.7$$

Rise time

$$t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\zeta'^2}}{\zeta'}}{\omega_n\sqrt{1-\zeta'^2}} = \frac{\pi - \tan^{-1}\frac{\sqrt{1-0.7^2}}{0.7}}{3.74\sqrt{1-0.7^2}} = 0.877 \text{ sec.}$$

Peak time $t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{3.74\sqrt{1-0.7^2}} = 1.17$ sec.

Maximum overshoot $M_P = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = e^{\frac{\pi 0.7}{\sqrt{1-0.7^2}}} \times 100 = 4.6\%$

Steady state error $= \frac{2\zeta}{\omega_n} + K_t = \frac{2 \times 0.7}{3.74} + 0.274 = 0.648$

11.11.4. PI Controller

In integral control action the actuating signal consists of proportional error signal with integral of the error signal. The block diagram of integral control is shown in fig. 11.23.

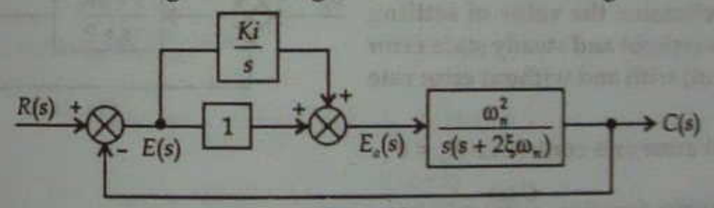


Fig. 11.23.

From the block diag. the closed loop transfer function will be

$$\frac{C(s)}{R(s)} = \frac{\left(1+\frac{K_i}{s}\right)\left(\frac{\omega_n^2}{s(s+2\zeta\omega_n)}\right)}{1+\left(1+\frac{k_i}{s}\right)\left(\frac{\omega_n^2}{s(s+2\zeta\omega_n)}\right)} \quad ...(11.31)$$

$$= \frac{(s+K_i)\omega_n^2}{s^3 + 2\zeta\omega_n s^2 + \omega^2{}_n s + K_i\omega_n^2}$$

The characteristic equation is the third order equation hence the system becomes third order system.

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)} = \frac{1}{1+\frac{(s+K_i)\omega_n^2}{s^2(s+2\zeta\omega_n)}}$$

$$E(s) = R(s)\frac{s^2(s+2\zeta\omega_n)}{s^3+2\zeta\omega_n s^2+\omega_n{}^2 s+K_i\omega_n{}^2}$$

From eqn. (11.32) if $R(s)$ is unit ramp, the steady state error will be zero.

If the input is parabolic input *i.e.* $R(s) = \frac{1}{s^3}$ then we have steady state error.

$$E(s) = \frac{1}{s^3}\frac{s^2(s+2\xi\omega_n)}{s^3+2\xi\omega_n s^2+\omega^2{}_n\, s+Ki\omega_n{}^2}$$

$$e_{ss} = \lim_{s\to 0} sE(s) = \frac{2\zeta\omega_n}{K_i\omega^2{}_n} = \frac{2\zeta}{K_i\omega_n}$$

11.11.5. Proportional Plus Derivative Plus Integral Control (PID)

In *PI D* control the actuating signal consists of proportional error signal added with derivative and integral of error signal.

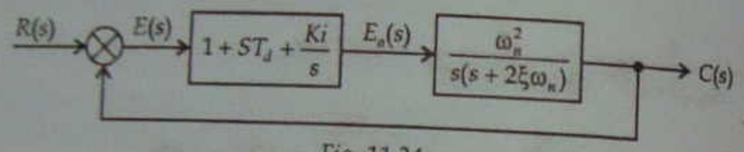


Fig. 11.24.

$$Ea(s) = E(s) + sT_d E(s) + \frac{K_i}{s}E(s) = E(s)\left[1+sT_d+\frac{K_i}{s}\right]$$

The block diag of *PI D* control is shwon in fig. 11.24

Example 11.3. Fig. 12.25 shows a system employing proportional plus error rate control. Determine the value of error rate factor K_e so that the damping ratio is 0.5. Determine the value of settling time, maximum overshoot and steady state error (for unit ramp input) with and without error rate control.

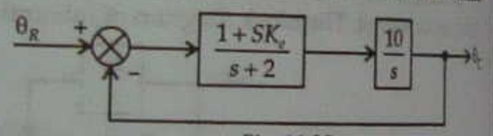


Fig. 11.25.

Solution : Without error rate control *i.e* $K_e = 0$

The overall transfer function $\frac{C(s)}{R(s)}$ will be

$$\frac{C(s)}{R(s)} = \frac{\frac{10}{s(s+2)}}{1+\frac{10}{s(s+2)}} = \frac{10}{s^2+2s+10}$$

characteristic equation $s^2 + 2s + 10 = 0$ compare without $s^2 + 2\zeta\omega_n s + \omega^2{}_n = 0$

$\omega_n{}^2 = 10 \quad \therefore \omega_n = 3.16$ rad/sec.

$2\zeta\omega_n = 2$

$2 \times \zeta \times 3.16 = 2 \quad \therefore \zeta = 0.316$

Maximum overshoot $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100 = e^{\frac{-\pi 0.316}{\sqrt{1-0.316^2}}} \times 100 = 35.14\%$

Settling time $t_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.316\times 3.16} = 4$ sec.

Steady state error $e_{ss} = \frac{2\zeta}{\omega_n} = \frac{2\times 0.316}{3.16} = 0.2$ rad.

With Error Control

The overall transfer function of fig. 11.25

$$\frac{C(s)}{R(s)} = \frac{\frac{10(1+sK_e)}{s(s+2)}}{1+\frac{10(1+sK_e)}{s(s+2)}} = \frac{10(1+sKe)}{s(s+2)+10(1+sKe)}$$

characteristic eq^n $s(s+2) + 10(1+sK_e) = 0$

or, $s^2 + s(2+10K_e) + 10 = 0$

compare with $s^2 + 2\zeta\omega_n s + \omega^2{}_n = 0$

$\omega^2{}_n = 10 \quad \therefore \omega_n = 3.16$ rad/sec.

$2\zeta\omega_n = 2 + 10\,k_e$

$2 \times 0.5 \times 3.16 = 2 + 10\,K_e \quad \therefore \boxed{K_e = 0.116}$

The characteristic equation with $K_e = 0.16$ is

$s^2 + 3.16s + 10 = 0$

Compare with $s^2 + 2\zeta\omega_n s + \omega^2{}_n = 0$, $\omega_n = 3.16$ rad/sec.

Maximum overshoot $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100$

$$M_p = e^{\frac{-\pi 0.5}{\sqrt{1-0.5^2}}} \times 100 = 16.3\%$$

Settling time $t_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.5\times 3.16} = 2.53$ sec.

Steady state error $e_{ss} = \lim_{s\to 0} sE(s)$

$$E(s) = \frac{1}{1+G(s)}\cdot R(s) \qquad \therefore H(s) = 1$$

$$e_{ss} = \lim_{s\to 0} s\cdot\frac{1}{s^2}\frac{s(s+2)}{10+1.16s+s(s+2)}$$

$e_{ss} = 0.2$ Ans.

Example 11.4. A unity feedback system is characterised by an open loop transfer function

$$G(s) = \frac{K}{s(s+10)}$$

Determine the gain K so that the system will have a damping ratio of 0.5. For this value of K determine setting time, peak overshoot and peak time for a unit step input.

Solution : The characteristic eq^n $1 + G(s)\,H(s) = 0$

$$1+\frac{K}{s(s+10)} = 0$$

or, $s^2 + 10s + K = 0$

Compare with $s^2 + 2\zeta\omega_n s + \omega^2{}_n = 0$

$\omega_n{}^2 = K \quad \therefore \omega_n = \sqrt{K}$

$2\zeta\omega_n = 10$

$\omega_n = \sqrt{100} = 10$ rad/sec.

$$2 \times 0.5\sqrt{K} = 10 \quad \therefore K = 100$$

Settling time $$t_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.5 \times 10} = 0.8 \text{ sec.}$$

Maxmum overshoot $$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100 = e^{-\pi 0.5/\sqrt{1-0.5^2}} \times 100 = 16.3\%$$

Peak time $$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{10\left(\sqrt{1-0.5^2}\right)} = 0.363 \text{ sec.}$$

Example 11.5. A feedback system employing output-rate damping shown in fig. 11.26. (*a*) In the absence of derivative feedback ($K_0 = 0$), determine the damping factor and natural frequency of the system. What is the steady state error resulting from unit ramp input?

(*b*) Determine the derivative feedback constant K_0, which will increase the damping factor of the system to 0.6. What is the steady state error resulting from unit ramp input with this setting of the derivative feedback constant ?

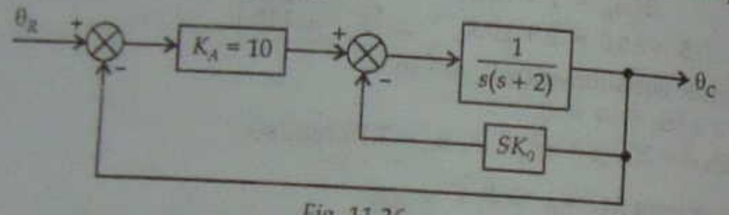


Fig. 11.26.

Solution : (*a*). When $K_0 = 0$

$$G(s) = \frac{10}{s(s+2)}$$

characteristic eq^n $1 + G(s)H(s) = 0$

$$1 + \frac{10}{s(s+2)} = 0$$

or, $s^2 + 2s + 10 = 0$ compare with $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$

$$\omega^2_n = 10 \quad \therefore \boxed{\omega_n = 3.16 \text{ rad/sec.}}$$
$$2\zeta\omega_n = 2$$
$$2 \times \zeta \times 3.16 = 2 \quad \boxed{\zeta = 0.316}$$

Steady state error $e_{ss} = \frac{2\zeta}{\omega_n} = \frac{2 \times 0.316}{3.16} = 0.2$ rad.

(*b*) The overall transfer function of fig. 11.26.

$$\frac{C(s)}{R(s)} = \frac{10}{s^2 + (2 + K_0)s + 10}$$

characteristic equation $s^2 + (2 + K_0)s + 10 = 0$ compare with $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$

$$\omega_n^2 = 10 \qquad \omega_n = 3.16 \text{ rad/sec.}$$
$$2\zeta\omega_n = 2 + K_0$$
$$2 \times 0.6 \times 3.16 = 2 + K_0 \quad \therefore \quad \boxed{K_0 = 1.792}$$

Steady state error $e_{ss} = \underset{s\to 0}{Lim}\, sE(s)$

$$E(s) = \frac{1}{1 + G(s)H(s)} R(s)$$

$$e_{ss} = \underset{s\to 0}{Lim}\, s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{10}{s(s+2)+1.8s}}$$

$$e_{ss} = \underset{s\to 0}{Lim}\, s \cdot \frac{1}{s^2} \frac{s(s+2)+1.8s}{s(s+2)+1.8s+10}$$

$$e_{ss} = \underset{s\to 0}{Lim} \frac{s+2+1.8}{10} = 0.38$$

$$\boxed{e_{ss} = 0.38 \text{ rad.}}$$

Example 11.6. Block diag. model of a position control system shown in fig 11.27.

(*a*) In the absence of derivative feedback ($K_t = 0$) determine the damping ratio of the system for amplifier gain $K_A = 5$. Also find the steady state error to unit ramp input.

(*b*) Find suitable values of the parameters K_A and K_t so that the damping ratio of the system is increased to 0.7 without affecting the steady state error as obtained in part (*a*). (GATE 1992)

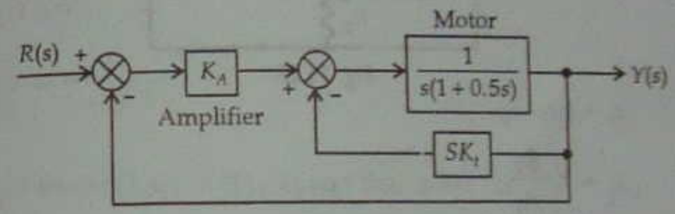


Fig. 11.27.

Solution :

(*a*) The solution of part '*a*' is same as in problem 11.5

$\zeta = 0.316$ and $e_{ss} = 0.2$ rad.

(*b*) The overall transfer function of fig 11.27

$$\frac{C(s)}{R(s)} = \frac{2K_A}{s^2 + s(2 + 2K_t) + 2K_A}$$

The characteristic eq^n. $s^2 + s(2 + 2K_t) + 2K_A = 0$ compare with $s^2 + 2\zeta\omega_n s + \omega^2_n = 0$

$$\omega^2_n = 2K_A$$
$$2\zeta\omega_n = 2 + 2K_t$$
$$2 \times 0.7\sqrt{2K_A} = 2 + 2K_t$$

Steady state error $e_{ss} = \underset{s\to 0}{Lim}\, s.E(s)$

$$e_{ss} = \underset{s\to 0}{Lim}\, s \cdot R(s) \frac{1}{1 + G(s)H(s)} = \underset{s\to 0}{Lim}\, s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{K_A}{s(1+0.5s) + sK_t}}$$

$$= \underset{s\to 0}{Lim} \frac{(1+0.5s)K_t}{s(1+0.5s) + sK_t + K_A} = \frac{1 + K_t}{K_A}$$

$e_{ss} = 0.2$ (as in part '*a*')

$$0.2 = \frac{1 + K_t}{K_A} \quad \text{or} \quad 1 + K_t = 0.2K_A$$

$$2 \times 0.7\sqrt{2K_A} = 2 + 2K_t = 2(1 + K_t) \quad \text{...(11.34)}$$

$$1 + K_t = 0.2K_A \quad \text{...(11.35)}$$

Put the value of $1 + K_t = 0.2\,K_A$ from (11.35) in (11.34)

$$1.98\sqrt{K_A} = 0.4\,K_A$$

or, $\boxed{K_A = 24.5}$ **Ans.**

from (11.35)

$$1 + K_t = 0.2\,(24.5)$$

$\boxed{K_t = 3.9}$ **Ans.**

11.12. ELECTRONIC CONTROLLERS

11.12.1. Proportional Controller

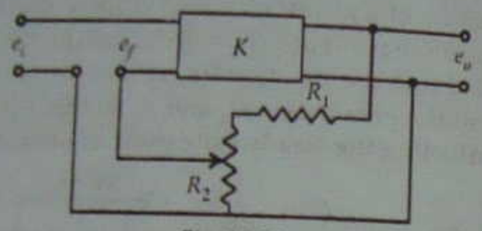


Fig. 11.28.

$$e_o = k(e_i - e_f)$$ ...(11.34)

$$e_f = e_o \frac{R_2}{R_1}$$ ...(11.35)

Laplace transform of (11.34) and (11.35)

$$E_o(s) = k[E_i(s) - E_f(s)]$$ ...(11.36)

$$E_f(s) = E_o(s) \cdot \frac{R_2}{R_1}$$ ...(11.37)

From equations (11.36) and (11.37)

$$E_o(s) = k\left[E_i(s) - E_o(s) \cdot \frac{R_2}{R_1}\right]$$

$$\frac{E_o(s)}{E_i(s)} = \frac{R_1}{R_2}$$

Put $\frac{R_1}{R_2} = K_p$

$$\therefore \quad \frac{E_o(s)}{E_i(s)} = K_p$$ ...(11.38)

11.12.2. PD Controller

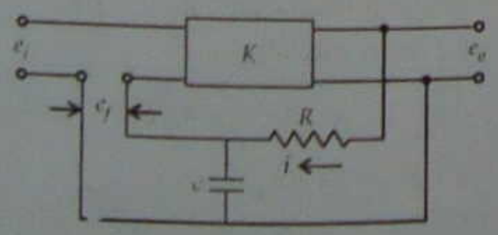


Fig. 11.29.

$$e_o = K(e_i - e_f)$$ ...(11.39)

$$e_o = iR + \frac{1}{C}\int i\,dt$$ ...(11.40)

Laplace transform of (11.39) and (11.40)

$$E_o(s) = k[E_i(s) - E_f(s)]$$ ...(11.41)

$$E_o(s) = RI(s) + \frac{1}{SC} I(s)$$ ...(11.42)

$$E_f = \frac{1}{C}\int i\,dt$$ ...(11.43)

Laplace transform of (11.43)

$$E_f(s) = \frac{1}{sC} I(s)$$ ...(11.44)

From (11.42) and (11.44)

$$\frac{E_f(s)}{E_o(s)} = \frac{1}{1 + RCs}$$ ...(11.45)

Put the value of $E_f(s)$ from (11.45) in (11.41) and then solve for $\frac{E_o(s)}{E_i(s)}$

$$\frac{E_o(s)}{E_i(s)} = \frac{K}{1 + \frac{K}{1 + RCS}}$$

If $\frac{K}{1 + RCS} >> 1$

then $$\frac{E_o(s)}{E_i(s)} = 1 + RCS$$ ...(11.46)

or $$\boxed{\frac{E_o(s)}{E_i(s)} = 1 + T_d S}$$ ...(11.47)

where $T_d = RC$

Thus, we get PD type control action.

11.12.3. PI Controller

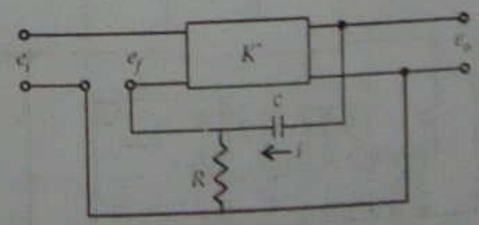


Fig. 11.30.

$$e_o = k(e_i - e_f)$$

Laplace transform of (11.48)

$$E_o(s) = k[E_i(s) - E_f(s)] \quad \text{...(11.49)}$$

$$e_o = Ri + \frac{1}{C}\int i\,dt \quad \text{...(11.50)}$$

Laplace transform of (11.50)

$$E_o(s) = RI(s) + \frac{1}{sc} I(s) \quad \text{...(11.51)}$$

$$E_o(s) = I(s)\left[\frac{RCs + 1}{Sc}\right]$$

$$E_f = Ri \quad \text{...(11.52)}$$

Laplace transform of (11.52)

$$E_f(s) = RI(s) \quad \text{...(11.53)}$$

From (11.51) and (11.53)

$$\frac{E_f(s)}{E_o(s)} = \frac{RCs}{1 + RCs} \quad \text{...(11.54)}$$

From (11.54) put E_f in (11.49) and solve for $\frac{E_o(s)}{E_i(s)}$

$$\frac{E_o(s)}{E_i(s)} = \frac{K}{1 + \frac{KRCS}{1 + RCS}}$$

If $\frac{KRCS}{1 + RCS} >> 1$ then

$$\frac{E_o(s)}{E_i(s)} = \frac{1 + RCs}{RCS} = \frac{1}{RCS} + 1 \quad \text{...(11.55)}$$

or $$\frac{E_o(s)}{E_i(s)} = \frac{1}{T_i S} + 1 \quad \text{...(11.56)}$$

where, $T_i = RC$

Thus, we get PI control action.

11.12.4. PID Controller

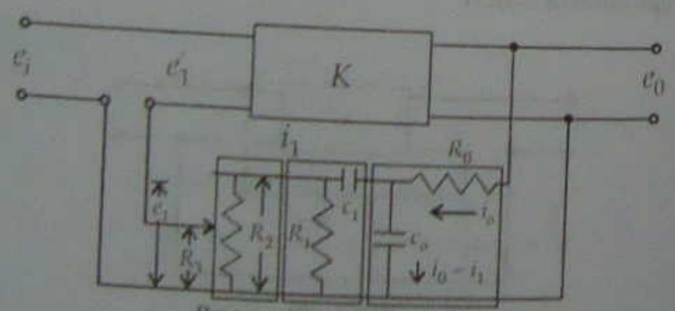


Fig. 11.31.

$$e_o = R_o i_o + \frac{1}{C_o}\int (i_o - i_1)\,dt \quad \text{...(11.57)}$$

Laplace transform

$$E_o(s) = R_o I_o(s) + \frac{1}{SC_o}(I_o(s) - I_1(s))$$

$$E_o(s) = I_o(s)\left[R_o + \frac{1}{sC_o}\right] - \frac{1}{sC_o} I_1(s) \quad \text{...(11.58)}$$

$$R_1 i_1 + \frac{1}{C_1}\int i_1\,dt - \frac{1}{C_o}\int (i_o - i_1)\,dt = 0 \quad \text{...(11.59)}$$

Laplace transform of (11.59)

$$R_1 I_1(s) + \frac{1}{SC_1} I_1(s) - \frac{1}{C_o}\left[I_o(s) - I_1(s)\right] = 0 \quad \text{...(11.60)}$$

Simplify equation (11.60)

or, $$I_o(s) = I_1(s)\left[\frac{R_1 C_1 C_o S^2 + SC_o + SC_1}{SC_1}\right] \quad \text{...(11.61)}$$

Put the value of $I_o(s)$ from (11.61) in (11.58) and simplify

$$E_o(s) = I_1(s)\left[\frac{R_1 C_1 R_o C_o S^2 + R_o C_o S + R_o C_1 S + R_1 C_1 S + 1}{SC_1}\right] \quad \text{...(11.62)}$$

Also, $$e_1 = R_1 i_1 \quad \text{...(11.63)}$$

Laplace transform of (11.63)

$$E_1(s) = R_1 I_1(s) \quad \text{...(11.64)}$$

From (11.62) and (11.64)

$$\frac{E_1(s)}{E_o(s)} = \frac{R_1 C_1 S}{R_1 C_1 R_o C_o S^2 + (R_o C_o + R_o C_1 + R_1 C_1) s + 1} \quad \text{...(11.63)}$$

Also, $$e_o = K(e_i - e_2) \quad \text{...(11.64)}$$

Laplace transform of (11.64)

$$E_o(s) = k[E_i(s) - E_2(s)] \quad \text{...(11.65)}$$

$$e_2 = e_1 \frac{R_3}{R_2} \quad \text{...(11.66)}$$

Laplace transform of (11.66)

$$E_2(s) = E_1(s) \cdot \frac{R_3}{R_2} \quad \text{...(11.67)}$$

Put the value of $E_2(s)$ from (11.67) to (11.65)

$$E_o(s) = \left[E_i(s) - E_i(s)\frac{R_3}{R_2}\right] K \quad \text{...(11.68)}$$

From (11.63) and (11.68) put the value of $E_1(s)$ from (11.63) in (11.68) and solve for $\frac{E_0(s)}{E_i(s)}$

$$\frac{E_0(s)}{E_i(s)} = \frac{K[R_1R_2C_1R_0C_0S^2 + R_2(R_0C_0 + R_0C_1 + R_1C_1)s + R_2]}{R_1R_2C_1R_0C_0S^2 + R_2(R_0C_0 + R_0C_1 + R_1C_1)s + R_2 + KR_1R_3C_1S} \quad \text{...(11.69)}$$

if $K >> 1$

$$\frac{E_0(s)}{E_i(s)} = K_1\left[sT_0 + \left(1 + \frac{R_0}{R_1} + \frac{T_0}{T_1}\right) + \frac{1}{sT_1}\right]$$

where, $K_1 = \frac{R_2}{R_3}$ ...(11.70)

$T_0 = R_0C_0$

$T_1 = R_1C_1$

or, $$\frac{E_0(s)}{E_i(s)} = K_1\alpha\left[1 + \frac{T_0}{\alpha}s + \frac{1}{\alpha T_1 s}\right]$$ ...(11.71)

where, $\alpha = 1 + \frac{R_0}{R_1} + \frac{T_0}{T_1}$

11.13. EFFECT OF INTEGRAL AND DERIVATIVE CONTROL ON THE SYSTEM PERFORMANCE

11.13.1. Integral Control Action

In proportional control of a plant whose transfer function does not possess an integrator (1/s) there is a steady-state error or an offset in the response to a step input. Such an offset can be eliminated if the integral control action is introduce in the controller.

The integral control action will removing offset or steady-state error may lead to oscillatory response of slowly decreasing amplitude or sometimes even increasing amplitude, both of which are usually undesirable.

11.13.2. Derivative Control Action

Derivative control action when added to proportional controller provides a means of obtaining a controller with high sensitivity and advantage of using derivative control action is that it responds to rate of change of actuating error can produces an efficient correction before the magnitude of actuating error becomes too large. Derivative control thus anticipate the actuating error, initiate an easily corrective action and leads to increase the stability of the system. Although derivative control does not effect the steady-state error directly, it adds damping to the system and thus permits the use of a larger value of gain 'K' which will result in an improvement in the steady-state accuracy. Because derivative control operates on the rate of change of actuating error, not as the actuating error itself, this mode is never used alone. It is always used in combination with proportional or proportional integral control.

SUMMARY

A controller accepts the error and made proper corrective action. The output of the controller is then applied to the control element.

In two position control the output of the controller is change to either a maximum or minimum value depending upon the control variable.

In proportion control action, the output of the controller is proportional to the error. When the error is zero, the controller output is constant. In this action the gain K_p and proportional band are inversely proportional to each other.

In integral control action, the output of the controller is change at a rate which is proportional to the actuating error signal $e(t)$. This type of action is also known as reset action. If the error is zero, the output remains at a fixed value to what is was, when the error become zero. In practice integral control action is never used alone.

In derivative control action the output of the controller depends on the time rate of change of the actual errors. When the error is zero or a constant, the derivative controller output is zero.

The characteristics of PI control action are (*i*) steady state accuracy improves (*ii*) rise time increases (*iii*) bandwidth decreases (*iv*) response is oscillatory.

The characteristics of PD control action are (*i*) rise time reduce (*ii*) improves the damping (*iii*) improves the bandwidth (*iv*) response is stable (*v*) overshoot reduces.

The characteristics of PID control action are (*i*) no oscillations (*ii*) improves the transient response (*iii*) improves the steady state response.

EXERCISE

1. What are the basic elements of an industrial automatic controller?
2. Explain Two position or ON-OFF control.
3. Write short note on proportional control action & integral control action.
4. Show that the steady state error increase by using the derivative feedback control.
5. Solve the example 11.1 for damping ratio 0.8.

SEMI-OBJECTIVE TYPE QUESTIONS

(*i*) Define mode of control. (Article 11.1)

(*ii*) With a suitable diagram show the elements of industrial automatic controller. (Article 11.2)

(*iii*) Classify the various type of controllers. (Article 11.3)

(*iv*) Write short note on two position control. (Article 11.4)

(*v*) Write short note on proportional control action. (Article 11.5)

(*vi*) Discuss in brief derivative control action. (Article 11.7)

(*vii*) What is PID controller? (Article 11.10)

(*viii*) Write short note on derivative and Integral control action.

(*ix*) Write short note on PD control action. (Summary)

(*x*) Write short note on PI control action. (Summary)

(*xi*) Write short note on PID control action. (Summary)

□□□

Chapter 12

Miscellaneous Problems

Example 12.1. Draw the electrical analogous circuit of the system shown in fig. 1.46 using *f-v* and *f - i* analogy and also write the equations.

Solution : *F - V* analogy :

$$V(s) = q_1 \frac{1}{C_1} + s^2 L_1 q_1 + \frac{1}{C_2}(q_1 - q_2)$$

$$\frac{1}{C_2}(q_2 - q_1) + s^2 L_2 q_2 + \frac{1}{C_3} q_2 = 0$$

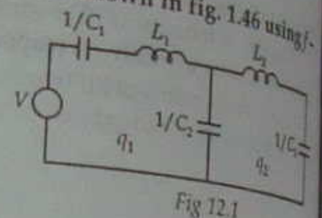


Fig 12.1

F-*i* Analogy

$$I(S) = \frac{1}{L_1}\phi_1 + s^2 C_1 \phi_1 + \frac{1}{L_2}(\phi_1 - \phi_2)$$

$$\frac{1}{L_2}(\phi_2 - \phi_1) + s^2 C_2 \phi_2 + \frac{1}{L_3}\phi_2 = 0$$

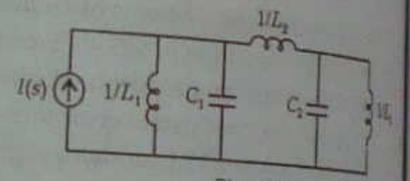


Fig. 12.2.

Example 12.2. Draw the mechanical equivalent network of the system shown in fig 1.35.

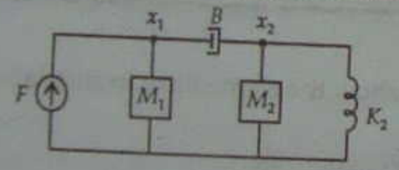


Fig 12.3

Example 12.3. Draw the mechanical equivalent network of the system shown in fig 1.53.

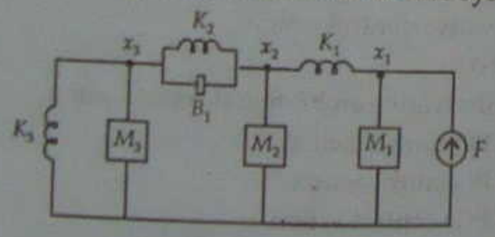


Fig. 12.4

Example 12.4. Draw the mechanical equivalent network of the system shown in fig. Also draw the electrical analogous circuit and write the set of equation using *f-v* and *f-i* analogy.

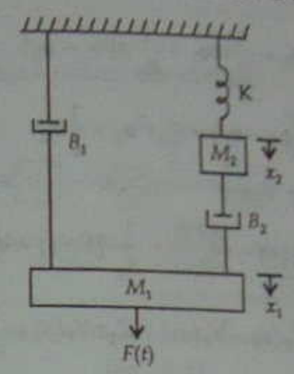


Fig. 12.5

Solution : (*a*) Mechanical equivalent network

$$F = M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + B_2 \frac{d}{dt}(x_1 - x_2)$$

$$0 = M_2 \frac{d^2 x_2}{dt^2} + Kx_2 + B_2 \frac{d}{dt}(x_2 - x_1)$$

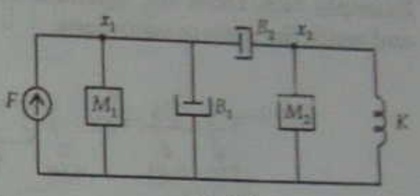


Fig. 12.6.

(*b*) *F-V* analogy

$$V = L_1 \frac{d^2}{dt^2} q_1 + R_1 \frac{dq_1}{dt} + R_2 \frac{d}{dt}(q_1 - q_2)$$

$$0 = R_2 \frac{d}{dt}(q_2 - q_1) + L_2 \frac{d^2}{dt^2} q_2 + \frac{1}{C} q_2$$

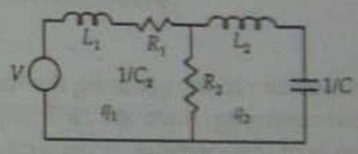


Fig. 12.7.

Take laplace transform of above equations

$$V(s) = L_1 s^2 q_1 + R_1 s q_1 + R_2 s(q_1 - q_2)$$

$$0 = R_2 s(q_2 - q_1) + L_2 s^2 q_2 + \frac{1}{C} q_2$$

since, $I = \frac{dq}{dt}$

or, $I(s) = sq$

or $q = \frac{I(s)}{s}$

then, $V(s) = L_1 s I_1(s) + R_1 I_1(s) + R_2[I_1(s) - I_2(s)]$

$$0 = R_2[I_2(s) - I_1(s)] + L_2 s I_2(s) + \frac{1}{sC} I_2(s)$$

(*c*) *F-i* analogy :

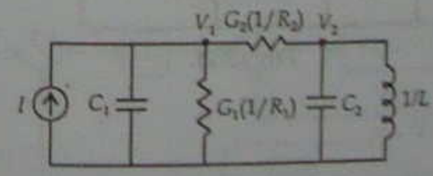


Fig. 12.8.

$$I(s) = C_1 s^2\phi_1 + \frac{1}{R_1}s\phi_1 + \frac{1}{R_2}s(\phi_1 - \phi_2)$$

$$0 = \frac{1}{R_2}s(\phi_2 - \phi_1) + C_2 s^2\phi_2 + \frac{1}{L}\phi_2$$

Since $V(s) = s\,\phi\,(s)$ then

$$I\,(s) = C_1 s V_1(s) + \frac{V_1(s)}{R_1} + \frac{1}{R_2}[V_1(s) - V_2(s)]$$

$$0 = \frac{1}{R_2}[V_2(s) - V_1(s)] + C_2 s\,V_2(s) + \frac{1}{sL}V_2(s)$$

Example 12.5. Draw the mechanical equivalent network of the following rotational system and write the system equations.

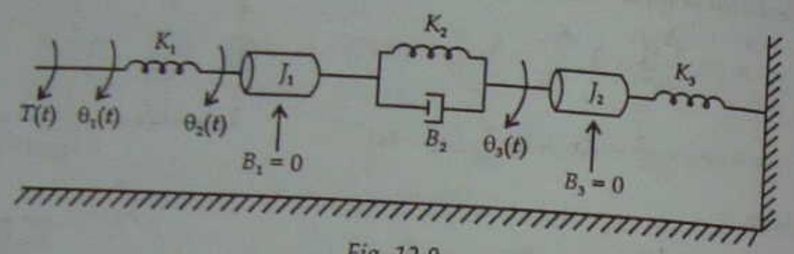


Fig. 12.9.

Solution : In the given fig the torque is applied to the spring so, there are two displacements at the ends of the spring. J_1 is under θ_2. K_2 and B_2 are in parallel between $\theta_2(t)$ and $\theta_3(t)$. J_2 and K_3 are under θ_3.

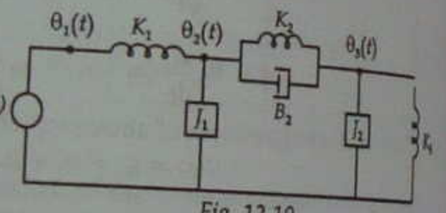


Fig. 12.10.

$$T(t) = K_1\,(\theta_1 - \theta_2)$$

$$K_1(\theta_2 - \theta_1) + J_1\frac{d^2\theta_2}{dt^2} + K_2(\theta_2 - \theta_3) + B_2\frac{d}{dt}(\theta_2 - \theta_3) = 0$$

$$J\frac{d^2\theta_3}{dt^2} + K_3\theta_3 + K_2(\theta_3 - \theta_2) + B_2\frac{d}{dt}(\theta_3 - \theta_2) = 0$$

Example 12.6. For Example (12.5) draw the electrical analogous circuit using *T-V* analogy.

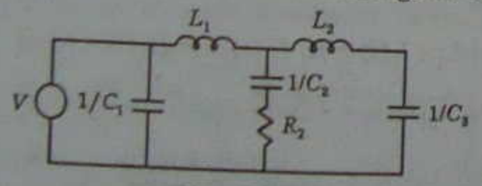


Fig. 12.11.

Example 12.7. Find $\frac{C(s)}{R(s)}$, using block reduction method.

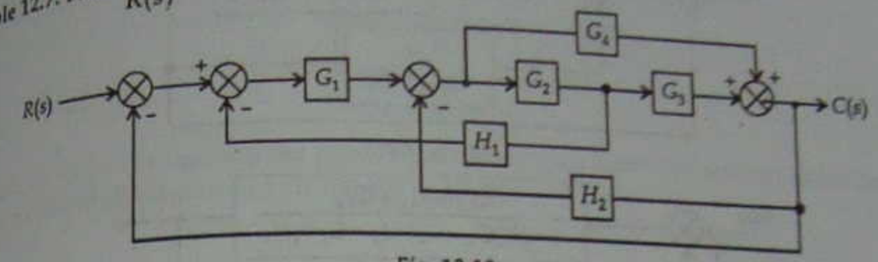


Fig. 12.12.

Solution :

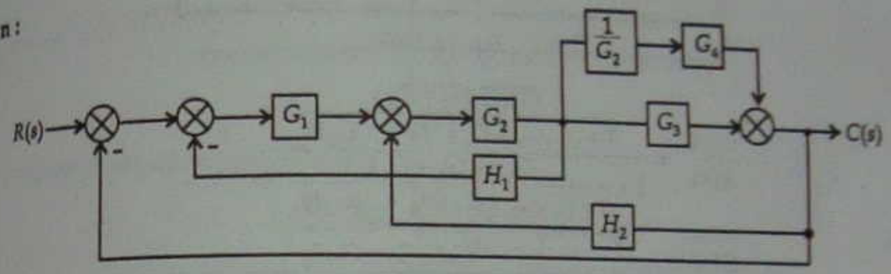


Fig. 12.12(a).

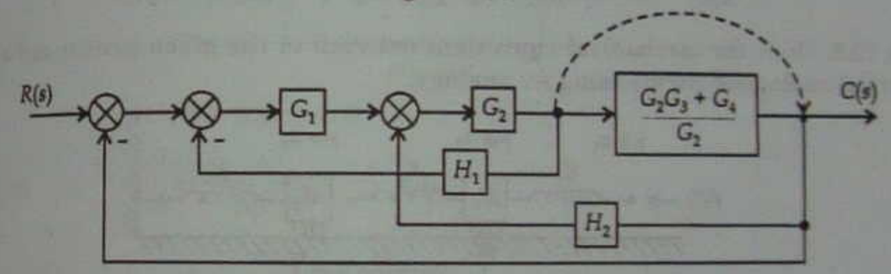


Fig. 12.12(b)

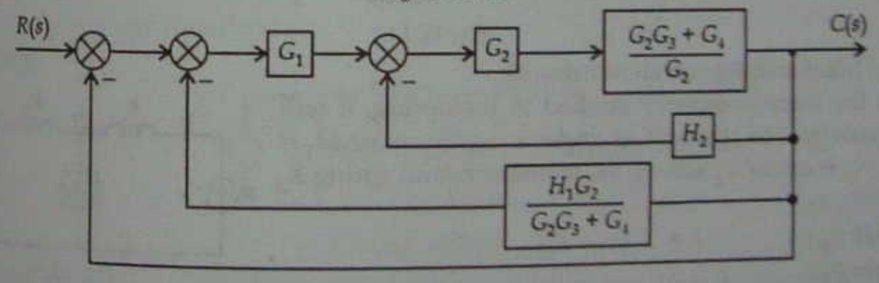


Fig. 12.12(c)

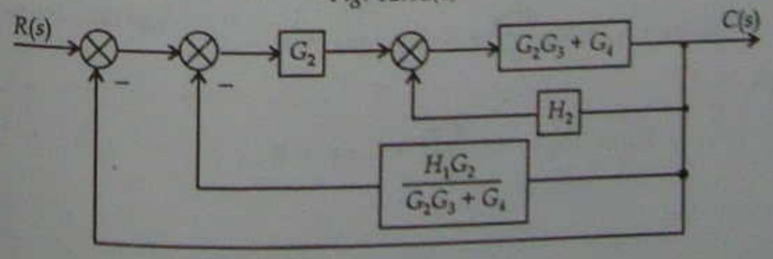


Fig. 12.12(d)

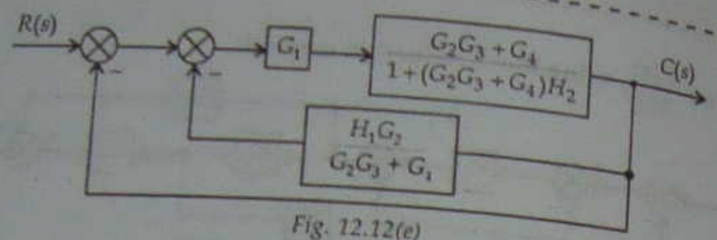


Fig. 12.12(e)

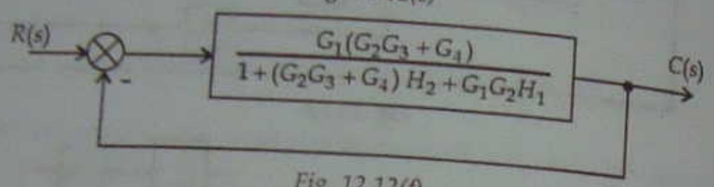


Fig. 12.12(f)

$$\frac{C(s)}{R(s)} = \frac{\dfrac{G_1(G_2G_3+G_4)}{1+(G_2G_3+G_4)H_2+G_1G_2H_1}}{1+\dfrac{G_1(G_2G_3+G_4)}{1+(G_2G_3+G_4)H_2+G_1G_2H_1}\cdot 1}$$

$$\frac{C(s)}{R(s)} = \frac{G_1G_2G_3+G_1G_4}{1+G_2G_3H_2+G_4H_2+G_1G_2H_1+G_1G_2G_3+G_1G_4}$$

Example 12.8. Draw the mechanical equivalent network of the given system and also draw the electrical analogous circuit using *f-v* analogy.

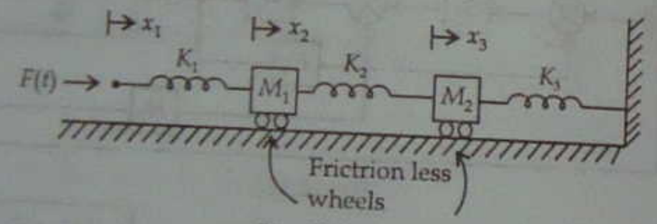


Fig. 12.13.

Solution : Mechanical equivalent network.

Since the force is directly applied to the spring, it will store the energy. The spring K_1 is under x_1 and x_2, mass M_1 is under x_2, K_2 is under x_2 and x_3, M_2 is under x_3 and spring K_3 is also under x_3.

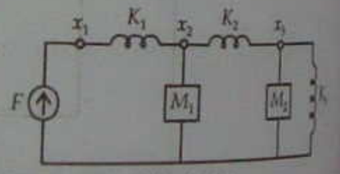


Fig. 12.14.

at node x_1 : $F = K_1(x_1 - x_2)$

at node x_2 :

$$K_1(x_2 - x_1) + M_1\frac{d^2x_2}{dt^2} + K_2(x_2 - x_3) = 0$$

at node x_3 :

$$K_2(x_3 - x_2) + M_2\frac{d^2x_3}{dt^2} + K_3x_3 = 0$$

f-v analogy :

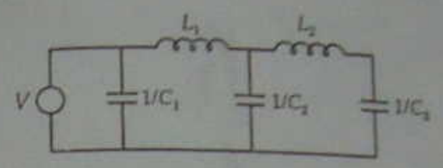


Fig. 12.15.

Example 12.9. Draw the mechanical equivalent circuit and write the system equations. Also draw the electrical analogous circuit use *f-v* analogy.

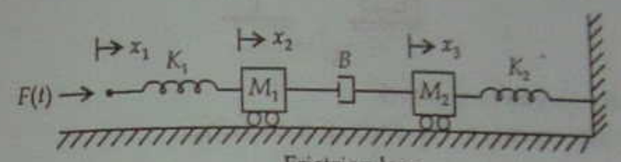


Fig. 12.16.

Solution : Mechanical equivalent network.

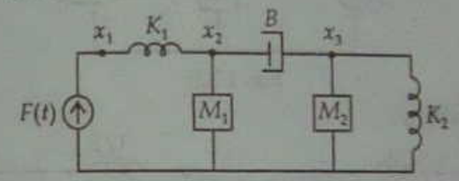


Fig. 12.17.

$$F(t) = K_1(x_1 - x_2)$$

$$K_1(x_2 - x_1) + M_1\frac{d^2x_2}{dt^2} + B\frac{d}{dt}(x_2 - x_3) = 0$$

$$B\frac{d}{dt}(x_3 - x_2) + M_2\frac{d^2x_3}{dt^2} + k_2x_3 = 0$$

These are the system equations.

f-v analogy

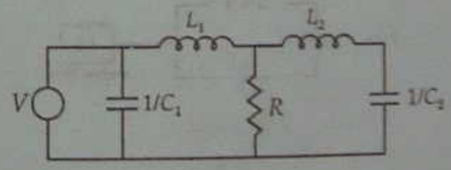


Fig. 12.18.

Example 12.10. Draw the mechanical network of given system and draw the electrical analogous circuit use *f-v* and *f-i* analogy.

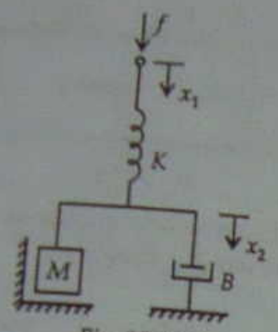


Fig. 12.19.

Solution : Mechanical equivalent network :

There are two displacements at the ends of the spring.

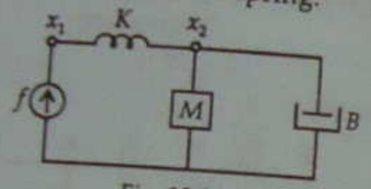
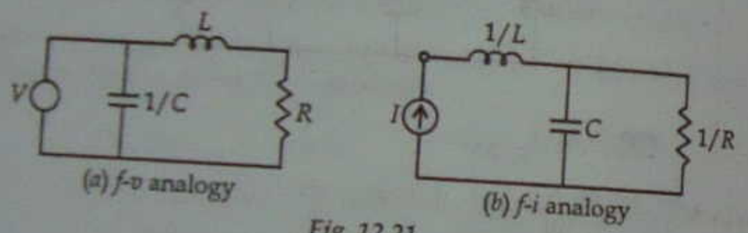


Fig. 12.20.

Electrical analogous circuits :



(a) *f-v* analogy (b) *f-i* analogy

Fig. 12.21.

Example 12.11 Draw the mechanical equivalent and write the system equations.

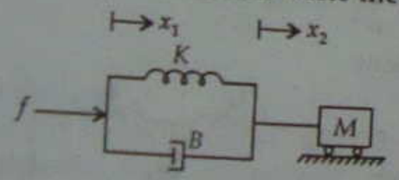


Fig. 12.22.

Solution :

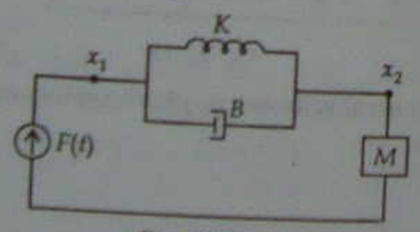


Fig. 12.23.

$$F(t) = K(x_1 - x_2) + B\frac{d}{dt}(x_1 - x_2)$$

$$K(x_2 - x_1) + B\frac{d}{dt}(x_2 - x_1) + M\frac{d^2x_2}{dt^2} = 0$$

Example 12.12. Draw the mechanical equivalent circuit and write the system equations.

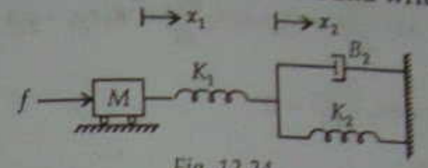


Fig. 12.24.

Solution :

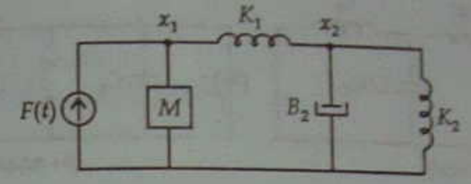
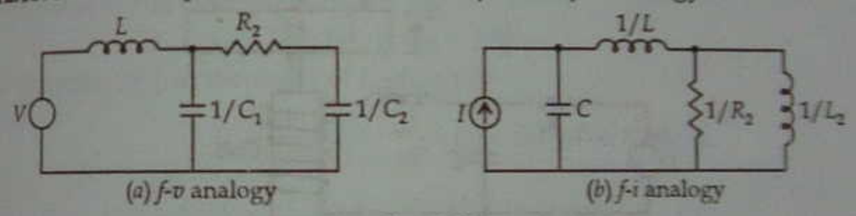


Fig. 12.25.

$$F(t) = M_1\frac{d^2x_1}{dt^2} + K_1(x_1 - x_2)$$

$$K_1(x_2 - x_1) + B_2\frac{d}{dt}x_2 + k_2x_2 = 0$$

Example 12.13. For above question no. 12.12 draw *f-v* and *f-i* analogy.



(a) *f-v* analogy (b) *f-i* analogy

Fig. 12.26.

Example 12.14. Write the differential equations governing the behaviour of the mechanical system shown in fig. Also obtain the analogous electrical circuit based on *f-v* and *f-i* analogy.

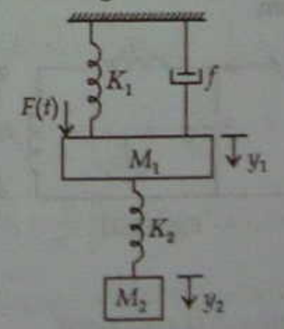


Fig. 12.27.

Solution : Mechanical equivalent network

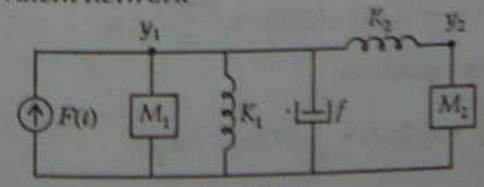
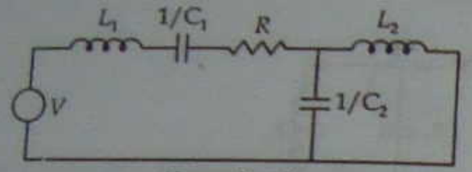


Fig. 12.28.

at node y_1 : $$F(t) = M_1 \frac{d^2y_1}{dt^2} + K_1 y_1 + f_1 \frac{dy_1}{dt} + K_2(y_1 - y_2)$$

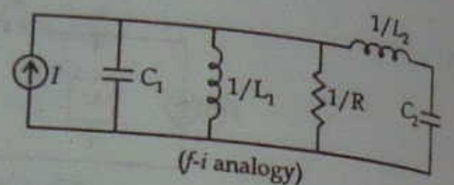
at node y_2 : $$K_2(y_2 - y_1) + M_2 \frac{d^2y_2}{dt^2} = 0$$

Analogous electrical circuits :



(*f*-*v* analogy)

Fig. 12.29.



(*f*-*i* analogy)

Fig. 12.30.

Example 12.15. Find the transfer function $\frac{X(s)}{E(s)}$ for the electromechanical system shown in fig 12.31.

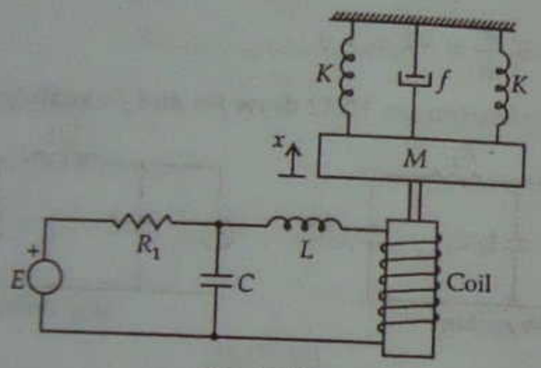


Fig. 12.31.

Solution : The given system consists of two parts (*a*) electrical (*b*) mechanical

First consider the electrical system.

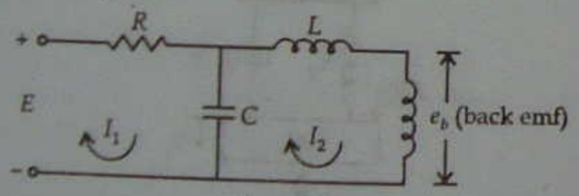


Fig. 12.32.

Apply KVL in both meshes :

$$E(t) = R_1 I_1(t) + \frac{1}{C}[I_1(t) - I_2(t)] \quad ...(1)$$

$$0 = \frac{1}{C}[I_2(t) - I_1(t)] + L\frac{dI_2(t)}{dt} + e_b \quad ...(2)$$

Laplace transform of both equations

$$E(s) = R_1 I_1(s) + \frac{1}{sC}[I_1(s) - I_2(s)] \quad ...(3)$$

$$0 = \frac{1}{sC}[I_2(s) - I_1(s)] + sL\, I_2(s) + E_b(s) \quad ...(4)$$

$$e_b \alpha \frac{dx}{dt}$$

Now,

$$e_b = K_1 \frac{dx}{dt} \quad ...(5)$$

$$E_b(s) = K_1 sX(s)$$

where K_1 = back e.m. f constant

Now consider the mechanical system :

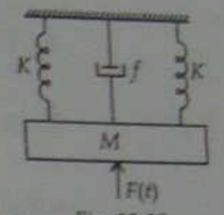


Fig. 12.33.

$$F(t) \alpha I_2(t)$$

$$F(t) = K_2 I_2(t)$$

$$F(s) = K_2 I_2(s) \quad ...(6)$$

From mechanical network

$$F(t) = M\frac{d^2x(t)}{dt^2} + f\frac{dx(t)}{dt} + 2kx(t)$$

$$F(s) = s^2 MX(s) + sfX(s) + 2KX(s)$$

$$F(s) = X(s)[s^2 M + sf + 2K] \quad ...(7)$$

Put the value of $F(s)$ from (6) in (7) we get

$$K_2 I_2(s) = X(s)[s^2 M + sf + 2K] \quad ...(8)$$

From equation (5) put the value of $E_b(s)$ in (4)

$$0 = \frac{1}{sC} I_2(s) - \frac{1}{sC} I_1(s) + sL\, I_2(s) + K_1 sX(s)$$

or $$\frac{1}{sC} I_1(s) = \frac{1}{sC} I_2(s) + sL\, I_2(s) + K_1 sX(s)$$

$$I_1(s) = I_2(s) + s^2 LCI_2(s) + s^2 XK_1 X(s) \quad ...(9)$$

Put the value of $I_1(s)$ from equation (9) in equation (3)

$$E(s) = R_1 I_1(s) + \frac{1}{sC} I_1(s) - \frac{1}{sC} I_2(s)$$

$$= \left[R_1 + \frac{1}{sC}\right] I_1(s) - \frac{1}{sC} I_2(s)$$

$$E(s) = \left[R_1 + \frac{1}{sC}\right][I_2(s) + s^2 LCI_2(s) + s^2 CK_1 X(s)] - \frac{1}{sC} I_2(s)$$

$$E(s) = R_1 I_2(s) + s^2 R_1 LCI_2(s) + s^2 R_1 CK_1 X(s) + \frac{1}{sC} I_2(s) + SLI_2(s) + SK_1 X(s) - \frac{1}{sC} I_2(s)$$

$$E(s) = I_2(s)[R_1 + s^2 R_1 LC + sL] + X(s)\, sK_1[1 + R_1 Cs]$$

$$I_2(s)[R_1 + s^2 R_1 LC + sL] = E(s) - X(s)\, sK_1(1 + R_1 Cs)$$

$$\therefore \quad I_2(s) = \frac{E(s) - X(s)\, sK_1(1 + R_1 Cs)}{R_1 + s^2 R_1 LC + sL} \quad ...(10)$$

Put the value of $I_2(s)$ from equation (10) in equation (8)

$$K_2\left[\frac{E(s)-X(s)\,sK_1(1+R_1Cs)}{R_1+s^2R_1LC+sL}\right] = X(s)[s^2M+sf+2K]$$

$$K_2E(s)-X(s)\,sK_1K_2(1+R_1Cs) = X(s)[R_1+s^2R_1LC+sL][s^2M+sf+2K]$$

$$K_2E(s)-X(s)\,sK_1K_2(1+R_1Cs) = X(s)[s^2R_1M+sR_1f+2KR_1 + s^4MR_1LC+s^3R_1LCf+2s^2R_1LCK+s^2ML+s^2fL+2KLs]$$

$$K_2E(s)-X(s)\,sK_1K_2(1+R_1Cs) = x(s)[s^4R_1LCM+s^3L(R_1Cf+M)+s^2R_1M+s^2fL+2s^2R_1LCK + 2R_1K+sR_1f+2KLs]$$

$$K_2E(s) = X(s)[s^4R_1LCM+s^3L(R_1Cf+M)+s^2R_1M+s^2fL+2s^2R_1LCK + 2KLs+sR_1f2R_1K+sK_1K_2+K_1K_2R_1Cs^2]$$

$$= X(s)[s^4R_1LCM+s^3L(R_1cf+M)+s^2(R_1M+fL+2R_1LCK + K_1K_2R_1C)+2KLS+2R_1K+sR_1f+sK_1K_2]$$

$$= X(s)[s^4R_1LCM+s^3L(R_1cf+M)+\{R_1M+Lf+R_1C(2LK+K_1K_2)\}s^2 + s(R_1f+2LK+K_1K_2)+2R_1K]$$

$$\therefore \frac{X(s)}{E(s)} = \frac{K_2}{\{s^4R_1LCM+s^3L(R_1cf+M)+s^2\{R_1M+Lf+R_1C(2LK+K_1K_2)\}+s(R_1f+2LK+K_1K_2)+2R_1K]}$$

This is the required answer.

Example 12.16. For mass-damper spring combination shown in fig 12.34 draw the mech. Equivalent network, write the system equations and draw *f-v*, *f-i* analog.

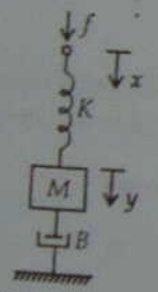


Fig. 12.34.

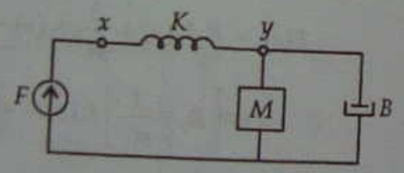


Fig. 12.35.

Solution: Mechanical network (Fig. 12.35)

at node x : $F = K(x-y)$

$$0 = K(y-x)+M\frac{d^2y}{dt^2}+B\frac{dy}{dt}$$

f-v analogy

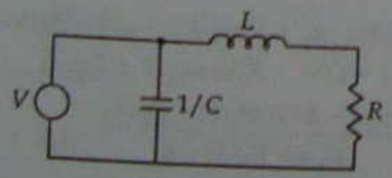


Fig. 12.36.

$$V = \frac{1}{C}(q_1-q_2)$$

$$\frac{1}{C}(q_2-q_1)+L\frac{d^2q_2}{dt^2}+R\frac{dq_2}{dt} = 0$$

$$I(t) = \frac{dq}{dt} \qquad \therefore I(s) = s\,q(s)$$

$$V = \frac{1}{sC}[I_1(s)-I_2(s)]$$

$$\therefore \frac{1}{sC}[I_2(s)-I_2(s)]+sLI_2(s)+RI_2(s) = 0$$

***f-i* analogy :**

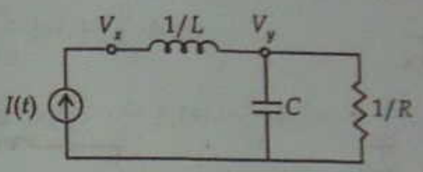


Fig. 12.37.

Example 12.17. For the mechanical system shown in fig 12.38(*a*) draw mechanical equivalent network (*b*) write the differential equations (*c*) draw the electrical analogous circuit using *f-v* and *f-i* analogy.

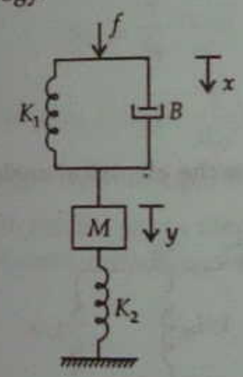


Fig. 12.38.

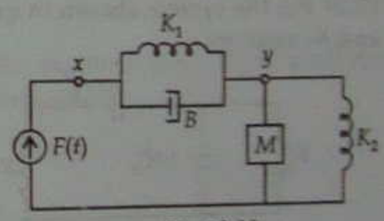


Fig. 12.39.

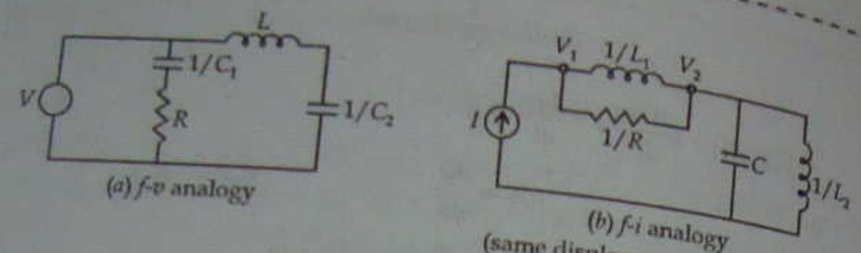
Solution : Mechanical network (Fig. 12.39):

(b) ***System equations :***

$$F(t) = K_1(x-y)+B\frac{d}{dt}(x-y)$$

$$K_1(y-x)+B\frac{d}{dt}(y-x)+M\frac{d^2y}{dt^2}+K_2y = 0$$

(c) Electrical analogous circuit :



(a) *f-v* analogy (b) *f-i* analogy (same displacement same voltage)

Fig. 12.40

Example 12.18. For the given mechanical system draw the mechanical equivalent network and write the system equations.

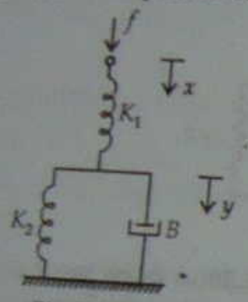


Fig. 12.41.

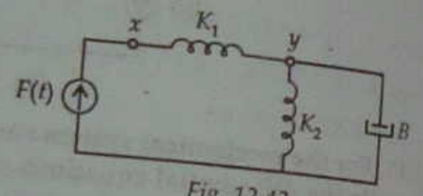


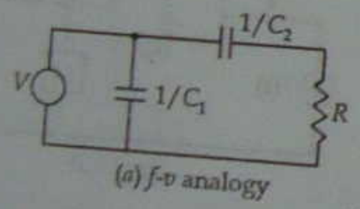
Fig. 12.42.

Solution : Mechanical equivalent network (Fig. 12.42)

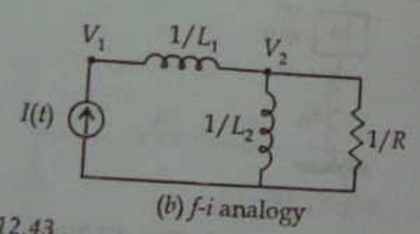
at node x : $F(t) = K_1 (x - y)$

at node y : $K_1(y - x) + K_2 y + B\frac{d}{dt} y = 0$

Example 12.19. For the system shown in question (12.18) draw the electrical analogous circuit using *f-v* and *f-i* analogy.



(a) *f-v* analogy



(b) *f-i* analogy

Fig. 12.43.

For *f-v* analogy :

$$V = \frac{1}{C_1}(q_1 - q_2)$$

$$0 = \frac{1}{C_1}(q_2 - q_1) + \frac{1}{C_2} q_2 + R\frac{d}{dt} q_2$$

Since $i = \frac{dq}{dt}$

or, $I(s) = sq(s)$

$$q(s) = \frac{I(s)}{s}$$

$$V = \frac{1}{sC_1}[I_1(s) - I_2(s)]$$

$$0 = \frac{1}{sC_1}[I_2(s) - I_1(s)] + \frac{1}{sC_2} I_2(s) + RI_2(s)$$

For *f-i* analogy :

laplace transform of the system equations given in question 12.18

$$F(s) = K_1 X(s) - K_1 Y(s)$$

$$K_1 Y(s) - K_1 X(s) + K_2 Y(s) + sBY(s) = 0$$

From *f-i* analogy circuit :

$$I(s) = \frac{1}{L_1}\phi_1(s) - \frac{1}{L_1}\phi_2(s)$$

$$\frac{1}{L_1}\phi_2(s) - \frac{1}{L_1}\phi_1(s) + \frac{1}{L_2}\phi_2(s) + s\cdot\frac{1}{R}\phi_2(s) = 0$$

since, $V(t) = \frac{d\phi}{dt}$ or, $V(s) = s\phi(s)$

replace $\phi(s)$ by $\frac{V(s)}{s}$

$$I(s) = \frac{1}{L_1}\frac{V_1(s)}{s} - \frac{1}{L_1}\frac{V_2(s)}{s} = \frac{1}{sL_1}[V_1(s) - V_2(s)]$$

$$\frac{1}{sL_1}[V_2(s) - V_1(s)] + \frac{1}{sL_2} V_2(s) + \frac{V_2(s)}{R} = 0$$

These are the required equations.

Example 12.20. For the given mechanical system draw (a) mechanical equivalent network (*b*) electrical analogous circuit (*c*) write the system equations.

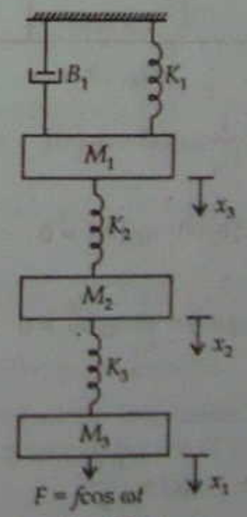


Fig. 12.44.

Solution : Mechanical equivalent network

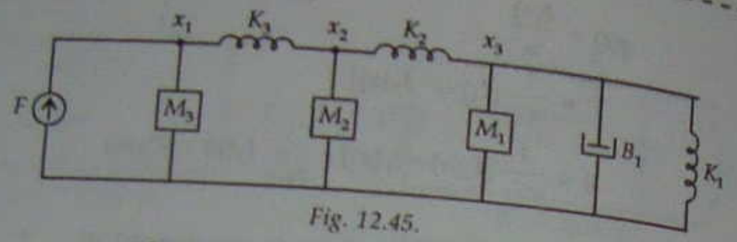


Fig. 12.45.

at node x_1 : $F = M_3 \frac{d^2x_1}{dt^2} + K_3(x_1 - x_2)$

at node x_2 : $K_3(x_2 - x_1) + M_2 \frac{d^2x_2}{dt^2} + K_2(x_2 - x_3) = 0$

at node x_3 : $K_2(x_3 - x_2) + M_1 \frac{d^2x_3}{dt^2} + B_1 \frac{dx_3}{dt} + K_1 x_3 = 0$

Laplace transform of above equations

$F = f\cos\omega t$ given

$$F(s) = \frac{f\omega}{s^2 + \omega^2}$$

$$\frac{f\omega}{s^2 + \omega^2} = s^2 M_3 X_1(s) + K_3 [X_1(s) - X_2(s)]$$

$$K_3[X_2(s) - X_1(s)] + s^2 M_2 X_2(s) + K_2]X_2(s) - X_3(s)] = 0$$

$$K_2[X_3(s) - X_2(s)] + s^2 M_1 X_3(s) + sB_1 X_3(s) + K_1 X_3(s) = 0$$

F-V analogy :

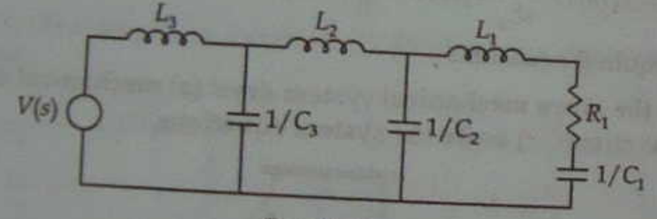


Fig. 12.46.

$$V(s) = s^2 L_3 q_1(s) + \frac{1}{C_3}[q_1(s) - q_2(s)]$$

$$\frac{1}{C_3}[q_2(s) - q_1(s)] + s^2 L_2 q_2(s) + \frac{1}{C_2}[q_2(s) - q_3(s)] = 0$$

$$\frac{1}{C_2}[q_3(s) - q_2(s)] + s^2 L_1 q_3(s) + sR_1 q_3(s) + \frac{1}{C_1} q_3(s) = 0$$

Since $i = \frac{dq}{dt}$

$I(s) = sq(s)$ or $q(s) = \frac{I(s)}{s}$

$$V(s) = s^2 L_3 \frac{I_1(s)}{s} + \frac{1}{sC_3}[I_1(s) - I_2(s)]$$

$$V(s) = sL_3 I_1(s) + \frac{1}{sC_3}[I_1(s) - I_2(s)]$$

$$\frac{1}{sC_3}[I_2(s) - I_1(s)] + sL_2 I_2(s) + \frac{1}{sC_2}[I_2(s) - I_3(s)] = 0$$

$$\frac{1}{sC_2}[I_3(s) - I_2(s)] + sL_1 I_3(s) + R_1 I_3(s) + \frac{1}{sC_1} R_3(s) = 0$$

F-i analogy :

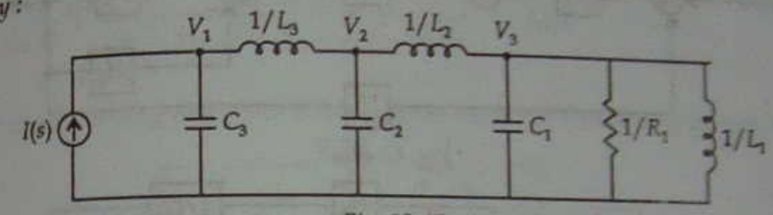


Fig. 12.47.

$$I(s) = s^2 C_3 \phi_1(s) + \frac{1}{L_3}[\phi_1(s) - \phi_2(s)]$$

$$\frac{1}{L_3}[\phi_2(s) - \phi_1(s)] + s^2 C_2 \phi_2(s) + \frac{1}{L_2}[\phi_2(s) - \phi_3(s)] = 0$$

$$\frac{1}{L_2}[\phi_3(s) - \phi_2(s)] + s^2 c_1 \phi_3(s) + s \cdot \frac{1}{R_1}\phi_3(s) + \frac{1}{L_1}\phi_3(s) = 0$$

Since $V(t) = \frac{d\phi}{dt}$

$V(s) = s\phi(s)$ or, $\phi(s) = \frac{V(s)}{s}$

$$I(s) = sC_3 V_1(s) + \frac{1}{sL_3}[V_1(s) - V_2(s)]$$

$$\frac{1}{sL_3}[V_2(s) - V_1(s)] + sC_2 V_2(s) + \frac{1}{sL_2}[V_2(s) - V_3(s)] = 0$$

$$\frac{1}{sL_2}[V_3(s) - V_2(s)] + sC_1 V_3(s) + \frac{V_3(s)}{R_1} + \frac{1}{sL_1} V_3(s) = 0$$

These are the required equations of the given system.

Example 12.21. Using block diagram reduction technique, find closed loop transfer function of the system whose block diagram is given in fig. 12.48 (*a*) when $R_1 = 0$ (*b*) when $R_2 = 0$.

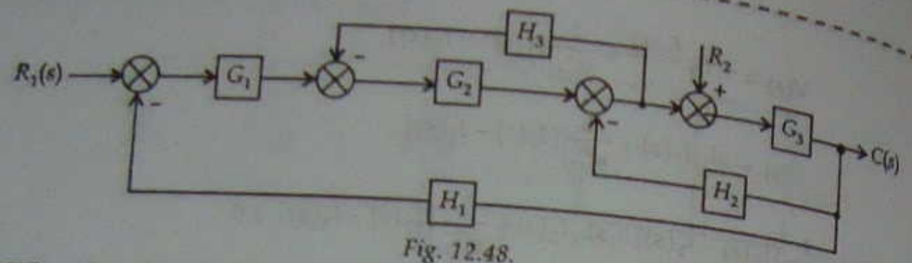


Fig. 12.48.

Solution : When $R_2 = 0$

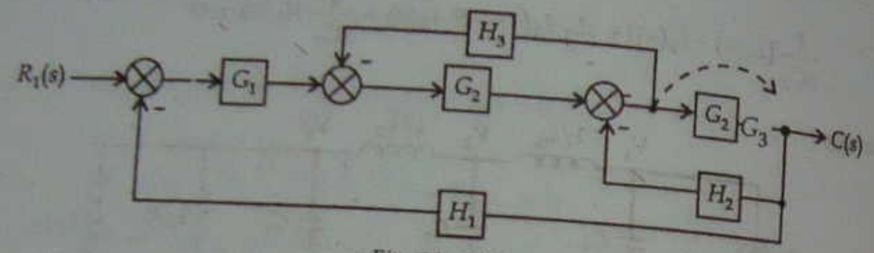


Fig. 12.48(a)

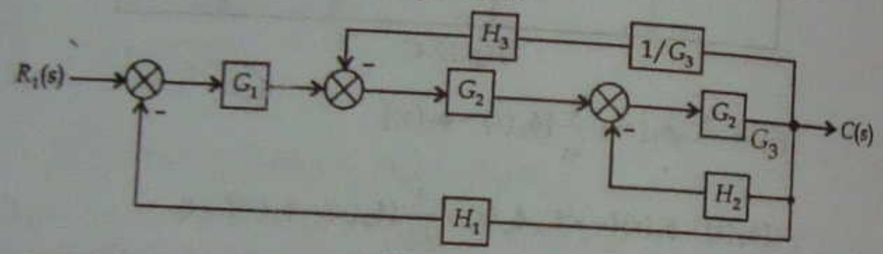


Fig. 12.48(b)

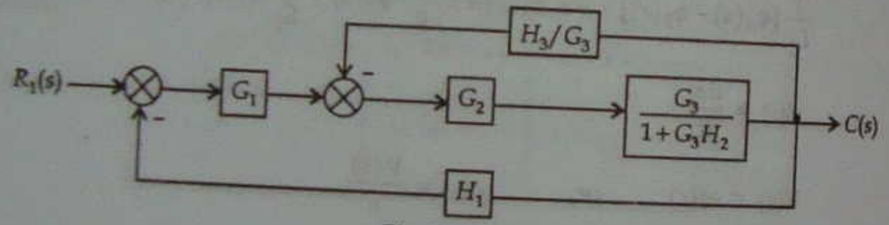


Fig. 12.48(c)

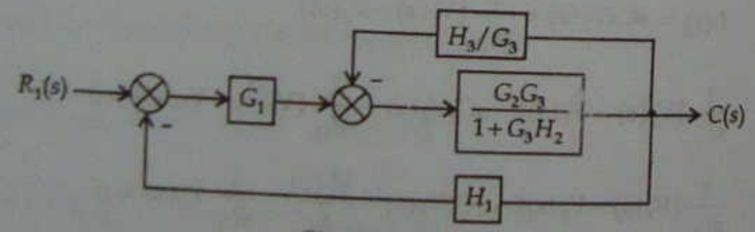


Fig. 12.48(d)

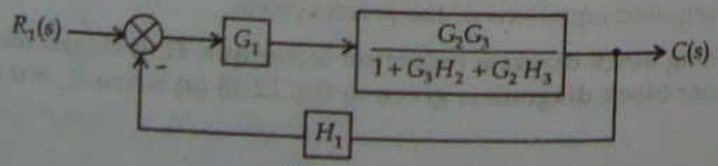


Fig. 12.48(e)

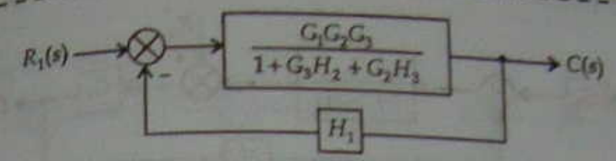


Fig. 12.48(f)

$$\frac{c(s)}{R_1(s)} = \frac{\dfrac{G_1 G_2 G_3}{1+G_3H_2+G_2H_3}}{1+\dfrac{G_1G_2G_3H_1}{1+G_3H_2+G_2H_3}}$$

$$\frac{C(s)}{R_1(s)} = \frac{G_1 G_2 G_3}{1+G_3 H_2+G_2 H_3+G_1 G_2 G_3H_1} \quad \textbf{Ans.}$$

(b) when $R_1 = 0$

Now remove the summing point since feedback is negative therefore put-sign with H_1

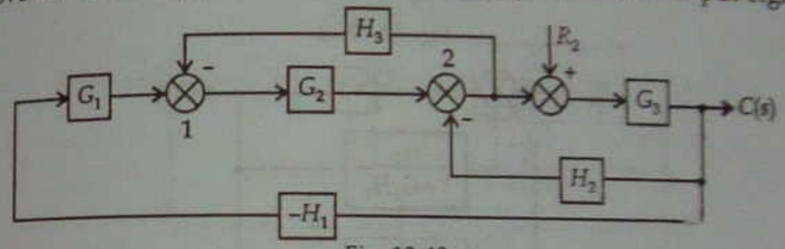


Fig. 12.49.

shift the summing point 1 after G_2 then

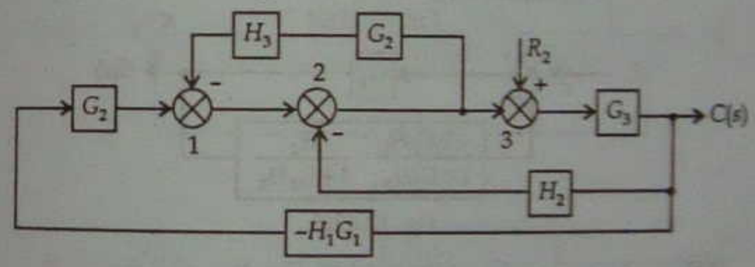


Fig. 12.49(a)

Now interchange the summing point 1 and 2

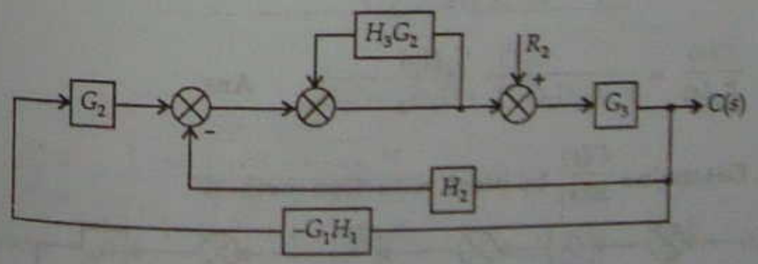


Fig. 12.49(b)

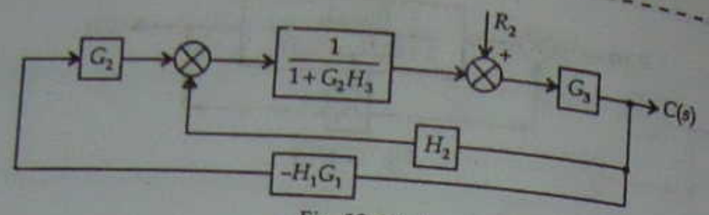


Fig. 12.49(c)

shift the summing point after the block

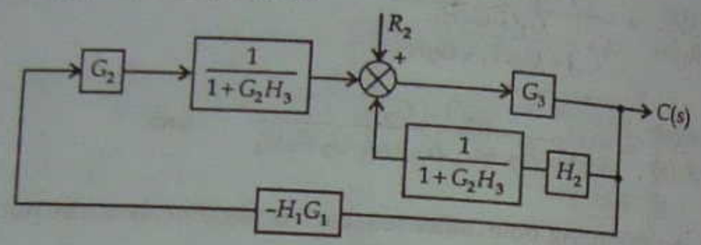


Fig. 12.49(d)

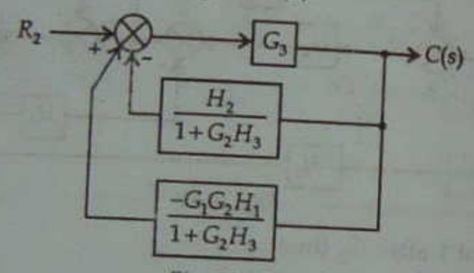


Fig. 12.49(e)

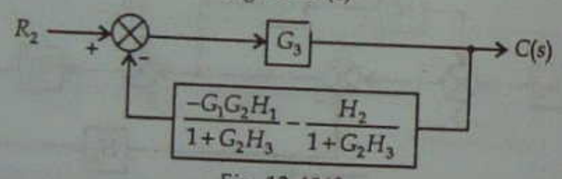


Fig. 12.49(f)

$$\frac{C(s)}{R_2(s)} = \frac{G_3}{1-\left[\frac{-G_1\,G_2\,H_1 - H_2}{1+G_2\,H_3}\right]G_3}$$

$$\frac{C(s)}{R_2(s)} = \frac{G_3(1+G_2H_3)}{1+G_1G_2G_3H_1+G_3H_2+G_2H_3}$$ **Ans.**

Example 12.22. Determine $\frac{C(s)}{R(s)}$ by block reduction method.

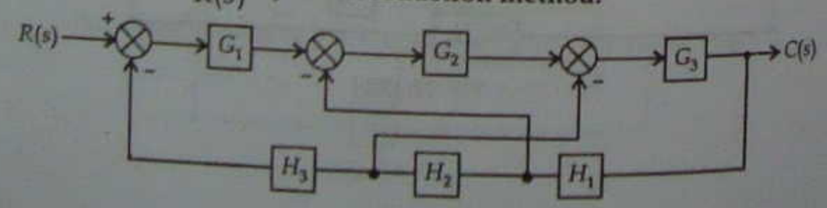


Fig. 12.50.

Solution : Rearrange the blocks

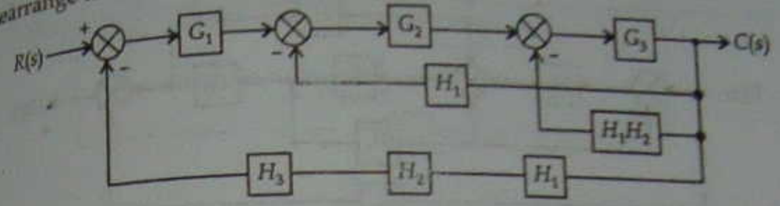


Fig. 12.50(a).

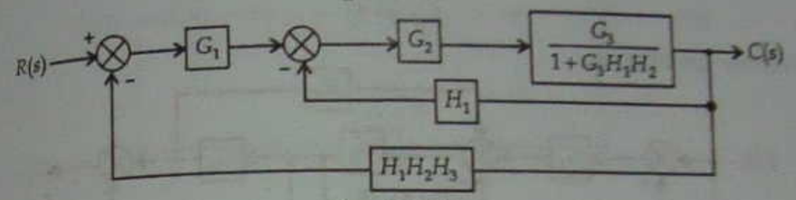


Fig. 12.50(b).

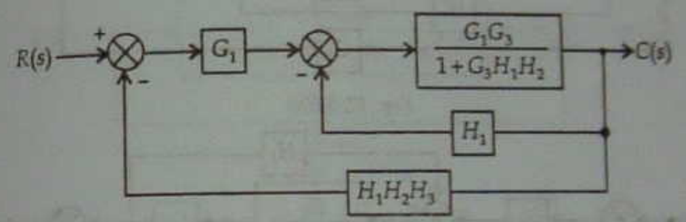


Fig. 12.50(c).

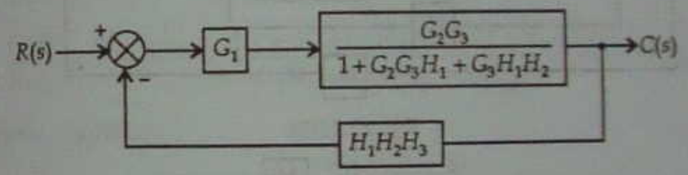


Fig. 12.50(d).

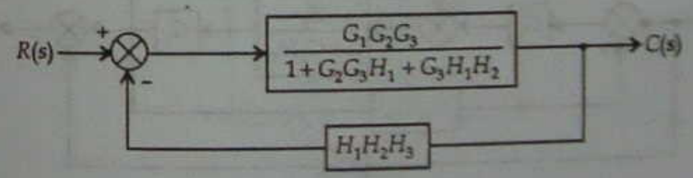


Fig. 12.50(e).

$$\frac{C(s)}{R(s)} = \frac{\frac{G_1\,G_2\,G_3}{1+G_3\,H_1H_2+G_2\,G_3\,H_1}}{1+\frac{G_1\,G_2G_3H_1H_2H_3}{1+G_3H_1H_2+G_2G_3H_1}}$$

$$\frac{C(s)}{R(s)} = \frac{G_1\,G_2\,G_3}{1+G_3H_1H_2+G_2\,G_3H_1+G_1G_2G_3H_1H_2H_3}$$

Example 12.23. Using block reduction technique, find the closed loop transfer function.

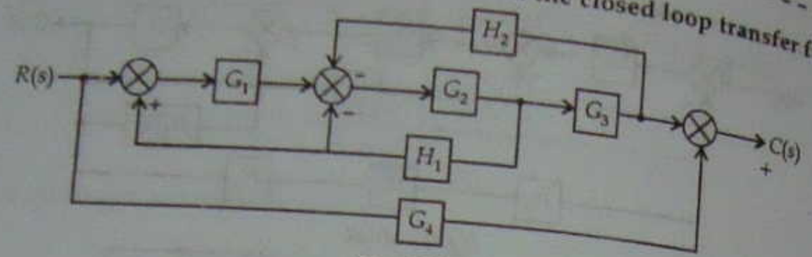


Fig. 12.51.

Solution : Redraw the diagram

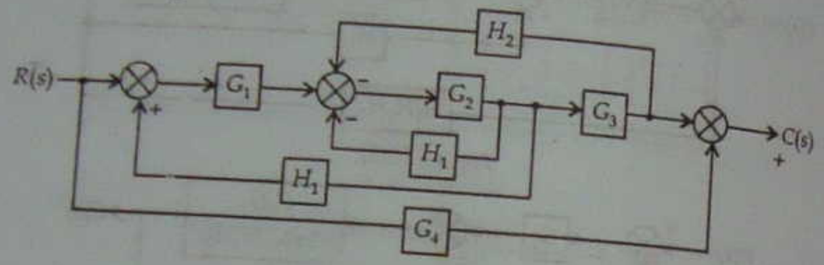


Fig. 12.51(a).

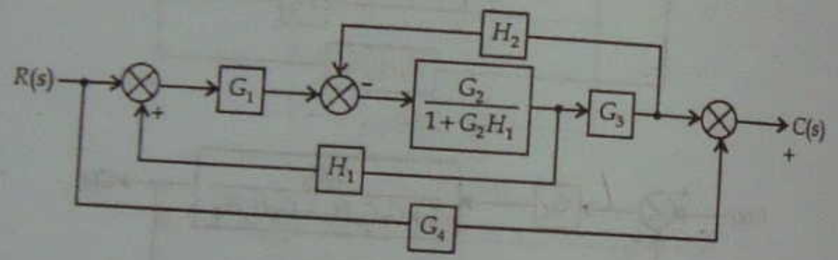


Fig. 12.51(b).

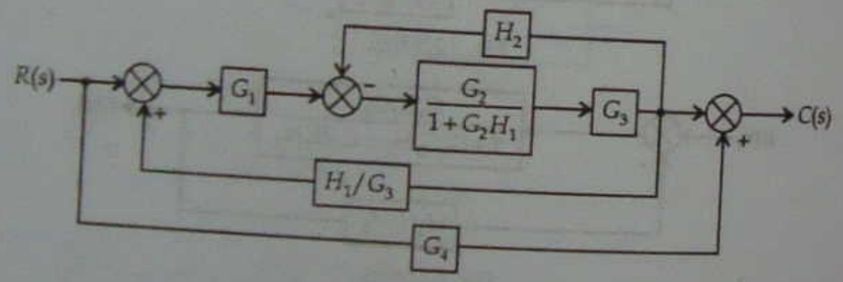


Fig. 12.51(c).

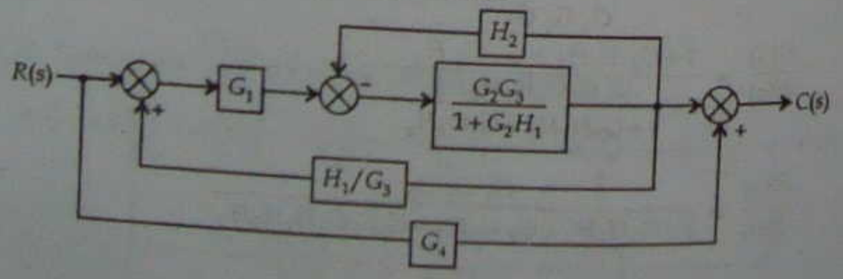


Fig. 12.51(d).

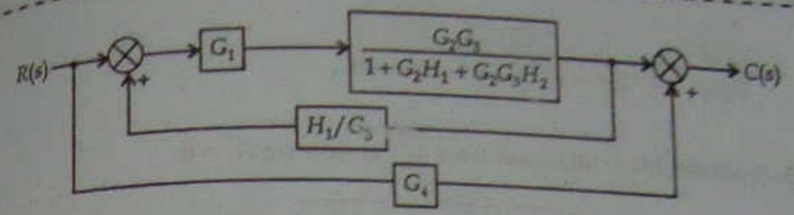


Fig. 12.51(e).

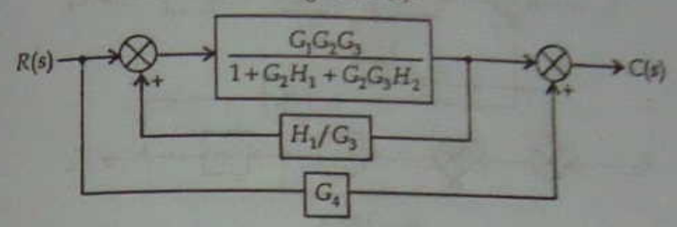


Fig. 12.51(f).

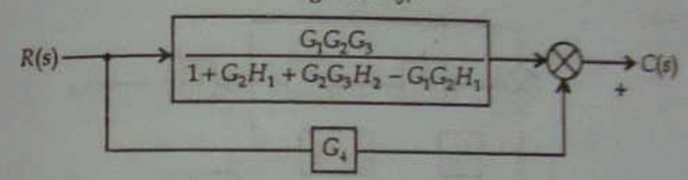


Fig. 12.51(g).

$$\frac{C(s)}{R(s)} = G_4 + \frac{G_1\,G_2\,G_3}{1+G_2H_1+G_2G_3H_2-G_1G_2H_1}$$

$$\frac{C(s)}{R(s)} = \frac{G_4+G_2\,G_4\,H_1+G_2G_3G_4H_2-G_1G_2G_4H_1+G_1G_2G_3}{1+G_2H_1+G_2G_3H_2-G_1G_2H_1}$$ **Ans.**

Example 12.24. Find $c(s)/R(s)$.

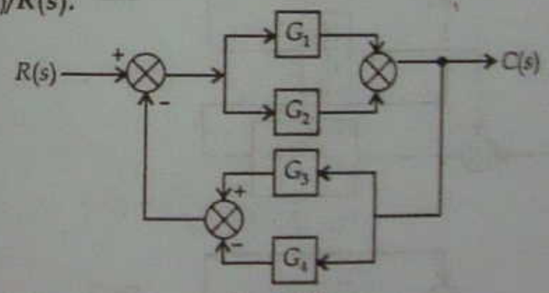


Fig. 12.52.

Solution : Block are in parallel

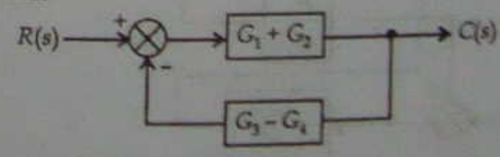


Fig. 12.52(a)

$$\frac{C(s)}{R(s)} = \frac{G_1+G_2}{1+(G_1+G_2)(G_3-G_4)}$$

$$\frac{C(s)}{R(s)} = \frac{G_1 + G_2}{1 + G_1G_3 - G_1G_4 + G_2G_3 - G_2G_4}$$

Example 12.25. From the block diagram find $\frac{C_1}{R_2}$ assuming $R_1 = 0$

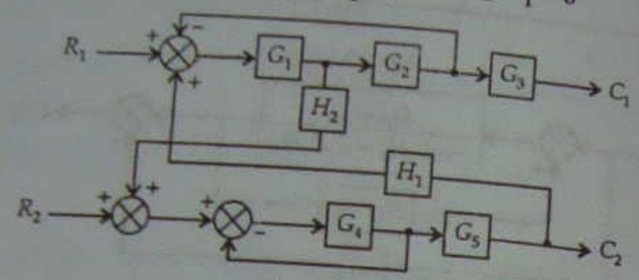


Fig. 12.53.

Solution : When $R_1 = 0$

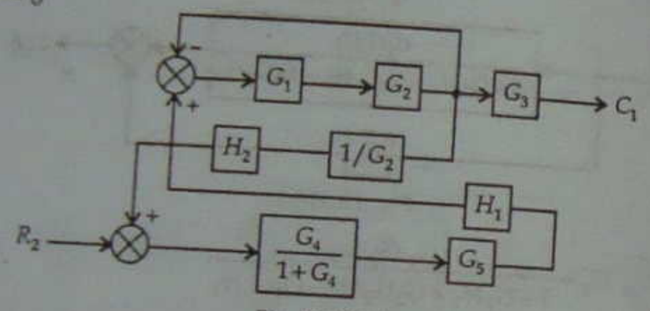


Fig. 12.53(a).

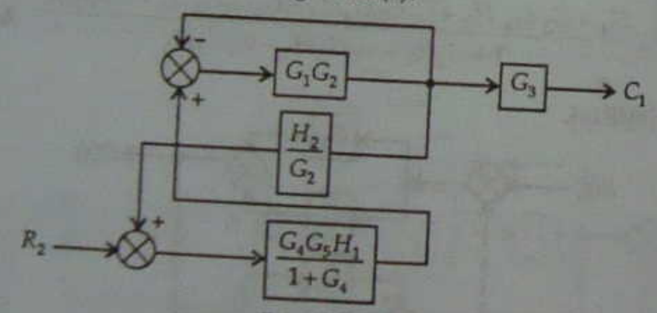


Fig. 12.53(b).

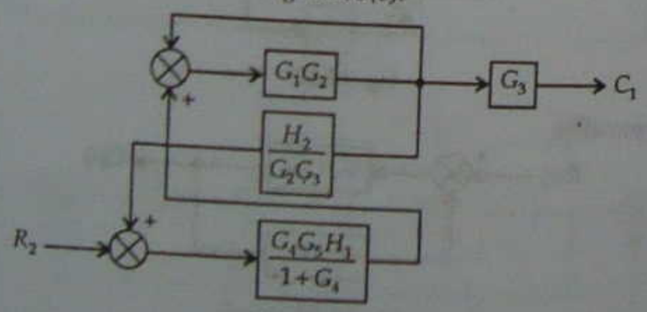


Fig. 12.53(c).

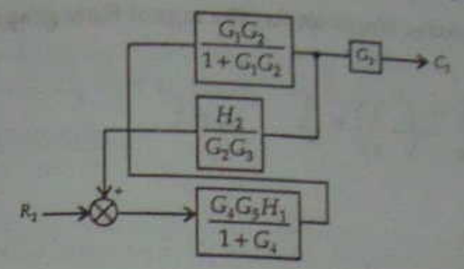


Fig. 12.53(d).

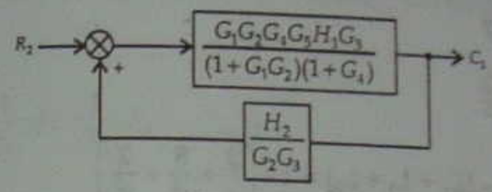


Fig. 12.53(e).

$$\frac{C_1(s)}{R_2(s)} = \frac{G_1\,G_2\,G_3\,G_4\,G_5\,H_1}{(1+G_1\,G_2)\,(1+G_4) - G_1G_4G_5H_1H_2}$$

$$\frac{c_1(s)}{R_2(s)} = \frac{G_1G_2G_3G_4G_5H_1}{1 + G_1G_2 + G_4 + G_1G_2G_4 - G_1G_4G_5H_1H_2} \quad \text{Ans.}$$

Example 12.26. Alternative method to determine C_1/R_1 and other ratio for example 1.34.

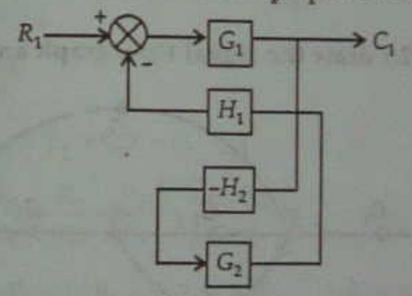


Fig. 11.54.

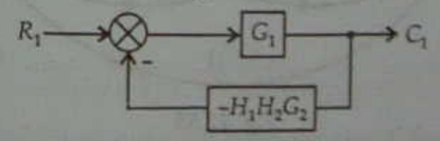


Fig. 12.54(a)

$$\frac{C_1}{R_1} = \frac{G_1}{1 + (-H_1H_2G_2)G_1}$$

$$\boxed{\frac{C_1}{R_1} = \frac{G_1}{1 - G_1G_2H_1H_2}}$$

Similarly, find other ratio.

Example 12.27. Find the transfer function for the signal flow graph shown in fig. 1.103c.

Solution : Forward path:

$$g_1 = -\frac{1}{s}\cdot\frac{1}{s}\cdot\left(-\frac{1}{s}\right) = \frac{1}{s^3} \qquad \Delta_1 = 1$$

Individual loops :

$$L_1 = -\frac{3}{s}$$

$$L_2 = \frac{-5}{s^2}$$

$$L_3 = \frac{-2}{s^2}$$

$$\Delta = 1 - [L_1 + L_2 + L_3] = 1 + \left[\frac{3}{s} + \frac{5}{s^2} + \frac{2}{s^3}\right]$$

$$\Delta = 1 + \left[\frac{3s^2+5s+2}{s^3}\right] = \frac{s^3+3s^2+5s+2}{s^3}$$

$$\frac{C(s)}{R(s)} = \frac{g_1\Delta_1}{\Delta} = \frac{\frac{1}{s^3}}{\frac{s^3+3s^2+5s+2}{s^3}} = \frac{1}{s^3+3s^2+5s+2}$$ **Ans.**

Example 12.28. For example 12.7 draw the signal flow graph and find the transfer function.

Solution :

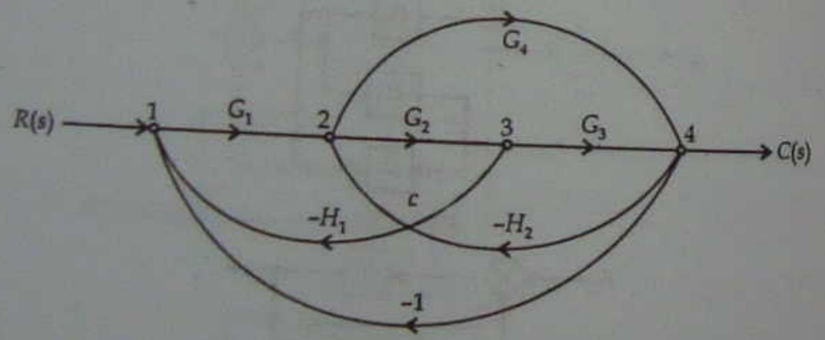


Fig. 12.55.

Forward paths :

$$g_1 = G_1 G_2 G_3 \qquad \Delta_1 = 1$$
$$g_2 = G_1 G_4 \qquad \Delta_2 = 1$$

Individual loops :

$$L_1 = -G_1 G_2 H_1 \quad ; \quad L_2 = -G_2 G_3 H_2$$
$$L_3 = -G_1 G_2 G_3 \quad ; \quad L_4 = -G_4 H_2$$
$$L_5 = -G_1 G_4$$

Two non-touching loops : None

$$\Delta = 1 - [-G_1 G_2 H_1 - G_2 G_3 H_2 - G_1 G_2 G_3 - G_4 H_2 - G_1 G_4]$$
$$= 1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 + G_4 H_2 + G_1 G_4$$

$$\frac{C(s)}{R(s)} = \frac{g_1\Delta_1 + g_2\Delta_2}{\Delta}$$

$$\frac{C(s)}{R(s)} = \frac{G_1G_2G_3 + G_1G_4}{1+G_1G_2H_1 + G_2G_3H_2 + G_1G_2G_3 + G_4H_2 + G_1G_4}$$

Example 12.29. Find the transfer function of the electrical circuit whose SFG is shown in fig 12.26.

Solution : Forward path :

$$g_1 = \frac{1}{R_1}\cdot R_3 \cdot \frac{1}{R_2}\cdot R_4 \qquad \Delta_1 = 1$$

Individual loops :

$$L_1 = -\frac{R_3}{R_1}$$

$$L_2 = -\frac{R_3}{R_2}$$

$$L_3 = -\frac{R_4}{R_2}$$

Two nontouching loops : $L_1\ L_3$

$$L_1 L_3 = \frac{R_3}{R_1}\cdot\frac{R_4}{R_2}$$

$$\Delta = 1 - [L_1 + L_2 + L_3] + [L_1 L_3]$$

$$= 1 + \left[\frac{R_3}{R_1} + \frac{R_3}{R_2} + \frac{R_4}{R_2}\right] + \left[\frac{R_3 R_4}{R_1 R_2}\right]$$

$$= 1 + \frac{R_2R_3 + R_3R_1 + R_1R_4 + R_3R_4}{R_1R_2} = \frac{R_1R_2 + R_2R_3 + R_3R_1 + R_1R_4 + R_3R_4}{R_1R_2}$$

$$\frac{C(s)}{R(s)} = \frac{g_1\Delta_1}{\Delta} = \frac{R_3R_4}{R_1R_2 + R_2R_3 + R_3R_1 + R_1R_4 + R_3R_4}$$ **Ans.**

Example 12.30. For the given electrical network draw the signal flow graph and find the transfer function by mason gain formula.

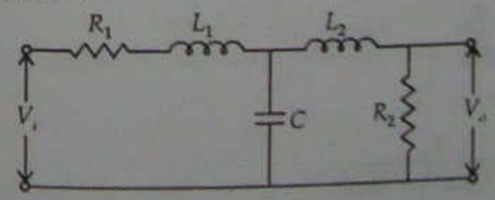


Fig. 12.56.

Solution : Draw the transform network

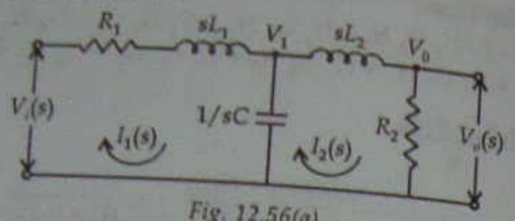


Fig. 12.56(a)

$$I_1(s) = \frac{V_i - V_1}{R_1 + sL_1}$$

$$V_1(s) = \frac{1}{sC}[I_1(s) - I_2(s)]$$

$$I_2(s) = \frac{V_1(s) - V_o(s)}{sL_2}$$

$$V_0(s) = R_2 I_2(s)$$

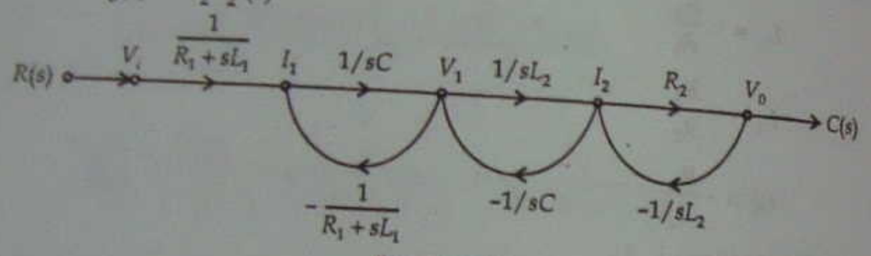


Fig. 12.56(b)

Forward path :

$$g_1 = \frac{1}{R_1 + sL_1} \cdot \frac{1}{sC} \cdot \frac{1}{sL_2} \cdot R_2$$

$$g_1 = \frac{R_2}{s^2 L_2 C(R_1 + sL_1)}, \qquad \Delta_1 = 1 - 0 = 1$$

Individual loops :

$$L_1 = -\frac{1}{sC(R_1 + sL_1)}$$

$$L_2 = -\frac{1}{s^2 L_2 C}$$

$$L_3 = -\frac{R_2}{sL_2}$$

Two nontouching loops : $L_1 L_3$

$$L_1 L_3 = \frac{R_2}{s^2 L_2 C(R_1 + sL_1)}$$

$$\Delta = 1 - [L_1 + L_2 + L_3] + L_1 L_3$$

$$\Delta = 1 + \frac{1}{sC(R_1 + sL_1)} + \frac{1}{s^2 L_2 C} + \frac{R_2}{sL_2} + \frac{R_2}{s^2 L_2 C(R_1 + sL_1)}$$

$$\frac{V_o}{V_i} = \frac{g_1 \Delta_1}{\Delta}$$

$$\frac{V_o}{V_i} = \frac{R_2}{s^2 L_2 C(R_1 + sL_1) + sL_2 + R_1 + sL_1 + sC(R_1 + sL_1) + R_2}$$ **Ans.**

Example 12.31. Draw the signal flow graph of series RLC circuit shown in fig. 1.10 and find out the transfer function.

Solution : Step 1: Draw the transform network

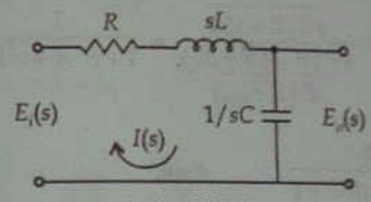


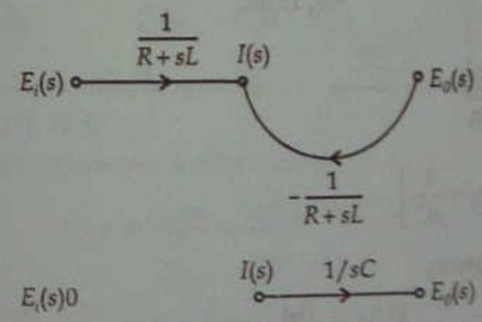
Fig. 12.57.

Step 2 : Write the equation

$$I(s) = \frac{E_i(s) - E_o(s)}{R + sL} = \frac{E_i(s)}{R + sL} - \frac{E_o(s)}{R + sL}$$

$$E_o(s) = I(s) \cdot \frac{1}{sC}$$

Step 3: For each equations obtained in step 2, draw SFG. Select the node variables as $I(s)$, $E_i(s)$ and $E_o(s)$.



Step 4: Combine the SFG. Obtained in step 3

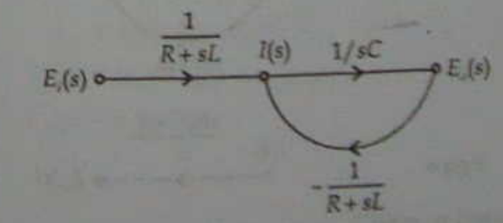


Fig. 12.57(a)

Step 5 : Forward path

$$g_1 = \frac{1}{R+sL}\cdot\frac{1}{sC} \qquad \Delta_1 = 1$$

Individual loop $L_1 = -\dfrac{1}{(R+sL)(sC)}$

Non touching loops: None

$$\Delta = 1 - L_1 = 1 + \frac{1}{(R+sL)(sC)} = \frac{(R+sL)(sC)+1}{(R+sL)(sC)}$$

$$\frac{C(s)}{R(s)} = \frac{g_1\Delta_1}{\Delta} = \frac{\dfrac{1}{sC(R+sL)}}{\dfrac{1+(sC)(R+sL)}{sC(R+sL)}} = \frac{1}{1+RCs+s^2L_C}$$

$$\frac{C(s)}{R(s)} = \frac{1}{s^2LC+sRC+1} \qquad \textbf{Ans.}$$

Example 12.32. Draw the SFG of lag network shown in fig 6.4 and find the transfer function.

Solution: Step 1 : Draw the transform network

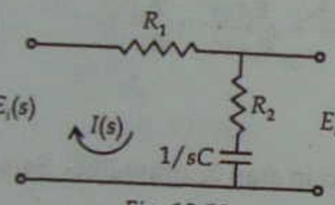


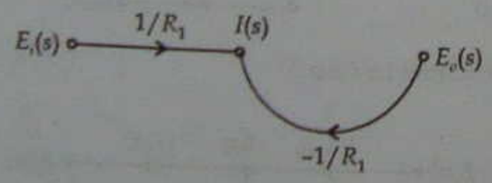
Fig. 12.58.

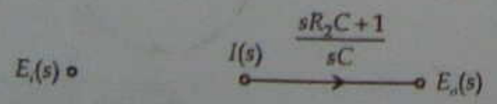
Step 2 : Write the equations

$$I(s) = \frac{E_i}{R_1} - \frac{E_o}{R_1}$$

$$E_o(s) = I(s)\left[R_2 + \frac{1}{sC}\right]$$

Step 3 : Draw SFG for each equation





Step 4 : Combine SFG obtained in step 3

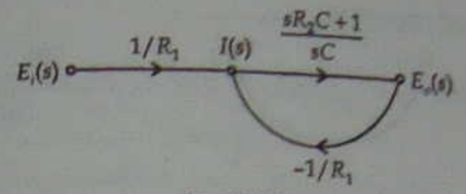


Fig. 12.58(a)

Step 5 : Forward path $g_1 = \dfrac{1+sR_2C}{sR_1C}$ $\Delta_1 = 0$

Individual loop $L_1 = -\dfrac{1+sR_2C}{sR_1C}$

$$\Delta = 1 - L_1 = 1 + \frac{1+sR_2C}{sR_1C}$$

$$\frac{E_o(s)}{E_i(s)} = \frac{C(s)}{R(s)} = \frac{g_1\Delta_1}{\Delta}$$

$$= \frac{1+sR_2C}{1+sC(R_1+R_2)} \qquad \textbf{Ans.}$$

Example 12.33. For the given circuit draw SFG and find out the transfer function.

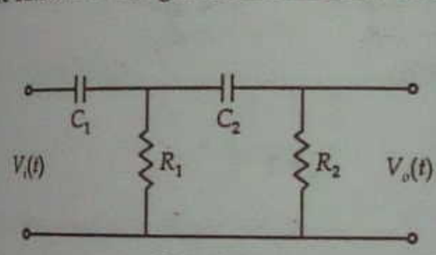


Fig. 12.59.

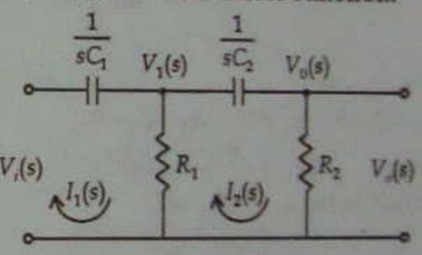


Fig. 12.60.

Solution: Draw the transform network (Fig. 12.60)

$$I_1(s) = \frac{V_i(s)}{\frac{1}{sC_1}} - \frac{V_1(s)}{\frac{1}{sC_1}}$$

$$= sC_1\,V_i(s) - sC_1\,V_1(s)$$

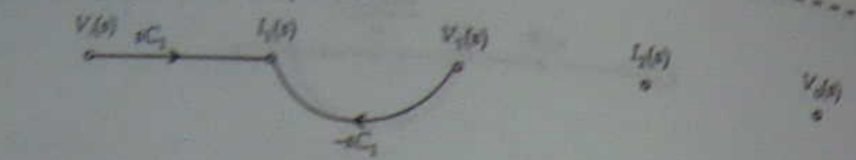
$$V_1(s) = [I_1(s) - I_2(s)]$$

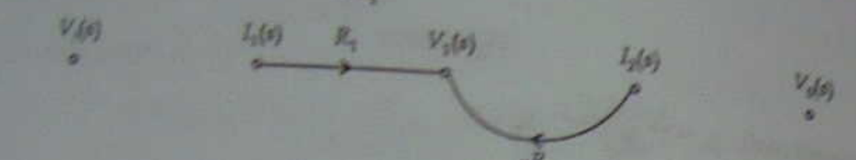
$$R_1 = I_1(s)\,R_1 - I_2(s)\,R_1$$

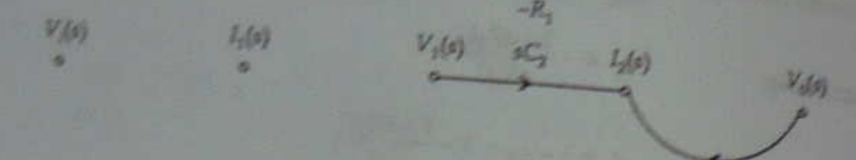
$$I_2(s) = sC_2\,V_1(s) - sC_2V_o(s)$$

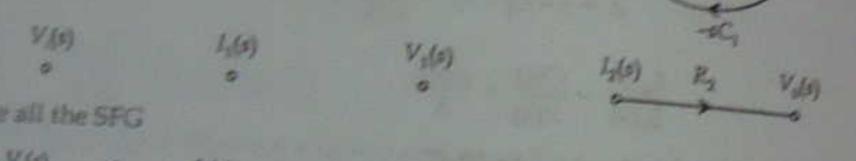
$$V_o(s) = I_2(s)\,R_2$$

Signal flow graph for each equation









Combine all the SFG

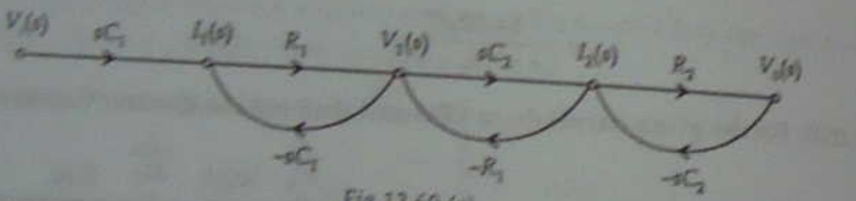


Fig 12.60 (a)

Forward path $g_1 = sC_1 R_1 \cdot sC_2 \cdot R_2$

$g_1 = s^2 R_1 R_2 C_1 C_2$ $\quad \Delta_1 = 1$

Individual loops $L_1 = -R_1 \cdot sC_1$

$L_1 = -sR_1 C_1$

$L_2 = -sR_1 C_2$

$L_3 = -sR_2 C_2$

Two non touching loops : $L_1 L_3$

$L_1 L_3 = s^2 R_1 R_2 C_1 C_2$

$\Delta = 1 - (L_1 + L_2 + L_3) + L_1 L_3$

$\Delta = 1 + sR_1 C_1 + sR_1 C_2 + sR_2 C_2 + s^2 R_1 R_2 C_1 C_2$

$$\frac{V_2(s)}{V_1(s)} = \frac{g_1 \Delta_1}{\Delta}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{s^2 R_1 R_2 C_1 C_2}{1 + s^2 R_1 R_2 C_1 C_2 + s(R_1 C_1 + R_1 C_2 + R_2 C_2)}$$

Example 12.34. For the given block diagram draw the SFG and find the ratio of $\frac{C(s)}{R(s)}$.

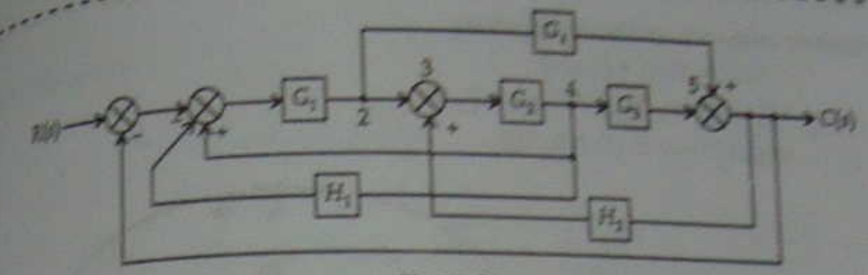


Fig. 12.61.

Solution : The required SFG is

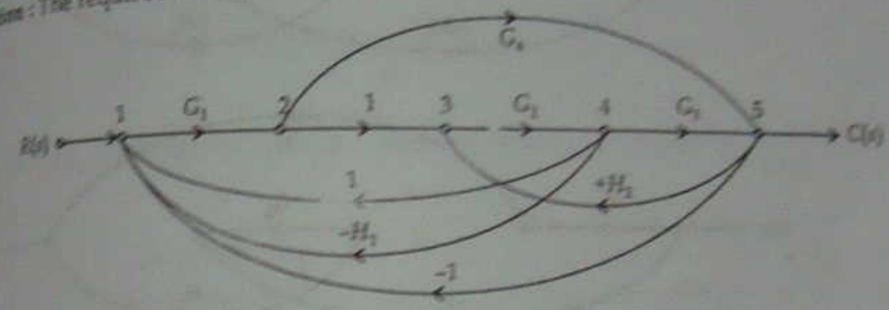


Fig. 12.61(a)

Forward paths $g_1 = G_1 G_2 G_3$ $\quad \Delta_1 = 1$

$g_2 = G_1 G_4$ $\quad \Delta_2 = 1$

Individual loops : $L_1 = G_1 G_2$ $\quad L_4 = -G_1 G_2 G_3$

$L_2 = -G_1 G_2 H_1$ $\quad L_5 = -G_1 G_4$

$L_3 = G_2 G_3 H_2$ $\quad *L_6 = -G_1 G_2 G_4 H_1 H_2$

$*L_7 = G_1 G_2 G_4 H_2$

Two non touching loops : None

$\Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6 + L_7)$

$= 1 - G_1 G_2 + G_1 G_2 H_1 - G_2 G_3 H_2 + G_1 G_2 G_3 + G_1 G_4 + G_1 G_2 H_1 H_2 - G_1 G_2 G_4 H_1$

$$\frac{C(s)}{R(s)} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta}$$

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 + G_1 G_4}{1 - G_1 G_2 + G_1 G_2 H_1 - G_2 G_3 H_2 + G_1 G_2 G_3 + G_1 G_4 + G_1 G_2 G_4 H_1 H_2 - G_1 G_2 G_4 H_2}$$

Example 12.35. For the system represented by the following equations, find the transfer function by SFG.

$$x = x_1 + \beta_3 u$$

$$\dot{x}_1 = -a_1 x_1 + x_2 + \beta_2 u$$

$$\dot{x}_2 = -a_2 x_1 + \beta_1 u$$

Solution : Let x = output

u = input

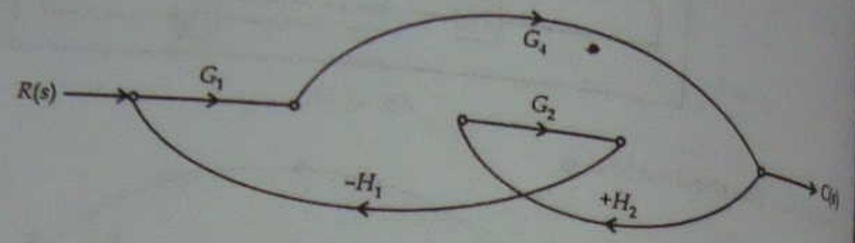
Take laplace transform of above equations

$$X(s) = X_1(s) + \beta_3\, u(s)$$

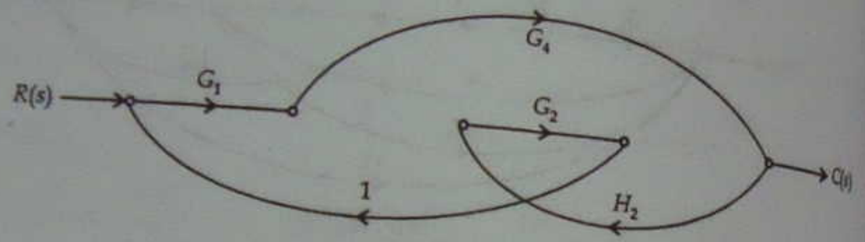
$$sX_1(s) = -a_1\, X_1(s) + X_2(s) + \beta_2\, u(s)$$

$$sX_2(s) = -a_2\, X_1(s) + \beta_1\, u(s)$$

* L_6 :



** L_7 :



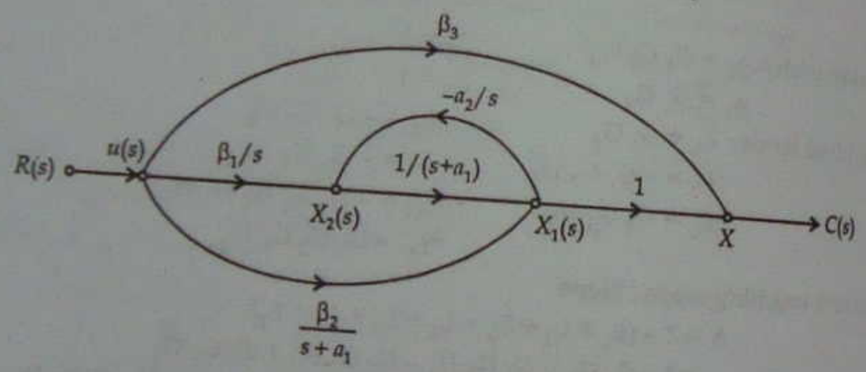


Fig. 12.62

or,

$$X(s) = X_1(s) + \beta_3\, u(s)$$

$$X_1(s) = \frac{1}{s+a_1} X_2(s) + \frac{\beta_1}{s+a_1} u(s)$$

$$X_2(s) = \frac{-a_2}{s} X_1(s) + \frac{\beta_1}{s} u(s)$$

Forward paths : $g_1 = \frac{\beta_1}{s} \cdot \frac{1}{s+a_1}$ $\quad \Delta_1 = 1$

$$g_2 = \frac{\beta_2}{s+a_1} \qquad \Delta_2 = 1$$

$$g_3 = \beta_3 \qquad \Delta_3 = 1 + \frac{a_2}{s(s+a_1)}$$

Individual loops : $L_1 = -\frac{a_2}{s(s+a)}$

$$\Delta = 1 - (L_1) = 1 + \frac{a_2}{s(s+a)} \qquad \because \; E_i(s) = Z_1(s)\, I(s)$$

$$E_o(s) = -Z_2\, I(s)$$

$$\frac{C(s)}{R(s)} = \frac{g_1\Delta_1 + g_2\Delta_2 + g_3\Delta_3}{\Delta} = \frac{\frac{\beta_1}{s(s+a_1)} + \frac{\beta_2}{(s+a_1)} + \beta_3\left(1 + \frac{a_2}{s(s+a_1)}\right)}{1 + \frac{a_2}{s(s+a_1)}}$$

$$\frac{C(s)}{R(s)} = \frac{\beta_1 + \beta_2 s + \beta_3(s^2 + a_1 s + a_2)}{s^2 + a_1 s + a_2} \qquad \text{Ans.}$$

Example 12.36. For the given op-amp circuit find the transfer function $\frac{E_o(s)}{E_i(s)}$.

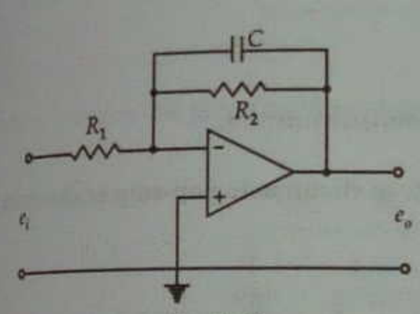


Fig. 12.63.

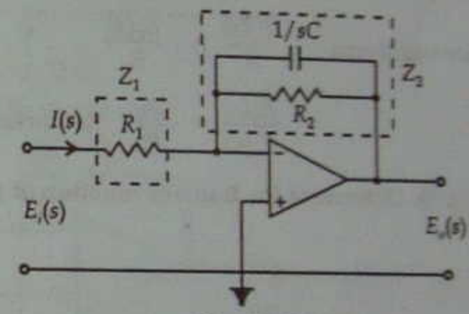


Fig. 12.63(a).

Solution : Transform network of the given circuit (Fig. 12.63).

$$Z_1(s) = R_1$$

$$Z_2(s) = \frac{R_2 \cdot \frac{1}{sC}}{R_2 + \frac{1}{sC}} = \frac{R_2}{1 + sR_2C}$$

$$\frac{E_o(s)}{E_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

$$\frac{E_o(s)}{E_i(s)} = -\frac{R_2}{R_1(1 + sR_2C)}$$

This is the required transfer function.

Example 12.37. Find the transfer function of inverting amplifier as shown in fig.

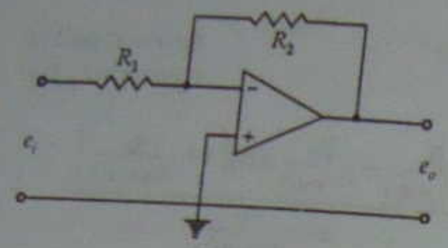


Fig. 12.64.

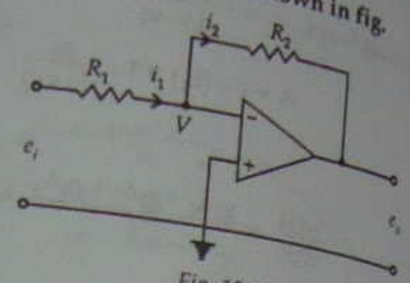


Fig. 12.64(a)

Solution :

$$i_1 = \frac{e_i - V}{R_1} \quad \text{and} \quad i_2 = \frac{V - e_o}{R_2}$$

Since $V = 0 \quad \therefore i_1 = i_2$

$$\frac{e_i}{R_1} = -\frac{e_o}{R_2}$$

Laplace transform $$\frac{E_i(s)}{R_1} = -\frac{E_o(s)}{R_2}$$

$$\frac{E_o(s)}{E_i(s)} = -\frac{R_2}{R_1}$$ Required transfer function

Example 12.38. Determine the transfer function of given circuit using op-amp as shown in fig. 12.65.

Solution :

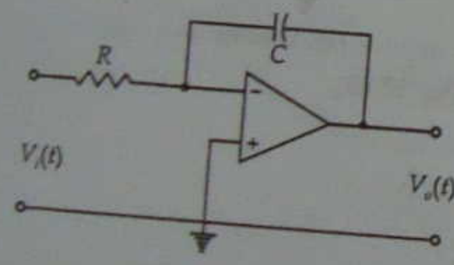


Fig. 12.65

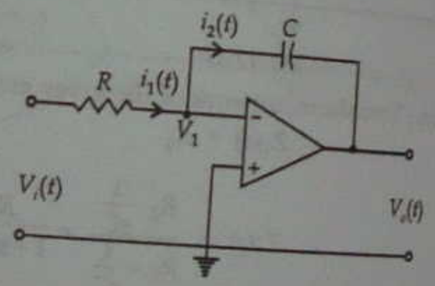


Fig. 12.65 (a)

$$i_1(t) = \frac{V_i(t) - V_1(t)}{R}$$

$$i_2(t) = C\frac{d}{dt}(V_1(t) - V_o(t))$$

Laplace transform of above equations

$$I_1(s) = \frac{1}{R}[V_i(s) - V_1(s)]$$

$$I_2(s) = sC[V_1(s) - V_o(s)]$$

Since, $V_1(s) \approx 0 \quad \therefore \quad I_1(s) = I_2(s)$

$$I_1(s) = \frac{V_i(s)}{R}$$

$$I_2(s) = -sCV_o(s)$$

$$\frac{V_i(s)}{R} = -sCV_o(s)$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{1}{sCR}$$

This is the required transfer function

Example 12.39. Find the transfer function of the given circuit shown in fig 12.66.

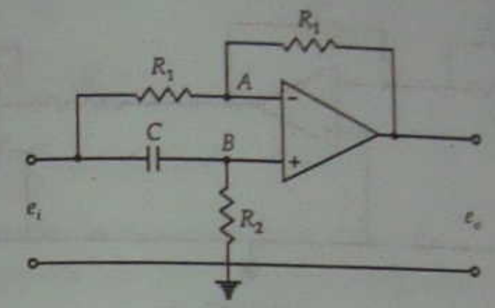


Fig. 12.66.

Solution : Redraw the given circuit as transform network.

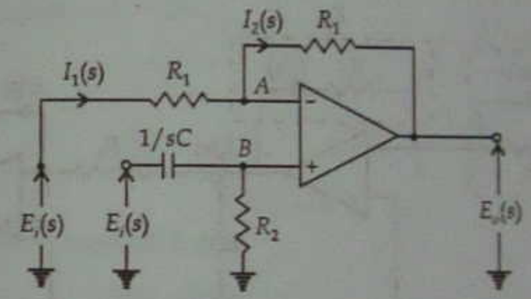


Fig. 12.66(a)

$$V_B(s) = \frac{R_2}{R_2 + \frac{1}{sC}} \cdot E_i(s)$$

$$V_A(s) = V_B(s) = \frac{R_2}{1/sC + R_2} E_i(s) = \frac{sR_2C}{1 + sR_2C} E_i(s)$$

$$I_1(s) = \frac{E_i(s) - V_A(s)}{R_1}$$

$$I_2(s) = \frac{V_A(s) - E_o(s)}{R_1}$$

Since $I_1(s) = I_2(s)$

$$\frac{E_i(s) - V_A(s)}{R_1} = \frac{V_A(s) - E_0(s)}{R_1}$$

or, $$2V_A(s) = E_i(s) + E_o(s)$$

or, $$E_o(s) = 2V_A(s) - E_i(s)$$

$$E_o(s) = 2\left[\frac{sR_2C}{1+sR_2C}E_i(s)\right] - E_i(s)$$

$$\frac{E_o(s)}{E_i(s)} = \frac{sCR_2 - 1}{sR_2 + 1}$$

This is the required transfer function. **Ans.**

Example 12.40. Find the transfer function of the circuit shown in fig. 12.67.

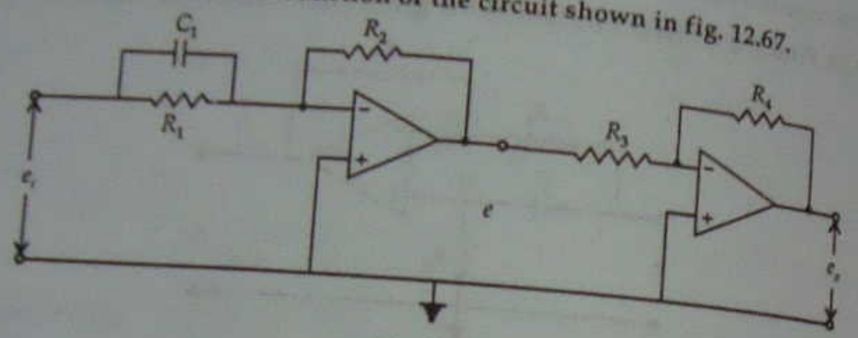


Fig. 12.67

Solution : Transform the network

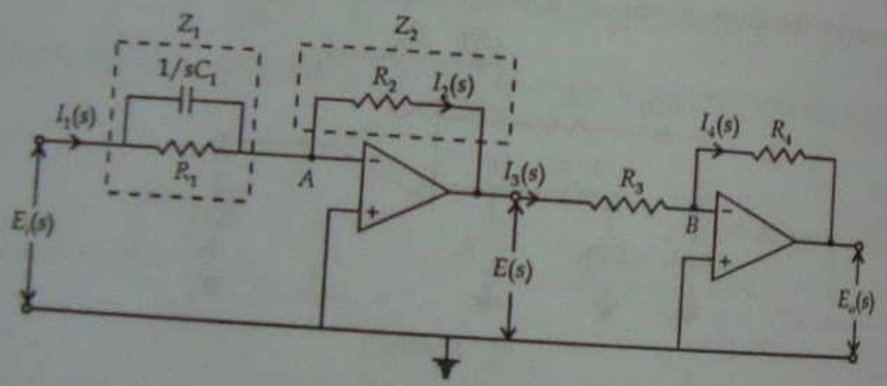


Fig 12.67 (a)

$$Z_1 = \frac{R_1}{1+sR_1C_1}$$

$$Z_2 = R_2$$

$$V_A = 0$$

$$I_1(s) = \frac{E_i(s) - V_A(s)}{Z_1} = \frac{1+sR_1C_1}{R_1}E_i(s)$$

$$I_2(s) = -\frac{E(s)}{R_2}$$

since, $I_1(s) = I_2(s)$

$$E(s) = -\frac{R_2(1+sR_1C_1)}{R_1}E_i(s) \quad \text{or,} \quad \boxed{\frac{E(s)}{E_i(s)} = -\frac{R_2(1+sR_1C_1)}{R_1}}$$

Also, $$V_B = 0$$

$$I_3(s) = \frac{E(s)}{R_3}$$

$$I_4 = -\frac{E_o(s)}{R_4}$$

and $$I_3(s) = I_4(s)$$

$$\frac{E(s)}{R_3} = -\frac{E_o(s)}{R_4}$$

$$\therefore \quad \boxed{\frac{E_o(s)}{E(s)} = -\frac{R_4}{R_3}}$$

$$\frac{E_0(s)}{E_i(s)} = \frac{E(s)}{E_i(s)} \cdot \frac{E_0(s)}{E(s)}$$

$$\frac{E_o(s)}{E_i(s)} = \frac{R_2 R_4}{R_1 R_3}(1+sR_1C_1)$$

∴

This is the required transfer function of the given circuit.

Example 12.41. Find the transfer function of the given circuit shown in fig. 12.68.

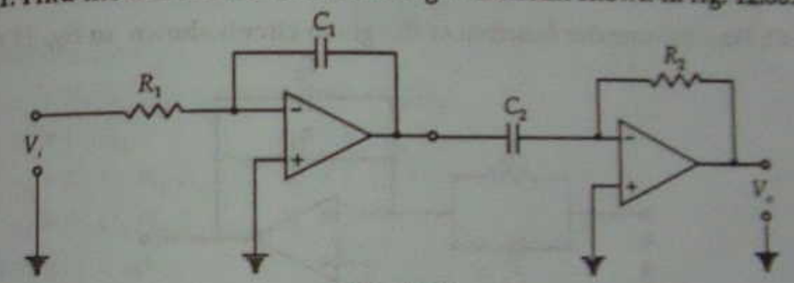


Fig. 12.68

Solution : Draw the transform network

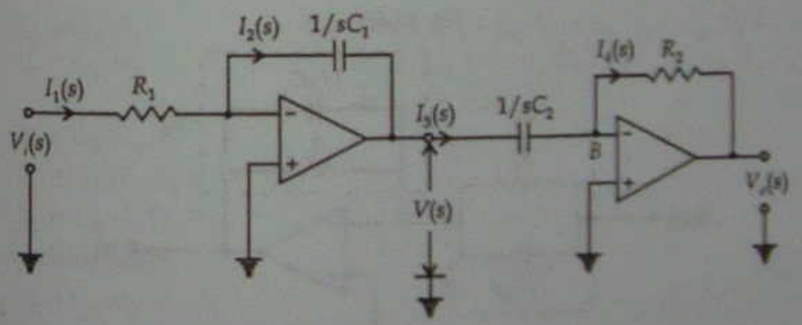


Fig. 12.68 (a)

$$I_1(s) = \frac{V_i(s)}{R_1}$$

$$I_2(s) = \frac{-V(s)}{1/sC_1} = -sC_1V(s)$$

since, $$I_1(s) = I_2(s)$$

$$\frac{V(s)}{V_i(s)} = \frac{-1}{sR_1C_1}$$

$$I_3(s) = \frac{V(s)}{1/sC_2} = sC_2V(s)$$

$$I_4(s) = -\frac{V_o(s)}{R_2}$$

$$I_3(s) = I_4(s)$$

$$\frac{V_o(s)}{V(s)} = -sR_2C_2$$

$$\therefore \quad \frac{V_o(s)}{V_i(s)} = \frac{V(s)}{V_i(s)} \cdot \frac{V_o(s)}{V(s)}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{R_2\,C_2}{R_1\,C_1}$$

This the required transfer function.

Example 12.42. Find the transfer function of the given circuit shown in fig. 12.69.

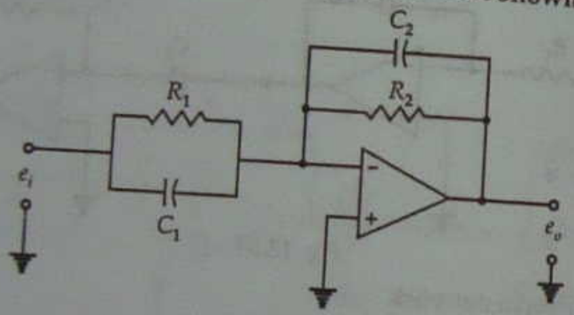


Fig. 12.69.

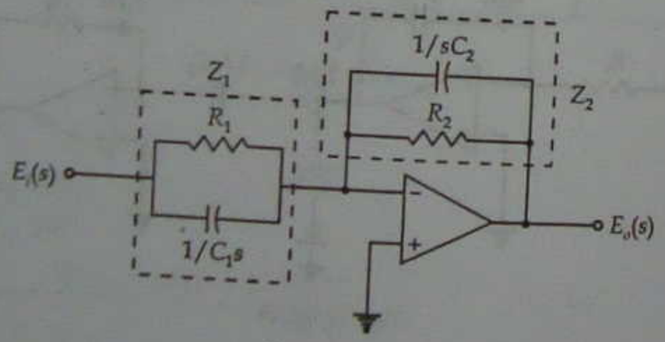


Fig. 12.69(a)

Solution: Transform the network show in fig 12.69 *a*

$$Z_1(s) = \frac{R_1 \cdot \frac{1}{C_1 s}}{R_1 + \frac{1}{sC_1}} = \frac{R_1}{1+sR_1C_1}$$

$$Z_2(s) = \frac{R_2}{1+sR_2C_2} \qquad \because \quad E_i(s) = Z_1(s)\,I(s)$$

$$\frac{E_0(s)}{E_i(s)} = -\frac{Z_2}{Z_1} \qquad E_o(s) = -Z_2(s)\,I(s)$$

$$\frac{E_o(s)}{E_i(s)} = -\frac{R_2(1+sR_1C_1)}{R_1(1+sR_2C_2)}$$

Example 12.43. For the given signal flow graph of a system find the expressions for the outputs C_1 and C_2. Also determine the condition that makes C_1 independent of R_2 and C_2 independent of R_1.

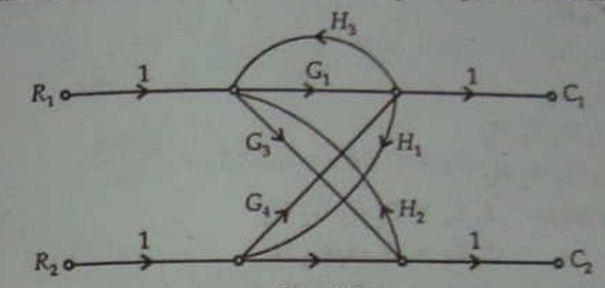


Fig. 12.70

Solution : Expression for C_1

$g_1 = R_1\,G_1$ $\qquad \Delta_1 = 1 - G_2\,H_4$

$g_2 = R_2\,G_4$ $\qquad \Delta_2 = 1 - G_3\,H_2$

$g_3 = R_1\,G_3\,H_4\,G_4$ $\qquad \Delta_3 = 1$

$g_4 = R_2\,G_2\,H_2\,G_1$ $\qquad \Delta_4 = 1$

Individual loops :

$L_1 = G_1\,H_3$ $\qquad \Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6) + (L_1\,L_4 + L_2\,L_3)$

$L_2 = G_4\,H_1$ $\qquad \Delta = 1 - (G_1\,H_3 + G_4\,H_1 + G_3\,H_2 + G_2\,H_4 + G_1\,G_2\,H_1\,H_2 + G_3\,G_4\,H_3\,H_4) + (G_1\,G_2\,H_3\,H_4 + G_3\,G_4\,H_1\,H_2)$

$L_3 = G_3\,H_2$

$L_4 = G_2\,H_4$

$L_5 = G_1\,H_1\,G_2\,H_2$

$L_6 = G_4\,H_3\,G_3\,H_4$

Two non-touching loops :

$L_1\,L_4 = G_1\,H_3\,G_2\,H_4$

$L_3\,L_2 = G_4\,H_1\,G_3\,H_2$

$$C_1 = \frac{g_1\Delta_1 + g_2\Delta_2 + g_3\Delta_3 + g_4\Delta_4}{\Delta}$$

$$C_1 = \frac{R_1G_1(1-G_2H_4) + R_2G_4(1-G_3H_2) + R_1G_3G_4H_4 + R_2G_1G_2H_2}{1-G_1H_3-G_4H_1-G_3H_2-G_2H_4-G_1G_2H_1H_2-G_3G_4H_3H_4 + G_1G_2H_3H_4 + G_3G_4H_1H_2}$$

Expression for C_2 :

$$g_1 = R_2 G_2 \qquad \Delta_1 = 1 - G_1 H_3$$
$$g_2 = R_2 G_4 H_3 G_3 \qquad \Delta_2 = 1$$
$$g_3 = R_1 G_3 \qquad \Delta_3 = 1 - G_4 H_1$$
$$g_4 = R_1 G_1 G_2 H_1 \qquad \Delta_4 = 1$$

Individual loops :

$$L_1 = G_1 H_3$$
$$L_2 = G_3 H_2$$
$$L_3 = G_4 H_1$$
$$L_4 = G_2 H_4$$
$$L_5 = G_4 H_3 G_3 H_4$$
$$L_6 = G_1 G_2 H_1 H_2$$

Two non-touching loops $L_1 L_4 = G_1 H_3 G_2 H_4$

$L_2 L_3 = G_3 H_2 G_4 H_1$

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6) + (L_1 L_4 + L_2 L_3)$$
$$\Delta = 1 - G_1 H_3 - G_3 H_2 - G_4 H_1 - G_2 H_4 - G_3 G_4 H_3 H_4 - G_1 G_2 H_1 H_2 + G_1 G_2 H_3 H_4 + G_3 G_4 H_1 H_2$$

$$C_2 = \frac{g_1\Delta_1 + g_2\Delta_2 + g_3\Delta_3 + g_4\Delta_4}{\Delta}$$

$$C_2 = \frac{R_2 G_1 (1 - G_1 H_3) + R_2 G_3 G_4 H_3 + R_1 G_3 (1 - G_4 H_1) + R_1 G_1 G_2 H_1}{1 - G_1 H_3 + G_3 H_2 + G_4 H_1 + G_2 H_4 + G_3 G_4 H_3 H_4 + G_1 G_2 H_1 H_2 - G_1 G_2 H_3 H_4 - G_3 G_4 H_1 H_2}$$

Condition C_1 Independent of R_2

From the expression of C_1

$$R_2 G_4 (1 - G_3 H_2) + 2 G_1 G_2 H_2 = 0$$
$$R_2 [G_4 - G_3 G_4 H_2 + G_1 G_2 H_2] = 0$$
$$G_4 - G_3 G_4 H_2 + G_1 G_2 H_2 = 0$$
$$-G_4 = (-G_3 G_4 + G_1 G_2) H_2$$
$$H_2 = \frac{-G_4}{G_1 G_2 - G_3 G_4}$$

Similarly C_2 independent of R_1 when

$$H_1 = \frac{-G_3}{G_1 G_2 - G_3 G_4}$$

Example 12.44. For the given circuit obtain the state equation. Take voltage across the capacitors $V_1(t)$ & $V_2(t)$ as two state variables and the current through the resistor as output variable.

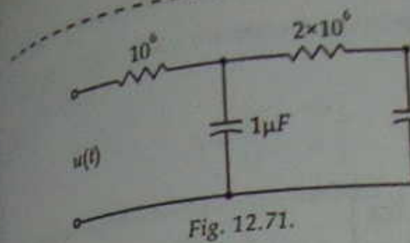


Fig. 12.71.

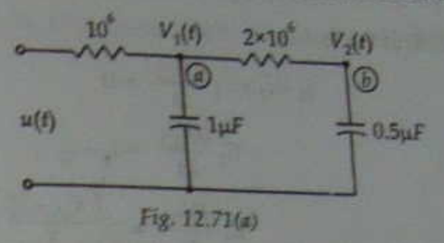


Fig. 12.71(a)

Solution :

Apply CKL at node '*a*'

$$\frac{u(t) - V_1(t)}{10^6} = 10^{-6}\frac{dV_1(t)}{dt} + \frac{V_1(t) - V_2(t)}{2\times10^6}$$

or,
$$\frac{dV_1(t)}{dt} = -1.5V_1(t) + 0.5V_2(t) + u(t)$$

KCL at node '*b*' :

$$\frac{V_1(t) - V_2(t)}{2\times10^6} = 0.5\times10^{-6}\frac{dV_2(t)}{dt}$$

or,
$$\frac{dV_2(t)}{dt} = V_1(t) - V_2(t)$$

output equation :

$$i(t) = \frac{V_1(t) - V_2(t)}{2\times10^6}$$
$$= 0.5 \times 10^{-6}\,[V_1(t) - V_2(t)]$$

$$\begin{bmatrix}\dot{V}_1(t)\\ \dot{V}_2(t)\end{bmatrix} = \begin{bmatrix}-1.5 & 0.5\\ 1 & -1\end{bmatrix}\begin{bmatrix}V_1(t)\\ V_2(t)\end{bmatrix} + \begin{bmatrix}1\\ 0\end{bmatrix}u(t)$$

$$i(t) = \begin{bmatrix}0.5\times10^{-6} & -0.5\times10^{-6}\end{bmatrix}\begin{bmatrix}V_1(t)\\ V_2(t)\end{bmatrix}$$

Example 12.45. Write the state equation for the circuit shown in fig 12.72.

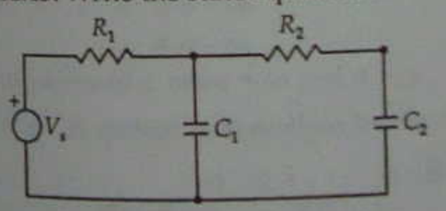


Fig. 12.72.

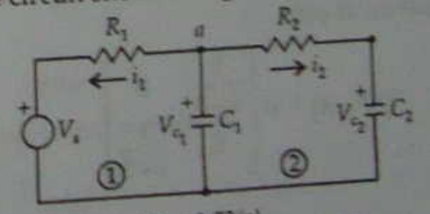


Fig 12.72(a)

Solution : KVL in mesh (1) : ...(1)

$$-i_1 R_1 - V_s + V_{C1} = 0$$

KVL in mech (2) : ...(2)

$$-i_2 R_2 - V_{C2} + V_{C1} = 0$$

KCL at node x :

$$i_1 + i_2 + C_1 \frac{dV_{C1}}{dt} = 0$$

or,
$$C_1 \frac{dV_{C1}}{dt} = i_1 - i_2$$

$$C_1 \frac{dV_{C1}}{dt} = -\left(\frac{V_{c1} - V_s}{R_1}\right) - \left(\frac{V_{c1} - V_{C2}}{R_2}\right)$$

$$\frac{dV_{C1}}{dt} = \left(-\frac{1}{R_1C_1} - \frac{1}{R_2C_1}\right) V_{C1} + \frac{1}{R_2C_1} V_{C2} + \frac{1}{R_1C_1} V_s$$

From eqn (2)

$$-C_2 \frac{dV_{C2}}{dt} R_2 - V_{C2} + V_{C1} = 0$$

$$\frac{dV_{C2}}{dt} = \frac{1}{R_2C_2} V_{C1} - \frac{1}{R_2C_2} V_{C2}$$

$$\begin{bmatrix} \dot{V}_{C1} \\ \dot{V}_{C2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1C_1} - \frac{1}{R_2C_1} & \frac{1}{R_2C_1} \\ \frac{1}{R_2C_2} & -\frac{1}{R_2C_2} \end{bmatrix} \begin{bmatrix} V_{C1} \\ V_{C2} \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1C_1} \\ 0 \end{bmatrix} [V_s]$$

Example 12.46. Find the state transition matrix $\phi(t)$ and the characteristic equation of A

$$A = \begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$$

Solution : The characteristic equation is defined as

$$|sI - A| = 0$$

$$sI - A = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} s+3 & 0 \\ 0 & s+3 \end{bmatrix}$$

$$|sI - A| = (s+3)(s+3) = s^2 + 6s + 9 = 0$$

$\therefore$ Required characteristic equation $\Rightarrow s^2 + 6s + 9 = 0$ **Ans.**

Calculation of $\phi(t)$:

$$[SI - A]^{-1} = \begin{bmatrix} \frac{1}{s+3} & 0 \\ 0 & \frac{1}{s+3} \end{bmatrix}$$

$$\phi(t) = \mathcal{L}^{-1}[sI - A]^{-1} = \begin{bmatrix} \mathcal{L}^{-1}\frac{1}{s+3} & 0 \\ 0 & \mathcal{L}^{-1}\frac{1}{s+3} \end{bmatrix}$$

$$\phi(t) = \begin{bmatrix} e^{-3t} & 0 \\ 0 & e^{-3t} \end{bmatrix}$$ **Ans.**

Example 12.47. Find the state transition matrix $\phi(t)$, the characteristic equation and the eigen values of A.

$$A = \begin{bmatrix} \sigma & -\omega \\ \omega & \sigma \end{bmatrix}$$

where σ and ω are real numbers.

Solution : $$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} \sigma & -\omega \\ \omega & \sigma \end{bmatrix} = \begin{bmatrix} s-\sigma & \omega \\ -\omega & s-\sigma \end{bmatrix}$$

characteristic equation $\Rightarrow |sI - A| = 0$

$$|sI - A| = \begin{vmatrix} s-\sigma & \omega \\ -\omega & s-\sigma \end{vmatrix}$$

$$= s^2 - 2\sigma s + \sigma^2 + \omega^2$$

Required characteristic equation is

$s^2 - (2\sigma)s + (\sigma^2 + \omega^2) = 0$ **Ans.**

$$[SI - A]^{-1} = \frac{1}{(s-\sigma)^2 + \omega^2} \begin{bmatrix} s-\sigma & -\omega \\ \omega & s-\sigma \end{bmatrix}$$

$$\phi(t) = \mathcal{L}^{-1}[SI - A]^{-1} = \mathcal{L}^{-1} \begin{bmatrix} \frac{s-\sigma}{(s-\sigma)^2 + \omega^2} & \frac{-\omega}{(s-\sigma)^2 + \omega^2} \\ \frac{\omega}{(s-\sigma)^2 + \omega^2} & \frac{s-\sigma}{(s-\sigma)^2 + \omega^2} \end{bmatrix}$$

$$\phi(t) = \begin{bmatrix} e^{\sigma t}\cos\omega t & -e^{\sigma t}\sin\omega t \\ e^{\sigma t}\sin\omega t & e^{\sigma t}\cos\omega t \end{bmatrix}$$ **Ans.**

The roots of the characteristic equation are

$$s = \frac{2\sigma \pm j2\omega}{2}$$

$$s_1 = \sigma + j\omega$$

$$s_2 = \sigma - j\omega$$

$\therefore$ eigen values of A are $\sigma + j\omega$ and $\sigma - j\omega$. **Ans.**

Example 12.48. A system is described by the following equations

$$\dot{x}(t) = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} x(t$$

Find the transfer function of the system. *(A.M.U., Aligrah B.E 89-90)*

Solution : We know that

$$G(s) = C[sI - A]^{-1} B$$

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s+1 & 1 \\ 0 & s+2 \end{bmatrix}$$

$$[sI - A]^{-1} = \frac{1}{s^2+3s+2}\begin{bmatrix} s+2 & 1 \\ 0 & s+1 \end{bmatrix}$$

$$C[sI - A]^{-1} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} s+2 & 1 \\ 0 & s+1 \end{bmatrix}\frac{1}{s^2+3s+2} = \frac{1}{s^2+3s+2}\begin{bmatrix} s+2 & 2s+3 \\ s+2 & 1 \\ s+2 & s+2 \end{bmatrix}$$

$$C[sI - A]^{-1} B = \frac{1}{s^2+3s+2}\begin{bmatrix} s+2 & 2s+3 \\ s+2 & 1 \\ s+2 & s+2 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$G(s) = \frac{s^3 - 3s^2 - 9s + 2}{s^2+3s+2}$$

This is the required transfer function **Ans.**

Example 12.49. The mechanical system shown in fig 12.5 choose x_1 and x_2 of mass M_1 and M_2 as state variables as well as output obtain the state model.

Solution : The system equations are

$$F = M_1 \frac{d^2x_1}{dt^2} + B_1 \frac{dx_1}{dt} + B_2 \frac{d}{dt}(x_1 - x_2) \quad ...(1)$$

$$0 = M_2 \frac{d^2x_2}{dt^2} + Kx_2 + B_2 \frac{d}{dt}(x_2 - x_1) \quad ...(2)$$

The above equations can be written as

$$F = M_1\ddot{x}_1 + B_1\dot{x}_1 + B_2\dot{x}_1 - B_2\dot{x}_2$$

$$F = M_1\ddot{x}_1 + (B_1 + B_2)\dot{x}_1 - B_2\dot{x}_2 \quad ...(3)$$

$$0 = M_2\ddot{x}_2 + Kx_2 + B_2\dot{x}_2 - B_2\dot{x}_1 \quad ...(4)$$

Select the state variable as

$x_1 = Z_1$ $\qquad$ $x_2 = Z_3$

$\dot{x}_1 = \dot{Z}_1 = Z_2$ $\qquad$ $\dot{x}_2 = \dot{Z}_3 = Z_4$

$\ddot{x}_1 = \dot{Z}_2$ $\qquad$ $\ddot{x}_2 = \dot{Z}_4$

equation (3) and (4) can be written as

$$F = M_1\dot{Z}_2 + (B_1 + B_2)Z_2 - B_2 Z_4$$

$$\dot{Z}_2 = \frac{F}{M_1} - \left(\frac{B_1 + B_2}{M_1}\right)Z_2 - \frac{B_2}{M_1}Z_4 \quad ...(5)$$

$$0 = M_2\dot{Z}_4 + KZ_3 + B_2Z_4 - B_2Z_2$$

$$\dot{Z}_4 = -\frac{K}{M_2}Z_3 - \frac{B_2}{M_2}Z_4 - \frac{B_2}{M_2}Z_2 \quad ...(6)$$

The state equations are

$$\dot{Z}_1 = Z_2$$

$$\dot{Z}_2 = \frac{F}{M_1} - \left(\frac{B_1 + B_2}{M_1}\right)Z_2 - \frac{B_2}{M_1}Z_4$$

$$\dot{Z}_3 = Z_4$$

$$\dot{Z}_4 = -\frac{K}{M_2}Z_3 - \frac{B_2}{M_2}Z_4 - \frac{B_2}{M_2}Z_2$$

$$\begin{bmatrix} \dot{Z}_1 \\ \dot{Z}_2 \\ \dot{Z}_3 \\ \dot{Z}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\left(\frac{B_1 + B_2}{M_1}\right) & 0 & -\frac{B_2}{M_1} \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{B_2}{M_2} & -\frac{K}{M_2} & -\frac{B_2}{M_2} \end{bmatrix}\begin{bmatrix} Z_1 \\ Z_2 \\ Z_3 \\ Z_4 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{M_1} \\ 0 \\ 0 \end{bmatrix}F$$

Select displacement of mass M_1 and M_2 as output variables

$\therefore$ $y_1 = x_1 = Z_1$

$y_2 = x_2 = Z_3$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} Z_1 \\ Z_3 \end{bmatrix}$$

Example 12.50. Find the characteristic equation when

$$A = \begin{bmatrix} -5 & 1 & 0 \\ 0 & -5 & 1 \\ 0 & 0 & -5 \end{bmatrix}$$

Solution : The characteristic equation $\Rightarrow |sI - A| = 0$

$$sI - A = s\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} -5 & 1 & 0 \\ 0 & -5 & 1 \\ 0 & 0 & -5 \end{bmatrix} = \begin{bmatrix} s+5 & -1 & 0 \\ 0 & s+5 & -1 \\ 0 & 0 & s+5 \end{bmatrix}$$

$$|sI - A| = s^3 + 15s^2 + 75s + 125$$

$\therefore$ Required characteristic equation

$s^3 + 15s^2 + 75s + 125 = 0$ **Ans.**

Example 12.51. Write short note characteristic equation for differential equation, transfer function and for state equation.

Solution: Characteristic Equation from Differential Equation

Consider the differential equation of linear time invariant system

$$\frac{d^n x(t)}{dt} + a_{n-1}\frac{d^{n-1}x(t)}{dt^{n-1}} + \ldots + a_0 x(t) = b_m \frac{d^m u(t)}{dt} + \ldots + b_0 u(t)$$

laplace transform of above equation

$$(s^n + a_{n-1}\, s_{n-1} + \ldots + a_0)\, X(s) = (b_m\, s^m + \ldots\, b_0)\, u(s)$$

The characteristic equation can be obtained by setting the homogeneous part to zero.

So, the characteristic equation will be

$$s^n + a_{n-1}\, s^{n-1} + \ldots + a_0 = 0$$

For example :

The characteristic equation of

$$\frac{d^4x(t)}{dt^4} + 3\frac{d^3x(t)}{dt^3} + 2\frac{d^2x(t)}{dt^2} + \frac{dx(t)}{dt} + 5x(t) = u(t) \text{ will be}$$

$$s^4 + 3s^3 + 2s^2 + s + 5 = 0$$

Characteristic Equation from transfer function

The characteristic equation from the transfer function can be obtained by equating the denominator polynomial to zero.

For example :

$$\frac{C(s)}{R(s)} = G(s) = \frac{b_m\, s^m + b_{m-1}s^{m-1} + \ldots b_o}{a_n s^n + \ldots + a_o}$$

The characteristic equation is

$$a_n\, s^n + \ldots + a_0 = 0$$

If

$$G(s) = \frac{K}{s(1+.4s)\,(1+0.25s)}$$

$$H(s) = 1$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)\,H(s)}$$

The characteristic equation

$$1 + G(s)\,H(s) = 0$$

$$1 + \frac{K}{s(1+0.4s)\,(1+0.25s)} = 0$$

$$s\,(1+0.4s)\,(1+0.25s) + K = 0$$

is the required characteristic equation.

Characteristic Equation From State Equation :

From equation 8.38

$$G(s) = C[sI - A]^{-1}B + D = C\frac{adj\,(sI-A)}{|sI-A|}B + D$$

$$G(s) = \frac{C\,adj\,(sI-A)\,B + |sI-A|D}{|sI-A|}$$

The characteristic equation can be obtained from transfer function by equating the denominator polynomial to zero.

Hence required characteristic equation is

$$|sI - A| = 0$$

Example 12.52. For the circuit shown in fig. 1.114, if $R_1 = R_2 = R$ and $C_1 = C_2 = C$, show that the system is always overdamped.

Solution : Transfer function of the given circuit

$$\frac{C(s)}{R(s)} = \frac{1}{1 + R_1 C_1 s + R_1 C_2 s + R_2 C_2 s + s^2 R_1 R_2 C_1 C_2}$$

Put $R_1 = R_2 = R$ and $C_1 = C_2 = C$

$$\frac{C(s)}{R(s)} = \frac{1}{s^2R^2C^2 + 3RCs + 1}$$

or,

$$\frac{C(s)}{R(s)} = \frac{\frac{1}{R^2C^2}}{s^2 + \frac{3}{RC}s + \frac{1}{R^2C^2}}$$

compare with

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n\, s + \omega_n^2}$$

$$\omega_n^2 = \frac{1}{R^2C^2} \quad \text{or} \quad \omega_n = \frac{1}{RC}$$

$$2\zeta\omega_n = \frac{3}{RC}$$

or,

$$2\zeta \cdot \frac{1}{RC} = \frac{3}{RC}$$

$$\therefore \quad \zeta = 1.5$$

Since, ζ is greater than one, hence the system is always overdamped.

Example 12.53. The differential equation of a control system is

$$\frac{d^2Y(t)}{dt^2} + 9\frac{dY(t)}{dt} + 20\,Y(t) = 20\,X(t)$$

Find the output response for unit step input $X(t)$ & $Y(t)$ are input and output respectively.

Solution : Given differential equation is

$$\frac{d^2Y(t)}{dt^2} + 9\frac{dY(t)}{dt} + 20Y(t) = 20\,X(t)$$

laplace transform of above equation

$$s^2\,Y(s) + 9\,s\,Y(s) + 20\,Y(s) = 20\,X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{20}{s^2 + 9s + 20}$$

$$X(s) = \frac{1}{s} \text{ (given)}$$

$$\therefore \quad Y(s) = \frac{20}{s(s^2+9s+20)}$$

or, $$Y(s) = \frac{1}{s} - \frac{5}{s+4} + \frac{4}{s+5}$$

Inverse laplace of above equation

$$\mathcal{L}^{-1} Y(s) = \mathcal{L}^{-1}\frac{1}{s} - \mathcal{L}^{-1}\frac{5}{s+4} + \mathcal{L}^{-1}\frac{4}{s+5}$$

$$y(t) = 1 - 5e^{-4t} + 4e^{-5t} \quad \text{Ans.}$$

Example 12.54. The output of a control system is

$$C(t) = 1 + 0.25e^{-50t} - 1.25e^{-10t}$$

(i) Obtain the expression for closed loop transfer function of the system.

(ii) Determine undamped natural frequency & damping ratio. Assume unit step input.

Solution : $C(t) = 1 + 0.25e^{-50t} - 1.25e^{-10t}$

laplace transform

$$C(s) = \frac{1}{s} + \frac{0.25}{s+50} - \frac{1.25}{s+10} = \frac{500}{s(s+10)(s+50)}$$

since, $R(s) = 1/s$ given

$$\therefore \quad \frac{C(s)}{R(s)} = \frac{500}{(s+10)(s+50)}$$

$\therefore$ closed loop transfer function $= \dfrac{500}{(s+10)(s+50)}$ **Ans.**

$$\frac{C(s)}{R(s)} = \frac{500}{s^2+60s+500}$$

compare with $\dfrac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

$\therefore \quad \omega_n^2 = 500$

or, $\omega_n = 22.36$ rad/sec.

$2\zeta\omega_n = 60$

$\therefore \quad 2\zeta 22.36 = 60$

$\zeta = 1.34$ **Ans.**

Example 12.55. Find impulse response of the given circuit. *(Mumbai university, 95)*

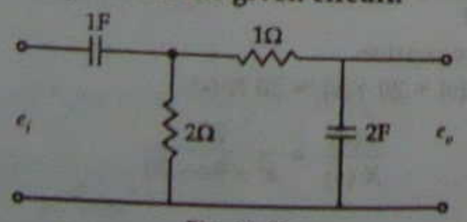


Fig. 12.73.

Solution : The transform network of the given circuit is shown in fig.

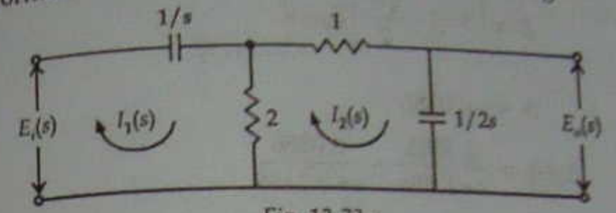


Fig. 12.73 a.

Apply KVL in first mesh:

$$E_i(s) = I_1(s)\left[\frac{1+2s}{s}\right] - 2I_2(s) \quad \text{...(1)}$$

KVL in second mesh

$$I_2(s)\left[3+\frac{1}{2s}\right] = 2I_1(s) \quad \text{...(2)}$$

$$I_1(s) = I_2(s)\left[\frac{6s+1}{2s}\right] \quad \text{...(3)}$$

Also, $$E_0(s) = \frac{1}{2s} I_2(s) \quad \text{...(4)}$$

Put the value of $I_1(s)$ from equation (3) in equation (1)

$$E_i(s) = I_2(s)\frac{4s^2+8s+1}{4s^2} \quad \text{...(5)}$$

from equation (4) and equation 5

$$\frac{E_0(s)}{E_i(s)} = \frac{2s}{4s^2+8s+1} = \frac{0.5s}{s^2+2s+1/4}$$

$E_i(s) = 1$ given

$$\therefore \quad E_0(s) = \frac{0.5s}{s^2+2s+1/4} = \frac{0.5s}{(s+1)^2+\frac{1}{4}-1}$$

$$E_0(s) = \frac{0.5s}{(s+1)^2-\left(\sqrt{3}/2\right)^2} = \frac{0.5s}{\left(s+1+\frac{\sqrt{3}}{2}\right)\left(s+1-\frac{\sqrt{3}}{2}\right)}$$

$$E_0(s) = \frac{A}{\left(s+1+\frac{\sqrt{3}}{2}\right)} + \frac{B}{\left(s+1-\frac{\sqrt{3}}{2}\right)}$$

$$A = \left(s+1+\frac{\sqrt{3}}{2}\right)E_0(s)\Big|_{s=-1-\frac{\sqrt{3}}{2}}$$

$$A = 0.5386$$

$$B = \left(s+1-\frac{\sqrt{3}}{2}\right)E_0(s)\Big|_{s=-1+\frac{\sqrt{3}}{2}}$$

$$B = -0.0386$$

$$E_0(s) = \frac{0.5386}{s+1.866} - \frac{0.0386}{s+0.1339}$$

inverse laplace

$$e_0(t) = 0.5386\, e^{-1.866t} - 0.0386\, e^{-0.1339t}$$ **Ans.**

Example 12.56. Obtain the step response of series RLC circuit with $R = 1\Omega$, $L = 1$ H and $C = 1$F output taken across 'R'. Assume input as a step of 10v. *(Pune university)*

Solution :

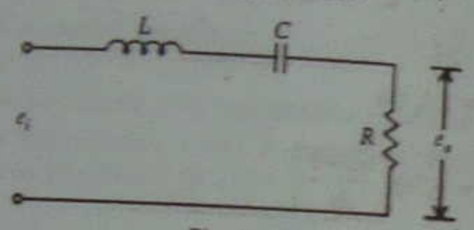


Fig. 12.74

The transform network is shown in fig.

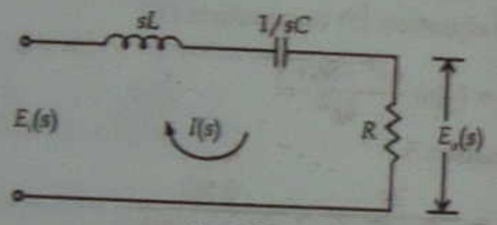


Fig. 12.74(a)

Apply KVL

$$E_i(s) = \left(R+sL+\frac{1}{sC}\right)I(s)$$

$$E_0(s) = RI(s)$$

$$\therefore \quad \frac{E_0(s)}{E_i(s)} = \frac{RCs}{s^2LC+sRC+1}$$

$$E_i(s) = \frac{10}{s} \text{ (given) \& } R=1, L=1, C=1$$

$$\therefore \quad E_0(s) = \frac{10}{s^2+s+1} = \frac{10}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{10}{\left(s+\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)\left(s+\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)}$$

$$E_0(s) = \frac{A}{\left(s+\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)} + \frac{B}{\left(s+\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)}$$

$$A = \left(s+\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)E_0(s)\Big|_{s=-\frac{1}{2}-j\frac{\sqrt{3}}{2}}$$

$$A = j\frac{10}{\sqrt{3}}$$

$$B = -j\frac{10}{\sqrt{3}}$$

$$E_0(s) = \frac{j10}{\sqrt{3}\left(s+\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)} - \frac{j10}{\sqrt{3}\left(s+\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)}$$

$$e_0(t) = \frac{j10}{\sqrt{3}}e^{\left(-\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)t} - \frac{j10}{\sqrt{3}}e^{\left(-\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)t} = \frac{j10}{\sqrt{3}}e^{-t/2}\left[e^{-j\frac{\sqrt{3}}{2}t} - e^{+j\frac{\sqrt{3}}{2}t}\right]$$

$$= \frac{j10\cdot 2j}{\sqrt{3}\cdot 2j}e^{-t/2}\left[e^{-j\frac{\sqrt{3}}{2}t} - e^{+j\frac{\sqrt{3}}{2}t}\right] = +\frac{20}{\sqrt{3}}e^{-t/2}\left[\frac{-e^{-j\frac{\sqrt{3}}{2}t}+e^{+j\frac{\sqrt{3}}{2}t}}{2j}\right]$$

$$= 11.54\, e^{-0.5t}\sin\frac{\sqrt{3}}{2}t$$

$$e_0(t) = 11.54\, e^{-0.5t}\sin 0.866\, t$$ **Ans.**

Example 12.57. For a control system shown in fig 12.75. Find the values of K and K_t so that the damping ratio (ζ) of system is 0.6 and setting time (t_s) is 0.1sec. Use $t_s = 3.2/\zeta\omega_n$. Assume unit step input.

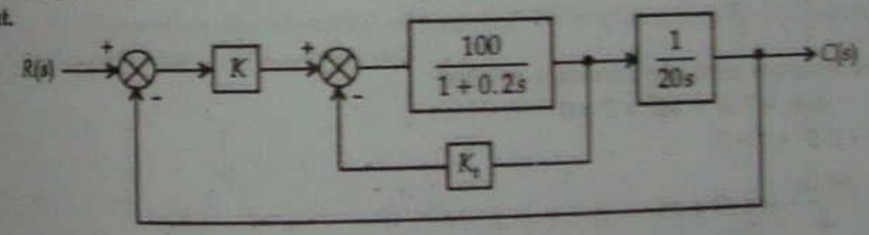
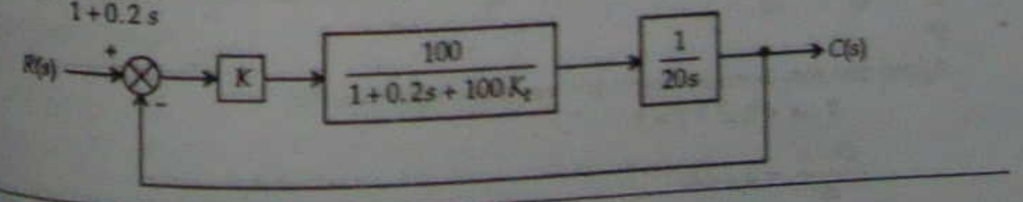


Fig. 12.75.

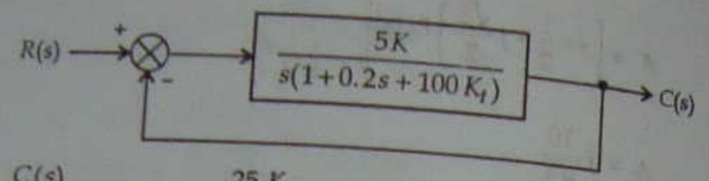
Solution : First solve inner closed loop.

$$\frac{\frac{100}{1+0.2s}}{1+\frac{100K_t}{1+0.2s}} = \frac{100}{1+0.2s+100K_t}$$



Take negative sign inside bracket

Now three blocks are in series



$$\therefore \quad \frac{C(s)}{R(s)} = \frac{25K}{s^2+5s(1+100K_t)+25K}$$

compare eq^n with $\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

$\omega_n^2 = 25k_t$ $\quad 2\zeta\omega_n = 5(1+100K_t)$

$\therefore \quad \omega_n = 5\sqrt{K}$

Given $t_s = \frac{3.2}{\zeta\omega_n}$

$$0.1 = \frac{3.2}{0.6\times 5\sqrt{K}}$$

$\therefore \quad K = 113.78$

$K = 113.78$

$\omega_n = 5\sqrt{K} = 5\sqrt{113.78} = 53.33$ rad/sec. $\quad \omega_n = 53.33$ rad/s **Ans.**

Example 12.58. The characteristic equation of a system is given by $S^6 + 3S^5 + 4S^4 + 6S^3 + 3S^2 + 3S + 2 = 0$. Comment on stability.

Solution :

S^6	1	4	5	2
S^5	3	6	3	-
S^4	2	4	2	
S^3	0	0	0	

one row becomes zero, the auxiliary eq^n $A(s)$

$$A(s) = 2S^4 + 4S^2 + 2 = 0$$

or, $S^4 + 2S^2 + 1 = 0$

$$\frac{dA(s)}{ds} = 4S^3 + 4S$$

S^6	1	4	5	2
S^5	3	6	3	
S^4	2	4	2	
S^3	4	4		
S^2	2	2		
S^1	0	0		
S^0				

Again one row becomes zero, the auxiliary equation $B(s)$

$$B(s) = 2s^2 + 2 = 0$$

$$\frac{dB(s)}{ds} = 4s = 0$$

S^6	1	4	5	2
S^5	3	6	3	
S^4	2	4	2	
S^3	4	4		
S^2	2	2		
S^1	4			
S^0	2			

All elements in first column are of same sign, therefore no root is in right half of s-plane, but there are two row of zeros, so check the system for marginally stable or unstable.

From first auxiliary equation

$$A(s) = S^4 + 2S^2 + 1$$

put $S^2 = x$

$\therefore \quad x^2 + 2x + 1 = 0$

$\therefore \quad x^2 + x + x + 1 = 0$

$(x+1)(x+1) = 0$

$x = -1, \quad x = -1$

$S^2 = -1, \quad S^2 = -1$

$S = \pm j, \quad S = \pm j$

$S_1 = +j, \quad S_3 = +j$

$S_2 = -j, \quad S_4 = -j$

From second auxiliary equation

$2S^2 + 2 = 0$

$S^2 = -1$

$S = \pm j$

$S_5 = +j \;\&\; S_6 = -j$

On imaginary axis the roots are repeated, hence system is unstable.

Example 12.59. Determine the stability of a system with characteristic equation.

$$S^5 + 4S^4 + 2S^3 + 8S^2 + S + 4 = 0$$

Solution :

S^5	1	2	1
S^4	4	8	4
S^3	0	0	
S^2			
S^1			
S^0			

One row becomes zero, use auxiliary equation

$$A(s) = 4s^4 + 8s^2 + 4 = 0 \qquad \therefore \quad \frac{dA(s)}{ds} = 16s^3 + 16s$$

or, $S^4 + 2S^2 + 1 = 0$

Put $S^2 = x$

$x^2 + 2x + 1 = 0$

$(x+1)(x+1) = 0$

	$x = -1$	,	$x = -1$
	$S^2 = -1$	,	$S^2 = -1$
	$S_1 = +j$		$S_3 = +j$
	$S_2 = -j$		$S_4 = -j$
S^5	1	2	1
S^4	4	8	4
S^3	16	16	
S^2	4	4	
S^1	0	0	
S^0			

Auxiliary equation

$$A(s) = 4s^2 + 4 = 0 \qquad \therefore \quad \frac{dA(s)}{ds} = 8s$$

$$S^2 = -1 \; ; \quad S = \pm j$$

Since, the roots are multiple roots on imaginary axis at $s = \pm j$ the system is unstable.

Example 12.60. Determine stability of a system having characteristic equation

$$S^6 + 5S^5 + 10S^4 + 24S^3 + 20S^2 + 15S + 10 = 0$$

Solution :

S^6	1	10	20	10
S^5	5	24	15	0
S^4	5.2	13.75	10	
S^3	10.78	-2.45		
S^2	14.93	10		
S^1	-9.67	0		
S^0	10			

No. of sign change = 2

Hence, system is unstable.

Example 12.61. Determine number of roots with positive real parts, zero real part and negative real part for the following polynomial equation using Routh criterion.

$$Q(s) = S^4 + 2S^3 + 8S^2 + 10S + 15$$ *(UPTU, B.Tech, C.O. Exam. 2003, Network and Systems)*

Solution :

S^4	1	8	15
S^3	2	10	
S^2	3	15	
S^1	0		
S^0			

Auxiliary equation

$$A(s) = 3s^2 + 15$$

$$\frac{dA(s)}{ds} = 6s$$

S^4	1	8	15
S^3	2	10	
S^2	3	15	
S^1	6		
S^0	15		

Since, no sign change in first column, the system may be stable. Solve auxiliary equation & determine the location of roots for stability.

$$3S^2 + 15 = 0$$

$$S_{1,2} = \pm j\sqrt{5}$$

$$S^2(S^2 + 5) + 2s(s^2 + 5) + 3(s^2 + 5) = 0$$

$$(S^2 + 5)(S^2 + 2S + 3) = 0$$

$$S^2 + 2S + 3 = 0$$

take $S_{3,4} = -1 \pm j1.415$

$$S_1 = 0 + j\sqrt{5}$$

$$S_2 = 0 - j\sqrt{5}$$

$$S_3 = -1 + j1.415$$

$$S_4 = -1 - j1.415$$

No. of roots with positive real part = 0

No. of roots with zero real part = 2

No. of roots with negative real part = 2

Example 12.62. Determine the number of roots with positive real part, no. of roots with zero real part and the number of roots with negative real part for the following equations, apply Routh-Hurwitz criterion.

(a) $2S^5 + S^4 + 6S^3 + 3S^2 + S + 1 = 0$

(b) $S^5 - S^4 - 2S^3 + 2S^2 - 8S + 8 = 0$

Solution : *(a)* $2S^5 + S^4 + 6S^3 + 3S^2 + S + 1 = 0$

S^5	2	6	1
S^4	1	3	1
S^3	0	-1	
S^2			
S^1			
S^0			

Multiply the characteristic equation by $(s + 1)$

$$(S + 1)(2S^5 + S^4 + 6S^3 + 3S^2 + S + 1) = 0$$

$$2S^6 + 3S^5 + 7S^4 + 9S^3 + 4S^2 + 2S + 1 = 0$$

S^6	2	7	4	1
S^5	3	9	2	
S^4	1	2.67	1	
S^3	1	-1		
S^2	3.67	1		
S^1	-1.27			
S^0	1			

No. of sign changes = 2

No. of roots with positive real part = 2

(b) $S^5 - S^4 - 2S^3 + 2S^2 - 8S + 8 = 0$

S^5	1	- 2	- 8
S^4	- 1	2	8
S^3	0	0	
S^2			
S^1			
S^0			

Auxiliary equation $A(s) = - S^4 + 2S^2 + 8 = 0$

$$\frac{dA(s)}{ds} = - 4S^3 + 4s$$

S^5	1	- 2	- 8
S^4	- 1	2	8
S^3	- 4	4	
S^2	1	8	
S^1	36		
S^0	8		

No. of sign changes = 2

∴ No. of roots with positive real part = 2

solve the auxiliary equation

$$S^4 - 2S^2 - 8 = 0$$

put $S^2 = x$

$$x^2 - 2x - 8 = 0$$

$$(x + 2)(x - 4) = 0$$

when $x + 2 = 0$ and when $x - 4 = 0$

$S^2 + 2 = 0$ $S^2 - 4 = 0$

$S = \pm j\sqrt{2}$ $(S + 2)(S - 2) = 0$

The given characteristic equation can be written in factored form

$(S - 1)(S^2 + 2)(S + 2)(S - 2) = 0$

No. of roots with positive real part = 2

No. of roots with zero real part = 2

No. of roots with negative real part = 1

Example 12.63. Apply Routh Hurwitz criterion to determine (1) number of roots with positive real part (2) the number of roots with zero real parts (3) the number of roots with negative real parts for the following equation.

(a) $S^4 + 2S^3 + 7S^2 + 10S + 10 = 0$

Solution :

S^4	1	7	10
S^3	2	10	
S^2	2	10	
S^1	0		
S^0			

Auxiliary equation $A(s) = 2S^2 + 10 = 0$

$$\frac{dA(s)}{ds} = 4s$$

		7	10
S^4	1	10	
S^3	2	10	
S^2	2		
S^1	4		
S^0	10		

No sign change in first column, system may be stable. Solve the auxiliary equation

$$2S^2 + 10 = 0$$

$$S = \pm j\sqrt{5}$$

Hence, roots are non-repeated, the system is marginally stable. The given equation can be written in factored form as

$(S^2 + 5)(S^2 + 2S + 2) = 0$

for $S^2 + 2S + 2 = 0$

$$S_{1,2} = - 1 \pm j1$$

for $S^2 + 5 = 0$

$$S_{3,4} = \pm j\sqrt{5}$$

No. of roots with positive real part = 0

No. of roots with zero real part = 2

No. of roots with negative real part = 2

$S_1 = - 1 + j1$

$S_2 = - 1 - j$

$S_3 = + j\sqrt{5}$

$S_4 = - j\sqrt{5}$

Example 12.64. Find the number of roots with positive real part, zero real part and negative real parts of the following characteristic equation

$S^4 + 5S^3 + 2S + 10 = 0$

Solution :

$S^4 + 5S^3 + 0\ S^2 + 2S + 10 = 0$

S^4	1	0	10
S^3	5	2	
S^2	-2/5	10	
S^1	127		
S^0	10		

No. of sign change = 2

No. of roots with positive real part = No. of sign change = 2

The given characteristic equation in factored form can be written as :

$(S + 5)(S^3 + 2)$

or, $(S + 5)(S + 1.26)(S^2 - 1.26\ S + 1.59)$

$\because$ $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$

or, $(S + 5)(S + 1.26)(S - 0.63 - j\ 1.09)(S - 0.63 + j\ 1.09) = 0$

$S_1 = -5$
$S_2 = -1.26$
$S_3 = 0.63 + j\,1.09$
$S_4 = 0.63 - j\,1.09$

No. of roots with positive real part = 2
No. of roots with zero real part = 0
No. of roots with negative real part = 2.

Example 12.65. Determine the values of K for open loop transfer function

$$G(s)\,H(s) = \frac{K}{s(1+0.1s)(1+s)}$$ **so that**

(a) the gain margin is 15 db and
(b) Phase margin is 60°.

Solution : Step 1 :

$$G(s)\,H(s) = \frac{K}{s(1+0.1s)(1+s)}$$

corner frequencies are

$$\omega_1 = \frac{1}{0.1} = 10 \text{ rad/sec.} \quad \text{and} \quad \omega_2 = \frac{1}{1} = 1 \text{ rad/sec.}$$

Step 2 : Draw the magnitude plot without K. Since there is a pole at origin, the initial slope of the magnitude plot will be - 20 db/dec.

(i) Draw a line having slope - 20db/dec. (parallel to the I slope marker on top of left corner) passing through $\omega = 1$ upto first corner frequency 1 rad/sec.

(ii) At $\omega = 1$, a simple pole occurs, the resultant slope of next line will be $-20 + (-20) = -40$ db/dec. Draw a line having slope - 40 db/dec. (parallel to II slope marker) upto II corner frequency 10 rad/sec.

(iii) At $\omega = 10$, a simple pole occurs, the resultant slope of next line will be $-40 + (-20) = -60$ db/dec. Draw a line having slope - 60 db/dec. from second corner frequency towards infinity as there is no pole or zero.

Step 3 : Draw phase plot

$$\phi = -90° - \tan^{-1} 0.1\,\omega - \tan^{-1} \omega$$

ω	$Arg(j\omega)$ ϕ_1	$Arg\,(1 + j\,0.1\,\omega)$ ϕ_2	$Arg\,(1 + j\omega)$ ϕ_3	*Resultant* $\phi = \phi_1 + \phi_2 + \phi_3$
0.2	- 90°	- 1.14°	- 11.3°	- 102.4°
0.4	- 90°	- 2.3°	- 21.8°	-114.1°
0.6	- 90°	- 3.43°	- 30.96°	- 124.39°
0.8	- 90°	- 4.6°	- 38.65°	- 133.25°
1.0	- 90°	- 5.7°	- 45°	- 140.7°
3.0	- 90°	- 16.69°	- 71.56°	- 178.25°
4.0	- 90°	- 21.8°	- 75.96°	- 187.76°

Step 4 : From Bode plot (plot 1), gain margin available = 21 db
Required gain margin = 15 db.

For required gain margin, shift the magnitude plot upwards by 21 - 15 = 6 db. 30, shift point A to point B = + 6 db (upward shift taken as positive and downward shift negative).

$$6 \text{ db} = 20 \log_{10} K$$
$$K = 1.99$$

Step 5 :

$$\text{P.M} = 180° + \angle G(j\omega)\,H(j\omega)\big|_{\omega=\omega_{pc}}$$

$$+ 60° = 180° + \angle G(j\omega)\,H(j\omega)$$

$$\therefore \quad \angle G(j\omega)\,H(j\omega) = 60° - 180 = -120°$$

Find this point on phase plot i.e., locate D at - 120°. Extend a line from D on magnitude plot this intersect at E, but this point should be at F for P.M (+ 60°). So shift the magnitude plot downwards.

i.e.,
$$EF = -6$$
$$-6 = 20 \log_{10} K$$
$$K = 0.501$$

Step 6 :
Value of K for gain margin 15 db = 1.99.
Value of K for phase margin 60° = 0.501.

Example 12.66. Determine the value of K for open loop transfer function

$$G(j\omega)\,H(j\omega) = \frac{K}{j\omega(1+j0.1\omega)(1+j0.05\omega)}$$

such that
(i) the gain margin is 20 db
(ii) the phase margin is 30°.

Solution : Corner frequencies are

$$\omega_1 = \frac{1}{0.1} = 10 \text{ rad/sec.}$$

$$\omega_2 = \frac{1}{0.05} = 20 \text{ rad/sec.}$$

Step 1 : Draw the magnitude plot without K. The procedure is same as in example 12.65.

Step 2 : Draw the phase plot

ω	$Arg\,(j\omega)$ ϕ_1	$Arg(1 + j\,0.1\,\omega)$ ϕ_2	$Arg\,(1 + j\,0.05\,\omega)$ ϕ_3	*Resultant* $\phi_1 + \phi_2 + \phi_3$
4.0	- 90°	- 21.8°	- 11.3	- 123.1
6.0	- 90°	- 30.96°	- 16.69°	- 137.65
8.0	- 90°	- 38.56°	- 21.8	- 150.36
10.0	- 90°	- 45°	- 26.56	- 161.56
12.0	- 90°	- 50.19°	- 30.96	- 171.46
14.0	- 90°	- 54.46°	- 35	- 179.46
18.0	- 90°	- 60.9°	- 42	- 192.9
20.0	- 90°	- 63.43°	- 45	- 198.43

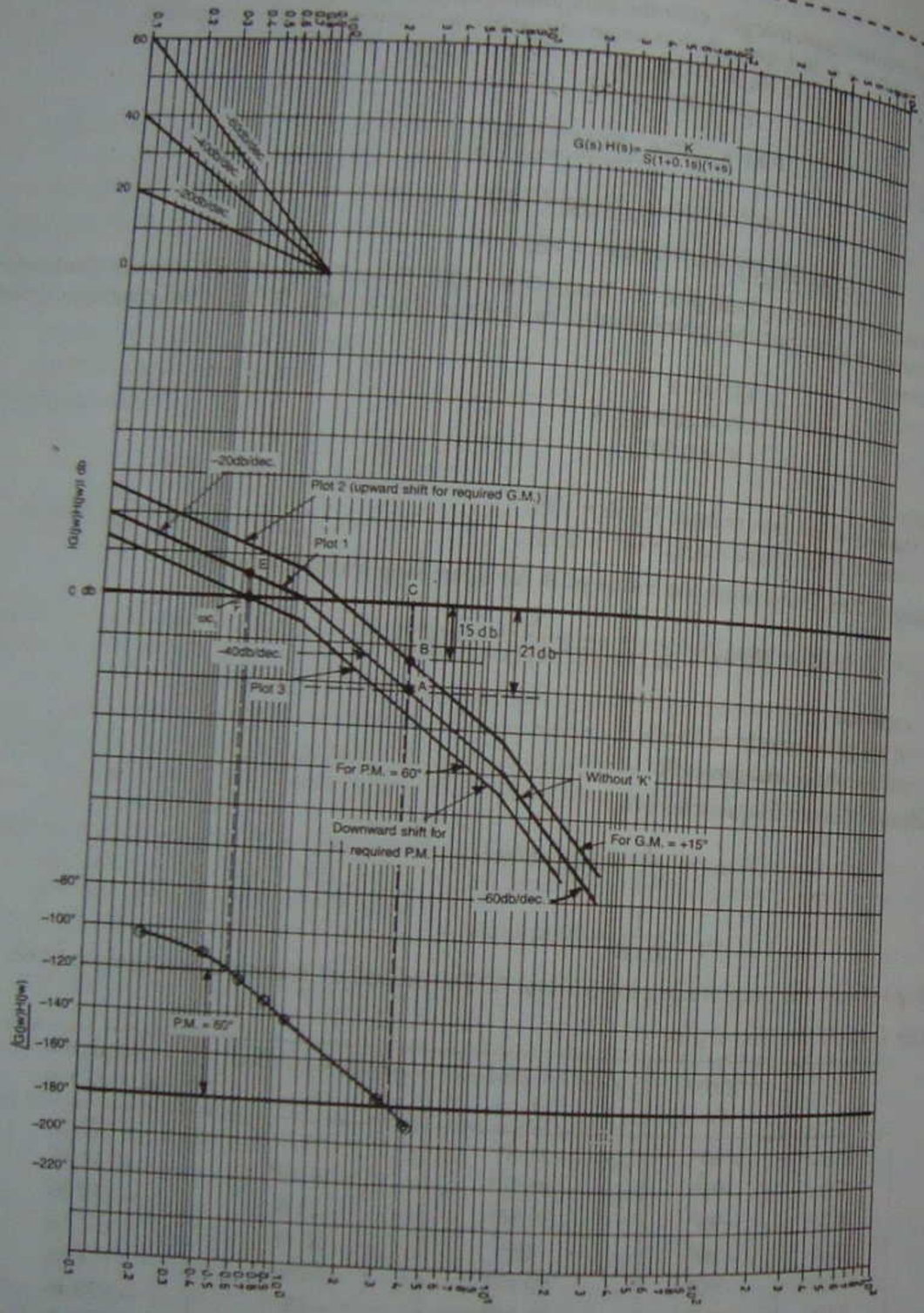


Fig. 12.76.

Step 4: From Bode plot

Gain margin available = 27 db

Required gain margin = 20 db

For required gain margin shift the magnitude plot upwards *i.e.*, shift the point *A* to *B* = + 7 db

$$7 \text{ db} = 20 \log_{10} K$$

$$\boxed{K = 2.23}$$

∴

Step 5 :

$$\text{P.M.} = 180° + \angle G(j\omega)\, H(j\omega)\Big|_{\omega = \omega_{c1}}$$

$$30° = 180° + \angle G(j\omega)\, H(j\omega)$$

$$\angle G(j\omega)\, H(j\omega) = 30 - 180 = -150°$$

∴ Find this point on phase plot *i.e.*, locate *D* at – 150°. Extend the line from *D* to the magnitude plot which intersects the magnitude plot at *E*. For required phase margin shift the magnitude plot upwards. For P.M. = 30° the magnitude plot should intersects 0db lione at *F*. So, the total shift is EF

$$EF = 19 \text{ db}$$

$$20 \log K = 19$$

$$\boxed{K = 8.91}$$

∴

Result :

(*i*) Value of K for gain margin 20 db = 2.23

(*ii*) Value of K for phase margin 30° = 8.91

Example 12.67. Determine the marginal value of K analytically for unity feedback system having

$$G(s)\, H(s) = \frac{K}{s(s+5)(s+10)}$$ **, so that the system will marginally stable.**

Solution : For marginally stable system

Phase margin = Gain margin = 0

Phase cross-over frequency = Gain crossover frequency.

We know that

$$\text{P.M.} = 180° + \angle G(j\omega)\, H(j\omega)$$

$$\text{P.M} = 0°$$

∴ $$0° = 180° + \angle G(j\omega)\, H(j\omega)$$

or, $$\angle G(j\omega)\, H(j\omega) = -180°$$

or, $$-90° - \tan^{-1} 0.2\,\omega - \tan^{-1} 0.1\,\omega = -180°$$

or $$\tan^{-1} 0.2\omega + \tan^{-1} 0.1\,\omega = 90°$$

taking tan on both sides

$$\tan(\tan^{-1} 0.2\,\omega + \tan^{-1} 0.1\,\omega) = \tan 90°$$

or, $$\frac{0.2\omega + 0.1\,\omega}{1 - (0.2\omega)(0.1\omega)} = \infty$$

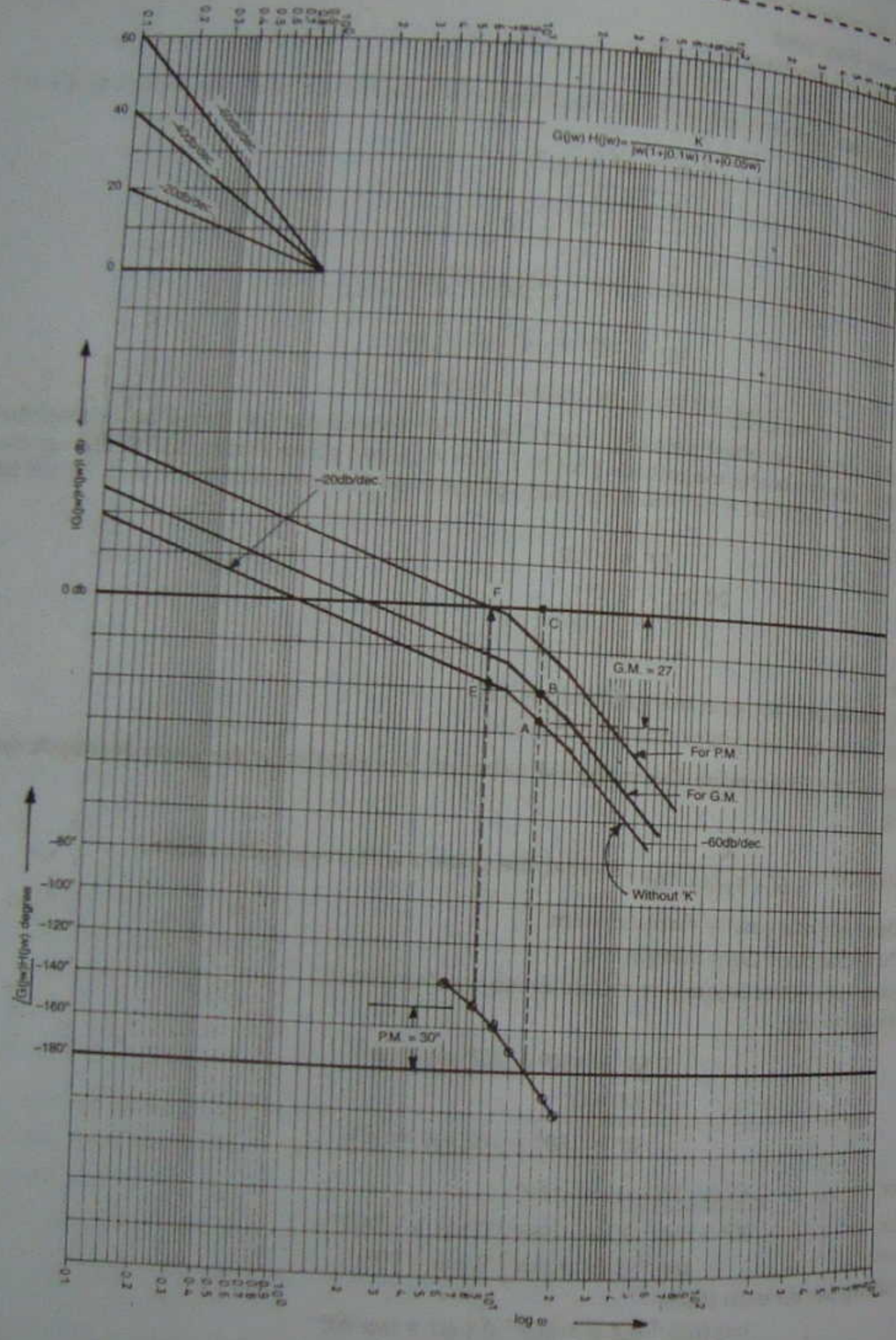


Fig. 12.77.

$$1 - 0.02\,\omega^2 = 0$$

$$\omega = 7.07 \text{ rad/sec.}$$

or,

or, This is the gain crossover frequency $\omega_{c_1} = 7.07$ rad/sec. At gain crossover frequency

$$|G(j\omega)\,H(j\omega)| = 1$$

$$\frac{K}{|j7.07||5+j7.07||10+j7.07|} = 1$$

$$\frac{K}{7.07 \times 8.66 \times 12.24} = 1$$

or,

$$\boxed{K = 749.82}$$

or,

$$\text{G.M} = 20 \log_{10} \frac{1}{|G(j\omega)\,H(j\omega)|_{\omega=\omega_{c_2}}}$$

put $|G(j\omega)\,H(j\omega)|_{\omega=\omega_{c_2}} = x$

$$\text{G.M.} = 20 \log_{10} \frac{1}{x}$$

$$0 = 20 \log_{10} \frac{1}{x}$$

$$\frac{1}{x} = 1 \qquad \text{or,} \qquad x = 1$$

$$G(j\omega)\,H(j\omega) = \frac{K}{(j\omega)(5+j\omega)(10+j\omega)}$$

Rationalize above function to find x, at point x $\angle G(j\omega)\,H(j\omega) = -180°$. i.e, imaginary part is zero.

$$G(j\omega)\,H(j\omega) = \frac{K(-j\omega)(5-j\omega)(10-j\omega)}{(j\omega)(5+j\omega)(10+j\omega)(5-j\omega)(10-j\omega)}$$

$$= \frac{15K}{(25+\omega^2)(100+\omega^2)} - j\frac{K\omega(50-\omega^2)}{\omega^2(25+\omega^2)(100+\omega^2)}$$

Equate the imaginary part to zero

$$50 - \omega^2 = 0$$

or,

$$\omega^2 = 50$$

$$\omega = 7.07 \text{ rad/sec.}$$

This is the phase crossover frequency $\omega_{c_2} = 7.07$ rad/sec. Put the value of ω_{c_2} in real part, the point x is

$$x = \frac{15\,K}{(25+50)(100+50)}$$

$$1 = \frac{15\,K}{11250}$$

$$\boxed{K = 750}$$

∴ For marginally stable system $K = 750$ **Ans.**

Example 12.68. The open loop transfer function with unity feedback is

$$G(s) = \frac{K}{s(s+2)(s+20)}.$$

Find analytically the value of 'K' for phase margin = 50°.

Solution : $G(s)\,H(s) = \dfrac{K}{s(s+2)(s+20)} = \dfrac{K/40}{s(1+0.5s)(1+0.05s)}$

put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{K/40}{j\omega(1+j0.5\omega)(1+j0.05\omega)}$$

We know that

$$P.M = 180° + \angle G(j\omega)\,H(j\omega)$$

$$\angle G(j\omega)\,H(j\omega) = -90° - \tan^{-1}0.5\,\omega - \tan^{-1}0.05\,\omega$$

$$\therefore \quad 50° = 180° + (-90° - \tan^{-1}0.5\,\omega - \tan^{-1}0.05\omega)$$

or, $\quad 40° = \tan^{-1}0.5\,\omega + \tan^{-1}0.05\,\omega$

taking tan on both sides

$$\tan 40° = \tan(\tan^{-1}0.5\,\omega + \tan^{-1}0.05\,\omega)$$

or, $\quad 0.839 = \dfrac{0.55\,\omega}{1-0.025\,\omega^2}$

or, $0.020975\,\omega^2 + 0.55\omega - 0.839 = 0$

or, $\quad \omega = 1.445$ rad/sec.

This is the gain cross-over frequency

$$\omega_g = 1.445 \text{ rad/sec.}$$

At gain cross over frequency $|G(j\omega)\,H(j\omega)| = 1$

$$\frac{K/40}{j1.445||1+j0.7225||1+j0.07225} = 1$$

$$K/40 = 1.7873$$

or $\quad \boxed{K = 71.49}$

Required value of $K = 71.49$ **Ans.**

Example 12.69. The open loop transfer function with unity feedback system is

$$G(s) = \frac{25}{s(s+1)(s+10)}$$

Determine phase margin and gain margin analytically.

Solution :

Step 1 : Calculation of Phase Crossover Frequency

$$G(j\omega)\,H(j\omega) = \frac{2.5}{j\omega(1+j\omega)(1+j0.1\omega)}$$

At phase cross-over frequency the angle is - 180°

$$\therefore \quad -180° = -90° - \tan^{-1}0.1\omega - \tan^{-1}\omega$$

or, $\quad 90° = \tan^{-1}0.1\omega + \tan^{-1}1\omega$

Take tan on both sides

$$\tan 90° = \tan(\tan^{-1}0.1\,\omega + \tan^{-1}\omega)$$

$$\infty = \frac{0.1\omega + \omega}{1-0.1\omega^2}$$

$$1 - 0.1\,\omega^2 = 0$$

or, $\quad \omega = 3.16$ rad/sec.

This is the phase crossover frequency ω_{c_2}

$$|G(j\omega)\,H(j\omega)|_{\omega_{c_2}} = \frac{2.5}{j3.16||1+j3.16||1+j\,0.316|}$$

$$= 0.2279$$

$$G.M = -20\log_{10}0.2279 = 12.84 \text{ db} \quad \textbf{Ans.}$$

∴ At gain crossover frequency $|G(j\omega)\,H(j\omega)| = 1$

$$\frac{2.5}{\omega\sqrt{1+\omega^2}\sqrt{1+0.01\omega^2}} = 1$$

or, $\quad \omega = 1.425$ rad/sec.

This is the gain crossover frequency ω_{c_1}

$$\angle G(j\omega)\,H(j\omega)\big|_{\omega=\omega c_1} = \frac{2.5}{j1.425(1+j1.425)(1+j0.1425)} = -153°$$

$$\therefore \quad P.M = 180° + (-153°)$$

$$P.M. = 27° \quad \textbf{Ans.}$$

Example 12.70. Draw Bode plot of transfer function given in example 12.69 and find P.M. & G.M.

Solution : $G(j\omega)\,H(j\omega) = \dfrac{2.5}{j\omega(1+j\omega)(1+j0.1\omega)}$

Step 1 : Draw the magnitude plot

(i) Draw a line having slope - 20 db/dec. upto 2.5 and mark I corner frequency 1 rad/sec. as there is one pole at origin.

(ii) At $\omega = 1$ rad/sec, there is a simple pole, so next slope will be $-20 + (-20) = -40$ db/sec. Draw a line having slope - 40 db/dec. from I corner frequency to II corner frequency 10 rad/sec.

(iii) At $\omega = 10$ rad/sec. there is a simple pole, so draw a line having slope $-20 + (-40) = -60$ db/dec. towards ∞.

Step 2 : Draw phase plot

$$\phi = -90° - \tan^{-1}\omega - \tan^{-1}0.1\,\omega$$

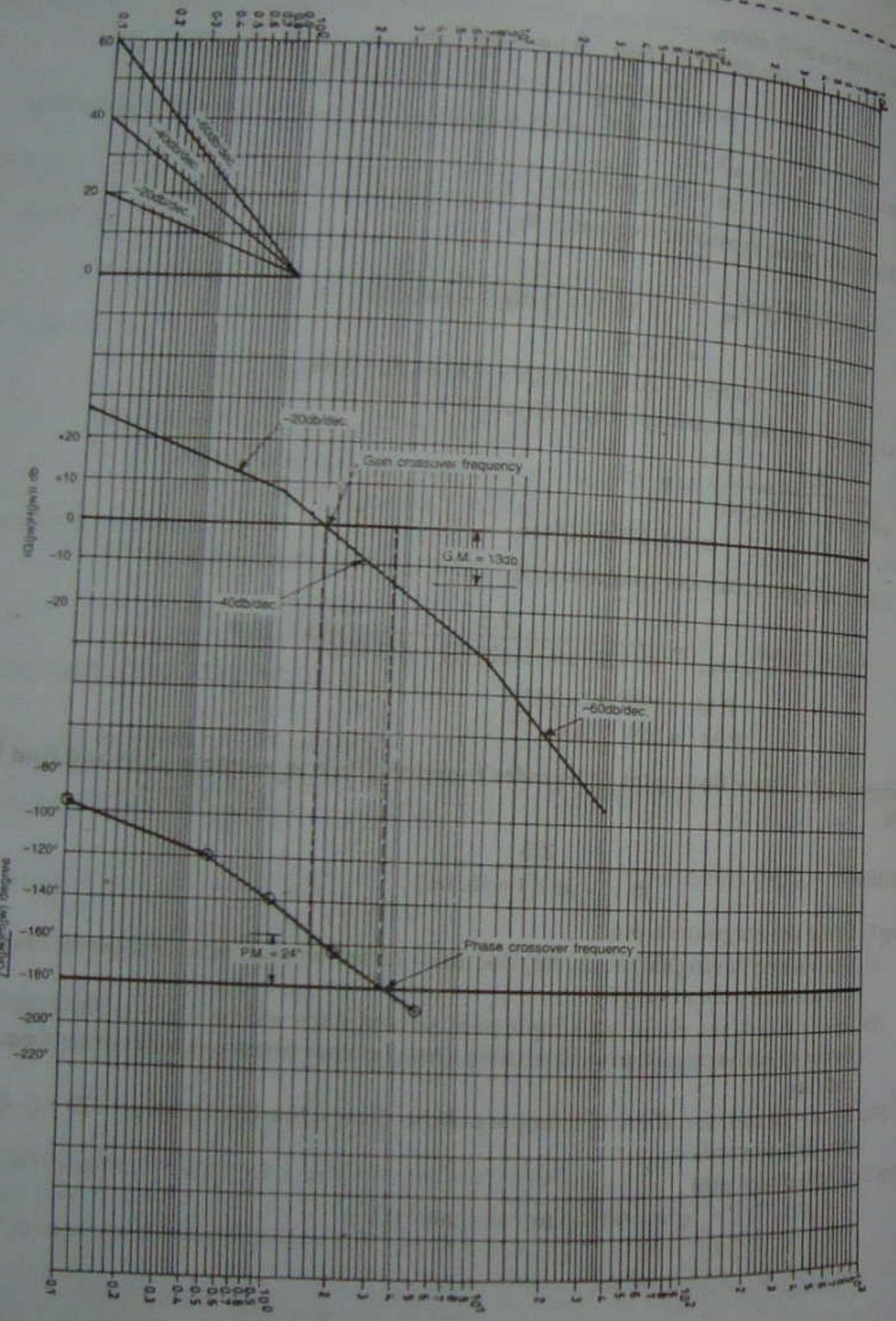


Fig. 12.78.

ω	Arg $(j\omega)$ ϕ_1	Arg $(1 + j\omega)$ ϕ_2	Arg $(1 + j\,0.1\,\omega)$ ϕ_3	Resultant $\phi = \phi_1 + \phi_2 + \phi_3$
0.1	- 90°	- 5.71°	- 0.573°	- 96.28°
0.5	- 90°	- 26.56°	- 2.86°	- 119.42°
1.0	- 90°	- 45°	- 5.71°	- 140.71°
1.8	- 90°	- 60.95°	- 10.20	- 161.15°
2.0	- 90°	- 63.43°	- 11.3°	- 164.73°
3.0	- 90°	- 71.56°	- 16.7°	- 178.26°
4.0	- 90°	- 75.96°	- 21.8°	- 187.76°

From Bode Plot

P.M. = 24°

G.M. = 13 db

Example 12.71. Determine the transfer function of the system using block diagram reduction of the system shown in fig. 12.80.

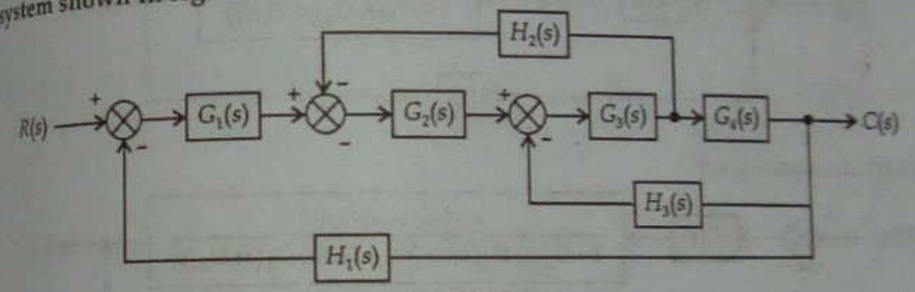
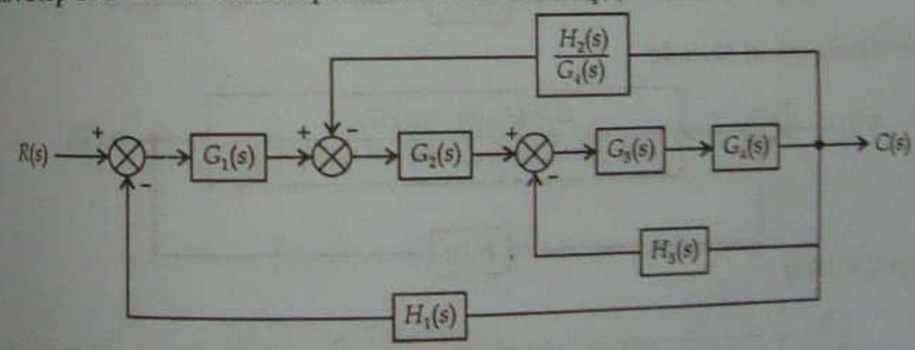
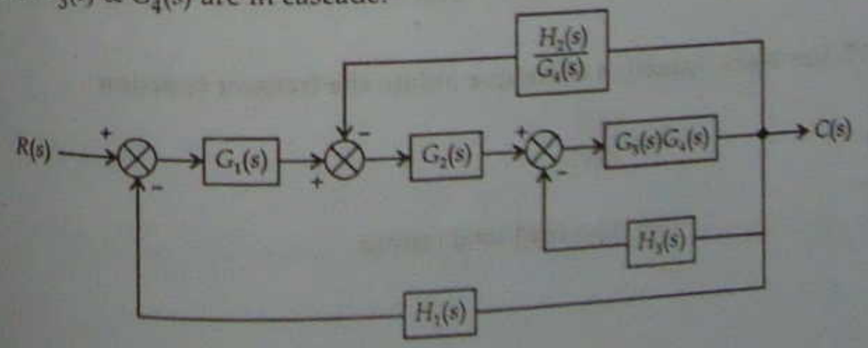


Fig. 12.79.

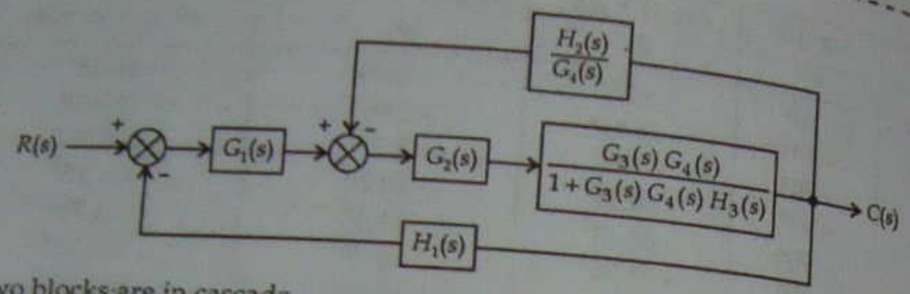
Solution : Step 1 : Shift the takeoff point after the block $G_4(s)$



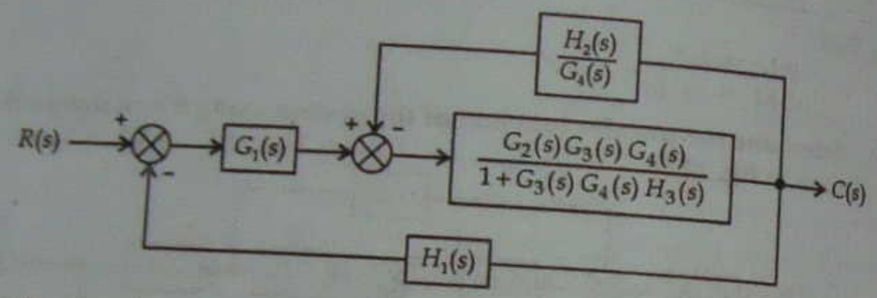
Step 2: Blocks $G_3(s)$ & $G_4(s)$ are in cascade.



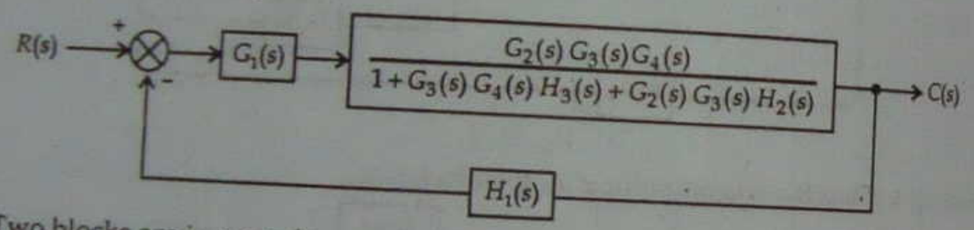
Step 3 : Solve the inner closed loop



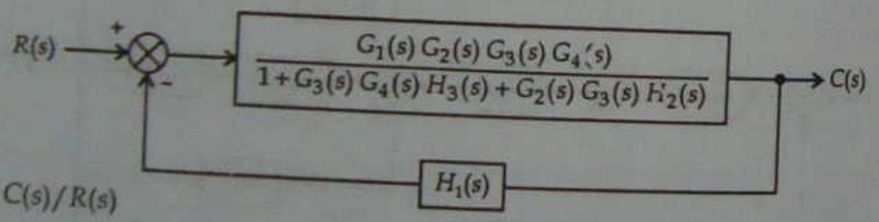
Step 4 : Two blocks are in cascade



Step 5 : Solve the closed loop



Step 6 : Two blocks are in cascade



Step 7 : Find $C(s)/R(s)$

$$\frac{C(s)}{R(s)} = \frac{G_1(s)\,G_2(s)\,G_3(s)\,G_4(s)}{1+G_3(s)\,G_4(s)\,H_3(s)+G_2(s)\,G_3(s)\,H_2(s)+G_1(s)G_2(s)G_3(s)G_4(s)H_1(s)}$$ Ans.

Example 12.72. Use block reduction technique obtain the transfer function

(*i*) $\frac{C(s)}{R(s)}$ (*ii*) $\frac{C(s)}{X(s)}$

(*iii*) $\frac{C(s)}{Y(s)}$ (*iv*) **Also find total output**

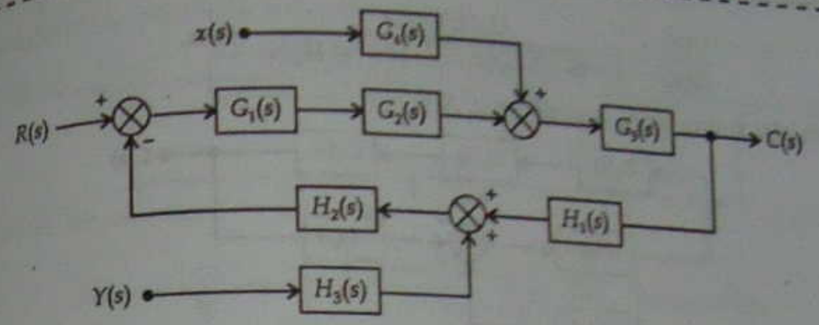
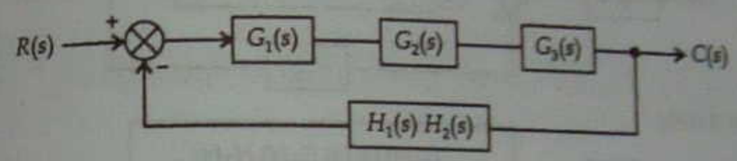
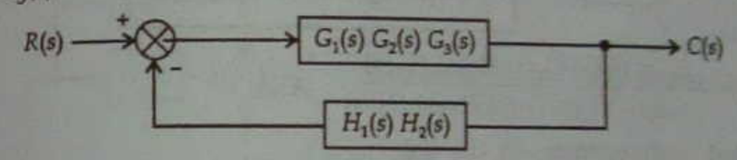


Fig. 12.80.

Solution: (*i*) When $R(s)$ acting alone and $x(s) = 0$, $Y(s) = 0$

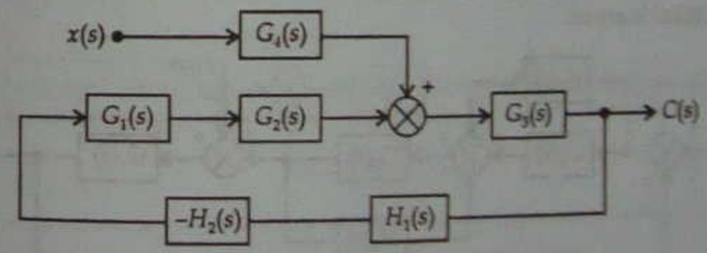


$G_1(s)$, $G_2(s)$, $G_3(s)$ are in cascade

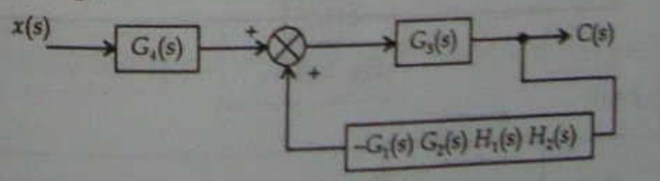


$$\frac{C(s)}{R(s)} = \frac{G_1(s)\,G_2(s)\,G_3(s)}{1+G_1(s)\,G_2(s)\,G_3(s)\,H_1(s)\,H_2(s)}$$ **Ans.** ...(*i*)

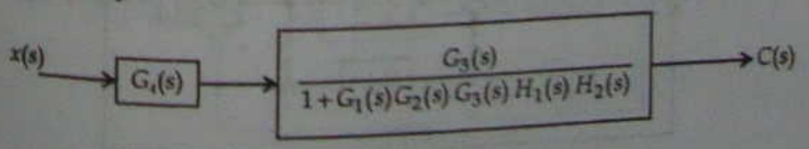
(*ii*) When $X(s)$ acting alone and $R(s) = 0$, $Y(s) = 0$



$G_1(s)$, $G_2(s)$, $H_1(s)$ & $-H_2(s)$ are in cascade

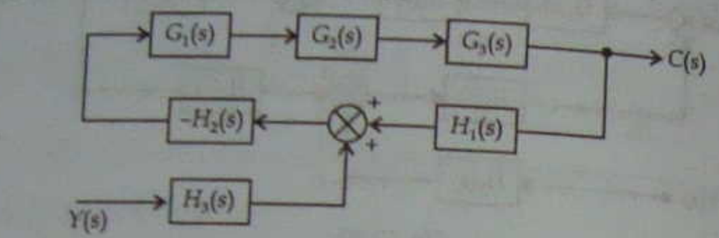


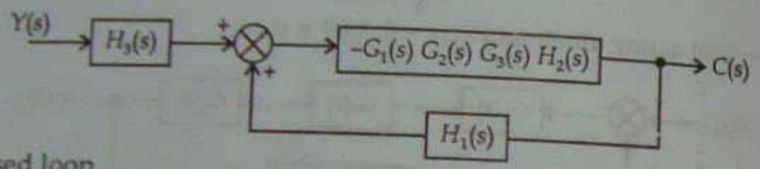
Solve the closed loop



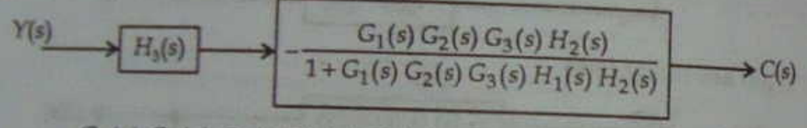
$$\frac{C(s)}{X(s)} = \frac{G_3(s)\,G_4(s)}{1 + G_1(s)\,G_2(s)\,G_3(s)\,H_1(s)\,H_2(s)}$$ Ans.

(iii) When Y(s) acting alone and $R(s) = 0$, $X(s) = 0$





Solve the closed loop



$$\frac{C(s)}{Y(s)} = -\frac{G_1(s)\,G_2(s)\,G_3(s)\,H_2(s)\,H_3(s)}{1 + G_1(s)\,G_2(s)\,G_3(s)\,H_1(s)\,H_2(s)}$$ Ans.

(iv) total output : from equtions (i), (ii), (iii)

$$C(s) = \frac{G_1(s)\,G_2(s)\,G_3(s)\,R(s) + G_3(s)\,G_4(s)\,X(s) - G_1(s)\,G_2(s)\,G_3(s)\,H_2(s)\,H_3(s)\,Y(s)}{1 + G_1(s)\,G_2(s)\,G_3(s)\,H_1(s)\,H_2(s)}$$ Ans.

Example 12.73. Using block reduction technique, find the transfer function from each input to the output C. Also total output.

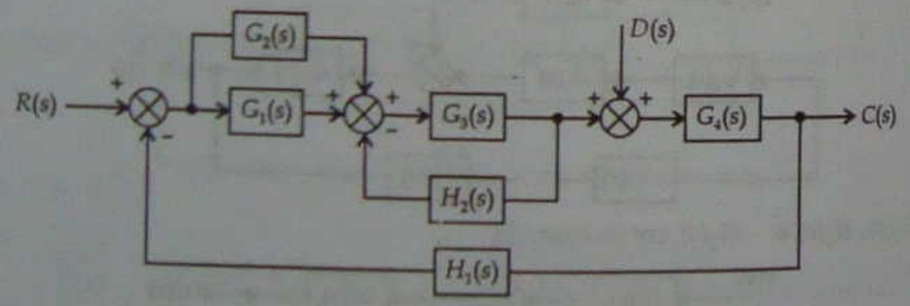
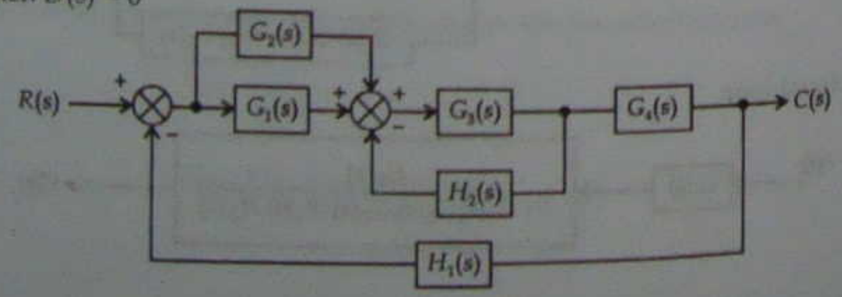
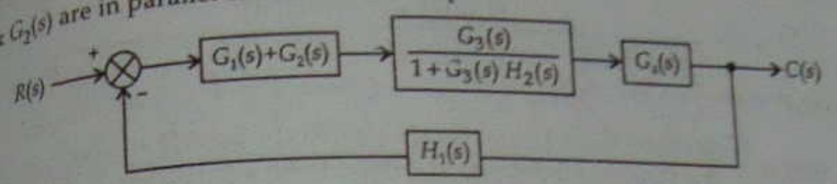


Fig. 12.81.

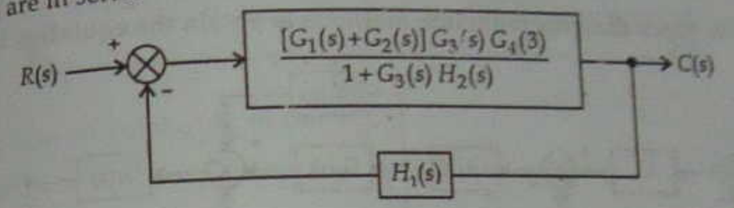
Solution : When $D(s) = 0$



$G_1(s)$ & $G_2(s)$ are in parallel & solve closed loop

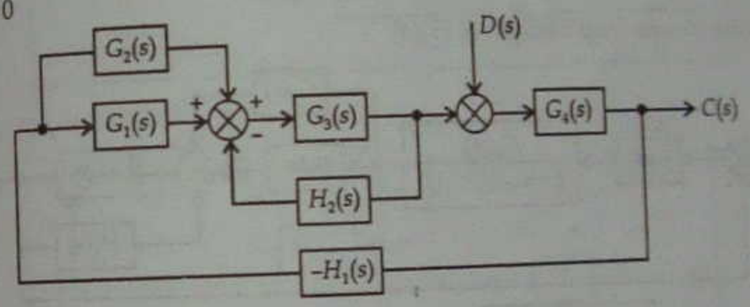


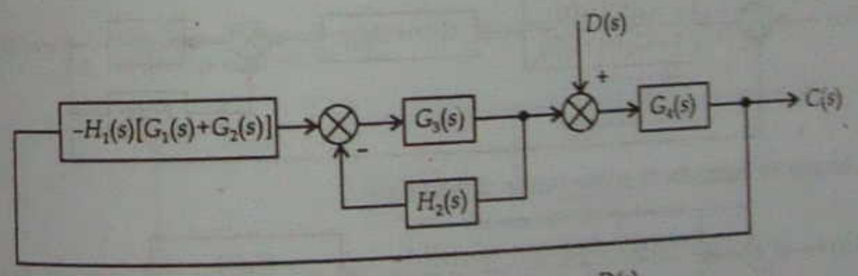
three blocks are in series

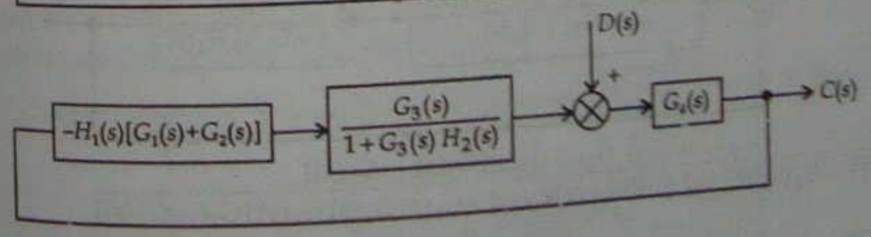


$$\frac{C(s)}{R(s)} = \frac{G_1(s)\,G_3(s)\,G_4(s) + G_2(s)\,G_3(s)\,G_4(s)}{1 + G_3(s)\,H_2(s) + G_1(s)\,G_3(s)\,G_4(s)\,H_1(s) + G_2(s)\,G_3(s)\,G_4(s)\,H_1(s)}$$ Ans.

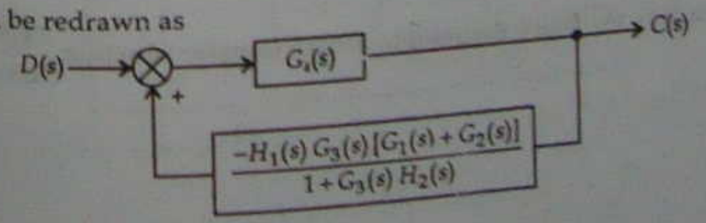
when $R(s) = 0$







Above fig. can be redrawn as



$$\frac{C(s)}{D(s)} = \frac{G_4(s)[1 + G_3(s)\,H_2(s)]}{1 + G_3(s)\,H_2(s) + G_3(s)\,G_4(s)\,G_1(s)\,H_1(s) + G_2(s)\,G_3(s)\,G_4(s)\,H_1(s)}$$ Ans.

Total output

$$C(s) = \frac{[G_1(s)\,G_3(s)\,G_4(s) + G_2(s)\,G_3(s)\,G_4(s)]\,R(s) + G_4(s)[1 + G_3(s)\,H_2(s)]\,D(s)}{1 + G_3(s)\,H_2(s) + G_1(s)\,G_3(s)\,G_4(s)\,H_1(s) + G_2(s)\,G_3(s)\,G_4(s)\,H_1(s)}$$ Ans.

Example 12.74. Use block diagram reduction methods to obtain the equivlent tranfer function from R to C.

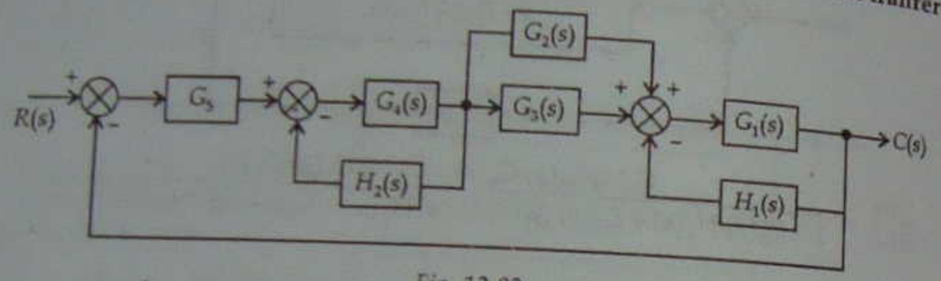
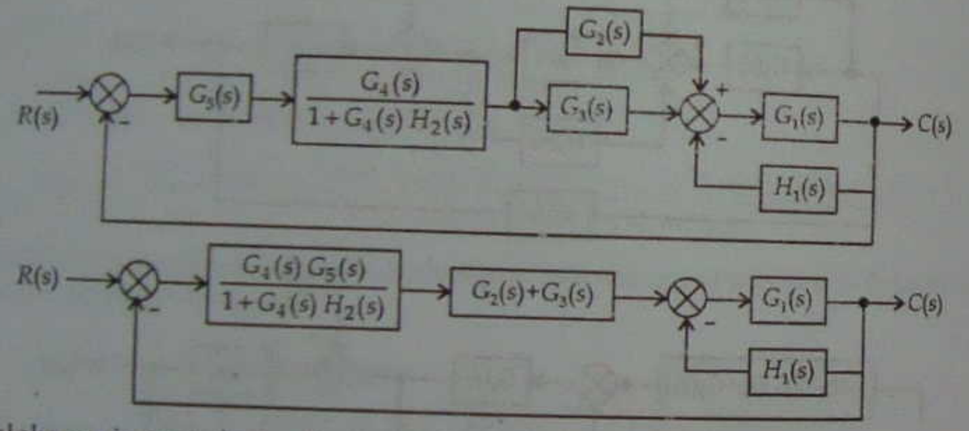
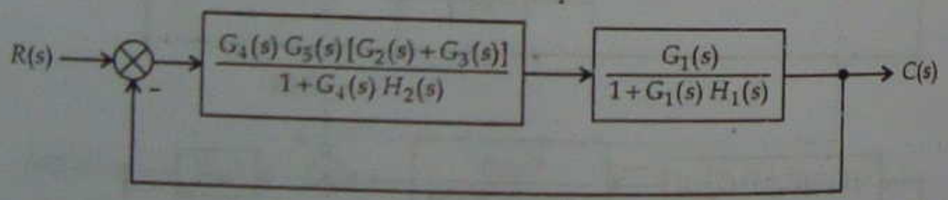


Fig. 12.82.

Solution : Solve the closed loop of $G_4(s)$ & $H_2(s)$



Two bloks are in cascade & solve inner closed loop



$$\frac{C(s)}{R(s)} = \frac{G_1(s)\;G_4(s)\,G_5(s)[G_2(s) + G_3(s)]}{[1 + G_4(s)\,H_2(s)]\;[1 + G_1(s)\,H_1(s)] + G_1(s)\,G_4(s)\,G_5(s)\,[G_2(s)\,G_3(s)]}$$

Example 12.75. Simplify the block diagram to obtain transfer function $\frac{C(s)}{R(s)}$

(UPTU 2003-2004)

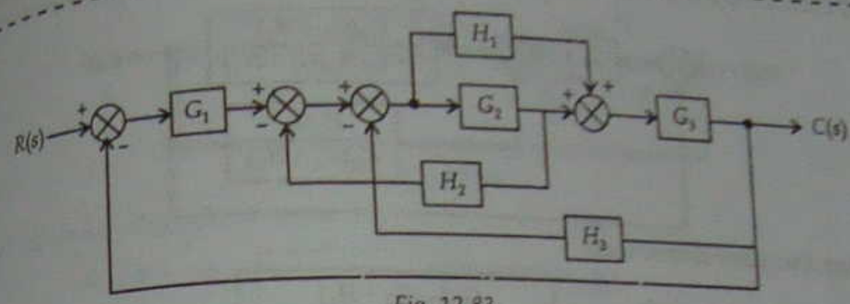
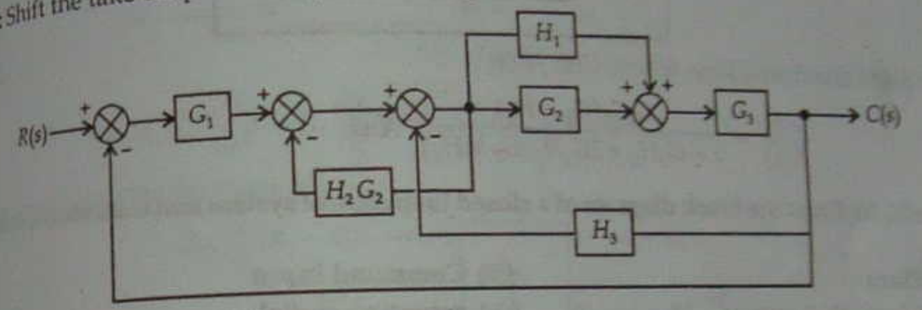
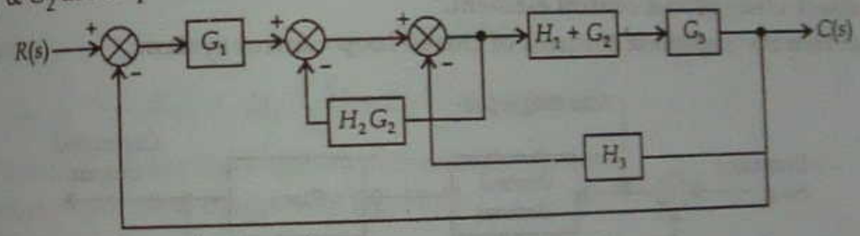


Fig. 12.83.

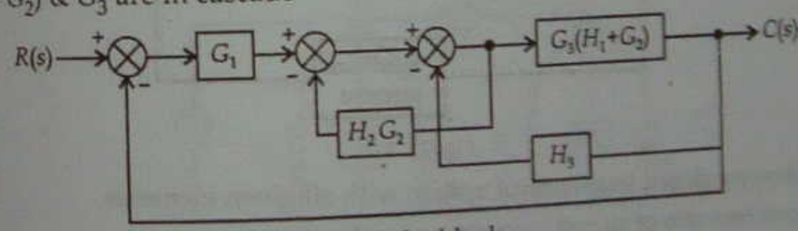
Step 1 : Shift the take off point of H_2 ahead of G_2.



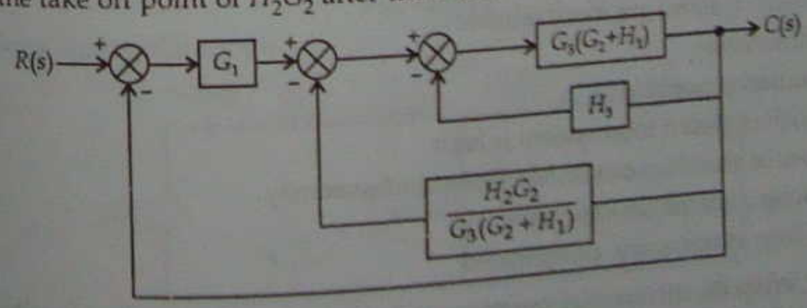
Step 2 : H_1 & G_2 are in parallel



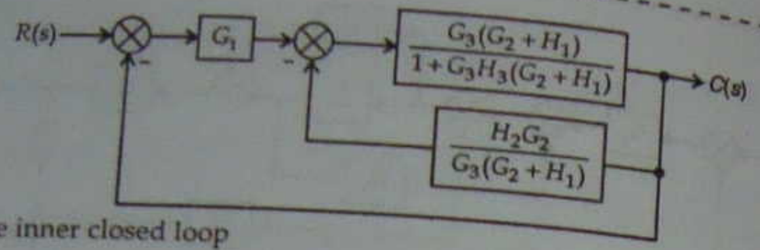
Step 3 : $(H_1 + G_2)$ & G_3 are in cascade



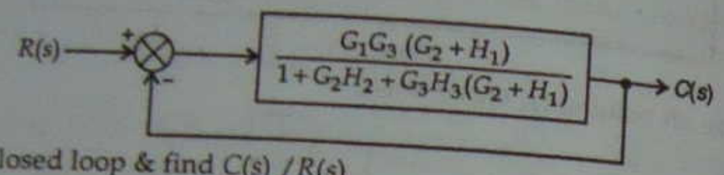
Step 4 : Shift the take off point of H_2G_2 after the block



Step 5 : Solve the innermost closed loop



Step 6 : reduce the inner closed loop



Step 7 : Solve the closed loop & find $C(s)/R(s)$

$$\frac{C(s)}{R(s)} = \frac{G_1G_3(G_2+H_1)}{1+G_2H_2+2G_3H_3(G_2+H_1)}$$ **Ans.**

Example 12.76. Draw the block diagram of a closed loop control system and indicate the following on it

(*i*) Plant **(*ii*) Command input**
(*iii*) Controlled output **(*iv*) actuating signal**
(*v*) feedback element and control element.
Also mention the important features of closed loop control system. *(UPTU 2003-2004)*

Solution :

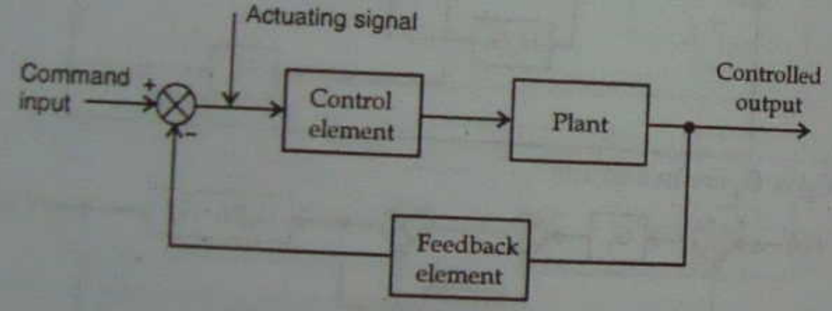


Fig. 12.84.

Fig. 12.85 shows a closed loop control system with all given elements.
The important features of closed loop control systems are

1. Closed loop systems are more reliable
2. They are accurate
3. Optimization is possible
4. Bandwidth of closed loop system is high.
5. A number of variables can be handled simultaneously.
6. Closed loop systems are expensive.
7. Closed loop systems are complicated.

Example 12.77. Write the differential equations and obtain ($f - v$) force-voltage analogous networks for the system shown below *(UPTU 2003-2004)*

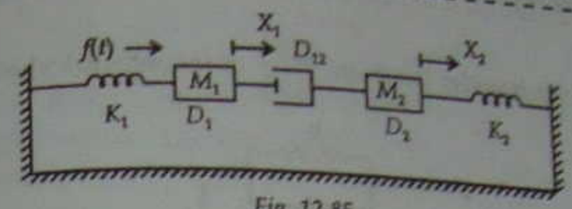


Fig. 12.85.

Solution : Freebody diagram of mass M_1

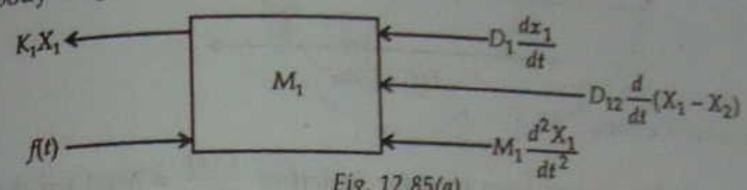


Fig. 12.85(a)

$$\therefore \quad f(t) = M_1\frac{d^2X_1}{dt^2} + D_1\frac{dX_1}{dt} + D_{12}\frac{d}{dt}(X_1 - X_2) + k_1X_1 \qquad ...(i)$$

Freebody diagram for mass M_2

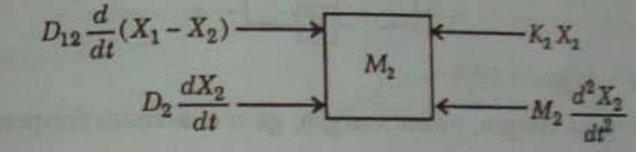


Fig. 12.85(b)

$$\therefore \quad D_{12}\frac{d}{dt}(X_1 - X_2) = M_2\frac{d^2X_2}{dt^2} + D_2\frac{dX_2}{dt} + k_2X_2 \qquad ...(ii)$$

Equations (*i*) & (*ii*) are the required differential equations.
Mechanical equivalent diagram of fig. 12.85.

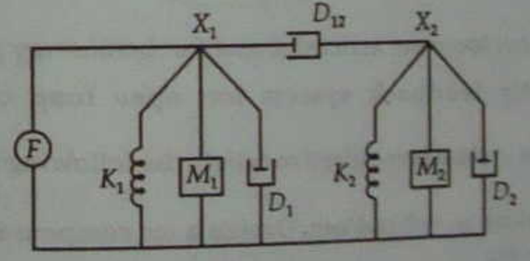


Fig. 12.85(c)

Force-voltage analogous circuit

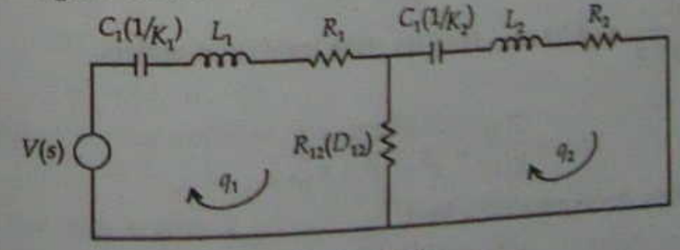


Fig. 12.85(d)

Fig. 12.86 (*d*) is the required $f - v$ analogous circuit.

Example 12.78. Derive the transfer function $\frac{E_o(s)}{E_i(s)}$ of the network shown below. *(UPTU 2002-2003)*

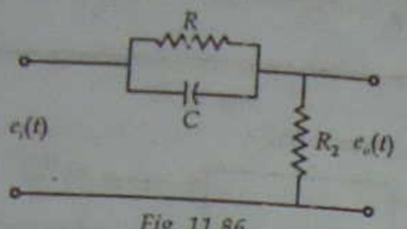
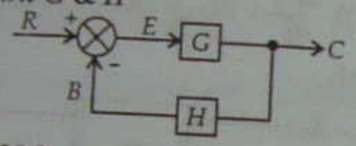


Fig. 11.86.

Solution : Refer solved examp 1 1.3

Example 12.79. Derive the closed loop transfer function $\frac{C(s)}{R(s)} = M(s)$ for the system shown below and find its sensitivity *w.r.t G & H* *(UPTU 2002-2003)*



Solution : Refer article 1.20 and 1.30.3.

Example 12.80. Define gain margin, phase margin, gain crossover frequency, phase crossover frequency in a polar plot.

Solution : Refer article 4.6

Example 12.81. State the rules for construction of root loci of $G(s)\,H(s)$. Find the breakaway points of $G(s)H(s) = \frac{k}{s(s+4)(s^2+4s+20)}$ *(UPTU 2002-2003)*

Solution :

For construction of root loci refer article 5.8 and for breakaway point refer example 5.33. *(UPTU 2002-2003)*

Example 12.82. A unity feedback system has open loop transfer function $G(s) = \frac{k}{s(0.1s+1)(0.2s+1)}$. The system is required to satisfy the following performance specifications $k_v = 30$, P.M ≥ 40 bandwidth $\omega_b = 5$ rad/sec. Design a lag compensator. *(UPTU 2003-2004)*

Solution : Refer example 6.4

Example 12.83. Obtain the state model of the system whose transfer function is given by $T(s) = \frac{s^2+3s+3}{s^3+2s^2+3s+1}$ and hence, find its state transition matrix. *(UPTU 2003-2004)*

Solution : Let $T(s) = \frac{C(s)}{R(s)} = \frac{s^2+3s+3}{s^3+2s^2+3s+1}$

Divide numerator & denominator by s^3

$$\frac{C(s)}{R(s)} = \frac{s^{-1}+3s^{-2}+3s^{-3}}{1+2s^{-1}+3s^{-1}+s^{-3}} \cdot \frac{Q(s)}{Q(s)}$$

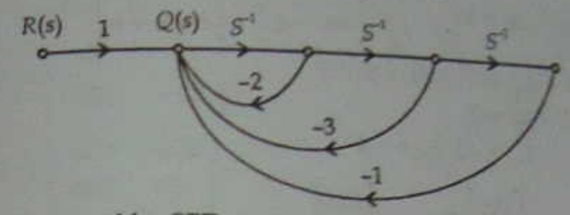
$$C(s) = Q(s)\,[s^{-1}+3s^{-2}+3s^{-3}]$$

$$R(s) = Q(s)\,[1+2s^{-1}+3s^{-2}+s^{-3}] \quad ...(i)$$

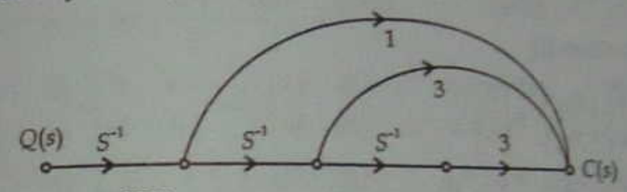
$$R(s) = Q(s) + Q(s)\,[2s^{-1}+3s^{-2}+s^{-3}] \quad ...(ii)$$

or, $$Q(s) = R(s) - Q(s)\,[2s^{-1}+3s^{-2}+s^{-3}] \quad ...(iii)$$

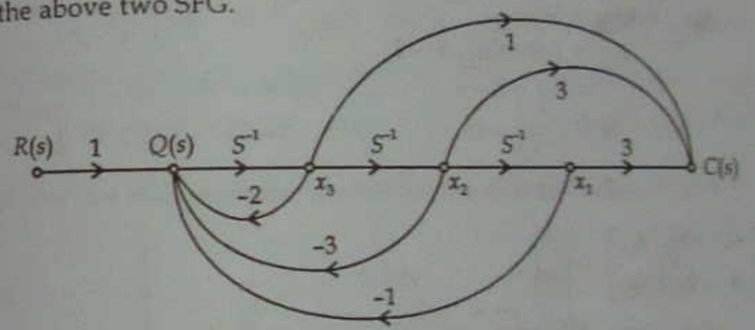
Equation (iii) can be depicted by signal flow diagram as



Equation (ii) can be depicted by SFD as



Combining the above two SFG.



The output of each integrator is a state variable, then we have

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = r(t) - x_1 - 3x_2 - 2x_3$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -3 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix} r(t)$$

$$C(t) = 3x_1 + 3x_2 + x_3$$

$$C(t) = [3 \quad 3 \quad 1]x$$

The above equations are controllable canonical form.

Example 12.84. For the state matrix $A = \begin{bmatrix}0 & 1\\ -8 & -6\end{bmatrix}$, find e^{At} by any two methods. *(UPTU 2002-2003)*

Solution : First method : Use of diagonal

$$A = \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix}$$

$$\lambda I - A = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix} = \begin{bmatrix} \lambda & -1 \\ 8 & \lambda + 6 \end{bmatrix}$$

$$|\lambda I - A| = \lambda^2 + 6\lambda + 8 = 0 \quad \text{...(i)}$$

$$(\lambda + 2)(\lambda + 4) = 0$$

$$\lambda_1 = -2, \lambda_2 = -4$$

let the eigen vector $p_1 = \begin{bmatrix} p_{11} \\ p_{21} \end{bmatrix}$ & $p_2 = \begin{bmatrix} p_{12} \\ p_{22} \end{bmatrix}$

Put $\lambda = -2$ in equation (*i*)

$$\begin{bmatrix} -2 & -1 \\ 8 & 4 \end{bmatrix} \begin{bmatrix} p_{11} \\ p_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-2p_{11} - p_{21} = 0$$

$$8p_{11} + 4p_{21} = 0$$

let $p_{11} = 1$ then $p_{21} = -2$

$$\therefore \quad p_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Put $\lambda = -4$

$$\begin{bmatrix} -4 & -1 \\ 8 & 2 \end{bmatrix} \begin{bmatrix} p_{12} \\ p_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-4p_{12} - p_{22} = 0$$

$$8p_{12} + 2p_{22} = 0$$

let $p_{12} = 1$ then $p_{22} = -4$

$$p_2 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

$$M = [p_1\ p_2] = \begin{bmatrix} 1 & 1 \\ -2 & -4 \end{bmatrix}$$

$$\therefore \quad M^{-1} = \begin{bmatrix} 2 & 0.5 \\ -1 & -0.5 \end{bmatrix} \quad \& \quad \wedge = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}$$

$$e^{At} = Me^{\wedge t} M^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-4t} \end{bmatrix} \begin{bmatrix} 2 & 0.5 \\ -1 & -0.5 \end{bmatrix}$$

$$\therefore \quad e^{At} = \begin{bmatrix} 2e^{-2t} - e^{-4t} & 0.5e^{-2t} - 0.5e^{-4t} \\ -4e^{-2t} + 4e^{-4t} & -e^{-2t} + 2e^{-4t} \end{bmatrix} \quad \textbf{Ans.}$$

SECOND METHOD : Series summation method.

$$A = \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -8 & -6 \\ 48 & 28 \end{bmatrix},$$

$$A^3 = \begin{bmatrix} 48 & 28 \\ -224 & -120 \end{bmatrix},$$

$$A^4 = \begin{bmatrix} -224 & -120 \\ 960 & 496 \end{bmatrix}$$

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \ldots.$$

$$e^{At} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix} t + \frac{t^2}{2} \begin{bmatrix} -8 & -6 \\ 48 & 28 \end{bmatrix} + \frac{t^3}{6} \begin{bmatrix} 48 & 28 \\ -224 & -120 \end{bmatrix} + \frac{t^4}{24} \begin{bmatrix} -224 & -120 \\ 960 & 496 \end{bmatrix} \ldots$$

$$e^{At} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & t \\ -8t & -6t \end{bmatrix} + \begin{bmatrix} -4t^2 & -3t^2 \\ 24t^2 & 14t^2 \end{bmatrix} + \begin{bmatrix} 8t^3 & 4.67t^3 \\ -37.3t^3 & -20t^3 \end{bmatrix} + \begin{bmatrix} -9.33t^4 & -5t^4 \\ 40t^4 & 20.67t^4 \end{bmatrix} \ldots$$

$$\therefore \quad e^{At} = \begin{bmatrix} t - 4t^2 + 8t^3 - 9.33t^4 \ldots\ldots & t - 3t^2 + 4.67t^3 - 5t^4 \ldots \\ -8t + 24t^2 - 37.3t^3 + 40t^4 \ldots & 1 - 6t + 14t^2 - 20t^3 + 20.67t^4 \ldots \end{bmatrix} \quad \textbf{Ans.}$$

Example 12.85. Derive the state variable model for the system shown below

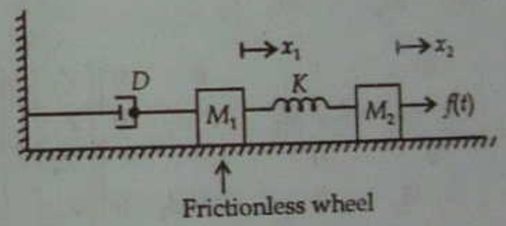


Fig. 12.87.

(UPTU 2002-2003)

Solution : Mechanical equivalent diagram

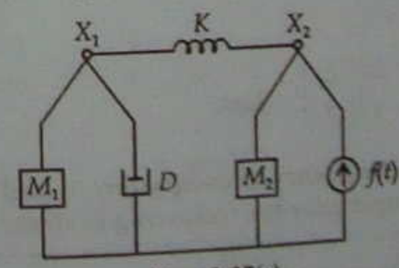


Fig. 12.87(a)

at node (*i*)

$$M_1\ddot{x}_1 + D\dot{x}_1 + k(x_1 - x_2) = 0$$

at node (*ii*)

$$f(t) = M_2\ddot{x}_2 + k(x_2 - x_1)$$

or, $M_1\ddot{x}_1 + D\dot{x}_1 + kx_1 - kx_2 = 0$

$$f(t) = M_2\ddot{x}_2 + kx_2 - kx_1$$

Select the state variable as

$$x_1 = z_1 \qquad x_2 = z_3 \qquad \text{...(i)}$$

$$\dot{x}_1 = \dot{z}_1 = z_2 \qquad \dot{x}_2 = \dot{z}_3 = z_4 \qquad \text{...(ii)}$$

$$\ddot{x}_1 = \dot{z}_2 \qquad \ddot{x}_2 = \dot{z}_4$$

$$f(t) = M_2\ddot{x}_2 + kx_2 - kx_1$$

$$= M_2\dot{z}_4 + kz_3 - kz_1$$

$$\therefore \quad \dot{Z}_4 = \frac{f(t)}{M_2} - \frac{k}{M_2}Z_3 - \frac{k}{M_2}Z_1$$

from equation (*i*)

$$\dot{Z}_2 = -\frac{k}{M_1}Z_1 - \frac{D}{M_1}Z_2 + \frac{k}{M_1}Z_3 \qquad \text{...(iii)}$$

state equations are

$$\dot{Z}_1 = Z_2 \qquad \text{...(iv)}$$

$$\dot{Z}_2 = -\frac{k}{M_1}Z_1 - \frac{D}{M_1}Z_2 + \frac{k}{M_1}Z_3$$

$$\dot{Z}_3 = Z_4$$

$$\dot{Z}_4 = \frac{f(t)}{M_2} - \frac{k}{M_2}Z_3 - \frac{k}{M_2}Z_1$$

$$\begin{bmatrix}\dot{Z}_1\\ \dot{Z}_2\\ \dot{Z}_3\\ \dot{Z}_4\end{bmatrix} = \begin{bmatrix}0 & 1 & 0 & 0\\ -k/M_1 & -D/M_1 & k/M_1 & 0\\ 0 & 0 & 0 & 1\\ -k/M_2 & 0 & -k/M_2 & 0\end{bmatrix}\begin{bmatrix}Z_1\\ Z_2\\ Z_3\\ Z_4\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 0\\ 1/M_2\end{bmatrix} f(t) \qquad \text{...(v)}$$

Select the displacement of mass M_1 & M_2 as output variables.

$$y_1 = x_1 = z_1$$

$$y_2 = x_2 = z_3$$

$$\begin{bmatrix}y_1\\ y_2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}Z_1\\ Z_3\end{bmatrix} \quad \text{Ans.} \qquad \text{...(vi)}$$

Example 12.86. Define natural frequency of oscillation ω_n and damping ratio of second order system. Find the nature of damping for the following systems. *(UPTU 2002-2003)*

(*i*) $R(s) \leftarrow \boxed{\dfrac{12}{S^2+8s+12}} \rightarrow C(s)$

(*ii*) $R(s) \leftarrow \boxed{\dfrac{16}{S^2+8s+16}} \rightarrow C(s)$

(*iii*) $R(s) \leftarrow \boxed{\dfrac{20}{S^2+8s+20}} \rightarrow C(s)$

Solution : Each and every system takes finite time to reach its steady state and during this period system oscillates or increases exponentially. The type & location of closed loop poles decide the behaviour of the system. Closed loop poles are depends upon the selection of parameters of the system. Every system has a tendency to oppose the oscillatory behaviour of the system which is called as damping. This damping is measured by a factor or ratio called damping ratio. This damping ratio is denoted by ξ (zeta). In some system it is low, such system oscillates but slowly i.e. with damped frequency. If it is zero, the system will oscillate with maximum frequency. As there is no opposition from system,system naturally & freely oscillates under such condition. Hence this frequency of oscillations at $\xi = 0$ called natural frequency of oscillation & denoted by ω_n red/sec.

(*i*) $R(s) \leftarrow \boxed{\dfrac{12}{S^2+8s+12}} \rightarrow C(s)$

$$\frac{C(s)}{R(s)} = \frac{12}{s^2+8s+12}$$

Compare with $\qquad \dfrac{C(s)}{R(s)} = \dfrac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} \qquad \text{...(i)}$

$$\omega_n^2 = 12 \quad \therefore \omega_n = 3.464 \text{ rad/sec.}$$

$$2\zeta\omega_n = 8 \quad \therefore \zeta = 1.15$$

Since, $\zeta > 1$, the system is overdamped.

(*ii*) $R(s) \leftarrow \boxed{\dfrac{16}{S^2+8s+16}} \rightarrow C(s)$

$$\frac{C(s)}{R(s)} = \frac{16}{s^2+8s+16}$$

Compare with equation (*i*)

$$\omega_n^2 = 16 \qquad \omega_n = 4 \text{ rad/sec.}$$

$$2\zeta\omega_n = 8 \qquad \zeta = 1$$

Since, $\zeta = 1$, the system is critically damped.

(*iii*) $R(s) \leftarrow \boxed{\dfrac{20}{S^2+8s+20}} \rightarrow C(s)$

$$\frac{C(s)}{R(s)} = \frac{20}{s^2+8s+20}$$

$$\omega_n^2 = 20 \quad \omega_n = 4.472 \text{ rad/sec.}$$

$$2\zeta\omega_n = 8 \quad \zeta = 0.894$$

Since, $\zeta < 1$, the system is underdamped.

Example 12.87. Find the damping factor ζ, natural frequency ω_n, peak time Tp and percentage overshoot for the system with

$$G(s) = \frac{1}{9s^2+2s+58}$$

Example 12.88. A closed loop servo system is represented by the differential equation

$$\frac{d^2C}{dt^2} + 8\frac{dC}{dt} = 64e$$

where C is the displacement of the output shaft, r is the displacement of the input shaft and $e = r - c$. Determine : (i) damping ratio (ii) damped natural frequency (iii) % M_p. (UPTU 2003-2004)

Solution :

$$\frac{d^2C}{dt^2} + 8\frac{dC}{dt} = 64\,(r - C)$$

$$s^2C(s) + 8sC(s) = 64\,[R(s) - C(s)]$$

$$C(s)\,[s^2 + 8s + 64] = 64\,R(s)$$

$$\frac{C(s)}{R(s)} = \frac{64}{S^2 + 8s + 64}$$

Compare with $$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$\omega_n 2 = 64$ $\qquad \omega_n = 8$ rad/sec.

$2\zeta\omega_n = 8$ $\qquad \therefore \zeta = 0.5$

$$\% M_p = 100\, e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$= 100\, e^{-\frac{\pi 0.5}{\sqrt{1-0.5^2}}}$$

$M_p = 16.3\ \%$ **Ans.**

Example 12.89. Sketch the root locus for the system having

$$G(s) = \frac{k}{(s^2 + 2s + 2)(s^2 + 2s + 5)}$$

Solution :

Step 1 : Root locus is symmetrical about real axis.

Step 2 : No. of poles = 4 at $s = -1 \pm j$ and $s = -1 \pm j\,2$ No. of zeros = 0

Step 3 : No. of root locii = No. of poles ($\because p > z$) = 4

Step 4 : Centroid of asymptotes

$$\sigma_A = \frac{-1 + j - 1 - j - 1 + j2 - 1 - j2}{4 - 0}$$

$$\sigma_A = -1$$

Step 5 : Angle of asymptotes

$$\phi = \frac{2k + 1}{p - z}180°$$

For $k = 0, 1, 2, 3$

$\phi_1 = 45°, \phi_2 = 135°, \phi_3 = 225°, \phi_4 = 315°$

Step 6 : Calculation of Breakaway point

The characteristic equation : $1 + G(s)\,H(s) = 0$

$$1 + \frac{k}{(s^2 + 2s + 2)(s^2 + 2s + 5)} = 0$$

$$k = -(s^4 + 4s^3 + 11s^2 + 14s + 10)$$

$$\frac{dk}{ds} = (4s^3 + 12s^2 + 22s + 14).$$

or, $$\frac{dk}{ds} = 2s^3 + 6s^2 + 11s + 7$$

Note : If the poles (and zeros) are symmetical *w.r.t* some vertical line (*i.e.* $s = -1$ line in this case) the equation of vertical line is one of the root of $\frac{dk}{ds} = 0$ (here $s = -1$ is one of the root of $\frac{dk}{ds} = 0$)

$$\therefore \quad \frac{dk}{ds} = (s + 1)(2s^2 + 4s + 7)$$

$\therefore$ Roots of $\frac{dk}{ds}$ are

$$s = -1, \qquad -1 \pm j\,1.581$$

$s = -1$ lies on real axis, this does not belong to root loucs so this is not the breakaway point.

At $s = -1 \pm j\,1.58$ angle criterion is satisfied hence it is the breakaway point.

Step 7: Real axis is not the part of root locus.

Step 8: Point of intersection with imaginary axis.

The characteristic equation is

$$s^4 + 4s^3 + 11s^2 + 14s + (10 + k) = 0$$

s^4	1	11	$10 + k$
s^3	4	14	
s^2	7.5	$10 + k$	
s^1	$\frac{65 - 4k}{7.5}$	0	
s^0	$10 + k$		

For roots on imaginary axis

$$65 - 4k = 0$$

or $$k = 16.25$$

Auxiliary equation

$$7.5s^2 + (10 + k) = 0$$

$$7.5s^2 + (10 + 16.25) = 0$$

$\therefore$ $$s = \pm j\,1.87$$

Step 9 : Calculation of angle of departure

$$\phi_d|_{s=-1+j2} = -90°$$

$\because$ $180 - (90° + 90° + 90°) = -90°$

$$\phi_d|_{s=-1+j} = 180 - (90° + 90° + 90°) = +90°$$

$\therefore$ angle of departure $|_{s=-1\pm j2} = \mp 90°$

angle of departure $|_{s=-1\pm j1} = \pm 90°$

The complete root locus is shown in fig. 12.88.

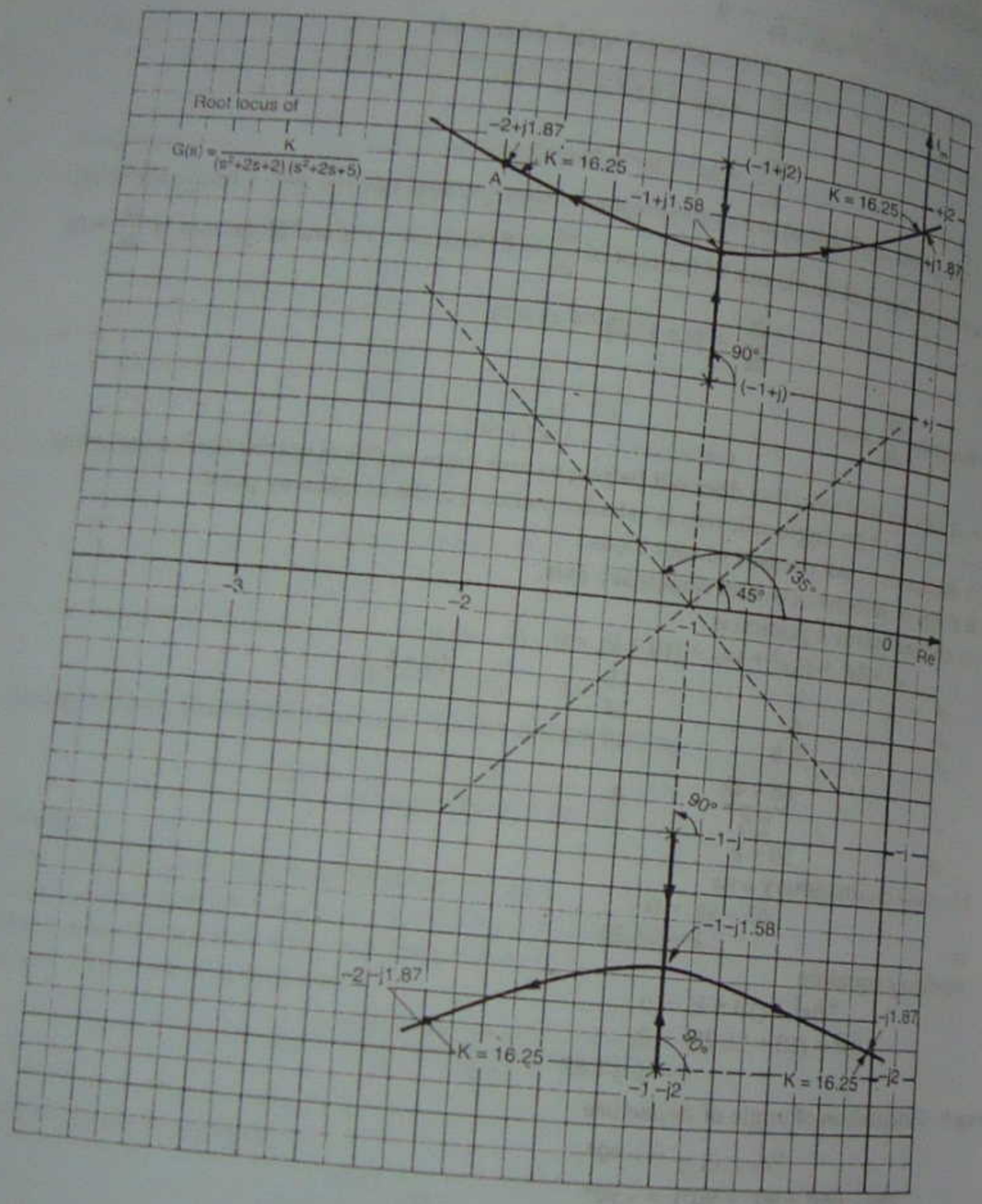


Fig. 12.88.

Example 12.90. Obtain the closed loop transfer function (in Example 12.89) such that the system exhibits oscillatory response.

Solution : Closed loop transfer for any gain k is

$$\frac{C(s)}{R(s)} = \frac{k\,[\text{closed loop zero factors}]}{[\text{closed loop pole factors}]}$$

The number of closed loop pole factors is same as the no. of root locus branches *i.e.* each closed loop pole factor corresponds to each root locus branch.

Closed loop zeros are same as the open loop zeros. There are four root locus branches where k is varying from zero to infinity. So, there must be four closed loop poles. Here we have to determine the closed loop transfer function for oscillatory response *i.e.* for that value of k for which roots lie on imaginary axis *i.e.* $k = 16.25$. Now identify four points on root loucs branches where $k = 16.25$ (either by trial & error or logically). These will be the four closed loop poles.

From the root locus shown in fig. 12.88 it is clear that the two closed loop poles on the two root locus branches for $k = 16.25$ are $S = +j1.87$ & $s = -j\,1.87$. Since root locus is symmetrical *w.r.t* vertical line $s = -1$, the remaining two closed loop poles on other two root locus branches can be obtained by projecting points $s = \pm j\,1.87$ on the other two root locus branches as shown at point A & B. Also by magnitude criterion for $k = 16.25$ the corresponding closed loop poles are $s = -2 \pm j\,1.87$.

Since, there are no open loop zeros, there will be no closed loop zeros.

Hence required closed loop transfer function for $k = 16.25$ is

$$\frac{C(s)}{R(s)} = \frac{16.25}{(s - j1.87)(s + j1.87)(s + 2 - j1.87)(s + 2 + j1.87)}$$

Example 12.91. For a rotational system shown in fig 12.89 draw electrical analogous circuit based on $T - i$ analogy.

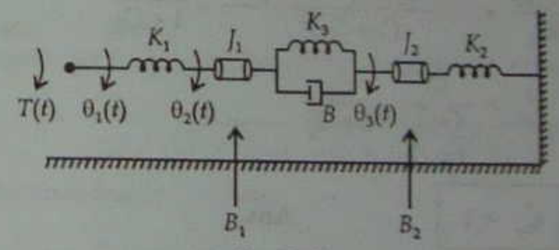


Fig. 12.89.

Solution : Draw the mechanical equivalent circuit

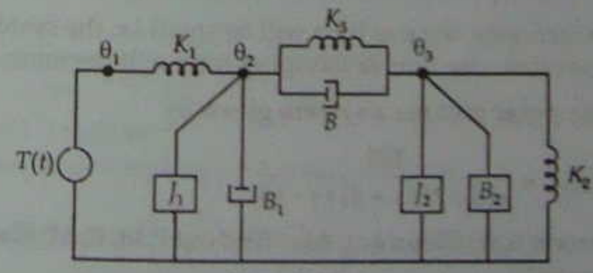


Fig. 12.89 (a)

Now electrical analogous circuit can be drawn as

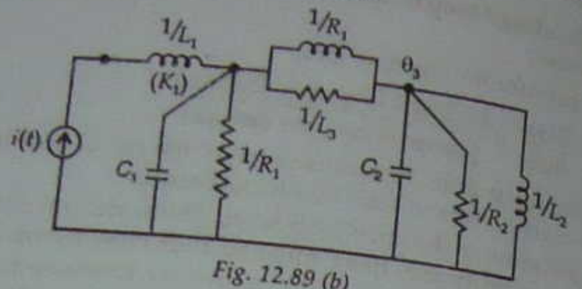


Fig. 12.89 (b)

Example 12.92. For the system shown in fig 12.90. find $s_{G_1}^T$.

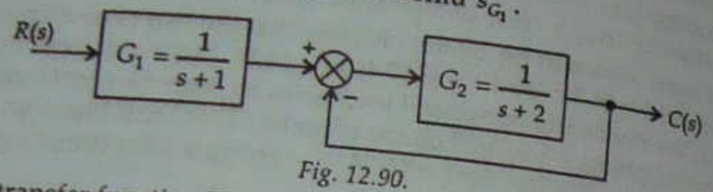


Fig. 12.90.

Solution : Overall transfer function $T(s)$ will be

$$T(s) = \frac{G_1 G_2}{1 + G_2}$$

$$s_{G_1}^T = \frac{G_1}{T} \cdot \frac{\partial T(s)}{\partial G_1}$$

$$\frac{\partial T(s)}{\partial G_1} = \frac{(1+G_2)G_2 - G_1 G_2 . 0}{(1+G_2)^2} = \frac{G_2}{1+G_2}$$

$$s_{G_1}^T = \frac{G_1}{\frac{G_1 G_2}{1+G_2}} \cdot \frac{G_2}{1+G_2}$$

$$\therefore \quad \boxed{s_{G_1}^T = 1} \qquad \textbf{Ans.}$$

Example 12.93. If bandwidth increases, then what will be the effect on damping ratio and rise time.

Solution : If bandwidth increases, the rise time will be small *i.e.* the system response will be fast. So, if bandwidth increases then rise time & damping ratio will decrease.

Example 12.94. Draw the polar plot for a system given by

$$G(s)\,H(s) = \frac{100}{s(s+2)(s+4)(s+8)}$$

Find whether the system is stable on not. Also find out P.M, G.M, phase cross-over frequency and gain crossover frequency.

Solution : Put $s = j\omega$

$$G(j\omega)\,H(j\omega) = \frac{100}{j\omega(2+j\omega)(4+j\omega)(8+j\omega)} \qquad ...(i)$$

$$|G(jw)\,H(jw)| = \frac{100}{\omega\sqrt{2^2+\omega^2}\sqrt{4^2+\omega^2}\sqrt{8^2+\omega^2}} \qquad ...(ii)$$

$$\angle G(j\omega)\,H(j\omega) = -90° - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{4} - \tan^{-1}\frac{\omega}{8} \qquad ...(iii)$$

A rough sketch of polar plot is shown in fig. 12.91.

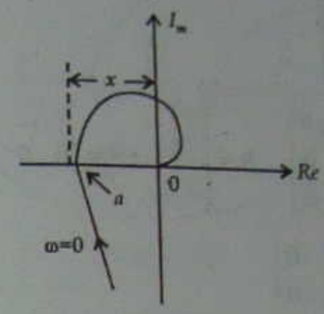


Fig. 12.91.

For stability determine the value of x *i.e.* distance from origin to 'a' the point 'a' is called phase cross-over point and frequency at this point is called phase cross-over frequency.

At point 'a' $\phi = -180°$

$$-180° = -90° - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{4} - \tan^{-1}\frac{\omega}{8}$$

$$90° = \tan^{-1}\frac{\omega}{2} + \tan^{-1}\frac{\omega}{4} + \tan^{-1}\frac{\omega}{8}$$

solve for ω.

Hence $\omega \approx 2.2$ rad/sec. **Ans.**

This is the phase cross-over frequency. Now find magnitude at this frequency.

$$|G(j\omega)\,H(j\omega)|_{\omega=2.2} = \frac{100}{(2.2)^2\sqrt{2^2+(2.2)^2}\sqrt{4^2+2.2^2}\sqrt{8^2+2.2^2}} = 0.403$$

Hence, $x < 1$, the system is stable **Ans.**

$$\text{G.M.} = \frac{1}{x} = \frac{1}{0.403} = 2.47 \qquad \textbf{Ans.}$$

Now calculate gain cross-over frequency. At this frequency $|GH| = 1$

$$1 = \frac{100}{\omega\sqrt{2^2+\omega^2}\sqrt{4^2+\omega^2}\sqrt{8^2+\omega^2}}$$

$$\omega \approx 1.2 \text{ rad/sec.}$$

This is the gain cross-over frequency = 1.2 rad/sec. **Ans.**

$$\therefore \quad \phi = -90° - \tan^{-1}\left(\frac{1.2}{2}\right) - \tan^{-1}\left(\frac{1.2}{4}\right) - \tan^{-1}\left(\frac{1.2}{8}\right)$$

$$= -146.18°$$

$$\text{P.M} = 180° + (-146.18°) = 33.82° \quad \textbf{Ans.}$$

Example 12.95. Obtain the transfer function of the given state equation

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ -1 & -1 & 0\\ 1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 1\\ 0\end{bmatrix}u$$

$$y = \begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Solution : Here, $A = \begin{bmatrix}0 & 1 & 0\\ -1 & -1 & 0\\ 1 & 0 & 0\end{bmatrix}$, $B = \begin{bmatrix}0\\ 1\\ 0\end{bmatrix}$, $C^T = \begin{bmatrix}0 & 0 & 1\end{bmatrix}$

$$SI - A = \begin{bmatrix}s & -1 & 0\\ 1 & s+1 & 0\\ -1 & 0 & s\end{bmatrix}$$

$$|SI - A| = s^3 + s + s$$

$$\text{Adj}\,(SI - A) = \begin{bmatrix}s^2+s & s & 0\\ -s & s^2 & 0\\ s+1 & 1 & s^2+s+1\end{bmatrix}$$

$$[SI - A]^{-1} = \frac{\text{Adj}\,(sI - A)}{|sI - A|} = \begin{bmatrix}\frac{s+1}{s^2+s+1} & \frac{1}{s^2+s+1} & 0\\ -\frac{1}{s^2+s+1} & \frac{s}{s^2+s+1} & 0\\ \frac{s+1}{s(s^2+s+1)} & \frac{1}{s(s^2+s+1)} & \frac{1}{s}\end{bmatrix}$$

$$C^T[SI - A]^{-1}B = \begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}\frac{s+1}{s^2+s+1} & \frac{1}{s^2+s+1} & 0\\ -\frac{1}{s^2+s+1} & \frac{s}{s^2+s+1} & 0\\ \frac{s+1}{s(s^2+s+1)} & \frac{1}{s(s^2+s+1)} & \frac{1}{s}\end{bmatrix}\begin{bmatrix}0\\ 1\\ 0\end{bmatrix} = \frac{1}{s^3+s^2+1}$$

This is the required transfer function.

Solution :

$$C(s) = Q(s)\,[10.4s^{-1} + 47s^{-2} + 160s^{-3}]$$

$$R(s) = Q(s)\,[1 + 14s^{-1} + 56s^{-2} + 160s^{-3}]$$

$$R(s) = Q(s) + Q(s)\,[14s^{-1} + 56s^{-1} + 160s^{-3}]$$

$$Q(s) = R(s) - Q(s)\,[14s^{-1} + 56s^{-2} + 160s^{-3}]$$

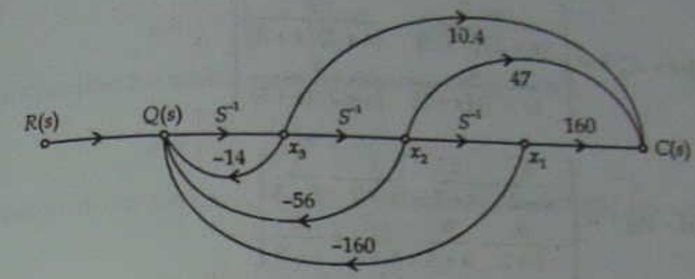


Fig. 12.92.

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = r(t) - 160x_1 - 56x_2 - 14x_3$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -160 & -56 & -14\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}r(t)$$

$$C(t) = 160x_1 + 47x_2 + 10.4x_3$$

$$= \begin{bmatrix}160 & 47 & 10.4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

The above equations are in controllable canonical form.

Example 12.97. Obtain the time response of the following system

$$\dot{x} = \begin{bmatrix}0 & 1\\ -6 & -5\end{bmatrix}\dot{x} + \begin{bmatrix}0\\ 1\end{bmatrix}u$$

and $$y = \begin{bmatrix}1 & 0\end{bmatrix}x$$

Where $u(t)$ is a unit step input and initial conditions $x_1(0) = 0$ and $x_2(0) = 0$

(R.M.L University Faizabad 2000 Network & Systems analysis)

Solution : Here, $A = \begin{bmatrix}0 & 1\\ -6 & -5\end{bmatrix}$, $B = \begin{bmatrix}0\\ 1\end{bmatrix}$

$$SI - A = \begin{bmatrix}s & 0\\ 0 & s\end{bmatrix} = \begin{bmatrix}s & -1\\ 6 & s+5\end{bmatrix}$$

$$sI - A = s^2 + 5s + 6 = (s + 2)\,(s + 3)$$

$$\text{cofactors of } (sI - A) = \begin{bmatrix}s+5 & -6\\ 1 & 5\end{bmatrix}$$

$$\text{Adj of } (sI - A) = \begin{bmatrix} s+5 & 1 \\ 6 & 5 \end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix} \dfrac{s+5}{(s+2)(s+3)} & \dfrac{1}{(s+2)(s+3)} \\ -\dfrac{6}{(s+2)(s+3)} & \dfrac{s}{(s+2)(s+3)} \end{bmatrix}$$

or,
$$[sI - A]^{-1} = \begin{bmatrix} \dfrac{3}{s+2} - \dfrac{2}{s+3} & \dfrac{1}{s+2} - \dfrac{1}{s+3} \\ -\dfrac{6}{s+2} + \dfrac{6}{s+3} & \dfrac{-2}{s+2} + \dfrac{3}{s+3} \end{bmatrix}$$

$$\mathcal{L}^{-1}[sI - A]^{-1} = \begin{bmatrix} 3e^{-2t} - 2e^{-3t} & e^{-2t} - e^{-3t} \\ -6e^{-2t} + 6e^{-3t} & -2e^{-2t} + 3e^{-3t} \end{bmatrix}$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} Bu(\tau)\, d\tau$$

$$x(t) = \begin{bmatrix} 3e^{-2t} - 2e^{-3t} & e^{-2t} - e^{-3t} \\ -6e^{-2t} + 6e^{-3t} & -2e^{-2t} + 3e^{-3t} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$+ \int_0^t \begin{bmatrix} 3e^{-2(t-\tau)} - 2e^{-3(t-\tau)} \\ -6e^{-2(t-\tau)} + 6e^{-3(t+\tau)} \end{bmatrix} \quad \begin{bmatrix} e^{-2(t-\tau)} - e^{-3(t-\tau)} \\ -2e^{-2(t-\tau)} + 3e^{-3(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} 1\, d\tau$$

$$x(t) = \int_0^t \begin{matrix} e^{-2(t-\tau)} - e^{-3(t-\tau)} \\ -2e^{-2(t-\tau)} + 3e^{-3(t-\tau)} \end{matrix} d\tau$$

$$x_1(t) = \int_0^t e^{-2(t-\tau)} - e^{-3(t-\tau)}\, d\tau$$

put $\quad -3(t-\tau) = x, \quad \tau = 0, \quad x = -3t$

or $\quad d\tau = \dfrac{1}{3}dx, \quad \tau = t, \quad x = 0$

$$\frac{1}{3}\int_{-3t}^{0} e^x\, dx = \frac{1}{3} - \frac{1}{3}e^{-3t}$$

Similarly calculate $\int_0^t e^{-2(t-\tau)}\, d\tau$

$\therefore$
$$x_1(t) = \frac{1}{6} - \frac{1}{2}e^{-2t} - \frac{1}{3}e^{-3t} \quad \textbf{Ans.}$$

and
$$x_2(t) = e^{-2t} - e^{-3t} \quad \textbf{Ans.}$$

output response

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \dfrac{1}{6} - \dfrac{1}{2}e^{-2t} - \dfrac{1}{3}e^{-3t} \\ e^{-2t} - e^{-3t} \end{bmatrix}$$

$$y = \frac{1}{6} - \frac{1}{2}e^{-2t} - \frac{1}{3}e^{-3t} \quad \textbf{Ans.}$$

Example 12.98. Consider the vector matrix differential equation as

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x$$

obtain the state transition matrix. *(R.M.L University Faizabad 2000 Network & systems analysis)*

Solution :

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$|sI - A| = (s+1)(s+2)$$

$$\phi(s) = (sI - A)^{-1}$$

$$= \begin{bmatrix} \dfrac{s+3}{(s+1)(s+2)} & \dfrac{1}{(s+1)(s+2)} \\ \dfrac{2}{(s+1)(s+2)} & \dfrac{s}{(s+1)(s+2)} \end{bmatrix}$$

$$\phi(t) = \mathcal{L}^{-1}\phi(s) = \mathcal{L}^{-1}[sI - A]^{-1}$$

$$= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \quad \textbf{Ans.}$$

Example 12.99. Determine the poles and zeros of the following system.

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r$$

$$C = \begin{bmatrix} -17 & -5 \end{bmatrix} x + [1]\, r \qquad \textit{(IES, 1996)}$$

Solution : The state equations are

$$\dot{x}(t) = Ax(t) + Bu(t) \quad \ldots(i)$$

$$y(t) = Cx(t) + Du(t) \quad \ldots(ii)$$

laplace transform of above equations

$$sx(s) = AX(s) + Bu(s) \quad \ldots(iii)$$

or
$$[sI - A]x(s) = Bu(s) \quad \ldots(iv)$$

$$x(s) = [SI - A]^{-1} Bu(s) \quad \ldots(v)$$

$$Y(s) = Cx(s) + Du(s)$$

put the value of $x(s)$ from (v)

$$Y(s) = C[sI - A]^{-1} Bu(s) + Du(s)$$

or, $$\frac{Y(s)}{u(s)} = C[sI - A]^{-1} B + D = C\frac{\text{Adj}(sI - A)}{|sI - A|}B + D$$

or, $$\frac{Y(s)}{u(s)} = \frac{C\,\text{Adj}(sI - A)\,B + |sI - A|\,D}{|sI - A|}$$

The denominator $|sI - A|$ is the characteristic equation. Hence we can calculate the poles —(vi)

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 \\ 20 & s+9 \end{bmatrix}$$

$$|sI - A| = s(s + 9) + 20$$

$$= s^2 + 9s + 20 = 0$$

or $$(s + 5)(s + 4) = 0$$

$$s = -5, -4$$

Hence, the poles are $s = -4, -5$

$$sI - A = \begin{bmatrix} s & -1 \\ 20 & s+9 \end{bmatrix}$$

cofactors of $(sI - A)$ can be calculated as

$$(-1)^{1+1}(s + 9) = s + 9$$

$$(-1)^{1+2}(20) = -20$$

$$(-1)^{2+1}(-1) = 1$$

$$(-1)^{2+2}(s) = s$$

$\therefore$ cofactor of $$(sI - A) = \begin{bmatrix} s+9 & -20 \\ 1 & s \end{bmatrix}$$

Adj of $$(sI - A) = \begin{bmatrix} s+9 & 1 \\ -20 & s \end{bmatrix}$$

Now equate the equation (vi) to zero we get

$$C\,\text{Adj}(sI - A)B + |sI - A|\,D = 0$$

$$\begin{bmatrix} -17 & -5 \end{bmatrix}\begin{bmatrix} s+9 & 1 \\ -20 & s \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} + (s^2 + 9s + 20)\,1 = 0$$

$$-17 - 5s + s^2 + 9s + 20 = 0$$

$$s^2 + 4s + 3 = 0$$

$$(s + 3)(s + 1) = 0$$

$\therefore$ zeros are at $s = -1, -3$ **Ans.**

Note : In this Example value of D is given.

Example 12.100. A close loop transfer function of a unity feedback control system is

$$\frac{C(s)}{R(s)} = \frac{20s^2}{(s+1)(s+3)(s+5)}$$

Determine the response of the system when the excitation is applied to the input terminal is

$$1 + 2t + \frac{3t^2}{2}$$

(IES 1999)

Solution : Given : $$r(t) = 1 + 2t + \frac{3t^2}{2}$$

Laplace transform of above equation

$$R(s) = \frac{1}{s} + \frac{2}{s^2} + \frac{3}{s^3} = \frac{s^2 + 2s + 3}{s^3}$$

Also, given

$$\frac{C(s)}{R(s)} = \frac{20s^2}{(s+1)(s+3)(s+5)}$$

$\therefore$ $$C(s) = R(s)\,\frac{20s^2}{(s+1)(s+3)(s+5)}$$

$$= \frac{s^2 + 2s + 3}{s^3} \cdot \frac{20s^2}{(s+1)(s+3)(s+5)} = \frac{20(s^2 + 2s + 3)}{(s+1)(s+3)(s+5)}$$

After partial fraction above equation can be written as

$$C(s) = \frac{5}{s+1} - \frac{30}{s+3} + \frac{45}{s+5}$$

Inverse Laplace of above equation gives

$$C(t) = 5e^{-t} - 30e^{-3t} + 45e^{-5t}$$ **Ans.**

Example 12.101. A unity feedback control system has a forward transfer function $\frac{25}{s(s+6)}$. Find the resonance peak and the corresponding frequency for the closed loop frequency response. Derive the formula used. (IES 1997)

Solution : $$\frac{C(s)}{R(s)} = \frac{\frac{25}{s(s+6)}}{1 + \frac{25}{s(s+6)}} = \frac{25}{s^2 + 6s + 25}$$

Compare with $$\frac{C(s)}{R(s)} = \frac{w_n^2}{s^2 + 2\xi w_n s + w_n^2}$$

$w_n^2 = 25$ $\quad \therefore w_n = 5$ rad/sec.

$2\xi w_n = 6$ $\quad \therefore \xi = 0.6$

$$w_r = w_n\sqrt{1 - 2\xi^2} = 5\sqrt{1 - 2(0.6)^2} = 2.65 \text{ rad/sec. Ans.}$$

$$M_r = \frac{1}{2\xi\sqrt{1 - \xi^2}} = \frac{1}{2 \times 0.6\sqrt{1 - (0.6)^2}} = 1.042 \text{ Ans.}$$

For derivation refer article no. 4.15.

Example 12.102. Draw the root locus diagram for the following control system and calculate the breakin and breakaway points. (IES 1992)

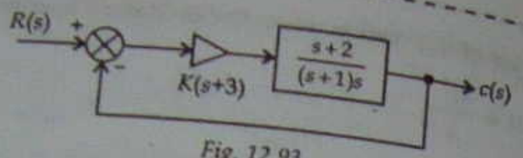


Fig. 12.93.

Solution : From the Fig. 12.93

$$G(s)\,H(s) = \frac{K(s+2)(s+3)}{s(s+1)}$$

Step 1 : Plot the Poles and Zeros

Poles : $s_1 = 0,\ s_2 = -1$

Zeros : $s_3 = -2,\ s_4 = -3$

Step 2 : No. of poles = 2

No. of zeros = 2

Since, no. of poles = no. of zeros, therefore, there will be no asymptotes.

Step 3 : Characteristic equation $1 + G(s)\,H(s) = 0$

$$1 + \frac{k(s+2)(s+3)}{s(s+1)} = 0$$

$$\therefore \qquad k = -\frac{s^2+s}{s^2+5s+6}$$

For break points

$$\frac{dk}{ds} = \frac{d}{ds}\left[-\frac{s^2+s}{s^2+5s+6}\right] = 0 \qquad \frac{dk}{ds} = 0$$

or $\qquad 2s^2 + 6s + 3 = 0$

$\therefore \qquad s = -0.634,\ -2.366$

As, $s = -0.634$ lies between two poles $s = 0$ and $s = -1$, therefore it is a breakaway point.

$s = -2.366$ lies between two zeros at $s = -2$ and $s = -3$ therefore it is breakin point.

Hence, root locus is a circle shown in Fig. 12.94 with centre at (– 1.5, 0) and passing through breakaway points.

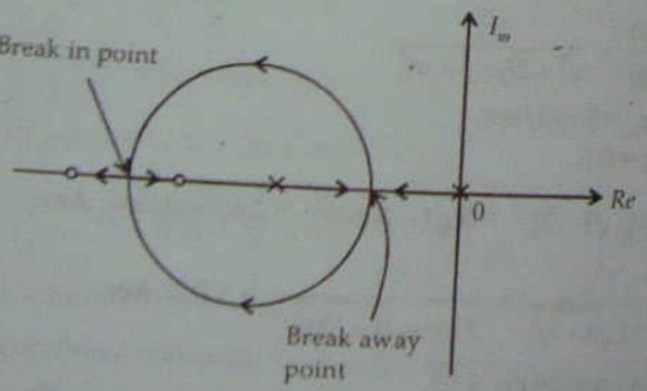


Fig. 12.94

Example 12.103. The system shown in Fig. 12.95 consists of a unity feedback loop containing a minor rate feed-back loop

(i) Without any rate feedback ($b = 0$) determine the damping factor, natural resonant frequency, overshoot of the system to a unit step input and steady state error resulting from a unit ramp input.

(ii) Determine the rate feedback constant which will increase the equivalent damping factor of the system to 0.8. Determine the overshoot to the system in this case to a unit step input and steady state error resulting from a unit ramp input. (IES 1994)

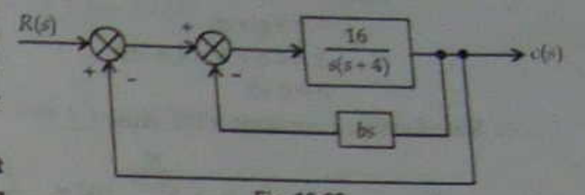


Fig. 12.95.

Solution : (i) When $b = 0$

$$\frac{C(s)}{R(s)} = \frac{\dfrac{16}{s(s+4)}}{1+\dfrac{16}{s(s+4)}.1} = \frac{16}{s^2+4s+16}$$

Compare with $\qquad \dfrac{C(s)}{R(s)} = \dfrac{w_n^2}{s^2+2\xi w_n s + w_n^2}$

$w_n^2 = 16 \qquad \therefore\ w_n = 4$ rad/sec.

$2\xi w_n = 4 \qquad \therefore\ \xi = 0.5$

Resonant frequency $\qquad w_r = w_n\sqrt{1-2\xi^2}$

$$w_r = 4\sqrt{1-2(0.5)^2} = 2.83 \text{ rad/sec. Ans.}$$

$$M_p = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}}\ 100 = e^{-\frac{\pi 0.5}{\sqrt{1-0.5^2}}}\ 100 = 16.32\%\ \textbf{Ans.}$$

Steady state error

$$e_{ss} = \lim_{s\to 0} s.\ \frac{R(s)}{1+G(s)\,H(s)}$$

Here $\qquad H(s) = 1$

$$e_{ss} = \lim_{s\to 0} s.\ \frac{R(s)}{1+G(s)\,H(s)} = \lim_{s\to 0} s.\ \frac{1/s^2}{1+\dfrac{16}{s(s+4)}} = \frac{1}{4}$$

$\therefore \qquad e_{ss} = 0.25$ **Ans.**

(ii) Consider feedback.

$$\frac{C(s)}{R(s)} = \frac{16}{s^2+(4+16b)s+16}$$

Compare with $\qquad \dfrac{C(s)}{R(s)} = \dfrac{w_n^2}{s^2+2\xi w_n s + w_n^2}$

$w_n^2 = 16$ $\quad \therefore \; w_n = 4$ rad/sec.

$2\xi w_n = 4 + 16b$

$\xi = 0.8$ given

$\therefore \quad 2 \times 0.8 \times 4 = 4 + 16b$

$\therefore \quad b = 0.15$

$\therefore$ Rate feedback constant = 0.5 **Ans.**

$$M_p = e^{\frac{-\pi\xi}{\sqrt{1-\xi^2}}} 100 = \frac{-\pi 0.8}{e^{\sqrt{1-(0.8)^2}}} 100$$

$M_p = 1.52\%$ **Ans.**

$$e_{ss} = \lim_{s \to 0} s \frac{1/s^2}{1 + \frac{16}{s^2 + 4s + 16bs}} = 0.75$$

$\therefore \quad e_{ss} = 0.75$ **Ans.**

Example 12.104. Find the transfer function draw the block diagram of the circuit shown in Fig. 12.96.

Solutoin : Draw the transform network

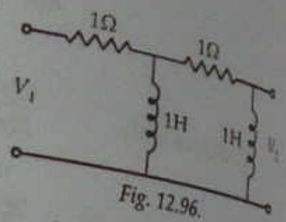


Fig. 12.96.

KCL at node (1)

$$\frac{V_1(s) - V(s)}{1} = \frac{V(s)}{s} + \frac{V(s) - V_2(s)}{1}$$

or $$V(s) = \frac{s}{(2s+1)} [V_1(s) + V_2(s)] \quad ...(1)$$

KCL at note (2)

$$\frac{V(s) - V_2(s)}{1} = \frac{V_2(s)}{s}$$

$$V(s) = V_2(s) \left[\frac{s+1}{s}\right] \quad ...(2)$$

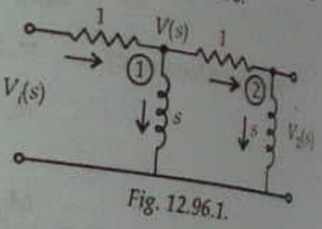


Fig. 12.96.1.

Equate the equations (1) and (2)

$$\frac{s}{2s+1} [V_1(s) + V_2(s)] = V_2(s) \left[\frac{s+1}{s}\right]$$

$$\therefore \quad \frac{V_2(s)}{V_1(s)} = \frac{s^2}{s^2 + 3s + 1}$$

For block diagram

$$I_1(s) = \frac{V_1(s) - V(s)}{1}$$

$$V(s) = s [I_1(s) - I_2(s)]$$

$$I_2(s) = \frac{V(s) - V_2(s)}{1}$$

$$V_2(s) = s I_2(s)$$

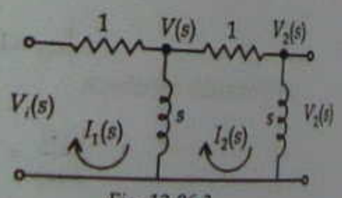


Fig. 12.96.2.

Block diagram for each above equation

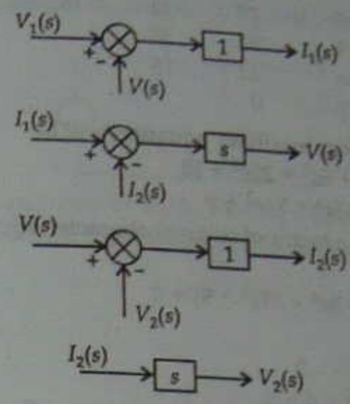


Fig. 12.96.3.

Combine all above diagram.

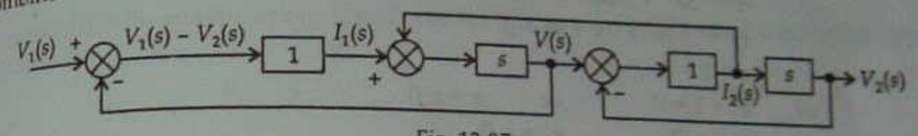


Fig. 12.97.

Example 12.105. Find the number of roots with positive real part, negative real part and zero real part of the following equations

(i) $s^6 + 1 = 0$

(ii) $s^8 + 3s^7 + 5s^6 + 9s^5 + 17s^4 + 33s^3 + 31s^2 + 27s + 18 = 0$

Solution : (i) $\quad s^6 + 1 = 0$

$$s^6 = -1$$

$$s^6 = (\cos \pi + j \sin \pi) \qquad (\because -1 = \cos \pi + j \sin \pi)$$

$$s = (\cos \pi + j \sin \pi)^{1/6}$$

$$= \cos \frac{2n\pi + \pi}{6} + j \sin \frac{2n\pi + \pi}{6}$$

$$= \cos \left(\frac{2n+1}{6}\right)\pi + j \sin \left(\frac{2n+1}{6}\right)\pi$$

for $\quad n = 0, \quad s_1 = \cos 30° + j \sin 30° = +0.866 + j0.5$

$n = 1, \quad s_2 = \cos 90° + j \sin 90° = 0 + j1$

$n = 2, \quad s_3 = \cos 150° + j \sin 150° = -0.866 + j0.5$

$n = 3, \quad s_4 = \cos 210° + j \sin 210° = -0.866 - j0.5$

$n = 4, \quad s_5 = \cos 270° + j \sin 270° = 0 - j1$

$n = 5, \quad s_6 = \cos 330° + j \sin 330° = +0.866 - j0.5$

$\therefore$ no. of roots with positive real part = 2

no. of roots with negative real parts = 2

no. of roots with zero real parts = 2

(ii) $s^8 + 3s^7 + 5s^6 + 9s^5 + 17s^4 + 33s^3 + 31s^2 + 27s + 18 = 0$

s^8	1	5	17	31	18
s^7	3	9	33	27	
s^6	2	6	22	18	
s^5	0	0	0		

Since, one row is zero, therefore auxiliary equation will be

$$A(s) = 2s^6 + 6s^4 + 22s^2 + 18$$

or $A(s) = s^6 + 3s^4 + 11s^2 + 9$

The auxiliary equation is also a factor of original characteristic equation, therefore characteristic equation can also be written as

$$(s^2 + 3s + 2)(s^6 + 3s^4 + 11s^2 + 9) = 0$$

Consider $s^2 + 3s + 2 = 0$

the roots are $s_1 = -1 + j0,\ s_2 = -2$

Now consider $s^6 + 3s^4 + 11s^2 + 9 = 0$

put $s^2 = x$

$\therefore\ x^3 + 3x^2 + 11x + 9 = 0$

$(x + 1)(x^2 + 2x + 9) = 0$

$x + 1 = 0$

or $s^2 + 1 = 0$ or $s = \pm j1$

$\therefore\ s_3 = \pm j1$ $\quad s_4 = -j1$

$x^2 + 2x + 9 = 0$

or $s^4 + 2s^2 + 9 = 0$

or $(s^2 + 2s + 3)(s^2 - 2s + 3) = 0$

$s^2 + 2s + 3 = 0$, the roots will be

$$s_5 = -1 + j\sqrt{2}$$

$$s_6 = -1 - j\sqrt{2}$$

$s^2 - 2s + 3 = 0$, the roots will be

$$s_7 = 1 + j\sqrt{2}$$

$$s_8 = 1 - j\sqrt{2}$$

$\therefore$ Required roots are

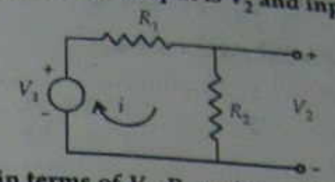
$$s_1 = -1 + j0$$
$$s_2 = -2 + j0$$
$$s_3 = 0 + j1$$
$$s_4 = 0 - j1$$
$$s_5 = -1 + j\sqrt{2}$$
$$s_6 = -1 - j\sqrt{2}$$
$$s_7 = 1 + j\sqrt{2}$$
$$s_8 = 1 - j\sqrt{2}$$

$\therefore$ No. of roots with positive real part = 2

No. of roots with negative real parts = 4

No. of roots with zero real parts = 2.

Example 12.106. For the given network, the output is V_2 and input is V_1



(a) Write an equation for V_2 in terms of V_1, R_1 and R_2 which yeilds an open loop system.

(b) Write an equation for V_2 which yields, a dosed loop system.

Solution: (a) $V_2 = iR_2$

$$i = \frac{V_1}{R_1 + R_2} \quad ...(1)$$

$$V_2 = V_1 \cdot \frac{R_2}{R_1 + R_2} \quad ...(2)$$

$\therefore$ ...(3)

Equation (3) is the required equation. Equation (3) can also be obtained directly by apply voltage divider rule.

(b) $$i = \frac{V_1 - V_2}{R_1} \quad ...(4)$$

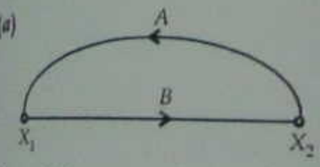
$$V_2 = iR_2 = \left(\frac{V_1 - V_2}{R_1}\right) R_2$$

$$= \frac{R_2}{R_1} V_1 - \frac{R_2}{R_1} V_2 \quad ...(5)$$

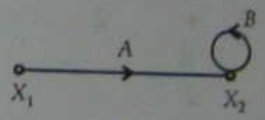
Equation (5) is the required equation. **Ans.**

Example 12.107. Simplify the signal flow graph.

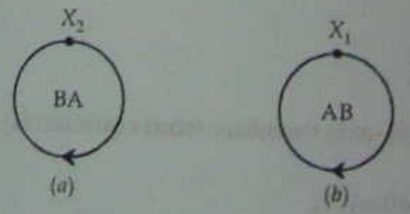
(a)



(b)



Solution : (a)



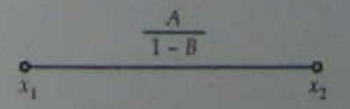
We have $X_2 = BX_1$ and $X_1 = X_2A$

$\therefore\ X_2 = BAX_2$ or $X_1 = ABX_1$

Yields Fig. (a) and (b).

(b) $$X_2 = X_1A + BX_2$$

or $$X_2 = \frac{AX_1}{1 - B}$$



Example 12.108. How many roots of the following polynomial are in the RHS plane and L.H.S plane and on $j\omega$-axis

$$P(s) = S^5 + 2S^4 + 2S^3 + 4S^2 + S + 2$$

Solution :

S^5	1	2	1
S^4	2	4	2
S^3	0	0	
S^2			
S^1			
S^0			

Third row having all elements zero.

∴ Auxiliary equation

$$A(s) = 2s^4 + 4s^2 + 2$$

$$\frac{dA(s)}{ds} = 8s^3 + 8s$$

S^5	1	2	1
S^4	2	4	2
S^3	8	8	
S^2	2	2	
S^1	0		
S^0			

(I)

again (V) row having all zero elements

$$A(s) = 2s^2 + 2$$

$$\frac{dA(s)}{ds} = 4s$$

S^5	1	2	1
S^4	2	4	2
S^3	8	8	
S^2	2	2	
S^1	4		
S^0	2		

Since no sign change in first column therefore

(*a*) no roots lies on right half of *s*-plane

(*b*) since auxiliary equation contains roots on $j\omega$-axis therefore from equation (1) 4 roots lies on $j\omega$-axis

(*c*) Now divide the given polynomial by equation (1)

$$2s^4 + 4s^2 + 2 \;\big)\; s^5 + 2s^4 + 2s^3 + 4s^2 + s + 2 \;\big(\; \frac{1}{2}s + 1$$

$$\underline{s^5 \quad + 2s^3 \quad + s}$$

$$2s^4 + 4s^2 + 2$$

$$\underline{2s^4 + 4s^2 + 2}$$

$$\times$$

$$\therefore s^5 + 2s^4 + 2s^3 + 4s^2 + s + 2 = \left(\frac{1}{2}s + 1\right)(2s^4 + 4s^2 + 2)$$

put $\quad \frac{1}{2}s + 1 = 0 \quad$ or $\quad s = -\frac{1}{2}$

we can say single root lie on L.H.S.

□□□

$$\therefore \text{Modal matrix} = \begin{bmatrix} 1 & 1 & 1 \\ -2 & -5 & -6 \\ 4 & 25 & 36 \end{bmatrix}$$

(ii) From the given circuit

For mesh (1)

$$L_1 \frac{di_1}{dt} + R_1 i_1 + e_T - V_c = 0$$

$$\frac{di_1}{dt} = -\frac{R_1}{L_1} i_1 - \frac{e_T}{L_1} + \frac{1}{L_1} V_c$$

For mesh (2)

$$\frac{di_2}{dt} = -\frac{R_2}{L_1} i_2 + \frac{1}{L_2} V_c$$

KCL at node (2)

$$i_1 + i_2 + C\frac{dV_c}{dt} = 0$$

$$\frac{dV_c}{dt} = -\frac{1}{C} i_1 - \frac{1}{C} i_2$$

$$\begin{bmatrix} \dot{i}_1 \\ \dot{i}_2 \\ \dot{V}_c \end{bmatrix} = \begin{bmatrix} -R_1/L_1 & 0 & 1/L_1 \\ 0 & -\frac{R_2}{L_2} & 1/L_2 \\ -1/C & -1/C & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ V_c \end{bmatrix} + \begin{bmatrix} -1/L_1 \\ 0 \\ 0 \end{bmatrix} e_T \quad \textbf{Ans.}$$

□□□

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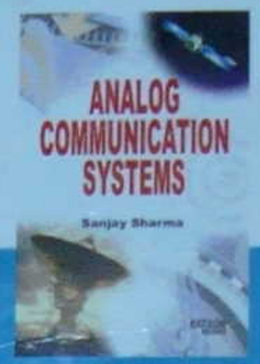

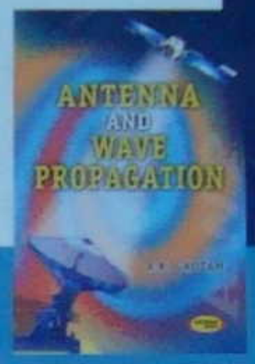

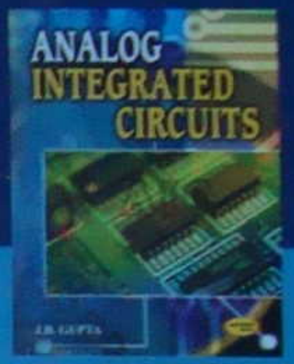

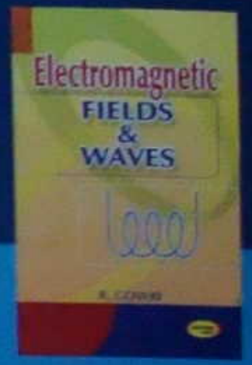

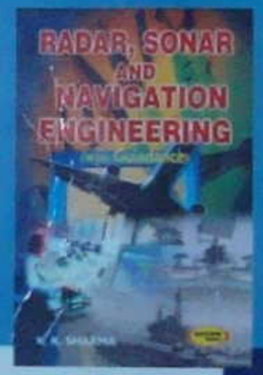

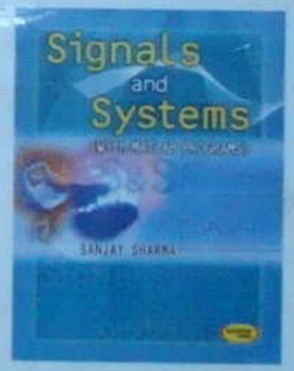

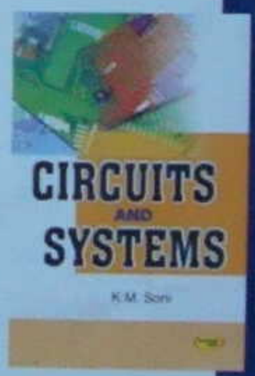

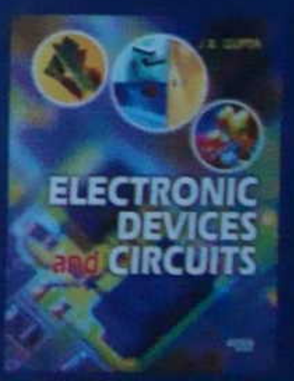

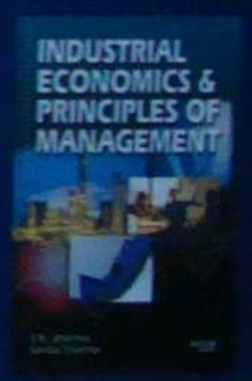

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